

EVERNORTH - ELGIN PHARMACY

For EVERNORTH

Construction Documents
24 April 2026
26136.0000

Project Manual

CORGAN
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Phoenix, AZ 85004
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DOCUMENT 000003

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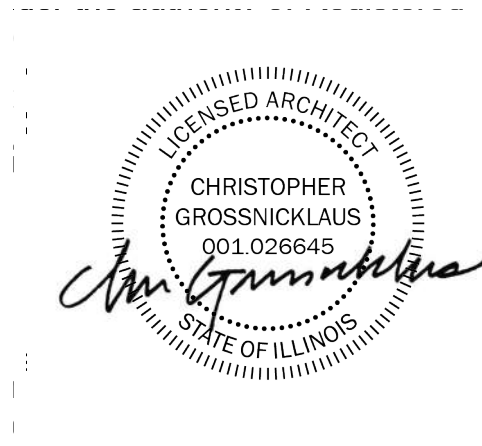
PROJECT:

Name: EVERNORTH - ELGIN PHARMACY
Location: 300 AIRPORT RD.
ELGIN, IL 60123
Project Number: 26136.0000

ARCHITECT OF RECORD

CORGAN ASSOCIATES, INC.
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Responsible for Sections marked with
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Architect of Record 4/24/26

SEALS PAGE (Continued)

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SEALS PAGE (Continued)

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Electrical Engineer of Record

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SUMMARY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Project Information
2. Work covered by Contract Documents
3. Phased construction
4. Work Contract Method.
5. Work by Owner.
6. Work under separate contracts.
7. Owner furnished products.
8. Access to site; Contractor use of site.
9. Work restrictions.
10. Specification and Drawing conventions.
11. Miscellaneous provisions.
12. Owner occupancy.

1.2 PROJECT INFORMATION

A. Project Information:

1. Project Identification: EVERNORTH - ELGIN PHARMACY located at 300 AIRPORT RD. ELGIN, IL 60123.
2. Code Requirements: This project is governed by the Codes as indicated on the Drawings, with local amendments applicable to ELGIN, IL. The Contractor shall comply with all requirements of the Codes and the Local Authority Having Jurisdiction.

1.3 WORK COVERED BY CONTRACT DOCUMENTS

- A.** The Work of Project is defined by the Contract Documents and consists of the following:
- B.** Alteration of an existing Building. Refer to the Drawings for Additional Information.

1.4 PHASED CONSTRUCTION

- A.** The Work may be conducted in phases, as the Contractor determines as appropriate to complete the Work by the established completion dates and as acceptable to the

Owner. The Contractor shall be responsible for all phasing coordination, planning and execution to complete all Work according to the agreed project schedule.

- B. Prior to commencing Work of each phase, submit an updated copy of Contractor's construction schedule showing the sequence, commencement and completion dates for each phase of the Work.

1.5 WORK CONTRACT METHOD

- A. Type of Contract: Single prime contract.
- B. Items noted "NIC" (Not in Contract) are provided by Owner.

1.6 WORK BY OWNER OR ALTERNATE CONTRACTS

- A. Cooperate fully with Owner so work may be carried out smoothly, without interfering with or delaying work under this Contract or work by Owner. Coordinate the work of this Contract with work performed by Owner.
- B. Preceding Work: The Owner reserves the right to perform certain construction operations at Project site. Those operations are scheduled to be substantially complete before work under this Contract begins. Cooperate and coordinate fully with the Owner to permit preceding Work to be completed without disruption.
- C. Concurrent Work: The Owner reserves the right to perform concurrent construction operations and Work at the Project site. Those operations may be conducted simultaneously with Work under this Contract. Cooperate and coordinate fully with the Owner to permit concurrent Work to be completed without disruption.
- D. Work under Separate Contracts: The Owner reserves the right to award separate contracts for portions of the Work. Cooperate fully with separate contractors so Work under those contracts may be carried out smoothly, without interfering with or delaying Work under this Contract or other contracts. Coordinate the Work of this Contract with Work performed under separate contracts.
- E. Purchase (Assign) Contracts: The Owner reserves the right to negotiate Purchase contracts with suppliers of material and equipment to be incorporated into the Work. Owner will assign the Purchase contracts to Contractor. Include costs for purchasing, receiving, handling, storage if required, and installation of material and equipment in the Contract Sum unless otherwise indicated.
 - 1. Contractor's responsibilities are same as if Contractor had negotiated Purchase contracts, including responsibility to renegotiate purchase and to execute final purchasing agreements.

1.7 OWNER FURNISHED PRODUCTS

- A. Owner furnished items shall be as indicated on the Drawings as Owner Furnished Contractor Installed (OFCI) items.

- B. Owner Responsibilities: Owner reserves the right to furnish certain products. The Work includes receiving, unloading, handling, storing, protecting, and installing Owner furnished products and making building services connections.
1. Owner will arrange for and deliver Shop Drawings, Product Data, and Samples to Contractor.
 2. Owner will arrange and pay for delivery of Owner furnished items according to Contractor's Construction Schedule.
 3. Upon delivery, Owner and Contractor will inspect delivered items for damage. Assist Owner with inspection.
 - a. If Owner furnished items are damaged, defective, or missing, Owner will arrange for replacement.
 4. Owner will arrange for manufacturer's field services and for delivery of manufacturer's warranties to Contractor.
 5. Owner will furnish Contractor a potential delivery date for Owner furnished products.
- C. Contractor Responsibilities: Using Owner furnished delivery dates, incorporate delivery dates of Owner furnished items in Contractor's Construction Schedule.
1. Review Shop Drawings, Product Data, and Samples and return to Architect noting discrepancies or anticipated problems in use of product.
 2. Receive, unload, and store Owner furnished items at site.
 3. Upon delivery, inspect delivered items with owner for damage.
 4. Protect Owner furnished items from damage during storage and handling, including damage from exposure to the elements.
 5. If Owner furnished items are damaged as a result of Contractor's operations, repair or replace items.
 6. Install and incorporate Owner furnished items into the Work.

1.8 CONTRACTOR'S USE OF PREMISES AND SITE

- A. Restricted Use: Contractor shall have limited use of Project site for construction operations as indicated on Drawings by the Contract limits and as indicated by requirements of this Section.
- B. Limits on Use of Site: Limit use of site to work in areas and areas within the Contract limits as indicated. Do not disturb portions of site beyond areas in which the work is indicated.
1. Limits on use of existing site and facilities: Confine construction operations to Work in Areas indicated and areas within the Contract Limits.
 2. Driveways, Walkways and Entrances: Keep driveways, all parking areas and garage, loading areas, and entrances, fire exits or lanes, or delivery routes serving premises clear and available to Owner, Owner's employees, tenants, the public and emergency vehicles. Do not use these areas for parking or for storage of materials.
 - a. Schedule deliveries to minimize use of driveways and entrances by construction operations.

- b. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.
 - 1) Schedule of Deliveries: Arrange through Building Management.
 - c. Access to Work: Arrange through Building Management and the Owner.
 - d. Maintain existing building in a weathertight condition throughout construction period. Repair damage caused by construction operations. Protect building and its occupants during construction period.
- C. Existing Utility Interruptions: Do not interrupt utilities serving facilities unless permitted under the following conditions and after providing temporary utility services according to requirements indicated:
 - 1. Notify Architect and Owner not less than two days in advance of proposed utility interruptions. Include length of service outages and intended scope of Work in all such notifications.
 - 2. Obtain Owner's written permission before proceeding with utility interruptions.
- D. Access routes for delivery of materials and equipment shall be as indicated on drawings by Contractor. Do not use access routes other than those indicated without permission of Owner and Contractor.
- E. Assume full responsibility for protection and safekeeping of products stored on project site, including but not limited to all salvaged items. Store materials and products in areas indicated for staging and protect from damage.
- F. Move stored products, under Contractor's control, which interfere with operations of Owner or as required by Architect. Do not unnecessarily encumber project site with materials and equipment.
- G. Store all salvaged items and protect from damage and weather conditions.
- H. Staging and material storage shall be limited to areas indicated. Obtain specific permission from Owner for use of other areas for storage and staging.
- I. Do not overload existing or new structures with weight that would compromise safety. Verify design loads for structure if necessary prior to loading structure.
- J. Obtain and pay for use of additional secured storage or Work areas needed for operations, material storage for new and salvaged items.
- K. Protect existing lawns, landscaping, sidewalks, pavements, curbs and utilities subject to damage by Work under this Contract. Repair or replace existing Work damaged by Contractor. Replace existing lawns damaged by Contractor's activities with hydromulched lawn seed to provide full stand of replacement grass.
- L. Parking areas for Contractor's personnel shall be as indicated by Owner.

1.9 COORDINATION WITH OCCUPANTS

- A. Partial Owner Occupancy: Owner will occupy the premises during construction period, with the exception of areas under construction. Cooperate with Owner during construction operations to minimize conflicts and facilitate Owner usage. Perform the Work so as not to interfere with Owner's operations. Maintain existing exits unless otherwise indicated.
 - 1. Maintain access to existing walkways, corridors, and other adjacent occupied or used facilities. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from Owner and authorities having jurisdiction.
 - 2. Provide not less than 72 hours' notice to Owner of activities that will affect Owner's operations.

1.10 PARTIAL OWNER OCCUPANCY

- A. Schedule construction operations for completion of portions of Work for any Owner's occupancy prior to Substantial Completion of entire Work.
- B. Contractor agrees to cooperate, coordinate and allow for occupancy of portion of Project by Owner prior to Substantial Completion of entire Project, provided Owner:
 - 1. Secures written consent of Contractor except when, in opinion of Architect, Contractor is chargeable with unwarranted delay in final completion of uncompleted items or other Contract requirements.
 - 2. Secures endorsement from insurance carrier and consent of surety permitting partial occupancy of Project or use of Project during remaining period of construction.
- C. Use and occupancy prior to Substantial Completion of entire Project does not relieve Contractor of responsibility to maintain specified insurance coverage on 100 percent basis for benefit of Owner, Contractor and subcontractors until Project is complete and accepted by Owner.
- D. After Owner occupancy, allow following:
 - 1. Access for Owner's personnel.
 - 2. Operation of heating, ventilating, air-conditioning and electrical systems.
- E. During partial occupancy, mutually acceptable arrangements shall be negotiated between Owner and Contractor regarding warranties and insurance requirements respecting portion of Work affected by partial occupancy and regarding operation and cost of building services so that costs attributable to partial occupancy shall be borne by Owner and costs attributable to performance of Work shall be borne by Contractor.
- F. Prior to occupancy, execute Certificate of Substantial Completion, designating areas to be occupied by Owner.

1.11 WORK SEQUENCE

- A. Construct Work in phases to accommodate Work to be completed on schedule and as agree with Owner; coordinate construction schedule and operations with Owner's representative.
- B. Construct Work to minimize public inconvenience. Do not close off public use travel paths until completion of one stage of construction will provide alternative usage.
- C. Owner may require certain Work to be performed after normal Working hours or on holidays or weekends. Negotiate all such Work with Owner prior to final Agreement.

1.12 WORK RESTRICTIONS

- A. Work Restrictions: Comply with restrictions on construction operations.
 - 1. Comply with limitations on use of public streets and with other requirements of authorities having jurisdiction.
- B. On-Site Work Hours: Limit work in the existing building to normal business working hours of 7:00 a.m. to 6:00 p.m., Monday through Friday, unless otherwise indicated or required by the Authority Having Jurisdiction.
 - 1. Weekend Hours: As permitted by Owner written authorization and in compliance with Local Ordinances.
 - 2. Early Morning Hours: Comply with time restrictions set forth by the Local Authority Having Jurisdiction for restrictions.
 - 3. Hours for Utility Shutdowns: Only during hours agreed to in writing by the Owner 2 weeks in advance.
 - 4. Hours for Core Drilling or other noisy activity: During times unrestricted by the Owner and in accordance with Local Ordinances.
- C. Noise, Vibration, and Odors: Coordinate operations that may result in high levels of noise and vibration, odors, or other disruption to Owner occupancy with Owner.
 - 1. Notify Owner not less than two days in advance of proposed disruptive operations.
 - 2. Obtain Owner's written permission before proceeding with disruptive operations.
- D. Nonsmoking Building: Smoking is not permitted within the building or within 25 feet of entrances, operable windows, or outdoor air intakes.
- E. Restricted Substances and Firearms: Use of tobacco products and other controlled substances and firearms on Project site is not permitted.
- F. Employee Identification: Owner may require identification tags for Contractor personnel Working on site. If so required, insure all personnel onsite use identification tags at all times.
- G. Comply with all Work Restrictions imposed by the Local Authority.
- H. Abide by all road cleaning and other requirements imposed by the Local Authority.

1.13 SPECIFICATION AND DRAWING CONVENTIONS

- A. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:
 - 1. Imperative mood and streamlined language are generally used in the Specifications. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.
 - 2. Specification requirements are to be performed by Contractor unless specifically stated otherwise.
- B. Division 1 General Requirements: Requirements of Sections in Division 1 apply to the Work of the specification sections.
- C. Drawing Coordination: Requirements for materials and products identified on Drawings are described in detail in the Specifications. One or more of the following are used on Drawings to identify materials and products:
 - 1. Terminology: Materials and products are identified by the typical generic terms used in the individual Specifications Sections.
 - 2. Abbreviations: Materials and products are identified by abbreviations and scheduled on Drawings.

PART 2 - PRODUCTS

2.1 HAZARDOUS MATERIALS - ASBESTOS

- A. Owner requires that products and materials used meet the following requirements:
 - 1. Contain not more than 1.0% of any kind or combination of asbestos, as determined by the Environmental Protection Agency (EPA) recommended test methods as listed in EPA/600R-93/116 July 1993 "Method for the Determination of Asbestos in Bulk Building Materials". This means one material component or a structure or layer of a material sample. Composite sample analysis is not allowed.
- B. Contractor, by executing this Contract, accepts responsibility for prohibiting materials or products containing asbestos from being used on this Project.
- C. Should Contractor furnish, use, or install products or materials containing asbestos, knowingly or otherwise, he has not met the requirements of this Contract.
- D. Asbestos Survey: Provide an asbestos survey, meeting the same criteria as required for new building permits, by an inspector licensed by the State of IL.
 - 1. Preliminary Survey: Contractor shall submit preliminary survey of materials installed at time of Substantial Completion with request for Substantial Completion to the Architect.

2. Final Survey: Contractor shall submit revised survey reflecting materials install after Substantial Completion to the Architect for transmittal to the Owner prior to release of Retainage.

2.2 HAZARDOUS MATERIALS – MOLD

- A. If mold is present on the inside of the building; determine source of moisture and correct prior to performing the following:
 1. Nonporous Surfaces: Mold growing on steel, aluminum or other nonporous materials may be cleaned using methods recommended by an Environmental Inspector licensed by the State of IL.
 2. Porous Surfaces: Entire mold contaminated surface must be removed and disposed of using methods recommended by an Environmental Inspector licensed by the State of IL.
 3. Following elimination of mold provide a report prepared by an Environmental Inspector licensed by the State of IL that the building is free from mold contamination.

PART 3 - EXECUTION
Not used.

END OF SECTION

SECTION 012200

UNIT PRICES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for unit prices.

1.2 DEFINITIONS

- A. Unit price is an amount incorporated into the Agreement, applicable during the duration of the Work as a price per unit of measurement for materials, equipment, or services, or a portion of the Work, added to or deducted from the Contract Sum by appropriate modification, if the scope of Work or estimated quantities of Work required by the Contract Documents are increased or decreased.

1.3 PROCEDURES

- A. Unit prices include all necessary material, plus cost for delivery, installation, insurance, applicable taxes, overhead, and profit.
- B. Measurement and Payment: See individual Specification Sections for work that requires establishment of unit prices. Methods of measurement and payment for unit prices are specified in those Sections.
- C. Owner reserves the right to reject Contractor's measurement of work-in-place that involves use of established unit prices and to have this work measured, at Owner's expense, by an independent surveyor acceptable to Contractor.
- D. List of Unit Prices: A schedule of unit prices is included in Part 3. Specification Sections referenced in the schedule contain requirements for materials described under each unit price.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

3.1 SCHEDULE OF UNIT PRICES

A. Unit Price No. 1: Cutting and patching of concrete floor slabs.

1. Description: Cutting of new or existing concrete floor slabs up to 6 inches thick, removal and excavation as required, and subsequent backfill, compaction, and patching of concrete in accordance with Section 03 30 53 "Miscellaneous Cast-In-Place Concrete" not otherwise indicated in the Contract Documents.
2. Unit of Measurement: Square feet of concrete removed and replaced.

B. Unit Price No. 2: Floor flatness and levelness of existing floor slabs.

1. Description:
 - a. Application of cementitious underlayment to new or existing floor slabs as required to achieve the floor flatness and floor levelness values specified in Section 02 21 35 "Floor Flatness and Levelness Surveys."
 - b. Preparation and installation of cementitious underlayment shall be completed in accordance with requirements of Section 03 54 16 "Hydraulic Cement Underlayment."
2. Unit of Measurement: Square feet of products installed.
 - a. For unit pricing purposes, assume an average application thickness of 1/4 inch.

C. Unit Price No. 3: Topical moisture vapor emission and alkalinity control of concrete floor slabs and cementitious underlayment installation in tile carpeted areas, wood flooring product areas, and resilient flooring product areas.

1. Description: Installation of topical moisture vapor emission and alkalinity control products and subsequent cementitious underlayment products to new or existing concrete floor slabs where the moisture vapor emissions and/or the alkalinity of the concrete floor slab exceeds the limits recommended by the manufacturer of the finished flooring product(s) to be installed. Preparation and installation of the topical moisture vapor emission and alkalinity control product and cementitious underlayment are to be completed in accordance with requirements of Section 09 61 05 "Moisture Vapor Emission and Alkalinity Control of Concrete Floor Slabs."
2. Unit of Measurement: Square feet of products installed.

D. Unit Price No. 4: Miscellaneous and structural steel.

1. Description: Miscellaneous lintels and other supports not otherwise indicated in the Contract Documents, in accordance with Section 05 50 00 "Metal Fabrications."
2. Unit of Measurement: Cost in place of pounds of fabricated steel as indicated on itemized invoice of steel supplier and verified by the Architect.

END OF SECTION

SECTION 012500

SUBSTITUTION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Administrative and procedural requirements for substitutions.

1.2 DEFINITIONS

A. Substitutions: Changes in products, materials, equipment, and methods of construction from those required by the Contract Documents, and proposed by Contractor, prior to award of Contract.

1. Substitutions for Cause: Changes proposed by Contractor that are required due to changed Project conditions, such as unavailability of product, regulatory changes, or unavailability of required warranty terms.
2. Substitutions for Convenience: Changes proposed by Contractor or Owner that are not required in order to meet other Project requirements but may offer advantage to Contractor or Owner.

B. Substitutions requested by Bidders during the bidding period, and accepted prior to award of Contract, are considered as included in the Contract Documents, and are not subject to requirements specified in this Section.

1.3 SUBMITTALS

A. Substitution Requests: Architect will consider only written requests from Contractor for any substitution of products in place of those specified, using form provided at the end of this Section.

1. Submit three copies of each request for consideration. Identify product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.
2. Documentation: Show compliance with requirements for substitutions and the following, as applicable:
 - a. Statement indicating why specified product or fabrication or installation cannot be provided, if applicable.
 - b. Coordination information, including a list of changes or revisions needed to other parts of the Work and to construction performed by Owner and separate contractors, that will be necessary to accommodate proposed substitution.

- c. Detailed comparison of significant qualities of proposed substitution with those of the Work specified. Include annotated copy of applicable Specification Section. Significant qualities may include attributes such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific features and requirements indicated. Indicate deviations, if any, from the Work specified.
 - d. Product Data, including drawings and descriptions of products and fabrication and installation procedures.
 - e. Samples, where applicable or requested.
 - f. Certificates and qualification data, where applicable or requested.
 - g. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners.
 - h. Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.
 - i. Research reports evidencing compliance with building code in effect for Project, from ICC-ES or other applicable code organization.
 - j. Detailed comparison of Contractor's construction schedule using proposed substitution with products specified for the Work, including effect on the overall Contract Time. If specified product or method of construction cannot be provided within the Contract Time, include letter from manufacturer, on manufacturer's letterhead, stating date of receipt of purchase order, lack of availability, or delays in delivery.
 - k. Cost information, including a proposal of change, if any, in the Contract Sum.
 - l. Contractor's certification that proposed substitution complies with requirements in the Contract Documents except as indicated in substitution request, is compatible with related materials, and is appropriate for applications indicated.
 - m. Contractor's waiver of rights to additional payment or time that may subsequently become necessary because of failure of proposed substitution to produce indicated results.
3. Architect's Action: If necessary, Architect will request additional information or documentation for evaluation within seven days of receipt of a request for substitution. Architect will notify Contractor of acceptance or rejection of proposed substitution within 15 days of receipt of request, or seven days of receipt of additional information or documentation, whichever is later.
- a. Forms of Acceptance: Change Order, Construction Change Directive, or Architect's Supplemental Instructions for minor changes in the Work.
 - b. Use product specified if Architect does not issue a decision on use of a proposed substitution within time allocated.

1.4 QUALITY ASSURANCE

- A. Compatibility of Substitutions: Investigate and document compatibility of proposed substitution with related products and materials. Engage a qualified testing agency to perform compatibility tests recommended by manufacturers.

1.5 PROCEDURES

- A. Coordination: Revise or adjust affected work as necessary to integrate work of the approved substitutions.

1.6 CONTRACTOR'S REPRESENTATION

- A. In making formal request for substitution Contractor represents that:
 - 1. He has investigated proposed product and has determined that it is equal to or superior in respects to that specified.
 - 2. He will provide same warranties or bonds for substitution as for product specified.
 - 3. He will coordinate installation of accepted substitution into Work and will make such changes as may be required for Work to be complete.
 - 4. He waives claims for additional costs caused by substitution which may subsequently become apparent.
 - 5. Cost data is complete and includes related costs under his Contract, but not costs under separate contracts.

1.7 ARCHITECT DUTIES

- A. Architect will determine acceptability of proposed substitutions.
- B. Review Contractor's requests for substitutions with reasonable promptness.
- C. Notify Contractor, in writing, of decision to accept or reject requested substitution.

PART 2 - PRODUCTS

2.1 SUBSTITUTIONS, GENERAL

- A. Properties including, but not limited to following, will be considered as applicable:
 - 1. Physical dimension requirements to satisfy space limitations.
 - 2. Static and dynamic weight limitations, structural properties.
 - 3. Audible noise levels.
 - 4. Vibration generation.
 - 5. Interchangeability of parts or components.
 - 6. Accessibility for maintenance, possible removal, or replacement.
 - 7. Colors, textures and compatibility with other materials, products, assemblies, and components.
 - 8. Equipment capacities and performance characteristics.
- B. Substitutions will not be considered for acceptance when:
 - 1. They are indicated or implied on Shop Drawings, Product Data Submittals, or Requests for Information (RFIs).
 - 2. They are requested directly by subcontractor or supplier.

3. Acceptance would require substantial revisions to Contract Documents or Contract time.
 4. Additional cost to Owner is involved.
- C. Substitute products shall not be ordered or installed without written acceptance of Architect. If proposed substitution is not accepted, provide specified product or materials.
- D. Burden of proof of merits of proposed substitute is upon proposer. Architect's decision of acceptance or rejection of proposed substitution will be final.
- E. Review of Architect, acceptance or failure to take exceptions to substitutions or other review documents, shall not relieve Contractor of his responsibility for compliance with performance or other requirements of Contract Documents.
- F. Contractor shall be responsible for and pay for any expenses incurred by Owner or Architect for changes to Contract Documents ultimately required by accepted Contractor requests for substitutions.

2.2 SUBSTITUTIONS

- A. Substitutions for Cause: Submit requests for substitution immediately on discovery of need for change, but not later than 15 days prior to time required for preparation and review of related submittals.
1. Conditions: Architect will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Architect will return requests without action, except to record noncompliance with these requirements:
 - a. Requested substitution is consistent with the Contract Documents and will produce indicated results.
 - b. Substitution request is fully documented and properly submitted.
 - c. Requested substitution will not adversely affect Contractor's construction schedule.
 - d. Requested substitution has received necessary approvals of authorities having jurisdiction.
 - e. Requested substitution is compatible with other portions of the Work.
 - f. Requested substitution has been coordinated with other portions of the Work.
 - g. Requested substitution provides specified warranty.
 - h. If requested substitution involves more than one contractor, requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all contractors involved.
- B. Substitutions for Convenience: Not allowed, unless otherwise indicated.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SUBSTITUTION REQUEST FORM

To Architect: _____

Project Name: _____

SPECIFIED ITEM: _____

Section

Page

Paragraph

Description

The undersigned Contractor requests consideration of the following:

PROPOSED SUBSTITUTION: _____

1. Attached data includes product description, specifications, drawings, photographs, performance and test data adequate for evaluation of the request; applicable portions of the data are clearly identified, both on the proposed substitution and the original specified product.
2. Attached data also includes description of changes to Contract Documents, which proposed Substitution will require for its proper installation.

The undersigned Contractor states that the following paragraphs, unless modified on attachments, are correct.

1. The proposed substitution does not affect dimensions on Drawings.
2. The undersigned Contractor will pay for changes to the building design, including architectural or engineering design, detailing, and construction costs caused by the requested substitution.
3. The proposed substitution will have no adverse effects on other trades, the construction schedule, or specified warranty requirements.
4. Maintenance and service parts will be locally available for the proposed substitution.

The Contractor further states that the function, appearance, and quality of the Proposed Substitution are equivalent or superior to the Specified Item. The Contractor further warrants that the intent of specification Section 01 25 00, Article 1.6 has been met.

1. Cost Reduction to the Owner: \$ _____

ACCEPTANCES:

1. Contractor Acceptance: _____ Date: _____ Representing: _____
2. Owner Acceptance: _____ Date: _____ Representing _____
3. Architect Acceptance: _____ Date: _____ Representing
Corgan Associates, Inc.

Accepted as Noted

Not Accepted

Received too late

Resubmit with complete information

SECTION 012600

CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for handling and processing Contract modifications.

1.2 MINOR CHANGES IN THE WORK

- A. Architect will issue supplemental instructions authorizing minor changes in the Work, not involving adjustment to the Contract Sum or the Contract Time, on AIA Document G710, in a Field Memorandum, or via web-based Project management software.
 - 1. Work Change Proposal Requests issued by Architect are not instructions either to stop work in progress or to execute the proposed change.

1.3 PROPOSAL REQUESTS

- A. Owner-Initiated Proposal Requests: Architect will issue a detailed description of proposed changes in the Work that may require adjustment to the Contract Sum or the Contract Time. If necessary, the description will include supplemental or revised Drawings and Specifications.
 - 1. Work Change Proposal Requests issued by Architect are not instructions either to stop work in progress or to execute the proposed change.
 - 2. Within time specified in Proposal Request after receipt of Proposal Request, submit a quotation estimating cost adjustments to the Contract Sum and the Contract Time necessary to execute the change.
 - a. Include a list of quantities of products required or eliminated and unit costs, with total amount of purchases and credits to be made. If requested, furnish survey data to substantiate quantities.
 - b. Indicate applicable taxes, delivery charges, equipment rental, and amounts of trade discounts.
 - c. Include costs of labor and supervision directly attributable to the change.
 - d. Include an updated Contractor's construction schedule that indicates the effect of the change, including, but not limited to, changes in activity duration, start and finish times, and activity relationship. Use available total float before requesting an extension of the Contract Time.

- B. Contractor-Initiated Proposals: If latent or changed conditions require modifications to the Contract, Contractor may initiate a claim by submitting a request for a change to Architect.
1. Include a statement outlining reasons for the change and the effect of the change on the Work. Provide a complete description of the proposed change. Indicate the effect of the proposed change on the Contract Sum and the Contract Time.
 2. Include a list of quantities of products required or eliminated and unit costs, with total amount of purchases and credits to be made. If requested, furnish survey data to substantiate quantities.
 3. Indicate applicable taxes, delivery charges, equipment rental, and amounts of trade discounts.
 4. Include costs of labor and supervision directly attributable to the change.
 5. Include an updated Contractor's construction schedule that indicates the effect of the change, including, but not limited to, changes in activity duration, start and finish times, and activity relationship. Use available total float before requesting an extension of the Contract Time.
 6. Comply with requirements in Section 01 25 00 "Substitution Procedures" if the proposed change requires substitution of one product or system for product or system specified.

1.4 CHANGE ORDER PROCEDURES

- A. On Owner's approval of a Work Change Proposal Request, Contractor will issue a Change Order for signatures of Owner and Contractor on AIA Document G701.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 012900
MEASUREMENT AND PAYMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Applications for Payment to Architect in accordance with schedule established by Conditions of Contract, and Agreement Between Owner and Contractor.
2. Schedule of Values allocated to various portions of Work in accordance with schedule established in this Section. Upon request of Architect, support values with data which substantiate correctness.
3. Unit Prices.
4. Change Order Procedures.

1.2 SCHEDULE OF VALUES

A. Preparation:

1. Coordinate preparation of Schedule of Values with preparation of Construction Progress Schedule.
2. Correlate line items in Schedule of Values with other required administrative schedules and forms, including following:
 - a. Construction Progress Schedule, in CPM or bar chart format.
 - b. Application for Payment form.
 - c. List of subcontractors.
 - d. Schedule of alternates.
 - e. List of products.
 - f. List of principal suppliers and fabricators.
 - g. Schedule of submittals.
3. Submit initial Schedule of Values to Architect within 14 days of Notice to Proceed.
4. Submit revised Schedule of Values at earliest possible date, but no later than minimum of 10 days before scheduled date of Initial Application for Payment.

B. Form and Content:

1. Arrange Schedule of Values on 8-1/2" x 11" tabular form on white paper in format acceptable to Architect.
2. Use bar chart or CPM chart as guide to establish format; compile information based on completed tasks.

3. Project Identification:
 - a. Project name and location.
 - b. Name of Architect.
 - c. Project number.
 - d. Contractor's name and address.
 - e. Date of submittal.
 - f. Owner's contract designation.
4. Provide columns to indicate following for each item listed:
 - a. Generic name.
 - b. Specific task description.
 - c. Change Orders numbers that have affected value.
 - d. Dollar value.
 - e. Percentage of Contract Sum to nearest one-hundredth percent, adjusted to total 100 percent.
5. Follow table of contents of Project Manual as format for listing component items; identify each line item with number and title of respective major section of specifications.
6. Provide breakdown of Contract Sum in sufficient detail to facilitate continued evaluation of Applications for Payment and progress reports.
7. Break principal subcontract amounts down into several line items by completed task in various locations.
8. For each major line item, list sub-values of major products or operations under item.
9. Round amounts off to nearest whole dollar, total of listed values shall equal total Contract Sum.
10. For each part of Work where Application for Payment includes materials or equipment purchased, fabricated, and stored, but not yet installed, include separate line items on Schedule of Values for initial cost of materials, for each subsequent stage of completion, and for total installed value of that part of Work.

C. Margins of Cost:

1. Show line items for indirect costs, and margins on actual costs, only to extent that such items will be listed individually in Applications for Payment.
2. Each item in Schedule of Values and Application for Payment shall be complete, including its total cost and proportionate share of general overhead and profit margin.
3. At Contractor's option, temporary facilities and other major cost items that are not direct costs of actual work-in-place may be shown as separate line items in Schedule of Values.
4. Furnish line-item cost for each of following general cost items:
 - a. Bonds.
 - b. Insurance.
 - c. Field supervision and layout.
 - d. Temporary facilities and controls.
 - e. Testing.

D. Resubmittal:

1. After review by Architect, revise and resubmit schedule as necessary.
2. Resubmit revised schedule monthly in same manner.

E. Subschedule of Unit Material Values:

1. Submit subschedule of unit costs and quantities for following:
 - a. Products specified under Unit Costs specified in Section 01 21 00.
 - b. Products on which progress payments will be requested for stored products.
2. Form and Content:
 - a. Form of submittal shall parallel that of Schedule of Values, with each item identified same as line item in Schedule of Values.
 - b. Unit quantity for bulk materials shall include an allowance for normal waste.
3. Unit values for materials shall be broken down as follows:
 - a. Cost of material, delivered and unloaded at Project Site, with taxes paid.
 - b. Installation costs, including overhead and profit.
4. Installed unit value multiplied by quantity listed shall equal cost of that item in Schedule of Values.
5. After review by Architect, revise and resubmit subschedules as necessary.
6. Resubmit revised subschedules monthly in same manner.

1.3 APPLICATIONS FOR PAYMENT

- A. Progress payments shall be made as Work proceeds at intervals stated in Contract.
- B. Work covered by Progress Payments shall, at time of payment, become property of Owner.
- C. Form of Application for Payment will be notarized AIA Document G702 - Application and Certification for Payment, supported by AIA Document G703 - Continuation Sheet, submitted in quadruplicate.
- D. Contractor shall submit to Architect within 15 days of execution of Owner -Contractor Agreement proposed sample of Lien Waiver and Bills Paid Affidavit forms for review and acceptance by Architect for use on this Contract.
- E. Conditions governing regular schedule for applications, payment, and retainage are as stated in Contract.
- F. Monthly Applications for Payment shall include Waivers of Liens for Work included in previous months' Application for Payment. Waiver of Liens for subcontractors and materialmen shall be total amount paid prior to previous month's Application for Payment.

- G. With each Application for Payment, Contractor shall certify that such Application for Payment represents just estimate of cost reimbursable to Contractor under terms of Contract, and shall also certify that there are no Mechanics' or Materialmens' Liens outstanding at date of Application for Payment, that due and payable bills with respect to Work have been paid to date or shall be paid from proceeds of Application for Payment, and that there is no known basis for filing of Mechanics' or Materialmens' Liens against surety in connection with Work, and that Waivers and Bills Paid Affidavit forms from subcontractors and materialmen have been, or will be, obtained in form specified in Contract.

1.4 UNIT PRICES

- A. Unit price is defined as amount stated in construction Contract as price for each unit of measurement for materials or services.

1.5 CHANGE ORDER PROCEDURES

- A. Submit to Architect within 10 days of execution of Owner-Contractor Agreement name of individual authorized to accept changes on behalf of Contractor and shall be responsible for informing others in Contractor's employ of changes in Work.
- B. Change Order forms will be furnished and issued by Architect.
- C. Contractor Documentation of Changes:
 - 1. Maintain detailed records of Work done on accounting basis acceptable to Architect and Owner.
 - 2. Provide full information required for evaluation of proposed changes.
 - 3. Document each quotation for a change in cost or time with sufficient data to allow evaluation of quotation.
 - 4. On request, provide additional data to support computations:
 - a. Quantities of products, labor and equipment.
 - b. Insurance and bonds.
 - c. Overhead and profit.
 - d. Justification for change in Contract Time.
 - e. Credit for deletions from Contract, similarly documented.
 - 5. Support each request for additional costs, and for Work proposed on time and material basis, with description of products, equipment, cost of labor and subcontracts, completely documented.
 - 6. Computation for changes in Work will be computed in one of manners described in Conditions of the Contract.
- D. Initiation of Changes:
 - 1. Architect may submit Proposal Request which includes detailed description of change with supplementary or revised Drawings and Specifications.

2. Contractor may initiate proposed change by submittal of request to Architect describing proposed change with statement of reason for change, and proposed effect on Contract Sum and Contract Time with full documentation, and statement of effect on Work of separate contractors.
3. Document requested substitutions in accordance with Section 01 25 00.
4. Submission of such requests and receipt of same by Architect does not mean acceptance or approval of proposed change.

E. Authorization:

1. Owner may request, through Architect, Construction Change Directive, in writing, instructing Contractor to proceed with changes in Work, for subsequent inclusion in Change Order that is pending.
2. Directive will propose basis for necessary adjustments, to Contract Sum or Time.
3. Changes that affect Contract Sum or Contract Time will require Change Order signed by Owner and Architect.
4. Contractor's signature indicates agreement.
5. Orders, written or oral, by Owner through Architect or by Architect shall be treated as Change Order only if Contractor gives Owner proper written notice as described in Conditions of Contract.
6. Promptly execute change in Work only upon receipt of approved Change Order or Owner's written Construction Change Directive.

F. Execution:

1. Architect will issue Change Orders for signatures of parties as provided in Conditions of Contract.
2. Promptly revise Schedule of Values and Application for Payment forms to record each authorized Change Order as separate line item and adjust Contract Sum as shown on Change Order.
3. Promptly revise Progress Schedules to reflect change in Contract Time, revise subschedules to adjust times for other items of Work affected by Change and resubmit Schedule.
4. Promptly enter Changes in Project Record Documents.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 013100
PROJECT COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Administrative provisions for coordinating construction operations on Project including, but not limited to, the following:
 - a. General coordination procedures.
 - b. Coordination meetings.
 - c. Digital project management procedures.
 - d. Coordination of Submittals.
 - e. Requests for Information (RFIs).
 - f. Coordination of space and installation.
 - g. Coordination of finishes.
 - h. Coordination of Contract closeout.

1.2 GENERAL COORDINATION PROCEDURES

A. Coordination: Coordinate construction operations included in different Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations included in different Sections that depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain the best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
2. Coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair.
3. Make adequate provisions to accommodate items scheduled for later installation.

B. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to, the following:

1. Preparation of Contractor's construction schedule.
2. Preparation of the schedule of values.
3. Installation and removal of temporary facilities and controls.
4. Delivery and processing of submittals.
5. Progress meetings.
6. Preinstallation conferences.

7. Project closeout activities.
 8. Startup and adjustment of systems.
- C. Communication Procedures: Communications to Owner from Contractor regarding Contract requirements shall be through Architect.

1.3 COORDINATION MEETINGS

- A. In addition to Progress Meetings scheduled in Section 01 31 19, Contractor shall hold coordination meetings and pre-installation meetings with Contractor's personnel, subcontractors, materialmen, and Architect, as necessary, to assure coordination of different trades and disciplines.
- B. Contractor shall schedule coordination and pre-installation meetings with Owner, Architect, Contractor and supplier to discuss hardware installation. Meeting shall initially be called within 30 days of Contract award.
- C. When necessary, prepare memoranda for distribution to each party outlining special procedures required for coordination. Include such items as required notices, reports, and attendance at meetings. Prepare similar memoranda for Owner and separate contractors when coordination of their work is required.
- D. Coordinate scheduling and timing of required administrative procedures with other construction activities to avoid conflicts and ensure orderly progress of Work. Such administrative activities include, but are not limited to, the following:
1. Preparation of schedules.
 2. Installation and removal of temporary facilities.
 3. Delivery and processing of submittals.
 4. Progress meetings.
 5. Project close-out procedures.

1.4 DIGITAL PROJECT MANAGEMENT PROCEDURES

- A. Use of Architect's Digital Data Files: Digital data files of Architect's BIM model are available from the Architect for the express purpose of preparing coordination drawings, Shop Drawings, and Project record drawings.
1. Digital Data Software Program: The Drawings are available in Revit RVT format, operating in Microsoft Windows operating system.
 2. Architect makes no representations as to the accuracy or completeness of digital data files as they relate to the Drawings.
 3. Execution of the Architect's Release Form is required to obtain the digital data files of the Drawings. Provision of the digital data files is subject to both the terms described herein and on the Release Form. It is offered at the Architect's sole discretion and may be withheld by the Architect for reasons deemed appropriate.
 - a. A copy of the Architect's Release Form is available upon request.

- B. Web-Based Project Software: Use Architect's web-based Project software site for purposes of hosting and managing Project communication and documentation until Final Completion.
1. Web-based Project software site includes, at a minimum, the following features:
 - a. Compilation of Project data, including Contractor, subcontractors, Architect, architect's consultants, Owner, and other entities involved in Project. Include names of individuals and contact information.
 - b. Access control for each entity for each workflow process, to determine entity's digital rights to create, modify, view, and print documents.
 - c. Document workflow planning, allowing customization of workflow between project entities.
 - d. Creation, logging, tracking, and notification for Project communications required in other Specification Sections, including, but not limited to, RFIs, submittals, Minor Changes in the Work, Construction Change Directives, and Change Orders.
 - e. Track status of each Project communication in real time, and log time and date when responses are provided.
 - f. Procedures for handling PDFs or similar file formats, allowing markups by each entity. Provide security features to lock markups against changes once submitted.
 - g. Processing and tracking of payment applications.
 - h. Processing and tracking of contract modifications.
 - i. Creating and distributing meeting minutes.
 - j. Document management for Drawings, Specifications, and coordination drawings, including revision control.
 - k. Management of construction progress photographs.
 - l. Mobile device compatibility, including smartphones and tablets.
 2. Provide up to twelve web-based Project software user licenses for use of Owner and Owner's Commissioning Authority. Provide eight hours of software training at Architect's office for web-based Project software users.
 3. At completion of Project, provide digital archive in format that is readable by common desktop software applications in format acceptable to Architect. Provide data in locked format to prevent further changes.
 4. Provide the following web-based Project software package under their current published licensing agreements:
 - a. Newforma, Inc.
- C. PDF Document Preparation: Where PDFs are required to be submitted to Architect, prepare as follows:
1. Assemble complete submittal package into a single indexed file incorporating submittal requirements of a single Specification Section and transmittal form with links enabling navigation to each item.
 2. Name file with submittal number or other unique identifier, including revision identifier.
 3. Certifications: Where digitally submitted certificates and certifications are required, provide a digital signature with digital certificate on where indicated.

1.5 COORDINATION OF SUBMITTALS

- A. Schedule and coordinate submittals.
- B. Coordinate Work of various trades having interdependent responsibilities for installing, connecting to, and placing in service, such equipment.
- C. Coordinate requests for substitutions to assure compatibility of space, of operating elements, effect on Work of other trades, and on Work scheduled for early completion.

1.6 REQUESTS FOR INFORMATION (RFIS)

- A. General: Immediately on discovery of the need for additional information or interpretation of the Contract Documents, Contractor shall prepare and submit an RFI in the form specified.
 - 1. Architect will return RFIs submitted to Architect by other entities controlled by Contractor with no response.
 - 2. Coordinate and submit RFIs in a prompt manner so as to avoid delays in Contractor's work or work of subcontractors.
- B. Content of the RFI: Include a detailed, legible description of item needing information or interpretation and the following:
 - 1. Project name.
 - 2. Project number.
 - 3. Date.
 - 4. Name of Contractor.
 - 5. Name of Architect.
 - 6. RFI number, numbered sequentially.
 - 7. RFI subject.
 - 8. Specification Section number and title and related paragraphs, as appropriate.
 - 9. Drawing number and detail references, as appropriate.
 - 10. Field dimensions and conditions, as appropriate.
 - 11. Contractor's suggested resolution. If Contractor's solution(s) impacts the Contract Time or the Contract Sum, Contractor shall state impact in the RFI.
 - 12. Contractor's signature.
 - 13. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.
 - 14. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.
- C. RFI Forms: Software-generated form with substantially the same content as indicated above, acceptable to Architect.

- D. Architect's Action: Architect and Construction Manager will review each RFI, determine action required, and respond. Allow five working days for Architect's response for each RFI. RFIs received by Architect after 1:00 p.m. will be considered as received the following working day.
1. The following RFIs will be returned without action:
 - a. Requests for approval of submittals.
 - b. Requests for approval of substitutions.
 - c. Requests for coordination information already indicated in the Contract Documents.
 - d. Requests for adjustments in the Contract Time or the Contract Sum.
 - e. Requests for interpretation of Architect's actions on submittals.
 - f. Incomplete RFIs or inaccurately prepared RFIs.
 2. Architect's action may include a request for additional information, in which case Architect's time for response will date from time of receipt of additional information.
 3. Architect's action on RFIs that may result in a change to the Contract Time or the Contract Sum may be eligible for Contractor to submit Change Proposal according to Section 01 29 00.
 - a. If Contractor believes the RFI response warrants change in the Contract Time or the Contract Sum, notify Architect in writing within 10 days of receipt of the RFI response.
- E. On receipt of Architect's action, update the RFI log and immediately distribute the RFI response to affected parties. Review response and notify Architect within seven days if Contractor disagrees with response.
- F. RFI Log: Prepare, maintain, and submit a tabular log of RFIs organized by the RFI number. Submit log weekly. Include the following:
1. Project name.
 2. Name and address of Contractor.
 3. Name and address of Architect.
 4. RFI number including RFIs that were dropped and not submitted.
 5. RFI description.
 6. Date the RFI was submitted.
 7. Date Architect's response was received.
 8. Identification of related Minor Change in the Work, Construction Change Directive, and Proposal Request, as appropriate.
 9. Identification of related Field Order, Work Change Directive, and Proposal Request, as appropriate.

1.7 COORDINATION OF SPACE AND INSTALLATION SEQUENCE

- A. Coordinate use of Project space and sequence of installation of equipment, elevators, escalators, walks, mechanical, electrical, plumbing, or other Work that is indicated diagrammatically on Drawings. Follow routings shown for tubes, pipes, ducts,

conduits, and other items as closely as practical, with due allowance for available physical space. Make runs parallel with lines of building, except where not feasible by construction. Utilize space efficiently to maximize accessibility for other installations, for Owner maintenance, and for repairs.

- B. Except as otherwise indicated in finished areas, conceal ducts, pipes, wiring, and other non-finish items within construction. Coordinate locations of concealed and exposed items with finish elements.
- C. Where availability of space is limited, coordinate installation of different components to ensure maximum accessibility for required maintenance, service and repair.
- D. Coordinate with architectural reflected ceiling plans exact location and dimensioning of exposed items and items which occur within hung ceilings. In event of conflict, request clarification from Architect prior to proceeding with fabrication or installation.
- E. Contractor shall be responsible for coordination of Work. Each subcontractor shall be responsible for coordination of their respective Work with the Work of the Contractor and other trades. Prepare coordination drawings in accordance with this Section.
- F. Where installation of one part of Work is dependent on installation of other components, either before or after its own installation, schedule construction activities in sequence required to obtain best results.
- G. Make adequate provisions to accommodate items scheduled for later installation, including accepted Bid alternates, Owner-supplied Contractor-installed items, work by others, and installation of products purchased with allowances.

1.8 COORDINATION OF FINISHES

- A. Identify each room by name and number as it appears on Finish Schedules by posting room identification sign outside each room at main entry to each room. Identification shall be clearly visible, legible, and attached without damaging surface.
- B. Post accepted finishes scheduled for each room on each door or frame in manner that does not damage or stain surface. Use copy of accepted finish schedule that clearly identifies each finish and location of finishes for that particular room or area.
- C. Room identification signs and finish schedules shall remain posted until permanent interior signage has been installed and finishes have been reviewed by Architect, unless otherwise directed.
- D. Where mounting heights are not indicated, refer decisions to Architect prior to installation.

1.9 COORDINATION OF CONTRACT CLOSEOUT

- A. Coordinate completion and cleanup of Work of separate phases and sections in preparation for Substantial Completion of portions of Work designated for Owner partial occupancy.
- B. After Owner occupancy of premises, coordinate access to site by requirements of individual Specification Sections regarding correction of defective Work and Work not in accordance with Contract Documents. Minimize disruption of Owner's operations.
- C. Assemble and coordinate Closeout submittals in accordance with Section 01 77 00.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 013119
PROJECT MEETINGS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Preconstruction conference.
2. Preinstallation conference.
3. Progress meetings.

1.2 PRECONSTRUCTION CONFERENCE

A. Preconstruction Conference: Architect will schedule and conduct a preconstruction conference before starting construction, at a time convenient to Owner and Architect, but no later than 15 days after execution of the Agreement.

1. Attendees: Authorized representatives of Owner Architect, and their consultants; Contractor and its superintendent; major subcontractors; suppliers; and other concerned parties shall attend the conference. Participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Discuss items of significance that could affect progress, including the following:
 - a. Responsibilities and personnel assignments.
 - b. Tentative construction schedule.
 - c. Phasing.
 - d. Critical work sequencing and long lead items.
 - e. Designation of key personnel and their duties.
 - f. Lines of communications.
 - g. Use of web-based Project software.
 - h. Procedures for processing field decisions and Change Orders.
 - i. Procedures for RFIs.
 - j. Procedures for testing and inspecting.
 - k. Procedures for processing Applications for Payment.
 - l. Distribution of the Contract Documents.
 - m. Submittal procedures.
 - n. Preparation of Record Documents.
 - o. Use of the premises and existing building.
 - p. Work restrictions.
 - q. Working hours.
 - r. Owner's occupancy requirements.
 - s. Responsibility for temporary facilities and controls.

- t. Procedures for moisture and mold control.
 - u. Procedures for disruptions and shutdowns.
 - v. Construction waste management and recycling.
 - w. Parking availability.
 - x. Office, work, and storage areas.
 - y. Equipment deliveries and priorities.
 - z. First aid.
 - aa. Security.
 - bb. Progress cleaning.
3. Minutes: Entity responsible for conducting meeting will record and distribute meeting minutes.

1.3 PREINSTALLATION CONFERENCE

- A. Preinstallation Conferences: Conduct a preinstallation conference at Project site before each construction activity when required by other sections and when required for coordination with other construction.

- 1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by the installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend the meeting. Advise Architect of scheduled meeting dates.
- 2. Agenda: Review progress of other construction activities and preparations for the particular activity under consideration, including requirements for the following:
 - a. Contract Documents.
 - b. Options.
 - c. Related RFIs.
 - d. Related Change Orders.
 - e. Purchases.
 - f. Deliveries.
 - g. Submittals.
 - h. Review of mockups.
 - i. Possible conflicts.
 - j. Compatibility requirements.
 - k. Time schedules.
 - l. Weather limitations.
 - m. Manufacturer's written instructions.
 - n. Warranty requirements.
 - o. Compatibility of materials.
 - p. Acceptability of substrates.
 - q. Temporary facilities and controls.
 - r. Space and access limitations.
 - s. Regulations of authorities having jurisdiction.
 - t. Testing and inspecting requirements.
 - u. Installation procedures.
 - v. Coordination with other work.
 - w. Required performance results.
 - x. Protection of adjacent work.

- y. Protection of construction and personnel.
- 3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
- 4. Reporting: Distribute minutes of the meeting to each party present and to other parties requiring information.
- 5. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of the Work and reconvene the conference at earliest feasible date.

1.4 PROGRESS MEETINGS

A. Progress Meetings: Conduct progress meetings at regular intervals.

- 1. Attendees: In addition to representatives of Owner and Architect, each contractor, subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented at these meetings. All participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
- 2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of Project.
 - a. Contractor's Construction Schedule: Review progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind schedule, in relation to Contractor's construction schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.
 - 1) Review schedule for next period.
 - b. Review present and future needs of each entity present, including the following:
 - 1) Interface requirements.
 - 2) Sequence of operations.
 - 3) Resolution of BIM component conflicts.
 - 4) Status of submittals.
 - 5) Deliveries.
 - 6) Off-site fabrication.
 - 7) Access.
 - 8) Site use.
 - 9) Temporary facilities and controls.
 - 10) Progress cleaning.
 - 11) Quality and work standards.
 - 12) Status of correction of deficient items.
 - 13) Field observations.
 - 14) Status of RFIs.

- 15) Status of Proposal Requests.
 - 16) Pending changes.
 - 17) Status of Change Orders.
 - 18) Pending claims and disputes.
 - 19) Documentation of information for payment requests.
3. Minutes: Entity responsible for conducting the meeting will record and distribute the meeting minutes to each party present and to parties requiring information.
- a. Schedule Updating: Revise Contractor's construction schedule after each progress meeting where revisions to the schedule have been made or recognized. Issue revised schedule concurrently with the report of each meeting.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 013200

CONSTRUCTION PROGRESS DOCUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:
 - 1. Startup construction schedule.
 - 2. Contractor's Construction Schedule.
 - 3. Construction schedule updating reports.
 - 4. Daily construction reports.
 - 5. Material location reports.
 - 6. Site condition reports.
 - 7. Unusual event reports.

1.2 DEFINITIONS

- A. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction Project. Activities included in a construction schedule consume time and resources.
 - 1. Critical Activity: An activity on the critical path that must start and finish on the planned early start and finish times.
 - 2. Predecessor Activity: An activity that precedes another activity in the network.
 - 3. Successor Activity: An activity that follows another activity in the network.

1.3 INFORMATIONAL SUBMITTALS

- A. Format for Submittals: Submit required submittals in the following format:
 - 1. PDF file.
 - 2. One paper copy of sufficient size to display entire period or schedule, as required.
- B. Startup construction schedule.
- C. Contractor's Construction Schedule: Initial schedule, of size required to display entire schedule for entire construction period.
- D. Construction Schedule Updating Reports: Submit with Applications for Payment.
- E. Daily Construction Reports: Submit at biweekly intervals.

- F. Material Location Reports: Submit at biweekly intervals.
- G. Site Condition Reports: Submit at time of discovery of differing conditions.
- H. Unusual Event Reports: Submit at time of unusual event.

1.4 COORDINATION

- A. Coordinate Contractor's Construction Schedule with the schedule of values, submittal schedule, progress reports, payment requests, and other required schedules and reports.
 - 1. Secure time commitments for performing critical elements of the Work from entities involved.
 - 2. Coordinate each construction activity in the network with other activities, and schedule them in proper sequence.

1.5 CONTRACTOR'S CONSTRUCTION SCHEDULE

- A. Computer Scheduling Software: Prepare schedules using current version of one of the following programs that has been developed specifically to manage construction schedules:
 - 1. Microsoft Project.
 - 2. Primavera.
 - 3. Meridian Prolog.
- B. Time Frame: Extend schedule from date established for the Notice to Proceed to date of Final Completion.
 - 1. Contract completion date shall not be changed by submission of a schedule that shows an early completion date, unless specifically authorized by Change Order.
- C. Activities: Treat each floor or separate area as a separate numbered activity for each main element of the Work. Comply with the following:
 - 1. Activity Duration: Define activities so no activity is longer than 14 days, unless specifically allowed by Architect.
 - 2. Temporary Facilities: Indicate start and completion dates for the following as applicable:
 - a. Securing of approvals and permits required for performance of the Work.
 - b. Temporary facilities.
 - c. Construction of mock-ups, prototypes and samples.
 - d. Owner interfaces and furnishing of items.
 - e. Interfaces with Separate Contracts.
 - f. Regulatory agency approvals.
 - g. Punch list.

3. Procurement Activities: Include procurement process activities for the following long lead-time items and major items, requiring a cycle of more than 60 days, as separate activities in schedule. Procurement cycle activities include, but are not limited to, submittals, approvals, purchasing, fabrication, and delivery.
 4. Submittal Review Time: Include review and resubmittal times indicated in Section 01 33 00 "Submittal Procedures" in schedule. Coordinate submittal review times in Contractor's Construction Schedule with submittal schedule.
 5. Mockups: Include construction, review, and approval times for mockups specified in individual specification sections.
 6. Startup and Testing Time: Include time for startup and testing.
 7. Commissioning Time: Include time for commissioning.
 8. Substantial Completion: Indicate completion in advance of date established for Substantial Completion and allow time for Architect's administrative procedures necessary for certification of Substantial Completion.
 9. Punch List and Final Completion: Include time for completion of punch list items and Final Completion.
- D. Constraints: Include constraints and work restrictions indicated in the Contract Documents and as follows in schedule and show how the sequence of the Work is affected.
1. Phasing: Arrange list of activities on schedule by phase.
 2. Work under More Than One Contract: Include a separate activity for each contract.
 3. Work by Owner: Include a separate activity for each portion of the Work performed by Owner.
 4. Products Ordered in Advance: Include a separate activity for each product. Delivery dates indicated stipulate the earliest possible delivery date.
 5. Owner-Furnished Products: Include a separate activity for each product. Delivery dates indicated stipulate the earliest possible delivery date.
 6. Work Restrictions: Show the effect of the following items on the schedule:
 - a. Coordination with existing construction.
 - b. Limitations of continued occupancies.
 - c. Uninterruptible services.
 - d. Partial occupancy before Substantial Completion.
 - e. Use-of-premises restrictions.
 - f. Provisions for future construction.
 - g. Seasonal variations.
 - h. Environmental control.
- E. Milestones: Include milestones indicated in the Contract Documents in schedule, including, but not limited to, the Notice to Proceed, Substantial Completion, and Final Completion.
- F. Upcoming Work Summary: Prepare summary report indicating activities scheduled to occur or commence prior to submittal of next schedule update. Summarize the following issues:
1. Unresolved issues.
 2. Unanswered Requests for Information.

3. Rejected or unreturned submittals.
 4. Notations on returned submittals.
 5. Pending modifications affecting the Work and the Contract Time.
- G. Contractor's Construction Schedule Updating: At monthly intervals, update schedule to reflect actual construction progress and activities. Issue schedule one week before each regularly scheduled progress meeting.
1. Revise schedule immediately after each meeting or other activity where revisions have been recognized or made. Issue updated schedule concurrently with the report of each such meeting.
 2. Include a report with updated schedule that indicates every change, including, but not limited to, changes in logic, durations, actual starts and finishes, and activity durations.
 3. As the Work progresses, indicate Final Completion percentage for each activity.
- H. Recovery Schedule: When periodic update indicates the Work is 14 or more calendar days behind the current approved schedule, submit a separate recovery schedule indicating means by which Contractor intends to regain compliance with the schedule. Indicate changes to working hours, working days, crew sizes, equipment required to achieve compliance, and date by which recovery will be accomplished.
- I. Distribution: Distribute copies of approved schedule to Architect, Owner, separate contractors, testing and inspecting agencies, and other parties identified by Contractor with a need-to-know schedule responsibility.
1. Post copies in Project meeting rooms and temporary field offices.
 2. When revisions are made, distribute updated schedules to the same parties and post in the same locations. Delete parties from distribution when they have completed their assigned portion of the Work and are no longer involved in performance of construction activities.

1.6 GANTT-CHART SCHEDULE REQUIREMENTS

- A. Gantt-Chart Schedule: Submit a comprehensive, fully developed, horizontal, Gantt-chart-type, Contractor's Construction Schedule within 30 days of date established for commencement of the Work.
1. Base schedule on the startup construction schedule and additional information received since the start of Project.
- B. Preparation: Indicate each significant construction activity separately. Identify first workday of each week with a continuous vertical line.
1. For construction activities that require three months or longer to complete, indicate an estimated completion percentage in 10 percent increments within time bar.

1.7 REPORTS

- A. Daily Construction Reports: Prepare a daily construction report recording the following information concerning events at Project site:
1. List of subcontractors at Project site.
 2. List of separate contractors at Project site.
 3. Approximate count of personnel at Project site.
 4. Equipment at Project site.
 5. Material deliveries.
 6. High and low temperatures and general weather conditions, including presence of rain or snow.
 7. Testing and inspection.
 8. Accidents.
 9. Meetings and significant decisions.
 10. Unusual events.
 11. Stoppages, delays, shortages, and losses.
 12. Meter readings and similar recordings.
 13. Emergency procedures.
 14. Orders and requests of authorities having jurisdiction.
 15. Change Orders received and implemented.
 16. Services connected and disconnected.
 17. Equipment or system tests and startups.
 18. Partial completions and occupancies.
 19. Substantial Completions authorized.
- B. Material Location Reports: At bi-weekly intervals, prepare and submit a comprehensive list of materials delivered to and stored at Project site. List shall be cumulative, showing materials previously reported plus items recently delivered. Include with list a statement of progress on and delivery dates for materials or items of equipment fabricated or stored away from Project site. Indicate the following categories for stored materials:
1. Material stored prior to previous report and remaining in storage.
 2. Material stored prior to previous report and since removed from storage and installed.
 3. Material stored following previous report and remaining in storage.
- C. Site Condition Reports: Immediately on discovery of a difference between site conditions and the Contract Documents, prepare and submit a detailed report. Submit with a Request for Information. Include a detailed description of the differing conditions, together with recommendations for changing the Contract Documents.
- D. Unusual Event Reports: When an event of an unusual and significant nature occurs at Project site, whether or not related directly to the Work, prepare and submit a special report. List chain of events, persons participating, responses by Contractor's personnel, evaluation of results or effects, and similar pertinent information. Advise Owner in advance when these events are known or predictable.
1. Submit unusual event reports directly to Owner within one day(s) of an occurrence. Distribute copies of report to parties affected by the occurrence.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 013233
PHOTOGRAPHIC DOCUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Photography
 - 2. Views and Techniques
 - 3. Images
 - 4. Media

1.2 PHOTOGRAPHY

- A. Take construction record photographs periodically during course of Work.
- B. Furnish construction photographs taken on commencement of Work and at monthly intervals.
- C. Submit prints and digital media monthly with each pay application.
- D. Photos may be incorporated into monthly construction report which should include schedule, progress of work, etc.
- E. Do not display photographs in publications, contests or other public or private forum without written consent of Owner and Architect.
- F. Parties requiring additional copies of digital images prints will order from and pay photographer directly.

PART 2 - PRODUCTS

2.1 DIGITAL IMAGES

- A. Provide color digital images, taken at 1028 x 768 minimum resolution.
- B. Identify each print listing time and date of exposure, location and orientation of view, project name and address.

2.2 DIGITAL MEDIA

- A. Digital/media (CD Rom) images will become property of Owner.

- B. Catalog and index digital images in chronological sequence. Provide typed table of contents. Place negatives in archive negative sheets and compiled in three-ring commercial quality binder.

PART 3 - EXECUTION

3.1 VIEWS REQUIRED

- A. Consult with Architect for instructions concerning views required at each specified visit to site.
- B. Photograph from locations to adequately illustrate condition of construction and state of progress.
- C. Minimum views and quantities required:
 - 1. At each specified time, take photographs from 12 different views.
 - 2. Architect will have right to request fewer photographs be taken at certain intervals so that more photographs may be taken at other times, providing that total number of photographs remains unchanged.

3.2 DELIVERY OF DIGITAL IMAGES

- A. Deliver digital images prints with each monthly pay application.
- B. Deliver one set of digital images to Architect, one to Owner, and place one set of digital images in Project Record file as specified in Section 01 78 39.

END OF SECTION

SECTION 013300

SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Submittal schedule requirements.
2. Administrative and procedural requirements for Submittals.

1.2 DEFINITIONS

- A. Action Submittals:** Written and graphic information and physical samples that require Architect's responsive action. Action submittals are those submittals indicated in individual Specification Sections as "action submittals."
- B. Informational Submittals:** Written and graphic information and physical samples that do not require Architect's responsive action. Submittals may be rejected for not complying with requirements. Informational submittals are those submittals indicated in individual Specification Sections as "informational submittals."

1.3 SUBMITTAL SCHEDULE

- A. Submittal Schedule:** Submit as an action submittal, arranged in chronological order by dates required by construction schedule. Include time required for review, ordering, manufacturing, fabrication, and delivery when establishing dates. Include additional time required for making corrections or revisions to submittals noted by Architect and additional time for handling and reviewing submittals required by those corrections.
1. Coordinate submittal schedule with list of subcontracts, the schedule of values, and Contractor's construction schedule.
 2. Initial Submittal: Submit concurrently with startup construction schedule. Include submittals required during the first 60 days of construction. List those submittals required to maintain orderly progress of the Work and those required early because of long lead time for manufacture or fabrication.
 3. Final Submittal: Submit concurrently with the first complete submittal of Contractor's construction schedule.
 - a. Submit revised submittal schedule to reflect changes in current status and timing for submittals.
 4. Format: Arrange the following information in a tabular format:
 - a. Scheduled date for first submittal.

- b. Specification Section number and title.
- c. Submittal category: Action; informational.
- d. Name of subcontractor.
- e. Description of the Work covered.
- f. Scheduled date for Architect's final release or approval.
- g. Scheduled date of fabrication.
- h. Scheduled dates for purchasing.
- i. Scheduled dates for installation.
- j. Activity or event number.

1.4 SUBMITTAL FORMATS

A. Submittal Information: Include the following information in each submittal:

- 1. Project name.
- 2. Date.
- 3. Name of Architect.
- 4. Name of Contractor.
- 5. Name of firm or entity that prepared submittal.
- 6. Names of subcontractor, manufacturer, and supplier.
- 7. Unique alphanumeric submittal number, including revision identifier, in the following format:
 - a. Include Specification Section number, followed by sequential alphanumeric identifier, followed by sequential alphanumeric suffix for resubmittals (e.g., 06 10 53 – 001 – 00).
 - b. Update sequential alphanumeric suffix for each subsequent resubmittal (e.g., 06 10 53 – 001 – 01).
- 8. Category and type of submittal.
- 9. Submittal purpose and description.
- 10. Number and title of Specification Section, with paragraph number and generic name for each of multiple items.
- 11. Drawing number and detail references, as appropriate.
- 12. Indication of full or partial submittal.
- 13. Location(s) where product is to be installed, as appropriate.
- 14. Other necessary identification.
- 15. Remarks.
- 16. Signature of review and transmitter.

B. Options: Identify options requiring selection by Architect.

- C. Deviations and Additional Information: On each submittal, clearly indicate deviations from requirements in the Contract Documents, including minor variations and limitations; include relevant additional information and revisions, other than those requested by Architect on previous submittals. Indicate by highlighting on each submittal or noting on attached separate sheet.
1. Contractor's responsibility for deviations in Submittals from requirements of Contract Documents is not relieved by Architect's review of Submittals unless Architect gives specific written acceptance for each specific deviation.
- D. Paper Submittal Requirements: Place a permanent label or title block on each submittal item for identification.
1. Provide a space approximately 6 by 8 inches on label or beside title block to record Contractor's review and approval markings and action taken by Architect.
 2. Action Submittals: Submit four paper copies of each submittal unless otherwise indicated. Architect will return two copies.
 3. Informational Submittals: Submit one paper copy of each submittal unless otherwise indicated. Architect will not return copies.
 4. Additional Copies: Unless additional copies are required for final submittal, and unless Architect observes noncompliance with provisions in the Contract Documents, initial submittal may serve as final submittal.
- E. Electronic Submittal Requirements:
1. Prepare submittals as PDF package, assembling complete submittal package into a single, indexed PDF file incorporating transmittal form and all submittal requirements of a Specification Section with internal bookmark links enabling for navigation to each item. Name PDF file with Submittal number.
 2. Provide electronic stamp to permanently record Contractor's review and approval markings and space for electronic stamp for action taken by Architect and Consultants as needed.
 3. Metadata: Include the following information as keywords in the electronic submittal file metadata:
 - a. Project name.
 - b. Number and title of appropriate Specification Section.
 - c. Manufacturer name.
 - d. Product name.
 4. Architect will return annotated file. Annotate and retain one copy of file as an electronic Project record document file.
- F. Transmittals for Submittals: Assemble each submittal individually and appropriately for transmittal and handling. Transmit each submittal using a transmittal form. Provide locations on form for the following information:
1. Project name.
 2. Date.
 3. Destination (To:).
 4. Source (From:).

5. Name and address of Architect.
6. Name of Construction Manager.
7. Name of Contractor.
8. Name of firm or entity that prepared submittal.
9. Names of subcontractor, manufacturer, and supplier.
10. Type of submittal.
11. Submittal purpose and description.
12. Specification Section number and title.
13. Specification paragraph number or drawing designation and generic name for each of multiple items.
14. Drawing number and detail references, as appropriate.
15. Indication of full or partial submittal.
16. Transmittal number, numbered consecutively.
17. Submittal and transmittal distribution record.
18. Remarks.
19. Signature of transmitter.

1.5 SUBMITTAL PROCEDURES

- A. Prepare and submit submittals required by individual Specification Sections. Types of submittals required are indicated in individual Specification Sections.
 1. Paper: Prepare submittals in paper form and deliver to Architect.
 2. Web-Based Project Software: Prepare submittals in PDF form, and upload to web-based Project software website. Enter required data in web-based software site to fully identify submittal.
- B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.
 1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
 2. Submit all submittal items required for each Specification Section concurrently unless partial submittals for portions of the Work are indicated on approved submittal schedule.
 3. Submit action submittals and informational submittals required by the same Specification Section as separate packages under separate transmittals.
 4. Coordinate transmittal of submittals for related parts of the Work specified in different Sections so processing will not be delayed because of need to review submittals concurrently for coordination.
 - a. Architect reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
 5. Make submittals for concurrent consultant review directly to consulting engineer with copy of transmittal sent to Architect.
- C. Processing Time: Allow time for submittal review, including time for resubmittals, as follows. Time for review shall commence on Architect's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit

submittals enough in advance of the Work to permit processing, including resubmittals. Architect will process submittals as quickly as possible but reserves the right to the following review periods.

1. Initial Review: Allow seven working days for initial review of each submittal. Submittals received by Architect after 1:00 p.m. will be considered as received the following working day. Allow additional time if coordination with subsequent submittals is required. Architect will advise Contractor when a submittal being processed must be delayed for coordination.
2. Intermediate Review: If intermediate submittal is necessary, process it in same manner as initial submittal.
3. Resubmittal Review: Allow seven working days for review of each resubmittal.
4. Sequential Review: Where sequential review of submittals by Architect's consultants, Owner, or other parties is indicated, allow 14 working days for initial review of each submittal.
5. Concurrent Consultant Review: Where the Contract Documents indicate that submittals may be transmitted simultaneously to Architect and to Architect's consultants, allow 10 working days for review of each submittal. Submittal will be returned to Architect before being returned to Contractor.

- D. Resubmittals: Make resubmittals in same form and number of copies as initial submittal.

1. Note date and content of previous submittal.
2. Note date and content of revision in label or title block and clearly indicate extent of revision.
3. Resubmit submittals until they are marked with approval notation from Architect's action stamp.

- E. Distribution: Furnish copies of final Submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities. Show distribution on transmittal forms.

1. Distribute accepted samples as directed by Architect.

- F. Use for Construction: Retain complete copies of submittals on Project site. Use only final Submittals that are marked with approval notation from Architect's action stamp.

1.6 SUBMITTAL REQUIREMENTS

- A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. If information must be specially prepared for submittal because standard published data are unsuitable for use, submit as Shop Drawings, not as Product Data.
2. Mark each copy of each submittal to show which products and options are applicable.

3. Include the following information, as applicable:
 - a. Manufacturer's catalog cuts.
 - b. Manufacturer's product specifications.
 - c. Standard color charts.
 - d. Statement of compliance with specified referenced standards.
 - e. Testing by recognized testing agency.
 - f. Application of testing agency labels and seals.
 - g. Notation of coordination requirements.
 - h. Availability and delivery time information.
 4. For equipment, include the following in addition to the above, as applicable:
 - a. Wiring diagrams that show factory-installed wiring.
 - b. Printed performance curves.
 - c. Operational range diagrams.
 - d. Clearances required to other construction, if not indicated on accompanying Shop Drawings.
 5. Submit Product Data before Shop Drawings, and before or concurrent with Samples.
- B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed manufacturer's data.
1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:
 - a. Identification of products.
 - b. Schedules.
 - c. Compliance with specified standards.
 - d. Notation of coordination requirements.
 - e. Notation of dimensions established or confirmed by field measurement.
 - f. Relationship and attachment to adjoining construction clearly indicated.
 - g. Seal and signature of professional engineer if specified.
 2. Paper Sheet Size: Except for templates, patterns, and similar full-size Drawings, submit Shop Drawings on sheets at least 11 by 17 inches, but no larger than 30 by 42 inches.
 - a. Three opaque copies of each submittal. Architect will retain two copies; remainder will be returned.
- C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other materials.
1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.

2. Identification: Permanently attach label on unexposed side of Samples that includes the following:
 - a. Project name and submittal number.
 - b. Generic description of Sample.
 - c. Product name and name of manufacturer.
 - d. Sample source.
 - e. Number and title of applicable Specification Section.
 - f. Specification paragraph number and generic name of each item.
3. Paper Transmittal: Include paper transmittal including complete submittal information indicated.
4. Web-Based Project Management Software: Prepare submittals in PDF form, and upload to web-based Project software website. Enter required data in web-based software site to fully identify submittal.
5. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 - a. Samples that may be incorporated into the Work are indicated in individual Specification Sections. Such Samples must be in an undamaged condition at time of use.
 - b. Samples not incorporated into the Work, or otherwise designated as Owner's property, are the property of Contractor.
6. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.
 - a. Number of Samples: Submit one full set of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Architect will return submittal with options selected.
7. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include, but are not limited to, the following: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
 - a. Number of Samples: Submit three sets of Samples. Architect will retain one Sample set; remainder will be returned. Mark up and retain one returned Sample set as a project record Sample.
 - 1) Submit a single Sample where assembly details, workmanship, fabrication techniques, connections, operation, and other similar characteristics are to be demonstrated.

- 2) If variation in color, pattern, texture, or other characteristic is inherent in material or product represented by a Sample, submit at least three sets of paired units that show approximate limits of variations.
- D. Product Schedule: As required in individual Specification Sections, prepare a written summary indicating types of products required for the Work and their intended location. Include the following information in tabular form:
1. Type of product. Include unique identifier for each product indicated in the Contract Documents or assigned by Contractor if none is indicated.
 2. Manufacturer and product name, and model number if applicable.
 3. Number and name of room or space.
 4. Location within room or space.
- E. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, contact information of architects and owners, and other information specified.
- F. Design Data: Prepare and submit written and graphic information indicating compliance with indicated performance and design criteria in individual Specification Sections. Include list of assumptions and summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Number each page of submittal.
- G. Certificates:
1. Certificates and Certifications Submittals: Submit a statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity. Provide a notarized signature where indicated.
 2. Installer Certificates: Submit written statements on manufacturer's letterhead certifying that Installer complies with requirements in the Contract Documents and, where required, is authorized by manufacturer for this specific Project.
 3. Manufacturer Certificates: Submit written statements on manufacturer's letterhead certifying that manufacturer complies with requirements in the Contract Documents. Include evidence of manufacturing experience where required.
 4. Material Certificates: Submit written statements on manufacturer's letterhead certifying that material complies with requirements in the Contract Documents.
 5. Product Certificates: Submit written statements on manufacturer's letterhead certifying that product complies with requirements in the Contract Documents.
 6. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements in the Contract Documents. Submit record of Welding Procedure Specification and Procedure Qualification Record on AWS forms. Include names of firms and personnel certified.

H. Test and Research Reports:

1. Compatibility Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
2. Field Test Reports: Submit written reports indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in the Contract Documents.
3. Material Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.
4. Preconstruction Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in the Contract Documents.
5. Product Test Reports: Submit written reports indicating that current product produced by manufacturer complies with requirements in the Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
6. Research Reports: Submit written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project. Include the following information:
 - a. Name of evaluation organization.
 - b. Date of evaluation.
 - c. Time period when report is in effect.
 - d. Product and manufacturers' names.
 - e. Description of product.
 - f. Test procedures and results.
 - g. Limitations of use.

1.7 DELEGATED DESIGN SERVICES

- A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.
1. If criteria indicated are insufficient to perform services or certification required, submit a written request for additional information to Architect.

- B. Delegated-Design Services Certification: In addition to Shop Drawings, Product Data, and other required submittals, submit digitally signed PDF file and one paper copy of certificate, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.
 - 1. Indicate that products and systems comply with performance and design criteria in the Contract Documents. Include list of codes, loads, and other factors used in performing these services.

1.8 CONTRACTOR'S REVIEW

- A. Action Submittals and Informational Submittals: Review each submittal and check for coordination with other Work of the Contract and for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to Architect.
 - 1. Submittals without Contractor's stamp and Submittals which, in the opinion of the Architect or Owner, are incomplete, contain numerous errors, have not been reviewed, or have only been reviewed superficially, will be returned without Architect's review or comments. Resultant schedule delays incurred are Contractor's responsibility.
- B. Contractor's Approval: Indicate Contractor's review and approval for each submittal with a uniform approval stamp. Include name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.
 - 1. Architect will not review submittals received from Contractor that do not have Contractor's stamp, review, and approval.

1.9 ARCHITECT'S REVIEW

- A. Architect's review of Submittals is for general design conformance and compliance with information provided in Contract Documents. Contractor is responsible for meeting requirements and characteristics of products and systems specified; dimensions and quantities to be confirmed and correlated at the Project site; and information that pertains solely to fabrication processes, installation techniques, and coordination of work required by other trades. Architect's review and indication of action to be taken shall not relieve Contractor of responsibility for deviation from requirements of Contract Documents.
- B. Action Submittals: Architect will review each submittal, indicate corrections or revisions required, and return it. Architect will indicate, via markup on each submittal, the appropriate action as follows:
 - 1. Final Unrestricted Release: When submittal is marked "No Exception Taken", that part of the Work covered by the submittal may proceed provided it complies

- with requirements of the Contract Documents; final acceptance will depend upon that compliance.
2. Final-But-Restricted Release: When submittal is marked "Comments Included", that part of the Work covered by the submittal may proceed provided it complies with both the Architect's notations and comments on the submittal and requirements of the Contract Documents; final acceptance will depend upon that compliance.
 3. Returned for Resubmittal: When submittal is marked "Revise and Submit", do not proceed with that part of the Work covered by the submittal, including purchasing, fabrication, delivery, or other activity. Revise or prepare a new submittal in accordance with the Architects notations or corrections; resubmit without delay.
 - a. Do not permit submittals marked "Revise and Submit" to be used at the Project site or elsewhere where construction is in progress.
 4. Rejected: When submittal is marked "Rejected" do not proceed with that part of the work as submission is either incomplete, not coordinated, or does not conform to requirements of Contract Documents. Resubmission of the same submittal will not be considered. Prepare and submit new submittal meeting requirements of Contract Documents.
 5. Not Reviewed: When submittal is marked "Not Reviewed. Submittal not required by Contract Documents" it has not been reviewed because the submittal was neither requested nor required by Contract Documents.
 6. Close-Out Submittal: When submittal is marked "Reviewed for Project Close-Out Requirements Only" it has only been reviewed only for compliance with close-out submission requirements. It has not been reviewed for accuracy or completeness. Burden of accuracy and completeness rests with the Contractor.
 7. Substitution Request: When a submittal is marked "Proposed Substitution. Resubmit. Comply with Section 01 25 00" submission does not conform to requirements of Contract Documents. Contractor may elect to resubmit under the provisions, conditions, and procedures outlined in Section 01 25 00 "Substitution Procedures," or prepare and submit new submittal meeting requirements of Contract Documents.
- C. Informational Submittals: Architect will review each submittal and will not return it, or will return it if it does not comply with requirements.
 - D. Partial Submittals prepared for a portion of the Work will be reviewed when use of partial Submittals has received prior approval from Architect.
 - E. Incomplete submittals are unacceptable, will be considered nonresponsive, and will be returned for resubmittal without review.
 - F. Architect will discard submittals received from sources other than Contractor.
 - G. Submittals not required by the Contract Documents will be returned by Architect without action.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 013516

ALTERATION PROJECT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Procedural requirements for alterations work.

1.2 DESCRIPTION

- A. Coordinate work of alterations and renovations with new construction.
- B. Procedures and administrative requirements of this Section apply to Sections of Specifications which are involved in alterations to existing building and equipment.
- C. In addition to demolition and removal specifically indicated, cut into or partially remove portions of existing building, finishes and equipment as necessary to allow for new construction:
 - 1. Cutting, moving or removal of items not shown to be cut, moved, or removed, but which must be cut, moved, or removed to allow new work to proceed.
 - 2. Work of items which are to remain in finished work shall be patched or reinstalled after their cutting, moving, or removal, and their joints and finishes made to match adjacent or similar work.
 - 3. Removal of existing surface finishes as needed to install new work and new finishes.
 - 4. Removal of abandoned items from project site.
 - 5. Removal of unsuitable or extraneous materials not marked for salvage.

1.3 QUALITY ASSURANCE

- A. Coordinate work of trades and schedule elements of alterations and renovation work by procedures and methods to expedite completion of Work.
- B. In addition to demolition work specified in Section 02 41 19 and that specifically shown, cut, move or remove items as necessary to provide access or to allow alterations and new work to proceed including such items as follows:
 - 1. Repair or removal of unsanitary conditions.
 - 2. Removal of abandoned items and items serving no useful purpose, such as abandoned piping, conduit, and wiring.
 - 3. Removal of unsuitable or extraneous materials not indicated for salvage, such as abandoned furnishings and equipment, and debris, such as rotted wood, rusted metals, and deteriorated concrete.

4. Cleaning of surfaces and removal of surface finishes as needed to install new work and finishes.
- C. Patch, repair and refinish existing items to remain, to specified condition for each material, with smooth transition to adjacent new items of construction.
- D. Assign work of moving, removal, cutting, and patching to trades qualified to perform work in manner to cause least damage to each type of work and provide means of returning surfaces to appearance of new work.
- E. Perform cutting and removal work to remove minimum necessary, and in manner to avoid damage to adjacent work. Cut finish surfaces such as masonry, tile, plaster, or metals by methods to terminate surfaces in straight line at natural point of division.
- F. Perform cutting and patching as specified in Section 01 73 29.
- G. Protect existing finishes, equipment and adjacent work which is scheduled to remain from damage. Protect existing and new work from weather and extremes of temperature. Maintain existing interior work above 60 degrees F. Provide weather protection, waterproofing, heat, and humidity control as needed to prevent damage to remaining existing work and to new work.
- H. Provide temporary enclosures to separate work areas from existing building and from areas occupied by Owner, and to provide weather protection.
- I. Discoveries of construction, furnishings and articles having historic or private value shall remain in possession of Owner.
 1. Promptly notify Architect of discovery.
 2. Protect discovery from damage.
 3. Architect will promptly transmit Owner's decision for disposition of discovery.
 4. Contractor shall store items to be retained by Owner in safe, dry place, on site and shall dispose of items which Owner releases.

1.4 SCHEDULING, ACCESS, AND SECURITY

- A. Submit separate detailed sub-schedule for alterations work, coordinated with Construction Schedule. Indicate following as a minimum:
 1. Each stage of work and dates of occupancy of areas.
 2. Date of Substantial Completion for each area of alterations work, as appropriate.
 3. Trades and subcontractors employed in each stage.
- B. Interruption of Services:
 1. Interruption of utilities and services is prohibited, except as specifically permitted in writing by Owner.
 2. Identify areas to be affected by interruption when requesting permission. Search out utility lines to determine effect of each outage upon operations within building.

3. Make requests at least three full days before time of proposed outage.
4. Warrantee interruption is prohibited.

C. Interruption of Fire Protection Sprinkler System:

1. Notify Owner in writing 72 hours prior to removing, relocating, or reworking sprinkler heads and prior to control valve closure in system.
2. Display red tags entitled "Fire Protection Out of Service" on each closed valve.
3. Notify Owner prior to restoring fire sprinkler system operation.
4. Interruption of existing fire sprinkler system will only be allowed between hours of 10:00 p.m. and 6:00 a.m., and only after written authorization from Owner.

D. Security:

1. When keys to locked areas are needed to perform Work, contact Owner representative for access to work area.
2. Maintain security of work areas during entire construction period.

E. Maintenance of Access of Operations:

1. During period of construction, Owner will continue to perform normal activities associated with facilities.
2. Maintain proper and safe access.
3. Keep access points free of construction debris.
4. Schedule demolition and remodeling operations with Owner in such manner as to allow existing operations to continue without interruptions.

F. Building Access:

1. Contractor shall access building at locations designated by Owner.
2. Access to construction areas within building shall be by passageways and corridors as designated by Owner.
3. Restrict construction traffic to designated work areas and to access areas specifically designated by Owner.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Salvage sufficient quantities of cut or removed material to replace damaged work of existing construction when material is not readily obtainable on current market. Store salvaged items in dry, secure place on site.
- B. Items not required for use in repair of existing work shall remain property of Owner.
- C. Do not incorporate salvaged or used material in new construction except with permission of Architect.

2.2 PRODUCTS FOR PATCHING, EXTENDING AND MATCHING

- A. Provide same products or types of construction as that in existing structure, as needed to patch, extend, or match existing work. Generally, Contract Documents will not define products or standards of workmanship present in existing construction. Contractor shall determine products by inspection and necessary testing and by use of existing as sample of comparison.
- B. Presence of product, finish, or type of construction, requires that patching, extending, or matching shall be performed as necessary to make Work complete and consistent to identical standards of quality.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Contractor shall accept site and existing building in condition in which they exist at time he is given access to begin Work.
- B. While Work under this Contract is in progress, protect existing buildings, grounds, contents, and occupants, including those on adjacent properties, whether private or public, from damage or harm due to Work under this Contract.

3.2 ALTERATIONS, CUTTING AND PROTECTION

- A. Perform cutting and removal work so as not to cut or remove more than is necessary and so as not to damage adjacent work or substrates.
- B. Responsibility and Assignment:
 - 1. Move, remove, cut, patch and repair and cause least damage to each type of work encountered.
 - 2. Patching of materials shall be assigned to mechanics skilled in work of finish trade involved.
- C. Protection:
 - 1. Protect remaining finishes, equipment, and adjacent work from damage caused by cutting, moving, removal and patching operations; protect surfaces which will remain part of finished work.
 - 2. During demolition, cutting, moving, removal and construction, provide positive dust-control including installation of temporary coverings, so as to prevent spread of dust and debris to adjacent Owner-occupied areas.
 - 3. Erect and maintain barricades and temporary controls as required to protect work; refer to Section 01 50 00.

D. Temporary Partitions:

1. Provide and maintain temporary and dust partitions to seal openings to Owner-occupied areas, to provide temporary security, and to provide protection to public.
2. Locate temporary partitions as indicated.
3. Extend partitions full height to structure or provide ceiling in manner to positively seal construction areas from Owner-occupied areas.
4. Framing:
 - a. Commercial softwood species, fire-retardant treated in accordance with AWPA C20, and bearing UL Label FR-S.
 - b. Provide continuous 2 x 4 top and bottom plates, 2 x 4 studs at 24 inches on center, and continuous 2 x 4 bridging 4 feet on center vertically.
 - c. At Contractor's option, drywall metal studs may be used.
 - d. Provide 3-1/2 inches wide studs 24 inches on center with continuous head and floor channels.
5. Covering:
 - a. 1/2 inch thick gypsum board.
 - b. Apply to one side of studs with vinyl foam tape and supplemental fasteners.
6. Doors: Single acting doors, opening out, with sturdy closer, closing against gasketed stops on frame to reduce passage of cover partitions; provide ample wood push bars and bump plates.
7. Sealing:
 - a. Seal perimeter of partitions and doors to prevent passage of dust.
 - b. Tape fastener depressions, horizontal joints between panels, and joints between panels and floors, ceilings, and columns with 2 inches wide pressure sensitive tape.
 - c. Install tape on construction side of partition.
 - d. Provide additional polyethylene sheeting where required to provide positive dust seal.
8. Mats: Provide mats at doors to reduce tracking of dust; replace or clean daily.

E. Demolition:

1. Cut and remove designated items in accordance with Section 02 41 19.
2. Do not cut or remove structural member.
3. Avoid excessive vibrations in demolition procedures that would be transmitted through existing structure and finish materials.
4. Schedule demolition in accordance with phasing requirements.

F. Debris:

1. Remove debris promptly from site each day.
2. Load removed material directly onto trucks for removal from property.
3. Dispose of removed material legally; do not burn.

4. Do not allow debris to enter sewers.
5. Do not let piled material endanger structure.
6. Prevent occurrence of unsanitary conditions, staining or damage due to debris.

3.3 PATCHING

- A. Quality of patched or extended work shall not be less than that specified for new work.
- B. When new work abuts or finishes flush with existing work, make smooth transition. Patched work shall match existing adjacent work in texture and appearance so that patch or transition is invisible at distance of 5 feet.
- C. When finished surfaces are cut in such manner that smooth transition with new work is not possible, terminate existing surface in straight line at natural line of division, and provide trim appropriate to finished surface.

3.4 ADJUSTING

- A. Where partitions are removed, patch floors, walls, and ceilings with finish materials to match existing. Where removal of partitions results in adjacent spaces becoming one, rework floors and ceilings to provide smooth planes breaks, steps or bulkheads.
- B. Where extreme change of plane of 2 inches or more occurs, request instructions from Architect as to method of making transition.
- C. Trim and refinish existing doors as necessary to clear new floors.
- D. Patch and replace portion of existing finished surface, which is found to be damaged, lifted, discolored or shows other imperfections with matching material.
- E. Provide adequate support of substrate prior to patching finish.
- F. Refinish patched portions of painted or coated surfaces in manner to produce uniform color and texture over entire surface. When existing surface finish cannot be matched, refinish entire surface to nearest intersections.

END OF SECTION

SECTION 014000

QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes Contractor's administrative and procedural requirements for quality assurance and quality control.

1.2 DEFINITIONS

- A. Quality-Assurance Services: Activities, actions, and procedures performed before and during execution of the Work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.
- B. Quality-Control Services: Tests, inspections, procedures, and related actions during and after execution of the Work to evaluate that actual products incorporated into the Work and completed construction comply with requirements. Services do not include contract enforcement activities performed by Architect or Construction Manager.
- C. Mockups: Full-size physical assemblies that are constructed on-site. Mockups are constructed to verify selections made under Sample submittals; to demonstrate aesthetic effects and, where indicated, qualities of materials and execution; to review coordination, testing, or operation; to show interface between dissimilar materials; and to demonstrate compliance with specified installation tolerances. Mockups are not Samples. Unless otherwise indicated, approved mockups establish the standard by which the Work will be judged.
- D. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.
 - 1. Use of trade-specific terminology in referring to a trade or entity does not require that certain construction activities be performed by accredited or unionized individuals, or that requirements specified apply exclusively to specific trade(s).
- E. Experienced: When used with an entity or individual, "experienced" means having successfully completed a minimum of five previous projects similar in nature, size, and extent to this Project; being familiar with special requirements indicated; and having complied with requirements of authorities having jurisdiction.

1.3 CONFLICTING REQUIREMENTS

- A. Referenced Standards: If compliance with two or more standards is specified and the standards establish different or conflicting requirements for minimum quantities or

quality levels, comply with the most stringent requirement. Refer conflicting requirements that are different, but apparently equal, to Architect for a decision before proceeding.

- B. Minimum Quantity or Quality Levels: The quantity or quality level shown or specified shall be the minimum provided or performed. The actual installation may comply exactly with the minimum quantity or quality specified, or it may exceed the minimum within reasonable limits. To comply with these requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Architect for a decision before proceeding.

1.4 SUBMITTALS

- A. Contractor's Quality-Control Plan: For quality-assurance and quality-control activities and responsibilities.

1.5 CONTRACTOR'S QUALITY-CONTROL PLAN

- A. Quality-Control Plan, General: Submit quality-control plan within five days of Notice to Proceed, and not less than five days prior to preconstruction conference. Submit in format acceptable to Architect. Identify personnel, procedures, controls, instructions, tests, records, and forms to be used to carry out Contractor's quality-assurance and quality-control responsibilities. Coordinate with Contractor's construction schedule.
- B. Quality-Control Personnel Qualifications: Engage qualified full-time personnel trained and experienced in managing and executing quality-assurance and quality-control procedures similar in nature and extent to those required for Project.
 - 1. Project quality-control manager may also serve as Project superintendent.
- C. Submittal Procedure: Describe procedures for ensuring compliance with requirements through review and management of submittal process. Indicate qualifications of personnel responsible for submittal review.
- D. Continuous Inspection of Workmanship: Describe process for continuous inspection during construction to identify and correct deficiencies in workmanship in addition to testing and inspection specified. Indicate types of corrective actions to be required to bring work into compliance with standards of workmanship established by Contract requirements and approved mockups.
- E. Monitoring and Documentation: Maintain testing and inspection reports including log of approved and rejected results. Include work Architect has indicated as nonconforming or defective. Indicate corrective actions taken to bring nonconforming work into compliance with requirements. Comply with requirements of authorities having jurisdiction.

1.6 REPORTS AND DOCUMENTS

- A. Manufacturer's Technical Representative's Field Reports: Prepare written information documenting manufacturer's technical representative's tests and inspections specified in other Sections. Include the following:
1. Name, address, and telephone number of technical representative making report.
 2. Statement on condition of substrates and their acceptability for installation of product.
 3. Statement that products at Project site comply with requirements.
 4. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
 5. Results of operational and other tests and a statement of whether observed performance complies with requirements.
 6. Statement whether conditions, products, and installation will affect warranty.
 7. Other required items indicated in individual Specification Sections.
- B. Factory-Authorized Service Representative's Reports: Prepare written information documenting manufacturer's factory-authorized service representative's tests and inspections specified in other Sections. Include the following:
1. Name, address, and telephone number of factory-authorized service representative making report.
 2. Statement that equipment complies with requirements.
 3. Results of operational and other tests and a statement of whether observed performance complies with requirements.
 4. Statement whether conditions, products, and installation will affect warranty.
 5. Other required items indicated in individual Specification Sections.
- C. Permits, Licenses, and Certificates: For Owner's records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of the Work.

1.7 QUALITY ASSURANCE

- A. General: Qualifications paragraphs in this article establish the minimum qualification levels required; individual Specification Sections specify additional requirements.
- B. Manufacturer Qualifications: A firm experienced in manufacturing products or systems similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- C. Fabricator Qualifications: A firm experienced in producing products similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- D. Installer Qualifications: A firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent to that indicated for this Project,

whose work has resulted in construction with a record of successful in-service performance.

- E. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of the system, assembly, or product that are similar in material, design, and extent to those indicated for this Project.
- F. Specialists: Certain Specification Sections require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for the activities indicated.
 - 1. Requirements of authorities having jurisdiction shall supersede requirements for specialists.
- G. Manufacturer's Technical Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to observe and inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.
- H. Factory-Authorized Service Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.
- I. Mockups: Before installing portions of the Work requiring mockups, build mockups for each form of construction and finish required to comply with the following requirements, using materials indicated for the completed Work:
 - 1. Build mockups in location and of size indicated or, if not indicated, as directed by Architect.
 - 2. Notify Architect seven days in advance of dates and times when mockups will be constructed.
 - 3. Employ supervisory personnel who will oversee mockup construction. Employ workers that will be employed during the construction at Project.
 - 4. Demonstrate the proposed range of aesthetic effects and workmanship.
 - 5. Obtain Architect's approval of mockups before starting work, fabrication, or construction.
 - a. Allow seven days for initial review and each re-review of each mockup.
 - 6. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
 - 7. Demolish and remove mockups when directed unless otherwise indicated.

1.8 QUALITY CONTROL

- A. Manufacturer's Field Services: Where indicated, engage a factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing as specified in Section 01 33 00 "Submittal Procedures."
- B. Manufacturer's Technical Services: Where indicated, engage a manufacturer's technical representative to observe and inspect the Work. Manufacturer's technical representative's services include participation in preinstallation conferences, examination of substrates and conditions, verification of materials, observation of Installer activities, inspection of completed portions of the Work, and submittal of written reports.
- C. Coordination: Coordinate sequence of activities to accommodate required testing, quality-assurance, and quality-control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspecting.
 - 1. Schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 014100

REGULATORY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: General regulatory requirements.

1.2 REFERENCES

- A. ASTM International.
 - 1. ASTM E 119: Test Methods for Fire Tests of Building Construction and Materials
- B. Underwriters Laboratories Inc.
 - 1. UL 263: Fire Tests of Building Construction and Materials.

1.3 GENERAL REQUIREMENTS

- A. General: Additional information with legal implications regarding applicable governing laws and jurisdictions can be found in Conditions of Contract.
- B. Codes:
 - 1. Where references are made on Drawings or Specifications to codes, they shall be considered an integral part of the Contract Documents as minimum standards. Nothing contained in Contract Documents shall be so construed as to be in conflict with law, bylaw or regulation of municipal, State, Federal or other authorities having jurisdiction.
 - 2. Perform Work in compliance with:
 - a. Codes listed on Drawings.
 - b. National, state and local barrier free codes, laws and ordinances.
 - c. NFPA 101 - Life Safety Code and other applicable NFPA Standards.
- C. Contractor shall, without additional expense to Owner, obtain necessary licenses and permits, and be responsible for complying with Federal, state, county, and municipal laws, codes, and regulations applicable to performance of Work, including, but not limited to, laws or regulations requiring use of licensed contractors to perform parts of Work.
- D. Occupancy Permit: The General Contractor shall be responsible for securing a Certificate of Occupancy permit at completion of project and shall deliver such permit to Owner. Final Payment shall be retained until permit has been received by Owner.

1.4 FIRE-RESISTANCE REQUIREMENTS

- A. Fire Resistance Ratings and Fire Tests: Fire-resistance ratings of building elements, components, and assemblies shall be determined only in accordance with the test procedures set forth in ASTM E 119 or UL 263, or by alternative methods approved by applicable authorities having jurisdiction.
1. Fire-resistance ratings shall be determined or listed based on fire tests performed by one of the following testing agencies, or other agencies acceptable to governing authorities having jurisdiction.
 - a. Factory Mutual Laboratories.
 - b. Intertek
 - c. Southwest Research Institute.
 - d. Underwriters Laboratories, Inc.
 2. Where reference is made to only one testing authority, equivalent fire ratings as determined or listed by another testing agency are acceptable if approved by applicable authorities having jurisdiction.
- B. Marking and Identification: Fire walls, fire barriers, fire partitions, smoke barriers, and smoke partitions, or any other walls required to have protected openings or penetrations, shall be permanently identified with signage or stenciling. Such identification shall:
1. Be located in accessible floor plenums, ceiling plenums, or attic spaces.
 2. Be repeated at intervals not exceeding 30 feet o.c., measured horizontally along the partition or wall.
 3. Include lettering not less than 0.5- inch in height, worded as follows: "FIRE AND/OR SMOKE BARRIER - PROTECT ALL OPENINGS AND PENETRATIONS."

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 014200

REFERENCES

PART 1 - GENERAL

1.1 DEFINITIONS

- A. "Acceptance", "acceptable", or words of similar import: Acceptance, acceptable or similar words shall be as determined by Architect.
- B. "Approved": When used to convey Architect's action on Contractor's submittals, applications, and requests, "approved" is limited to Architect's duties and responsibilities as stated in the Conditions of the Contract. Approval does not release Contractor from responsibility to fulfill Contract Document requirements.
- C. "At no extra cost to Owner", "With no extra compensation to Contractor", "At Contractor's own expense", or words of similar import: Terms shall be understood to mean that Contractor shall perform or provide specified operation of Work at no increase to Contract Sum stated in executed Contract.
- D. "Directed": A command or instruction by Architect. Other terms including "requested," "authorized," "selected," "required," and "permitted" have the same meaning as "directed."
- E. "Furnish": Supply and deliver to Project site, ready for unloading, unpacking, assembly, installation, and similar operations.
- F. "Indicated": Requirements expressed by graphic representations or in written form on Drawings, in Specifications, and in other Contract Documents. Other terms including "shown," "noted," "scheduled," and "specified" have the same meaning as "indicated." No limitation on location is intended except as specifically noted.
- G. "Install": Operations at Project site including unloading, temporarily storing, unpacking, assembling, erecting, placing, anchoring, applying, working to dimension, finishing, curing, protecting, cleaning, and similar operations.
- H. "Installer" is entity engaged by Contractor, either as employee, subcontractor or sub-subcontractor for performance of particular construction activity, including installation, erection, application and similar operations. Installers are required to be experienced in operations they are engaged to perform. Term "experienced", when used with term "installer", means having minimum five previous projects similar in size and scope to this Project, and familiar with precautions required, and has complied with requirements of authority having jurisdiction.
- I. "NIC": Work of this Project which is not being performed or provided as part of Contract; term shall mean "Not in This Contract" or "Not Part of Work to be Performed"

or Provided by Contractor". "NIC" work is indicated as aid to Contractor in scheduling amount of time and materials necessary for completion of Contract.

- J. "Other acceptable manufacturer", "equal", "acceptable equal", "equivalent", or words of similar import: It shall be understood that words are followed by expression "at sole discretion of Architect" even though words may not appear in print.
- K. "Perform": Contractor, at his own expense, shall perform operations necessary to complete Work, including furnishing of necessary labor, tools and equipment, and further including and installing of materials indicated, specified or required to complete performance.
- L. "Project Site" is space available to Contractor for performance of Work, either exclusively or in conjunction with others performing construction as part of Project. Extent of Project Site is shown on Contract Drawings and may or may not be identical with description of land upon which Project is to be built.
- M. "Provide": Contractor, at his own expense, shall furnish and install Work complete in place and ready for use, including furnishing of necessary labor, materials, tools, equipment and transportation. Definitions apply same to future, present and past tenses, except word "provided" may mean "contingent upon" where context is apparent.
- N. "Regulations": Laws, ordinances, statutes, and lawful orders issued by authorities having jurisdiction, and rules, conventions, and agreements within the construction industry that control performance of the Work.
- O. "Require" and words of similar import: As required to complete Work and as required by Architect.
- P. "Testing Laboratory" is independent entity engaged to perform specific inspections or tests, either at Project Site or elsewhere, and to report on, and if required, to interpret results of those inspections or tests.

1.2 SPECIFICATION SENTENCE STRUCTURE

- A. Specifications are written in modified brief style. In general, words "the", "a", "an", "shall", "shall be", and "all" are not used. Requirements indicated and specified apply to work of same kind, class, and type even though word "all" is not stated.
- B. Simple imperative mood of sentence structure is used in Specification Sections which places verb as first word in sentence. Where "perform", "provide", "install", "erect", "furnish", "connect", "test", or words of similar import are used, it shall be understood that words include meanings of phrase "Contractor Shall..." before words.
- C. Standard paragraph titles and other identifications of subject matter in Specifications are intended as aid in locating and recognizing various requirements in Specifications. Titles do not define, limit or otherwise restrict Specifications text. Capitalizing of words in text does not signify or mean that words convey special or unique meanings having

precedence over other parts of Contract Documents. Specification text shall govern over titling and shall be understood to be interpreted as a whole.

1.3 DOCUMENT ORGANIZATION

- A. Organization of Project Manual and Contract Drawings are not intended to control or to lessen responsibility of Contractor in dividing Work among his subcontractors, or in establishing extent of Work to be performed by each trade.

1.4 SYMBOLS

- A. Graphic symbols used in Contract Documents are those symbols recognized in construction industry for indicated purposes. Where not otherwise noted, symbols are those defined in "Architectural Graphics Standards", published by John Wiley & Sons, Inc., Eighth Edition.
- B. Graphic symbols used on mechanical and electrical drawings are generally aligned with symbols recommended by ASHRAE. Where appropriate, mechanical and electrical symbols are supplemented by more specific symbols recommended by technical associations including ASME, ASPE, IEEE, and similar organizations. Request clarification from Architect if symbols are unfamiliar.

1.5 REFERENCE STANDARDS

- A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into the Contract Documents to the extent referenced. Such standards are made a part of the Contract Documents by reference.
 - 1. When conflict exists between requirements of reference standards and Contract Documents, request clarification from Architect before proceeding.
- B. Publication Dates: Comply with standards in effect as of date of the Contract Documents unless otherwise indicated.
- C. Copies of Standards: Each entity engaged in construction on Project should be familiar with industry standards applicable to its construction activity. Copies of applicable standards are not bound with the Contract Documents.
 - 1. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.
 - 2. Make reference standards available as requested or required by Architect or Owner. Maintain copies of standard at project site throughout construction period.

1.6 ABBREVIATIONS AND ACRONYMS

- A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. This information is subject to change and is believed to be accurate as of the date of the Contract Documents.

1. AABC - Associated Air Balance Council; www.aabc.com.
2. AAMA - American Architectural Manufacturers Association; www.aamanet.org.
3. AATCC - American Association of Textile Chemists and Colorists; www.aatcc.org.
4. ACI - American Concrete Institute; (Formerly: ACI International); www.abma.com.
5. AEIC - Association of Edison Illuminating Companies, Inc. (The); www.aeic.org.
6. AF&PA - American Forest & Paper Association; www.afandpa.org.
7. AHAM - Association of Home Appliance Manufacturers; www.aham.org.
8. AHRI - Air-Conditioning, Heating, and Refrigeration Institute (The); www.ahrinet.org.
9. AIA - American Institute of Architects (The); www.aia.org.
10. AISC - American Institute of Steel Construction; www.aisc.org.
11. AISI - American Iron and Steel Institute; www.steel.org.
12. AITC - American Institute of Timber Construction; www.aitc-glulam.org.
13. AMCA - Air Movement and Control Association International, Inc.; www.amca.org.
14. ANSI - American National Standards Institute; www.ansi.org.
15. APA - APA - The Engineered Wood Association; www.apawood.org.
16. APA - Architectural Precast Association; www.archprecast.org.
17. ARI - Air-Conditioning & Refrigeration Institute; (See AHRI).
18. ARI - American Refrigeration Institute; (See AHRI).
19. ASCE - American Society of Civil Engineers; www.asce.org.
20. ASCE/SEI - American Society of Civil Engineers/Structural Engineering Institute; (See ASCE).
21. ASHRAE - American Society of Heating, Refrigerating and Air-Conditioning Engineers; www.ashrae.org.
22. ASME - ASME International; (American Society of Mechanical Engineers); www.asme.org.
23. ASSE - American Society of Safety Engineers (The); www.asse.org.
24. ASSE - American Society of Sanitary Engineering; www.asse-plumbing.org.
25. ASTM - ASTM International; www.astm.org.
26. ATIS - Alliance for Telecommunications Industry Solutions; www.atis.org.
27. AWI - Architectural Woodwork Institute; www.awinet.org.
28. AWMAC - Architectural Woodwork Manufacturers Association of Canada; www.awmac.com.
29. AWWA - American Water Works Association; www.awwa.org.
30. AWS - American Welding Society; www.aws.org.
31. AWWA - American Water Works Association; www.awwa.org.
32. BHMA - Builders Hardware Manufacturers Association; www.buildershardware.com.
33. BIA - Brick Industry Association (The); www.gobrick.com.
34. BIFMA - BIFMA International; (Business and Institutional Furniture Manufacturer's Association); www.bifma.org.

35. CEA - Consumer Electronics Association; www.ce.org.
36. CFFA - Chemical Fabrics and Film Association, Inc.; www.chemicalfabricsandfilm.com.
37. CFSEI - Cold-Formed Steel Engineers Institute; www.cfsei.org.
38. CISCA - Ceilings & Interior Systems Construction Association; www.cisca.org.
39. CISPI - Cast Iron Soil Pipe Institute; www.cispi.org.
40. CLFMI - Chain Link Fence Manufacturers Institute; www.chainlinkinfo.org.
41. CRI - Carpet and Rug Institute (The); www.carpet-rug.org.
42. CRSI - Concrete Reinforcing Steel Institute; www.crsi.org.
43. CSI - Construction Specifications Institute (The); www.csinet.org.
44. CTI - Cooling Technology Institute; (Formerly: Cooling Tower Institute); www.cti.org.
45. CWC - Composite Wood Council; (See CPA).
46. DASMA - Door and Access Systems Manufacturers Association; www.dasma.com.
47. DHI - Door and Hardware Institute; www.dhi.org.
48. ECA - Electronic Components Association; (See ECIA).
49. ECAMA - Electronic Components Assemblies & Materials Association; (See ECIA).
50. ECIA - Electronic Components Industry Association; www.eciaonline.org.
51. EIA - Electronic Industries Alliance; (See TIA).
52. EJMA - Expansion Joint Manufacturers Association, Inc.; www.ejma.org.
53. ESD - ESD Association; (Electrostatic Discharge Association); www.esda.org.
54. FM Approvals - FM Approvals LLC; www.fmglobal.com.
55. GA - Gypsum Association; www.gypsum.org.
56. GANA - Glass Association of North America; www.glasswebsite.com.
57. GS - Green Seal; www.greenseal.org.
58. HMMA - Hollow Metal Manufacturers Association; (See NAAMM).
59. HPVA - Hardwood Plywood & Veneer Association; www.hpva.org.
60. HPW - H. P. White Laboratory, Inc.; www.hpwhite.com.
61. IAS - International Accreditation Service; www.iasonline.org.
62. IAS - International Approval Services; (See CSA).
63. ICBO - International Conference of Building Officials; (See ICC).
64. ICC - International Code Council; www.iccsafe.org.
65. IEEE - Institute of Electrical and Electronics Engineers, Inc. (The); www.ieee.org.
66. IES - Illuminating Engineering Society; (Formerly: Illuminating Engineering Society of North America); www.ies.org.
67. IESNA - Illuminating Engineering Society of North America; (See IES).
68. IEST - Institute of Environmental Sciences and Technology; www.iest.org.
69. IGMA - Insulating Glass Manufacturers Alliance; www.igmaonline.org.
70. ILI - Indiana Limestone Institute of America, Inc.; www.ili.ai.com.
71. Intertek - Intertek Group; (Formerly: ETL SEMCO; Intertek Testing Service NA); www.intertek.com.
72. ISFA - International Surface Fabricators Association; (Formerly: International Solid Surface Fabricators Association); www.isfanow.org.
73. ISO - International Organization for Standardization; www.iso.org.
74. ISSFA - International Solid Surface Fabricators Association; (See ISFA).
75. ITU - International Telecommunication Union; www.itu.int/home.
76. KCMA - Kitchen Cabinet Manufacturers Association; www.kcma.org.
77. LMA - Laminating Materials Association; (See CPA).
78. LPI - Lightning Protection Institute; www.lightning.org.

79. MCA - Metal Construction Association; www.metalconstruction.org.
80. MFMA - Maple Flooring Manufacturers Association, Inc.; www.maplefloor.org.
81. MFMA - Metal Framing Manufacturers Association, Inc.;
www.metalframingmfg.org.
82. MIA - Marble Institute of America; www.marble-institute.com.
83. MMPA - Moulding & Millwork Producers Association; www.wmmpa.com.
84. MPI - Master Painters Institute; www.paintinfo.com.
85. NAAMM - National Association of Architectural Metal Manufacturers;
www.naamm.org.
86. NADCA - National Air Duct Cleaners Association; www.nadca.com.
87. NAIMA - North American Insulation Manufacturers Association; www.naima.org.
88. NBGQA - National Building Granite Quarries Association, Inc.; www.nbgqa.com.
89. NBI - New Buildings Institute; www.newbuildings.org.
90. NCMA - National Concrete Masonry Association; www.ncma.org.
91. NECA - National Electrical Contractors Association; www.necanet.org.
92. NeLMA - Northeastern Lumber Manufacturers Association; www.nelma.org.
93. NEMA - National Electrical Manufacturers Association; www.nema.org.
94. NETA - InterNational Electrical Testing Association; www.netaworld.org.
95. NFPA - National Fire Protection Association; www.nfpa.org.
96. NFPA - NFPA International; (See NFPA).
97. NFRC - National Fenestration Rating Council; www.nfrc.org.
98. NHLA - National Hardwood Lumber Association; www.nhla.com.
99. NLGA - National Lumber Grades Authority; www.nlga.org.
100. NOFMA - National Oak Flooring Manufacturers Association; (See NWFA).
101. NOMMA - National Ornamental & Miscellaneous Metals Association;
www.nomma.org.
102. NRMCA - National Ready Mixed Concrete Association; www.nrmca.org.
103. NSF - NSF International; www.nsf.org.
104. NSPE - National Society of Professional Engineers; www.nspe.org.
105. NTMA - National Terrazzo & Mosaic Association, Inc. (The); www.ntma.com.
106. NWFA - National Wood Flooring Association; www.nwfa.org.
107. PLASA - PLASA; (Formerly: ESTA - Entertainment Services and Technology
Association); www.plasa.org.
108. RFCI - Resilient Floor Covering Institute; www.rfci.com.
109. SAE - SAE International; www.sae.org.
110. SDI - Steel Deck Institute; www.sdi.org.
111. SDI - Steel Door Institute; www.steeldoors.org.
112. SEI/ASCE - Structural Engineering Institute/American Society of Civil Engineers;
(See ASCE).
113. SMACNA - Sheet Metal and Air Conditioning Contractors' National Association;
www.smacna.org.
114. SPFA - Spray Polyurethane Foam Alliance; www.sprayfoam.org.
115. SPIB - Southern Pine Inspection Bureau; www.spib.org.
116. SSINA - Specialty Steel Industry of North America; www.ssina.com.
117. SSPC - SSPC: The Society for Protective Coatings; www.sspc.org.
118. SWI - Steel Window Institute; www.steelwindows.com.
119. TCNA - Tile Council of North America, Inc.; www.tileusa.com.
120. TIA - Telecommunications Industry Association (The); (Formerly: TIA/EIA -
Telecommunications Industry Association/Electronic Industries Alliance);
www.tiaonline.org.

121. TIA/EIA - Telecommunications Industry Association/Electronic Industries Alliance; (See TIA).
 122. TMS - The Masonry Society; www.masonrysociety.org.
 123. UL - Underwriters Laboratories Inc.; www.ul.com.
 124. WCLIB - West Coast Lumber Inspection Bureau; www.wclib.org.
 125. WCMA - Window Covering Manufacturers Association; www.wcmanet.org.
 126. WDMA - Window & Door Manufacturers Association; www.wdma.com.
 127. WI - Woodwork Institute; www.wicnet.org.
 128. WWPA - Western Wood Products Association; www.wwpa.org.
- B. Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. This information is believed to be accurate as of the date of the Contract Documents.
1. DIN - Deutsches Institut fur Normung e.V.; www.din.de.
 2. IAPMO - International Association of Plumbing and Mechanical Officials; www.iapmo.org.
 3. ICC - International Code Council; www.iccsafe.org.
 4. ICC-ES - ICC Evaluation Service, LLC; www.icc-es.org.
- C. Federal Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. Information is subject to change and is up to date as of the date of the Contract Documents.
1. COE - Army Corps of Engineers; www.usace.army.mil.
 2. CPSC - Consumer Product Safety Commission; www.cpsc.gov.
 3. DOC - Department of Commerce; National Institute of Standards and Technology; www.nist.gov.
 4. DOD - Department of Defense; www.quicksearch.dla.mil.
 5. DOE - Department of Energy; www.energy.gov.
 6. EPA - Environmental Protection Agency; www.epa.gov.
 7. GSA - General Services Administration; www.gsa.gov.
 8. LBL - Lawrence Berkeley National Laboratory; Environmental Energy Technologies Division; www.eetd.lbl.gov.
 9. OSHA - Occupational Safety & Health Administration; www.osha.gov.
 10. SD - Department of State; www.state.gov.
 11. USDA - Department of Agriculture; Rural Utilities Service; www.usda.gov.
 12. USDJ - Department of Justice; Office of Justice Programs; National Institute of Justice; www.ojp.usdoj.gov.
 13. USPS - United States Postal Service; www.usps.com.
- D. Standards and Regulations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the standards and regulations in the following list. This information is subject to change and is believed to be accurate as of the date of the Contract Documents.
1. CFR - Code of Federal Regulations; Available from Government Printing Office; www.gpo.gov/fdsys.

2. DOD - Department of Defense; Military Specifications and Standards; Available from DLA Document Services; www.quicksearch.dla.mil.
 3. DSCC - Defense Supply Center Columbus; (See FS).
 4. FED-STD - Federal Standard; (See FS).
 5. FS - Federal Specification; Available from DLA Document Services; www.quicksearch.dla.mil.
 - a. Available from Defense Standardization Program; www.dsp.dla.mil.
 - b. Available from General Services Administration; www.gsa.gov.
 - c. Available from National Institute of Building Sciences/Whole Building Design Guide; www.wbdg.org/ccb.
 6. MILSPEC - Military Specification and Standards; (See DOD).
 7. USAB - United States Access Board; www.access-board.gov.
- E. State Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. This information is subject to change and is believed to be accurate as of the date of the Contract Documents.
1. CDHS; California Department of Health Services; (See CDPH).
 2. CDPH; California Department of Public Health; Indoor Air Quality Program; www.cal-iaq.org.
 3. SCAQMD; South Coast Air Quality Management District; www.aqmd.gov.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 014339

MOCKUPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Administrative requirements for mockups specified in other Sections.

1.2 DEFINITIONS

A. Mockups: Physical assemblies of portions of the Work constructed to establish the standard by which the Work will be judged. Mockups are not Samples.

1. Mockups are used for one or more of the following:
 - a. Verify selections made under Sample submittals.
 - b. Demonstrate aesthetic effects.
 - c. Demonstrate the qualities of products and workmanship.
 - d. Demonstrate successful installation of interfaces between components and systems.
 - e. Perform preconstruction testing to determine system performance.
2. Product Mockups: Mockups that may include multiple products, materials, or systems specified in a single Section.
3. In-Place Mockups: Mockups constructed on-site in their actual final location as part of permanent construction.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Meet with Owner, Architect, testing and inspecting agency representative, and installers of whose Work is included in mockups.
2. Review locations and extent of mockups.
3. Review and finalize schedule for mockups, and verify availability of materials, personnel, equipment, and facilities needed to complete mockups and maintain schedule for the Work.

1.4 QUALITY ASSURANCE

- A. Build mockups to do the following:
 - 1. Verify selections made under Sample submittals.
 - 2. Demonstrate aesthetic effects.
 - 3. Demonstrate the qualities of products and workmanship.
 - 4. Demonstrate acceptable coordination between components and systems.
- B. Fabrication: Use materials and installation methods as required for the Work.
 - 1. Build mockups of size indicated.
 - 2. Build mockups in location indicated or, if not indicated, as directed by Architect.
 - 3. Employ supervisory personnel who will oversee mockup construction. Employ workers who will be employed to perform same tasks during the construction at Project.
 - 4. Demonstrate the proposed range of aesthetic effects and workmanship.
 - 5. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
- C. Notifications:
 - 1. Notify Architect seven days in advance of the dates and times when mockups will be constructed.
 - 2. Allow three days for initial review and each re-review of each mockup.
- D. Approval: Obtain Architect's approval of mockups before starting fabrication or construction of corresponding Work.
 - 1. Unless otherwise indicated, approved mockups establish the standard by which the Work will be judged.
 - 2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
 - 3. Subject to compliance with requirements and unless otherwise directed, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion
 - a. Demolish and remove mockups where directed in other Sections.

1.5 COORDINATION

- A. Coordinate schedule for construction of mockups, so construction and review of mockups do not impact Project schedule.

PART 2 - PRODUCTS

2.1 MOCKUPS

- A. Construct mockups to demonstrate constructability, coordination of trades, and sequencing of Work; and to ensure materials, components, subassemblies, assemblies, and interfaces integrate into a system complying with indicated performance and aesthetic requirements.
- B. Use specified products that have been approved by Architect. Coordinate installation of materials and products specified in individual Specification Sections that include Work included in mockups.
- C. Photographic Documentation: Document construction of integrated exterior mockups with photographs in accordance with Section 01 32 33 "Photographic Documentation." Provide photographs showing details of interface of different materials and assemblies.
- D. Retain approved mockups constructed in place. Incorporate fully into the Work.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 014536
CONCRETE IMAGING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Procedural requirements for the nondestructive examination (NDE) of concrete, three-dimensionally locating, mapping and identifying embedments in the concrete in advance of drilling, coring, cutting or breaking activities.
 - 1. NDE services include inspections, tests, and related actions, including reports performed by an approved, independent testing agency.
 - 2. NDE services are required to verify compliance with requirements specified or indicated in the contract documents. These services do not relieve Contractor of responsibility for compliance with Contract Document requirements.
- B. Requirements for the Contractor to provide other quality-control services required by Architect, Owner, or authorities having jurisdiction are not limited by provisions of this Section.
- C. Related Work: The following Sections contain requirements that relate to this Section:
 - 1. Section 01 73 29 "Cutting and Patching" specifies requirements for repair and restoration of construction disturbed by inspection and testing activities.
 - 2. Section 01 33 00 "Submittal Procedures" specifies requirements for development of a schedule of required tests and inspections.

1.2 RESPONSIBILITIES

- A. Contractor Responsibilities: Unless otherwise indicated as the responsibility of another identified entity, Contractor shall provide all NDE inspections, tests, and other quality-control services specified elsewhere in the Contract Documents and required by authorities having jurisdiction. Costs for these services are included in the Contract Sum.
- B. Retesting: The Contractor is responsible for retesting where results of NDE inspections, tests, or other quality-control services prove unsatisfactory and indicate noncompliance with Contract Document requirements, regardless of whether the original test was Contractor's responsibility.
- C. Associated Services: Cooperate with agencies performing required inspections, tests, and similar services, and provide reasonable auxiliary services as requested. Notify the NDE agency sufficiently in advance of operations to permit assignment of personnel. Auxiliary services required include, but are not limited to, the following:
 - 1. Provide access to the Work.

2. Furnish incidental labor and facilities necessary to facilitate inspections and tests.
 3. Provide the NDE agency with available as-built and project design documentation as requested by the agency.
- D. Duties of the NDE Testing Agency: The independent agency engaged to perform concrete NDE inspections and surveys shall cooperate with the Architect and the Contractor in performance of the agency's duties. The testing agency shall provide qualified personnel to perform required inspections and tests.
1. The agency shall notify the Architect and the Contractor promptly of potential clashes between proposed penetrations of the subject concrete and embedments observed in the Work during performance of its services.
 2. The agency is not authorized to release, revoke, alter, or enlarge requirements of the Contract Documents or approve or accept any portion of the Work.
 3. The agency shall not perform any duties of the Contractor.
- E. Coordination: Coordinate the sequence of activities to accommodate required NDE services with a minimum of delay. Coordinate activities to avoid the necessity of removing and replacing construction to accommodate inspections and tests.
- F. The Contractor is responsible for scheduling times for inspections, tests, taking samples, and similar activities.

1.3 SUBMITTALS

- A. The independent NDE testing agency shall submit a written report, in duplicate, of each inspection, test, survey, or similar service to the Architect. If the Contractor is responsible for the service, submit a certified written report, in duplicate, of each inspection, test, or similar service through the Contractor.
1. Submit additional copies of each written report directly to the governing authority, when the authority so directs.
 2. Report Data: Written reports of each inspection, test, or similar service include, but are not limited to, the following:
 - a. Date of issue.
 - b. Project title and number.
 - c. Name, address, and telephone number of testing agency.
 - d. Dates and locations of tests or inspection
 - e. Names of individuals making the inspection or test.
 - f. Designation of the Work and test method(s).
 - g. Complete NDE test data with annotations.
 - h. Survey results and an interpretation of the results.
 - i. Site conditions at the time of NDE testing.
 - j. Comments or professional opinions on whether proposed concrete penetrations clash or interfere with observed embedments in the concrete.
 - k. Name and signature of NDE report preparer.
 - l. Recommendations on retesting.

1.4 QUALITY ASSURANCE

A. Qualifications for NDE Service Agencies:

1. NDE agency, shall possess a minimum of ten (10) years demonstrated successful experience in the use of ground penetrating radar (GPR) for the three-dimensional locating, mapping and identification of concrete embedments for similar, mission-critical, facilities.
2. GPR field technicians and analysts shall possess a minimum of five years (5) demonstrated successful experience in the use of ground penetrating radar (GPR) for the three-dimensional locating, mapping and identification of concrete embedments for similar, mission-critical, facilities.
3. Demonstrated ability to accurately, dimensionally-locate concrete survey and data collection areas.
4. Ability to archive all project reports, collected data, and all related documents for a period of not less than five (5) years from completion of the NDE Work.
5. Professional Liability Insurance ("Errors and Omissions") with minimum limits of \$2 million per occurrence / \$2 million annual aggregate.
6. NDE Work shall be performed in accordance with the applicable sections of ASTM D6432, CSDA Best Practice BP-007, and as amended herein.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

3.1 NONDESTRUCTIVE EXAMINATION (NDE) OF CONCRETE

- A. Ground penetrating radar (GPR) shall be the primary NDE investigative tool deployed for the three-dimensional (3D) imaging of concrete.
- B. Prior to any drilling, coring, cutting, or breaking of concrete, the area shall be imaged utilizing GPR by an approved NDE agency and evaluated by the agency for potential clashes with embedded elements in the concrete (conduit, piping, reinforcing bar, post tensioning tendons, etc.).

3.2 GROUND PENETRATING RADAR (GPR)

- A. GPR system shall utilize a ground coupled antenna system with minimum center frequency of 1000 MHz.
- B. Data shall be collected utilizing defined, bi-directional grids with supplemental line scans being performed as deemed necessary by the operator. Data shall be collected on a grid basis, unless physical constraints restrict the data collection process to only line scans.

- C. Grid locations shall be dimensionally located relative to fixed building or site elements. The grid origin position and grid axis orientation shall be identified and recorded.
- D. Data shall be initially processed on-site, with observed embedments "real-time" marked on the concrete surface. Marking methods and materials shall be approved by the Owner in advance of the NDE services commencing. Marks shall be documented using digital photography.
- E. Data shall be digitally recorded and post-processed utilizing the system manufacturer's analytical software. Line scan images and GPR plan map images shall be annotated to convey the nature and three-dimensional location of observed embedments. Annotated images shall be included as part of each written report.

3.3 RESTRICTION ON LOCATION OF PROPOSED CONCRETE PENETRATIONS

- A. Concrete penetrations shall not be made within four (4) inches of any fixed obstruction (wall, etc.) due to the physical limitation presented by the typical GPR antenna dimensions.

END OF SECTION

SECTION 015000

TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

1.2 USE CHARGES

- A. Installation and removal of and use charges for temporary facilities shall be included in the Contract Sum unless otherwise indicated. Allow other entities engaged in the Project to use temporary services and facilities without cost, including, but not limited to, Owner's construction forces, Architect, occupants of Project, testing agencies, and authorities having jurisdiction.
- B. Sewer Service: Pay sewer service use charges for sewer usage by all entities for construction operations. Provide connections and extensions of services as required for construction operations.
- C. Water Service: Pay water service use charges for water used by all entities for construction operations. Provide connections and extensions of services and metering as required for construction operations.
- D. Electric Power Service: Pay electric power-service use charges for electricity used by all entities for construction operations. Provide connections and extensions of services and metering as required for construction operations.

1.3 INFORMATIONAL SUBMITTALS

- A. Site Utilization Plan: Show temporary facilities, temporary utility lines and connections, staging areas, construction site entrances, vehicle circulation, and parking areas for construction personnel.
- B. Project Identification and Temporary Signs: Show fabrication and installation details, including plans, elevations, details, layouts, typestyles, graphic elements, and message content.
- C. Fire-Safety Program: Show compliance with requirements of NFPA 241 and authorities having jurisdiction. Indicate Contractor personnel responsible for management of fire-prevention program.
- D. Moisture- and Mold-Protection Plan: Describe procedures and controls for protecting materials and construction from water absorption and damage and mold. Describe

delivery, handling, storage, installation, and protection provisions for materials subject to water absorption or water damage.

1. Indicate procedures for discarding water-damaged materials, protocols for mitigating water intrusion into completed Work, and requirements for replacing water-damaged Work.
2. Indicate sequencing of work that requires water, such as sprayed fire-resistive materials, plastering, and terrazzo grinding, and describe plans for dealing with water from these operations. Show procedures for verifying that wet construction has dried sufficiently to permit installation of finish materials.
3. Indicate methods to be used to avoid trapping water in finished work.

1.4 QUALITY ASSURANCE

- A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.
- B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.
- C. Accessible Temporary Egress: Comply with applicable provisions in the United States Access Board's ADA-ABA Accessibility Guidelines and ICC/ANSI A117.1.

1.5 PROJECT CONDITIONS

- A. Temporary Use of Permanent Facilities: Engage Installer of each permanent service to assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before Owner's acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 TEMPORARY FACILITIES

- A. Field Offices: Prefabricated, mobile units, or designated portion of building with serviceable finishes, temperature controls, and foundations adequate for normal loading.
- B. Common Use Field Office: Of sufficient size to accommodate needs of Owner, Architect, and construction personnel office activities and to accommodate Project meetings specified in other Division 1 Sections. Keep office clean and orderly. Furnish and equip offices as follows:
 1. Furniture required for Project site documents including file cabinets, plan tables, plan racks, and bookcases.
 2. Conference room of sufficient size to accommodate meetings of 15 individuals. Provide electrical power service and 120-V ac duplex receptacles, with no fewer

- than one receptacle on each wall. Furnish room with conference table, chairs, and 4 foot (1.2 m) square tack and marker boards.
 - 3. Drinking water and private toilet.
 - 4. Heating and cooling equipment necessary to maintain a uniform indoor temperature of 68 degrees F to 72 degrees F (20 degrees C to 22 degrees C).
 - 5. Lighting fixtures capable of maintaining average illumination of 20 fc (215 lx) at desk height.
- C. Storage and Fabrication Sheds: Provide sheds sized, furnished, and equipped to accommodate materials and equipment for construction operations.
- 1. Store combustible materials apart from building.

2.2 EQUIPMENT

- A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.
- B. HVAC Equipment: Use of Owner's existing permanent HVAC system is permitted to the extent it is available during the work.
- 1. Use of gasoline burning space heaters, open flame heaters, or salamander type heating units is prohibited.
 - 2. Heating Units: Listed and labeled for type of fuel being consumed, by a qualified testing agency acceptable to authorities having jurisdiction and marked for intended location and application.
 - 3. Permanent HVAC System: If Owner authorizes use of permanent HVAC system for temporary use during construction, provide filter with minimum MERV of 8 at each return air grille in system, remove at end of construction, and clean HVAC system as required in Section 01 77 00.
- C. Air-Filtration Units: Primary and secondary HEPA-filter-equipped portable units with four-stage filtration. Provide single switch for emergency shutoff. Configure to run continuously.

PART 3 - EXECUTION

3.1 TEMPORARY FACILITIES, GENERAL

- A. Conservation: Coordinate construction and use of temporary facilities with consideration given to conservation of energy, water, and materials. Coordinate use of temporary utilities to minimize waste.

3.2 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.3 TEMPORARY UTILITY INSTALLATION

- A. Install temporary service or connect to existing service.
 - 1. Arrange with utility company, Owner, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.
 - 2. When tapping into Owner's existing services, coordinate with Owner's Representative before tapping into existing utility services available in the facility.
- B. Electric Power Service: Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.
 - 1. Install electric power service underground unless otherwise indicated.
 - 2. Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.
 - 3. Connect temporary service to Owner's existing power source as directed by Owner.
- C. Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.
 - 1. Connect temporary sewers to municipal system as directed by authorities having jurisdiction.
- D. Water Service: Install water service and distribution piping in sizes and pressures adequate for construction.
- E. Sanitary Facilities: Provide temporary toilets, wash facilities, and drinking water for use of construction personnel. Comply with requirements of authorities having jurisdiction for type, number, location, operation, and maintenance of fixtures and facilities.
 - 1. Toilets: Use of Owner's existing toilet facilities will be permitted, provided facilities are cleaned and maintained in a sanitary condition acceptable to Owner. At Substantial Completion, restore these facilities to condition existing before initial use.
 - a. Provide toilet paper, paper towels, and related accessories and maintenance items.
 - b. Thoroughly clean and disinfect toilet rooms daily.
 - c. At the end of the work, thoroughly clean and sanitize, disinfect, and restore toilet rooms to like new condition. Replace plumbing fixtures and toilet accessories broken, damaged, or removed by construction personnel. If necessary, repaint paintable surfaces.
- F. Temporary Heating and Cooling: Provide temporary heating and cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select

equipment that will not have a harmful effect on completed installations or elements being installed.

1. Provide temporary dehumidification systems when required to reduce ambient and substrate moisture levels to level required to allow installation or application of finishes and their proper curing or drying.
- G. Isolation of Work Areas in Occupied Facilities: Prevent dust, fumes, and odors from entering occupied areas.
1. Prior to commencing work, isolate the HVAC system in area where work is to be performed.
 - a. Disconnect supply and return ductwork in work area from HVAC systems servicing occupied areas.
 - b. Maintain negative air pressure within work area using HEPA equipped air filtration units, starting with commencement of temporary partition construction, and continuing until removal of temporary partitions is complete.
 2. Maintain dust partitions during the work. Use vacuum collection attachments on dust producing equipment. Isolate limited work within occupied areas using portable dust containment devices.
 3. Perform daily construction cleanup and final cleanup using approved, HEPA filter equipped vacuum equipment.
- H. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.
1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.
- I. Telephone Service: Provide temporary telephone service in common use facilities for use by all construction personnel. Install Wi-Fi cell phone access equipment and at least one land-based telephone line(s) for each field office.
1. Provide additional telephone lines as determined by usage.
 2. At each telephone, post a list of important telephone numbers.
 - a. Police and fire departments.
 - b. Ambulance service.
 - c. Contractor's home office.
 - d. Contractor's emergency afterhours telephone number.
 - e. Architect's office.
 - f. Engineers' offices.
 - g. Owner's office.
 - h. Principal subcontractors' field and home offices.
 3. Include costs for installation, maintenance and removal, and service charges for local calls. Toll charges shall be paid by party who places call.

- J. Electronic Communication Service: Provide a desktop computer in the primary field office adequate for use by Architect and Owner to access Project electronic documents and maintain electronic communications. Equip computer with not less than the following:
1. Processor: Intel Core i5 or i7.
 2. Memory: 16 gigabytes.
 3. Disk Storage: 1 terabyte hard disk drive and combination DVD-RW/CD-RW drive.
 4. Display: 24-inch (610 mm) LCD monitor with 256-Mb dedicated video RAM.
 5. Full size keyboard and mouse.
 6. Network Connectivity: 10/100BaseT Ethernet or Gigabit.
 7. Operating System: Microsoft Windows 10 Professional.
 8. Productivity Software:
 - a. Microsoft Office Professional, 2013 or higher, including Word, Excel, and Outlook.
 - b. Adobe Reader 11.0 or higher.
 - c. WinZip 10.0 or higher.
 9. Printer: All in one unit equipped with printer server, combining color printing, photocopying, scanning, and faxing, or separate units for each of these three functions.
 10. Internet Service: Broadband modem, router and ISP, equipped with hardware firewall, providing minimum 10.0 Mbps upload and 15 Mbps download speeds at each computer.
 11. Internet Security: Integrated software, providing software firewall, virus, spyware, phishing, and spam protection in a combined application.
 12. Backup: External hard drive, minimum 2 terabyte, with automated backup software providing daily backups.

3.4 SUPPORT FACILITIES INSTALLATION

- A. Install and maintain support facilities until Substantial Completion inspection. Remove prior to Substantial Completion. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Owner.
- B. Temporary Facilities and Services:
1. Provide construction for temporary offices, shops, and sheds located within construction area or within 30 feet (9 m) of building lines that is noncombustible according to ASTM E 136. Comply with NFPA 241.
 2. Maintain support facilities until Architect schedules Substantial Completion inspection. Remove before Substantial Completion. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Owner.
- C. Parking: Provide temporary parking areas for construction personnel.
1. Parking is limited. Provide temporary parking unless Owner designates parking areas for construction personnel.

- D. Project Signs: Provide Project signs as indicated. Unauthorized signs are not permitted.
 - 1. Identification Signs: Provide Project identification signs as indicated on Drawings.
 - 2. Temporary Signs: Provide other signs indicated and as required to inform public and individuals seeking entrance to Project.
 - a. Provide temporary, directional signs for construction personnel and visitors.
 - 3. Maintain and touch up signs so they are legible at all times.
- E. Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.
 - 1. Truck cranes and similar devices used for hoisting materials are considered tools and equipment and are not temporary facilities.
- F. Existing Elevator Use: Use of Owner's existing elevators will be permitted, provided elevators are cleaned and maintained in a condition acceptable to Owner. At Substantial Completion, restore elevators to condition existing before initial use, including replacing worn cables, guide shoes, and similar items of limited life.
 - 1. Do not load elevators beyond their rated weight capacity.
 - 2. Provide protective coverings, barriers, devices, signs, or procedures to protect elevator car and entrance doors and frame. If, despite protection, elevators become damaged, engage elevator Installer to restore damaged work so no evidence remains of correction work. Return items that cannot be refinished in field to the shop, make required repairs and refinish entire unit, or provide new units as required.
- G. Temporary Stairs: Until permanent stairs are available, provide temporary stairs where ladders are not adequate.
- H. Existing Stair Usage: Use of Owner's existing stairs will be permitted, provided stairs are cleaned and maintained in a condition acceptable to Owner. Coordinate use of existing stairs with Owner to avoid overloading or interfering with Owner use. At Substantial Completion, restore stairs to condition existing before initial use.
 - 1. Provide protective coverings, barriers, devices, signs, or other procedures to protect stairs and to maintain means of egress. If stairs become damaged, restore damaged areas so no evidence remains of correction work.
 - 2. Limit access from existing stairs to project site to construction personnel.
- I. Temporary Use of Permanent Stairs: Use of new stairs for construction traffic will be permitted, provided stairs are protected and finishes restored to new condition at time of Substantial Completion.

3.5 SECURITY AND PROTECTION FACILITIES INSTALLATION

- A. Protection of Existing Facilities: Protect existing vegetation, equipment, structures, utilities, and other improvements at Project site and on adjacent properties, except those indicated to be removed or altered. Repair damage to existing facilities.
 - 1. Where access to adjacent properties is required in order to affect protection of existing facilities, obtain written permission from adjacent property owner to access property for that purpose.
- B. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction as required to comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.
 - 1. Comply with specified work restrictions.
- C. Pest Control: Engage pest control service to recommend practices to minimize attraction and harboring of rodents, roaches, and other pests and to perform extermination and control procedures at regular intervals so Project will be free of pests and their residues at Substantial Completion. Perform control operations lawfully, using materials approved by authorities having jurisdiction.
- D. Site Enclosure Fence: Before construction operations begin, furnish and install site enclosure fence in a manner that will prevent people from easily entering site except by entrance gates.
 - 1. Extent of Fence: Required to enclose entire Project site or portion determined sufficient to accommodate construction operations.
 - 2. Maintain security by limiting number of keys and restricting distribution to authorized personnel.
- E. Security Enclosure and Lockup: Install temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security. Lock entrances at end of each workday.
- F. Barricades, Warning Signs, and Lights: Comply with requirements of authorities having jurisdiction for erecting structurally adequate barricades, including warning signs and lighting.
- G. Temporary Egress: Maintain temporary egress from existing occupied facilities as indicated and as required by authorities having jurisdiction.
- H. Covered Walkway: Erect protective, covered walkway for passage of individuals through or adjacent to Project site. Coordinate with entrance gates, other facilities, and obstructions. Comply with regulations of authorities having jurisdiction.
 - 1. Provide overhead decking, protective enclosure walls, handrails, barricades, warning signs, exit signs, lights, safe and well drained walkways, and similar provisions for protection and safe passage.

2. Paint and maintain appearance of walkway for duration of the work.
- I. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.
 1. Where heating or cooling is needed and permanent enclosure is incomplete, insulate temporary enclosures.
- J. Temporary Partitions: Provide floor to ceiling dustproof partitions to limit dust and dirt migration and to separate areas occupied by Owner from fumes and noise.
 1. Construct dustproof partitions with gypsum wallboard with joints taped on occupied side, and fire-retardant treated plywood on construction operations side.
 2. Construct dustproof partitions with two layers of 6 mil (0.14 mm) polyethylene sheet on each side. Cover floor with two layers of 6 mil (0.14 mm) polyethylene sheet, extending sheets 18 inches (460 mm) up the sidewalls. Overlap and tape full length of joints. Cover floor with fire retardant treated plywood.
 - a. Construct vestibule and airlock at each entrance through temporary partition with not less than 48 inches (1219 mm) between doors. Maintain water dampened foot mats in vestibule.
 3. Where fire resistance rated temporary partitions are indicated or are required by authorities having jurisdiction, construct partitions according to the rated assemblies.
 4. Insulate partitions to control noise transmission to occupied areas.
 5. Seal joints and perimeter. Equip partitions with gasketed dustproof doors and security locks where openings are required.
 6. Protect air handling equipment.
 7. Provide walk off mats at each entrance through temporary partition.
- K. Temporary Fire Protection: Install and maintain temporary fire protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241; manage fire prevention program.
 1. Prohibit smoking in construction areas. Comply with additional limits on smoking specified in other Sections.
 2. Supervise welding operations, combustion type temporary heating units, and similar sources of fire ignition according to requirements of authorities having jurisdiction.
 3. Develop and supervise an overall fire prevention and protection program for personnel at Project site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
 4. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

3.6 MOISTURE AND MOLD CONTROL

- A. Contractor's Moisture Protection Plan: Describe delivery, handling, storage, installation, and protection provisions for materials subject to water absorption or water damage.
 - 1. Indicate procedures for discarding water damaged materials, protocols for mitigating water intrusion into completed work, and replacing water damaged Work.
 - 2. Indicate sequencing of work that requires water, such as sprayed fire resistive materials, plastering, and terrazzo grinding, and describe plans for dealing with water from these operations. Show procedures for verifying that wet construction has dried sufficiently to permit installation of finish materials.
 - 3. Indicate methods to be used to avoid trapping water in finished work.
- B. Exposed Construction Period: Before installation of weather barriers, when materials are subject to wetting and exposure and to airborne mold spores, protect as follows:
 - 1. Protect porous materials from water damage.
 - 2. Protect stored and installed material from flowing or standing water.
 - 3. Keep porous and organic materials from coming into prolonged contact with concrete.
 - 4. Remove standing water from decks.
 - 5. Keep deck openings covered or dammed.
- C. Partially Enclosed Construction Period: After installation of weather barriers but before full enclosure and conditioning of building, when installed materials are still subject to infiltration of moisture and ambient mold spores, protect as follows:
 - 1. Do not load or install drywall or other porous materials or components, or items with high organic content, into partially enclosed building.
 - 2. Keep interior spaces reasonably clean and protected from water damage.
 - 3. Periodically collect and remove waste containing cellulose or other organic matter.
 - 4. Discard or replace water damaged material.
 - 5. Do not install material that is wet.
 - 6. Discard and replace stored or installed material that begins to grow mold.
 - 7. Perform work in a sequence that allows wet materials adequate time to dry before enclosing the material in gypsum board or other interior finishes.
- D. Controlled Construction Period: After completing and sealing of the building enclosure but prior to the full operation of permanent HVAC systems, maintain as follows:
 - 1. Control moisture and humidity inside building by maintaining effective dry in conditions.
 - 2. Use temporary or permanent HVAC system to control humidity within ranges specified for installed and stored materials.
 - 3. Comply with manufacturer's written instructions for temperature, relative humidity, and exposure to water limits.
 - a. Hygroscopic materials that may support mold growth, including wood and gypsum based products, that become wet during the course of construction

and remain wet for 48 hours are considered defective and require replacing.

- b. Measure moisture content of materials that have been exposed to moisture during construction operations or after installation. Record readings beginning at time of exposure and continuing daily for 48 hours. Identify materials containing moisture levels higher than allowed. Report findings in writing to Architect.
- c. Remove and replace materials that cannot be completely restored to their manufactured moisture level within 48 hours.

3.7 OPERATION, TERMINATION, AND REMOVAL

- A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.
- B. Maintenance: Maintain facilities in good operating condition until removal.
 - 1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.
- C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.
- D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.
 - 1. Materials and facilities that constitute temporary facilities are property of Contractor. Owner reserves right to take possession of Project identification signs.
 - 2. At Substantial Completion, repair, renovate, and clean permanent facilities used during construction period. Comply with final cleaning requirements specified in Section 01 77 00 "Closeout Procedures."

END OF SECTION

SECTION 016000

PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Administrative and procedural requirements governing the selection of products for use in the project.
2. Product delivery, storage, and handling.
3. Manufacturers' standard and special warranties on products.
4. Manufacturers' instructions and certifications.

1.2 DEFINITIONS

- A. Products:** Items obtained for incorporating into the Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes the terms "material," "equipment," "system," and terms of similar intent.
1. **Named Products:** Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature, that is current as of date of the Contract Documents.
 2. **New Products:** Items that have not previously been incorporated into another project or facility. Products salvaged or recycled from other projects are not considered new products.
 3. **Comparable Product:** Product that is demonstrated and approved through submittal process to have the indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.
- B. Basis-of-Design Product Specification:** A specification in which a specific manufacturer's product is named and accompanied by the words "basis-of-design product," including make or model number or other designation, to establish the significant qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics for purposes of evaluating comparable products of additional manufacturers named in the specification.

1.3 SUBMITTALS

- A. Proposed Products List:** Within period of 30 days after award of Contract, submit to Architect five (5) copies of complete list of major Products, which are proposed for installation.
1. Tabulate products by Specification Section number, title, and Article number.

2. For products specified only by reference standards, list for each such product:
 - a. Name and address of manufacturer.
 - b. Trade name.
 - c. Model or catalogue designation.
 - d. Manufacturer's Data including reference standards and performance test data.
3. Architect will reply promptly in writing stating whether there is reasonable objection to any listed items. Failure to object to a listed item shall not constitute waiver of requirements of Contract Documents.

1.4 QUALITY ASSURANCE

- A. Materials specified are to define a standard of quality or performance and to establish basis for evaluation of proposals and substitutions.
- B. Comply with individual Specification Sections and referenced standards as minimum requirements.
- C. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, select product compatible with products previously selected, even if previously selected products were also options.
 1. Each contractor is responsible for providing products and construction methods compatible with products and construction methods of other contractors.
 2. If a dispute arises between contractors over concurrently selectable but incompatible products, Architect will determine which products shall be used.
- D. Source Limitations: Components required to be supplied in quantity within Specification Section shall be of same manufacturer and shall be interchangeable.
- E. Nameplates and Labels: Except for required labels and operating data, do not attach or imprint manufacturer's or producer's nameplates or trademarks on surfaces of products which will be exposed to view in occupied spaces or on the exterior:
 1. Labels: Locate required product labels and stamps on a concealed surface or, where required for observation after installation, on an accessible surface approved by Architect and governing authorities.
 2. Nameplates: Provide a permanent nameplate on each item of service-connected or power-operated equipment. Locate on an easily accessible surface which is inconspicuous in occupied spaces.

1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle products using means and methods that will prevent damage, deterioration, and loss, including theft and vandalism. Comply with manufacturer's written instructions.

B. Delivery and Handling:

1. Schedule delivery to minimize long-term storage at Project site and to prevent overcrowding of construction spaces.
2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
3. Deliver products to Project site in an undamaged condition in manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
4. Inspect products on delivery to determine compliance with the Contract Documents and to determine that products are undamaged and properly protected.

C. Storage:

1. Store products to allow for inspection and measurement of quantity or counting of units.
2. Store materials in a manner that will not endanger Project structure.
3. Store products that are subject to damage by the elements, under cover in a weathertight enclosure above ground, with ventilation adequate to prevent condensation.
4. Protect foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
5. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
6. Protect stored products from damage and liquids from freezing.
7. Provide a secure location and enclosure at Project site for storage of materials and equipment by Owner's construction forces. Coordinate location with Owner.

1.6 MANUFACTURER'S INSTRUCTIONS

A. When Contract Documents require that installation of work shall comply with manufacturers printed instructions, obtain and distribute copies of instructions to parties involved in installation, including two copies to Architect, prior to commencing work.

1. Maintain one set of complete instructions at job site during installation and until complete.
2. Maintain one set of complete instructions for Project Record Documents.

B. Handle, install, connect, clean, condition and adjust products in strict accord with manufacturer's instructions and in conformity with specified requirements.

1. Should job conditions or specified requirements conflict with manufacturer's instructions, notify Architect in writing for further instructions.
2. Do not proceed with work without clear instructions.

- C. Perform Work in accordance with manufacturer's instructions. Do not omit preparatory steps of installation procedures unless specifically modified or exempted by Contract Documents.

1.7 MANUFACTURER CERTIFICATION

- A. Prior to Final Acceptance of Work, for items designated in Specification Sections, an authorized representative of each manufacturer of materials or equipment installed under work of that Section shall examine installation and operation of his materials, system and equipment to determine they are correctly installed and operating properly.
- B. Examination and testing shall be accomplished for work which will be concealed during execution of Work, after completion of installation and prior to concealment and for Work which will not be concealed, at completion of Work.
- C. Each representative shall submit signed statement to Owner, through Contractor, certifying to his examination and to correct installation and proper operation of materials, systems or equipment. Certification shall list items included.
- D. Contractor shall transmit certifications to Architect at or prior to Final Acceptance Inspection. Transmittal shall include list of certifications included.

1.8 MAINTENANCE

- A. For mechanical and electrical equipment in long-term storage, provide manufacturer's service instructions shown on exterior of package.
- B. Service equipment on a regular basis as recommended by manufacturer. Maintain log of maintenance services. Submit log as Project Record Document in accordance with requirements of Section 01 78 39.

1.9 PRODUCT WARRANTIES

- A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by the Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of the Contract Documents.
 - 1. Manufacturer's Warranty: Written warranty furnished by individual manufacturer for a particular product and specifically endorsed by manufacturer to Owner.
 - 2. Special Warranty: Written warranty required by the Contract Documents to provide specific rights for Owner.
- B. Special Warranties: Prepare a written document that contains appropriate terms and identification, ready for execution.
 - 1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.

2. Specified Form: When specified forms are included with the Specifications, prepare a written document using indicated form properly executed.
3. See other Sections for specific content requirements and particular requirements for submitting special warranties.

C. Submittal Time: Comply with requirements in Section 01 77 00 "Closeout Procedures."

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION

- A. General Product Requirements: Provide products that comply with the Contract Documents, are undamaged and, unless otherwise indicated, are new at time of installation.
1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
 2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types that have been produced and used successfully in similar situations on other projects.
 3. Owner reserves the right to limit selection to products with warranties not in conflict with requirements of the Contract Documents.
 4. Where products are accompanied by the term "as selected," Architect will make selection.
 5. Descriptive, performance, and reference standard requirements in the Specifications establish salient characteristics of products.
 6. Or Equal: For products specified by name and accompanied by the term "or equal," or "or approved equal," or "or approved," comply with requirements in "Comparable Products" Article to obtain approval for use of an unnamed product.
- B. Product Selection Procedures: Product selection is governed by the Contract Documents and governing regulations, not by previous Project experience.
1. Product: Where Specifications name a single manufacturer and product, provide the named product that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
 2. Manufacturer/Source: Where Specifications name a single manufacturer or source, provide a product by the named manufacturer or source that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
 3. Products:
 - a. Restricted List: Where Specifications include a list of names of both manufacturers and products, provide one of the products listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered unless Section 01 25 00 - "Substitutions" is referenced.
 - b. Nonrestricted List: Where Specifications include a list of names of both available manufacturers and products, provide one of the products listed,

or an unnamed product, that complies with requirements. Comply with requirements in "Comparable Products" Article for consideration of an unnamed product.

4. Manufacturers:
 - a. Restricted List: Where Specifications include a list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered unless Section 01 25 00 - "Substitutions" is referenced.
 - b. Nonrestricted List: Where Specifications include a list of available manufacturers, provide a product by one of the manufacturers listed, or a product by an unnamed manufacturer, that complies with requirements. Comply with requirements in "Comparable Products" Article for consideration of an unnamed manufacturer's product.
5. Basis-of-Design Product: Where Specifications name a product, or refer to a product indicated on Drawings, and include a list of manufacturers, provide the specified or indicated product or a comparable product by one of the other named manufacturers. Drawings and Specifications indicate sizes, profiles, dimensions, and other characteristics that are based on the product named. Comply with requirements in "Comparable Products" Article for consideration of an unnamed product by one of the other named manufacturers.
- C. Performance Specifications Requirements: Where specifications require compliance with performance requirements, provide products that comply with these requirements, and are recommended by the manufacturer for the application indicated.
- D. Compliance with Standards, Codes and Regulations: Where specifications require compliance with an imposed code, standard or regulation, select a product that complies with the code, standard or regulations specified.
- E. Visual Matching Specification: Where Specifications require "match Architect's sample", provide a product that complies with requirements and matches Architect's sample. Architect's decision will be final on whether a proposed product matches.
 1. If no product available within specified category matches and complies with other specified requirements, comply with requirements in Section 01 25 00 "Substitution Procedures" for proposal of product.
- F. Visual Selection Specification: Where Specifications include the phrase "as selected by Architect from manufacturer's full range" or similar phrase, select a product that complies with requirements. Architect will select color, gloss, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 COMPARABLE PRODUCTS

- A. Conditions for Consideration: Architect will consider Contractor's request for comparable product when the following conditions are satisfied. If the following conditions are not satisfied, Architect may return requests without action, except to record noncompliance with these requirements:
 - 1. Evidence that the proposed product does not require revisions to the Contract Documents, that it is consistent with the Contract Documents and will produce the indicated results, and that it is compatible with other portions of the Work.
 - 2. Detailed comparison of significant qualities of proposed product with those named in the Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
 - 3. Evidence that proposed product provides specified warranty.
 - 4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
 - 5. Samples, if requested.

2.3 SUBSTITUTIONS

- A. Refer to Section 01 25 00 - "Substitution Procedures."

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Make periodic examinations of stored materials to verify that products are maintained under specified conditions and are free from damage or deterioration.
- B. Verify that storage facilities comply with manufacturer's product storage requirements.
- C. Verify that manufacturer required environmental conditions are maintained continually.
- D. Verify that surfaces of products exposed to elements are not adversely affected and that weathering of finishes is within acceptable tolerances under requirements of Contract Documents.

3.2 PROTECTION

- A. Furnish protection against weather. Cover building openings to protect interior of building from weather.
- B. Maintain work, materials, apparatus and fixtures free from damage.
- C. Protect items having factory finish to prevent damage to finish and equipment.

- D. At end of day's work, cover new work likely to be damaged or otherwise protect as necessary.
- E. After installation, secure substantial coverings as necessary to protect installed products from damage from traffic and subsequent construction operations.
- F. Remove protection when no longer needed. Upon completion of work, remove storage facilities from site.

3.3 ADJUSTING

- A. Do not use materials in work which have deteriorated, become damaged or are otherwise unfit for use.
- B. Replace stored items damaged by inadequate protection or environmental controls.

END OF SECTION

SECTION 017329
CUTTING AND PATCHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Requirements and limitations of cutting and patching of Work.
2. Execute cutting, fitting and patching, including attendant excavation and backfill, required to complete Work or to:
 - a. Make its several parts fit together properly.
 - b. Uncover portions of the Work to provide for installation of ill-timed work.
 - c. Remove and replace defective work.
 - d. Remove and replace work not conforming to requirements of Contract Documents.
 - e. Remove samples of installed work as specified for testing.
 - f. Make routine penetrations of non-structural surfaces for installation of mechanical, electrical and plumbing Work.
 - g. Uncover work that has been covered prior to Architect's required observation.

1.2 SUBMITTALS

A. Submit written request to Architect in advance of executing cutting or alteration, other than required by Contract Documents, which affects:

1. Work of Owner or separate contractor.
2. Structural value or integrity of any element of Project.
3. Integrity or effectiveness of weather-exposed or moisture-resistant elements or systems.
4. Efficiency, operational life, maintenance or safety of operational elements.
5. Visual qualities of sight-exposed elements.

B. Request shall include:

1. Identification of Project.
2. Location and description of affected work.
3. Necessity for cutting, alteration, or excavation.
4. Effect on work of Owner or separate contractor, or on structural or weatherproof integrity of Project.
5. Description of proposed work:
 - a. Scope of cutting, patching, alteration, or excavation.
 - b. Trades who will execute work.

- c. Products proposed to be used.
 - d. Extent of refinishing to be done.
 - e. Cost proposal when applicable.
- 6. Alternatives to cutting and patching.
- 7. Written permission of separate contractor whose work will be affected.
- C. Should conditions of Work or schedule indicate change of products from original installation, Contractor shall submit request for substitution as specified in Section 01 25 00.
- D. Submit written notice to Architect designating date and time work will be uncovered or altered.

1.3 COORDINATION

- A. Where warranties are in force for existing work, coordinate cutting and patching work with manufacturer and installer of warranted product to avoid voiding warranty.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Comply with specifications and standards for each specific product involved.
- B. Should conditions of work or schedule indicate change of products from original installation, submit request for substitution as specified in Section 01 25 00.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine existing conditions of Project, including elements subject to damage or to movement during cutting, patching, excavating, and backfilling.
- B. After uncovering work, examine conditions affecting installation of Products, or performance of work.
- C. Report unsatisfactory or questionable conditions to Architect in writing. Do not proceed with work until Architect has provided further instructions.

3.2 PREPARATION

- A. Provide adequate temporary support as necessary to assure structural value or integrity of affected portion of Work.

- B. Provide devices and methods to protect other portions of Project from damage.
- C. Provide protection from elements for that portion of the Project which may be exposed by cutting and patching work.
- D. Maintain excavations free from water.

3.3 DUST CONTROL

- A. Provide positive methods of dust control and apply dust control materials to minimize raising dust from cutting and patching operations.

3.4 PERFORMANCE

- A. Execute cutting and demolition by methods which will prevent damage to other work and will provide proper surfaces to receive installation of repairs.
- B. Execute fitting and adjustment of products to provide finished installation to comply with specified products, functions, tolerances and finishes.
- C. Restore work which has been cut or removed. Install new products to provide Work in accordance with requirements of Contract Documents.
- D. Fit work airtight to pipes, sleeves, ducts, conduit and other penetrations through surfaces. Where fire-rated separations are penetrated, fill space around pipe or insert with material with physical characteristics equivalent to fire-resistance requirement of penetrated surface.
- E. Refinish entire surfaces as necessary to provide even finish to match adjacent finishes:
 - 1. For continuous surfaces, refinish to nearest intersection.
 - 2. For assembly, refinish entire unit.
- F. Employ original installer or fabricator of work performed under this Contract to perform cutting and patching for:
 - 1. Weather-exposed or moisture-resistant elements.
 - 2. Sight-exposed finished surfaces.

END OF SECTION

SECTION 017413
CONSTRUCTION CLEANING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Cleaning and disposal of waste materials, debris, and rubbish during construction.

1.2 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Hazard Control: Store volatile waste in covered metal containers and remove from premises daily.
 - 2. Pollution Control: Conduct cleaning and disposal operations to comply with local codes, ordinances, and anti-pollution laws.

PART 2 - PRODUCTS

2.1 EQUIPMENT

- A. Provide acceptable covered containers for deposit of waste materials, debris and rubbish.
- B. Use only cleaning materials which will not create hazards to health and property, and which will not damage surfaces.
- C. Use only those cleaning materials recommended by manufacturer of surface to be cleaned.
- D. Use cleaning materials only on surfaces recommended by cleaning material manufacturer.

PART 3 - EXECUTION

3.1 CLEANING DURING CONSTRUCTION

- A. Execute daily "broom-clean" construction cleaning to keep Work, site and adjacent properties free from accumulations of waste materials, rubbish and debris. Maintain site in a clean and orderly condition including the following:
 - 1. Remove debris and rubbish from pipe chases, plenums, attics, and other closed or remote spaces, prior to enclosing the space.
 - 2. Broom and vacuum clean interior areas prior to start of surface finishing and continue cleaning to eliminate conditions that could affect final finishing.
 - 3. Terminate closed chutes into appropriate containers with lids. Open free-fall chutes are not permitted.
 - 4. Provide on-site dump containers for collection of waste materials, rubbish and debris.
 - 5. Do not allow waste materials, rubbish and debris to accumulate and become an unsightly and hazardous condition. Provide additional collection and dispose of debris whenever periodic schedule is inadequate to prevent accumulation.
 - 6. At not less than every week during progress of Work, cleanup site and access, and legally dispose of waste materials, rubbish, and debris away from site.
 - 7. Do not overload trucks to prevent spillage on access and haul routes. Periodically examine traffic areas and maintain clear routes.

3.2 DUST CONTROL

- A. Sprinkle dusty debris with water.
- B. Vacuum clean interior building areas, including floors, ledges, pipes, ducts, and other places where dust can accumulate, in conjunction with gypsum board installation and sanding operations, and when ready to receive finish painting.
- C. Continue vacuum cleaning on an as-needed basis until building is ready for acceptance or occupancy.
- D. Schedule operations so that dust and other contaminants resulting from cleaning process will not fall on wet and newly painted surfaces.
- E. Clean floors prior to installation of final floor coverings.

3.3 DISPOSAL

- A. Remove waste materials, debris, and rubbish from site daily and dispose off-site.
- B. Do not dispose of volatile wastes such as mineral spirits, oil or paint thinner in storm or sanitary sewer. Comply with requirements of NFPA 241 for removal of combustible waste material and debris. Enforce requirements strictly.

- C. Do not hold materials more than 7 days during normal weather or 3 days when temperature is expected to rise above 80 degrees F.
- D. Handle hazardous, dangerous, or unsanitary waste materials separately from other waste by containerizing properly. Dispose of material in lawful manner.
- E. Do not dispose of wastes into streams or waterways. Do not burn or bury rubbish and waste material on Project site.
- F. Disposal of waste in Owner's containers on site is not permitted.

END OF SECTION

SECTION 017423

FINAL CLEANING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Administrative and procedural requirements for final cleaning at Substantial Completion.

1.2 PROJECT CONDITIONS

- A. Environmental Requirements: Conduct cleaning and waste disposal operations in full compliance with federal and local environmental and anti-pollution regulations, ordinances and laws.
- B. Do not dispose of volatile wastes such as mineral spirits, oil or paint thinner in storm or sanitary drains.
- C. Burning or burying of debris, rubbish or other waste material on the premises will not be permitted.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Cleaning Agents:
 - 1. Use cleaning materials and agents recommended by the manufacturer or fabricator of surface to be cleaned.
 - 2. Do not use cleaning agents that are potentially hazardous to health or property, or that might damage finished surfaces.

PART 3 - EXECUTION

3.1 FINAL CLEANING AT SUBSTANTIAL COMPLETION

- A. General:
 - 1. Complying with manufacturer's instructions clean each surface or unit of Work to the condition expected from a commercial building cleaning and maintenance program using experienced workers or professional cleaners.

2. Complete cleaning operations and conduct an examination of Work areas with Owner and Architect before requesting inspection for Certification of Substantial Completion for the entire Project or a portion of the Project.
- B. Remove grease, petro-chemical spills, mastic, adhesives, dust, dirt, stains, fingerprints, labels, and other foreign materials from sight exposed interior and exterior surfaces.
- C. Remove debris and surface dust from limited access spaces, including plenums, shafts, trenches, equipment vaults, manholes, attics and similar spaces.
- D. Broom clean concrete floors in unoccupied spaces.
- E. Wash and shine glazing and mirrors.
- F. Remove labels that are not permanent labels.
- G. Clean plumbing fixtures to a sanitary condition, free of stains.
- H. Polish glossy surfaces to a clear shine.
- I. Clean resilient flooring, paver flooring, ceramic tile flooring, and carpeting as recommended by manufacturers.
- J. Repair, patch and touch up marred surfaces to match adjacent surfaces. Replace broken or scratched glass.
- K. Broom clean exterior paved surfaces; rake clean other surfaces of grounds.
- L. Heating, Ventilating, and Air Conditioning Systems:
 1. Clean permanent filters and replace disposable filters if units were operated during construction.
 2. Clean exposed surfaces of diffusers, registers and grills.
 3. Clean ducts, blowers, and coils if units were operated without filters during construction.
 4. Clean exposed ductwork after other cleaning has been completed.
- M. Clean light fixtures, lamps, gloves, reflectors and replace burned out bulbs and defective starters.
- N. Clean food service equipment to a sanitary condition, ready and acceptable for its intended use.
- O. Clean roof areas of debris; flush roof drainage system with water until clear.
- P. Leave Project clean and ready for occupancy.

END OF SECTION

SECTION 017700

CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Administrative and procedural requirements for project closeout, including but not limited to:
 - 1. Substantial Completion.
 - 2. Inspection procedures.
 - 3. Closeout submittals.
 - 4. Payments and release of liens.
 - 5. Post construction examinations.

1.2 SUBSTANTIAL COMPLETION

- A. When Contractor considers Work is substantially complete, submit to Architect:
 - 1. Written certification that Work, or designated portion thereof, is substantially complete.
 - 2. List of items to be completed or corrected, value of incomplete construction and reasons the Work is not complete.
- B. Within seven calendar days after receipt of such certificate, Architect will make examination to determine status of completion.
- C. Should Architect determine that Work is not substantially complete:
 - 1. Architect will promptly notify Contractor in writing, giving reasons.
 - 2. Contractor shall remedy deficiencies in Work and send second written notice of substantial completion to Architect.
 - 3. Architect will re-examine Work.
- D. When Architect concurs that Work is substantially complete, he will:
 - 1. Prepare Certificate of Substantial Completion on AIA Form G704, accompanied by Contractor's list of items to be completed or corrected, as verified and amended by Architect.
 - 2. Submit Certificate to Owner and Contractor for their written acceptance of responsibilities assigned to them in Certificate.
- E. After Work is substantially complete, Contractor shall:
 - 1. Allow Owner occupancy of Project under provisions stated in Certificate of Substantial Completion.

2. Obtain and submit Certificate of Occupancy, operating certificates and similar releases enabling the Owner unrestricted use of the Work.
3. Complete work listed for completion or correction within designated form.
4. Advise Owner of pending insurance change-over requirements.
5. Perform final cleaning in accordance with 01 74 23.
6. Make final change-over of permanent locks and transmit keys to Owner.

1.3 FINAL INSPECTION

- A. When Contractor considers Work is complete, submit written certification that:
 1. Contract Documents have been reviewed.
 2. Work has been examined for compliance and completed in accordance with Contract Documents.
 3. Equipment and systems have been tested in presence of Owner's representative and are operational.
 4. Work is completed and ready for final examination.
- B. Architect will make examination to verify status of completion within seven calendar days after receipt of such certification.
- C. Should Architect consider that Work is incomplete or defective:
 1. Architect will promptly notify Contractor in writing, listing incomplete or defective work.
 2. Contractor shall take immediate steps to remedy stated deficiencies and send second written certification to Architect that Work is complete.
 3. Architect will re-examine Work.
- D. When Architect finds that Work is acceptable under Contract Documents, he shall request Contractor to make closeout submittals.

1.4 REINSPECTION FEES

- A. Should Architect perform re-examinations due to failure of Work to comply with claims of status of completion made by Contractor:
 1. Owner will compensate Architect for such additional services.
 2. Owner will deduct amount of such compensation from final payment to Contractor.

1.5 CLOSEOUT SUBMITTALS

- A. Evidence of compliance with requirements of governing authorities:
 1. Certificate of Occupancy.
 2. Certificates of Inspection: Mechanical and Electrical systems as required by respective sections.

- B. Project Record Documents: Comply with Section 01 78 39.
- C. Operating and Maintenance Data: Comply with Section 01 78 23.
- D. Warranties and Bonds: Comply with Section 01 78 30.
- E. Spare Parts and Maintenance Material:
 - 1. Provide products, spare parts, and maintenance materials in quantities specified in each specification section in addition to that required for completion of Work.
 - 2. Coordinate with Owner, deliver to Project site, store properly, and obtain receipt prior to final payment.
- F. Certificate of Insurance for Products and Completed Operations.

1.6 EVIDENCE OF PAYMENTS AND RELEASE OF LIENS

- A. Contractor's Affidavit of Payment of Debts and Claims: AIA G706.
- B. Contractor's Affidavit of Release of Liens: AIA G706A.
- C. Attachment to Contractor's Affidavit of Release of Liens:
 - 1. Consent of Surety to Final Payment: AIA G707.
 - 2. Contractor's Release or Waiver of Liens.
 - 3. Separate releases of waivers of liens from subcontractors, suppliers and others with lien rights against property of Owner, together with list of those parties.
- D. Submittals shall be duly executed before delivery to Owner.

1.7 FINAL ADJUSTMENT OF ACCOUNTS

- A. Submit final statement of accounting to Architect.
- B. Statement shall reflect adjustments to Contract Sum:
 - 1. Original Contract Sum.
 - 2. Additions and deductions resulting from:
 - 3. Previous Change Orders.
 - 4. Allowances.
 - 5. Unit Prices.
 - 6. Deductions for uncorrected Work.
 - 7. Penalties and Bonuses.
 - 8. Deductions for liquidated damages.
 - 9. Deductions for re-examination payments.
 - 10. Other adjustments.
 - 11. Total Contract Sum, as adjusted.
 - 12. Previous payments.
 - 13. Sum remaining due.

- C. Architect will prepare final Change Order, reflecting approved adjustments to Contract Sum which were not previously made by Change Orders.

1.8 FINAL APPLICATION FOR PAYMENT

- A. Contractor shall submit final Application for Payment in accordance with procedures and requirements stated in Conditions of the Contract.

1.9 ADDITIONAL ADJUSTMENT

- A. No adjustments to Contract requested by Contractor will be allowed if asserted after execution of Final Payment of Contract.

1.10 POST-CONSTRUCTION EXAMINATION

- A. Prior to expiration of one year from Date of Substantial Completion, Owner will make visual examination of Project in company of Contractor to determine whether further correction of Work is required in accordance with provisions of Contract.
- B. Owner will promptly notify Contractor, in writing, of observed deficiencies.
- C. Contractor will contact Owner to arrange time and establish schedule for correction of deficiencies.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 017823
OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Administrative and procedural requirements the preparation and submittal for operating and maintenance manuals including the following:
 - 1. Operating and maintenance manuals for building systems or equipment.
 - 2. Instruction manual covering the care, preservation and maintenance of architectural products and finishes.
 - 3. Instruction of Owner's operating personnel in operation and maintenance of building systems and equipment.

1.2 FORM OF SUBMITTALS

- A. Prepare instructional manuals and data bound in commercial quality 3-ring binders:
 - 1. Organize with index tabs according to sequence of Specification Sections.
 - 2. Identify each volume with type or printed title as instructed by Architect.

1.3 CONTENT OF MANUALS

- A. Arrange typewritten table of contents for each volume, in systematic order:
 - 1. List of each product required to be included with name, address, and telephone number of:
 - a. Subcontractor or installer.
 - b. Maintenance contractor, as appropriate.
 - c. Local source of supply for parts and replacement.
 - 2. Identifying each product by product name and other identifying symbols.
- B. Product Data:
 - 1. Include only those sheets which are pertinent to specific product with product clearly identified.
 - 2. Delete references to inapplicable information.
 - 3. Annotate each sheet to clearly identify specific product or part installed, and data applicable to installation.

- C. Drawings:
 - 1. Supplement product data with drawings as necessary to clearly illustrate relations of component parts of equipment and systems and control and flow diagrams.
 - 2. Coordinate drawings with information in Project Record Documents to assure correct illustration of completed installation.
- D. Written Text: As required to supplement product data for particular installation to provide logical sequence of instructions for each procedure, organized in a consistent format and in logical sequence of instructions for each procedure.
- E. Recommended Spare Parts: Furnish a list of recommended spare parts for each equipment item that will be needed to support that item of equipment for a 12-month period. Spare parts list shall contain the following information:
 - 1. Parts Descriptions.
 - 2. Manufacturer's Part Number.
 - 3. Shelf Life.
 - 4. Recommended Quantity.
 - 5. Unit Price.
 - 6. Name and address of the part manufacturer.
 - 7. Name and address of a local supplier for the part.

1.4 EQUIPMENT AND SYSTEMS MANUAL REQUIREMENTS

- A. Submit three copies of completed manuals in final form.
- B. Content, for each unit of equipment and system, as appropriate:
 - 1. Description of unit and component parts.
 - a. Function, normal operating characteristics, and limiting conditions.
 - b. Performance curves, engineering data and tests.
 - c. Complete nomenclature and commercial number of replaceable parts.
 - 2. Operating procedures:
 - a. Start-up, break-in, routine and normal operating instructions.
 - b. Regulation, control, stopping, shutdown and emergency instructions.
 - c. Summer and winter operating instructions.
 - d. Special operating instructions.
 - 3. Maintenance Procedures:
 - a. Routine operations.
 - b. Guide to "troubleshooting".
 - c. Disassembly, repair and reassembly.
 - d. Alignment, adjusting and checking.

4. Servicing and lubrication schedule, including list of lubricants required.
 5. Manufacturer's printed operating and maintenance instructions.
 6. Description of sequence of operation by control manufacturer.
 7. Original manufacturer's parts list, price lists, illustrations, assembly drawings and diagrams required for maintenance, predicted life of parts subject to wear and items recommended to be stocked as spare parts.
 8. As-installed control diagrams by controls manufacturer.
 9. Each subcontractor's coordination drawings including as-installed color-coded piping diagrams.
 10. Charts of valve tag numbers, with location and function of each valve.
 11. Water treatment procedures and tests.
 12. Final balancing reports for mechanical systems.
 13. List of original manufacturer's spare parts, manufacturer's current prices, and recommended quantities to be maintained in storage.
 14. Other data as required under pertinent sections of specifications.
- C. Content, for each electric and electronic system, as appropriate:
1. Description of system and component parts.
 - a. Function, normal operating characteristics, and limiting conditions.
 - b. Performance curves, engineering data and tests.
 - c. Complete nomenclature and commercial number of replaceable parts.
 2. Circuit directories of panelboards.
 - a. Electrical service.
 - b. Controls.
 - c. Communications.
 3. As-installed color-coded wiring diagrams.
 4. Operating procedures:
 - a. Routine and normal operating instructions.
 - b. Sequences required.
 - c. Special operating instructions.
 5. Maintenance procedures:
 - a. Routine operations.
 - b. Guide to "troubleshooting".
 - c. Disassembly, repair and reassembly.
 - d. Adjustment and checking.
 6. Manufacturer's printed operating and maintenance instructions.
 7. List of original manufacturer's spare parts, manufacturer's current prices, and recommended quantities to be maintained in storage.
 8. Other data as required under pertinent sections of specifications.
- D. Include warnings of detrimental maintenance practices.

- E. Prepare and include additional data when need for such data becomes apparent during instruction of Owner's personnel or as required under pertinent Specification Section.
- F. Refer to individual Sections of Project Manual for additional requirements for operating and maintenance data.
- G. Provide complete information for products and equipment specified in:
 - 1. Division 21: Fire Suppression
 - 2. Division 22: Plumbing
 - 3. Division 23: Heating, Ventilating, and Air-Conditioning (HVAC)
 - 4. Division 26: Electrical Systems.
 - 5. Division 27: Communications
 - 6. Division 28: Electronic Safety and Security

1.5 ARCHITECTURAL PRODUCTS MANUAL REQUIREMENTS

- A. Submit three copies of complete manual in final form.
- B. Refer to individual Sections of Project Manual for submittal requirements.
- C. Content: Manufacturer's data, giving full information on products, catalog numbers, sizes, and composition; and finish designations.
- D. Information required for re-ordering.
- E. Instructions for care and maintenance.
 - 1. Manufacturer's recommended lubricants.
 - 2. Manufacturer's recommendations for types of cleaning agents and methods.
 - 3. Cautions against cleaning agents and methods which are detrimental to product.
 - 4. Recommended maintenance and cleaning schedule.
- F. List of original manufacturer's spare parts, manufacturer's current prices, and recommended quantities to be maintained in storage.

1.6 ARCHITECTURAL FINISHES MANUAL REQUIREMENTS

- A. Submit three copies of complete manual in final form.
- B. Contractor is to provide finish information and samples of actual materials installed in Project. Provide separate line item for each type of finish installed. Attach small sample of each finish that matches finish installed. Provide information on manufacturer, catalog number, finish designation, or other information necessary for Owner to match finish in the future.
- C. Refer to individual Sections of Project Manual for submittal requirements.

- D. Contractor will provide heavy, cardboard pages printed in required page format, approved by Architect, to attach samples and information to for assembly into notebook format.
- E. Contractor shall compile information and samples for Manual on Architectural Finishes throughout construction period to simplify assembly at Project close-out.

1.7 SUBMITTAL SCHEDULE

- A. Submit one copy of completed data in final form 30 days prior to demonstrations of equipment.
- B. Copy will be returned approved or with comments for revisions.
- C. Submit specified number of copies of approved data in final form within 10 days prior to equipment demonstrations and prior to final inspection or acceptance.

1.8 INSTRUCTIONS OF OWNER'S PERSONNEL

- A. Prior to final inspection, instruct the Owner's personnel in operation, adjustment, and maintenance of products equipment and systems. Provide instruction at mutually agreed upon times.
 - 1. For equipment that requires seasonal operation, provide similar instruction during other seasons.
 - 2. Use operation and maintenance manuals for each piece of equipment or system as the basis of instruction. Review contents in detail to explain aspects of operation and maintenance.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 017830

WARRANTIES AND BONDS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: General administrative and procedural requirements for warranties and bonds required by the Contract Documents, including manufacturer's standard warranties on products and special warranties.
 - 1. Compile specified warranties and bonds.
 - 2. Compile specified service and maintenance contracts.
 - 3. Co-execute submittals when so specified.
 - 4. Review submittals to verify compliance with Contract Documents.
 - 5. Submit to Architect for review and transmittal to Owner.

1.2 DEFINITIONS

- A. Standard Product Warranties: Reprinted written warranties published by individual manufacturers for particular products and are specifically endorsed by the manufacturer to the Owner.
- B. Special Warranties: Written warranties required by the Contract Documents, either to extend time limits provided standard product warranties or to provide greater rights for the Owner.
- C. Emergency Repairs: Owner reserves right to make emergency repairs as required to keep equipment or materials in operation or to prevent damage to persons or property without voiding Contractor's warranty or bond, or relieving Contractor of his responsibilities during contract, warranty or bond periods.

1.3 WARRANTY REQUIREMENTS

- A. Related Damages and Losses: When correcting warranted Work that has failed, remove and replace other Work that has been damaged as a result of such failure or that must be removed and replaced to provide access for correction of warranted Work.
 - 1. Reinstatement of Warranty: When Work covered by a warranty by written endorsement.
- B. Reinstated warranty shall be equal to the original warranty with an equitable adjustment for depreciation but not less than 50% of the original warranty period of time.

- C. Replacement Cost: Upon determination that Work covered by a warranty has failed, replace or rebuild the Work to an acceptable condition complying with requirements of Contract Documents.
 - 1. Cost of replacing or rebuilding defective Work regardless of whether the Owner has benefited from use of the Work through its anticipated useful service life is the Contractor's responsibility.
- D. Rejection of Warranties: Owner reserves the right to reject warranties and to limit selections to products with warranties not in conflict with requirements of the Contract Documents.
- E. Acceptance of Work: Owner reserves the right to refuse to accept Work for the Project where a special warranty is required on such Work or part of the Work, until evidence is presented that entities required to countersign such commitments are willing to do so.

1.4 SUBMITTAL REQUIREMENTS

- A. Assemble warranties, bonds, service contracts and maintenance contracts, executed by each of respective manufacturers, suppliers, and subcontractors.
- B. Number of original signed copies required: Two each.
- C. Table of Contents: Neatly typed, in orderly sequence and provide complete information for each item.
 - 1. Product or work item.
 - 2. Firm, with name of principal, address and telephone number.
 - 3. Scope.
 - 4. Date of beginning of each warranty, bond or service and maintenance contract.
 - 5. Duration of warranty, bond or service maintenance contract.
 - 6. Provide information for Owner's personnel:
 - a. Proper procedure in case of failure.
 - b. Instances which might affect validity of warranty or bond.
 - 7. Contractor, name of responsible principal, address and telephone number.

1.5 FORM OF SUBMITTALS

- A. Prepare in duplicate packets.
- B. Format:
 - 1. Size 8-½ inches by 11 inches, punch sheets for standard 3-ring binder.
 - 2. Fold larger sheets to fit into binders.
 - 3. Cover: Identify each packet with typed or printed title "WARRANTIES AND BONDS".

- 4. List:
 - a. Title of Project.
 - b. Name of Contractor.

- C. Binders: Commercial quality, 3-ring, with durable and cleanable plastic covers.

1.6 TIME OF SUBMITTALS

- A. Submit warranties to Architect prior to date certified for Substantial Completion or on date specified in Architect Certificate of Substantial Completion:
 - 1. When a designated portion of the Work is completed and occupied or used by the Owner submit properly executed warranties to Architect within fifteen days of completion of that designated portion of the Work.
 - 2. For items of work, where acceptance is delayed materially beyond Date of Substantial Completion, provide updated submittal within ten days after acceptance, listing date of acceptance as start of warranty period.

1.7 SUBMITTALS REQUIRED

- A. Submit warranties, bonds, service contracts and maintenance contracts as specified in each respective Specification Section.
- B. Refer to each individual Section of Project Manual for specific warranty and bond submittal requirements.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 017839
PROJECT RECORD DOCUMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Administrative and procedural requirements for Project Record Documents including but not limited to the following:
 - 1. Marked-up copies of Contract Documents including field records for variable and concealed conditions.
 - 2. Marked-up copies of submittals, shop drawings, product data, and samples.
 - 3. Newly prepared drawings.

1.2 MAINTENANCE OF DOCUMENTS AND SAMPLES

- A. For duration of Project, maintain at job site following:
 - 1. One copy of Drawings, Specifications, Addenda, shop drawings, product data, miscellaneous requested submittal data, change orders and other modifications to Contract, field orders, field tests or written instructions.
 - 2. One copy of transmittal letters.
 - 3. One set of samples.
- B. Store documents and samples in Contractor's field office apart from documents used for construction.
 - 1. Provide files and racks for storage of documents.
 - 2. Provide locked cabinet or secure storage space for storage of samples.
 - 3. Do not permit record documents to be used for construction purposes.
- C. File documents and samples in accordance with CSI MasterFormat 2020.
- D. Maintain documents in a clean, dry, legible condition and in good order. Do not use record documents for construction purposes.
- E. Make documents and samples available for examination by Architect.
- F. Incomplete or out of order documents and samples will be grounds for not approving application for payment.
- G. Provide felt tip marking pens for recording information in color code designated by Architect.

- H. Label each document "PROJECT RECORD" in neat large, printed letters. Keep record documents current. Record information concurrently with construction progress. Do not conceal any work until required information is recorded.
 - 1. Prior to creation of electronic copies of record documents, remove Architect's and Engineer's seals from all documents as directed by the Architect.

1.3 CLOSEOUT SUBMITTALS

- A. Record Drawings: Comply with the following:
 - 1. Number of Copies: Submit copies of record Drawings as follows:
 - a. Initial Submittal: Submit one paper copy set of marked-up record prints. Architect will indicate whether general scope of changes, additional information recorded, and quality of drafting are acceptable.
 - b. Final Submittal: Submit PDF electronic files of marked-up record prints. Print each Drawing, whether or not changes and additional information were recorded.
- B. Record Specifications: Submit annotated PDF electronic files of Project's Specifications, including addenda and contract modifications.
- C. Record Product Data: Submit annotated PDF electronic files and directories of each submittal.
 - 1. Where record Product Data are required as part of operation and maintenance manuals, submit duplicate marked-up Product Data as a component of manual.
- D. Miscellaneous Record Submittals: Refer to other Specification Sections for miscellaneous record-keeping requirements and submittals in connection with various construction activities. Submit annotated PDF electronic files and directories of each submittal.

1.4 BURDEN OF ACCURACY

- A. Contractor shall bear costs of damages incurred by Owner due to inaccuracies or incompleteness of submitted Record Documents for a period of time following Substantial Completion as defined by Conditions of the Contract.

PART 2 - PRODUCTS

2.1 RECORD DRAWINGS

- A. Record Prints: Maintain one set of marked-up paper copies of the Contract Drawings and Shop Drawings.
1. Preparation: Mark record prints to show the actual installation where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to provide information for preparation of corresponding marked-up record prints.
 - a. Give particular attention to information on concealed elements that would be difficult to identify or measure and record later.
 - b. Accurately record information in an acceptable drawing technique.
 - c. Record data as soon as possible after obtaining it.
 - d. Record and check the markup before enclosing concealed installations.
 - e. Cross-reference record prints to corresponding archive photographic documentation.
 2. Content: Types of items requiring marking include, but are not limited to, the following:
 - a. Dimensional changes to Drawings.
 - b. Revisions to details shown on Drawings.
 - c. Depths of foundations below first floor.
 - d. Locations and depths of underground utilities.
 - e. Revisions to routing of piping and conduits.
 - f. Revisions to electrical circuitry.
 - g. Actual equipment locations.
 - h. Duct size and routing.
 - i. Locations of concealed internal utilities.
 - j. Changes made by Change Order or Construction Change Directive.
 - k. Changes made following Architect's written orders.
 - l. Details not on the original Contract Drawings.
 - m. Field records for variable and concealed conditions.
 - n. Record information on the Work that is shown only schematically.
 3. Mark the Contract Drawings and Shop Drawings completely and accurately. Utilize personnel proficient at recording graphic information in production of marked-up record prints.
 4. Mark record sets with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at same location.
 5. Mark important additional information that was either shown schematically or omitted from original Drawings.
 6. Note Construction Change Directive numbers, alternate numbers, Change Order numbers, and similar identification, where applicable

- B. Record Digital Data Files: Immediately before inspection for Certificate of Substantial Completion, review marked-up record prints with Architect. When authorized, prepare a full set of corrected digital data files of the Contract Drawings, as follows:
 - 1. Formats:
 - a. Annotated PDF electronic file with comment function enabled.
 - 2. Incorporate changes and additional information previously marked on record prints. Delete, redraw, and add details and notations where applicable.
 - 3. Refer instances of uncertainty to Architect for resolution.
 - 4. Architect will furnish Contractor one set of digital data files of the Contract Drawings for use in recording information.
 - a. Refer to Section 01 31 00 "Project Coordination" for requirements related to use of Architect's digital data files.
- C. Newly Prepared Record Drawings: Prepare new Drawings instead of preparing record Drawings where Architect determines that neither the original Contract Drawings nor Shop Drawings are suitable to show actual installation.
 - 1. New Drawings may be required when a Change Order is issued as a result of accepting an alternate, substitution, or other modification.
 - 2. Consult Architect for proper scale and scope of detailing and notations required to record the actual physical installation and its relation to other construction. Integrate newly prepared record Drawings into record Drawing sets; comply with procedures for formatting, organizing, copying, binding, and submitting.
- D. Format: Identify and date each record Drawing; include the designation "PROJECT RECORD DRAWING" in a prominent location.
 - 1. Format: Annotated PDF electronic file with comment function enabled.
 - 2. Record Digital Data Files: Organize digital data information into separate electronic files that correspond to each sheet of the Contract Drawings. Name each file with the sheet identification. Include identification in each digital data file.
 - 3. Identification: As follows:
 - a. Project name.
 - b. Date.
 - c. Designation "PROJECT RECORD DRAWINGS."
 - d. Name of Architect.
 - e. Name of Contractor.

2.2 RECORD SPECIFICATIONS

- A. Preparation: Mark Specifications to indicate the actual product installation where installation varies from that indicated in Specifications, addenda, and contract modifications.
 - 1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
 - 2. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.
 - 3. Record the name of manufacturer, supplier, Installer, and other information necessary to provide a record of selections made.
 - 4. Note related Change Orders, record Product Data, and record Drawings where applicable.
- B. Format: Submit record Specifications as scanned PDF electronic file(s) of marked up paper copy of Specifications.

2.3 RECORD PRODUCT DATA

- A. Preparation: Mark Product Data to indicate the actual product installation where installation varies substantially from that indicated in Product Data submittal.
 - 1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
 - 2. Include significant changes in the product delivered to Project site and changes in manufacturer's written instructions for installation.
 - 3. Note related Change Orders, record Specifications, and record Drawings where applicable.
- B. Format: Submit record Product Data as scanned PDF electronic file(s) of marked up paper copy of Product Data.
- C. Include record Product Data directory organized by specification section number and title, electronically linked to each item of record Product Data.

2.4 MISCELLANEOUS RECORD SUBMITTALS

- A. Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference.
- B. Format: Submit miscellaneous record submittals as scanned PDF electronic file(s) of marked up miscellaneous record submittals.
 - 1. Include miscellaneous record submittals directory organized by specification section number and title, electronically linked to each item of miscellaneous record submittals.

PART 3 - EXECUTION

3.1 RECORDING

- A. Recording: Maintain one copy of each submittal during the construction period for project record document purposes. Post changes and modifications to the Documents as they occur. Do not wait until the end of the project. Architect will periodically review record documents to assure compliance.
- B. Maintenance of Record Documents and Samples: Store record documents and Samples in the field office apart from the Contract Documents used for construction. Do not use project record documents for construction purposes. Maintain record documents in good order and in a clean, dry, legible condition, protected from deterioration and loss. Provide access to project record documents for Architect's reference during normal working hours

END OF SECTION

SECTION 017846
EXTRA MATERIALS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Administrative and procedural requirements for stocking of extra material.
- B. Divisions 2 - 41: Specific quantities of extra material required by individual specifications sections.

1.2 PRODUCTS REQUIRED

- A. Provide quantities of extra materials specified in individual specification sections to Owner, in addition to quantities required for completion of Work.
- B. Products to be identical to those installed in Work. Include quantities in original purchase from supplier or manufacturer to avoid variations in manufacture.

1.3 STORAGE AND MAINTENANCE

- A. Temporarily store extra materials with products to be installed in Work, under provision of Section 01 60 00, or in other location acceptable to Owner.
- B. When adequate, secure storage facilities available at Site, capable of maintaining conditions required for storage and not required for Contract Work or storage; extra materials may be stored in available space.
- C. Maintain extra materials in manufacturer's unopened original containers with labels intact and legible, until delivery to Owner.

1.4 DELIVERY

- A. Coordinate final delivery of extra materials with Owner prior to Substantial Completion.
- B. Deliver, unload, store, and account for specified quantities of extra materials in presence of Owner's representative.
- C. Owner will indicate final placement in building of extra materials.
- D. Obtain written acceptance from Owner's representative of receipt of specified quantities of extra materials.

- E. For portions of Work accepted and occupied by Owner prior to Substantial Completion, deliver proportional quantity of spare parts and maintenance materials if requested by Owner. Record quantities delivered with Owner's representative.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

Not Used.

END OF SECTION

SECTION 017900

DEMONSTRATION AND TRAINING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Administrative and procedural requirements for instructing Owner's personnel, including the following:
 - 1. Demonstration of operation of systems, subsystems, and equipment.
 - 2. Training in operation and maintenance of systems, subsystems, and equipment.

1.2 SUBMITTALS

- A. Instruction Program: Prior to the execution of any demonstrations or training, submit outline of instructional program, including a list of training modules and a schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module.

1.3 QUALITY ASSURANCE

- A. Instructor Qualifications: A factory-authorized service representative, experienced in operation and maintenance procedures and training.
- B. Preconstruction Conference: Conduct conference at Project site with Owner's personnel. Review methods and procedures related to demonstration and training including, but not limited to, the following:
 - 1. Inspect and discuss locations and other facilities required for instruction.
 - 2. Review and finalize instruction schedule and verify availability of educational materials, instructors' personnel, audiovisual equipment, and facilities needed to avoid delays.
 - 3. Review required content of instruction.
 - 4. For instruction that must occur outside, review weather and forecasted weather conditions and procedures to follow if conditions are unfavorable.

1.4 COORDINATION

- A. Coordinate instruction schedule with Owner's operations. Adjust schedule as required to minimize disrupting Owner's operations and to ensure availability of Owner's personnel.
- B. Coordinate instructors, including providing notification of dates, times, length of instruction time, and course content.

- C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals.

PART 2 - PRODUCTS

2.1 INSTRUCTION PROGRAM

- A. Program Structure: Develop an instruction program that includes individual training modules for each system and for equipment not part of a system, as required by individual Specification Sections.
- B. Training Modules: Develop a learning objective and teaching outline for each module. Include a description of specific skills and knowledge that participant is expected to master. For each module, include instruction for the following as applicable to the system, equipment, or component:
 - 1. Basis of System Design, Operational Requirements, and Criteria: Include the following:
 - a. System, subsystem, and equipment descriptions.
 - b. Performance and design criteria if Contractor is delegated design responsibility.
 - c. Operating standards.
 - d. Regulatory requirements.
 - e. Equipment function.
 - f. Operating characteristics.
 - g. Limiting conditions.
 - h. Performance curves.
 - 2. Documentation: Review the following items in detail:
 - a. Emergency manuals.
 - b. Operations manuals.
 - c. Maintenance manuals.
 - d. Project record documents.
 - e. Identification systems.
 - f. Warranties and bonds.
 - g. Maintenance service agreements and similar continuing commitments.
 - 3. Emergencies: Include the following, as applicable:
 - a. Instructions on meaning of warnings, trouble indications, and error messages.
 - b. Instructions on stopping.
 - c. Shutdown instructions for each type of emergency.
 - d. Operating instructions for conditions outside of normal operating limits.
 - e. Sequences for electric or electronic systems.
 - f. Special operating instructions and procedures.

4. Operations: Include the following, as applicable:
 - a. Startup procedures.
 - b. Equipment or system break-in procedures.
 - c. Routine and normal operating instructions.
 - d. Regulation and control procedures.
 - e. Control sequences.
 - f. Safety procedures.
 - g. Instructions on stopping.
 - h. Normal shutdown instructions.
 - i. Operating procedures for emergencies.
 - j. Operating procedures for system, subsystem, or equipment failure.
 - k. Seasonal and weekend operating instructions.
 - l. Required sequences for electric or electronic systems.
 - m. Special operating instructions and procedures.
5. Adjustments: Include the following:
 - a. Alignments.
 - b. Checking adjustments.
 - c. Noise and vibration adjustments.
 - d. Economy and efficiency adjustments.
6. Troubleshooting: Include the following:
 - a. Diagnostic instructions.
 - b. Test and inspection procedures.
7. Maintenance: Include the following:
 - a. Inspection procedures.
 - b. Types of cleaning agents to be used and methods of cleaning.
 - c. List of cleaning agents and methods of cleaning detrimental to product.
 - d. Procedures for routine cleaning
 - e. Procedures for preventive maintenance.
 - f. Procedures for routine maintenance.
 - g. Instruction on use of special tools.
8. Repairs: Include the following:
 - a. Diagnosis instructions.
 - b. Repair instructions.
 - c. Disassembly; component removal, repair, and replacement; and reassembly instructions.
 - d. Instructions for identifying parts and components.
 - e. Review of spare parts needed for operation and maintenance.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Assemble educational materials necessary for instruction, including documentation and training module. Assemble training modules into a training manual organized in coordination with requirements in Section 01 78 23 "Operation and Maintenance Data."
- B. Set up instructional equipment at instruction location.

3.2 INSTRUCTION

- A. Engage qualified instructors to instruct Owner's personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.
- B. Scheduling: Provide instruction at mutually agreed on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.
 - 1. Schedule training with Owner with at least seven days' advance notice.
- C. Training Location and Reference Material: Conduct training on-site in the completed and fully operational facility using the actual equipment in-place. Conduct training using final operation and maintenance data submittals.
- D. Evaluation: At conclusion of each training module, assess and document each participant's mastery of module by use of a demonstration performance-based test.
- E. Cleanup: Collect used and leftover educational materials and give to Owner. Remove instructional equipment. Restore systems and equipment to condition existing before initial training use.

END OF SECTION

SECTION 024119

SELECTIVE DEMOLITION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Demolition and removal of selected portions of building or structure.
2. Demolition and removal of selected site elements.
3. Salvage of existing items to be reused.

1.2 DEFINITIONS

- A. Remove:** Detach items from existing construction and legally dispose of them off-site unless indicated to be removed and salvaged or removed and reinstalled.
- B. Remove and Salvage:** Carefully detach from existing construction, in a manner to prevent damage, and deliver to Owner ready for reuse.
- C. Remove and Reinstall:** Detach items from existing construction, prepare for reuse, and reinstall where indicated.
- D. Existing to Remain:** Existing items of construction that are not to be permanently removed and that are not otherwise indicated to be removed, removed and salvaged, or removed and reinstalled.

1.3 PREINSTALLATION MEETINGS

- A. Predemolition Conference:** Conduct conference at Project site.

1.4 CLOSEOUT SUBMITTALS

- A. Landfill Records:** Indicate receipt and acceptance of hazardous wastes by a landfill facility licensed to accept hazardous wastes.

1.5 FIELD CONDITIONS

- A. Conditions existing at time of inspection for bidding purpose will be maintained by Owner as far as practical.**
- B. Notify Architect of discrepancies between existing conditions and Drawings before proceeding with selective demolition.**

- C. Hazardous Materials: It is not expected that hazardous materials will be encountered in the Work.
 - 1. If suspected hazardous materials are encountered, do not disturb; immediately notify Architect and Owner. Hazardous materials will be removed by Owner under a separate contract.
- D. Storage or sale of removed items or materials on-site is not permitted.
- E. Utility Service: Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.
 - 1. Maintain fire-protection facilities in service during selective demolition operations.

1.6 WARRANTY

- A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during selective demolition, by methods and with materials so as not to void existing warranties.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Regulatory Requirements: Comply with governing EPA notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.
- B. Standards: Comply with ANSI/ASSE A10.6 and NFPA 241.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that utilities have been disconnected and capped before starting selective demolition operations.
- B. Survey existing conditions and correlate with requirements indicated to determine extent of selective demolition required.
- C. When unanticipated mechanical, electrical, or structural elements that conflict with intended function or design are encountered, investigate and measure the nature and extent of conflict. Promptly submit a written report to Architect.
- D. Perform an engineering survey of condition of building to determine whether removing any element might result in structural deficiency or unplanned collapse of any portion of structure or adjacent structures during selective building demolition operations.

- E. Survey of Existing Conditions: Record existing conditions by use of measured drawings, preconstruction photographs, preconstruction videotapes, and templates.

- 1. Comply with requirements specified in Section 01 32 33 "Photographic Documentation."

3.2 UTILITY SERVICES AND MECHANICAL/ELECTRICAL SYSTEMS

- A. Existing Services/Systems to Remain: Maintain services/systems indicated to remain and protect them against damage.
- B. Existing Services/Systems to Be Removed, Relocated, or Abandoned: Locate, identify, disconnect, and seal or cap off indicated utility services and mechanical/electrical systems serving areas to be selectively demolished.
 - 1. Building manager will arrange to shut off indicated services/systems when requested by Contractor.
 - 2. Arrange to shut off indicated utilities with utility companies.
 - 3. If services/systems are required to be removed, relocated, or abandoned, provide temporary services/systems that bypass area of selective demolition and that maintain continuity of services/systems to other parts of building.
 - 4. Disconnect, demolish, and remove fire-suppression systems, plumbing, and HVAC systems, equipment, and components indicated to be removed.
 - a. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
 - b. Piping to Be Abandoned in Place: Drain piping and cap or plug piping with same or compatible piping material.
 - c. Equipment to Be Removed: Disconnect and cap services and remove equipment.
 - d. Equipment to Be Removed and Reinstalled: Disconnect and cap services and remove, clean, and store equipment; when appropriate, reinstall, reconnect, and make equipment operational.
 - e. Equipment to Be Removed and Salvaged: Disconnect and cap services and remove equipment and deliver to Owner.
 - f. Ducts to Be Removed: Remove portion of ducts indicated to be removed and plug remaining ducts with same or compatible ductwork material.
 - g. Ducts to Be Abandoned in Place: Cap or plug ducts with same or compatible ductwork material.
- C. Refrigerant: Remove refrigerant from mechanical equipment to be selectively demolished according to 40 CFR 82 and regulations of authorities having jurisdiction.

3.3 PREPARATION

- A. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

- B. Temporary Facilities: Provide temporary barricades and other protection required to prevent injury to people and damage to adjacent buildings and facilities to remain.
- C. Temporary Shoring: Provide and maintain shoring, bracing, and structural supports as required to preserve stability and prevent movement, settlement, or collapse of construction and finishes to remain, and to prevent unexpected or uncontrolled movement or collapse of construction being demolished.

3.4 SELECTIVE DEMOLITION, GENERAL

- A. General: Demolish and remove existing construction only to the extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:
 - 1. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage construction to remain or adjoining construction. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping, to minimize disturbance of adjacent surfaces. Temporarily cover openings to remain.
 - 2. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.
 - 3. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space before starting flame-cutting operations. Maintain fire watch and portable fire-suppression devices during flame-cutting operations.
 - 4. Locate selective demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.
 - 5. Dispose of demolished items and materials promptly.
- B. Removed and Salvaged Items:
 - 1. Clean salvaged items.
 - 2. Pack or crate items after cleaning. Identify contents of containers.
 - 3. Store items in a secure area until delivery to Owner.
 - 4. Transport items to Owner's storage area designated by Owner.
 - 5. Protect items from damage during transport and storage.
- C. Removed and Reinstalled Items:
 - 1. Clean and repair items to functional condition adequate for intended reuse.
 - 2. Pack or crate items after cleaning and repairing. Identify contents of containers.
 - 3. Protect items from damage during transport and storage.
 - 4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.
- D. Existing Items to Remain: Protect construction indicated to remain against damage and soiling during selective demolition. When permitted by Architect, items may be removed to a suitable, protected storage location during selective demolition and

cleaned and reinstalled in their original locations after selective demolition operations are complete.

3.5 DISPOSAL OF DEMOLISHED MATERIALS

- A. General: Except for items or materials indicated to be reused, salvaged, reinstalled, or otherwise indicated to remain Owner's property, remove demolished materials from Project site and legally dispose of them in an EPA-approved landfill.
 - 1. Do not allow demolished materials to accumulate on-site.
 - 2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
 - 3. Remove debris from elevated portions of building by chute, hoist, or other device that will convey debris to grade level in a controlled descent.
- B. Burning: Do not burn demolished materials.
- C. Disposal: Transport demolished materials off Owner's property and legally dispose of them.

3.6 CLEANING

- A. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

END OF SECTION

SECTION 035416
HYDRAULIC CEMENT UNDERLAYMENT

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes polymer-modified, self-leveling, hydraulic cement underlayment for application below interior floor coverings.

1.2 PRICE AND PAYMENT PROCEDURES

- A. Refer to Section 01 21 00 "Allowances," for description of work under this Section affected by Allowances.

1.3 UNIT PRICES

- A. Refer to Section 01 22 00 "Unit Prices," for description of work under this Section affected by Unit Prices.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.5 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications: Installer who is approved by manufacturer for application of underlayment products required for this Project.

1.7 FIELD CONDITIONS

- A. Environmental Limitations: Comply with manufacturer's written instructions for substrate temperature, ventilation, ambient temperature and humidity, and other conditions affecting underlayment performance.
 - 1. Place hydraulic cement underlayments only when ambient temperature and temperature of substrates are between 50 and 80 deg F.

PART 2 - PRODUCTS

2.1 HYDRAULIC CEMENT UNDERLAYMENTS

- A. Hydraulic Cement Underlayment: Polymer-modified, self-leveling, hydraulic cement product that can be applied in minimum uniform thickness of 1/4 inch and that can be feathered at edges to match adjacent floor elevations.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX GmbH.
 - b. Dayton Superior.
 - c. Hacker Industries, Inc.
 - d. MAPEI Corporation.
 - e. Maxxon Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.
 - g. United States Gypsum Company.
 - 2. Cement Binder: ASTM C 150/C 150M, portland cement, or hydraulic or blended hydraulic cement as defined by ASTM C 219.
 - 3. Compressive Strength: Not less than 5000 psi at 28 days when tested according to ASTM C 109/C 109M.
 - 4. Underlayment Additive: Resilient-emulsion product of underlayment manufacturer, formulated for use with underlayment when applied to substrate and conditions indicated.
- B. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch; or coarse sand as recommended by underlayment manufacturer.
 - 1. Provide aggregate when recommended in writing by underlayment manufacturer for underlayment thickness required.
- C. Water: Potable and at a temperature of not more than 70 deg F.
- D. Primer: Product of underlayment manufacturer recommended in writing for substrate, conditions, and application indicated.
- E. Surface Sealer: Designed to reduce porosity as recommended by manufacturer for type of floor covering to be applied to underlayment.

PART 3 - EXECUTION

3.1 PREPARATION

- A. General: Prepare and clean substrate according to manufacturer's written instructions.
 - 1. Treat nonmoving substrate cracks according to manufacturer's written instructions to prevent cracks from telegraphing (reflecting) through underlayment.
 - 2. Fill substrate voids to prevent underlayment from leaking.
- B. Concrete Substrates: Mechanically remove, according to manufacturer's written instructions, laitance, glaze, efflorescence, curing compounds, form-release agents, dust, dirt, grease, oil, and other contaminants that might impair underlayment bond.
 - 1. Moisture Testing: Perform anhydrous calcium chloride test, ASTM F 1869. Proceed with installation only after substrates do not exceed a maximum moisture-vapor-emission rate of 3 lb of water/1000 sq. ft. in 24 hours.
- C. Nonporous Substrates: For ceramic tile, quarry tile, and terrazzo substrates, remove waxes, sealants, and other contaminants that might impair underlayment bond, and prepare surfaces according to manufacturer's written instructions.
- D. Adhesion Tests: After substrate preparation, test substrate for adhesion with underlayment according to manufacturer's written instructions.

3.2 APPLICATION

- A. General: Mix and apply underlayment components according to manufacturer's written instructions.
 - 1. Close areas to traffic during underlayment application and for time period after application recommended in writing by manufacturer.
 - 2. Coordinate application of components to provide optimum adhesion to substrate and between coats.
 - 3. At substrate expansion, isolation, and other moving joints, allow joint of same width to continue through underlayment.
- B. Apply primer over prepared substrate at manufacturer's recommended spreading rate.
- C. Apply underlayment to produce uniform, level surface.
 - 1. Apply a final layer without aggregate to product surface.
 - 2. Feather edges to match adjacent floor elevations.
- D. Cure underlayment according to manufacturer's written instructions. Prevent contamination during application and curing processes.
- E. Do not install floor coverings over underlayment until after time period recommended in writing by underlayment manufacturer.

- F. Apply surface sealer at rate recommended by manufacturer.
- G. Remove and replace underlayment areas that evidence lack of bond with substrate, including areas that emit a "hollow" sound when tapped.

END OF SECTION

SECTION 04 22 00
CONCRETE UNIT MASONRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Concrete masonry units.
2. Mortar and grout.
3. Steel reinforcing bars.
4. Masonry joint reinforcement.
5. Embedded flashing.
6. Miscellaneous masonry accessories.

1.2 DEFINITIONS

- A. CMU(s):** Concrete masonry unit(s).
- B. Reinforced Masonry:** Masonry containing reinforcing steel in grouted cells.

1.3 ACTION SUBMITTALS

- A. Product Data:** Technical data for each type of product.
- B. Shop Drawings:** For the following:
1. Masonry Units: Show sizes, profiles, coursing, and locations of special shapes.
 2. Reinforcing Steel: Detail bending, lap lengths, and placement of unit masonry reinforcing bars. Comply with ACI 315. Show elevations of reinforced walls.
 3. Fabricated Flashing: Detail corner units, end dam units, and special applications.
- C. Samples:** Submit each type and color of the following:
1. Flashing and accessories intended for CMU.

1.4 INFORMATIONAL SUBMITTALS

- A. Qualification Data:** Submit data for testing agency.
- B. Material Certificates:** Submit certificates for each type and size of the following:
1. Masonry units.

- a. Include data on material properties and material test reports substantiating compliance with requirements.
 - b. For masonry units used in structural masonry, include data and calculations establishing average net-area compressive strength of units.
- 2. Cementitious materials. Include name of manufacturer, brand name, and type.
- 3. Mortar admixtures.
- 4. Preblended, dry mortar mixes. Include description of type and proportions of ingredients.
- 5. Grout mixes. Include description of type and proportions of ingredients.
- 6. Reinforcing bars.
- 7. Joint reinforcement.
- 8. Anchors, ties, and metal accessories.
- C. Mix Designs: For each type of mortar and grout. Include description of type and proportions of ingredients.
 - 1. Include test reports for mortar mixes required to comply with property specification. Test according to ASTM C 109/C 109M for compressive strength, ASTM C 1506 for water retention, and ASTM C 91/C 91M for air content.
 - 2. Include test reports, according to ASTM C 1019, for grout mixes required to comply with compressive strength requirement.
- D. Statement of Compressive Strength of Masonry: For each combination of masonry unit type and mortar type, provide statement of average net-area compressive strength of masonry units, mortar type, and resulting net-area compressive strength of masonry determined according to TMS 602/ACI 530.1/ASCE 6.
- E. Cold Weather and Hot Weather Procedures: Detailed description of methods, materials, and equipment to be used to comply with requirements.

1.5 QUALITY ASSURANCE

- A. Source Limitations for Masonry Units: Obtain exposed masonry units of a uniform texture and color, or a uniform blend within the ranges accepted for these characteristics, from single source from single manufacturer for each product required.
- B. Source Limitations for Mortar Materials: Obtain mortar ingredients of a uniform quality, including color for exposed masonry, from single manufacturer for each cementitious component and from single source or producer for each aggregate.
- C. Testing Agency Qualifications: Qualified according to ASTM C 1093 for testing indicated.
- D. Mockups: Build mockups to verify selections, to demonstrate aesthetic effects, and to set quality standards for materials and execution.
 - 1. Build mockups for each type of exposed unit masonry construction in sizes approximately 72 inches (1800 mm) long by 72 inches (1800 mm) high by full thickness, including face and backup wythes and accessories.

- a. Include a sealant filled joint at least 16 inches (400 mm) long in each mockup.
 - b. Include through wall flashing installed for a 24 inch (600 mm) length in corner of exterior wall mockup approximately 16 inches (400 mm) down from top of mockup, with a 12 inch (300 mm) length of flashing left exposed to view (omit masonry above half of flashing).
 2. Protect accepted mockups from the elements with weather resistant membrane.
 3. Approval of mockups is for color, texture, and blending of masonry units; relationship of mortar and sealant colors to masonry unit colors; tooling of joints; and aesthetic qualities of workmanship.
 - a. Approval of mockups is for other material and construction qualities specifically approved by Architect in writing.
 - b. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
 4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.
- E. Preinstallation Conference: Conduct conference at site.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Store masonry units on elevated platforms in a dry location. If units are not stored in an enclosed location, cover tops and sides of stacks with waterproof sheeting, securely tied. If units become wet, do not install until they are dry.
- B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.
- C. Store aggregates where grading and required characteristics can be maintained and contamination avoided.
- D. Deliver preblended, dry mortar mix in moisture-resistant containers. Store preblended, dry mortar mix in delivery containers on elevated platforms in a dry location or in covered weatherproof dispensing silos.
- E. Store masonry accessories, including metal items, to prevent corrosion and accumulation of dirt and oil.

1.7 FIELD CONDITIONS

- A. Protection of Masonry: During construction, cover tops of walls, projections, and sills with waterproof sheeting at end of each day's work. Cover partially completed masonry when construction is not in progress.

1. Extend cover a minimum of 24 inches (600 mm) down both sides of walls, and hold cover securely in place.
- B. Do not apply uniform floor or roof loads for at least 12 hours and concentrated loads for at least three days after building masonry walls or columns.
- C. Stain Prevention: Prevent grout, mortar, and soil from staining the face of masonry to be left exposed or painted. Immediately remove grout, mortar, and soil in contact with masonry.
 1. Protect base of walls from rain splashed mud and from mortar splatter by spreading coverings on ground and over wall surface.
 2. Protect sills, ledges, and projections from mortar droppings.
 3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.
 4. Turn scaffold boards near the wall on edge at the end of each day to prevent rain from splashing mortar and dirt onto completed masonry.
- D. Cold Weather Requirements: Do not use frozen materials or materials mixed or coated with ice or frost. Do not build on frozen substrates. Remove and replace unit masonry damaged by frost or by freezing conditions. Comply with cold-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.
 1. Cold Weather Cleaning: Use liquid cleaning methods only when air temperature is 40 degrees F (4 degrees C) and higher and will remain so until masonry has dried, but not less than seven days after completing cleaning.
- E. Hot Weather Requirements: Comply with hot-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Provide unit masonry that develops indicated net-area compressive strengths at 28 days.
 1. Determine net-area compressive strength of masonry from average net-area compressive strengths of masonry units and mortar types (unit-strength method) according to TMS 602/ACI 530.1/ASCE 6.
 2. Determine net-area compressive strength of masonry by testing masonry prisms according to ASTM C 1314.

2.2 UNIT MASONRY

- A. Masonry Standard: Comply with TMS 602/ACI 530.1/ASCE 6 except as modified by requirements in the Contract Documents.

- B. Defective Units: Referenced masonry unit standards may allow a certain percentage of units to contain chips, cracks, or other defects exceeding limits stated. Do not use units where such defects are exposed in the completed work.
- C. Fire Resistance Ratings: Comply with requirements for fire resistance rated assembly designs indicated.
 - 1. Where fire resistance rated construction is indicated, units shall be listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction.

2.3 CONCRETE MASONRY UNITS

- A. Shapes: Provide shapes indicated and as follows, with exposed surfaces matching exposed faces of adjacent units unless otherwise indicated.
 - 1. Provide special shapes for lintels, corners, jambs, sashes, movement joints, headers, bonding, and other special conditions.
 - 2. Provide bullnose units for outside corners unless otherwise indicated.
- B. CMUs: ASTM C 90 Normal weight unless otherwise indicated.
 - 1. Size (Width): Manufactured to dimensions 3/8 inch (10 mm) less-than-nominal dimensions.
 - 2. Exposed Faces: Provide color and texture matching the range represented by Architect's sample.
- C. Concrete Building Brick: ASTM C 55.
 - 1. Density Classification: Normal weight.
 - 2. Size (Actual Dimensions): 3-5/8 inches (92 mm) wide by 2-1/4 inches (57 mm) high by 7-5/8 inches (194 mm) long.

2.4 MORTAR AND GROUT MATERIALS

- A. Portland Cement: ASTM C 150/C 150M, Type I or II, except Type III may be used for cold weather construction. Provide natural color or white cement as required to produce mortar color indicated.
 - 1. Alkali content shall not be more than 0.1 percent when tested according to ASTM C 114.
- B. Hydrated Lime: ASTM C 207, Type S.
- C. Portland Cement-Lime Mix: Packaged blend of portland cement and hydrated lime containing no other ingredients.
- D. Masonry Cement: ASTM C 91/C 91M.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- a. Cemex S.A.B. de C.V.
 - b. Essroc.
 - c. Holcim (US) Inc.
 - d. Lafarge North America Inc.
 - e. Lehigh Hanson; HeidelbergCement Group.
- E. Mortar Cement: ASTM C 1329/C 1329M.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Lafarge North America Inc.
- F. Aggregate for Mortar: ASTM C 144.
 - 1. For mortar that is exposed to view, use washed aggregate consisting of natural sand or crushed stone.
 - 2. For joints less than 1/4 inch (6 mm) thick, use aggregate graded with 100 percent passing the No. 16 (1.18-mm) sieve.
 - 3. White Mortar Aggregates: Natural white sand or crushed white stone.
 - 4. Colored Mortar Aggregates: Natural sand or crushed stone of color necessary to produce required mortar color.
- G. Aggregate for Grout: ASTM C 404.
- H. Cold Weather Admixture: Nonchloride, noncorrosive, accelerating admixture complying with ASTM C 494/C 494M, Type C, and recommended by manufacturer for use in masonry mortar of composition indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. BASF Corp. - Construction Chemicals.
 - b. Euclid Chemical Company (The); an RPM company.
 - c. GCP Applied Technologies Inc. (formerly Grace Construction Products).
- I. Water Repellent Admixture: Liquid water repellent mortar admixture intended for use with CMUs containing integral water repellent from same manufacturer.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ACM Chemistries.
 - b. BASF Corp. - Construction Chemicals.
 - c. Euclid Chemical Company (The); an RPM company.
 - d. GCP Applied Technologies Inc. (formerly Grace Construction Products).
- J. Water: Potable.

2.5 REINFORCEMENT

- A. Uncoated Steel Reinforcing Bars: ASTM A 615/A 615M or ASTM A 996/A 996M, Grade 60 (Grade 420).
- B. Reinforcing Bar Positioners: Wire units designed to fit into mortar bed joints spanning masonry unit cells and to hold reinforcing bars in center of cells. Units are formed from 0.148 inch (3.77-mm) steel wire, hot dip galvanized after fabrication. Provide units designed for number of bars indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Dur-O-Wal; a Hohmann & Barnard company.
 - b. Heckmann Building Products, Inc.
 - c. Hohmann & Barnard, Inc.
 - d. Lock Rite.
 - e. Wire-Bond.
- C. Masonry Joint Reinforcement: Ladder type complying with ASTM A 951/A 951M.
 - 1. Interior Walls: Hot dip galvanized carbon steel.
 - 2. Exterior Walls: Hot dip galvanized carbon steel.
 - 3. Wire Size for Side Rods: 0.187 inch (4.76 mm) diameter.
 - 4. Wire Size for Cross Rods: 0.187 inch (4.76 mm) diameter.
 - 5. Spacing of Cross Rods: Not more than 16 inches (407 mm) o.c.
 - 6. Provide in lengths of not less than 10 feet (3 m), with prefabricated corner and tee units.

2.6 TIES AND ANCHORS

- A. Ties and anchors shall extend at least 1-1/2 inches (38 mm) into masonry but with at least a 5/8 inch (16 mm) cover on outside face.
- B. Materials: Provide ties and anchors specified in this article that are made from materials that comply with the following unless otherwise indicated:
 - 1. Mill Galvanized, Carbon Steel Wire: ASTM A 82/A 82M, with ASTM A 641/A 641M, Class 1 coating.
 - 2. Hot Dip Galvanized, Carbon Steel Wire: ASTM A 82/A 82M, with ASTM A 153/A 153M, Class B-2 coating.
 - 3. Stainless Steel Wire: ASTM A 580/A 580M, Type 316.
 - 4. Galvanized Steel Sheet: ASTM A 653/A 653M, Commercial Steel, G60 (Z180) zinc coating.
 - 5. Steel Sheet, Galvanized after Fabrication: ASTM A 1008/A 1008M, Commercial Steel, with ASTM A 153/A 153M, Class B coating.
 - 6. Stainless Steel Sheet: ASTM A 240/A 240M or ASTM A 666, Type 304.
 - 7. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.

- C. Adjustable Anchors for Connecting to Structural Steel Framing: Provide anchors that allow vertical or horizontal adjustment but resist tension and compression forces perpendicular to plane of wall.
1. Anchor Section for Welding to Steel Frame: Crimped 1/4 inch (6.35 mm) diameter, stainless steel wire. Mill galvanized wire may be used at interior walls unless otherwise indicated.
 2. Tie Section: Triangular shaped wire tie made from 0.25 inch (6.35 mm) diameter, stainless steel wire. Mill galvanized wire may be used at interior walls unless otherwise indicated.
- D. Adjustable Anchors for Connecting to Concrete: Provide anchors that allow vertical or horizontal adjustment but resist tension and compression forces perpendicular to plane of wall.
1. Connector Section: Dovetail tabs for inserting into dovetail slots in concrete and attached to tie section; formed from 0.109 inch (2.78 mm) thick, stainless steel sheet.
 - a. 0.108 inch (2.74 mm) thick, galvanized steel sheet may be used at interior walls unless otherwise indicated.
 2. Tie Section: Triangular shaped wire tie made from 0.25 inch (6.35 mm) diameter, stainless steel wire. Mill galvanized wire may be used at interior walls unless otherwise indicated.
 3. Corrugated Metal Ties: Metal strips not less than 7/8 inch (22 mm) wide with corrugations having a wavelength of 0.3 to 0.5 inch (7.6 to 12.7 mm) and an amplitude of 0.06 to 0.10 inch (1.5 to 2.5 mm) made from 0.105 inch (2.66 mm) thick steel sheet, galvanized after fabrication with dovetail tabs for inserting into dovetail slots in concrete.
- E. Partition Top Anchors: 0.105 inch (2.66 mm) thick metal plate with a 3/8 inch (9.5 mm) diameter metal rod 6 inches (152 mm) long welded to plate and with closed end plastic tube fitted over rod that allows rod to move in and out of tube. Fabricate from steel, hot dip galvanized after fabrication.
- F. Rigid Anchors: Fabricate from steel bars 1-1/2 inches (38 mm) wide by 1/4 inch (6.35 mm) thick by 24 inches (610 mm) long, with ends turned up 2 inches (51 mm) or with cross pins unless otherwise indicated bent to configuration indicated.
1. Corrosion Protection: Hot dip galvanized to comply with ASTM A 153/A 153M.

2.7 EMBEDDED FLASHING MATERIALS

- A. Metal Flashing: Provide metal flashing complying with SMACNA Architectural Sheet Metal Manual and as follows:
1. Stainless Steel: ASTM A 240/A 240M or ASTM A 666, Type 304, 0.016 inch (0.40 mm) thick.

2. Fabricate continuous flashings in sections 96 inches (2400 mm) long minimum, but not exceeding 12 feet (3.7 m). Provide splice plates at joints of formed, smooth metal flashing.
3. Fabricate through wall metal flashing embedded in masonry from stainless steel, with ribs at 3 inch (76 mm) intervals along length of flashing to provide an integral mortar bond.
 - a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1) Cheney Flashing Company.
 - 2) Hohmann & Barnard, Inc.
 - 3) Keystone Flashing Company, Inc.
4. Fabricate through wall flashing with snaplock receiver on exterior face where indicated to receive counterflashing.
5. Fabricate through wall flashing with drip edge unless otherwise indicated. Fabricate by extending flashing 1/2 inch (13 mm) out from wall, with outer edge bent down 30 degrees and hemmed.
6. Fabricate through wall flashing with sealant stop where indicated. Fabricate by bending metal back on itself 3/4 inch (19 mm) at exterior face of wall and down into joint 1/4 inch (6 mm) to form a stop for retaining sealant backer rod.
7. Fabricate metal drip edges from stainless steel. Extend at least 3 inches (76 mm) into wall and 1/2 inch (13 mm) out from wall, with outer edge bent down 30 degrees and hemmed.
8. Fabricate metal sealant stops from stainless steel. Extend at least 3 inches (76 mm) into wall and out to exterior face of wall. At exterior face of wall, bend metal back on itself for 3/4 inch (19 mm) and down into joint 1/4 inch (6 mm) to form a stop for retaining sealant backer rod.
9. Fabricate metal expansion joint strips from stainless steel to shapes indicated.
10. Solder metal items at corners.

B. Flexible Flashing:

1. Butyl Rubber Flashing: Composite, self adhesive, flashing product consisting of a pliable, butyl rubber compound, bonded to a high density polyethylene film, aluminum foil, or spunbonded polyolefin to produce an overall thickness of not less than 0.040 inch (1.02 mm).
 - a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1) DuPont Safety & Construction.
 - 2) GCP Applied Technologies Inc. (formerly Grace Construction Products).
 - 3) Protecto Wrap Company.
 - 4) Raven Industries, Inc.
 - 5) Wire-Bond.
 - b. Accessories: Provide preformed corners, end dams, other special shapes, and seaming materials produced by flashing manufacturer.

- C. Application: Unless otherwise indicated, use the following:
 - 1. Where flashing is indicated to receive counterflashing, use metal flashing.
 - 2. Where flashing is indicated to be turned down at or beyond the wall face, use metal flashing.
 - 3. Where flashing is partly exposed and is indicated to terminate at the wall face, use flexible flashing with a metal drip edge.
 - 4. Where flashing is fully concealed, use flexible flashing.
 - 5. Provide stainless steel flashing at all through wall conditions indicated.
- D. Single Wythe CMU Flashing System: System of CMU cell flashing pans and interlocking CMU web covers made from UV-resistant, high-density polyethylene. Cell flashing pans have integral weep spouts designed to be built into mortar bed joints and that extend into the cell to prevent clogging with mortar.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Mortar Net Solutions.
- E. Solder and Sealants for Sheet Metal Flashings: Specified in Section 07 62 00.
 - 1. Solder for Stainless Steel: ASTM B 32, Grade Sn60, with acid flux of type recommended by stainless steel sheet manufacturer.
 - 2. Elastomeric Sealant: ASTM C 920, chemically curing sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and remain watertight and as recommended by the flexible flashing manufacturer.
- F. Adhesives, Primers, and Seam Tapes for Flashings: Flashing Recommended by flashing manufacturer for bonding flashing sheets to each other and to substrates.

2.8 MISCELLANEOUS MASONRY ACCESSORIES

- A. Compressible Filler: Premolded filler strips complying with ASTM D 1056, Grade 2A1; compressible up to 35 percent; of width and thickness indicated; formulated from neoprene.
- B. Preformed Control Joint Gaskets: Made from styrene-butadiene-rubber compound, complying with ASTM D 2000, Designation M2AA-805 and designed to fit standard sash block and to maintain lateral stability in masonry wall; size and configuration as indicated.
- C. Bond Breaker Strips: Asphalt saturated felt complying with ASTM D 226/D 226M, Type I (No. 15 asphalt felt).

2.9 MORTAR AND GROUT MIXES

- A. Do not use admixtures, including pigments, air entraining agents, accelerators, retarders, water repellent agents, antifreeze compounds, or admixtures unless otherwise indicated.
 - 1. Do not use calcium chloride in mortar or grout.
 - 2. Use portland cement-lime mortar unless otherwise indicated.
 - 3. Add cold weather admixture (if used) at same rate for mortar that will be exposed to view, regardless of weather conditions, to ensure that mortar color is consistent.
- B. Preblended, Dry Mortar Mix: Furnish dry mortar ingredients in form of a preblended mix. Measure quantities by weight to ensure accurate proportions, and thoroughly blend ingredients before delivering to Project site.
- C. Mortar for Unit Masonry: Comply with ASTM C 270, Proportion Specification. Provide the following types of mortar for applications stated unless another type is indicated or needed to provide required compressive strength of masonry.
 - 1. For masonry below grade or in contact with earth, use Type M.
 - 2. For reinforced masonry, use Type S.
 - 3. For mortar parge coats, use Type S or Type N.
 - 4. For exterior, above grade, load bearing and nonload bearing walls and parapet walls; for interior load bearing walls; for interior nonload bearing partitions; and for other applications where another type is not indicated, use Type N.
 - 5. For interior nonload bearing partitions, Type O may be used instead of Type N.
- D. Grout for Unit Masonry: Comply with ASTM C 476.
 - 1. Use grout of type indicated or, if not otherwise indicated, of type (fine or coarse) that will comply with TMS 602/ACI 530.1/ASCE 6 for dimensions of grout spaces and pour height.
 - 2. Proportion grout in accordance with ASTM C 476, paragraph 4.2.2 for specified 28-day compressive strength indicated, but not less than 2000 psi (14 MPa).
 - 3. Provide grout with a slump of 8 to 11 inches (200 to 280 mm) as measured according to ASTM C 143/C 143M.
- E. Epoxy Pointing Mortar: Mix epoxy pointing mortar to comply with mortar manufacturer's written instructions.
 - 1. Application: Use epoxy pointing mortar for exposed mortar joints with pre-faced CMUs.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions, for compliance with requirements for installation tolerances and other conditions affecting performance of the work.
 - 1. For the record, prepare written report listing conditions detrimental to performance of the Work.
 - 2. Verify that foundations are within tolerances specified.
 - 3. Verify that reinforcing dowels are properly placed.
 - 4. Verify that substrates are free of substances that would impair mortar bond.
- B. Before installation, examine rough-in and built-in construction for piping systems to verify actual locations of piping.
- C. Proceed with installation after correcting unsatisfactory conditions.

3.2 INSTALLATION

- A. Build chases and recesses to accommodate items specified in this and other Sections.
- B. Leave openings for equipment to be installed before completing masonry. After installing equipment, complete masonry to match construction immediately adjacent to opening.
- C. Use full size units without cutting if possible. If cutting is required to provide a continuous pattern or to fit adjoining construction, cut units with motor driven saws; provide clean, sharp, unchipped edges. Allow units to dry before laying unless wetting of units is specified. Install cut units with cut surfaces and, where possible, cut edges concealed.

3.3 TOLERANCES

- A. Dimensions and Locations of Elements:
 - 1. For dimensions in cross section or elevation, do not vary by more than plus 1/2 inch (12 mm) or minus 1/4 inch (6 mm).
 - 2. For location of elements in plan, do not vary from that indicated by more than plus or minus 1/2 inch (12 mm).
 - 3. For location of elements in elevation, do not vary from that indicated by more than plus or minus 1/4 inch (6 mm) in a story height or 1/2 inch (12 mm) total.
- B. Lines and Levels:
 - 1. For bed joints and top surfaces of bearing walls, do not vary from level by more than 1/4 inch in 10 feet (6 mm in 3 m), or 1/2 inch (12 mm) maximum.

2. For conspicuous horizontal lines, such as lintels, sills, parapets, and reveals, do not vary from level by more than 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2 inch (12 mm) maximum.
3. For vertical lines and surfaces do not vary from plumb by more than 1/4 inch in 10 feet (6 mm in 3 m), 3/8 inch in 20 feet (9 mm in 6 m), or 1/2 inch (12 mm) maximum.
4. For conspicuous vertical lines, such as external corners, door jambs, reveals, and expansion and control joints, do not vary from plumb by more than 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2 inch (12 mm) maximum.
5. For lines and surfaces, do not vary from straight by more than 1/4 inch in 10 feet (6 mm in 3 m), 3/8 inch in 20 feet (9 mm in 6 m), or 1/2 inch (12 mm) maximum.
6. For vertical alignment of exposed head joints, do not vary from plumb by more than 1/4 inch in 10 feet (6 mm in 3 m), or 1/2-inch (12 mm) maximum.
7. For faces of adjacent exposed masonry units, do not vary from flush alignment by more than 1/16 inch (1.5 mm).

C. Joints:

1. For bed joints, do not vary from thickness indicated by more than plus or minus 1/8 inch (3 mm), with a maximum thickness limited to 1/2 inch (12 mm).
2. For exposed bed joints, do not vary from bed joint thickness of adjacent courses by more than 1/8 inch (3 mm).
3. For head and collar joints, do not vary from thickness indicated by more than plus 3/8 inch (9 mm) or minus 1/4 inch (6 mm).
4. For exposed head joints, do not vary from thickness indicated by more than plus or minus 1/8 inch (3 mm).

3.4 LAYING MASONRY WALLS

- A. Lay out walls in advance for accurate spacing of surface bond patterns with uniform joint thicknesses and for accurate location of openings, movement-type joints, returns, and offsets. Avoid using less than half size units, particularly at corners, jambs, and, where possible, at other locations.
- B. Bond Pattern for Exposed Masonry: Unless otherwise indicated, lay exposed masonry in running bond; do not use units with less than nominal 4 inch (100 mm) horizontal face dimensions at corners or jambs.
- C. Lay concealed masonry with all units in a wythe in running bond or bonded by lapping not less than 4 inches (100 mm). Bond and interlock each course of each wythe at corners. Do not use units with less than nominal 4 inch (100 mm) horizontal face dimensions at corners or jambs.
- D. Stopping and Resuming Work: Stop work by stepping back units in each course from those in course below; do not tooth. When resuming work, clean masonry surfaces that are to receive mortar, remove loose masonry units and mortar, and wet brick if required before laying fresh masonry.

- E. Built-in Work: As construction progresses, build in items specified in this and other Sections. Fill in solidly with masonry around built-in items.
- F. Fill space between steel frames and masonry solidly with mortar unless otherwise indicated.
- G. Where built-in items are to be embedded in cores of hollow masonry units, place a layer of metal lath, wire mesh, or plastic mesh in the joint below, and rod mortar or grout into core.
- H. Fill cores in hollow CMUs with grout 24 inches (600 mm) under bearing plates, beams, lintels, posts, and similar items unless otherwise indicated.
- I. Build nonload bearing interior partitions full height of story to underside of solid floor or roof structure above unless otherwise indicated.
 - 1. Install compressible filler in joint between top of partition and underside of structure above.
 - 2. Fasten partition top anchors to structure above and build into top of partition. Grout cells of CMUs solidly around plastic tubes of anchors and push tubes down into grout to provide 1/2 inch (13 mm) clearance between end of anchor rod and end of tube. Space anchors 48 inches (1200 mm) o.c. unless otherwise indicated.
 - 3. Wedge nonload bearing partitions against structure above with small pieces of tile, slate, or metal. Fill joint with mortar after dead load deflection of structure above approaches final position.
 - 4. At fire rated partitions, treat joint between top of partition and underside of structure above to comply with Section 07 84 43.

3.5 MORTAR BEDDING AND JOINTING

- A. Lay hollow CMUs as follows:
 - 1. Bed face shells in mortar and make head joints of depth equal to bed joints.
 - 2. Bed webs in mortar in all courses of piers, columns, and pilasters.
 - 3. Bed webs in mortar in grouted masonry, including starting course on footings.
 - 4. Fully bed entire units, including areas under cells, at starting course on footings where cells are not grouted.
- B. Lay solid CMUs with filled bed and head joints; butter ends with mortar to fill head joints and shove into place. Do not deeply furrow bed joints or slush head joints.
- C. Set cast stone trim units in full bed of mortar with full vertical joints. Fill dowel, anchor, and similar holes.
 - 1. Clean soiled surfaces with fiber brush and soap powder and rinse thoroughly with clear water.
 - 2. Wet joint surfaces thoroughly before applying mortar.
 - 3. Rake out mortar joints for pointing with sealant.

- D. Rake out mortar joints at prefaced CMUs to a uniform depth of 1/4 inch (6 mm) and point with epoxy mortar to comply with epoxy mortar manufacturer's written instructions.
- E. Tool exposed joints slightly concave when thumbprint hard, using a jointer larger than joint thickness unless otherwise indicated.
- F. Cut joints flush for masonry walls to receive plaster or other direct-applied finishes (other than paint) unless otherwise indicated.
- G. Cut joints flush where indicated to receive waterproofing unless otherwise indicated.

3.6 MASONRY JOINT REINFORCEMENT

- A. Install entire length of longitudinal side rods in mortar with a minimum cover of 5/8 inch (16 mm) on exterior side of walls, 1/2 inch (13 mm) elsewhere. Lap reinforcement a minimum of 6 inches (150 mm).
 - 1. Space reinforcement not more than 16 inches (406 mm) o.c.
 - 2. Space reinforcement not more than 8 inches (203 mm) o.c. in foundation walls and parapet walls.
 - 3. Provide reinforcement not more than 8 inches (203 mm) above and below wall openings and extending 12 inches (305 mm) beyond openings in addition to continuous reinforcement.
- B. Interrupt joint reinforcement at control and expansion joints unless otherwise indicated.
- C. Provide continuity at wall intersections by using prefabricated T-shaped units.
- D. Provide continuity at corners by using prefabricated L-shaped units.
- E. Cut and bend reinforcing units as directed by manufacturer for continuity at corners, returns, offsets, column fireproofing, pipe enclosures, and other special conditions.

3.7 ANCHORING MASONRY TO STRUCTURAL STEEL AND CONCRETE

- A. Anchor masonry to structural steel and concrete, where masonry abuts or faces structural steel or concrete, to comply with the following:
 - 1. Provide an open space not less than 1 inch (25 mm) wide between masonry and structural steel or concrete unless otherwise indicated. Keep open space free of mortar and other rigid materials.
 - 2. Anchor masonry with anchors embedded in masonry joints and attached to structure.
 - 3. Space anchors as indicated, but not more than 24 inches (610 mm) o.c. vertically and 36 inches (915 mm) o.c. horizontally.

3.8 CONTROL AND EXPANSION JOINTS

- A. Install control and expansion joint materials in unit masonry as masonry progresses. Do not allow materials to span control and expansion joints without provision to allow for in plane wall or partition movement.
- B. Form control joints in concrete masonry as follows:
 - 1. Install preformed control joint gaskets designed to fit standard sash block.
 - 2. Install temporary foam plastic filler in head joints and remove filler when unit masonry is complete for application of sealant.

3.9 FLASHING

- A. General: Install embedded flashing at ledges and other obstructions to downward flow of water in wall where indicated.
- B. Install flashing as follows unless otherwise indicated:
 - 1. Prepare masonry surfaces so they are smooth and free from projections that could puncture flashing. Where flashing is within mortar joint, place through wall flashing on sloping bed of mortar and cover with mortar. Before covering with mortar, seal penetrations in flashing with adhesive, sealant, or tape recommended by flashing manufacturer.
 - 2. At lintels, extend flashing a minimum of 6 inches (150 mm) into masonry at each end. At heads and sills, extend flashing 6 inches (150 mm) at ends and turn up not less than 2 inches (50 mm) to form end dams.
 - 3. Interlock end joints of ribbed sheet metal flashing by overlapping ribs not less than 1-1/2 inches (38 mm) or as recommended by flashing manufacturer, and seal lap with elastomeric sealant complying with requirements in Section 07 92 00 for application indicated.
 - 4. Install metal drip edges and sealant stops with ribbed sheet metal flashing by interlocking hemmed edges to form hooked seam. Seal seam with elastomeric sealant complying with requirements in Section 07 92 00 for application indicated.
 - 5. Install metal drip edges beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch (13 mm) back from outside face of wall, and adhere flexible flashing to top of metal drip edge.
 - 6. Install metal flashing termination beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch (13 mm) back from outside face of wall and adhere flexible flashing to top of metal flashing termination.
 - 7. Cut flexible flashing off flush with face of wall after masonry wall construction is completed.
- C. Install single wythe CMU flashing system in bed joints of CMU walls where indicated to comply with manufacturer's written instructions. Install CMU cell pans with upturned edges located below face shells and webs of CMUs above and with weep spouts aligned with face of wall. Install CMU web covers so that they cover upturned edges of CMU cell pans at CMU webs and extend from face shell to face shell.

- D. Install reglets and nailers for flashing and other related construction where they are shown to be built into masonry.

3.10 REINFORCED UNIT MASONRY INSTALLATION

- A. Temporary Formwork and Shores: Construct formwork and shores as needed to support reinforced masonry elements during construction.
 - 1. Construct formwork to provide shape, line, and dimensions of completed masonry as indicated. Make forms sufficiently tight to prevent leakage of mortar and grout. Brace, tie, and support forms to maintain position and shape during construction and curing of reinforced masonry.
 - 2. Do not remove forms and shores until reinforced masonry members have hardened sufficiently to carry their own weight and other loads that may be placed on them during construction.
- B. Placing Reinforcement: Comply with requirements in TMS 602/ACI 530.1/ASCE 6.
- C. Grouting: Do not place grout until entire height of masonry to be grouted has attained enough strength to resist grout pressure.
 - 1. Comply with requirements in TMS 602/ACI 530.1/ASCE 6 for cleanouts and for grout placement, including minimum grout space and maximum pour height.
 - 2. Limit height of vertical grout pours to not more than 60 inches (1520 mm).

3.11 FIELD QUALITY CONTROL

- A. Testing and Inspecting: Owner will engage special inspectors to perform tests and inspections and prepare reports. Allow inspectors access to scaffolding and work areas as needed to perform tests and inspections. Retesting of materials that fail to comply with specified requirements shall be done at Contractor's expense.
- B. Inspections: Special inspections according to Level indicated by the Structural Engineer's Special Inspection statement and in accordance with TMS 402/ACI 530/ASCE 5.
 - 1. Begin masonry construction only after inspectors have verified proportions of site prepared mortar.
 - 2. Place grout only after inspectors have verified compliance of grout spaces and of grades, sizes, and locations of reinforcement.
 - 3. Place grout only after inspectors have verified proportions of site prepared grout.
- C. Testing Prior to Construction: One set of tests.
- D. Testing Frequency: One set of tests for each 5000 sq. ft. (464 sq. m) of wall area or portion thereof.
- E. Concrete Masonry Unit Test: For each type of unit provided, according to ASTM C 140 for compressive strength.

- F. Mortar Aggregate Ratio Test (Proportion Specification): For each mix provided, according to ASTM C 780.
- G. Mortar Test (Property Specification): For each mix provided, according to ASTM C 780. Test mortar for mortar air content and compressive strength.
- H. Grout Test (Compressive Strength): For each mix provided, according to ASTM C 1019.
- I. Prism Test: For each type of construction provided, according to ASTM C 1314 at 7 days and at 28 days.

3.12 PARGING

- A. Parge exterior faces of below grade masonry walls, where indicated, in two uniform coats to a total thickness of 3/4 inch (19 mm). Dampen wall before applying first coat, and scarify first coat to ensure full bond to subsequent coat.
- B. Use a steel-trowel finish to produce a smooth, flat, dense surface with a maximum surface variation of 1/8 inch per foot (3 mm per 300 mm). Form a wash at top of parging and a cove at bottom.
- C. Damp cure parging for at least 24 hours and protect parging until cured.

3.13 REPAIRING, POINTING, AND CLEANING

- A. Remove and replace masonry units that are loose, chipped, broken, stained, or otherwise damaged or that do not match adjoining units. Install new units to match adjoining units; install in fresh mortar, pointed to eliminate evidence of replacement.
- B. Pointing: During the tooling of joints, enlarge voids and holes, except weep holes, and completely fill with mortar. Point up joints, including corners, openings, and adjacent construction, to provide a neat, uniform appearance. Prepare joints for sealant application, where indicated.
- C. In-Progress Cleaning: Clean unit masonry as work progresses by dry brushing to remove mortar fins and smears before tooling joints.
- D. Final Cleaning: After mortar is thoroughly set and cured, clean exposed masonry as follows:
 - 1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
 - 2. Test cleaning methods on sample wall panel; leave one-half of panel uncleaned for comparison purposes. Obtain Architect's approval of sample cleaning before proceeding with cleaning of masonry.
 - 3. Protect adjacent stone and nonmasonry surfaces from contact with cleaner by covering them with liquid strippable masking agent or polyethylene film and waterproof masking tape.

4. Wet wall surfaces with water before applying cleaners; remove cleaners promptly by rinsing surfaces thoroughly with clear water.
5. Clean concrete masonry by applicable cleaning methods indicated in NCMA TEK 8-4A.

3.14 MASONRY WASTE DISPOSAL

- A. Salvageable Materials: Unless otherwise indicated, excess masonry materials are Contractor's property. At completion of unit masonry work, remove from Project site.
- B. Waste Disposal as Fill Material: Dispose of clean masonry waste, including excess or soil contaminated sand, waste mortar, and broken masonry units, by crushing and mixing with fill material as fill is placed.
 1. Crush masonry waste to less than 4 inches (100 mm) in each dimension.
 2. Mix masonry waste with at least two parts of specified fill material for each part of masonry waste. Fill material is specified in Section 31 20 00.
 3. Do not dispose of masonry waste as fill within 18 inches (450 mm) of finished grade.
- C. Masonry Waste Recycling: Return broken CMUs not used as fill to manufacturer for recycling.
- D. Excess Masonry Waste: Remove excess clean masonry waste that cannot be used as fill, as described above or recycled, and other masonry waste, and legally dispose of off Owner's property.

END OF SECTION

SECTION 055000
METAL FABRICATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Steel framing and supports for countertops.
2. Steel framing and supports for mechanical and electrical equipment.
3. Steel framing and supports for applications where framing and supports are not specified in other Sections.
4. Slotted channel framing.
5. Welded steel security cage and gate assemblies.

1.2 COORDINATION

- A.** Coordinate installation of metal fabrications that are anchored to or that receive other work. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

1.3 ACTION SUBMITTALS

- A.** Shop Drawings: Show fabrication and installation details. Include plans, elevations, sections, and details of metal fabrications and their connections. Show anchorage and accessory items.

1.4 QUALITY ASSURANCE

- A.** Welding Qualifications: Qualify procedures and personnel in accordance with the following:
1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."

1.5 FIELD CONDITIONS

- A.** Field Measurements: Verify actual locations of walls, floor slabs, decks, and other construction contiguous with metal fabrications by field measurements before fabrication.

PART 2 - PRODUCTS

2.1 METALS

- A. Metal Surfaces, General: Provide materials with smooth, flat surfaces unless otherwise indicated. For metal fabrications exposed to view in the completed Work, provide materials without seam marks, roller marks, rolled trade names, or blemishes.
- B. Steel Plates, Shapes, and Bars: ASTM A36/A36M.
- C. Steel Tubing: ASTM A500/A500M, cold-formed steel tubing.
- D. Steel Pipe: ASTM A53/A53M, Standard Weight (Schedule 40) unless otherwise indicated.
- E. Slotted Channel Framing: Cold-formed metal box channels (struts) complying with MFMA-4.
 - 1. Size of Channels: 1-5/8 by 1-5/8 inches.
 - 2. Cold-Rolled Steel: ASTM A1008/A1008M, structural steel, Grade 33; 0.0677-inch minimum thickness; coated with rust-inhibitive, baked-on, acrylic enamel.

2.2 FASTENERS

- A. Steel Bolts and Nuts: Regular hexagon-head bolts, ASTM A307, Grade A; with hex nuts, ASTM A563; and, where indicated, flat washers.
- B. High-Strength Bolts, Nuts, and Washers: ASTM F3125/F3125M, Grade A325, Type 3, heavy-hex steel structural bolts; ASTM A563, Grade DH3, heavy-hex carbon-steel nuts; and where indicated, flat washers.
- C. Post-Installed Anchors: Torque-controlled expansion anchors or chemical anchors.
 - 1. Material: Carbon-steel components zinc plated to comply with ASTM B633 or ASTM F1941/F1941M, Class Fe/Zn 5, unless otherwise indicated.
- D. Slotted-Channel Inserts: Cold-formed, hot-dip galvanized-steel box channels (struts) complying with MFMA-4, 1-5/8 by 7/8 inches by length indicated with anchor straps or studs not less than 3 inches long at not more than 8 inches o.c. Provide with temporary filler and tee-head bolts, complete with washers and nuts, all zinc-plated to comply with ASTM B633, Class Fe/Zn 5, as needed for fastening to inserts.

2.3 MISCELLANEOUS MATERIALS

- A. Shop Primers: Provide primers that comply with Section 09 91 23 "Interior Painting."
- B. Shrinkage-Resistant Grout: Factory-packaged, nonmetallic, nonstaining, noncorrosive, nongaseous grout complying with ASTM C1107/C1107M. Provide grout specifically recommended by manufacturer for interior and exterior applications.

2.4 FABRICATION, GENERAL

- A. Shop Assembly: Preassemble items in the shop to greatest extent possible. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.
- B. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.
- C. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.
- D. Form exposed work with accurate angles and surfaces and straight edges.
- E. Weld corners and seams continuously to comply with the following:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
- F. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners or welds where possible. Where exposed fasteners are required, use Phillips flat-head (countersunk) fasteners unless otherwise indicated. Locate joints where least conspicuous.
- G. Fabricate seams and other connections that are exposed to weather in a manner to exclude water. Provide weep holes where water may accumulate.
- H. Cut, reinforce, drill, and tap metal fabrications as indicated to receive finish hardware, screws, and similar items.
- I. Provide for anchorage of type indicated; coordinate with supporting structure. Space anchoring devices to secure metal fabrications rigidly in place and to support indicated loads.

2.5 MISCELLANEOUS FRAMING AND SUPPORTS

- A. General: Provide steel framing and supports not specified in other Sections as needed to complete the Work.
- B. Fabricate units from steel shapes, plates, and bars of welded construction unless otherwise indicated. Fabricate to sizes, shapes, and profiles indicated and as necessary to receive adjacent construction.

1. Fabricate units from slotted channel framing where indicated.
 2. Furnish inserts for units installed after concrete is placed.
- C. Fabricate supports for operable partitions from continuous steel beams of size indicated, or if not indicated, as recommended by partition manufacturer with attached bearing plates, anchors, and braces as recommended by partition manufacturer. Drill or punch bottom flanges of beams to receive partition track hanger rods; locate holes where indicated on operable partition Shop Drawings.
- D. Prime miscellaneous framing and supports with primer specified in Section 09 91 23 "Interior Painting.

2.6 SECURITY CAGE AND GATE ASSEMBLIES

- A. Provide welded steel security cage systems consisting of posts, rails, infill panels, and operable gate assemblies complete with anchorage, hardware reinforcement, and finishes.
1. Materials: ASTM A500 steel tubing; posts 3" x 3" (11 ga), rails 2" x 2" (16 ga), gate frame 2" x 3" (16 ga); perforated steel infill (min. 11 ga).
 2. Fabrication: Welded construction; grind exposed welds smooth; reinforce at hinges and hardware; provide anti-sag design.
 3. Performance: Resist 5 psf lateral load minimum; maintain alignment and operability under use.
 4. Anchorage: 1/4" base plates minimum; 1/2" anchors minimum; design for imposed loads.
 5. Finish: Powder coat; color as selected.

2.7 MISCELLANEOUS STEEL TRIM

- A. Unless otherwise indicated, fabricate units from steel shapes, plates, and bars of profiles shown with continuously welded joints and smooth exposed edges. Miter corners and use concealed field splices where possible.
- B. Provide cutouts, fittings, and anchorages as needed to coordinate assembly and installation with other work.
1. Provide with integrally welded steel strap anchors for embedding in concrete or masonry construction.
- C. Prime miscellaneous steel trim and supports with primer specified in Section 09 91 23 "Interior Painting.

2.8 GENERAL FINISH REQUIREMENTS

- A. Finish metal fabrications after assembly.
- B. Finish exposed surfaces to remove tool and die marks and stretch lines, and to blend into surrounding surface.

2.9 STEEL AND IRON FINISHES

- A. Shop prime iron and steel items unless they are to be embedded in concrete, sprayed-on fireproofing, or masonry, or unless otherwise indicated.
 - 1. Shop prime with primers specified in Section 09 91 23 "Interior Painting."
- B. Preparation for Shop Priming: Prepare surfaces to comply with requirements indicated below:
 - 1. SSPC-SP 3, "Power Tool Cleaning."
- C. Shop Priming: Apply shop primer to comply with SSPC-PA 1, "Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel," for shop painting.
 - 1. Stripe paint corners, crevices, bolts, welds, and sharp edges.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation; with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.
- B. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations.
- C. Field Welding: Comply with the following requirements:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
- D. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction. Provide threaded fasteners for use with concrete and masonry inserts, toggle bolts, through bolts, lag screws, wood screws, and other connectors.

3.2 INSTALLATION OF MISCELLANEOUS FRAMING AND SUPPORTS

- A. General: Install framing and supports to comply with requirements of items being supported, including manufacturers' written instructions and requirements indicated on Shop Drawings.
- B. Anchor supports for ceiling hung toilet partitions, operable partitions, overhead doors, and overhead grilles securely to, and rigidly brace from, building structure.

3.3 REPAIRS

- A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas. Paint uncoated and abraded areas with same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.
 - 1. Apply by brush or spray to provide a minimum 2.0-mil dry film thickness.

END OF SECTION

SECTION 057500
DECORATIVE FORMED METAL

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes decorative formed metal for interior applications:
 - 1. Closures and trim.
 - 2. Filler panels between dissimilar construction.
 - 3. Metal base.
 - 4. Miscellaneous decorative formed metal items made primarily from sheet metal.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product, including finishing materials.
- B. Shop Drawings: Show fabrication and installation details for decorative formed metal.
 - 1. Include plans, elevations, component details, and attachment details.
 - 2. Indicate materials and profiles of each decorative formed metal member, fittings, joinery, finishes, fasteners, anchorages, and accessory items.
- C. Samples: For each type of exposed finish required, prepared on 6-inch-square Samples of metal of same thickness and material indicated for the Work.

1.3 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: For decorative formed metal elements that house items specified in other Sections. Show dimensions of housed items, including locations of housing penetrations and attachments, and necessary clearances.
- B. Evaluation Reports: For post-installed anchors, from ICC-ES.

1.4 QUALITY ASSURANCE

- A. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for fabrication and installation.
 - 1. Build mockups of each type of type of decorative formed metal.
 - 2. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

PART 2 - PRODUCTS

2.1 SHEET METAL

- A. General: Fabricate products from sheet metal without pitting, seam marks, roller marks, stains, discolorations, or other imperfections where exposed to view on finished units.
- B. Aluminum Sheet: Flat sheet complying with ASTM B 209, alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with strength and durability properties of not less than Alloy 5005-H32.
- C. Galvanized-Steel Sheet: ASTM A 653/A 653M, G90 coating, either commercial steel or forming steel.
- D. Steel Sheet: Uncoated, cold-rolled, ASTM A 1008/A 1008M, commercial steel, exposed.
- E. Stainless-Steel Sheet: ASTM A 240/A 240M or ASTM A 666, Type 304, stretcher-leveled standard of flatness.

2.2 MISCELLANEOUS MATERIALS

- A. Gaskets: As required to seal joints in decorative formed metal and remain airtight; as recommended in writing by decorative formed metal manufacturer.
 - 1. ASTM D 1056, Type 1, Class A, grade as recommended by gasket manufacturer to obtain seal for application indicated.
 - 2. Closed-cell polyurethane foam, adhesive on two sides, release paper protected.
- B. Sealants, Interior: Nonsag, paintable, nonstaining, latex sealant complying with ASTM C 834; of type and grade required to seal joints in decorative formed metal; and as recommended in writing by decorative formed metal manufacturer.
 - 1. Sealants shall have a VOC content of not more than 250 g/L when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
- C. Filler Metal and Electrodes: Provide type and alloy of filler metal and electrodes as necessary for strength, corrosion resistance, and compatibility in fabricated items.
 - 1. Use filler metals that will match the color of metal being joined.
- D. Fasteners: Fabricated from same basic metal and alloy as fastened metal unless otherwise indicated.
 - 1. Provide tamper-resistant flat-head machine screws for exposed fasteners unless otherwise indicated.
- E. Anchors: Provide fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC193 or ICC-ES AC308.

F. Anchor Materials:

1. Material for Interior Locations: Carbon-steel components zinc plated to comply with ASTM B 633 or ASTM F 1941, Class Fe/Zn 5, unless otherwise indicated.

G. Sound-Deadening Materials:

1. Insulation: Unfaced, mineral-fiber blanket insulation complying with ASTM C 665, Type I, and passing ASTM E 136 test.
2. Mastic: Cold-applied asphalt emulsion complying with ASTM D 1187/D 1187M.

H. Laminating Adhesive: Adhesive recommended by metal fabricator that will fully bond metal to metal and is noncombustible after curing.

1. Metal-to-Metal Adhesive: VOC content of not more than 30 g/L when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

I. Isolation Coating: Manufacturer's standard alkali-resistant coating, bituminous paint, or epoxy coating.

2.3 PAINTS AND COATINGS

- A. Etching Cleaner for Galvanized Metal: Complying with MPI#25.
- B. Shop Primers: Comply with Section 09 91 23 "Interior Painting."
- C. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D 1187/D 1187M.

2.4 FABRICATION, GENERAL

- A. Shop Assembly: Preassemble decorative formed metal items in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation.
- B. Form decorative formed metal to required shapes and sizes, true to line and level with true curves and accurate angles and surfaces. Finish exposed surfaces to be smooth, well-defined lines and arrises.
- C. Fold back exposed edges of unsupported sheet metal to form a 1/2-inch- (12-mm-) wide hem on the concealed side, or ease edges to a radius of approximately 1/32 inch (1 mm) and support with concealed stiffeners.
- D. Increase metal thickness or reinforce with concealed stiffeners, backing materials, or both, as needed to provide surface flatness and sufficient strength for indicated use.
 1. Support joints with concealed stiffeners as needed to hold exposed faces of adjoining sheets in flush alignment.

- E. Where welding is indicated, weld joints and seams continuously. Grind, fill, and dress to produce smooth, flush, exposed surfaces in which joints are not visible after finishing is completed.

2.5 CLOSURES AND TRIM

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Fry Reglet Corporation.
 - 2. Pittcon Industries.
- B. Form closures and trim from metal of type and thickness indicated below. Fabricate to fit tightly to adjoining construction.
 - 1. Aluminum Sheet: 0.063 inch (1.60 mm).
 - a. Finish: As indicated on the Drawings.

2.6 FILLER PANELS

- A. Form from two sheets of metal of type and thickness indicated below, separated by channels formed from the same material, producing a panel of same thickness as partitions, mullions, or adjacent assembly unless otherwise indicated. Incorporate reveals, trim, and concealed anchorages for attaching to adjacent surfaces.
 - 1. Steel Sheet: 0.060 inch (1.52 mm).
 - a. Finish: As indicated on the Drawings.
- B. Fill interior of panel with sound-deadening insulation permanently attached to inside panel faces.
- C. Adhesively attach gaskets to filler panel edges where they abut mullions or glazing.

2.7 METAL BASE

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Fry Reglet Corporation.
 - 2. Pittcon Industries.
- B. Form metal base from metal of type and thickness indicated below:
 - 1. Aluminum Sheet: 0.063 inch (1.60 mm).
 - a. Finish: As indicated on the Drawings.

2. Stainless-Steel Sheet: 0.050 inch (1.27 mm).
 - a. Finish: As indicated on the Drawings.

2.8 MISCELLANEOUS DECORATIVE FORMED METAL ITEMS

- A. Custom fabricate units of metal and finish indicated, to comply with design and details indicated. Coordinate with adjacent finishes to provide integrated, closely fitted assemblies.

2.9 GENERAL FINISH REQUIREMENTS

- A. Complete mechanical finishes of flat sheet metal surfaces before fabrication where possible. After fabrication, finish all joints, bends, abrasions, and other surface blemishes to match sheet finish.
- B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
- C. Apply organic and anodic finishes to formed metal after fabrication unless otherwise indicated.
- D. Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable if they are within one-half of the range of approved Samples. Noticeable variations in the same piece are not acceptable. Variations in appearance of other components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

2.10 ALUMINUM FINISHES

- A. Clear Anodic Finish: AAMA 611, AA-M12C22A41, Class I, 0.018 mm or thicker.
- B. Color Anodic Finish: AAMA 611, AA-M12C22A42/A44, Class I, 0.018 mm or thicker.
 1. Color: As indicated on the Drawings.
- C. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils (0.04 mm). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.
 1. Color and Gloss: As indicated on the Drawings.

2.11 STEEL SHEET FINISHES

- A. Surface Preparation: Clean surfaces to comply with SSPC-SP 1, "Solvent Cleaning," to remove dirt, oil, grease, or other contaminants that could impair paint bond. Remove mill scale and rust, if present, from uncoated steel, complying with SSPC-SP 5/NACE No. 1, "White Metal Blast Cleaning," or with SSPC-SP 8, "Pickling."

- B. Pretreatment: Immediately after cleaning, apply a conversion coating of type suited to organic coating.
- C. Factory Priming for Field-Painted Finish: Where field painting after installation is indicated, apply shop primer to prepared surfaces of items unless otherwise indicated. Comply with requirements in SSPC-PA 1, "Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel," for shop painting.
- D. Baked-Enamel Finish: Immediately after cleaning and pretreating, apply manufacturer's standard two-coat, baked-enamel finish consisting of prime coat and thermosetting topcoat. Comply with coating manufacturer's written instructions for applying and baking to achieve a minimum dry film thickness of 2 mils (0.05 mm).
 - 1. Color and Gloss: As indicated on the Drawings.
- E. Powder-Coat Finish: Immediately after cleaning and pretreating, apply manufacturer's standard thermosetting polyester or acrylic urethane powder coating with cured-film thickness not less than 1.5 mils (0.04 mm). Prepare, treat, and coat metal to comply with resin manufacturer's written instructions.
 - 1. Color and Gloss: As indicated on the Drawings.

2.12 STAINLESS-STEEL FINISHES

- A. Surface Preparation: Remove tool and die marks and stretch lines, or blend into finish.
- B. Polished Finishes: Grind and polish surfaces to produce uniform finish, free of cross scratches.
 - 1. Run grain of directional finishes with long dimension of each piece.
- C. Bright, Cold-Rolled, Unpolished Finish: No. 2B.
- D. Directional Satin Finish: No. 4.
- E. Dull Satin Finish: No. 6.
- F. Satin, Reflective, Directional Polish: No. 7.
- G. Mirrorlike Reflective, Nondirectional Polish: No. 8 finish.
- H. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Locate and place decorative formed metal items level and plumb and in alignment with adjacent construction. Perform cutting, drilling, and fitting required to install decorative formed metal.
- B. Use concealed anchorages where possible. Provide brass or lead washers fitted to screws where needed to protect metal surfaces and to make a weathertight connection.
- C. Form tight joints with exposed connections accurately fitted together. Provide reveals and openings for sealants and joint fillers as indicated.
- D. Corrosion Protection: Apply bituminous paint or other permanent separation materials on concealed surfaces where metals would otherwise be in direct contact with substrate materials that are incompatible or could result in corrosion or deterioration of either material or finish.
- E. Restore finishes damaged during installation and construction period so no evidence remains of correction work. Return items that cannot be refinished in the field to the shop; make required alterations and refinish entire unit or provide new units.

END OF SECTION

SECTION 061053
MISCELLANEOUS ROUGH CARPENTRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wood blocking and nailers.
2. Wood furring and grounds.
3. Wood sleepers.
4. Utility shelving.
5. Plywood backing panels.

1.2 INFORMATIONAL SUBMITTALS

A. Evaluation Reports: For the following, from ICC-ES:

1. Preservative-treated wood.
2. Fire-retardant-treated wood.
3. Power-driven fasteners.

PART 2 - PRODUCTS

2.1 WOOD PRODUCTS, GENERAL

- A. Lumber:** DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, provide lumber that complies with the applicable rules of any rules-writing agency certified by the ALSC Board of Review. Provide lumber graded by an agency certified by the ALSC Board of Review to inspect and grade lumber under the rules indicated.
1. Factory mark each piece of lumber with grade stamp of grading agency.
 2. Provide dressed lumber, S4S, unless otherwise indicated.
- B. Maximum Moisture Content of Lumber:** 15 percent for 2-inch nominal thickness or less, 19 percent for more than 2-inch nominal thickness unless otherwise indicated.

2.2 WOOD-PRESERVATIVE-TREATED MATERIALS

- A. Preservative Treatment by Pressure Process:** AWP A U1; Use Category UC2 for interior construction not in contact with the ground, Use Category UC3b for exterior

construction not in contact with the ground, and Use Category UC4a for items in contact with the ground.

1. Preservative Chemicals: Acceptable to authorities having jurisdiction and containing no arsenic or chromium. Do not use inorganic boron (SBX) for sill plates.
- B. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent. Do not use material that is warped or does not comply with requirements for untreated material.
- C. Mark lumber with treatment quality mark of an inspection agency approved by the ALSC Board of Review.
- D. Application: Treat items indicated on Drawings, and the following:
 1. Wood sills, sleepers, blocking, furring, and similar concealed members in contact with masonry or concrete.
 2. Wood framing and furring attached directly to the interior of below-grade exterior masonry or concrete walls.
 3. Wood framing members that are less than 18 inches above the ground in crawl spaces or unexcavated areas.
 4. Wood floor plates that are installed over concrete slabs-on-grade.

2.3 FIRE-RETARDANT-TREATED MATERIALS

- A. Fire-Retardant-Treated Lumber and Plywood by Pressure Process: Products with a flame spread index of 25 or less when tested according to ASTM E 84, and with no evidence of significant progressive combustion when the test is extended an additional 20 minutes, and with the flame front not extending more than 10.5 feet beyond the centerline of the burners at any time during the test.
 1. Interior Type A: Treated materials shall have a moisture content of 28 percent or less when tested according to ASTM D 3201 at 92 percent relative humidity. Use where exterior type is not indicated.
 2. Design Value Adjustment Factors: Treated lumber shall be tested according ASTM D 5664, and design value adjustment factors shall be calculated according to ASTM D 6841.
- B. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent. Kiln-dry plywood after treatment to a maximum moisture content of 15 percent.
- C. Identify fire-retardant-treated wood with appropriate classification marking of testing and inspecting agency acceptable to authorities having jurisdiction.
- D. Application: Treat items indicated on Drawings, and the following:
 1. Framing for raised platforms.
 2. Concealed blocking.
 3. Plywood backing panels.

2.4 MISCELLANEOUS LUMBER

- A. General: Provide miscellaneous lumber indicated and lumber for support or attachment of other construction, including the following:
 - 1. Blocking.
 - 2. Nailers.
 - 3. Furring.
 - 4. Grounds.
 - 5. Utility shelving.
- B. For items of dimension lumber size, provide Construction or No. 2 grade lumber of any species.
- C. For utility shelving, provide lumber with 15 percent maximum moisture content of eastern white pine, Idaho white, lodgepole, ponderosa, or sugar pine; Premium or No. 2 Common (Sterling) grade; NeLMA, NLGA, WCLIB, or WWPA.
- D. For concealed boards, provide lumber with 19 percent maximum moisture content and any of the following species and grades:
 - 1. Mixed southern pine, No. 2 grade; SPIB.
 - 2. Eastern softwoods, No. 2 Common grade; NELMA.
 - 3. Northern species, No. 2 Common grade; NLGA.
 - 4. Western woods, Construction or No. 2 Common grade; WCLIB or WWPA.

2.5 PLYWOOD BACKING PANELS

- A. Equipment Backing Panels: DOC PS 1, Exterior, AC, fire-retardant treated, in thickness indicated or, if not indicated, not less than 3/4-inch nominal thickness.

2.6 FASTENERS

- A. General: Provide fasteners of size and type indicated that comply with requirements specified in this article for material and manufacture.
 - 1. Where carpentry is in ground contact, pressure-preservative treated, or in area of high relative humidity, provide fasteners of Type 304 stainless steel.
- B. Power-Driven Fasteners: NES NER-272.
- C. Screws for Fastening to Metal Framing: ASTM C 1002 or ASTM C 954, as appropriate to framing members, and length as recommended by screw manufacturer for material being fastened.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Set carpentry to required levels and lines, with members plumb, true to line, cut, and fitted. Fit carpentry to other construction; scribe and cope as needed for accurate fit.
- B. Where wood-preservative-treated lumber is installed adjacent to metal decking, install continuous flexible flashing separator between wood and metal decking.
- C. Framing Standard: Comply with AF&PA's WCD 1, "Details for Conventional Wood Frame Construction," unless otherwise indicated.
- D. Install plywood backing panels by fastening to studs; coordinate locations with utilities requiring backing panels. Install fire-retardant treated plywood backing panels with classification marking of testing agency exposed to view.
- E. Do not splice structural members between supports unless otherwise indicated.
- F. Comply with AWP A M4 for applying field treatment to cut surfaces of preservative-treated lumber.
- G. Securely attach carpentry work to substrate by anchoring and fastening as indicated, complying with the following:
 - 1. NES NER-272 for power-driven fasteners.
 - 2. Table 2304.9.1, "Fastening Schedule," in ICC's International Building Code.

3.2 PROTECTION

- A. Protect wood that has been treated with inorganic boron (SBX) from weather. If, despite protection, inorganic boron-treated wood becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

END OF SECTION

SECTION 062513
FLEXIBLE WOOD SLAT PANELS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Solid and Veneer flexible wood slat panels.
- B. Related Requirements:
 - 1. Section 09 21 16 - Gypsum Board Assemblies for substrate.

1.2 DEFINITIONS

- A. MDF: Medium-density fiberboard.

1.3 ACTION SUBMITTALS

- A. Product Data: Product data for each type of product.
- B. Shop Drawings: Submit diagram showing layout of pattern and configuration, including details of perimeter conditions, and mounting.
- C. Samples for Initial Selection: For each type of product involving selection of colors, profiles, or textures.
- D. Samples for Verification:
 - 1. For each finish system and color of lumber and panel products with factory-applied finish, 50 sq. in. for lumber and 8 by 10 inches for panels.

1.4 QUALITY ASSURANCE

- A. Manufacturer: Minimum of 5 years experience manufacturing similar products.
- B. Installer: Minimum of 2 years experience installing similar products.
- C. Field Measurements: To the greatest extent practical, take field measurements prior to fabrication.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver interior finish carpentry materials only when environmental conditions comply with requirements specified for installation areas. If interior finish carpentry materials must be stored in other than installation areas, store only where environmental conditions comply with requirements specified for installation areas.
- B. Deliver materials and products in unopened factory labeled packages. Store and handle in strict compliance with manufacturer's instructions and recommendations.

1.6 FIELD CONDITIONS

- A. Environmental Limitations: Do not deliver or install flexible wood slat panels until building is enclosed and weatherproof, wet-work in space is completed and nominally dry, and HVAC system is operating and maintaining temperature and relative humidity at occupancy levels during the remainder of the construction period.
- B. Do not install flexible wood slat panels that are wet, moisture damaged, or mold damaged.
 - 1. Indications that materials are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
 - 2. Indications that materials are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

PART 2 - PRODUCTS

2.1 MATERIALS, GENERAL

- A. Lumber: DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, comply with applicable rules of any rules-writing agency certified by the American Lumber Standard Committee's (ALSC) Board of Review. Grade lumber by an agency certified by the ALSC's Board of Review to inspect and grade lumber under the rules indicated.
 - 1. Factory mark each piece of lumber with grade stamp of grading agency.
 - 2. For exposed lumber, mark grade stamp on end or back of each piece.
- B. MDF: ANSI A208.2, Grade 130.

2.2 MANUFACTURER

- A. Manufacturer: Subject to compliance with requirements provide products by the following:
 - 1. Surfacing Solutions; Tambour.
 - a. Business Phone: 952.448.6566

b. Website: www.surfacingsolution.com.

- B. Source Limitations: Obtain flexible wood slat panels from single source from single manufacturer.

2.3 SOLID WOOD FLEXIBLE WOOD PANELS

- A. Product: Subject to compliance with requirements, provide the following:
1. Surfacing Solutions; Solid Wood Tambour Panels.
- B. Hardwood Slat Panels for Transparent Finish (Stain or Clear Finish):
1. Species: As indicated on Drawings.
 2. Grade: Clear; NHLA.
 3. Maximum Moisture Content: 10 percent.
- C. Hardwood Slat Panels for Opaque Finish (Painted Finish):
1. Species and Grade: Poplar; A Finish; NHLA.
 2. Maximum Moisture Content: 10 percent.
- D. Sheet Lengths: As indicated on Drawings.
- E. Sheet Width: 12 inches (305 mm).
- F. Slat Profiles:
1. T693: Slat width of 1-1/2 inches (38 mm) and height of 3/4 inches (19 mm).
- G. Field Finish: As selected by Architect from the following options:
1. Stain and clear coat polyurethane.
 2. Clear.
 3. Natural oil.
 4. Paint.

2.4 MISCELLANEOUS MATERIALS

- A. Fasteners for flexible wood slat panels: Provide manufacturer's recommended mounting devices of type, size, material, and finish required for application indicated to provide secure attachment, concealed where possible.
- B. Paneling Adhesive: Comply with paneling manufacturer's written instructions for adhesives.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine flexible wood slat panels before installation. Reject materials that are wet, moisture damaged, and mold damaged.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Clean substrates of projections and substances detrimental to application.
- B. Before installing flexible wood slat panels, condition materials to average prevailing humidity in installation areas for a minimum of 24 hours.

3.3 INSTALLATION, GENERAL

- A. Do not use materials that are unsound; warped; improperly treated or finished; inadequately seasoned; too small to fabricate with proper jointing arrangements; or with defective surfaces, sizes, or patterns.
- B. Install flexible wood slat panels level, plumb, true, and aligned with adjacent materials.
 - 1. Use concealed shims where necessary for alignment.
 - 2. Scribe and cut flexible wood slat panels to fit adjoining work. Refinish and seal cuts as recommended by manufacturer.
 - 3. Where face fastening is unavoidable, countersink fasteners, fill surface flush, and sand unless otherwise indicated.
 - 4. Install to tolerance of 1/8 inch in 96 inches for level and plumb. Install adjoining flexible wood slat panels with 1/32-inch maximum offset for flush installation and 1/16-inch maximum offset for reveal installation.
 - 5. Coordinate flexible wood slat panels with materials and systems in or adjacent to it. Provide cutouts for mechanical and electrical items that penetrate interior finish carpentry.

3.4 INSTALLATION OF FLEXIBLE WOOD PANELS

- A. Install products in strict accordance with manufacturer's instructions, approved submittals and in proper relationship to adjacent construction.
 - 1. Clean substrate of dirt and bond breaking substances prior to beginning installation.
 - 2. Acclimatize panels at the installation site for a minimum of 48 hours prior to installation.

3. Roll tambour panels in accordance with manufacturer's instructions to properly space slats.
4. Follow manufacturer's recommendations for adhesives and mounting devices.
5. Replace damaged or defaced products prior to Substantial Completion.

3.5 ADJUSTING

- A. Replace flexible wood slat panels that is damaged or does not comply with requirements.
 1. flexible wood slat panels may be repaired or refinished if work complies with requirements and shows no evidence of repair or refinishing.
- B. Adjust joinery for uniform appearance.

3.6 CLEANING

- A. Clean flexible wood slat panels on exposed and semi-exposed surfaces.
- B. Restore damaged or soiled areas and touch up factory-applied finishes if any.

3.7 PROTECTION

- A. Protect installed products from damage from weather and other causes during construction.
- B. Remove and replace flexible wood slat panels that are wet, moisture damaged, and mold damaged.
 1. Indications that materials are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
 2. Indications that materials are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

END OF SECTION

SECTION 064116

PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Plastic-laminate-clad architectural cabinets.
2. Cabinet hardware and accessories.
3. Wood furring, blocking, shims, and hanging strips for installing plastic-laminate-clad architectural cabinets that are not concealed within other construction.

1.2 DEFINITIONS

A. Exposed Surfaces:

1. Surfaces visible when doors and drawers are closed.
2. Bottoms of cases more than 4'-0" above finish floor.
3. Back and edges of hinged doors exposed when opened.

B. Semi-Exposed Surfaces:

1. Surfaces that become visible when drawers and doors are open.
2. Tops of cases 6'-0" or more above finish floor.

C. Concealed Surfaces: Surfaces not visible after installation.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product, including panel products, high-pressure decorative laminates, adhesive for bonding plastic laminate and cabinet hardware and accessories.

B. Shop Drawings: For architectural cabinets.

1. Include plans, elevations, sections, and attachment details.
2. Show large-scale details.
3. Show locations and sizes of furring, blocking, and hanging strips, including concealed blocking and reinforcement specified in other Sections.
4. Show locations and sizes of cutouts and holes for items installed in architectural cabinets.

C. Samples:

1. Plastic laminates, 12 by 12 inches, for each color, pattern, and surface finish, with one sample applied to core material and edge material applied to one edge.

1.4 QUALITY ASSURANCE

- A. AWI Catalog: Catalog numbers indicated on the drawings and in the specifications are for the convenience of identifying specific cabinet types. Unless modified by notation on the drawings or otherwise specified, current description for indicated number, together with indicated or specified options or accessories, constitutes requirements for each cabinet.
1. Catalog numbers and specific requirements indicated on the drawings and in the specification are given for the purpose of establishing standard design and quality of materials, construction, and workmanship.
 2. Catalog numbers noted on the Drawings are based upon AWI Architectural Woodwork Standards, Design Ideas.
- B. Fabricator's Qualifications: Employs skilled workers who custom fabricate products similar to those required for this Project and whose products have a record of successful in-service performance.
- C. Installer Qualifications: Fabricator of products.
- D. Mockups: Build mockups to verify selections, to demonstrate aesthetic effects, and to set quality standards for materials and execution.
1. Build mockups of typical architectural cabinets and banquetts as shown on Drawings in locations as selected by Architect.
 2. Subject to compliance with requirements, approved mockups may become part of the completed work if undisturbed at time of Substantial Completion.

1.5 FIELD CONDITIONS

- A. Environmental Limitations with Humidity Control: Do not deliver or install cabinets until building is enclosed, wet-work is complete, and HVAC system is operating and maintaining temperature between 60 and 90 deg F and relative humidity between 25 and 55 percent during the remainder of the construction period.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Protect plastic-laminate-faced architectural cabinets with appropriate heavy duty wrapping materials at factory prior to shipment. Mark each unit with appropriate identification required for installation.
- B. Protect architectural woodwork during handling, transit and storage to prevent damage and deterioration. Store in conditioned space complying with environmental limitations described above.

PART 2 - PRODUCTS

2.1 PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS

- A. Quality Standard: Unless otherwise indicated, comply with the AWI Architectural Woodwork Standards for grades of architectural cabinets indicated for construction, finishes, installation, and other requirements.
 - 1. The Contract Documents may contain selections chosen from options in the quality standard and additional requirements beyond those of the quality standard. Comply with those selections and requirements in addition to the quality standard.
- B. Architectural Woodwork Standards Grade: Custom.
- C. Type of Construction: Frameless.
- D. Door and Drawer-Front Style: Flush overlay.
- E. High-Pressure Decorative Laminate: ISO 4586-3, grades as indicated or if not indicated, as required by quality standard.
 - 1. Products: As indicated on Drawings.
- F. Laminate Cladding for Exposed Surfaces:
 - 1. Horizontal Surfaces: Grade HGS.
 - 2. Vertical Surfaces: Grade HGS.
 - 3. Edges: Grade HGS.
 - 4. Pattern Direction: Vertically for drawer fronts, doors, and fixed panels, unless indicated otherwise on the Drawings.
- G. Cladding for Semi-Exposed Surfaces:
 - 1. Surfaces Other Than Drawer Bodies: High-pressure decorative laminate, ISO 4586-3, Grade VGS.
 - 2. Drawer Sides and Backs: Solid-hardwood lumber.
 - 3. Drawer Bottoms: Hardwood plywood.
- H. Concealed Backs of Panels with Exposed Plastic-Laminate Surfaces: High-pressure decorative laminate, ISO 4583-3, Grade BKL.
- I. Drawer Construction: Fabricate with exposed fronts fastened to subfront with mounting screws from interior of body.
 - 1. Join subfronts, backs, and sides with glued rabbeted joints supplemented by mechanical fasteners or glued dovetail joints.
- J. Dust Panels: 1/4-inch plywood or tempered hardboard above compartments and drawers unless located directly under tops.

- K. Colors, Patterns, and Finishes: As indicated on the Drawings.

2.2 WOOD MATERIALS

- A. Wood Products: Provide materials that comply with requirements of referenced quality standard for each type of architectural cabinet and quality grade specified unless otherwise indicated.
1. Wood Moisture Content:
 - a. 5 to 10 percent: Typical, except:
 - 1) 8 to 13 percent: For projects in damp, coastal areas.
 - 2) 4 to 9 percent: For projects dry, arid areas.
- B. Composite Wood Products: Provide materials that comply with requirements of referenced quality standard for each type of architectural cabinet and quality grade specified unless otherwise indicated.
1. Medium-Density Fiberboard: ANSI A208.2, Grade 130, made with binder containing no urea formaldehyde.
 2. Particleboard: ANSI A208.1, Grade M-2, made with binder containing no urea formaldehyde.
 3. Softwood Plywood: DOC PS 1.

2.3 CABINET HARDWARE AND ACCESSORIES

- A. General: Provide cabinet hardware and accessory materials associated with architectural cabinets except for items specified in Section 08 71 00 "Door Hardware."
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following unless otherwise indicated:
 - a. Accuride International Inc.
 - b. Grass America.
 - c. Häfele America Co.
 - d. Julius Blum & Co., Inc.
 - e. Knappe & Vogt Manufacturing Company
 - f. Richilieu Hardware.
 - g. Mockett.
- B. Back-Mounted Pulls: Products as indicated on Drawings.
- C. Frameless Concealed Hinges (European Type): BHMA A156.9, B01602, 100, 135, or 170 degrees of opening (provide maximum opening as permitted by adjacent wall condition), soft-closing.
- D. Catches: Push-in magnetic catches, BHMA A156.9, B03131.

- E. Adjustable Shelf Standards and Supports: ANSI/BHMA A156.9, B04071; with shelf rests, B04081.
- F. Shelf Rests: ANSI/BHMA A156.9, B04013:
 - 1. Exposed Locations: Metal.
 - 2. Sem-Exposed Locations: Two-pin plastic with shelf hold-down clip.
- G. Drawer Slides: BHMA A156.9.
 - 1. Standard Duty (Grade 1 and Grade 2): Side mount.
 - 2. Heavy Duty (Grade 1HD-100 and Grade 1HD-200): Side mount.
 - a. Type: Full extension.
 - b. Material: Zinc-plated ball bearing slides.
 - c. Motion Feature: Push to open and self-closing mechanism.
 - 3. For drawers not more than 3 inches high and not more than 24 inches wide, provide Grade 2.
 - 4. For drawers more than 3 inches high but not more than 6 inches high and not more than 24 inches wide, provide Grade 1.
 - 5. For drawers more than 6 inches high or more than 24 inches wide, provide Grade 1HD-100.
 - 6. For trash bins, provide Grade 1HD-200 KV wood frame, single waste bin pullout slides and 35-quart waste bin
 - a. Subject to compliance with requirements, provide Hafele; 503.12 Series.
- H. Slides for Sliding Glass Doors: ANSI/BHMA A156.9, B07063; aluminum.
- I. Door Locks: BHMA A156.11, E07121.
- J. Drawer Locks: BHMA A156.11, E07041.
- K. Door and Drawer Silencers: BHMA A156.16, L03011.
- L. Trash Grommets: ASTM A240/A240M, Type 304, 0.062 inch thick by 3 inch deep by 10-inch diameter stainless steel sheet ring, formed with 5/16 inch top flange for mounting.
 - 1. Acceptable Product: TM10 by Doug Mockett & Company, Inc.
- M. Exposed Hardware Finishes: Satin Stainless Steel: BHMA 630, unless indicated otherwise on the Drawings.
 - 1. For other exposed hardware finishes indicated, provide finish that complies with BHMA A156.18 for BHMA finish numbers as follows:
 - a. Dark, Oxidized, Satin Bronze, Oil Rubbed: BHMA 613 for bronze base; BHMA 640 for steel base; match Architect's sample.
 - b. Bright Brass, Clear Coated: BHMA 605 for brass base; BHMA 632 for steel base.

- c. Satin Brass, Blackened, Bright Relieved, Clear Coated: BHMA 610 for brass base; BHMA 636 for steel base.
- d. Satin Chromium Plated: BHMA 626 for brass or bronze base; BHMA 652 for steel base.
- e. Bright Chromium Plated: BHMA 625 for brass or bronze base; BHMA 651 for steel base.

2.4 OTHER ACCESSORIES

- A. Banquette Springs: Sinuous, two-dimensional, s-shaped steel wire coils fastened to seat and back frames with no-sag clips.
 - 1. Seats: "Firm" grade, 8 gauge steel wire.
 - 2. Backs: "Soft" grade, 11 gauge steel wire.
- B. Banquette Cushion Materials: Flexible polyurethane foam complying with the following:
 - 1. Seatback Cushion Polymer Density: 1.4 pcf.
 - 2. Seat Cushion Polymer Density: 5 inch high density base layer with 2 inch medium density face layer.
 - 3. Indentation Force Deflection (IFD): 80 lb.
 - 4. Support Factor: 3.0.
- C. Banquette Upholstery Fabric: Provide products indicated on Drawings.
- D. Work Surface Support Bracket:
 - 1. Stamped steel or extruded aluminum construction.
 - 2. Size: Nominally 24 inches high by 18 inches deep.
 - 3. Capacity: Minimum 400 lb. capacity per pair, spaced at a maximum of 24 inches on center.
 - 4. Counter depth: Maximum of 25 inches.
 - 5. Finish: Powder Coated; color as selected by Architect.
 - 6. Acceptable Product:
 - a. Type 1 (Exposed): SWS4 by Doug Mockett & Company, Inc.
 - b. Type 2 (Concealed): EH-1818 by Rakks/Rangine Corporation.
- E. Coat Hooks:
 - 1. Finish: Burnish cast aluminum double hook.
 - 2. Acceptable Product: No. 580 by Ives (if double hook).
- F. Hanging Rods:
 - 1. Closet Shelf and Rod Supports:
 - a. Steel; painted white.
 - b. 11" L x 10" H
 - c. 1195 by Knape & Vogt.

2. Center Support:
 - a. 24" Length; 1-1/16" od tubing; anochrome finish.
 - b. 760 by Knappe and Vogt.
3. Tubing: Stainless Steel; 1-1/16" diameter; 660 by Knappe & Vogt.
4. Tubing Flange: 1-1/16" o.d. mounted with two 5mm pins.
5. End Cap: 730 by Knappe and Vogt.

2.5 MISCELLANEOUS MATERIALS

- A. Furring, Blocking, Shims, and Hanging Strips: Fire-retardant-treated softwood lumber, kiln-dried to less than 15 percent moisture content.
- B. Anchors: Select material, type, size, and finish required for each substrate for secure anchorage. Provide metal expansion sleeves or expansion bolts for post-installed anchors. Use nonferrous-metal or hot-dip galvanized anchors and inserts at inside face of exterior walls and at floors.
- C. Adhesives: Do not use adhesives that contain urea formaldehyde.
- D. Adhesive for Bonding Plastic Laminate: Contact cement.

2.6 FABRICATION

- A. Complete fabrication, including assembly and hardware application, to maximum extent possible before shipment to Project site. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.
- B. Shop-cut openings to maximum extent possible to receive hardware, appliances, electrical work, and similar items. Locate openings accurately and use templates or roughing-in diagrams to produce accurately sized and shaped openings. Sand edges of cutouts to remove splinters and burrs.
- C. Install glass to comply with applicable requirements in Section 08 80 00 "Glazing" and in GANA's "Glazing Manual."
 1. For glass in frames, secure glass with removable stops.
 2. For exposed glass edges, polish and grind smooth.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Before installation, condition cabinets to humidity conditions in installation areas for not less than 72 hours.

3.2 INSTALLATION

- A. Grade: Install cabinets to comply with quality standard grade of item to be installed.
- B. Anchor cabinets to anchors or blocking built in or directly attached to substrates. Secure with wafer-head cabinet installation screws.
- C. Install cabinets level, plumb, and true in line to a tolerance of 1/8 inch in 96 inches using concealed shims.
 - 1. Scribe and cut cabinets to fit adjoining work, refinish cut surfaces, and repair damaged finish at cuts.
 - 2. Install cabinets without distortion so doors and drawers fit openings and are accurately aligned. Adjust hardware to center doors and drawers in openings and to provide unencumbered operation. Complete installation of hardware and accessory items as indicated.
 - 3. Fasten wall cabinets through back, near top and bottom, and at ends not more than 16 inches o.c. with No. 10 wafer-head screws sized for not less than 1-1/2-inch penetration into wood framing, blocking, or hanging strips or No. 10 wafer-head sheet metal screws through metal backing or metal framing behind wall finish.

END OF SECTION

SECTION 066400
PLASTIC PANELING (FRP)

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes plastic sheet paneling ("FRP").

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For plastic paneling and trim accessories.

1.3 QUALITY ASSURANCE

- A. Testing Agency: Acceptable to authorities having jurisdiction.

PART 2 - PRODUCTS

2.1 PLASTIC SHEET PANELING

- A. Glass-Fiber-Reinforced Plastic Paneling: Gelcoat-finished, glass-fiber-reinforced plastic panels complying with ASTM D 5319.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Crane Composites, Inc.
 - b. Glasteel.
 - c. Marlite.
 - d. Nudo Products, Inc.
 - 2. Surface-Burning Characteristics: As follows when tested by a qualified testing agency according to ASTM E 84. Identify products with appropriate markings of applicable testing agency.
 - a. Flame-Spread Index: 25 or less.
 - b. Smoke-Developed Index: 450 or less.
 - 3. Nominal Thickness: Not less than 0.09 inch.
 - 4. Surface Finish and Color: As indicated on Drawings.

2.2 ACCESSORIES

- A. Trim Accessories: Manufacturer's standard one-piece vinyl extrusions designed to retain and cover edges of panels. Provide division bars, inside corners, outside corners, and caps as needed to conceal edges.
 - 1. Color: Match panels.
- B. Adhesive: As recommended by plastic paneling manufacturer and with a VOC content of 50 g/L or less.
- C. Sealant: Mildew-resistant, single-component, neutral-curing silicone sealant recommended by plastic paneling manufacturer and complying with requirements in Section 07 92 00 "Joint Sealants."

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean substrates of substances that could impair adhesive bond, including oil, grease, dirt, and dust.
- B. Condition panels by unpacking and placing in installation space before installation according to manufacturer's written recommendations.
- C. Lay out paneling before installing. Locate panel joints to provide equal panels at ends of walls not less than half the width of full panels.

3.2 INSTALLATION

- A. Install plastic paneling according to manufacturer's written instructions.
- B. Install panels in a full spread of adhesive.
- C. Install trim accessories with adhesive and nails or staples. Do not fasten through panels.
- D. Fill grooves in trim accessories with sealant before installing panels, and bed inside corner trim in a bead of sealant.
- E. Maintain uniform space between panels and wall fixtures. Fill space with sealant.
- F. Remove excess sealant and smears as paneling is installed. Clean with solvent recommended by sealant manufacturer and then wipe with clean dry cloths until no residue remains.

END OF SECTION

SECTION 072116.13
BLANKET INSULATION, UNFACED

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Glass-fiber blanket insulation, unfaced.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

- A. Product test reports.
- B. Research reports.

PART 2 - PRODUCTS

2.1 GLASS-FIBER BLANKET

- A. Glass-Fiber Blanket, Unfaced: ASTM C 665, Type I; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively, per ASTM E 84; passing ASTM E 136 for combustion characteristics.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. CertainTeed Corporation.
 - b. Guardian Building Products, Inc.
 - c. Johns Manville; a Berkshire Hathaway company.
 - d. Knauf Insulation.
 - e. Owens Corning.

2.2 ACCESSORIES

- A. Insulation for Miscellaneous Voids:

1. Glass-Fiber Insulation: ASTM C 764, Type II, loose fill; with maximum flame-spread and smoke-developed indexes of 5, per ASTM E 84.
- B. Insulation Anchors, Spindles, and Standoffs: As recommended by manufacturer.
- C. Adhesive for Bonding Insulation: Product compatible with insulation and air and water barrier materials, and with demonstrated capability to bond insulation securely to substrates without damaging insulation and substrates.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Comply with insulation manufacturer's written instructions applicable to products and applications.
- B. Install insulation that is undamaged, dry, and unsoiled and that has not been left exposed to ice, rain, or snow at any time.
- C. Extend insulation to envelop entire area to be insulated. Fit tightly around obstructions and fill voids with insulation. Remove projections that interfere with placement.
- D. Provide sizes to fit applications and selected from manufacturer's standard thicknesses, widths, and lengths. Apply single layer of insulation units to achieve indicated R-value.

3.2 INSTALLATION OF INSULATION IN FRAMED CONSTRUCTION

- A. Blanket Insulation: Install in cavities formed by framing members according to the following requirements:
 1. Use insulation widths and lengths that fill the cavities formed by framing members. If more than one length is required to fill the cavities, provide lengths that will produce a snug fit between ends.
 2. Place insulation in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.
 3. Maintain 3-inch clearance of insulation around recessed lighting fixtures not rated for or protected from contact with insulation.
 4. For metal-framed wall cavities where cavity heights exceed 96 inches, support unfaced blankets mechanically.
- B. Miscellaneous Voids: Install insulation in miscellaneous voids and cavity spaces where required to prevent gaps in insulation using the following materials:
 1. Glass-Fiber Insulation: Compact to approximately 40 percent of normal maximum volume equaling a density of approximately 2.5 lb/cu. ft.

END OF SECTION

SECTION 078413
PENETRATION FIRESTOPPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Penetrations in fire-resistance-rated walls.
2. Penetrations in horizontal assemblies.
3. Penetrations in smoke barriers.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Product Schedule: For each penetration firestopping system. Include location and design designation of qualified testing and inspecting agency.

1. Where Project conditions require modification to a qualified testing and inspecting agency's illustration for a particular penetration firestopping condition, submit illustration, with modifications marked, approved by penetration firestopping manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly.

1.3 INFORMATIONAL SUBMITTALS

A. Installer Certificates: From Installer indicating penetration firestopping has been installed in compliance with requirements and manufacturer's written recommendations.

B. Product test reports.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: A firm that has been approved by FM Global according to FM Global 4991, "Approval of Firestop Contractors," or been evaluated by UL and found to comply with its "Qualified Firestop Contractor Program Requirements."

B. Fire-Test-Response Characteristics: Penetration firestopping shall comply with the following requirements:

1. Penetration firestopping tests are performed by UL or a qualified testing agency acceptable to authorities having jurisdiction.

2. Penetration firestopping is identical to those tested per testing standard referenced in "Penetration Firestopping" Article. Provide rated systems bearing marking of qualified testing and inspection agency.
- C. Preinstallation Conference: Conduct conference at Project site.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. Hilti, Inc.
 2. Specified Technologies Inc.
 3. 3M Fire Protection Products.

2.2 PENETRATION FIRESTOPPING

- A. Provide penetration firestopping that is produced and installed to resist spread of fire according to requirements indicated, resist passage of smoke and other gases, and maintain original fire-resistance rating of construction penetrated. Penetration firestopping systems shall be compatible with one another, with the substrates forming openings, and with penetrating items if any.
- B. Penetrations in Fire-Resistance-Rated Walls: Ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg.
1. F-Rating: Not less than the fire-resistance rating of constructions penetrated.
- C. Penetrations in Horizontal Assemblies: Ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg.
1. F-Rating: At least 1 hour, but not less than the fire-resistance rating of constructions penetrated.
 2. T-Rating: At least 1 hour, but not less than the fire-resistance rating of constructions penetrated except for floor penetrations within the cavity of a wall.
- D. Penetrations in Smoke Barriers: Provide penetration firestopping with ratings determined per UL 1479.
1. L-Rating: Not exceeding 5.0 cfm/sq. ft. of penetration opening at 0.30-inch wg at both ambient and elevated temperatures.
- E. Exposed Penetration Firestopping: Provide products with flame-spread and smoke-developed indexes of less than 25 and 450, respectively, as determined per ASTM E 84.

- F. VOC Content: Penetration firestopping sealants and sealant primers shall comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
 - 1. Sealants: 250 g/L.
 - 2. Sealant Primers for Nonporous Substrates: 250 g/L.
 - 3. Sealant Primers for Porous Substrates: 775 g/L.
- G. Accessories: Provide components for each penetration firestopping system that are needed to install fill materials and to maintain ratings required. Use only those components specified by penetration firestopping manufacturer and approved by qualified testing and inspecting agency for firestopping indicated.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Examine substrates and conditions, with Installer present, for compliance with requirements for opening configurations, penetrating items, substrates, and other conditions affecting performance of the Work.
- B. Install penetration firestopping to comply with manufacturer's written installation instructions and published drawings for products and applications indicated.
- C. Install forming materials and other accessories of types required to support fill materials during their application and in the position needed to produce cross-sectional shapes and depths required to achieve fire ratings indicated.
 - 1. After installing fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of firestopping.
- D. Install fill materials for firestopping by proven techniques to produce the following results:
 - 1. Fill voids and cavities formed by openings, forming materials, accessories, and penetrating items as required to achieve fire-resistance ratings indicated.
 - 2. Apply materials so they contact and adhere to substrates formed by openings and penetrating items.
 - 3. For fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.2 IDENTIFICATION

- A. Identify penetration firestopping with preprinted metal or plastic labels. Attach labels permanently to surfaces adjacent to and within 6 inches of firestopping edge so labels will be visible to anyone seeking to remove penetrating items or firestopping. Use mechanical fasteners or self-adhering-type labels with adhesives capable of

permanently bonding labels to surfaces on which labels are placed. Include the following information on labels:

1. The words "Warning - Penetration Firestopping - Do Not Disturb. Notify Building Management of Any Damage."
2. Contractor's name, address, and phone number.
3. Designation of applicable testing and inspecting agency.
4. Date of installation.
5. Manufacturer's name.
6. Installer's name.

3.3 FIELD QUALITY CONTROL

- A. Owner will engage a qualified testing agency to perform tests and inspections.
- B. Where deficiencies are found or penetration firestopping is damaged or removed because of testing, repair or replace penetration firestopping to comply with requirements.
- C. Proceed with enclosing penetration firestopping with other construction only after inspection reports are issued and installations comply with requirements.

3.4 PENETRATION FIRESTOPPING SCHEDULE

- A. Where UL-classified systems are indicated, they refer to system numbers in UL's "Fire Resistance Directory" under product Category XHEZ.
- B. Where Intertek ETL SEMKO-listed systems are indicated, they refer to design numbers in Intertek ETL SEMKO's "Directory of Listed Building Products" under "Firestop Systems."
- C. Where FM Global-approved systems are indicated, they refer to design numbers listed in FM Global's "Building Materials Approval Guide" under "Wall and Floor Penetration Fire Stops."

END OF SECTION

SECTION 078446
FIRE-RESISTIVE JOINT SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Joints in or between fire-resistance-rated constructions.
2. Joints at exterior curtain-wall/floor intersections.
3. Joints in smoke barriers.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Product Schedule: For each fire-resistive joint system. Include location and design designation of qualified testing agency.

1. Where Project conditions require modification to a qualified testing agency's illustration for a particular fire-resistive joint system condition, submit illustration, with modifications marked, approved by fire-resistive joint system manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly.

1.3 INFORMATIONAL SUBMITTALS

A. Installer Certificates: From Installer indicating fire-resistive joint systems have been installed in compliance with requirements and manufacturer's written recommendations.

B. Product test reports.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: A firm that has been approved by FM Global according to FM Global 4991, "Approval of Firestop Contractors," or been evaluated by UL and found to comply with UL's "Qualified Firestop Contractor Program Requirements."

B. Fire-Test-Response Characteristics: Fire-resistive joint systems shall comply with the following requirements:

1. Fire-resistive joint system tests are performed by UL or a qualified testing agency acceptable to authorities having jurisdiction.

- C. Preinstallation Conference: Conduct conference at Project site.

PART 2 - PRODUCTS

2.1 FIRE-RESISTIVE JOINT SYSTEMS

- A. Where required, provide fire-resistive joint systems that are produced and installed to resist spread of fire according to requirements indicated, resist passage of smoke and other gases, and maintain original fire-resistance rating of assemblies in or between which fire-resistive joint systems are installed. Fire-resistive joint systems shall accommodate building movements without impairing their ability to resist the passage of fire and hot gases.
- B. Joints in or between Fire-Resistance-Rated Construction: Ratings determined per ASTM E 1966 or UL 2079:
1. Fire-Resistance Rating: Equal to or exceeding the fire-resistance rating of construction they will join.
 2. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Hilti, Inc.
 - b. Specified Technologies Inc.
 - c. 3M Fire Protection Products.
- C. Joints at Exterior Curtain-Wall/Floor Intersections: Rating determined by ASTM E 119 based on testing at a positive pressure differential of 0.01-inch wg or ASTM E 2307.
1. Fire-Resistance Rating: Equal to or exceeding the fire-resistance rating of the floor assembly.
 2. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Hilti, Inc.
 - b. Specified Technologies Inc.
 - c. 3M Fire Protection Products.
 - d. Thermafiber, Inc.
 - e. USG Corporation.
- D. Joints in Smoke Barriers: Ratings determined per UL 2079.
1. L-Rating: Not exceeding 5.0 cfm/ft of joint at 0.30 inch wg at both ambient and elevated temperatures.
 2. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

- a. Hilti, Inc.
 - b. Specified Technologies Inc.
 - c. 3M Fire Protection Products.
- E. Exposed Fire-Resistive Joint Systems: Provide products with flame-spread and smoke-developed indexes of less than 25 and 450, respectively, as determined per ASTM E 84.
- F. VOC Content: Fire-resistive joint system sealants shall comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
 1. Architectural Sealants: 250 g/L.
 2. Sealant Primers for Nonporous Substrates: 250 g/L.
 3. Sealant Primers for Porous Substrates: 775 g/L.
- G. Accessories: Provide components of fire-resistive joint systems, including primers and forming materials, that are needed to install fill materials and to maintain ratings required. Use only components specified by fire-resistive joint system manufacturer and approved by the qualified testing agency for systems indicated.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Examine substrates and conditions, with Installer present, for compliance with requirements for joint configurations, substrates, and other conditions affecting performance of the Work.
- B. Install fire-resistive joint systems to comply with manufacturer's written installation instructions and published drawings for products and applications indicated.
- C. Install forming materials and other accessories of types required to support fill materials during their application and in position needed to produce cross-sectional shapes and depths required to achieve fire ratings indicated.
 1. After installing fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of fire-resistive joint system.
- D. Install fill materials for fire-resistive joint systems by proven techniques to produce the following results:
 1. Fill voids and cavities formed by joints and forming materials as required to achieve fire-resistance ratings indicated.
 2. Apply fill materials so they contact and adhere to substrates formed by joints.
 3. For fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.2 IDENTIFICATION

- A. Identify fire-resistive joint systems with preprinted metal or plastic labels. Attach labels permanently to surfaces adjacent to and within 6 inches of joint edge so labels will be visible to anyone seeking to remove or penetrate joint system. Use mechanical fasteners or self-adhering-type labels with adhesives capable of permanently bonding labels to surfaces on which labels are placed. Include the following information on labels:
 - 1. The words "Warning - Fire-Resistive Joint System - Do Not Disturb. Notify Building Management of Any Damage."
 - 2. Contractor's name, address, and phone number.
 - 3. Designation of applicable testing agency.
 - 4. Date of installation.
 - 5. Manufacturer's name.
 - 6. Installer's name.

3.3 FIELD QUALITY CONTROL

- A. Inspecting Agency: Owner will engage a qualified testing agency to perform tests and inspections.
- B. Where deficiencies are found or fire-resistive joint systems are damaged or removed due to testing, repair or replace fire-resistive joint systems so they comply with requirements.
- C. Proceed with enclosing fire-resistive joint systems with other construction only after inspection reports are issued and installations comply with requirements.

END OF SECTION

SECTION 079200
JOINT SEALANTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Interior joint sealants.

1.2 ACTION SUBMITTALS

- A. Product Data: For each joint-sealant product.
- B. Samples: For each kind and color of joint sealant required.
- C. Joint-Sealant Schedule: Include the following information:
 - 1. Joint-sealant application, joint location, and designation.
 - 2. Joint-sealant manufacturer and product name.
 - 3. Joint-sealant formulation.
 - 4. Joint-sealant color.

1.3 INFORMATIONAL SUBMITTALS

- A. Product test reports.

1.4 WARRANTY

- A. Special Installer's Warranty: Installer agrees to repair or replace joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.
 - 1. Warranty Period: Two years from date of Substantial Completion.
- B. Special Manufacturer's Warranty: Manufacturer agrees to furnish joint sealants to repair or replace those joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.
 - 1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 JOINT SEALANTS, GENERAL

- A. VOC Content of Interior Sealants: Sealants and sealant primers used inside the weatherproofing system shall comply with the following:
 - 1. Architectural sealants shall have a VOC content of 250 g/L or less.
 - 2. Sealants and sealant primers for porous substrates shall have a VOC content of 250 g/L or less.
 - 3. Sealants and sealant primers for nonporous substrates shall have a VOC content of 775 g/L or less.
- B. Colors of Exposed Joint Sealants: As selected by Architect from manufacturer's full range.

2.2 URETHANE JOINT SEALANTS

- A. Urethane, S, NS, 25, NT – Type JS-1:
 - 1. Single-component, nonsag, nontraffic-use, plus 25 percent and minus 25 percent movement capability, urethane joint sealant; ASTM C 920, Type S, Grade NS, Class 25, Use NT.
 - 2. Products: Subject to compliance with requirements, provide one of the following:
 - a. BASF Construction Chemicals, LLC, Building Systems; Sonalastic TX1.
 - b. Pecora Corporation; Dynatrol I-XL.
 - c. Sika Corporation U.S.; Sikaflex Textured Sealant.
 - d. Tremco Incorporated; Dymonic.
- B. Urethane, S, NS, 25, T, NT – Type JS-2:
 - 1. Single-component, nonsag, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, urethane joint sealant; ASTM C 920, Type S, Grade NS, Class 25, Uses T and NT.
 - 2. Products: Subject to compliance with requirements, provide the following:
 - a. LymTal International, Inc.; Iso-Flex 330.
- C. Urethane, S, P, 25, T, NT – Type JS-3:
 - 1. Single-component, pourable, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, urethane joint sealant; ASTM C 920, Type S, Grade P, Class 25, Uses T and NT.
 - 2. Products: Subject to compliance with requirements, provide one of the following:
 - a. BASF Construction Chemicals, LLC, Building Systems; Sonolastic SL 1.
 - b. Pecora Corporation; NR-201.
 - c. Sika Corporation U.S.; Sikaflex 1c SL.
 - d. Tremco Incorporated; Vulkem 45.

D. Urethane, M, NS, 25, T, NT – Type JS-4:

1. Multicomponent, nonsag, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, urethane joint sealant; ASTM C 920, Type M, Grade NS, Class 25, Uses T and NT.
2. Products: Subject to compliance with requirements, provide one of the following:
 - a. Bostik, Inc.; Chem-Calk 505.
 - b. LymTal International, Inc.; Iso-Flex 881.
 - c. Sika Corporation U.S.; Sikaflex - 2c NS EZ Mix.

2.3 MILDEW-RESISTANT JOINT SEALANTS

A. Mildew-Resistant Joint Sealants: Formulated for prolonged exposure to humidity with fungicide to prevent mold and mildew growth.

B. Silicone, Mildew Resistant, Acid Curing, S, NS, 25, NT – Type JS-5:

1. Mildew-resistant, single-component, nonsag, plus 25 percent and minus 25 percent movement capability, nontraffic-use, acid-curing silicone joint sealant; ASTM C 920, Type S, Grade NS, Class 25, Use NT.
2. Products: Subject to compliance with requirements, provide one of the following:
 - a. Dow Corning Corporation; 786-M White.
 - b. GE Construction Sealants; SCS1700 Sanitary.
 - c. May National Associates, Inc., a subsidiary of Sika Corporation U.S.; Bondaflex Sil 100 WF.
 - d. Tremco Incorporated; Tremsil 200.

2.4 BUTYL JOINT SEALANTS

A. Butyl-Rubber-Based Joint Sealant, ASTM C 1311 – Type JS-6:

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Bostik, Inc; Chem-Calk 300.
 - b. Pecora Corporation; BC-158.

2.5 LATEX JOINT SEALANTS

A. Acrylic Latex – Type JS-7:

1. Acrylic latex or siliconized acrylic latex, ASTM C 834, Type OP, Grade NF.
2. Products: Subject to compliance with requirements, provide one of the following:
 - a. BASF Construction Chemicals, LLC, Building Systems; Sonolac.
 - b. May National Associates, Inc., a subsidiary of Sika Corporation U.S.; Bondaflex 600 or Bondaflex Sil-A 700.
 - c. Pecora Corporation; AC-20.

- d. Sherwin-Williams Company (The); 850A or 950A.
- e. Tremco Incorporated; Tremflex 834.

2.6 JOINT-SEALANT BACKING

- A. Cylindrical Sealant Backings: ASTM C 1330, Type C (closed-cell material with a surface skin) or Type B (bicellular material with a surface skin), and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.
 - 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. BASF Construction Chemicals, LLC, Building Systems.
 - b. Construction Foam Products, a division of Nomaco, Inc.
- B. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer.

2.7 MISCELLANEOUS MATERIALS

- A. Primer: Material recommended by joint-sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint-sealant-substrate tests and field tests.
- B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials.
- C. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Surface Cleaning of Joints: Clean out joints immediately before installing joint sealants to comply with joint-sealant manufacturer's written instructions and the following requirements:
 - 1. Remove laitance and form-release agents from concrete.
 - 2. Clean nonporous joint substrate surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion.
- B. Joint Priming: Prime joint substrates where recommended by joint-sealant manufacturer or as indicated by preconstruction joint-sealant-substrate tests or prior experience.

- C. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces.

3.2 INSTALLATION OF JOINT SEALANTS

- A. General: Comply with ASTM C 1193 and joint-sealant manufacturer's written installation instructions for products and applications indicated, unless more stringent requirements apply.
- B. Install sealant backings of kind indicated to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.
- C. Install bond-breaker tape behind sealants where sealant backings are not used between sealants and backs of joints.
- D. Install sealants using proven techniques that comply with the following and at the same time backings are installed:
 - 1. Place sealants so they directly contact and fully wet joint substrates.
 - 2. Completely fill recesses in each joint configuration.
 - 3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.
- E. Tooling of Nonsag Sealants: Immediately after sealant application and before skinning or curing begins, tool sealants to form smooth, uniform beads of configuration indicated. Use tooling agents that are approved in writing by sealant manufacturer and that do not discolor sealants or adjacent surfaces.
 - 1. Provide concave joint profile per Figure 8A in ASTM C 1193 unless otherwise indicated.

3.3 JOINT-SEALANT SCHEDULE

- A. Joint-Sealant Application: Interior joints in horizontal traffic surfaces.
 - 1. Joint Locations:
 - a. Isolation joints in cast-in-place concrete slabs.
 - b. Control and expansion joints in stone flooring.
 - c. Control and expansion joints in tile flooring.
 - d. Other joints as indicated on Drawings.
 - 2. Joint Sealant: Type JS-2, JS-3, or JS-4.
 - 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

- B. Joint-Sealant Application: Interior joints in vertical surfaces and horizontal nontraffic surfaces.
1. Joint Locations:
 - a. Control and expansion joints on exposed interior surfaces of exterior walls.
 - b. Tile control and expansion joints.
 - c. Vertical joints on exposed surfaces of unit masonry and concrete walls and partitions.
 2. Joint Sealant: Type JS-1.
 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.
- C. Joint-Sealant Application: Interior joints in vertical surfaces and horizontal nontraffic surfaces not subject to significant movement.
1. Joint Locations:
 - a. Control joints on exposed interior surfaces of exterior walls.
 - b. Perimeter joints between interior wall surfaces and frames of interior doors, windows, and elevator entrances.
 - c. Other joints as indicated on Drawings.
 2. Joint Sealant: Type JS-7.
 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.
- D. Joint-Sealant Application: Mildew-resistant interior joints in vertical surfaces and horizontal nontraffic surfaces.
1. Joint Locations:
 - a. Joints between plumbing fixtures and adjoining walls, floors, and counters.
 - b. Other joints as indicated on Drawings.
 2. Joint Sealant: Type JS-5.
 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.
- E. Joint-Sealant Application: Concealed mastics.
1. Joint Locations:
 - a. Aluminum thresholds.
 - b. Sill plates.
 2. Joint Sealant: Type JS-6.
 3. Joint-Sealant Color: As indicated by manufacturer's designations.

END OF SECTION

SECTION 08 11 13
HOLLOW METAL DOORS AND FRAMES

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Interior hollow metal steel doors and frames.
2. Exterior hollow metal doors and frames.
3. Hollow metal opening framing.
4. Hollow metal panels.
5. Integration of electrified hardware, access control systems, and security systems into door and frame assemblies.

1.2 DEFINITIONS

- A. Minimum Thickness:** Minimum thickness of base metal without coatings according to NAAMM-HMMA 803 or SDI A250.8.

1.3 ACTION SUBMITTALS

- A. Product Data:** Technical data including construction detail, material descriptions, core descriptions, label compliance, sound and fire resistance ratings, temperature rise ratings and finishes for each type of door and frame specified.
- B. Provide technical data on all doors and frames to indicate thermal and other Performance Requirements compliance.**
- C. Shop Drawings:** Submit drawings using same designations in door and frame schedule on the Drawings. Manufacturers published details may NOT replace shop drawings for this project. Provide the following at minimum:
1. Opening size(s), handing of doors.
 2. Frame throat dimensions
 3. Details of each frame type.
 4. Elevations and profiles of door design types.
 5. Details of door and frame construction including vertical and horizontal edge details and metal thickness.
 6. Location and installation requirements of door hardware and reinforcements, hardware group numbers.
 7. Details of anchorage, joint, field splice, and connections, details of moldings, removable stops, and glazing.
 8. Fire label requirements including fire rating time duration, maximum temperature rise requirements, and smoke label requirements.

9. Indicate routing of electrical conduit and dimensions and locations of cutouts in doors and frames to accept electric hardware devices, signals, and control systems.
- D. Schedule: Provide schedule of steel doors and frames, prepared by or under the supervision of supplier, using same reference numbers for details and openings as those on Drawings. Coordinate with final door hardware schedule.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: Submit test reports for each type of steel door and frame assembly for tests performed by a qualified testing agency.
- B. Certificate of Compliance for Fire Rated Doors: Provide copies of Certificate of Compliance for fire rated door assemblies, smoke and draft control door assemblies, and temperature rise rated door assemblies.
 1. Oversize Construction Certification: For assemblies required to be fire rated and exceeding limitations of labeled assemblies.

1.5 QUALITY ASSURANCE

- A. Regulatory Requirements:
 1. Steel Door and Frame Standard: Comply with applicable provisions of Hollow Metal Manufacturers Association (HMMA) Div. of National Association of Architectural Metal Manufacturers (NAAMM):
 - a. HMMA Hollow Metal Manual.
 - b. HMMA 861 Guide Specifications for Commercial Hollow Metal Doors and Frames.
- B. Manufacturer Qualifications: Firm having minimum 5 years documented experience in manufacturing steel doors and frames, with sufficient production capacity to produce required units.
- C. Source Limitations: Obtain steel doors and frames from single source from single manufacturer.
- D. Preinstallation Conference: Conduct conference at site.

1.6 COORDINATION

- A. Coordinate anchorage installation for hollow metal frames. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors. Deliver to site in time for installation.
- B. Coordinate requirements for installation of door hardware, electrified door hardware, and access control and security systems.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver steel doors and frames palletized, packaged, or crated to provide protection during transit and site storage. Do not use nonvented plastic. Provide additional protection to prevent damage to factory finished units.
- B. Deliver welded frames with two removable spreader bars across bottom of frames, tack welded to jambs and mullions.
- C. Store hollow metal work vertically under cover at site with head up. Place on minimum 4 inch (102 mm) high wood blocking. Provide minimum 1/4 inch (6 mm) space between each stacked door to permit air circulation.
- D. Inspect doors and frames, on delivery, for damage. Tool marks, rust, blemishes, and other damage on exposed surfaces is not acceptable. Remove and replace damaged items directed by Owner. Store doors and frames at building site in dry location and off ground to prevent deterioration.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Fire Rated Assemblies: Complying with NFPA 80 and listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction for fire protection ratings and temperature rise limits indicated, based on testing at positive pressure according to NFPA 252 or UL 10C. Provide metal labels permanently fastened on each door and frame assembly within size limitations established by the labeling authority having jurisdiction.
 - 1. Smoke and Draft Control Assemblies: Provide assemblies with gaskets listed and labeled for smoke and draft control by a qualified testing agency acceptable to authorities having jurisdiction, based on testing according to UL 1784 and installed in compliance with NFPA 105.
 - 2. Oversize Fire Rated Door Assemblies: For units exceeding sizes of tested assemblies, provide certification by a qualified testing agency that doors comply with standard construction requirements for tested and labeled fire-rated door assemblies except for size.
- B. Fire Rated, Borrowed Lite Assemblies: Complying with NFPA 80 and listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction, for fire protection ratings indicated, based on testing according to NFPA 257 or UL 9.
- A. Thermally Rated Door Assemblies: Fabricate doors and frames for thermal insulating assemblies and tested in accordance with ASTM C 518.
 - 1. Provide door assemblies with U-factor of not more than 0.34 degrees Btu/F x h x sq. ft. when tested according to ASTM C 518.

- B. Sound Rated (Acoustical) Assemblies: Provide door and frame assemblies fabricated as sound reducing assemblies, tested in accordance with ASTM E 1408, and classified in accordance with ASTM E 413.
 - 1. Provide acoustical assemblies with sound ratings of STC 36 or better.
- C. Accessibility Requirements: Comply with applicable requirements.
 - 1. U.S. Architectural and Transportation Barriers Compliance Board Americans with Disabilities Act Accessibility Guidelines for Buildings and Facilities (ADAAG) 2010.
 - 2. ICC/ANSI A117.1 Accessible and Useable Building and Facilities..

2.2 MATERIALS

- A. Manufacturers: Subject to compliance with requirements, provide products complying with requirements of one of the following:
 - 1. Amweld International, LLC.
 - 2. Ceco Door Products; an Assa Abloy Group company.
 - 3. Curries Company; an Assa Abloy Group company.
 - 4. Mesker Door Inc.
 - 5. Republic Doors and Frames.
 - 6. Rocky Mountain Metals, Inc.
 - 7. Steelcraft; an Allegion company.
- B. Cold Rolled Steel Sheet: ASTM A 1008/A 1008M, Commercial Steel (CS), Type B; suitable for exposed applications.
- C. Hot Rolled Steel Sheet: ASTM A 1011/A 1011M, Commercial Steel (CS), Type B; free of scale, pitting, or surface defects; pickled and oiled.
- D. Metallic Coated Steel Sheet: ASTM A 653/A 653M, Commercial Steel (CS), Type B.
- E. Inserts, Bolts, and Fasteners: Hot dip galvanized according to ASTM A 153/A 153M.
- F. Frame Anchors: ASTM A 879/A 879M, Commercial Steel (CS), 04Z (12G) coating designation; mill phosphatized.
 - 1. For anchors built into exterior walls, steel sheet complying with ASTM A 1008/A 1008M or ASTM A 1011/A 1011M, hot dip galvanized according to ASTM A 153/A 153M, Class B.
- G. Power Actuated Fasteners in Concrete: Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with clips or other accessory devices for attaching hollow-metal frames of type indicated.
- H. Mineral Fiber Insulation: ASTM C 665, Type I (blankets without membrane facing); consisting of fibers manufactured from slag or rock wool; with maximum flame spread

and smoke developed indexes of 25 and 50, respectively; passing ASTM E 136 for combustion characteristics.

- I. Grout: ASTM C 476, except with a maximum slump of 4 inches (102 mm), as measured according to ASTM C 143/C 143M.
- J. Glazing: Comply with requirements in Section 08 80 00.
- K. Bituminous Coating: Cold applied asphalt mastic, compounded for 15 mil (0.4 mm) dry film thickness per coat. Provide inert type noncorrosive compound free of asbestos fibers, sulfur components, and other deleterious impurities.

2.3 EXTERIOR STEEL DOORS AND FRAMES

- A. Construct hollow metal doors and frames to comply with standards indicated for materials, fabrication, hardware locations, hardware reinforcement, tolerances, and clearances, and as specified.
- B. Maximum Duty Doors and Frames: SDI A250.8, Level 4; SDI A250.4, Level A..
 - 1. Doors:
 - a. Type: Indicated in the Door Schedule.
 - b. Thickness: 1-3/4 inches (44.5 mm).
 - c. Face: Metallic coated steel sheet, minimum thickness of 0.067 inch (1.7 mm), with minimum A60 (ZF180) coating.
 - d. Edge Construction: Model 2, Seamless.
 - e. Edge Bevel: Bevel lock edge 1/8 inch in 2 inches (3.2 mm in 51 mm).
 - f. Top Edge Closures: Close top edges of doors with flush closures of same material as face sheets. Seal joints against water penetration.
 - g. Bottom Edges: Close bottom edges of doors with end closures or channels of same material as face sheets. Provide weep hole openings in bottoms of exterior doors to permit moisture to escape.
 - h. Core: Polyisocyanurate insulation to meet U value indicated with Vertical steel stiffener.
 - i. Fire Rated Core: Vertical steel stiffener with insulation core for fire rated doors.
 - j. Achieve an operable U-Factor of 0.34 to meet all Performance Requirements.
 - k. Basis of Design: Provide Trio-E Energy Efficient Steel Stiffened Doors by Ceco Door.
 - 2. Frames:
 - a. Materials: Metallic coated steel sheet, minimum thickness of 0.067 inch (1.7 mm), with minimum A60 (ZF180) coating.
 - b. Construction: Full profile welded.
 - 3. Exposed Finish: Prime.

2.4 INTERIOR STANDARD STEEL DOORS AND FRAMES

- A. Construct hollow metal doors and frames to comply with standards indicated for materials, fabrication, hardware locations, hardware reinforcement, tolerances, and clearances, and as specified.
- B. Extra Heavy Duty Doors and Frames: SDI A250.8, Level 3; SDI A250.4, Level A..
 - 1. Doors:
 - a. Type: Indicated in the Door Schedule.
 - b. Thickness: 1-3/4 inches (44.5 mm).
 - c. Face: Metallic coated steel sheet, minimum thickness of 0.053 inch (1.3 mm).
 - d. Edge Construction: Model 2, Seamless.
 - e. Edge Bevel: Bevel lock edge 1/8 inch in 2 inches (3.2 mm in 51 mm).
 - f. Core: Vertical steel stiffener.
 - g. Fire Rated Core: laminated mineral board core for fire rated and temperature rise rated doors.
 - 2. Frames:
 - a. Materials: Metallic coated steel sheet, minimum thickness of 0.053 inch (1.3 mm).
 - b. Sidelite and Transom Frames: Fabricated from same thickness material as adjacent door frame.
 - c. Construction: Full profile welded.
 - 3. Exposed Finish: Prime.

2.5 BORROWED LITES

- A. Fabricate of Metallic coated steel sheet, minimum thickness of 0.053 inch (1.3 mm).
- B. Construction: Full profile welded.
- C. Fabricate in one piece except where handling and shipping limitations require multiple sections. Where frames are fabricated in sections due to shipping or handling limitations, provide alignment plates or angles at each joint, fabricated of metal of same or greater thickness as metal as frames.
- D. Provide countersunk, flat or oval head exposed screws and bolts for exposed fasteners unless otherwise indicated.

2.6 HOLLOW METAL PANELS

- A. Provide hollow metal panels of same materials, construction, and finish as adjacent door assemblies.

2.7 FRAME ANCHORS

A. Jamb Anchors:

1. Type: Provide anchors of minimum size and type required by applicable door and frame standard, and suitable for performance level indicated.
2. Quantity: Minimum of three anchors per jamb, with one additional anchor for frames with no floor anchor. Provide one additional anchor for each 24 inches (610 mm) of frame height above 7 feet (2.1 m).
3. Postinstalled Expansion Anchor: Minimum 3/8 inch (9.5 mm) diameter bolts with expansion shields or inserts, with pipe spacer.

B. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor.

C. Floor Anchors for Concrete Slabs with Underlayment: Adjustable type anchors with extension clips, allowing not less than 2 inch (51 mm) height adjustment. Terminate bottom of frames at top of underlayment.

2.8 FABRICATION

A. Fabricate steel doors and frames rigid and free from defects, warp, or buckle. Accurately form metal to required sized and profiles. Take field measurements as required for coordination with adjoining work. Where possible, fit and assemble units in manufacturer's plant. Clearly identify work which cannot be permanently factory assembled before shipment to assure proper assembly at site.

1. Metallic Core Construction: Weld cores to both door face sheets.
2. Nonmetallic Core Construction: Laminate core material to both door face sheets with waterproof adhesive.

B. Remove tool marks and surface imperfections; dress smooth exposed faces of welded joints. Use of metallic filler to conceal manufacturing defects is not acceptable.

C. Exposed Fasteners: Provide countersunk Phillips heads for exposed screws and bolts.

D. Door Astragals: Provide overlapping astragal on one leaf of pairs of doors where required by NFPA 80 for fire performance rating or where indicated. Extend minimum 3/4 inch (19 mm) beyond edge of door on which astragal is mounted or as required to comply with published listing of qualified testing agency.

E. Hollow Metal Frames: Fabricate frames in one piece in compliance with the reference standard and the following:

1. Sidelite and Transom Bar Frames: Provide closed tubular members with no visible face seams or joints, fabricated from same material as door frame. Fasten members at crossings and to jambs by welding.
2. Provide countersunk, flat or oval head exposed screws and bolts for exposed fasteners unless otherwise indicated.
3. Grout Guards: Weld guards to frame at back of hardware mortises in grouted frames.

4. Door Silencers: Except on weather stripped frames, drill stops to receive door silencers as follows. Keep holes clear during construction.
 - a. Single Door Frames: Drill stop in strike jamb to receive three door silencers.
 - b. Double Door Frames: Drill stop in head jamb to receive two door silencers.
- F. Jamb Anchors:
 1. Masonry Type: Locate anchors not more than 16 inches (400 mm) from top and bottom of frame. Space anchors not more than 32 inches (825 mm) o.c., to match coursing:
 - a. Two anchors per jamb up to 60 inches (1530 mm) high.
 - b. Three anchors per jamb from 60 to 90 inches (1530 mm to 2290 mm) high.
 - c. Four anchors per jamb from 90 to 120 inches (2290 mm to 3060 mm) high.
 - d. Four anchors per jamb plus one additional anchor per jamb for each 24 inches (610 mm) or fraction thereof above 120 inches (3060 mm) high.
 2. Stud Wall Type: Locate anchors not more than 18 inches (460 mm) from top and bottom of frame. Space anchors not more than 32 inches (815 mm) o.c.:
 - a. Three anchors per jamb up to 60 inches (1530 mm) high.
 - b. Four anchors per jamb from 60 to 90 inches (1530 mm to 2290 mm) high.
 - c. Five anchors per jamb from 90 to 96 inches (2290 mm to 2440 mm) high.
 - d. Five anchors per jamb plus one additional anchor per jamb for each 24 inches (610 mm) or fraction thereof above 96 inches (2440 mm) high.
 3. Compression Type: Not less than two anchors in each frame.
 4. Postinstalled Expansion Type: Locate anchors not more than 6 inches (150 mm) from top and bottom of frame. Space anchors not more than 26 inches (660 mm) o.c.
- G. Head Anchors: Two anchors per head for frames more than 42 inches (1065 mm) wide and mounted in metal stud partitions.
- H. Hardware Preparation: Factory prepare hollow metal doors and frames to receive templated mortised hardware, and electrical wiring; include cutouts, reinforcement, mortising, drilling, and tapping according to SDI A250.6, the Door Hardware Schedule, and templates.
 1. Reinforce doors and frames to receive nontemplated, mortised, and surface mounted door hardware. Offset reinforcement so faces of mortised hardware items are flush with door surfaces. Secure reinforcement by spot welding.
 2. Comply with BHMA A156.115 and SDI 250.6 for preparing hollow metal doors and frames for hardware.
 - a. Hinges and Pivots: 0.167 inch (4.2 mm) thick by 1-1/2 inches (38 mm) wide by 9 inches (229 mm).
 - b. Lock Front, Strike, and Flushbolt Reinforcement: 0.093 inch (2.3 mm) thick by size as required by hardware manufacturer.

- c. Lock Reinforcement Units: 0.067 inch (1.7 mm) thick by size as required by hardware manufacturer.
 - d. Closer Reinforcements: 0.093 inch (2.3 mm) thick one piece channel by size as required by hardware manufacturer.
 - 1) For concealed overhead door closers, provide space, cutouts, reinforcing, and provisions for fastening in top rail of doors or head of frames as applicable.
 - e. Other Hardware Reinforcements: Required for adequate strength and anchorage; in lieu of reinforcement specified, hardware manufacturers recommended reinforcing units may be used.
 - f. Exit Device Reinforcements: 0.250 inch (6.35 mm) thick by 10 inches (245 mm) high by 4 inches (101 mm) wide centered on exit device case body, unless otherwise recommended by exit device manufacturer.
- I. Glazed Lites: Provide stops and moldings around glazed lites and louvers where indicated. Form corners of stops and moldings with mitered hairline joints.
 - 1. Provide stops and moldings flush with face of door, and with square stops unless otherwise indicated.
 - 2. Multiple Glazed Lites: Provide fixed and removable stops and moldings so each glazed lite is capable of being removed independently.
 - 3. Weld fixed frame moldings on outside of exterior and on secure side of interior doors and frames.
 - 4. Provide loose stops and moldings on inside of hollow metal doors and frames.
 - 5. Coordinate rabbet width between fixed and removable stops with glazing and installation types indicated.
 - 6. Provide stops for installation with countersunk flat or oval head machine screws spaced uniformly not more than 9 inches (230 mm) o.c. and not more than 2 inches (51 mm) o.c. from each corner.
- J. Mullions and Transom Bars: Join to adjacent members by welding or rigid mechanical anchors.
- K. Plaster and Grout Guards and Removable Access Plates: Provide 0.016 inch (0.4mm) thick plaster guards or dust cover boxes of same material as frame, welded to frame at back of hardware cutouts to close off interior of openings and prevent mortar or other materials from obstructing hardware operation. Provide removable access plates in the heads of frames to receive overhead concealed door closers.
- L. Electrical Requirements: Coordinate provisions with installation of electrical items including electronic hardware, security system components, and similar items having electrical requirements; arrange so wiring can be readily removed and replaced.
 - 1. Provide cutouts and reinforcements required for steel doors and frames to accept security system components.
 - 2. Doors with Electric Hinges and Pivots: Provide with metal conduit or raceway to permit wiring from electric hinge or pivot to other electric door hardware.

- a. Hinge Location: Center for doors less than 90 inches (2286 mm) tall or second hinge from door bottom for doors greater than 90 inches (2286 mm); top or bottom electric hinge locations shall not be permitted.

2.9 ACCESSORIES

- A. Rubber Door Silencers: Except on weatherstripped and smoke-gasketed doors, drill stop in strike jamb to receive three silencers on single-door frames and drill head jamb stop to receive four silencers on double-door frames. Install plastic plugs to keep holes clear during construction.
- B. Shim Space Insulation: loose or batt mineral fiber or foam-in-place type unless otherwise indicated on the Drawings.

2.10 STEEL FINISHES

- A. Surface Preparation: Clean, treat and prime surfaces of fabricated steel door and frame work, inside and out, whether exposed or concealed in the construction.
 - 1. Clean surfaces with nonpetroleum solvent so surfaces are free of dirt, oil, grease, and contaminants that could impair paint bond.
 - 2. Remove mill scale and rust, if present, from uncoated steel, complying with SSPC-SP 3 *Power Tool Cleaning* or SSPC-SP 6/NACE No. 3 *Commercial Blast Cleaning*.
 - 3. After cleaning, apply conversion coating suited to organic coating applied over surface.
 - 4. Clean welds, mechanical connections, and abraded areas, and apply galvanizing repair paint complying with ASTM A 780.
 - 5. Galvanizing Repair Paint: High zinc dust content paint for regalvanizing welds in steel, complying with SSPC-Paint 20.
- B. Factory Priming for Field Painted Finish: Apply shop primer immediately after surface preparation and pretreatment. Apply fast curing, corrosion inhibiting, lead and chromate free, universal primer complying with ANSI A224.1; compatible with substrate and field applied finish paint system indicated; providing sound foundation for field applied topcoats despite prolonged exposure. Apply primer coat of a smooth, even consistency of provide uniform dry film thickness of not less than 0.7 mils (0.02 mm). Touch up surfaces having runs, smears, or bare spots.
 - 1. Finish Coats: Spray finish coats complying with Sections governing interior and exterior painting as appropriate.
- C. Factory Finish: Clean, pretreat, and apply two coat, baked on finish consisting of prime coat and thermosetting topcoat, complying with SDI A250.3.
 - 1. Color and Gloss: Selected by Architect.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions for compliance with requirements for installation tolerances and conditions affecting performance of the work. Examine roughing in for embedded and built-in anchors to verify actual locations before frame installation.
- B. Existing Steel Frames and Doors: Examine existing frames and doors for indications of hinge stress, improperly fitting and operating hardware, excessive wear, rust or corrosion, dents and patches, out of plumb, damaged core, fire rated opening clearances, delamination, forced entry and vandalism and similar wear indicators.
- C. Prepare written report listing conditions detrimental to performance of the work.
- D. Proceed with installation after correcting unsatisfactory conditions.

3.2 PREPARATION

- A. Remove welds in shipping spreaders installed at factory. Restore exposed finish by grinding, filling, and dressing, as required to make repaired area smooth, flush, and invisible on exposed faces. Touch up factory applied finishes where spreaders are removed.
- B. Prior to installation and with installation spreaders in place, adjust and securely brace hollow metal door frames for squareness, alignment, twist, and plumb to the following tolerances:
 - 1. Squareness: Plus or minus 1/16 inch (1.6 mm), measured at door rabbet on a line 90 degrees from jamb, and perpendicular to frame head.
 - 2. Alignment: Plus or minus 1/16 inch (1.6 mm), measured at jambs on a horizontal line parallel to plane of wall.
 - 3. Twist: Plus or minus 1/16 inch (1.6 mm), measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.
 - 4. Plumbness: Plus or minus 1/16 inch (1.6 mm), measured at jambs on a perpendicular line from head to floor.
- C. Drill and tap doors and frames to receive nontemplated, mortised, and surface mounted door hardware.

3.3 INSTALLATION

- A. Install hollow metal doors and frames plumb, rigid, properly aligned, and securely fastened in place. Comply with approved Shop Drawings and with manufacturer's written instructions.
- B. Hollow Metal Frames: Comply with SDI A250.11.

1. Set frames accurately in position; plumbed, aligned, and braced securely until permanent anchors are set. After wall construction is complete, remove temporary braces without damage to completed work.
 - a. Extend frame anchorages below fills and finishes. Coordinate the installation of built in anchors for wall and partition construction required with work.
 - b. Where frames are fabricated in sections, field splice at approved locations by welding face joint continuously; grind, fill, dress, and make splice smooth, flush, and invisible on exposed faces. Touch up finishes.
 - c. Install frames with removable stops located on secure side of opening.
 - d. Install door silencers in frames before grouting.
 - e. Field apply corrosion resistant coating to backs of frames in contact with grout or plaster containing antifreezing agents.
2. Welded Frames: Set anchorage devices where required for securing frames to in place concrete or masonry construction. Set anchorage devices opposite each anchor location as specified and anchorage device manufacturer's written instructions. Leave drilled holes rough, not reamed, and free of dust and debris.
3. Fire Rated Openings: Install frames according to NFPA 80.
4. Floor Anchors: Install floor anchors for each jamb and mullion that extends to floor, and secure with postinstalled expansion anchors.
 - a. Floor anchors may be set with power actuated fasteners in lieu of postinstalled expansion anchors if indicated and approved on Shop Drawings.
5. Metal Stud Partition: Solidly pack mineral fiber insulation inside frames.
6. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with grout or mortar.
7. Concrete Walls: Solidly pack mineral fiber insulation inside frames; Coordinate installation of frames to allow for solidly filling space between frames and concrete with grout or mortar where indicated on the Drawings.
8. In Place Concrete or Masonry Construction: Secure frames in place with postinstalled expansion anchors. Stainless Steel Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.
9. Installation Tolerances: Adjust hollow metal frames to specified tolerances:
 - a. Squareness: Plus or minus 1/16 inch (1.6 mm), measured at door rabbet on a line 90 degrees from jamb perpendicular to frame head.
 - b. Alignment: Plus or minus 1/16 inch (1.6 mm), measured at jambs on a horizontal line parallel to plane of wall.
 - c. Twist: Plus or minus 1/16 inch (1.6 mm), measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.
 - d. Plumbness: Plus or minus 1/16 inch (1.6 mm), measured at jambs at floor.
10. Exterior Frames:
 - a. Coordinate installation with all other Work, including but not limited to sheet metal flashing and trim and sealant.

- 1) Provide appropriate mineral fiber or foam-in-place shim space fill for sealant joint backing.
 - 2) Ensure sufficient frame fill is in place to permit proper installation of backer rod and sealant for all units to achieve a 2 point adhesion for all exterior frames.
 - b. Install exterior frames in accordance with manufacturer's requirements to achieve U value required by Performance Requirements.
 - c. Provide stainless steel drip edge flashing at all frame heads.
 - d. Clean all frames prior to installation of sealant to ensure 2 point adhesion of sealant assembly.
- C. Hollow Metal Doors: Fit and adjust hollow metal doors accurately in frames, within clearances specified.
 1. Nonfire Rated Steel Doors: Comply with SDI A250.8.
 - a. Nonfire Rated Steel Doors:
 - 1) Between Door and Frame Jambs and Head: 1/8 inch (3 mm) plus or minus 1/32 inch (0.8 mm).
 - 2) Between Edges of Pairs of Doors: 1/8 inch to 1/4 inch (3 mm to 6 mm) plus or minus 1/32 inch (0.8 mm).
 - 3) At Bottom of Door:
 - a) No Threshold or Carpet: 3/8 inch (9 mm) plus or minus 1/32 inch (0.8 mm).
 - b) Threshold or Carpet: 1/8 inch (3 mm) plus or minus 1/32 inch (0.8 mm).
 - 4) Between Door Face and Stop: 1/16 inch to 1/8 inch (1.6 mm to 3 mm) plus or minus 1/32 inch (0.8 mm).
 2. Rated Assemblies: Place doors in frames with clearances specified in NFPA Standard No. 80:
 - a. Doors, Frames, and Meeting Edges, Pairs of Doors: 1/8 inch.
 - b. Bottom: Maximum 3/8 inch between door and raised noncombustible sill.
 - c. Bottom: Maximum 3/4 inch above finished floor between door and floor where there is no sill.
 - d. Bottom: Maximum 1/2 inch between door and nominal floor covering surface.
 - e. Astragals: Comply with NFPA 80.
 3. Smoke Control Doors: Install doors according to NFPA 105.
 4. Install exterior doors in accordance with manufacturer's requirements to achieve U value required by Performance Requirements.
- D. Glazing: Comply with installation requirements in Section 08 80 00 and with hollow metal manufacturer's written instructions.

1. Secure stops with countersunk flat or oval head machine screws spaced uniformly not more than 9 inches (230 mm) o.c. and not more than 2 inches (50 mm) o.c. from each corner.
- E. Hardware: Apply hardware in accordance with hardware manufacturer's instructions and Section 08 71 00. Drill and tap for machine screws as required. Do not use self tapping sheet metal screws. Adjust door installation to provide uniform clearance at head and jams, and to contact stops uniformly. Adjust hardware items just prior to final inspection. Leave work in complete and proper operating condition.
- F. Repair of Existing Doors and Frames:
 1. Hinge Damage: Remove the door and hinge. Drill welding holes in area where hinge reinforcement is located. Clamp hinge reinforcement in its original position and plug weld. Grind smooth, prime, and finish paint. Rehang door ensuring it operates smoothly and no longer sags.
 2. Holes: Patch weld the hole. For fire rated doors, comply with NFPA requirements.
 3. Dents: If dent has not penetrated the face, fill dent with recommended patching compound, sand smooth, and refinish the door.

3.4 ADJUSTING

- A. Final Adjustments: Test and adjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow metal work that is warped, bowed, or unacceptable.

3.5 CLEANING AND TOUCHUP

- A. Remove grout and other bonding material from hollow-metal work immediately after installation.
- B. Prime Coat Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat and apply touchup of compatible air drying, rust inhibitive primer.
- C. Metallic Coated Surface Touchup: Clean abraded areas and repair with galvanizing repair paint according to manufacturer's written instructions.
- D. Factory Finish Touchup: Clean abraded areas and repair with same material used for factory finish according to manufacturer's written instructions.
- E. Touchup Painting: Cleaning and touchup painting of abraded areas of paint are specified in painting Sections.
- F. Institute protective measures required throughout the remainder of the construction period to ensure that steel doors and frames will be without damage or deterioration, at time of substantial completion.

END OF SECTION

SECTION 087100
DOOR HARDWARE

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes:
 - 1. Mechanical door hardware.
 - 2. Cylinders for door hardware specified in other Sections.
 - 3. Electrified door hardware.
- B. Products furnished, but not installed, under this Section include the products listed below. Coordinating and scheduling the purchase and delivery of these products remain requirements of this Section.
 - 1. Lock cylinders to be installed under other Sections.
 - 2. Permanent lock cores to be installed by Owner.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings: Details of electrified door hardware.
- C. Samples: For each exposed product and for each color and texture specified.
- D. Other Action Submittals:
 - 1. Door Hardware Schedule: Prepared by or under the supervision of Installer, detailing fabrication and assembly of door hardware, as well as installation procedures and diagrams. Coordinate final door hardware schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.
 - a. Format: Use same scheduling sequence and format and use same door numbers as in the Contract Documents.
 - b. Content: Include the following information:
 - 1) Identification number, location, hand, fire rating, size, and material of each door and frame.
 - 2) Locations of each door hardware set, cross-referenced to Drawings on floor plans and to door and frame schedule.
 - 3) Complete designations, including name and manufacturer, type, style, function, size, quantity, function, and finish of each door hardware product.

- 4) Description of electrified door hardware sequences of operation and interfaces with other building control systems.
2. Keying Schedule: Prepared by or under the supervision of Installer, detailing Owner's final keying instructions for locks.

1.3 QUALITY ASSURANCE

- A. Installer Qualifications: Supplier of products and an employer of workers trained and approved by product manufacturers and an Architectural Hardware Consultant who is available during the course of the Work to consult with Contractor, Architect, and Owner about door hardware and keying.
- B. Architectural Hardware Consultant Qualifications: A person who is experienced in providing consulting services for door hardware installations that are comparable in material, design, and extent to that indicated for this Project and who is currently certified by DHI as follows:
 1. For door hardware, an Architectural Hardware Consultant (AHC).
- C. Source Limitations: Provide electrified door hardware from same manufacturer as mechanical door hardware, unless otherwise indicated. Manufacturers that perform electrical modifications and that are listed by a testing and inspecting agency acceptable to authorities having jurisdiction are acceptable.
- D. Fire-Rated Door Assemblies: Where fire-rated door assemblies are indicated, provide door hardware rated for use in assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency, for fire-protection ratings indicated, based on testing at positive pressure according to NFPA 252 or UL 10C, unless otherwise indicated.
- E. Smoke- and Draft-Control Door Assemblies: Where smoke- and draft-control door assemblies are required, provide door hardware that meet requirements of assemblies tested according to UL 1784 and installed in compliance with NFPA 105.
 1. Air Leakage Rate: Maximum air leakage of 0.3 cfm/sq. ft. at the tested pressure differential of 0.3-inch wg of water.
- F. Electrified Door Hardware: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction.
- G. Means of Egress Doors: Latches do not require more than 15 lbf to release the latch. Locks do not require use of a key, tool, or special knowledge for operation.
- H. Accessibility Requirements: For door hardware on doors in an accessible route, comply with the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines and ICC/ANSI A117.1.
 1. Provide operating devices that do not require tight grasping, pinching, or twisting of the wrist and that operate with a force of not more than 5 lbf.

2. Comply with the following maximum opening-force requirements:
 - a. Interior, Non-Fire-Rated Hinged Doors: 5 lbf applied perpendicular to door.
 - b. Sliding or Folding Doors: 5 lbf applied parallel to door at latch.
 - c. Fire Doors: Minimum opening force allowable by authorities having jurisdiction.
3. Bevel raised thresholds with a slope of not more than 1:2. Provide thresholds not more than 1/2 inch high.
4. Adjust door closer sweep periods so that, from an open position of 70 degrees, the door will take at least 3 seconds to move to a point 3 inches from the latch, measured to the leading edge of the door.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Deliver keys to manufacturer of key control system for subsequent delivery to Owner.

1.5 WARRANTY

- A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of door hardware that fail in materials or workmanship within specified warranty period.
 1. Warranty Period: Three years from date of Substantial Completion, unless otherwise indicated.
 - a. Electromagnetic Locks: Five years from date of Substantial Completion.
 - b. Exit Devices: Two years from date of Substantial Completion.
 - c. Manual Closers: 10 years from date of Substantial Completion.
 - d. Concealed Floor Closers: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SCHEDULED DOOR HARDWARE

- A. Provide products for each door that comply with requirements indicated in Part 2 and door hardware schedule.

2.2 HINGES

- A. Hinges: BHMA A156.1. Provide template-produced hinges for hinges installed on hollow-metal doors and hollow-metal frames.
 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. McKinney Products Company; an ASSA ABLOY Group company.

- b. Hager Companies.
 - c. Stanley Commercial Hardware; Div. of The Stanley Works.
 - d. IVES Hardware; an Allegion plc company.
- B. Full Mortise Hinges: Five knuckle, ANSI A156.1, with non-rising pins, anti-friction bearings on all hinges, heavy weight construction where noted in Schedule.
- C. Non-Removable Pins: Provide butts with set screw in barrel making hinge non-removable when door is in closed position for exterior and interior doors with locks, and doors with security controls (credential reader and electromechanical/electromagnetic hardware items).
- D. Reverse Spring Hinges: Full mortise, where noted in schedule.
- E. Sound Control Hinges: Cam-lift hinges, where noted in schedule.
- F. Applications:

Type	McKinney	Hager	Stanley	Ives
Type 2	TA2714	BB1279	FBB179	5BB1
Type 4	T4A3786	BB1168	FBB168	5BB1HW

- 1. Interior doors with closers: Type 2 or 4.
 - 2. Interior doors over 36 inches wide: Type 4.
 - 3. Interior doors 36 inches or less without closer: Type 2.
 - 4. Provide NRP (non-removable pins) at out-swinging lockable doors.
- G. Sizes: 1-3/4-inch Doors: 4-1/2 inch by 4-1/2 inch
- H. Quantities:
 - 1. Two hinges per leaf for openings through 60 inches high.
 - 2. One additional hinge per leaf for each additional 30 inches in height or fraction thereof.

2.3 SELF-CLOSING HINGES AND PIVOTS

- A. Self-Closing Hinges and Pivots: BHMA A156.17.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Bommer Industries, Inc.
 - b. Hager Companies.
 - c. IVES Hardware; an Allegion plc company.
 - d. McKinney Products Company; an ASSA ABLOY Group company.
 - e. Stanley Commercial Hardware; Div. of The Stanley Works.

2.4 CENTER-HUNG AND OFFSET PIVOTS

A. Center-Hung and Offset Pivots: BHMA A156.4.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. DORMA Architectural Hardware; Member of The DORMA Group North America.
 - b. IVES Hardware; an Allegion plc company.
 - c. Rixson Specialty Door Controls; an ASSA ABLOY Group company.

2.5 CONTINUOUS HINGES

- A. Continuous Hinges: BHMA A156.26; minimum 0.120-inch-thick, hinge leaves with minimum overall width of 4 inches; fabricated to full height of door and frame and to template screw locations; with components finished after milling and drilling are complete.
- B. Continuous, Gear-Type Hinges: Extruded-aluminum or stainless steel, pinless, geared hinge leaves joined by a continuous extruded-aluminum or stainless steel channel cap; with concealed, self-lubricating thrust bearings.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. IVES Hardware; an Allegion plc company.
 - b. McKinney Products Company; an ASSA ABLOY Group company.
 - c. Pemko Manufacturing Co.; an ASSA ABLOY Group company.
 - d. Roton - Hager Companies
 - e. Stanley Commercial Hardware; Div. of The Stanley Works.

C. Applications:

	Pemko	Roton	Stanley	Ives
	FM83SLFHD	780-112HD	661HD	112HD
	-	790-900	651	700

- D. Provide sufficient number of concealed wires to accommodate electric function of specified hardware, supply as three-piece hinge (removable electronic wire section in middle leaf).
- E. Continuous hinges: Heavy duty, at exterior doors and doors with exit devices.
- F. Materials: Aluminum hinges at storefront doors and electrified openings; stainless steel at all other locations indicated.

2.6 MECHANICAL LOCKS AND LATCHES

- A. Mortise Locks: BHMA A156.13; Operational Grade 1; stamped steel case with steel or brass parts; Series 1000.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Corbin Russwin Architectural Hardware; an ASSA ABLOY Group Company; ML2000 Series.
 - b. SARGENT Manufacturing Company; an ASSA ABLOY Group company; 8200 Series.
 - c. Schlage Commercial Lock Division; an Allegion plc company; L9000 Series.
- B. Construction:
1. Fronts: 8" x 1-1/4", adjustable to 1/8" in 2" with 2-3/4" backset.
 2. Minimum projection of latch bolt: 3/4".
 3. Minimum throw of dead bolt: 1".
 4. Beveled, rounded or rabbeted faces where required.
 5. Where lock stiles are too narrow for backsets of locks specified, furnish special backsets.
 6. Electronic functions supplied with factory-installed request-to-exit monitor switch, security monitor switch, latch bolt monitor switch, and electrified trim components, as scheduled.
- C. Lever Handles and Escutcheons: Of style indicated on Drawings, cast of forged brass or bronze material, levers supported by internal spring.
1. On doors into hazardous areas that are accessible to physically handicapped persons, provide knurled lever contact surfaces. Abrasive coating is not permitted
- D. Strikes:
1. Furnish locks and latches with wrought box strikes.
 2. On single swing doors, provide latch strike plates with minimum lip projection necessary to project from trim.
 3. On pair of doors with or without astragal, lip projection of latch strike plates shall not extend beyond face of lock style of inactive leaf.
 4. Size: 4-7/8" x 1-1/4" x 3/32".
- E. Dummy Trim: Match to lock specified on pair of doors, provide complete set both sides of doors.

2.7 ELECTROMAGNETIC LOCKS

- A. Electromagnetic Locks: BHMA A156.23; electrically powered; with electromagnet attached to frame and armature plate attached to door; full-exterior or full-interior type, as required by application indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Schlage Commercial Lock Division; an Allegion plc company.
 - b. Securitron Magnalock Corporation; an ASSA ABLOY Group company.

2.8 ELECTROMECHANICAL LOCKS

- A. Electromechanical Locks: BHMA A156.25; Grade 1; motor or solenoid driven; mortise latchbolt; with strike that suits frame.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Corbin Russwin Architectural Hardware; an ASSA ABLOY Group Company; ML2000 Series.
 - b. SARGENT Manufacturing Company; an ASSA ABLOY Group company; 8200 Series.
 - c. Schlage Commercial Lock Division; an Allegion plc company; L9000 Series.

2.9 SELF-CONTAINED ELECTRONIC LOCKS

- A. Self-Contained Electronic Locks: BHMA A156.25, mortise; with internal, battery-powered, self-contained electronic locks; consisting of complete lockset, motor-driven lock mechanism, and actuating device; enclosed in zinc-dichromate-plated, wrought-steel case, and strike that suits frame. Provide key override, low-battery detection and warning, LED status indicators, and ability to program at the lock.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. SARGENT Manufacturing Company; an ASSA ABLOY Group company.
 - b. Schlage Commercial Lock Division; an Allegion plc company.

2.10 SURFACE BOLTS

- A. Surface Bolts: BHMA A156.16.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Burns Manufacturing Incorporated.

- b. Door Controls International, Inc.
- c. IVES Hardware; an Allegion plc Company.
- d. Rockwood Manufacturing Company.
- e. Trimco.

2.11 MANUAL FLUSH BOLTS

- A. Manual Flush Bolts: BHMA A156.16; minimum 3/4-inch throw; designed for mortising into door edge.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Burns Manufacturing Incorporated.
 - b. Door Controls International, Inc.
 - c. IVES Hardware; an Allegion plc Company.
 - d. Rockwood Manufacturing Company.
 - e. Trimco.

2.12 AUTOMATIC AND SELF-LATCHING FLUSH BOLTS

- A. Automatic and Self-Latching Flush Bolts: BHMA A156.16; minimum 3/4-inch throw; designed for mortising into door edge.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Door Controls International, Inc.
 - b. IVES Hardware; an Allegion plc Company.
 - c. Trimco.

2.13 EXIT DEVICES AND AUXILIARY ITEMS

- A. Exit Devices and Auxiliary Items: BHMA A156.3.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. SARGENT Manufacturing Company; an ASSA ABLOY Group company; 19-43-80 Series X with Free-Wheeling Trim x Guarded Latchbolt (rim devices)
 - b. Von Duprin; an Allegion plc company; 35 & 98 Series.

- B. Description:

1. Listed under "Fire Exit Hardware" in accident equipment list of Underwriters Laboratories.
2. Chassis mounted unit construction with removable covers, identical wall thickness on all types of exit devices.
3. Minimum top and bottom rods of vertical rod devices: 3/8" diameter.

4. Stainless steel main arm pivot and springs.
5. Stainless steel surface for exit device touch pad.
6. Base metal stainless steel, brass, or bronze.
7. Heavy-duty pull trim, or lever trim to match locksets.
8. Electronic functions supplied with factory-installed request-to-exit monitor switch(s), latch bolt retraction system, latch bolt monitor switch, electrified trim components, as scheduled.

2.14 LOCK CYLINDERS

- A. Lock Cylinders: Tumbler type, constructed from brass or bronze, stainless steel, or nickel silver.
 1. Manufacturer: Match Building Standard.
- B. Construction Master Keys: Provide cylinders with feature that permits voiding of construction keys without cylinder removal. Provide 10 construction master keys.
- C. Construction Cores: Provide construction cores that are replaceable by permanent cores. Provide 10 construction master keys.

2.15 KEYING

- A. Keying System: Factory registered, complying with guidelines in BHMA A156.28, Appendix A. Incorporate decisions made in keying conference.
 1. Existing System:
 - a. Master key or grand master key locks to Owner's existing system.
- B. Keys: Brass.
 1. Quantity: In addition to one extra key blank for each lock, provide the following:
 - a. Cylinder Change Keys: Two.
 - b. Master Keys: Three.
 - c. Grand Master Keys: Three.
 - d. Great-Grand Master Keys: Three.

2.16 ACCESSORIES FOR PAIRS OF DOORS

- A. Coordinators: BHMA A156.3; consisting of active-leaf, hold-open lever and inactive-leaf release trigger; fabricated from steel with nylon-coated strike plates; with built-in, adjustable safety release.
- B. Astragals: BHMA A156.22.

2.17 SURFACE CLOSERS

- A. Surface Closers: BHMA A156.4; comply with manufacturer's written recommendations for size of door closers depending on size of door, exposure to weather, and anticipated frequency of use.
 - 1. Products: Subject to compliance with requirements, provide one of the following products:
 - a. LCN Closers; an Allegion plc company; 4040 Series.
 - b. SARGENT Manufacturing Company; an ASSA ABLOY Group company; 281 Series with no PRV.
- B. Description:
 - 1. Cast iron body, rack and pinion construction with compression spring, fully hydraulic.
 - 2. Closing speed, latching speed and backcheck controlled by independently operated concealed key valves.
 - 3. Intensity of backcheck feature to be adjustable.
 - 4. Equipped with spring adjustment allowing adjustment of spring power to suit individual door conditions.
 - 5. Suitable for mounting on 1-3/4" minimum top rail of door.
 - 6. Size as recommended by manufacturer for door size and weight.
 - 7. Provide mounting plates, sex nuts and bolts.
 - 8. No graphics allowed on cover.
 - 9. Provide parallel arms for exterior doors, hall doors, and outswinging interior doors.
 - 10. Provide stop arms for exterior door closers with parallel arms.
 - 11. For doors in areas accessible to physically handicapped persons, provide doors with adjustable opening force and delayed closing actions.
 - 12. Mount closers to doors with thru-bolts.
- C. Arm Finish: Painted, aluminum enamel.
- D. Closer Cover Finish: Sprayed enamel, color selected by Architect.

2.18 CONCEALED CLOSERS

- A. Concealed Closers: BHMA A156.4; rack-and-pinion hydraulic type with adjustable sweep and latch speeds controlled by key-operated valves. Comply with manufacturer's written recommendations for size of door closers depending on size of door, exposure to weather, and anticipated frequency of use. Provide factory-sized closers, adjustable to meet field conditions and requirements for opening force.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. DORMA Architectural Hardware; Member of The DORMA Group North America.

- b. LCN Closers; an Allegion plc company.
- c. Rixson Specialty Door Controls; an ASSA ABLOY Group company.
- d. SARGENT Manufacturing Company; an ASSA ABLOY Group company.

2.19 CLOSER HOLDER RELEASE DEVICES

- A. Closer Holder Release Devices: BHMA A156.15; Grade 1; closer connected with separate or integral releasing and fire- or smoke-detecting devices. Door shall become self-closing on interruption of signal to release device. Automatic release is activated by loss of power.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. LCN Closers; an Allegion plc company.
 - b. Rixson Specialty Door Controls; an ASSA ABLOY Group company.

2.20 MECHANICAL STOPS AND HOLDERS

- A. Wall- and Floor-Mounted Stops: BHMA A156.16.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Glynn-Johnson; (for overhead stops; surface applied and concealed application).
 - b. IVES Hardware; an Allegion plc company.
 - c. Rockwood Manufacturing Company.
 - d. Rixson; an ASSA ABLOY Group company.
 - e. Trimco.

- B. Wall- and Floor-Mounted Stop Types:

1. Wall Stops: Convex gray rubber bumper and brass, bronze or steel with concealed fasteners.
2. Floor Stops: Dome type with plated brass, bronze or steel stop.
3. Floor Stop and Holder: Two-piece floor mounted stop with door holder.

2.21 ELECTROMAGNETIC STOPS AND HOLDERS

- A. Electromagnetic Door Holders: BHMA A156.15, Grade 1; wall- or floor-mounted electromagnetic single unit with strike plate attached to swinging door; coordinated with fire detectors and interface with fire alarm system for labeled fire-rated door assemblies.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. LCN Closers; an Allegion plc company.

- b. Rixson Specialty Door Controls; an ASSA ABLOY Group company.

2.22 OVERHEAD STOPS AND HOLDERS

A. Overhead Stops and Holders: BHMA A156.8.

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Glynn-Johnson; an Allegion plc company.
 - b. Rockwood Manufacturing Company.
 - c. SARGENT Manufacturing Company; an ASSA ABLOY Group company.

2.23 DOOR GASKETING

A. Door Gasketing: BHMA A156.22; air leakage not to exceed 0.50 cfm per foot of crack length for gasketing other than for smoke control, as tested according to ASTM E 283; with resilient or flexible seal strips that are easily replaceable and readily available from stocks maintained by manufacturer.

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. National Guard Products.
 - b. Pemko Manufacturing Co.; an ASSA ABLOY Group company.
 - c. Reese Enterprises, Inc.
 - d. Zero International; an Allegion, plc. company.

B. Sound Control Gasketing:

- a. National Guard Products.
 - 1) Head and Jamb gasket: 107.
 - 2) Door Bottom: 112.
- b. Pemko Manufacturing Co.; an ASSA ABLOY Group company.
 - 1) Head and Jamb gasket: 379.
 - 2) Door Bottom: 379.
- c. Zero International; an Allegion, plc. Company.
 - 1) Head and Jamb gasket: 770.
 - 2) Door Bottom: 52

2.24 THRESHOLDS

- A. Thresholds: BHMA A156.21; fabricated to full width of opening indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. National Guard Products.
 - b. Pemko Manufacturing Co.; an ASSA ABLOY Group company.
 - c. Reese Enterprises, Inc.
 - d. Zero International; an Allegion, plc. company.
 - 2. Sound Control Thresholds:
 - a. National Guard Products, 411.
 - b. Pemko Manufacturing Co.; an ASSA ABLOY Group company; 151.

2.25 SLIDING DOOR HARDWARE

- A. Sliding Door Hardware: BHMA A156.14; consisting of complete sets including rails, hangers, supports, bumpers, floor guides, and accessories indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Cox, Arthur, & Sons, Inc.
 - b. Hager Companies.
 - c. Henderson, PC Inc.
 - d. Johnson, L. E., Products, Inc.
 - e. Stanley Commercial Hardware; Div. of The Stanley Works.

2.26 METAL PROTECTIVE TRIM UNITS

- A. Metal Protective Trim Units: BHMA A156.6; fabricated from 0.050-inch-thick stainless steel; with manufacturer's standard machine or self-tapping screw fasteners.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. IVES Hardware; an Ingersoll-Rand company.
 - b. Rockwood Manufacturing Company.
 - c. Trimco.

2.27 AUXILIARY DOOR HARDWARE

A. Auxiliary Hardware: BHMA A156.16.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Rockwood Manufacturing Company.
 - b. Stanley Commercial Hardware; Div. of The Stanley Works.
 - c. Trimco.

2.28 AUXILIARY ELECTRIFIED DOOR HARDWARE

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. GE Security, Inc.
2. Schlage Commercial Lock Division; an Allegion plc company.
3. Securitron Magnalock Corporation; an ASSA ABLOY Group company.

2.29 FABRICATION

A. Fasteners: Provide door hardware manufactured to comply with published templates prepared for machine, wood, and sheet metal screws. Provide screws that comply with commercially recognized industry standards for application intended, except aluminum fasteners are not permitted. Provide Phillips flat-head screws with finished heads to match surface of door hardware, unless otherwise indicated.

1. Concealed Fasteners: For door hardware units that are exposed when door is closed, except for units already specified with concealed fasteners. Do not use through bolts for installation where bolt head or nut on opposite face is exposed unless it is the only means of securely attaching the door hardware. Where through bolts are used on hollow door and frame construction, provide sleeves for each through bolt.
2. Fire-Rated Applications:
 - a. Wood or Machine Screws: For the following:
 - 1) Hinges mortised to doors or frames; use threaded-to-the-head wood screws for wood doors and frames.
 - 2) Strike plates to frames.
 - 3) Closers to doors and frames.
 - b. Steel Through Bolts: For the following unless door blocking is provided:
 - 1) Surface hinges to doors.
 - 2) Closers to doors and frames.
 - 3) Surface-mounted exit devices.
3. Spacers or Sex Bolts: For through bolting of hollow-metal doors.

4. Fasteners for Wood Doors: Comply with requirements in DHI WDHS.2, "Recommended Fasteners for Wood Doors."
5. Gasketing Fasteners: Provide noncorrosive fasteners for exterior applications and elsewhere as indicated.

2.30 FINISHES

- A. Provide finishes complying with BHMA A156.18 as indicated in door hardware schedule.
- B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Steel Doors and Frames: For surface applied door hardware, drill and tap doors and frames according to ANSI/SDI A250.6.
- B. Wood Doors: Comply with DHI WDHS.5 "Recommended Hardware Reinforcement Locations for Mineral Core Wood Flush Doors."
- C. Mounting Heights: Mount door hardware units at heights to comply with the following unless otherwise indicated or required to comply with governing regulations.
 1. Standard Steel Doors and Frames: ANSI/SDI A250.8.
 2. Custom Steel Doors and Frames: HMMA 831.
 3. Wood Doors: DHI WDHS.3, "Recommended Locations for Architectural Hardware for Wood Flush Doors."
- D. Install each door hardware item to comply with manufacturer's written instructions. Where cutting and fitting are required to install door hardware onto or into surfaces that are later to be painted or finished in another way, coordinate removal, storage, and reinstallation of surface protective trim units with finishing work. Do not install surface-mounted items until finishes have been completed on substrates involved.
 1. Set units level, plumb, and true to line and location. Adjust and reinforce attachment substrates as necessary for proper installation and operation.
 2. Drill and countersink units that are not factory prepared for anchorage fasteners. Space fasteners and anchors according to industry standards.
- E. Hinges: Install types and in quantities indicated in door hardware schedule but not fewer than the number recommended by manufacturer for application indicated or one hinge for every 30 inches of door height, whichever is more stringent, unless other equivalent means of support for door, such as spring hinges or pivots, are provided.

- F. Intermediate Offset Pivots: Where offset pivots are indicated, provide intermediate offset pivots in quantities indicated in door hardware schedule but not fewer than one intermediate offset pivot per door and one additional intermediate offset pivot for every 30 inches of door height greater than 90 inches.
- G. Lock Cylinders: Install construction cores to secure building and areas during construction period.
 - 1. Replace construction cores with permanent cores as directed by Owner.
 - 2. Furnish permanent cores to Owner for installation.
- H. Key Control System: Tag keys and place them on markers and hooks in key control system cabinet, as determined by final keying schedule.
- I. Boxed Power Supplies: Locate power supplies as indicated or, if not indicated, above accessible ceilings. Verify location with Architect.
 - 1. Configuration: Provide one power supply for each door opening with electrified door hardware.
- J. Thresholds: Set thresholds for exterior doors and other doors indicated in full bed of sealant complying with requirements specified in Section 07 92 00 "Joint Sealants."
- K. Stops: Provide floor stops for doors unless wall or other type stops are indicated in door hardware schedule. Do not mount floor stops where they will impede traffic.
- L. Perimeter Gasketing: Apply to head and jamb, forming seal between door and frame.
- M. Meeting Stile Gasketing: Fasten to meeting stiles, forming seal when doors are closed.
- N. Door Bottoms: Apply to bottom of door, forming seal with threshold when door is closed.
- O. Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

3.2 HARDWARE SETS

Abbreviation	Name
HES	Hes Inc
IVE	H.B. Ives
LCN	LCN Commercial Division
SCH	Schlage Lock Company

Legend:

-  Link to catalog cut sheet
-  Electrified Opening

HARDWARE SET: 01

EACH TO HAVE:

1	EA	ENTRANCE LOCK	ND53PD SPA		626	SCH
			BALANCE OF HARDWARE EXISTING TO REMAIN			

HARDWARE GROUP NO. 02

Provide each SGL door(s) with the following:

3	EA	HINGE	5BB1 4.5 X 4.5 NRP	652	IVE
1	EA	STOREROOM LOCK	ND80P6D RHO	626	SCH
1	EA	ELECTRIC STRIKE	8000C (FSE) X 2005M3 SMART PAC III	630	HES
1	EA	SURFACE CLOSER	4040XP RW/PA	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407CCV	630	IVE
1	EA	GASKETING	188SBK PSA	BK	ZER
1	EA	CREDENTIAL READER	BY SECURITY VENDOR	BLK	HID
1	EA	REQUEST TO EXIT	BY SECURITY VENDOR		
1	EA	DOOR POSITION SWITCH	BY SECURITY VENDOR		
1		LOW VOLTAGE POWER	BY SECURITY VENDOR		

Door normally closed and locked.

Present credential to card reader for entry or by key override.

Request to exit to shunt door position switch upon egress.

Free egress at all times.

Door to remain locked, fail secure during power failure.

HARDWARE SET: 03

EACH TO HAVE:

1	EA	PASSAGE SET	ND10S SPA		626	SCH
			BALANCE OF HARDWARE EXISTING TO REMAIN			

HARDWARE SET: 04

EACH TO HAVE:

3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	STOREROOM LOCK	ND80PD SPA		626	SCH
1	EA	ELECTRIC STRIKE	8000C X 2005M3	 ⚡	630	HES
1	EA	SURFACE CLOSER	4040XP		689	LCN
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	SET	SEALS	BY ALUMINUM FRAME MANUFACTURER			
1	EA	CARD READER	BY ACCESS CONTROL INTEGRATOR	⚡		
1	EA	RX MOTION SENSOR	BY ACCESS CONTROL INTEGRATOR	⚡		
1	EA	DOOR CONTACT	BY ACCESS CONTROL INTEGRATOR	⚡		
1	EA	POWER SUPPLY	BY ACCESS CONTROL INTEGRATOR	⚡		

DOOR NORMALLY CLOSED AND LOCKED.
ENTRY BY VALID CREDENTIAL AT CARD READER OR BY KEY AT LOCK.
RX MOTION SENSOR SHUNTS DOOR FORCED OPEN IN ACCESS CONTROL SYSTEM.
KEY OVER-RIDE WILL CAUSE DOOR FORCED ALARM IN ACCESS CONTROL SYSTEM.
FREE EGRESS AT ALL TIMES.

END OF SECTION

SECTION 088000

GLAZING

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Glass for interior doors, borrowed lites, and framed storefronts.
2. Glazing sealants and accessories.

1.2 COORDINATION

- A.** Coordinate glazing channel dimensions to provide necessary bite on glass, minimum edge and face clearances, and adequate sealant thicknesses, with reasonable tolerances.

1.3 ACTION SUBMITTALS

- A.** Product Data: For each type of product.
- B.** Glass Samples: For each type of glass product other than clear monolithic vision glass; 12 inches square.
- C.** Glazing Schedule: List glass types and thicknesses for each size opening and location. Use same designations indicated on Drawings.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A.** Basis-of-Design Glass Product: Subject to compliance with requirements, provide product indicated on Drawings or comparable product by one of the following:
1. AGC Glass Company North America, Inc.
 2. Berkowitz, JE, LP.
 3. Cardinal Glass Industries.
 4. Cristacurva Glass.
 5. Guardian Industries Corp.
 6. Oldcastle BuildingEnvelope.
 7. Pilkington North America Inc.
 8. PPG Industries, Inc.
 9. Saint-Gobain Corporation.

10. Schott North America, Inc.
11. Trulite Glass & Aluminum Solutions.
12. Viracon, Inc.
13. Vitro Architectural Glass.

2.2 PERFORMANCE REQUIREMENTS

- A. Structural Performance: Glass installed adjacent to walking surfaces shall withstand the following design loads within limits and under conditions indicated:
 1. Differential deflection of adjacent unsupported edges shall not exceed glass thickness when subjected to 50 lbf/ft. (730 N/m) applied horizontally to one panel at any point up to 42 inches (1067 mm) above the adjacent walking surface.
 2. Base design on thickness at thinnest part of the glass.
- B. Safety Glazing: Where safety glazing is indicated, provide glazing that complies with 16 CFR 1201, Category II.

2.3 GLASS PRODUCTS, GENERAL

- A. Glazing Publications: Comply with published recommendations of glass product manufacturers and organizations below unless more stringent requirements are indicated. See these publications for glazing terms not otherwise defined in this Section or in referenced standards.
 1. GANA Publications: "Laminated Glazing Reference Manual" and "Glazing Manual."
- B. Safety Glazing Labeling: Where safety glazing is indicated, permanently mark glazing with certification label of the SGCC or another certification agency acceptable to authorities having jurisdiction. Label shall indicate manufacturer's name, type of glass, thickness, and safety glazing standard with which glass complies.
- C. Thickness: Where glass thickness is indicated, it is a minimum. Provide glass that complies with performance requirements and is not less than the thickness indicated.
 1. Minimum Glass Thickness: 1/4 inch (6 mm).
- D. Strength: Where annealed float glass is indicated, provide annealed float glass or fully tempered float glass. Where fully tempered float glass is indicated, provide fully tempered float glass.

2.4 GLASS PRODUCTS

- A. Clear Annealed Float Glass: ASTM C 1036, Type I, Class 1 (clear), Quality-Q3.

- B. Ultraclear Float Glass: ASTM C 1036, Type I, Class I (clear), Quality-Q3; and with visible light transmission of not less than 91 percent.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. AGC Glass Company North America, Inc.; Krystal Klear.
 - b. Guardian Industries Corp.; UltraWhite.
 - c. Pilkington North America; Optiwhite.
 - d. PPG Industries, Inc.; Starphire.
 - e. Saint-Gobain Corporation; Diamant.
- C. Fully Tempered Float Glass: ASTM C 1048, Kind FT (fully tempered), Condition A (uncoated) unless otherwise indicated, Type I, Class 1 (clear) or Class 2 (tinted) as indicated, Quality-Q3.
 - 1. Fabrication Process: By horizontal (roller-hearth) process with roll-wave distortion parallel to bottom edge of glass as installed unless otherwise indicated.

2.5 LAMINATED GLASS

- A. Laminated Glass: ASTM C 1172. Use materials that have a proven record of no tendency to bubble, discolor, or lose physical and mechanical properties after fabrication and installation.
 - 1. Construction: Laminate glass with polyvinyl butyral interlayer to comply with interlayer manufacturer's written instructions.
 - 2. Interlayer Thickness: Provide thickness not less than that indicated and as needed to comply with requirements.
 - 3. Interlayer Color: Clear unless otherwise indicated.

2.6 GLAZING SEALANTS

- A. General:
 - 1. Compatibility: Compatible with one another and with other materials they contact, including glass products, seals of insulating-glass units, and glazing channel substrates, under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.
 - 2. Suitability: Comply with sealant and glass manufacturers' written instructions for selecting glazing sealants suitable for applications indicated and for conditions existing at time of installation.
 - 3. Colors of Exposed Glazing Sealants: As indicated on Drawings or as selected by Architect from manufacturer's full range.

- B. Glazing Sealant: Neutral-curing silicone glazing sealant complying with ASTM C 920, Type S, Grade NS, Class 100/50, Use NT.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Dow Corning Corporation; 790.
 - b. GE Advanced Materials - Silicones; SilPruf LM SCS2700.
 - c. Pecora Corporation; 890NST.
 - d. Sika Corporation U.S.; Sikasil WS-290.
 - e. Tremco Incorporated; Spectrem 1.
- C. Glazing Sealant: Neutral-curing silicone glazing sealant complying with ASTM C 920, Type S, Grade NS, Class 50, Use NT.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Dow Corning Corporation; 791.
 - b. GE Advanced Materials - Silicones; SilGlaze II SCS2800.
 - c. Pecora Corporation; 896.
 - d. Sika Corporation U.S.; Sikasil WS-295.
 - e. Tremco Incorporated; Spectrem 2.

2.7 GLAZING TAPES

- A. Back-Bedding Mastic Glazing Tapes: Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with ASTM C 1281 and AAMA 800 for products indicated below:
 - 1. AAMA 804.3 tape, where indicated.
 - 2. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
 - 3. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.
- B. Expanded Cellular Glazing Tapes: Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800 for the following types:
 - 1. AAMA 810.1, Type 1, for glazing applications in which tape acts as the primary sealant.
 - 2. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

2.8 MISCELLANEOUS GLAZING MATERIALS

- A. Cleaners, Primers, and Sealers: Types recommended by sealant or gasket manufacturer.

- B. Setting Blocks: Elastomeric material with a Shore, Type A durometer hardness of 85, plus or minus 5.
- C. Spacers: Elastomeric blocks or continuous extrusions of hardness required by glass manufacturer to maintain glass lites in place for installation indicated.
- D. Edge Blocks: Elastomeric material of hardness needed to limit glass lateral movement (side walking).
- E. Cylindrical Glazing Sealant Backing: ASTM C 1330, Type O (open-cell material), of size and density to control glazing sealant depth and otherwise produce optimum glazing sealant performance.

PART 3 - EXECUTION

3.1 GLAZING, GENERAL

- A. Comply with combined written instructions of manufacturers of glass, sealants, gaskets, and other glazing materials, unless more stringent requirements are indicated, including those in referenced glazing publications.
- B. Protect glass edges from damage during handling and installation. Remove damaged glass from Project site and legally dispose of off Project site. Damaged glass includes glass with edge damage or other imperfections that, when installed, could weaken glass, impair performance, or impair appearance.
- C. Apply primers to joint surfaces where required for adhesion of sealants, as determined by preconstruction testing.
- D. Install setting blocks in sill rabbets, sized and located to comply with referenced glazing publications, unless otherwise required by glass manufacturer. Set blocks in thin course of compatible sealant suitable for heel bead.
- E. Do not exceed edge pressures stipulated by glass manufacturers for installing glass lites.
- F. Provide spacers for glass lites where length plus width is larger than 50 inches.
- G. Provide edge blocking where indicated or needed to prevent glass lites from moving sideways in glazing channel, as recommended in writing by glass manufacturer and according to requirements in referenced glazing publications.
- H. Safety Glazing:
 - 1. Install tempered glass in the following locations:
 - a. Doors.
 - b. Operable or fixed lites adjacent to and within the same wall plane as a door whose nearest vertical edge is within 12 inches of the door in closed

- position and whose bottom edge is less than 60 inches above the floor or walking surface.
- c. Fixed lites which have a glazed area in excess of 9 sq. ft. and lowest edge is less than 18 inches above finished floor or walking surface within 36 inches of such glazing where panels are not protected with a horizontal member not less than 1-1/2 inches wide located between 24 and 36 inches above the floor or walking surface.
 - d. Other locations as required by Building Code.

3.2 TAPE GLAZING

- A. Position tapes on fixed stops so that, when compressed by glass, their exposed edges are flush with or protrude slightly above sightline of stops.
- B. Install tapes continuously, but not necessarily in one continuous length. Do not stretch tapes to make them fit opening.
- C. Cover vertical framing joints by applying tapes to heads and sills first, then to jambs. Cover horizontal framing joints by applying tapes to jambs, then to heads and sills.
- D. Place joints in tapes at corners of opening with adjoining lengths butted together, not lapped. Seal joints in tapes with compatible sealant approved by tape manufacturer.
- E. Center glass lites in openings on setting blocks and press firmly against tape by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings.
- F. Apply cap bead of elastomeric sealant over exposed edge of tape.

3.3 GASKET GLAZING (DRY)

- A. Cut compression gaskets to lengths recommended by gasket manufacturer to fit openings exactly, with allowance for stretch during installation.
- B. Insert soft compression gasket between glass and frame or fixed stop so it is securely in place with joints miter cut and bonded together at corners.
- C. Installation with Drive-in Wedge Gaskets: Center glass lites in openings on setting blocks and press firmly against soft compression gasket by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings. Compress gaskets to produce a weathertight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended by gasket manufacturer.
- D. Installation with Pressure-Glazing Stops: Center glass lites in openings on setting blocks and press firmly against soft compression gasket. Install dense compression gaskets and pressure-glazing stops, applying pressure uniformly to compression gaskets. Compress gaskets to produce a weathertight seal without developing bending

stresses in glass. Seal gasket joints with sealant recommended by gasket manufacturer.

- E. Install gaskets so they protrude past face of glazing stops.

3.4 SEALANT GLAZING (WET)

- A. Install continuous spacers, or spacers combined with cylindrical sealant backing, between glass lites and glazing stops to maintain glass face clearances and to prevent sealant from extruding into glass channel and blocking weep systems until sealants cure. Secure spacers or spacers and backings in place and in position to control depth of installed sealant relative to edge clearance for optimum sealant performance.
- B. Force sealants into glazing channels to eliminate voids and to ensure complete wetting or bond of sealant to glass and channel surfaces.
- C. Tool exposed surfaces of sealants to provide a substantial wash away from glass.

3.5 CLEANING AND PROTECTION

- A. Immediately after installation remove nonpermanent labels and clean surfaces.
- B. Protect glass from contact with contaminating substances resulting from construction operations. Examine glass surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for buildup of dirt, scum, alkaline deposits, or stains.
 - 1. If, despite such protection, contaminating substances do come into contact with glass, remove substances immediately as recommended in writing by glass manufacturer. Remove and replace glass that cannot be cleaned without damage to coatings.
- C. Remove and replace glass that is damaged during construction period.

3.6 MONOLITHIC GLASS SCHEDULE

- A. Clear Glass: Clear annealed or fully tempered float glass.
 - 1. Minimum Thickness: As required by performance requirements, but not less than 6 mm.
 - 2. Provide safety glazing as required.
- B. Ultraclear Glass: Ultraclear annealed or fully tempered float glass.
 - 1. Minimum Thickness: As required by performance requirements, but not less than 6 mm.
 - 2. Provide safety glazing as required.

3.7 LAMINATED GLASS SCHEDULE

- A. Clear Laminated Glass: Clear laminated glass with two plies of annealed or fully tempered float glass.
 - 1. Minimum Thickness of Each Glass Ply: As required to meet performance requirements, but not less than 3 mm.
 - 2. Interlayer Thickness: Not less than 0.030 inch.

END OF SECTION

SECTION 088300

MIRRORS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes the following types of silvered flat glass mirrors:

1. Annealed monolithic glass mirrors.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings: Include mirror elevations, edge details, mirror hardware, and attachment details.

1.3 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For mirrors to include in maintenance manuals.

1.4 WARRANTY

- A. Special Warranty: Manufacturer agrees to replace mirrors that deteriorate within specified warranty period. Deterioration of mirrors is defined as defects developed from normal use that are not attributed to mirror breakage or to maintaining and cleaning mirrors contrary to manufacturer's written instructions. Defects include discoloration, black spots, and clouding of the silver film.
1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. Avalon Glass and Mirror Company.
 2. Binswanger Glass.
 3. Glasswerks LA, Inc.

4. Guardian Industries Corp.
5. Independent Mirror Industries, Inc.
6. National Glass Industries.
7. Trulite Glass & Aluminum Solutions.
8. Virginia Mirror Company, Inc.
9. Walker Glass Co., Ltd.

2.2 SILVERED FLAT GLASS MIRRORS

- A. Mirrors, General: ASTM C 1503.
- B. Annealed Monolithic Glass Mirrors: Mirror Glazing Quality, ultraclear (low-iron) float glass with a minimum 91 percent visible light transmission.
 1. Nominal Thickness: 6.0 mm.

2.3 MISCELLANEOUS MATERIALS

- A. Setting Blocks: Elastomeric material with a Shore, Type A durometer hardness of 85, plus or minus 5.
- B. Edge Sealer: Coating approved by mirror manufacturer.
- C. Mirror Mastic: An adhesive setting compound, asbestos-free, produced specifically for setting mirrors.
 1. Verify adhesives have a VOC content of 70 g/L or less.

2.4 MIRROR HARDWARE

- A. Aluminum J-Channels: As indicated on Drawings.
 1. Bottom and Side Trim: J-channels formed with front leg and back leg not less than 3/8 and 7/8 inch in height, respectively, and a thickness of not less than 0.04 inch.
 2. Top Trim: J-channels formed with front leg and back leg not less than 5/8 and 1 inch in height, respectively, and a thickness of not less than 0.04 inch.
 3. Finish: Clear bright anodized.
- B. Mirror Bottom Clips: As indicated on Drawings.
- C. Mirror Top Clips: As indicated on Drawings.
- D. Fasteners: Fabricated of same basic metal and alloy as fastened metal and matching it in finished color and texture where fasteners are exposed.

2.5 FABRICATION

- A. Fabricate cutouts for notches and holes in mirrors without marring visible surfaces. Locate and size cutouts so they fit closely around penetrations in mirrors.
- B. Mirror Edge Treatment: Flat polished. Seal edges of mirrors with edge sealer.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, over which mirrors are to be mounted, with Installer present, for compliance with installation tolerances, substrate preparation, and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected and surfaces are dry.

3.2 PREPARATION

- A. Comply with mastic manufacturer's written installation instructions for preparation of substrates, including coating substrates with mastic manufacturer's special bond coating where applicable.

3.3 INSTALLATION

- A. General: Install mirrors to comply with mirror manufacturer's written instructions and with referenced GANA publications. Mount mirrors accurately in place in a manner that avoids distorting reflected images.
- B. Install mirrors with mastic and mirror hardware. Attach mirror hardware securely to mounting surfaces with mechanical fasteners installed with anchors or inserts as applicable. Install fasteners so heads do not impose point loads on backs of mirrors.
 - 1. Apply mastic to comply with mastic manufacturer's written instructions for coverage and to allow air circulation between back of mirrors and face of mounting surface.
- C. Clean exposed surface of mirrors not more than four days before date scheduled for inspections that establish date of Substantial Completion. Clean mirrors as recommended in writing by mirror manufacturer.

END OF SECTION

SECTION 092116
GYPSUM BOARD ASSEMBLIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Non-load-bearing steel framing systems for interior partitions.
2. Suspension systems for interior ceilings and soffits.
3. Gypsum board.
4. Tile backing panels.
5. Acoustical accessories.
6. Trim accessories.
7. Joint treatment materials.

1.2 ACTION SUBMITTALS

A. Product Data: For the following:

1. Non-load-bearing steel framing systems for interior partitions.
2. Suspension systems for interior ceilings and soffits.
3. Grid suspension systems for gypsum board ceilings.
4. Gypsum board products.
5. Acoustical accessories.
6. Trim accessories.
7. Joint treatment materials.

B. Shop Drawings:

1. When studs must be spliced to obtain vertical span required, provide engineered drawing detail, signed and sealed by the responsible design professional, indicating total height of wall, splice length, and type and frequency of fasteners.

C. Samples: For the following products:

1. Trim Accessories: Full-size Sample in 12-inch-long length for each trim accessory indicated.

1.3 INFORMATIONAL SUBMITTALS

A. Product Certificates: For each type of code-compliance certification for studs and tracks.

- B. Evaluation Reports: For embossed, high-strength steel studs and firestop tracks from ICC-ES or other qualified testing agency acceptable to authorities having jurisdiction.

1.4 QUALITY ASSURANCE

- A. Mockups: Build mockups of at least 100 sq. ft. in surface area to demonstrate aesthetic effects and to set quality standards for materials and execution.
 - 1. Build mockups for each level of gypsum board finish indicated for use in exposed locations.
 - 2. Apply or install final decoration indicated, including painting and wallcoverings, on exposed surfaces for review of mockups.
 - 3. Simulate finished lighting conditions for review of mockups.
 - 4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Store materials inside under cover and keep them dry and protected against weather, condensation, direct sunlight, construction traffic, and other potential causes of damage. Stack panels flat and supported on risers on a flat platform to prevent sagging.

1.6 FIELD CONDITIONS

- A. Environmental Limitations: Comply with ASTM C840 requirements or gypsum board manufacturer's written instructions, whichever are more stringent.
- B. Do not install paper-faced gypsum panels until installation areas are enclosed and conditioned.
- C. Do not install panels that are wet, moisture damaged, and mold damaged.
 - 1. Indications that panels are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
 - 2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Fire-Resistance-Rated Assemblies: For fire-resistance-rated assemblies, provide materials identical to those tested in UL design assemblies indicated on Drawings according to ASTM E119 by an independent testing agency.

- B. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E90 and classified according to ASTM E413 by an independent testing agency.
- C. Horizontal Deflection: For composite wall assemblies, limited to 1/240 of the wall height based on horizontal loading of 5 lbf/sq. ft.

2.2 FRAMING SYSTEMS

- A. Framing Members, General: Comply with ASTM C754 for conditions indicated.
 - 1. Steel Sheet Components: Comply with ASTM C645 requirements for steel unless otherwise indicated.
 - 2. Protective Coating: Coating with equivalent corrosion resistance of ASTM A653/A653M, G40, hot-dip galvanized unless otherwise indicated.
- B. Studs and Tracks: ASTM C645. Use either conventional steel studs and tracks or embossed, high-strength steel studs and tracks.
 - 1. Steel Studs and Tracks:
 - a. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:
 - 1) CEMCO; California Expanded Metal Products Co.
 - 2) ClarkDietrich.
 - 3) Custom Stud.
 - 4) MarinoWARE.
 - 5) MBA Building Supplies.
 - 6) Phillips Manufacturing Co.
 - 7) SCAFCO Steel Stud Company.
 - 8) Steel Construction Systems.
 - 9) Telling Industries.
 - 10) The Steel Network, Inc.
 - b. Minimum Base-Steel Thickness: As indicated on Drawings, in correlation with vertical unbraced span and horizontal deflection performance requirements.
 - c. Depth: As indicated on Drawings
 - d. Spacing: As indicated on Drawings.
 - e. Application: Permitted for use in all applications except STC-rated assemblies.

2. Embossed, High Strength Steel Studs and Tracks: Roll-formed and embossed with surface deformations to stiffen the framing members so that they are structurally comparable to conventional ASTM C645 steel studs and tracks.
 - a. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:
 - 1) CEMCO; California Expanded Metal Products Co.
 - 2) ClarkDietrich.
 - 3) MarinoWARE.
 - 4) MBA Building Supplies.
 - 5) Phillips Manufacturing Co.
 - 6) SCAFCO Steel Stud Company.
 - 7) Steel Construction Systems.
 - 8) Telling Industries.
 - 9) The Steel Network, Inc.
 - b. Minimum Base-Steel Thickness: As indicated on Drawings, in correlation with vertical unbraced span and horizontal deflection performance requirements.
 - c. Depth: As indicated on Drawings.
 - d. Spacing:
 - 1) 24-inches on center unless otherwise indicated.
 - e. Application:
 - 1) STC-rated assemblies.
 - 2) Permitted for all other applications except partitions scheduled to receive ceramic, glass, or stone tile.
3. High-Performing Acoustical Partition Wall Studs: Factory-assembled 3-5/8 inches (92 mm), 4 inches (101.6 mm), or 6 inches (152.4 mm) sound isolating double stud using closed cell foam isolators creating sound dampening air gap. Use standard track.
 - a. Basis-of-Design Product: Subject to compliance with requirements, provide SoundGuard Stud by one of the following:
 - 1) MarinoWARE
 - 2) SCAFCO
 - b. Minimum Base-Metal Thickness: As required by horizontal deflection performance requirements, but not less than 0.019 inch (0.483 mm), (formerly 25 gage).
 - c. Depth: As indicated on Drawings.

- C. Slip-Type Head Joints: Provide one or more of the following as required to accommodate anticipated floor or roof structure deflection:
1. Single Long-Leg Track System: ASTM C645 top track with 4 inches deep flanges in thickness not less than indicated for studs, installed with studs friction fit into top track and with continuous bridging located within 12 inches of the top of studs to provide lateral bracing.
 2. Double-Track System: ASTM C645 top outer tracks, inside track with 2-inch-deep flanges in thickness not less than indicated for studs and fastened to studs, and outer track sized to friction-fit over inner track.
 3. Deflection Track: Steel sheet top track manufactured to prevent cracking of finishes applied to interior partition framing resulting from deflection of structure above and allowing 1-1/2 inches minimum vertical movement; in thickness not less than indicated for studs and in width to accommodate depth of studs.
 - a. Products: Subject to compliance with requirements, provide one of the following:
 - 1) CEMCO; California Expanded Metal Products Co.; CST Slotted Deflection Track.
 - 2) ClarkDietrich; MaxTrak Slotted Deflection Track.
 - 3) MarinoWARE; Slotted Track.
 - 4) Perfect Wall, Inc.; The System Slotted Deflection Track.
 - 5) SCAFCO Steel Stud Company; SDLT-Slotted Leg Track System.
 - 6) Steel Construction Systems; SDLT-Slotted Leg Track System.
 - 7) Telling Industries; Interior Slotted Track.
- D. Firestop Tracks: Provide one or more of the following as required to accommodate anticipated floor or roof structure deflection:
1. Top track manufactured to allow partition heads to expand and contract with movement of structure while maintaining continuity of fire-resistance-rated assembly indicated and allowing up to 2 inches vertical movement; in thickness not less than indicated for studs and in width to accommodate depth of studs.
 - a. Products: Subject to compliance with requirements, provide one of the following:
 - 1) CEMCO; California Expanded Metal Products Co.; FAS Track.
 - 2) MarinoWARE; FAS Track 1000.
 - 3) Perfect Wall, Inc.; The System Slotted Deflection Track.
 - 4) SCAFCO Steel Stud Company; SCAFCO Slotted Leg Track System.
 - 5) Steel Construction Systems; Steel-Con Slotted Leg Track System.
- E. Acoustical Tracks: For STC-rated assemblies, provide the following as required to accommodate anticipated floor or roof structure deflection:
1. Top track manufactured with a 5/8-inch or 1-1/4" offset shoulder to allow partition heads to expand and contract with movement of structure while maintaining continuity of STC-rated assembly indicated and allowing up to 4 inches vertical

movement; in thickness not less than indicated for studs and in width to accommodate depth of studs.

- a. Product: Subject to compliance with requirements, provide one of the following:
 - 1) ClarkDietrich; BlazeFrame RipTRAK attached to studs with RTC-33-RipTRAK Clips.
 - 2) Fire Trak Corp; Fire Trak Shadowline attached to studs with Fire Trak PosiKlip.
- F. Flat Strap and Backing Plate: Steel sheet for blocking and bracing in length and width indicated.
 - 1. Minimum Base-Steel Thickness: 0.0538-inch.
- G. Cold-Rolled Channel Bridging: Steel, 0.0538-inch minimum base-steel thickness, with minimum 1/2-inch-wide flanges.
 - 1. Depth: 1-1/2 inches.
 - 2. Clip Angle: Not less than 1-1/2 by 1-1/2 inches, 0.068-inch-thick, galvanized steel.
- H. Hat-Shaped, Rigid Furring Channels: ASTM C645.
 - 1. Minimum Base-Steel Thickness: 0.0296 inch.
 - 2. Depth: 7/8 inch.
- I. Cold-Rolled Furring Channels: 0.053-inch uncoated-steel thickness, with minimum 1/2-inch-wide flanges.
 - 1. Depth: 3/4 inch.
 - 2. Furring Brackets: Adjustable, corrugated-edge-type steel sheet with minimum uncoated-steel thickness of 0.0329 inch.
 - 3. Tie Wire: ASTM A641/A641M, Class 1 zinc coating, soft temper, 0.062-inch-diameter wire, or double strand of 0.048-inch-diameter wire.
- J. Z-Shaped Furring: With slotted or nonslotted web, face flange of 1-1/4 inches, wall attachment flange of 7/8 inch, minimum uncoated-steel thickness of 0.0179 inch, and depth required to fit insulation thickness indicated.

2.3 SUSPENSION SYSTEMS

- A. Tie Wire: ASTM A641/A641M, Class 1 zinc coating, soft temper, 0.062-inch-diameter wire, or double strand of 0.048-inch-diameter wire.

B. Hanger Attachments to Concrete:

1. Post-Installed Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01 AC193 AC58 or AC308 as appropriate for the substrate.
 - a. Material: Carbon-steel components zinc-plated to comply with ASTM B633 or ASTM F1941, Class Fe/Zn 5, unless otherwise indicated.
2. Power-Actuated Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.

C. Wire Hangers: ASTM A641/A641M, Class 1 zinc coating, soft temper, 0.16 inch in diameter.

D. Carrying Channels (Main Runners): Cold-rolled, commercial-steel sheet with a base-steel thickness of 0.0538 inch and minimum 1/2-inch-wide flanges.

1. Depth: 2 inches.

E. Furring Channels (Furring Members):

1. Cold-Rolled Channels: 0.0538-inch uncoated-steel thickness, with minimum 1/2-inch-wide flanges, 3/4 inch deep.
2. Steel Studs and Tracks: ASTM C645.
 - a. Minimum Base-Steel Thickness: 0.0296 inch.
 - b. Depth: As indicated on Drawings.
3. Embossed, High-Strength Steel Studs and Tracks: ASTM C645.
 - a. Minimum Base-Steel Thickness: 0.0190 inch.
 - b. Depth: As indicated on Drawings.
4. Hat-Shaped, Rigid Furring Channels: ASTM C645, 7/8 inch deep.
 - a. Minimum Base-Steel Thickness: 0.0296 inch.

F. Grid Suspension System for Gypsum Board Ceilings: ASTM C645, direct-hung system composed of main beams and cross-furring members that interlock.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Armstrong Ceiling & Wall Solutions; Drywall Grid Systems.
 - b. Rockfon (Rockwool International); 640/660 Drywall Ceiling Suspension.
 - c. USG Corporation; Drywall Suspension System.

2.4 GYPSUM BOARD, GENERAL

- A. Size: Provide maximum lengths and widths available that will minimize joints in each area and that correspond with support system indicated.

2.5 INTERIOR GYPSUM BOARD

A. Gypsum Board: ASTM C1396/C1396M.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. American Gypsum.
 - b. CertainTeed Gypsum.
 - c. Georgia-Pacific Gypsum LLC.
 - d. National Gypsum Company.
 - e. PABCO Gypsum.
 - f. USG Corporation.
2. Core: 5/8-inch, Type X, unless otherwise indicated.
3. Long Edges: Tapered.

B. Flexible Gypsum Board: ASTM C1396/C1396M. Manufactured to bend to fit radii and to be more flexible than standard regular-type gypsum board of same thickness.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. CertainTeed Gypsum.
 - b. Georgia-Pacific Gypsum LLC.
 - c. National Gypsum Company.
 - d. PABCO Gypsum.
 - e. USG Corporation.
2. Thickness: 1/4 inch.
3. Long Edges: Tapered.

C. Mold-Resistant Gypsum Board: ASTM C1396/C1396M. With moisture- and mold-resistant core and paper surfaces.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. American Gypsum; M-Bloc® Type X with Mold & Moisture Resistance.
 - b. CertainTeed Gypsum; CertainTeed M2Tech Mold and Moisture Resistant Type X Gypsum Board.
 - c. National Gypsum Company; Gold Bond® XP® Fire-Shield® Gypsum Board.
 - d. PABCO Gypsum; MOLD CURB® Plus.
 - e. USG Corporation; USG Sheetrock® Brand Mold Tough® Firecode® X Panels.
2. Core: 5/8-inch, Type X.
3. Long Edges: Tapered.
4. Mold Resistance: ASTM D3273, score of 10 as rated according to ASTM D3274.

- D. Acoustically Enhanced Gypsum Board: ASTM C1396/C1396M. Multilayer products constructed of two layers of gypsum boards sandwiching a viscoelastic sound-absorbing polymer core.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. CertainTeed Gypsum; CertainTeed SilentFX QuickCut Type X Acoustical Gypsum Board.
 - b. National Gypsum Company; Gold Bond® SoundBreak® XP® Fire Shield® Gypsum Board.
 - c. PABCO Gypsum; QuietRock® ES.
 2. Core: 5/8 inch (15.9 mm), Type X.
 3. Long Edges: Tapered.

2.6 TILE BACKING PANELS

- A. Glass-Mat, Water-Resistant Backing Board: ASTM C1178/C1178M, with manufacturer's standard edges.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. CertainTeed Corporation; GlasRoc Tile Backer.
 - b. Georgia-Pacific Gypsum LLC; DensShield Tile Backer.
 - c. National Gypsum Company; Gold Bond® eXP® Tile Backer.
 - d. USG Corporation; USG Durock™ Glass-Mat Tile Backerboard.
 2. Core: 5/8 inch, Type X.
 3. Mold Resistance: ASTM D3273, score of 10 as rated according to ASTM D3274.
- B. Cementitious Backer Units: ANSI A118.9 and ASTM C1288 or ASTM C1325, with manufacturer's standard edges.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. FinPan, Inc; Util-A-Crete Concrete Backer Board.
 - b. National Gypsum Company; PermaBase® Cement Board.
 - c. USG Corporation; DUROCK Cement Board.
 2. Thickness: 5/8 inch.
 3. Mold Resistance: ASTM D3273, score of 10 as rated according to ASTM D3274.

2.7 ACOUSTICAL ACCESSORIES

- A. Sound-Attenuation Blankets: ASTM C665, Type I (blankets without membrane facing) produced by combining thermosetting resins with mineral fibers manufactured from glass, slag wool, or rock wool.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. CertainTeed Corporation; NoiseReducer™.
 - b. Johns Manville; a Berkshire Hathaway company; Formaldehyde-free™ Unfaced thermal and acoustical batts.
 - c. Knauf Insulation; EcoBatt® Insulation.
 - d. Owens Corning; EcoTouch® Sonobatts Insulation.
 2. Flame-Spread Index: Not more than 25 when tested in accordance with ASTM E84.
 3. Smoke-Developed Index: Not more than 50 when tested in accordance with ASTM E84.
- B. Electrical J-Box Putty Pads: Preformed, moldable putty pads formulated to maintain the performance of acoustically rated wall assemblies by sealing penetrations including common electrical outlets boxes, phone outlet boxes, electrical switches, HVAC ducts, and plumbing connections.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. ATS Acoustics; ATS Acoustics Putty Pads.
 - b. The Soundproofing Company Inc.; Acoustical Putty Pads.
 - c. QuietRock; QuietPutty.
- C. Acoustical Sealant for Exposed and Concealed Joints: Manufacturer's standard nonsag, paintable, nonstaining latex acoustical sealant complying with ASTM C834.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. GE Construction Sealants; Momentive Performance Materials Inc.; RCS20 Acoustical.
 - b. Hilti, Inc.; CP 506 Smoke and Acoustical Sealant.
 - c. QuietRock; QuietSeal Pro.
 - d. Specified Technologies, Inc.; SpecSeal Smoke 'N' Sound Sealant.
 - e. Tremco Incorporated; Tremco Acoustical Sealant.
 - f. USG Corporation; SHEETROCK Acoustical Sealant.
 2. Colors of Exposed Acoustical Joint Sealants: White.

- D. Resilient Sound Isolation Clips: Isolation clips consisting of a steel and rubber component, designed to receive standard 7/8-inch, hat-shaped, rigid furring channels, and isolate framing members from metal-to-metal or other rigid contact.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Kinetics Noise Control; IsoMax.
 - b. PAC International, LLC; RSIC-1.
 - c. Pliteq, Inc.; GenieClip RST.

2.8 TRIM ACCESSORIES

- A. Interior Trim: ASTM C1047, galvanized or aluminum-coated steel sheet, rolled zinc, or paper-faced galvanized-steel sheet.

1. Edge Shapes:
 - a. Cornerbead.
 - b. Bullnose bead.
 - c. LC-Bead: J-shaped; exposed long flange receives joint compound.
 - d. L-Bead: L-shaped; exposed long flange receives joint compound.
 - e. U-Bead: J-shaped; exposed short flange does not receive joint compound.
 - f. Curved-Edge Cornerbead: With notched or flexible flanges.
2. Expansion (Control) Joints.
3. Fire-Rated Expansion (Control) Joints: Composite control joint with intumescent tape factory applied to back of control joint on one side.
 - a. Products: Subject to compliance with requirements, provide the following:
 - 1) ClarkDietrich; FAS-093X Fire Rated Control Joint.

- B. Base-of-Wall PVC Moisture Barrier Trim: Extruded PVC, 1/2 inch high.

1. Products: Subject to compliance with requirements, provide the following:
 - a. Waterguard; Waterguard.

- C. Extruded Aluminum Trim Accessories: Aluminum: Alloy and temper with not less than the strength and durability properties of ASTM B221, Alloy 6063-T5, of profiles and dimensions indicated.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Fry Reglet Corp.
 - b. Gordon, Inc.
 - c. Pittcon Industries.
2. Finish and Profile: As indicated on Drawings.

- D. Fire-Rated Reveal Backer: 0.0359 inch thick, ASTM A653/A653M, hot-dip galvanized, ASTM C645 flat steel strap backer plate with an affixed cured intumescent strip to maintain fire ratings behind architectural reveal moldings in fire-rated partitions.
 - 1. Products: Subject to compliance with requirements, provide the following:
 - a. ClarkDietrich; BlazeFrame® "FSB" (Flat Strap Backer).

2.9 AUXILIARY MATERIALS

- A. Provide auxiliary materials that comply with referenced installation standards and manufacturer's written instructions.
- B. Floor Track Seal: Provide the following where indicated:
 - 1. Foam Gasket: Adhesive-backed, closed-cell vinyl foam strips that allow fastener penetration without foam displacement, 1/8 inch thick, in width to suit steel stud size.
- C. Fasteners for Steel Framing: Of type, material, size, corrosion resistance, holding power, and other properties required to fasten steel members to substrates.
- D. Steel Drill Screws for Gypsum Board: ASTM C1002 unless otherwise indicated.
 - 1. Use screws complying with ASTM C954 for fastening panels to steel members from 0.033 to 0.112 inch thick.
 - 2. For fastening cementitious backer units, use screws of type and size recommended by panel manufacturer.
- E. Laminating Adhesive: Adhesive or joint compound recommended for directly adhering gypsum panels to continuous substrate.

2.10 JOINT TREATMENT MATERIALS

- A. General: Comply with ASTM C475/C475M.
- B. Joint Tape:
 - 1. Interior Gypsum Board: Paper.
 - 2. Tile Backing Panels: As recommended by panel manufacturer.

- C. Joint Compound for Interior Gypsum Board: For each coat, use formulation that is compatible with other compounds applied on previous or for successive coats.
1. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use setting-type taping compound.
 - a. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to the following:
 - 1) National Gypsum Company; Proform Quick Set 210 Setting Compound.
 - 2) USG Corporation; Sheetrock Durabond 210 Setting-Type Joint Compound.
 2. Fill Coat: For second coat, use setting-type, sandable topping compound.
 - a. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to the following:
 - 1) National Gypsum Company; Proform Quick Set 90 Setting Compound.
 - 2) USG Corporation; Sheetrock Durabond 90 Setting-Type Joint Compound.
 3. Finish Coat: For third coat, use drying-type, all-purpose compound.
 - a. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to the following:
 - 1) National Gypsum Company; Proform All Purpose Joint Compound, Ready-Mixed.
 - 2) USG Corporation; Sheetrock Brand All Purpose Joint Compound, Ready-Mixed.
 4. Skim Coat: For final coat of Level 5 finish, use drying-type, all-purpose compound.
 - a. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to the following:
 - 1) National Gypsum Company; Proform Lite Joint Compound, Ready-Mixed.
 - 2) USG Corporation; Sheetrock Brand Lightweight Joint Compound, Ready-Mixed.

- D. Joint Compound for Tile Backing Panels:
 - 1. Glass-Mat, Water-Resistant Backing Panel: As recommended by backing panel manufacturer.
 - 2. Cementitious Backer Units: As recommended by backer unit manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and substrates, with Installer present, and including welded hollow-metal frames, cast-in anchors, and support framing, for compliance with requirements and other conditions affecting performance of the Work.
- B. Examine panels before installation. Reject panels that are wet, moisture damaged, and mold damaged.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

- A. Installation Standard: ASTM C754.
 - 1. Gypsum Board Assemblies: Also comply with requirements in ASTM C 840 that apply to framing installation.
- B. Install framing and accessories plumb, square, and true to line, with connections securely fastened.
- C. Install supplementary framing, and blocking to support fixtures, equipment services, heavy trim, grab bars, toilet accessories, furnishings, or similar construction.
- D. Install bracing at terminations in assemblies.
- E. Do not bridge building control and expansion joints with non-load-bearing steel framing members. Frame both sides of joints independently.

3.3 INSTALLING FRAMED ASSEMBLIES

- A. Install framing system components according to spacings indicated, but not greater than spacings required by referenced installation standards for assembly types.
- B. Where studs are installed directly against exterior masonry walls or dissimilar metals at exterior walls, install isolation strip between studs and exterior wall.
- C. Install studs so flanges within framing system point in same direction.

- D. Install tracks at floors and overhead supports. Extend framing full height to structural supports or substrates above suspended ceilings except where partitions are indicated to terminate at suspended ceilings. Continue framing around and above ducts that penetrate partitions above ceiling.
1. Slip-Type Head Joints: Where framing extends to overhead structural supports, install to produce joints at tops of framing systems that prevent axial loading of finished assemblies.
 2. Door Openings: Screw vertical studs at jambs to jamb anchor clips on door frames; install track section (for cripple studs) at head and secure to jamb studs.
 - a. Install two studs at each jamb unless otherwise indicated.
 - b. Install cripple studs at head adjacent to each jamb stud, with a minimum 1/2-inch clearance from jamb stud to allow for installation of control joint in finished assembly.
 - c. Extend both jamb studs through suspended ceilings and attach to underside of overhead structure.
 3. Other Framed Openings: Frame openings other than door openings the same as required for door openings unless otherwise indicated. Install framing below sills of openings to match framing required above door heads.
 4. Fire-Resistance-Rated Partitions: Install framing to comply with fire-resistance-rated assembly indicated and support closures and to make partitions continuous from floor to underside of solid structure.
 - a. Firestop Track: Install to maintain continuity of fire-resistance-rated assembly indicated.
 5. Sound-Rated Partitions: Install framing to comply with sound-rated assembly indicated.
 - a. Acoustical Tracks: Install to maintain continuity of STC-rated assembly indicated.
 - b. Double Stud Walls and High-Performing Acoustical Partition Wall Studs:
 - 1) Do not acoustically short-circuit independent rows of studs by bracing together with rigid framing members.
 - 2) Do not allow rigid piping within chase walls to be connect to framing members with metal or other rigid pipe connectors. Use resilient pipe connectors.
 6. Curved Partitions:
 - a. Bend track to uniform curve and locate straight lengths so they are tangent to arcs.
 - b. Begin and end each arc with a stud, and space intermediate studs equally along arcs. On straight lengths of no fewer than two studs at ends of arcs, place studs 6 inches o.c.

E. Direct Furring:

1. Attach to concrete or masonry with stub nails, screws designed for masonry attachment, or powder-driven fasteners spaced 24 inches o.c.

F. Z-Shaped Furring Members:

1. Erect insulation, specified in Division 07, vertically and hold in place with Z-shaped furring members spaced 24 inches o.c.
2. Except at exterior corners, securely attach narrow flanges of furring members to wall with concrete stub nails, screws designed for masonry attachment, or powder-driven fasteners spaced 24 inches o.c.
3. At exterior corners, attach wide flange of furring members to wall with short flange extending beyond corner; on adjacent wall surface, screw-attach short flange of furring channel to web of attached channel. At interior corners, space second member no more than 12 inches from corner and cut insulation to fit.

G. Installation Tolerance: Install each framing member so fastening surfaces vary not more than 1/8 inch from the plane formed by faces of adjacent framing.

3.4 INSTALLING CEILING SUSPENSION SYSTEMS

A. Install suspension system components according to spacings indicated, but not greater than spacings required by referenced installation standards for assembly types.

1. Hangers: 48 inches o.c.
2. Carrying Channels (Main Runners): 48 inches o.c.
3. Furring Channels (Furring Members): 16 inches o.c.

B. Isolate suspension systems from building structure where they abut or are penetrated by building structure to prevent transfer of loading imposed by structural movement.

C. Suspend hangers from building structure as follows:

1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structural or suspension system.
 - a. Splay hangers only where required to miss obstructions and offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
2. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with locations of hangers required to support standard suspension system members, install supplemental suspension members and hangers in the form of trapezes or equivalent devices.
 - a. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by ASTM C 754.

3. Wire Hangers: Secure by looping and wire tying, either directly to structures or to inserts, eye screws, or other devices and fasteners that are secure and appropriate for substrate, and in a manner that will not cause hangers to deteriorate or otherwise fail.
 4. Do not attach hangers to steel roof deck.
 5. Do not attach hangers to permanent metal forms. Furnish cast-in-place hanger inserts that extend through forms.
 6. Do not attach hangers to rolled-in hanger tabs of composite steel floor deck.
 7. Do not connect or suspend steel framing from ducts, pipes, or conduit.
- D. Fire-Resistance-Rated Assemblies: Wire tie furring channels to supports.
- E. Grid Suspension Systems: Attach perimeter wall track or angle where grid suspension systems meet vertical surfaces. Mechanically join main beam and cross-furring members to each other and butt-cut to fit into wall track.
- F. Installation Tolerances:
1. Install suspension systems that are level to within 1/8 inch in 12 feet measured lengthwise on each member that will receive finishes and transversely between parallel members that will receive finishes.
 2. Where sprinkler heads, diffusers, and speakers are arranged in alignment, variation from exact alignment shall not vary more than 1/2-inch either side of centerline through various element openings.

3.5 INSTALLATION AND FINISHING OF PANELS, GENERAL

- A. Comply with ASTM C840.
- B. Install ceiling panels across framing to minimize the number of abutting end joints and to avoid abutting end joints in central area of each ceiling. Stagger abutting end joints of adjacent panels not less than one framing member.
- C. Install panels with face side out. Butt panels together for a light contact at edges and ends with not more than 1/16 inch of open space between panels. Do not force into place.
- D. Locate edge and end joints over supports, except in ceiling applications where intermediate supports or gypsum board back-blocking is provided behind end joints. Do not place tapered edges against cut edges or ends. Stagger vertical joints on opposite sides of partitions. Do not make joints other than control joints at corners of framed openings.
- E. Form control and expansion joints with space between edges of adjoining gypsum panels.

- F. Cover both faces of support framing with gypsum panels in concealed spaces (above ceilings, etc.), except in chases braced internally.
 - 1. Unless concealed application is indicated or required for sound, fire, air, or smoke ratings, coverage may be accomplished with scraps of not less than 8 sq. ft. in area.
 - 2. Fit gypsum panels around ducts, pipes, and conduits.
 - 3. Where partitions intersect structural members projecting below underside of floor/roof slabs and decks, cut gypsum panels to fit profile formed by structural members; allow 1/4- to 3/8-inch-wide joints to install sealant.
- G. Isolate perimeter of gypsum board applied to non-load-bearing partitions at structural abutments. Provide 1/4- to 1/2-inch-wide spaces at these locations and trim edges with edge trim where edges of panels are exposed. Seal joints between edges and abutting structural surfaces with acoustical sealant.
- H. Attachment to Steel Framing: Attach panels so leading edge or end of each panel is attached to open (unsupported) edges of stud flanges first.
- I. STC-Rated Assemblies:
 - 1. At top-of-wall conditions below metal floor or roof decks, castle-cut gypsum panels to match profile deck.
 - 2. Seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical sealant. Install acoustical sealant at both faces of partitions at perimeters and through penetrations.
 - 3. Comply with ASTM C919 and with manufacturer's written instructions for locating edge trim and closing off sound-flanking paths around or through assemblies, including sealing partitions above acoustical ceilings.
- J. Install sound attenuation blankets in STC-rated assemblies before installing gypsum panels unless blankets are readily installed after panels have been installed on one side.

3.6 INSTALLATION OF INTERIOR GYPSUM BOARD

- A. Install interior gypsum board in the following locations:
 - 1. Type X: All framed surfaces unless otherwise indicated.
 - 2. Flexible Type: Apply in double layer at curved assemblies.
 - 3. Mold-Resistant Type: At all framed interior surfaces of exterior walls and all framed surfaces where construction will not be fully protected from potential weather exposure at time of installation.
 - 4. Acoustically Enhanced Type: As indicated on Drawings.
- B. Single-Layer Application:
 - 1. On ceilings, apply gypsum panels before wall/partition board application to greatest extent possible and at right angles to framing unless otherwise indicated.

2. On partitions/walls, apply gypsum panels vertically (parallel to framing) unless otherwise indicated or required by fire-resistance-rated assembly, and minimize end joints.
 - a. Stagger abutting end joints not less than 24 inches in alternate courses of panels.
 - b. At stairwells and other high walls, install panels horizontally unless otherwise indicated or required by fire-resistance-rated assembly.
3. On Z-shaped furring members, apply gypsum panels vertically (parallel to framing) with no end joints. Locate edge joints over furring members.
4. Fastening Methods: Apply gypsum panels to supports with steel drill screws.

C. Multilayer Application:

1. On ceilings, apply gypsum board indicated for base layers before applying base layers on walls/partitions; apply face layers in same sequence. Apply base layers at right angles to framing members and offset face-layer joints one framing member, 16 inches minimum, from parallel base-layer joints, unless otherwise indicated or required by fire-resistance-rated assembly.
2. On partitions/walls, apply gypsum board indicated for base layers and face layers vertically (parallel to framing) with joints of base layers located over stud or furring member and face-layer joints offset at least one stud or furring member with base-layer joints unless otherwise indicated or required by fire-resistance-rated assembly. Stagger joints on opposite sides of partitions.
3. On Z-shaped furring members, apply base layer vertically (parallel to framing) and face layer either vertically (parallel to framing) or horizontally (perpendicular to framing) with vertical joints offset at least one furring member. Locate edge joints of base layer over furring members.
4. Fastening Methods: Fasten base layers with screws; fasten face layers with adhesive and supplementary fasteners.

- D. Laminating to Substrate: Where gypsum panels are indicated as directly adhered to a substrate (other than studs, joists, furring members, or base layer of gypsum board), comply with gypsum board manufacturer's written instructions and temporarily brace or fasten gypsum panels until fastening adhesive has set.

E. Curved Surfaces:

1. Install panels horizontally (perpendicular to supports) and unbroken, to extent possible, across curved surface plus 12-inch-long straight sections at ends of curves and tangent to them.
2. For double-layer construction, fasten base layer to studs with screws 16 inches o.c. Center gypsum board face layer over joints in base layer, and fasten to studs with screws spaced 12 inches o.c.

3.7 INSTALLATION OF TILE BACKING PANELS

- A. Install tile backing panels in the following locations:
 - 1. All framed surfaces indicated to receive tile.
- B. Glass-Mat, Water-Resistant Backing Panels: Comply with manufacturer's written installation instructions and install at locations indicated to receive tile less than or equal to 1.4 sq. ft. in area or 5 lb./sq. ft. in weight. Install with 1/4-inch gap where panels abut other construction or penetrations.
- C. Cementitious Backer Units: ANSI A108.11, at locations indicated to receive tile greater than 1.4 sq. ft. in area or 5 lb./sq. ft. in weight.
- D. Where tile backing panels abut other types of panels in same plane, shim framed surfaces to produce a uniform plane across panel surfaces.

3.8 INSTALLATION OF ACOUSTICAL ACCESSORIES

- A. Place acoustical insulation in STC-rated partitions tight within spaces, around cut openings, behind and around electrical and mechanical items within or behind partitions, and tight to items passing through partitions.
- B. Install electrical J-box putty pads at STC-rated partitions in accordance with manufacturer's instructions.
- C. Install resilient sound isolation clips at STC-rated partitions indicated to receive rigid furring channels to avoid metal-to-metal or other rigid substrate contact.
- D. Install acoustical sealant at STC-rated partitions in accordance with manufacturer's instructions.

3.9 INSTALLATION OF TRIM ACCESSORIES

- A. General: For trim with back flanges intended for fasteners, attach to framing with same fasteners used for panels. Otherwise, attach trim according to manufacturer's written instructions.
- B. Control Joints: Maintain fire ratings of assemblies at control joints. Install control joints according to ASTM C840 and as follows:
 - 1. Partitions: Install control joints:
 - a. no greater than 30 feet on center where a partition runs in an uninterrupted straight plane;
 - b. at both corners of openings in wall planes, above and below opening, where width of opening is 6 feet or greater, or where ratio of width to height of wall area above or below opening exceeds 4:1;
 - c. at all spliced joints of vertical studs;
 - d. at other locations indicated on Drawings.

2. Ceilings: Install control joints:
 - a. no greater than 30 feet on center and with total area not to exceed 900 sq. ft.;
 - b. where sections of "A", "L", "O", "U", "T" and "X" shaped ceiling areas or furr-down areas intersect;
 - c. at other locations indicated on Drawings.

C. Interior Trim: Install in the following locations:

1. Cornerbead: Use at outside corners unless otherwise indicated.
2. Bullnose Bead: Use where indicated on Drawings.
3. LC-Bead: Use at exposed panel edges.
4. L-Bead: Use where edge trim can only be installed after gypsum board is installed.
5. U-Bead: Use at exposed panel edges not intended to receive joint compound.
6. Curved-Edge Cornerbead: Use at curved openings.

D. Aluminum Trim: Maintain fire ratings of assemblies at aluminum trim. Install in locations indicated on Drawings.

3.10 FINISHING GYPSUM BOARD

- A. General: Treat gypsum board joints, interior angles, edge trim, control joints, penetrations, fastener heads, surface defects, and elsewhere as required to prepare gypsum board surfaces for decoration. Promptly remove residual joint compound from adjacent surfaces.
- B. Prefill open joints, rounded or beveled edges, and damaged surface areas.
- C. Apply joint tape over gypsum board joints, except for trim products specifically indicated as not intended to receive tape.
- D. Gypsum Board Finish Levels:
 1. Level 0: No taping, finishing, or accessories required.
 2. Level 1:
 - a. Joints: Tape set in joint compound.
 - b. Interior Angles: Tape set in joint compound.
 - c. Surface: Tool marks and ridges acceptable. Surface free of excess joint compound.
 3. Level 2:
 - a. Joints: Tape embedded in joint compound and wiped with a joint knife, leaving a thin coat of compound over tape.
 - b. Interior Angles: Tape embedded in joint compound and wiped with a joint knife, leaving a thin coat of compound over tape.
 - c. Accessories: Shall be covered to one separate coat of joint compound.

- d. Fasteners: Shall be covered by one separate coat of joint compound.
 - e. Surface: Surface shall be free of excess joint compound. Tool marks and ridges acceptable. Joint compound applied over the body of the tape at the time of tape embedment shall be considered a separate coat of joint compound and shall satisfy the conditions of this level.
- 4. Level 3:
 - a. Joints: Taped as in Level 2, then covered with one separate coat of joint compound.
 - b. Interior Angles: Taped as in Level 2, then covered with one separate coat of joint compound.
 - c. Accessories: Shall be covered by two separate coats of joint compound.
 - d. Fasteners: Shall be covered by two separate coats of joint compound.
 - e. Surface: Joint compound shall be smooth and free of tool marks and ridges.
- 5. Level 4:
 - a. Joints: Taped as in Level 2, then covered with two separate coats of joint compound.
 - b. Interior Angles: Taped as in Level 2, then covered with one separate coat of joint compound.
 - c. Accessories: Shall be covered by three separate coats of joint compound.
 - d. Fasteners: Shall be covered by three separate coats of joint compound.
 - e. Surface: Joint compound shall be smooth and free of tool marks and ridges.
- 6. Level 5:
 - a. Joints: Taped as in Level 2, then covered with two separate coats of joint compound.
 - b. Interior Angles: Taped as in Level 2, then covered with one separate coat of joint compound.
 - c. Accessories: Shall be covered by three separate coats of joint compound.
 - d. Fasteners: Shall be covered by three separate coats of joint compound.
 - e. Surface: A thin skin coat of joint compound, or a material manufactured especially for this purpose, shall be applied to the entire surface. The surface shall be smooth and free of tool marks and ridges.
- E. Gypsum Board Finish Level Schedule: Finish panels to levels indicated below and according to ASTM C840:
 - 1. Level 0: Temporary construction or where final finishes have not yet been determined. Do not use in areas where fire ratings or smoke control are required.
 - 2. Level 1: Ceiling plenum areas, concealed areas, service corridors and other areas not normally exposed to view.
 - 3. Level 2: Panels that are substrates for tile.

4. Level 3: Areas which are to receive heavy or medium, spray or troweled, texture finishes before final painting and areas where heavy grade wall coverings are to be applied as the final finish.
5. Level 4: At panel surfaces that will be exposed to view unless otherwise indicated.
 - a. Primer and its application to surfaces are specified in Section 09 91 23 "Interior Painting."
6. Level 5: At panel surfaces flooded with natural or artificial light and scheduled to receive semi-gloss or gloss paint finish.
 - a. Primer and its application to surfaces are specified in Section 09 91 23 "Interior Painting."
- F. Glass-Mat Faced Panels: Finish according to manufacturer's written instructions.
- G. Cementitious Backer Units: Finish according to manufacturer's written instructions.

3.11 PROTECTION

- A. Protect adjacent surfaces from drywall compound and promptly remove from floors and other non-drywall surfaces. Repair surfaces stained, marred, or otherwise damaged during drywall application.
- B. Protect installed products from damage from weather, condensation, direct sunlight, construction, and other causes during remainder of the construction period.
- C. Remove and replace panels that are wet, moisture damaged, and mold damaged.
 1. Indications that panels are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
 2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

END OF SECTION

SECTION 093013

CERAMIC TILING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Ceramic mosaic tile.
2. Quarry tile.
3. Unglazed porcelain floor or wall tile.
4. Glazed or unglazed wall tile.
5. Gauged tile panels
6. Waterproof membrane.
7. Crack isolation membrane.
8. Metal edge strips.

1.2 DEFINITIONS

- A. General:** Definitions in the ANSI A108 series of tile installation standards and in ANSI A137.1 apply to Work of this Section unless otherwise specified.
- B. High-Performance Tile Grout:** A factory-prepared grouting material mixture of cement and other ingredients, including a redispersible latex/polymer powder, to which only water is added at the jobsite, or a liquid latex additive.
- C. Improved Modified Dry-Set Cement Mortar (Thinset):** Modified dry-set mortar with a minimum bond strength of 300 psi to impervious ceramic tile.
- D. Improved Modified Dry-Set Cement Mortar (Thin-Bed):** A factory-prepared setting material mixture of cement and other ingredients, including a redispersible latex/polymer powder, to which only water is added at the jobsite, or a liquid latex additive.
- E. Large and Heavy Tile (LHT):** Any tile material weighing 5 lbs./sq. ft. or greater, or any tile with a least horizontal dimension of 15 inches or more.
- F. Gauged Tile or Tile Panel/Slab:** A tile panel/slab that has been manufactured to a specific nominal thickness, and whose thickness characteristic is largely associated with how it is installed and/or used in wall or floor applications depending on manufacturer's recommendations.
- G. Tile Panel/Slab:** A ceramic tile of size greater than or equal to 1 sq. m. (112 sq. ft.). A product less than 1 sq. m. (11 sq. ft.) shall be regarded as "tile."

- H. Tile: A ceramic surfacing unit, thin in relation to facial area, having either a glazed or unglazed face and fired above red heat in the course of manufacture to a temperature sufficiently high to product specific physical properties and characteristics.
- I. Modified Dry-Set Mortar for Large and Heavy Tile (LHT): Formerly "Medium Bed Mortar," is a modified dry-set mortar formulated to have a bond coat thickness between 3/32 and 1/2 inch after tile embedment and declared as an "LHT" setting material by the manufacturer based on these characteristics.
- J. Modified Dry-Set Mortar (Thinset): A factory-prepared setting material mixture of cement and other ingredients, including a redispersible latex/polymer powder, to which only water is added at the jobsite, or a liquid latex additive.

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.
 - 1. Review requirements in ANSI A108.01 for substrates and for preparation by other trades.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings:
 - 1. Show locations of each type of tile and tile pattern.
 - 2. Show widths, details, and locations of expansion, contraction, control, and isolation joints in tile substrates and finished tile surfaces.
 - 3. Show installation details for ceramic tiles at shower enclosures.
 - a. Include plans, elevations, component details, and trim details.
 - b. Indicate materials and profiles of each item.
 - c. Show fittings, joinery, penetrations, fasteners, anchorages, plumbing fixtures, and other accessory items.
- C. Samples:
 - 1. Each type and composition of tile and for each color and finish required. For ceramic mosaic tile in color blend patterns, provide samples of each color blend.

1.5 QUALITY ASSURANCE

- A. Mockups: Build mockups to verify selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Build mockup of each type of floor tile installation.
 - 2. Build mockup of each type of wall tile installation.

3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

PART 2 - PRODUCTS

2.1 PRODUCTS, GENERAL

- A. ANSI Ceramic Tile Standard: Provide Standard-grade tile that complies with ANSI A137.1 and ANSI A137.3 for types, compositions, and other characteristics indicated.
- B. ANSI Standards for Tile Installation Materials: Provide materials complying with ANSI A108.02, ANSI standards referenced in other Part 2 articles, ANSI standards referenced by TCNA installation methods specified in tile installation schedules, and other requirements specified.
- C. Factory Blending: For tile exhibiting color variations within ranges, blend tile in factory and package so tile units taken from one package show same range in colors as those taken from other packages and match approved Samples.

2.2 TILE PRODUCTS

- A. Factory-Mounted Ceramic Mosaic Tile:
 1. Products: Subject to compliance with requirements, provide product indicated on Drawings.
 2. Composition: Impervious natural clay or porcelain.
 3. Certification: Tile certified by the Porcelain Tile Certification Agency.
 4. Module Size: As indicated by product designations on Drawings.
 5. Thickness: 1/4 inch.
 6. Face: As indicated by product designations on Drawings.
 7. Dynamic Coefficient of Friction: For products on floor surfaces, not less than 0.42.
 8. Finish: As indicated by product designations on Drawings.
 9. Tile Color and Pattern: As indicated on Drawings.
 10. Grout Color: As indicated on Drawings.
 11. Trim Units: Coordinated with sizes and coursing of adjoining flat tile where applicable and matching characteristics of adjoining flat tile. Provide the following coordinating shapes where indicated on Drawings:
 - a. Base: Coves.
 - b. Base Caps: Surface bullnose.
 - c. Wainscot Caps: Surface bullnose.
 - d. External Corners: Bullnose.
 - e. Internal Corners: Coves.

B. Square-Edged Quarry Tile:

1. Products: Subject to compliance with requirements, provide product indicated on Drawings.
2. Face Size: As indicated by product designations on Drawings.
3. Thickness: Not less than 1/2 inch.
4. Wearing Surface: As indicated by product designations on Drawings.
5. Dynamic Coefficient of Friction: Not less than 0.42.
6. Finish: As indicated by product designations on Drawings.
7. Tile Color and Pattern: As indicated on Drawings.
8. Grout Color: As indicated on Drawings.
9. Trim Units: Coordinated with sizes and coursing of adjoining flat tile where applicable and matching characteristics of adjoining flat tile. Provide shapes as follows, selected from manufacturer's standard shapes:
 - a. Base: Coved or straight, as indicated.
 - b. Wainscot Cap: Surface bullnose.
 - c. Wainscot Cap for Flush Conditions: Regular flat tile for conditions where tile wainscot is shown flush with wall surface above it, same size as adjoining flat tile.

C. Unglazed Porcelain Floor or Wall Tile:

1. Products: Subject to compliance with requirements, provide product indicated on Drawings.
2. Certification: Tile certified by the Porcelain Tile Certification Agency.
3. Face: As indicated by product designations on Drawings.
4. Face Size Variation: Rectified.
5. Thickness: As indicated by product designations on Drawings, but not less than 3/8 inch.
6. Face: As indicated by product designations on Drawings.
7. Dynamic Coefficient of Friction: For products on floor surfaces, not less than 0.42.
8. Tile Color and Pattern: As indicated on Drawings.
9. Grout Color: As indicated on Drawings.
10. Trim Units: Coordinated with sizes and coursing of adjoining flat tile where applicable and matching characteristics of adjoining flat tile. Provide the following coordinating shapes where indicated on Drawings:
 - a. Base Cap: Surface bullnose, module as indicated on Drawings.
 - b. Wainscot Cap: Surface bullnose, module size as indicated on Drawings.
 - c. External Corners: Surface bullnose, module size as indicated on Drawings.

D. Glazed or Unglazed Wall Tile:

1. Products: Subject to compliance with requirements, provide product indicated on Drawings.
2. Module Size: As indicated by product designations on Drawings.
3. Face Size Variation: Rectified.
4. Thickness: 5/16 inch.
5. Face: As indicated by product designations on Drawings.

6. Finish: As indicated by product designations on Drawings.
7. Tile Color and Pattern: As indicated on Drawings.
8. Grout Color: As indicated on Drawings.
9. Trim Units: Coordinated with sizes and coursing of adjoining flat tile where applicable and matching characteristics of adjoining flat tile. Provide the following coordinating shapes where indicated on Drawings:
 - a. Base: Coved or straight as indicated.
 - b. Wainscot Cap: Bullnose cap.
 - c. External Corners: Bullnose, same size as adjoining flat tile.
 - d. Internal Corners: Field-buttet square corners. For coved base and cap use angle pieces designed to fit with stretcher shapes.

E. Gauged Porcelain Tile or Gauged Porcelain Tile Panel/Slab:

1. Products: Subject to compliance with requirements, provide product indicated on Drawings.
2. Module Size: As indicated by product designations on Drawings.
3. Face Size Variation: Rectified.
4. Thickness: Manufacturer's standard.
5. Face: As indicated by product designations on Drawings.
6. Finish: As indicated by product designations on Drawings.
7. Tile Color and Pattern: As indicated on Drawings.
8. Grout Color: As indicated on Drawings.
9. Trim Units: Coordinated with sizes and coursing of adjoining gauged tile or tile panel/slab where applicable and matching characteristics of adjoining gauged tile or tile panel/slab.

2.3 MANUFACTURED SHOWER RECEPTOR SYSTEM

- A. Manufactured Shower Receptor System: Provide products from Manufacturer's standard integrated Shower System for a complete shower or curbless shower receptor installation.
1. Source Limitation: Include all products for complete Shower System from same source.
 2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Shower Systems by Schluter Systems.
 - b. Pre-Slopes and Shower Bases by Noble Company.

2.4 WATERPROOF MEMBRANE

- A. General: Manufacturer's standard product that complies with ANSI A118.10 and is recommended by the manufacturer for the application indicated. Include reinforcement and accessories recommended by manufacturer.

B. Floor Surfaces:

1. Fabric-Reinforced, Fluid-Applied Membrane: System consisting of liquid-latex rubber or elastomeric polymer and continuous fabric reinforcement.
 - a. Products: Subject to compliance with requirements, provide one of the following:
 - 1) Bostik, Inc; Hydroment Blacktop 90210.
 - 2) Custom Building Products; 9240 Waterproofing and Anti-Fracture Membrane.
 - 3) Laticrete International, Inc; Laticrete 9235 Waterproof Membrane.
 - 4) MAPEI Corporation; Mapelastic™ 400 or Mapelastic™ HPG

C. Wall Surfaces:

1. Fluid-Applied Membrane: Liquid-latex rubber or elastomeric polymer.
 - a. Products: Subject to compliance with requirements, provide one of the following:
 - 1) Bostik, Inc; Bostik GoldPlus or Durabond D-222 Duraguard Membrane.
 - 2) Custom Building Products; Custom 9240 or RedGard Waterproofing and Crack Prevention Membrane.
 - 3) Laticrete International, Inc; Laticrete Hydro Ban or Laticrete Hydro Barrier.
 - 4) MAPEI Corporation; Mapelastic HPG or Mapelastic™ AquaDefense.

D. Manufactured Shower Receptor Systems:

1. Water-proofing Membrane:
 - a. Products: Subject to compliance with requirements, provide one of the following products to be used in conjunction with Manufactured shower receptor systems.
 - 1) Noble Company (The); Nobleseal CIS.
 - 2) Schluter Systems; Schluter-Kerdi Membrane sheet product.

2.5 CRACK ISOLATION MEMBRANE

- A. General: Manufacturer's standard product, selected from the following, that complies with ANSI A118.12 for high performance and is recommended by the manufacturer for the application indicated. Include reinforcement and accessories recommended by manufacturer.

- B. Chlorinated Polyethylene Sheet: Nonplasticized, chlorinated polyethylene faced on both sides with nonwoven polyester fabric; 0.030-inch nominal thickness.
 - 1. Products: Subject to compliance with requirements, provide the following:
 - a. Noble Company (The); Nobleseal CIS.
- C. Fabric-Reinforced, Modified-Bituminous Sheet: Self-adhering, modified-bituminous sheet with fabric reinforcement facing; 0.040-inch nominal thickness.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Custom Building Products; Crack Buster Pro-Crack Prevention Mat Underlayment.
 - b. MAPEI Corporation; Mapeguard™ 2or Mapesonic 2.
- D. Fabric-Reinforced, Fluid-Applied Membrane: System consisting of liquid-latex rubber or elastomeric polymer and fabric reinforcement.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Bostik, Inc; Hydroment Blacktop 90210.
 - b. Custom Building Products; 9240 Waterproofing and Anti-Fracture Membrane.
 - c. Laticrete International, Inc; Laticrete Blue 92 Anti-Fracture Membrane.
 - d. MAPEI Corporation; Mapelastic HPG with MAPEI Fiberglass Mesh.

2.6 SETTING MATERIALS

- A. Portland Cement Mortar (Thickset) Installation Materials: ANSI A108.02.
- B. Standard Dry-Set Cement Mortar (Thinset): ANSI A118.1.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX Americas.
 - b. Bostik, Inc.
 - c. Custom Building Products.
 - d. Laticrete International, Inc.
 - e. MAPEI Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.
- C. Modified Dry-Set Cement Mortar (Thinset): ANSI A118.4.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX Americas.
 - b. Bostik, Inc.
 - c. Custom Building Products.

- d. Laticrete International, Inc.
 - e. MAPEI Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.
 - 2. Provide prepackaged, dry-mortar mix to which only water must be added at Project site.
 - 3. For wall applications, provide nonsagging mortar.
- D. Modified Dry-Set Cement Mortar for Large and Heavy Tile (LHT): Comply with requirements in ANSI A118.4. Provide product that is approved by manufacturer for application thickness of 5/8 inch.
- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX Americas.
 - b. Bostik, Inc.
 - c. Custom Building Products.
 - d. Laticrete International, Inc.
 - e. MAPEI Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.
 - 2. Provide prepackaged, dry-mortar mix to which only water must be added at Project site.
 - 3. For wall applications, provide nonsagging mortar.
- E. Improved Modified Dry-Set Cement Mortar (Thin-Bed): ANSI A118.15.
- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX Americas.
 - b. Bostik, Inc.
 - c. Custom Building Products.
 - d. Laticrete International, Inc.
 - e. MAPEI Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.
 - 2. Provide prepackaged, dry-mortar mix to which only water must be added at Project site.
 - 3. For wall applications, provide nonsagging mortar.

2.7 GROUT MATERIALS

- A. Sand-Portland Cement Grout: ANSI A108.10, consisting of white or gray cement and white or colored aggregate as required to produce color indicated.

B. Standard Cement Grout: ANSI A118.6.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX Americas.
 - b. Bostik, Inc.
 - c. Custom Building Products.
 - d. Laticrete International, Inc.
 - e. MAPEI Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.
2. Where joints size is 1/8 inch or less, use unsanded grout. Where joints are greater than 1/8 inch, use sanded grout.

C. High-Performance Tile Grout: ANSI A118.7.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX Americas.
 - b. Bostik, Inc.
 - c. Custom Building Products.
 - d. Laticrete International, Inc.
 - e. MAPEI Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.
2. Polymer Type: Dry, redispersible form, prepackaged with other dry ingredients.
3. Where joints size is 1/8 inch or less, use unsanded grout. Where joints are greater than 1/8 inch, use sanded grout.

D. Water-Cleanable Epoxy Grout: ANSI A118.3, with a VOC content of 65 g/L or less.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. ARDEX Americas.
 - b. Bostik, Inc.
 - c. Custom Building Products.
 - d. Laticrete International, Inc.
 - e. MAPEI Corporation.
 - f. TEC; H.B. Fuller Construction Products Inc.

2.8 MISCELLANEOUS MATERIALS

- A. Trowelable Underlayments and Patching Compounds: Latex-modified, portland cement-based formulation provided or approved by manufacturer of tile-setting materials for installations indicated.

- B. Metal Edge and Trim Strips: Angle or L-shape, height to match tile and setting-bed thickness, designed specifically for application indicated; stainless-steel, ASTM A 666, 300 Series exposed-edge material.
 - 1. Products: Subject to compliance with requirements, provide products indicated on Drawings.
- C. Floor Sealer: Manufacturer's standard product for sealing grout joints and that does not change color or appearance of grout.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Bonsal American, an Oldcastle company.
 - b. Custom Building Products.
 - c. Summitville Tiles, Inc.
 - d. TEC; H.B. Fuller Construction Products Inc.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions where tile will be installed, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
 - 1. Verify that substrates for setting tile are firm; dry; clean; free of coatings that are incompatible with tile-setting materials, including curing compounds and other substances that contain soap, wax, oil, or silicone; and comply with flatness tolerances required by ANSI A108.01 for installations indicated.
 - 2. Verify that concrete substrates for tile floors installed with bonded mortar bed or thinset mortar comply with surface finish requirements in ANSI A108.01 for installations indicated.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Fill cracks, holes, and depressions in concrete substrates for tile floors installed with thinset mortar with trowelable leveling and patching compound specifically recommended by tile-setting material manufacturer.
- B. Where indicated, prepare substrates to receive waterproofing by applying a reinforced mortar bed that complies with ANSI A108.1A and is sloped 1/4 inch per foot toward drains.
- C. Blending: For tile exhibiting color variations, verify that tile has been factory blended and packaged so tile units taken from one package show same range of colors as

those taken from other packages and match approved Samples. If not factory blended, either return to manufacturer or blend tiles at Project site before installing.

3.3 CERAMIC TILE INSTALLATION

- A. Comply with TCNA's "Handbook for Ceramic, Glass, and Stone Tile Installation" for TCNA installation methods specified in tile installation schedules. Comply with parts of the ANSI A108 series "Specifications for Installation of Ceramic Tile" that are referenced in TCNA installation methods, specified in tile installation schedules, and apply to types of setting and grouting materials used.
 - 1. For the following installations, follow procedures in the ANSI A108 series of tile installation standards for providing 95 percent mortar coverage:
 - a. Tile floors in wet areas.
 - b. Tile floors consisting of tiles 8 by 8 inches or larger.
 - c. Tile floors consisting of rib-backed tiles.
- B. Extend tile work into recesses and under or behind equipment and fixtures to form complete covering without interruptions unless otherwise indicated. Terminate work neatly at obstructions, edges, and corners without disrupting pattern or joint alignments.
- C. Accurately form intersections and returns. Perform cutting and drilling of tile without marring visible surfaces. Carefully grind cut edges of tile abutting trim, finish, or built-in items for straight aligned joints. Fit tile closely to electrical outlets, piping, fixtures, and other penetrations so plates, collars, or covers overlap tile.
- D. Provide manufacturer's standard trim shapes where necessary or indicated to eliminate exposed tile edges.
- E. Where accent tile differs in thickness from field tile, vary setting bed thickness so that tiles are flush.
- F. Jointing Pattern: Lay tile in grid pattern unless otherwise indicated. Lay out tile work and center tile fields in both directions in each space or on each wall area. Lay out tile work to minimize the use of pieces that are less than half of a tile. Provide uniform joint widths unless otherwise indicated.
- G. Joint Widths: Install tile with joint widths recommended by the manufacturer for products indicated.
- H. Lay out tile wainscots to dimensions indicated or to next full tile beyond dimensions indicated.

- I. Expansion Joints: Provide expansion joints and other sealant-filled joints, including control, contraction, and isolation joints where indicated but not greater than 30 feet on center.
 - 1. Form joints during installation of setting materials, mortar beds, and tile. Do not saw-cut joints after installing tiles.
 - 2. Where joints occur in concrete substrates, locate joints in tile surfaces directly above them.
 - 3. Install control and expansion joints in accordance with TCNA Handbook Method No. EJ171.
- J. Metal Edge Strips: Install at locations indicated. If not indicated, install where exposed edge of tile flooring meets carpet, wood, or other flooring that finishes flush with top of tile.
- K. Grout Sealer: Apply grout sealer to cementitious grout joints in tile floors according to grout-sealer manufacturer's written instructions. As soon as floor has penetrated grout joints, remove excess sealer and sealer from tile faces by wiping with soft cloth.
- L. Install tile backing panels and treat joints according to ANSI A108.11 and manufacturer's written instructions for type of application indicated. Use modified dry-set mortar for bonding material unless otherwise directed in manufacturer's written instructions.
- M. Install waterproofing to comply with ANSI A108.13 and manufacturer's written instructions to produce waterproof membrane of uniform thickness that is bonded securely to substrate.
- N. Install crack isolation membrane to comply with ANSI A108.17 and manufacturer's written instructions to produce membrane of uniform thickness that is bonded securely to substrate.

3.4 WATERPROOFING INSTALLATION

- A. Install waterproofing to comply with ANSI A108.13 and manufacturer's written instructions to produce waterproof membrane of uniform thickness that is bonded securely to substrate.
- B. Allow waterproofing to cure and verify by testing that it is watertight before installing tile or setting materials over it.

3.5 CRACK ISOLATION MEMBRANE INSTALLATION

- A. Install crack isolation membrane to comply with ANSI A108.17 and manufacturer's written instructions to produce membrane of uniform thickness that is bonded securely to substrate.
- B. Allow crack isolation membrane to cure before installing tile or setting materials over it.

3.6 ADJUSTING AND CLEANING

- A. Remove and replace tile that is damaged or that does not match adjoining tile. Provide new matching units, installed as specified and to eliminate evidence of replacement.
- B. Cleaning: On completion of placement and grouting, clean ceramic tile surfaces so they are free of foreign matter.
 - 1. Remove grout residue from tile as soon as possible.
 - 2. Clean grout smears and haze from tile according to tile and grout manufacturer's written instructions but no sooner than 10 days after installation. Use cleaners recommended by tile and grout manufacturers and after determining that cleaners are safe to use by testing on samples of tile and other surfaces to be cleaned. Protect metal surfaces and plumbing fixtures from effects of cleaning. Flush surfaces with clean water before and after cleaning.

3.7 PROTECTION

- A. Protect installed tile work with kraft paper or other heavy covering during construction period to prevent staining, damage, and wear. If recommended by tile manufacturer, apply coat of neutral protective cleaner to completed tile walls and floors.
- B. Prohibit foot and wheel traffic from tiled floors for at least seven days after grouting is completed.
- C. Before final inspection, remove protective coverings and rinse neutral protective cleaner from tile surfaces.

3.8 THICKSET INTERIOR CERAMIC TILE INSTALLATION SCHEDULE

- A. Floor Installations, Concrete Subfloor:
 - 1. Commercial Kitchen Floors: TCNA F114 and ANSI A108.1A; cement mortar bed (thickset) with cleavage membrane; epoxy grout.
 - a. Bond Coat for Cured-Bed Method: Improved modified dry-set mortar.
 - b. Grout: Water-cleanable epoxy grout.
- B. Shower Receptor Installations:
 - 1. Ceramic Tile Installation: TCNA B414 and ANSI A108.1A; cement mortar bed (thickset) on waterproof membrane bonded to concrete.
 - a. Bond Coat for Cured-Bed Method: Modified dry-set mortar.
 - b. Grout: High-performance grout.

C. Curbless Shower Receptor Installation:

1. Ceramic Tile Installation: TCNA B421C and ANSI A118.4 or better; cementitious bond coat on waterproof membrane bonded to concrete.
 - a. Bond Coat for Cured-Bed Method: Modified dry-set mortar.
 - b. Grout: High-performance grout.

D. Shower Receptor Installation with Manufactured Shower-System:

1. Manufacturer's standard shower-system installation method for complete shower enclosure.

3.9 THINSET INTERIOR CERAMIC TILE INSTALLATION SCHEDULE

A. Floor Installations, Concrete Subfloor:

1. Slabs-On-Grade: Unless otherwise indicated, TCNA F113; thinset mortar.
 - a. Thinset Mortar: Modified dry-set mortar. Use modified dry-set mortar approved for LHT applications for large and heavy tile.
 - b. Grout: High-performance grout.
2. Restroom and Served Floors on Slabs-On-Grade: TCNA F113; thinset mortar.
 - a. Thinset Mortar: Modified dry-set mortar. Use modified dry-set mortar approved for LHT applications for large and heavy tile.
 - b. Grout: Water-cleanable epoxy grout.
3. Elevated Slabs: Unless otherwise indicated, TCNA F125-Full; thinset mortar on crack isolation membrane.
 - a. Thinset Mortar: Modified dry-set mortar. Use modified dry-set mortar approved for LHT applications for large and heavy tile.
 - b. Grout: High-performance grout.
4. Restroom and Served Area Floors on Elevated Slabs: TCNA F125-Full; thinset mortar on crack isolation membrane.
 - a. Thinset Mortar: Modified dry-set mortar. Use modified dry-set mortar approved for LHT applications for large and heavy tile.
 - b. Grout: Water-cleanable epoxy grout.
5. Gauged Porcelain Tile Panels/Slabs on Slab-On-Grade: TCNA F113 and ANSI A108.19; thin-bed mortar.
 - a. Thin-Bed Mortar: Improved modified dry-set cement mortar. Use improved modified dry-set cement mortar approved for gauged porcelain tile panels/slabs applications.
 - b. Grout: High-performance grout.

6. Gauged Porcelain Tile Panels/Slabs on Elevated Slabs: TCNA F125-Full and ANSI A108.19; thin-bed mortar on crack isolation membrane.
 - a. Thin-Bed Mortar: Improved modified dry-set cement mortar. Use improved modified dry-set cement mortar approved for gauged porcelain tile panels/slabs applications.
 - b. Grout: High-performance grout.
- B. Stair Installations:
 1. Concrete-Filled Metal Pan Stairs: Thinset mortar on crack isolation membrane.
 - a. Thin-Bed Mortar: Improved modified dry-set cement mortar.
 - b. Grout: High-performance grout.
- C. Wall Installations:
 1. Dry Locations: Unless otherwise indicated, TCNA W244C or TCNA W244F; thinset mortar on cementitious backer units or fiber-cement backer board.
 - a. Thinset Mortar: Modified dry-set mortar. Use modified dry-set mortar approved for LHT applications for large and heavy tile.
 - b. Grout: High-performance grout.
 2. Gauged Porcelain Tile Panels/Slabs: TCNA W244C and ANSI A108.19; thin-bed mortar on cementitious backer units.
 - a. Thin-Bed Mortar: Improved Modified dry-set cement mortar. Use improved modified dry-set cement mortar approved for gauged porcelain tile panels/slabs applications.
 - b. Grout: High-performance grout.
- D. Shower Wall Installations:
 1. Wet Locations: TCNA B415; thinset mortar on waterproof membrane over cementitious backer units or fiber-cement backer board.
 - a. Thinset Mortar: Modified dry-set mortar. Use modified dry-set mortar approved for LHT applications for large and heavy tile.
 - b. Grout: High-performance grout.

END OF SECTION

SECTION 095113
ACOUSTICAL PANEL CEILINGS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes acoustical panels and exposed suspension systems for ceilings.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples for Verification: For each component indicated and for each exposed finish required, prepared on Samples of size indicated below.
 - 1. Acoustical Panel: Set of 6-inch-square Samples of each type, color, pattern, and texture.
 - 2. Exposed Suspension-System Members, Moldings, and Trim: Set of 6-inch-long Samples of each type, finish, and color.

1.3 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For finishes to include in maintenance manuals.

1.4 QUALITY ASSURANCE

- A. Mockups: Build mockups to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Build mockup of typical ceiling area as shown on Drawings.
 - 2. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver acoustical panels, suspension-system components, and accessories to Project site in original, unopened packages and store them in a fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.
- B. Before installing acoustical panels, permit them to reach room temperature and a stabilized moisture content.

- C. Handle acoustical panels carefully to avoid chipping edges or damaging units in any way.

1.6 FIELD CONDITIONS

- A. Environmental Limitations: Do not install acoustical panel ceilings until spaces are enclosed and weatherproof, wet work in spaces is complete and dry, work above ceilings is complete, and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.
 - 1. Pressurized Plenums: Operate ventilation system for not less than 48 hours before beginning acoustical panel ceiling installation.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Seismic Performance: Acoustical ceiling shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
- B. Surface-Burning Characteristics: Comply with ASTM E 84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
 - 1. Flame-Spread Index: Comply with ASTM E 1264 for Class A materials.
 - 2. Smoke-Developed Index: 50 or less.

2.2 ACOUSTICAL PANELS, GENERAL

- A. Glass-Fiber-Based Panels: Made with binder containing no urea formaldehyde.
- B. Acoustical Panel Standard: Provide manufacturer's standard panels of configuration indicated that comply with ASTM E 1264 classifications as designated by types, patterns, acoustical ratings, and light reflectances unless otherwise indicated.

2.3 ACOUSTICAL PANELS

- A. Basis-of-Design Product: Subject to compliance with requirements, provide products indicated on Drawings.

2.4 METAL SUSPENSION SYSTEMS, GENERAL

- A. Metal Suspension-System Standard: Provide manufacturer's standard direct-hung metal suspension systems of types, structural classifications, and finishes indicated that comply with applicable requirements in ASTM C 635/C 635M.
 - 1. High-Humidity Finish: Comply with ASTM C 635/C 635M requirements for "Coating Classification for Severe Environment Performance" where high-humidity finishes are indicated.
- B. Attachment Devices: Size for five times the design load indicated in ASTM C 635/C 635M, Table 1, "Direct Hung," unless otherwise indicated. Comply with seismic design requirements.
 - 1. Anchors in Concrete: Anchors of type and material indicated below, with holes or loops for attaching hangers of type indicated and with capability to sustain, without failure, a load equal to five times that imposed by ceiling construction, as determined by testing according to ASTM E 488 or ASTM E 1512 as applicable, conducted by a qualified testing and inspecting agency.
 - a. Type: Postinstalled expansion or bonded anchors.
 - b. Corrosion Protection: Carbon-steel components zinc plated to comply with ASTM B 633, Class Fe/Zn 5 (0.005 mm) for Class SC 1 service condition.
 - 2. Power-Actuated Fasteners in Concrete: Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with clips or other accessory devices for attaching hangers of type indicated and with capability to sustain, without failure, a load equal to 10 times that imposed by ceiling construction, as determined by testing according to ASTM E 1190, conducted by a qualified testing and inspecting agency.
- C. Wire Hangers, Braces, and Ties: Provide wires complying with the following requirements:
 - 1. Zinc-Coated, Carbon-Steel Wire: ASTM A 641/A 641M, Class 1 zinc coating, soft temper.
 - 2. Stainless-Steel Wire: ASTM A 580/A 580M, Type 304, nonmagnetic.
 - 3. Size: Select wire diameter so its stress at three times hanger design load (ASTM C 635/C 635M, Table 1, "Direct Hung") will be less than yield stress of wire but provide not less than 0.106-inch-diameter wire.
- D. Flat Hangers: Mild steel, zinc coated or protected with rust-inhibitive paint.
- E. Angle Hangers: Angles with legs not less than 7/8 inch wide; formed with 0.04-inch-thick, galvanized-steel sheet complying with ASTM A 653/A 653M, G90 coating designation; with bolted connections and 5/16-inch-diameter bolts.
- F. Seismic Stabilizer Bars: Manufacturer's standard perimeter stabilizers designed to accommodate seismic forces.

- G. Seismic Struts: Manufacturer's standard compression struts designed to accommodate seismic forces.
- H. Seismic Clips: Manufacturer's standard seismic clips designed and spaced to secure acoustical panels in place.

2.5 METAL SUSPENSION SYSTEM

- A. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on Drawings.
- B. Steel Suspension System: Main and cross runners roll formed from cold-rolled steel sheet; prepainted, electrolytically zinc coated, or hot-dip galvanized according to ASTM A 653/A 653M, not less than G30 coating designation; with prefinished flanges.
 - 1. Structural Classification: Intermediate-duty system.
 - 2. End Condition of Cross Runners: Override (stepped) or butt-edge type.

2.6 METAL EDGE MOLDINGS AND TRIM

- A. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on Drawings.
- B. Extruded-Aluminum Edge Moldings and Trim: Where indicated, provide manufacturer's extruded-aluminum edge moldings and trim of profile indicated or referenced by manufacturer's designations, including splice plates, corner pieces, and attachment and other clips, complying with seismic design requirements and the following:
 - 1. Aluminum Alloy: Alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with not less than the strength and durability properties of aluminum extrusions complying with ASTM B 221 for Alloy and Temper 6063-T5.
 - 2. Clear Anodic Finish: AAMA 611, AA-M12C22A31, Class II, 0.010 mm or thicker.
 - 3. Baked-Enamel or Powder-Coat Finish: Minimum dry film thickness of 1.5 mils. Comply with ASTM C 635/C 635M and coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

2.7 ACOUSTICAL SEALANT

- A. Acoustical Sealant: Manufacturer's standard sealant complying with ASTM C 834 and effective in reducing airborne sound transmission through perimeter joints and openings in building construction as demonstrated by testing representative assemblies according to ASTM E 90.
 - 1. Exposed and Concealed Joints: Nonsag, paintable, nonstaining latex sealant.
 - 2. Concealed Joints: Nondrying, nonhardening, nonskinning, nonstaining, gunnable, synthetic-rubber sealant.

3. Acoustical sealant shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, including structural framing to which acoustical panel ceilings attach or abut, with Installer present, for compliance with requirements specified in this and other Sections that affect ceiling installation and anchorage and with requirements for installation tolerances and other conditions affecting performance of acoustical panel ceilings.
- B. Examine acoustical panels before installation. Reject acoustical panels that are wet, moisture damaged, or mold damaged.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Measure each ceiling area and establish layout of acoustical panels to balance border widths at opposite edges of each ceiling. Avoid using less-than-half-width panels at borders and comply with layout shown on reflected ceiling plans.

3.3 INSTALLATION

- A. General: Install acoustical panel ceilings to comply with ASTM C 636/C 636M and seismic design requirements indicated, according to manufacturer's written instructions and CISCA's "Ceiling Systems Handbook."
 1. Fire-Rated Assembly: Install fire-rated ceiling systems according to tested fire-rated design.
- B. Suspend ceiling hangers from building's structural members and as follows:
 1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structure or of ceiling suspension system.
 2. Splay hangers only where required to miss obstructions; offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
 3. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with location of hangers at spacings required to support standard suspension-system members, install supplemental suspension members and hangers in form of trapezes or equivalent devices.
 4. Secure wire hangers to ceiling-suspension members and to supports above with a minimum of three tight turns. Connect hangers directly either to structures or to inserts, eye screws, or other devices that are secure and appropriate for

- substrate and that will not deteriorate or otherwise fail due to age, corrosion, or elevated temperatures.
5. Secure flat, angle, channel, and rod hangers to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices that are secure and appropriate for both the structure to which hangers are attached and the type of hanger involved. Install hangers in a manner that will not cause them to deteriorate or fail due to age, corrosion, or elevated temperatures.
 6. Do not support ceilings directly from permanent metal forms or floor deck. Fasten hangers to postinstalled mechanical or adhesive anchors, or power-actuated fasteners that extend through forms into concrete.
 7. When steel framing does not permit installation of hanger wires at spacing required, install carrying channels or other supplemental support for attachment of hanger wires.
 8. Do not attach hangers to steel deck tabs.
 9. Do not attach hangers to steel roof deck. Attach hangers to structural members.
 10. Space hangers not more than 48 inches o.c. along each member supported directly from hangers unless otherwise indicated; provide hangers not more than 8 inches from ends of each member.
 11. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced standards and publications.
- C. Install edge moldings and trim of type indicated at perimeter of acoustical ceiling area and where necessary to conceal edges of acoustical panels.
1. Apply acoustical sealant in a continuous ribbon concealed on back of vertical legs of moldings before they are installed.
 2. Screw attach moldings to substrate at intervals not more than 16 inches o.c. and not more than 3 inches from ends, leveling with ceiling suspension system to a tolerance of 1/8 inch in 12 feet. Miter corners accurately and connect securely.
 3. Do not use exposed fasteners, including pop rivets, on moldings and trim.
- D. Install suspension-system runners so they are square and securely interlocked with one another. Remove and replace dented, bent, or kinked members.
- E. Install acoustical panels with undamaged edges and fit accurately into suspension-system runners and edge moldings. Scribe and cut panels at borders and penetrations to provide a neat, precise fit.
1. For square-edged panels, install panels with edges fully hidden from view by flanges of suspension-system runners and moldings.
 2. For reveal-edged panels on suspension-system runners, install panels with bottom of reveal in firm contact with top surface of runner flanges.
 3. For reveal-edged panels on suspension-system members with box-shaped flanges, install panels with reveal surfaces in firm contact with suspension-system surfaces and panel faces flush with bottom face of runners.
 4. Paint cut edges of panel remaining exposed after installation; match color of exposed panel surfaces using coating recommended in writing for this purpose by acoustical panel manufacturer.
 5. Protect lighting fixtures and air ducts to comply with requirements indicated for fire-resistance-rated assembly.

3.4 CLEANING

- A. Clean exposed surfaces of acoustical panel ceilings, including trim, edge moldings, and suspension-system members. Comply with manufacturer's written instructions for cleaning and touchup of minor finish damage. Remove and replace ceiling components that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage.

END OF SECTION

SECTION 096105

MOISTURE VAPOR EMISSION AND ALKALINITY CONTROL

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Fluid-applied, resin-based, membrane-forming systems that control the moisture-vapor-emission rate of high-moisture, interior concrete slabs to prepare them for floor covering installation.

B. Related Requirements:

1. Pre-installation testing methods and quantities for each unique flooring product are specified within each individual flooring product Section in Division 09.

1.2 UNIT PRICES

- A. Refer to Section 01 22 00 "Unit Prices," for description of work under this Section affected by Unit Prices.**

1.3 DEFINITIONS

- A. MVE: Moisture vapor emission.**
- B. MVER: Moisture vapor emission rate.**

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site. Review methods and procedures related to installation including, but not limited to, the following:**
1. Review substrate conditions, moisture and pH test results, manufacturer's installation instructions, and warranty requirements.
 2. Document proceedings, including required corrective measures.

1.5 ACTION SUBMITTALS

- A. Product Data: For each type of product used in moisture vapor emission control system.**

- B. Product Schedule: For all floor areas to receive moisture vapor emission and alkalinity control system products. Use same room label and numbering designations indicated on Drawings.
 - 1. Distinguish on schedule between those areas required to receive cementitious surfacing over moisture vapor emission and alkalinity control system and those areas where not required.

1.6 INFORMATIONAL SUBMITTALS

- A. Qualification Data: Certificates indicating Installer of vapor emission control treatments is trained and certified or employed by treatment manufacturer.
- B. Product Test Reports: For each MVE-control system, for tests performed by a qualified testing agency.
- C. Preinstallation Test Reports: For alkalinity, calcium chloride, and relative humidity of concrete slabs.
- D. Sample Warranty: For manufacturer's warranty for moisture vapor emission control coating system and certificate of underwriter's coverage of manufacturer's warranty.

1.7 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Minimum five years' experience in manufacturing moisture vapor emission control systems.
- B. Installer Qualifications: Manufacturer of products or an entity that employs installers and supervisors who are trained and certified by manufacturer. Installer must have a minimum of five years' experience installing vapor emissions control systems.
- C. Testing Agency Qualifications: Moisture and pH testing shall be performed by an International Concrete Repair Institute (ICRI), Certified Concrete Slab Moisture Testing Technician – Grade 1.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in manufacturer's unopened containers fully identified with brand, type, grade, class and all other qualifying information.
- B. Deliver materials in accordance with manufacturer's written instructions and recommendations.
- C. Store materials in a dry, well-ventilated area at minimum 50 deg F and maximum 90 deg F.

1.9 COORDINATION AND SEQUENCING

- A. Coordinate testing agency to test concrete slabs not less than one week or more than 5 weeks prior to scheduled flooring installation.
 - 1. Apply treatment to areas with moisture vapor emission or relative humidity rates which exceed floor covering manufacturer's written limits, as determined by ASTM F 1869 and ASTM F 2170 testing.
- B. Coordinate with installation of floor coverings. Ensure flooring installation complies with vapor emission control system manufacturer's warranty requirements.

1.10 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in original packages and containers, with seals unbroken, bearing manufacturer's labels indicating directions for storage and mixing with other components.

1.11 FIELD CONDITIONS

- A. Environmental Limitations: Comply with MVE-control system manufacturer's written instructions for substrate and ambient temperatures, humidity, ventilation, and other conditions affecting system installation.
 - 1. Store system components in a temperature-controlled environment and protected from weather and at ambient temperature of not less than 65 deg F and not more than 85 deg F at least 48 hours before use.
 - 2. Maintain ambient temperature and relative humidity in installation areas within range recommended in writing by MVE-control system manufacturer, but not less than 65 deg F or more than 85 deg F and not less than 40 or more than 60 percent relative humidity, for 48 hours before installation, during installation, and for 48 hours after installation unless longer period is recommended in writing by manufacturer.
 - 3. Install MVE-control systems where concrete surface temperatures will remain a minimum of 5 deg F higher than the dew point for ambient temperature and relative humidity conditions in installation areas for 48 hours before installation, during installation, and for 48 hours after installation unless longer period is recommended in writing by manufacturer.

1.12 WARRANTY

- A. Manufacturer's Special Warranty: Manufacturer agrees to repair or replace components of treatment system, cementitious surfacings, floor covering materials, adhesives, and installation labor for same period resulting from moisture vapor emission related failure that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: 10 years from date of Substantial Completion.

- B. Warranty shall be underwritten by product liability insurance carrier having a minimum "A" rating from Best or equivalent rating system in the amount of \$1,000,000 per occurrence.
- C. Warranty shall guarantee moisture vapor and alkalinity emission rates to be at or below published requirements of floor covering manufacturers.
- D. Warranty shall not exclude concrete slabs containing silica or silicate compounds.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. MVE-Control System Capabilities: Capable of suppressing MVE without failure where installed on concrete that exhibits the following conditions:
 - 1. MVER: Maximum 25 lb of water/1000 sq. ft. when tested according to ASTM F1869.
 - 2. Relative Humidity: Maximum 100 percent when tested according to ASTM F2170 using in situ probes.
- B. Water-Vapor Transmission: Through MVE-control system, maximum 0.10 perm when tested according to ASTM E96/E96M.
- C. Tensile Bond Strength: For MVE-control system, greater than 200 psi with failure in the concrete according to ASTM D7234

2.2 MOISTURE VAPOR EMISSION AND ALKALINITY CONTROL SYSTEMS

- A. Products: Subject to compliance with requirements, provide one of the following:
 - 1. ARDEX Americas; ARDEX MC™ RAPID One-Coat Moisture Control System.
 - 2. Floor Seal Technology, Inc.; MES 100 Remedial Treatment.
 - 3. KOSTER American Corporation; VAP I 2000 UFS.
 - 4. Schönox; HPS North America, Inc.; Schönox EPA RAPID.
- B. MVE-Control System: ASTM F3010-qualified, fluid-applied, two-component, epoxy-resin, membrane-forming system; formulated for application on concrete substrates to reduce MVER to level required for installation of floor coverings indicated and acceptable to manufacturers of floor covering products indicated, including adhesives.
 - 1. Substrate Primer: Provide MVE-control system manufacturer's concrete-substrate primer if required for system indicated by substrate conditions.
 - 2. Cementitious Underlayment Primer: If required for subsequent installation of cementitious underlayment products, provide MVE-control system manufacturer's primer to ensure adhesion of products to MVE-control system.
 - 3. VOC Content: Provide coating with VOC content of 100 g/L or less.

2.3 ACCESSORIES

- A. Patching and Leveling Material: Moisture-, mildew-, and alkali-resistant product recommended in writing by MVE-control system manufacturer and with minimum of 3000-psi compressive strength after 28 days when tested according to ASTM C109/C109M.
- B. Crack-Filling Material: Resin-based material recommended in writing by MVE-control system manufacturer for sealing concrete substrate crack repair.
- C. Cementitious Underlayment: If required to maintain manufacturer's warranty, provide MVE-control system manufacturer's gypsum or hydraulic cement-based underlayment.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Installer present, for compliance with requirements for maximum moisture content, installation tolerances, and other conditions affecting performance of the Work
- B. Prepare written report, endorsed by Installer, listing conditions detrimental to performance.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.
 - 1. Installation of system indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Preinstallation Testing:
 - 1. Testing Agency: Engage a qualified testing agency to perform tests.
 - 2. Alkalinity Testing: Perform pH testing according to ASTM F710. Install MVE-control system in areas where pH readings are less than 7.0 and in areas where pH readings are greater than 8.5.
 - 3. Moisture Testing: Perform tests so that each test area does not exceed 1000 sq. ft., and perform no fewer than three tests in each installation area and with test areas evenly spaced in installation areas.
 - a. Anhydrous Calcium Chloride Test: ASTM F1869. Install MVE-control system in locations where concrete substrate MVER exceeds 3 lb of water/1000 sq. ft. in 24 hours.
 - b. Internal Relative Humidity Test: Using in situ probes, ASTM F2170. Install MVE-control system in locations where concrete substrates exhibit relative humidity level greater than 75 percent.
 - 4. Tensile-Bond-Strength Testing: For typical locations indicated to receive installation of MVE-control system, install minimum 100-sq. ft. area of MVE-

control system to prepared concrete substrate and test according to ASTM D7234.

- a. Proceed with installation only where tensile bond strength is greater than 200 psi with failure in the concrete.
- B. Concrete Substrates: Prepare and clean substrates according to MVE-control system manufacturer's written instructions to ensure adhesion of system to concrete.
1. Remove coatings and other substances that are incompatible with MVE-control system and that contain soap, wax, oil, or silicone, using mechanical methods recommended in writing by MVE-control system manufacturer. Do not use solvents.
 2. Provide concrete surface profile complying with ICRI 310.2R CSP 3 by shot blasting using apparatus that abrades the concrete surface with shot, contains the dispensed shot within the apparatus, and recirculates the shot by vacuum pickup.
 3. After shot blasting, repair damaged and deteriorated concrete according to MVE-control system manufacturer's written instructions.
 4. Protect substrate voids and joints to prevent resins from flowing into or leaking through them.
 5. Fill surface depressions and irregularities with patching and leveling material.
 6. Fill surface cracks, grooves, control joints, and other nonmoving joints with crack-filling material.
 7. Allow concrete to dry, undisturbed, for period recommended in writing by MVE-control system manufacturer after surface preparation, but not less than 24 hours.
 8. Before installing MVE-control systems, broom sweep and vacuum prepared concrete.
- C. Protect walls, floor openings, electrical openings, door frames, and other obstructions during installation.

3.3 INSTALLATION

- A. Install MVE-control system according to ASTM F3010 and manufacturer's written instructions to produce a uniform, monolithic surface free of surface deficiencies such as pin holes, fish eyes, and voids.
1. Install primers as required to comply with manufacturer's written instructions.
- B. Do not apply MVE-control system across substrate expansion, isolation, and other moving joints.
- C. Apply system, including component coats if any, in thickness recommended in writing by MVE-control system manufacturer for MVER indicated by preinstallation testing.
- D. Cure MVE-control system components according to manufacturer's written instructions. Prevent contamination or other damage during installation and curing processes.

- E. After curing, examine MVE-control system for surface deficiencies. Repair surface deficiencies according to manufacturer's written instructions.
- F. Install cementitious underlayment over cured membrane if required to maintain manufacturer's warranty and in thickness required to maintain the warranty.

3.4 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform installation inspections.
- B. Installation Inspections: Inspect substrate preparation and installation of system components to ensure compliance with manufacturer's written instructions and to ensure that a complete MVE-control system is installed without deficiencies.
 - 1. Verify that surface preparation meets requirements.
 - 2. Verify that component coats and complete MVE-control-system film thicknesses comply with manufacturer's written instructions.
 - 3. Verify that MVE-control-system components and installation areas that evidence deficiencies are repaired according to manufacturer's written instructions.
- C. MVE-control system will be considered defective if it does not pass inspections.

3.5 PROTECTION

- A. Protect MVE-control system from damage, wear, dirt, dust, and other contaminants before floor covering installation. Use protective methods and materials, including temporary coverings, recommended in writing by MVE-control system manufacturer.
- B. Do not allow subsequent preinstallation examination and testing for floor covering installation to damage, puncture, or otherwise compromise the MVE-control system membrane.

END OF SECTION

SECTION 096513.13
RUBBER BASE AND ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Rubber base.
 - 2. Rubber molding accessories.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For each exposed product and for each color and texture specified, not less than 12 inches long.

PART 2 - PRODUCTS

2.1 RUBBER BASE

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. AB; American Biltrite.
 - 2. Allstate Rubber Corp.
 - 3. Armstrong World Industries, Inc.
 - 4. Burke Mercer Flooring Products, Division of Burke Industries Inc.
 - 5. Flexco Flooring.
 - 6. Johnsonite; A Tarkett Company.
 - 7. Mondo Rubber International, Inc.
 - 8. Nora Systems, Inc.
 - 9. Roppe Corporation, USA.
 - 10. VPI, LLC, Floor Products Division.
- B. Product Standard: ASTM F 1861, Type TP (rubber, thermoplastic).
 - 1. Group: I (solid, homogeneous).
 - 2. Style and Location:
 - a. Style A, Straight: Provide in areas indicated.
 - b. Style B, Cove: Provide in areas indicated.
 - c. Style C, Butt to: Provide in areas indicated.

d. Style D, Sculptured: Provide in areas indicated.

1) Profile: As indicated.

- C. Thickness: 0.125 inch.
- D. Height: As indicated on Drawings.
- E. Lengths: Coils in manufacturer's standard length.
- F. Outside Corners: Preformed.
- G. Inside Corners: Job formed.
- H. Colors: As indicated on Drawings.

2.2 RUBBER MOLDING ACCESSORY

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Roppe Corporation, USA.
 - 2. VPI, LLC, Floor Products Division.
 - 3. Johnsonite, A Tarkett Company.
 - 4. Flexco Flooring.
- B. Profile and Dimensions: As indicated on Drawings.
- C. Locations: Provide rubber molding accessories in areas indicated.
- D. Colors and Patterns: As indicated on Drawings.

2.3 INSTALLATION MATERIALS

- A. Adhesives: Water-resistant type recommended by resilient-product manufacturer for resilient products and substrate conditions indicated.
 - 1. Adhesives shall have a VOC content of 50 g/L or less.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Prepare substrates according to manufacturer's written instructions to ensure adhesion of resilient products.

- B. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound; remove bumps and ridges to produce a uniform and smooth substrate.
- C. Do not install resilient products until they are the same temperature as the space where they are to be installed.
- D. Immediately before installation, sweep and vacuum clean substrates to be covered by resilient products.

3.2 RUBBER BASE INSTALLATION

- A. Comply with manufacturer's written instructions for installing resilient base.
- B. Apply resilient base to walls, columns, pilasters, casework and cabinets in toe spaces, and other permanent fixtures in rooms and areas where base is required.
- C. Install resilient base in lengths as long as practical without gaps at seams and with tops of adjacent pieces aligned.
- D. Tightly adhere resilient base to substrate throughout length of each piece, with base in continuous contact with horizontal and vertical substrates.
- E. Do not stretch resilient base during installation.
- F. On masonry surfaces or other similar irregular substrates, fill voids along top edge of resilient base with manufacturer's recommended adhesive filler material.
- G. Preformed Corners: Install preformed corners before installing straight pieces.
- H. Job-Formed Corners:
 - 1. Inside Corners: Use straight pieces of maximum lengths possible and form with returns not less than 4 inches in length.
 - a. Miter or cope corners to minimize open joints.

3.3 RUBBER ACCESSORY INSTALLATION

- A. Comply with manufacturer's written instructions for installing resilient accessories.
- B. Resilient Stair Accessories:
 - 1. Use stair-tread-nose filler to fill nosing substrates that do not conform to tread contours.
 - 2. Tightly adhere to substrates throughout length of each piece.
 - 3. For treads installed as separate, equal-length units, install to produce a flush joint between units.

- C. Resilient Molding Accessories: Butt to adjacent materials and tightly adhere to substrates throughout length of each piece. Install reducer strips at edges of floor covering that would otherwise be exposed.

3.4 CLEANING AND PROTECTION

- A. Comply with manufacturer's written instructions for cleaning and protecting resilient products.

END OF SECTION

SECTION 096519.23
SOLID VINYL TILE FLOORING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Solid vinyl floor tile.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For each type of floor tile. Include floor tile layouts, edges, columns, doorways, enclosing partitions, built-in furniture, cabinets, and cutouts.

1. Show details of special patterns.

C. Samples: Full-size units of each color and pattern of floor tile required.

1.3 CLOSEOUT SUBMITTALS

A. Maintenance data.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics: For resilient tile flooring, as determined by testing identical products according to ASTM E 648 or NFPA 253 by a qualified testing agency.

1. Critical Radiant Flux Classification: Class I, not less than 0.45 W/sq. cm.

2.2 SOLID VINYL FLOOR TILE

A. Products: Subject to compliance with requirements, provide products indicated on Drawings.

- B. Tile Standard: ASTM F 1700.
 - 1. Class: As indicated by product designations.
 - 2. Type: As indicated by product designations.
- C. Thickness: As indicated by product designations.
- D. Size: As indicated on Drawings.
- E. Colors and Patterns: As indicated on Drawings.

2.3 INSTALLATION MATERIALS

- A. Trowelable Leveling and Patching Compounds: Latex-modified, portland cement based or blended hydraulic-cement-based formulation provided or approved by floor tile manufacturer for applications indicated.
- B. Adhesives: Water-resistant type recommended by floor tile and adhesive manufacturers to suit floor tile and substrate conditions indicated.
 - 1. Adhesives shall have a VOC content of 50 g/L or less.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Prepare substrates according to floor tile manufacturer's written instructions to ensure adhesion of resilient products.
- B. Concrete Substrates: Prepare according to ASTM F 710.
 - 1. Verify that substrates are dry and free of curing compounds, sealers, and hardeners.
 - 2. Remove substrate coatings and other substances that are incompatible with adhesives and that contain soap, wax, oil, or silicone, using mechanical methods recommended by floor tile manufacturer. Do not use solvents.
 - 3. Alkalinity and Adhesion Testing: Perform tests recommended by floor tile manufacturer. Proceed with installation only after substrate alkalinity falls within range on pH scale recommended by manufacturer in writing, but not less than 5 or more than 9 pH.
 - 4. Moisture Testing: Proceed with installation only after substrates pass testing according to floor tile manufacturer's written recommendations, but not less stringent than the following:
 - a. Perform anhydrous calcium chloride test according to ASTM F 1869. Proceed with installation only after substrates have maximum moisture-vapor-emission rate of 3 lb of water/1000 sq. ft. in 24 hours.

- b. Perform relative humidity test using in situ probes according to ASTM F 2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level.
- C. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound; remove bumps and ridges to produce a uniform and smooth substrate.
- D. Do not install floor tiles until they are the same temperature as the space where they are to be installed.
- E. Immediately before installation, sweep and vacuum clean substrates to be covered by resilient floor tile.

3.2 FLOOR TILE INSTALLATION

- A. Comply with manufacturer's written instructions for installing floor tile.
- B. Lay out floor tiles from center marks established with principal walls, discounting minor offsets, so tiles at opposite edges of room are of equal width. Adjust as necessary to avoid using cut widths that equal less than one-half tile at perimeter.
 - 1. Lay tiles square with room axis.
- C. Match floor tiles for color and pattern by selecting tiles from cartons in the same sequence as manufactured and packaged, if so numbered. Discard broken, cracked, chipped, or deformed tiles.
 - 1. Lay tiles in pattern of colors and sizes indicated.
- D. Scribe, cut, and fit floor tiles to butt neatly and tightly to vertical surfaces and permanent fixtures including built-in furniture, cabinets, pipes, outlets, and door frames.
- E. Extend floor tiles into toe spaces, door reveals, closets, and similar openings. Extend floor tiles to center of door openings.
- F. Maintain reference markers, holes, and openings that are in place or marked for future cutting by repeating on floor tiles as marked on substrates. Use chalk or other nonpermanent marking device.
- G. Install floor tiles on covers for telephone and electrical ducts, building expansion-joint covers, and similar items in finished floor areas. Maintain overall continuity of color and pattern between pieces of tile installed on covers and adjoining tiles. Tightly adhere tile edges to substrates that abut covers and to cover perimeters.
- H. Adhere floor tiles to flooring substrates using a full spread of adhesive applied to substrate to produce a completed installation without open cracks, voids, raising and puckering at joints, telegraphing of adhesive spreader marks, and other surface imperfections.

3.3 CLEANING AND PROTECTION

- A. Comply with manufacturer's written instructions for cleaning and protecting floor tile.
- B. Protect solid vinyl tile flooring from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period.

END OF SECTION

SECTION 096536
STATIC-CONTROL RESILIENT FLOORING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Static-control, solid vinyl floor tile.

1.2 ACTION SUBMITTALS

- A. Product Data:** For each type of product.
- B. Shop Drawings:** For static-control, solid vinyl floor tile. Include floor-covering layouts, edges, columns, doorways, enclosing partitions, built-in furniture, cabinets, and cutouts.
- C. Samples:** For static-control, solid vinyl floor tile and in each color, pattern, and texture required.

1.3 INFORMATIONAL SUBMITTALS

- A. Product test reports.**
- B. Field quality-control reports.**

1.4 CLOSEOUT SUBMITTALS

- A. Maintenance data.**

1.5 QUALITY ASSURANCE

- A. Installer Qualifications:** A qualified installer who employs workers for this Project who are competent in installation techniques required by manufacturer for specified static-control resilient flooring.
1. Engage an installer who employs workers for this Project who are trained or certified by manufacturer for installation techniques required for specified products.

1.6 PROJECT CONDITIONS

- A. Maintain ambient temperatures in spaces to receive static-control resilient flooring within range recommended by manufacturer for period recommended in writing before installation, during installation, and after installation.
- B. Close spaces to traffic during static-control resilient flooring installation.
- C. Close spaces to traffic for period recommended in writing by manufacturer after static-control resilient flooring installation.
- D. Install static-control resilient flooring after other finishing operations, including painting, have been completed.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Static-Dissipative Properties: Provide static-control resilient flooring with static-control properties indicated as determined by testing identical products per test method indicated by an independent testing and inspecting agency.
 - 1. Electrical Resistance: Test per ASTM F 150 with 100-V applied voltage or ESD-STM-7.1.
 - a. Average greater than 1 megohm and less than or equal to 1000 megohms when installed floor coverings are tested surface to ground.
 - 2. Static Generation: Less than 300 V when tested per AATCC TM134 at 20 percent relative humidity with static-control footwear.
 - 3. Static Decay: 5000 to 0 V in less than 0.25 seconds when tested in accordance with FED-STD-101C, Method 4046.1.
- B. Critical Radiant Flux: 0.45 W/sq. cm or greater when tested in accordance with ASTM E648 or NFPA 253.

2.2 STATIC CONTROL, SOLID VINYL FLOOR TILE

- A. Static-Control, Solid Vinyl Floor Tile: Manufacturer's standard, high-density vinyl resin with smooth surface.
 - 1. Products: Subject to compliance with requirements, provide products indicated on Drawings.
 - 2. Thickness: Manufacturer's standard, but not less than 0.08 inch.
 - 3. Size: As indicated by product designations on Drawings.
 - 4. Colors and Patterns: As indicated on Drawings.
 - 5. Maintenance Floor Tiles: Special floor tiles inscribed "CONDUCTIVE FLOOR. DO NOT WAX."

2.3 INSTALLATION MATERIALS

- A. Trowelable Leveling and Patching Compounds: Latex-modified portland cement or blended hydraulic-cement-based formulation provided or approved by manufacturer for applications indicated.
- B. Static-Control Adhesive: Provided or approved by manufacturer; type that maintains electrical continuity of floor-covering system to ground connection.
 - 1. Adhesives shall have a VOC content of 50 g/L or less.
- C. Grounding Strips: Provided or approved by manufacturer; type and size that maintains electrical continuity of floor-covering system to ground connection.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Prepare substrates in accordance with manufacturer's written instructions and with oversight by manufacturer's representative to ensure successful installation of static-control resilient flooring and electrical continuity of floor-covering systems.
- B. Concrete Substrates: Prepare in accordance with ASTM F710.
 - 1. Verify that substrates are dry and free of curing compounds, sealers, and hardeners.
 - 2. Remove substrate coatings and other substances that are incompatible with floor-covering adhesives and that contain soap, wax, oil, or silicone, using mechanical methods recommended by manufacturer. Do not use solvents.
 - 3. Alkalinity and Adhesion Testing: Perform tests recommended in writing by manufacturer. Proceed with installation only after substrate alkalinity is not less than 6 or more than 8 pH unless otherwise recommended in writing by flooring manufacturer.
 - 4. Moisture Testing: Perform tests so that each test area does not exceed 1000 sq. ft., and perform no fewer than three tests in each installation area and with test areas evenly spaced in installation areas.
 - a. Anhydrous Calcium Chloride Test: ASTM F1869. Proceed with installation only after substrates have maximum moisture-vapor-emission rate of 3 lb of water/1000 sq. ft. in 24 hours.
 - b. Relative Humidity Test: Using in-situ probes, ASTM F2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement.
- C. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound and remove bumps and ridges to produce a uniform and smooth substrate.

- D. Do not install static-control resilient flooring until it is same temperature as space where it is to be installed.
 - 1. Move static-control resilient flooring and installation materials into spaces where they will be installed at least 48 hours in advance of installation.
- E. Sweep and vacuum substrates to be covered by static-control resilient flooring immediately before installation.

3.2 INSTALLATION, GENERAL

- A. Install static-control resilient flooring in accordance with manufacturer's written instructions and with oversight by manufacturer's representative.
- B. Extend grounding strips beyond perimeter of static-control resilient floor-covering surfaces to ground connections.
 - 1. For adhesively installed flooring, embed grounding strips in static-control adhesive.
- C. Scribe, cut, and fit static-control resilient flooring to butt neatly and tightly to vertical surfaces and permanent fixtures including built-in furniture, cabinets, pipes, outlets, and door frames.
 - 1. Extend static-control resilient flooring below built-in items and permanent, but movable, items that allow for a flexible layout where indicated on Drawings.
- D. Extend static-control resilient flooring into toe spaces, door reveals, closets, and similar openings.
- E. Extend static-control resilient flooring to center of door openings where flooring or color transitions occur.
- F. Maintain reference markers, holes, and openings that are in place or marked for future cutting by repeating on static-control resilient flooring as marked on substrates. Use chalk or other nonpermanent, nonstaining marking device.
- G. Adhesive Installation: Adhere static-control resilient flooring to substrates using a full spread of static-control adhesive applied to substrate to produce a completed installation without open cracks, voids, raising and puckering at joints, telegraphing of adhesive spreader marks, and other surface imperfections.

3.3 INSTALLATION OF FLOOR TILE

- A. Comply with manufacturer's written instructions for installing floor tile.

- B. Lay out floor tiles from center marks established with principal walls, discounting minor offsets, so floor tiles at opposite edges of room are of equal width. Adjust as necessary to avoid using cut widths that equal less than one-half floor tile at perimeter.
 - 1. Lay floor tiles square with room axis.
- C. Match floor tiles for color and pattern by selecting floor tiles from cartons in same sequence as manufactured and packaged if so numbered. Discard broken, cracked, chipped, or deformed floor tiles.
- D. In each space where conductive, solid vinyl floor tile is installed, install maintenance floor tile identifying conductive floor tile in locations approved by Architect.

3.4 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to test electrical resistance of static-control resilient flooring in accordance with ASTM F150 or ESD STM7.1 for compliance with requirements.
 - 1. Arrange for testing after the following:
 - a. Static-control adhesives have fully cured.
 - b. Static-control resilient flooring has stabilized to ambient conditions.
 - c. Ground connections are completed.
- B. Static-control resilient flooring will be considered defective if it does not pass tests and inspections.
- C. Prepare test and inspection reports.

3.5 CLEANING AND PROTECTION

- A. Comply with manufacturer's written instructions for cleaning and protection of static-control resilient flooring.
- B. Protect static-control resilient flooring from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period.
- C. Cover static-control resilient flooring and protect from rolling loads until Substantial Completion.

END OF SECTION

SECTION 096813

TILE CARPETING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes modular carpet tile.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings: For carpet tile installation, plans showing the following:
 - 1. Columns, doorways, enclosing walls or partitions, built-in cabinets, and locations where cutouts are required in carpet tiles.
 - 2. Carpet tile type, color, and dye lot.
 - 3. Type of subfloor.
 - 4. Type of installation.
 - 5. Pattern of installation.
 - 6. Pattern type, location, and direction.
 - 7. Pile direction.
 - 8. Type, color, and location of insets and borders.
 - 9. Type, color, and location of edge, transition, and other accessory strips.
 - 10. Transition details to other flooring materials.
- C. Samples: For each exposed product and for each color and texture required.

1.3 INFORMATIONAL SUBMITTALS

- A. Product test reports.
- B. Sample warranty.

1.4 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Certified by the International Certified Floorcovering Installers Association at the Commercial II certification level.

- B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for fabrication and installation.
 - 1. Build mockups of typical flooring area as indicated on Drawings.
 - 2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Owner specifically approves such deviations by Change Order.
 - 3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.6 WARRANTY

- A. Special Warranty for Carpet Tiles: Manufacturer agrees to repair or replace components of carpet tile installation that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 CARPET TILE

- A. Products: Subject to compliance with requirements, provide products indicated on Drawings.
- B. Color: As indicated on Drawings.
- C. Pattern: As indicated on Drawings.
- D. Primary Backing/Backcoating: Manufacturer's standard composite materials.
- E. Secondary Backing: Manufacturer's standard material.
- F. Module Size: As indicated on Drawings.
- G. Applied Treatments:
 - 1. Soil-Resistance Treatment: Manufacturer's standard treatment.
 - 2. Antimicrobial Treatment: Manufacturer's standard treatment that protects carpet tiles as follows:
 - a. Antimicrobial Activity: Not less than 2-mm halo of inhibition for gram-positive bacteria, not less than 1-mm halo of inhibition for gram-negative bacteria, and no fungal growth, according to AATCC 174.

H. Performance Characteristics:

1. Appearance Retention Rating: Heavy traffic, 3.0 minimum according to ASTM D 7330.
2. Critical Radiant Flux Classification: Not less than 0.45 W/sq. cm according to NFPA 253.
3. Dry Breaking Strength: Not less than 100 lbf according to ASTM D 2646.
4. Tuft Bind: Not less than 6.2 lbf according to ASTM D 1335.
5. Delamination: Not less than 3.5 lbf/in. according to ASTM D 3936.
6. Dimensional Tolerance: Within 1/32 inch of specified size dimensions, as determined by physical measurement.
7. Dimensional Stability: 0.2 percent or less according to ISO 2551 (Aachen Test).
8. Colorfastness to Crocking: Not less than 4, wet and dry, according to AATCC 165.
9. Colorfastness to Light: Not less than 4 after 60 AFU (AATCC fading units) according to AATCC 16, Option E.
10. Electrostatic Propensity: Less than 3.5 kV according to AATCC 134.

2.2 INSTALLATION ACCESSORIES

- A. Trowelable Leveling and Patching Compounds: Latex-modified, hydraulic-cement-based formulation provided or recommended by carpet tile manufacturer.
- B. Adhesives: Water-resistant, mildew-resistant, nonstaining, pressure-sensitive type to suit products and subfloor conditions indicated, that comply with flammability requirements for installed carpet tile, and are recommended by carpet tile manufacturer for releasable installation.
 1. Adhesives shall have a VOC content of 50 g/L or less.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Concrete Slabs:
 1. Moisture Testing: Perform tests so that each test area does not exceed 1000 sq. ft., and perform no fewer than three tests in each installation area and with test areas evenly spaced in installation areas.
 - a. Anhydrous Calcium Chloride Test: ASTM F 1869. Proceed with installation only after substrates have maximum moisture-vapor-emission rate of 3 lb of water/1000 sq. ft. in 24 hours.
 - b. Relative Humidity Test: Using in situ probes, ASTM F 2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement.

- c. Perform additional moisture tests recommended in writing by adhesive and carpet tile manufacturers. Proceed with installation only after substrates pass testing.

3.2 PREPARATION

- A. General: Comply with CRI's "CRI Carpet Installation Standards" and with carpet tile manufacturer's written installation instructions for preparing substrates indicated to receive carpet tile.
- B. Use trowelable leveling and patching compounds, according to manufacturer's written instructions, to fill cracks, holes, depressions, and protrusions in substrates. Fill or level cracks, holes and depressions 1/8 inch wide or wider, and protrusions more than 1/32 inch unless more stringent requirements are required by manufacturer's written instructions.
- C. Concrete Substrates: Remove coatings, including curing compounds, and other substances that are incompatible with adhesives and that contain soap, wax, oil, or silicone, without using solvents. Use mechanical methods recommended in writing by adhesive and carpet tile manufacturers.
- D. Metal Substrates: Clean grease, oil, soil and rust, and prime if recommended in writing by adhesive manufacturer. Rough sand painted metal surfaces and remove loose paint. Sand aluminum surfaces, to remove metal oxides, immediately before applying adhesive.
- E. Broom and vacuum clean substrates to be covered immediately before installing carpet tile.

3.3 INSTALLATION

- A. General: Comply with CRI's "CRI Carpet Installation Standard," Section 18, "Modular Carpet" and with carpet tile manufacturer's written installation instructions.
- B. Installation Method: As recommended in writing by carpet tile manufacturer.
- C. Maintain dye-lot integrity. Do not mix dye lots in same area.
- D. Cut and fit carpet tile to butt tightly to vertical surfaces, permanent fixtures, and built-in furniture including cabinets, pipes, outlets, edgings, thresholds, and nosings. Bind or seal cut edges as recommended by carpet tile manufacturer.
- E. Extend carpet tile into toe spaces, door reveals, closets, open-bottomed obstructions, removable flanges, alcoves, and similar openings.
- F. Maintain reference markers, holes, and openings that are in place or marked for future cutting by repeating on carpet tile as marked on subfloor. Use nonpermanent, nonstaining marking device.
- G. Install pattern parallel to walls and borders.

- H. Protect carpet tile against damage from construction operations and placement of equipment and fixtures during the remainder of construction period. Use protection methods indicated or recommended in writing by carpet tile manufacturer.

END OF SECTION

SECTION 099123
INTERIOR PAINTING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes surface preparation and the application of paint systems on the following interior substrates:
 - 1. Concrete.
 - 2. Concrete masonry units (CMU).
 - 3. Steel.
 - 4. Galvanized metal.
 - 5. Aluminum (not anodized or otherwise coated).
 - 6. Wood.
 - 7. Gypsum board.
 - 8. Cotton or canvas insulation covering.

1.2 DEFINITIONS

- A. Gloss Level 1: Not more than 5 units at 60 degrees and 10 units at 85 degrees, according to ASTM D 523.
- B. Gloss Level 2: Not more than 10 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.
- C. Gloss Level 3: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.
- D. Gloss Level 4: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.
- E. Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D 523.
- F. Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D 523.
- G. Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D 523.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product. Include preparation requirements and application instructions.
 - 1. Include Printout of current "MPI Approved Products List" for each product category specified, with the proposed product highlighted.

2. Indicate VOC content.

- B. Product List: Cross-reference to paint system and locations of application areas. Use same designations indicated on Drawings and in schedules. Include color designations.

1.4 QUALITY ASSURANCE

- A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

1. Architect will select one surface to represent surfaces and conditions for application of each paint system specified in Part 3.

- a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft..
- b. Other Items: Architect will designate items or areas required.

2. Final approval of color selections will be based on mockups.

- a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Architect at no added cost to Owner.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F.

- 1. Maintain containers in clean condition, free of foreign materials and residue.
- 2. Remove rags and waste from storage areas daily.

1.6 FIELD CONDITIONS

- A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F.

- B. Do not apply paints when relative humidity exceeds 85 percent; at temperatures less than 5 deg F above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- 1. Benjamin Moore & Co.

2. Coronado Paint; Benjamin Moore Company.
3. Dunn-Edwards Corporation.
4. Frazee Paint; Comex Group.
5. Glidden Professional.
6. Kelly-Moore Paint Company Inc.
7. Kwal Paint; Comex Group.
8. Parker Paint; Comex Group.
9. PPG Architectural Finishes, Inc.
10. Pratt & Lambert.
11. Sherwin-Williams Company (The).

2.2 PAINT, GENERAL

- A. MPI Standards: Provide products that comply with MPI standards indicated and that are listed in its "MPI Approved Products List."
- B. Material Compatibility:
 1. Provide materials for use within each paint system that are compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
 2. For each coat in a paint system, provide products recommended in writing by manufacturers of topcoat for use in paint system and on substrate indicated.
- C. VOC Content: Products shall comply with VOC limits of authorities having jurisdiction and, for interior paints and coatings applied at Project site, the following VOC limits, exclusive of colorants added to a tint base, when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 1. Flat Paints and Coatings: 50 g/L.
 2. Nonflat Paints and Coatings: 100 g/L.
 3. Nonflat – High-gloss coatings: 150 g/L.
 4. Dry-Fog Coatings: 150 g/L.
 5. Primers, Sealers, and Undercoaters: 100 g/L.
 6. Anticorrosive and Antirust Paints Applied to Ferrous Metals: 250 g/L.
 7. Zinc-Rich Industrial Maintenance Primers: 340 g/L.
 8. Pretreatment Wash Primers: 420 g/L.
 9. Floor Coatings: 100 g/L.
 10. Shellacs, Clear: 730 g/L.
 11. Shellacs, Pigmented: 550 g/L.
- D. Colors: As indicated in a color schedule on Drawings.

2.3 BLOCK FILLERS

- A. Block Filler, Latex, Interior/Exterior: MPI #4.

2.4 PRIMERS/SEALERS

- A. Primer Sealer, Latex, Interior: MPI #50.
- B. Primer, Alkali Resistant, Water Based: MPI #3.
- C. Primer Sealer, Interior, Institutional Low Odor/VOC: MPI #149.
- D. Primer, Latex, for Interior Wood: MPI #39.
- E. Wood-Knot Sealer: Sealer recommended in writing by topcoat manufacturer for use in paint systems indicated.

2.5 METAL PRIMERS

- A. Primer, Rust-Inhibitive, Water Based: MPI #107.
- B. Primer, Galvanized, Water Based: MPI #134.
- C. Primer, Quick Dry, for Aluminum: MPI #95.

2.6 WATER-BASED PAINTS

- A. Latex, Interior, Flat, (Gloss Level 1): MPI #53.
- B. Latex, Interior, (Gloss Level 3): MPI #52.
- C. Latex, Interior, (Gloss Level 4): MPI #43.
- D. Latex, Interior, Semi-Gloss, (Gloss Level 5): MPI #54.
- E. Latex, Interior, Gloss, (Gloss Level 6, except minimum gloss of 65 units at 60 degrees): MPI #114.
- F. Latex, Interior, Institutional Low Odor/VOC, Flat (Gloss Level 1): MPI #143.
- G. Latex, Interior, Institutional Low Odor/VOC, (Gloss Level 3): MPI #145.
- H. Latex, Interior, Institutional Low Odor/VOC, Semi-Gloss (Gloss Level 5): MPI #147.
- I. Latex, Interior, High Performance Architectural, (Gloss Level 4): MPI #140.
- J. Latex, Interior, High Performance Architectural, Semi-Gloss (Gloss Level 5): MPI #141.
- K. Light Industrial Coating, Interior, Water Based (Gloss Level 3): MPI #151.
- L. Light Industrial Coating, Interior, Water Based, Semi-Gloss (Gloss Level 5): MPI #153.

2.7 TEXTURED COATING

- A. Textured Coating, Latex, Flat: MPI #42.

2.8 DRY FOG/FALL COATINGS

- A. Dry Fall, Latex, Flat: MPI #118.
- B. Dry Fall, Water Based, for Galvanized Steel, Flat (Gloss Level 1): MPI #133.

2.9 FLOOR COATINGS

- A. Stain, Interior, for Concrete Floors: MPI #58.
- B. Floor Paint, Latex, Low Gloss (Maximum Gloss Level 3): MPI #60.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:
 - 1. Concrete: 12 percent.
 - 2. Masonry (Clay and CMU): 12 percent.
 - 3. Wood: 15 percent.
 - 4. Gypsum Board: 12 percent.
- C. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.
- D. Proceed with coating application only after unsatisfactory conditions have been corrected.
 - 1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions and recommendations in "MPI Manual" applicable to substrates indicated.
- B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of

size or weight of item, provide surface-applied protection before surface preparation and painting.

1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.
- C. Clean substrates of substances that could impair bond of paints, including dust, dirt, oil, grease, and incompatible paints and encapsulants.
1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce paint systems indicated.

3.3 APPLICATION

- A. Apply paints according to manufacturer's written instructions and to recommendations in "MPI Manual."
- B. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

3.4 CLEANING AND PROTECTION

- A. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.
- B. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.5 INTERIOR PAINTING SCHEDULE

- A. Concrete Substrates, Nontraffic Surfaces – MPI INT 3.1A:
1. Latex System – Eggshell Finish:
 - a. Prime Coat: Primer, alkali resistant, water based, MPI #3.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, (Gloss Level 3), MPI #52.
 2. Latex System – Semi-Gloss Finish:
 - a. Prime Coat: Primer, alkali resistant, water based, MPI #3.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, semi-gloss, (Gloss Level 5), MPI #54.

B. Concrete Substrates, Traffic Surfaces – MPI INT 3.2A:

1. Latex Floor Enamel System:
 - a. Prime Coat: Floor paint, latex, low gloss (maximum Gloss Level 3), MPI #60.
 - b. Intermediate Coat: Floor paint, latex, low gloss (maximum Gloss Level 3), MPI #60.
 - c. Topcoat: Floor paint, latex, low gloss (maximum Gloss Level 3), MPI #60.
2. Concrete Stain System:
 - a. First Coat: Stain, interior, for concrete floors, MPI #58.
 - b. Topcoat: Stain, interior, for concrete floors, MPI #58.
3. Clear Concrete Sealer System – Traffic-Exposed:
 - a. Preparation: Prepare concrete surfaces in accordance with MPI requirements for concrete floors, including removal of curing compounds, sealers, laitance, and contaminants; provide surface profile appropriate for sealer type.
 - b. Sealer: Clear, non-yellowing system, one of the following:
 - 1) Penetrating/Densifier: Silane, siloxane, or lithium/sodium silicate type, breathable, MPI #36.
 - 2) Film-Forming Sealer: Acrylic or polyurethane, clear, MPI #39.
 - c. Number of Coats: As recommended in writing by manufacturer to achieve uniform coverage and performance.
 - d. Finish: Clear, matte to satin unless otherwise indicated.
 - e. Performance Requirements:
 - 1) Suitable for light to moderate foot traffic.
 - 2) Resist dusting, staining, and moisture intrusion.
 - 3) Non-yellowing.
 - 4) VOC compliant.
 - 5) Maintain slip resistance; provide slip-resistant additive where required.

C. Concrete Substrates, Deck or Ceiling Above – MPI INT 3.1N

1. Latex Aggregate System – Flat Finish:
 - a. Prime Coat: As recommended in writing by topcoat manufacturer.
 - b. Intermediate Coat: As recommended in writing by topcoat manufacturer.
 - c. Topcoat: Textured coating, latex, flat, MPI #42.

D. CMU Substrates – MPI INT 4.2A:

1. Latex System – Eggshell Finish:
 - a. Block Filler: Block filler, latex, interior/exterior, MPI #4.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, (Gloss Level 3), MPI #52.
2. Latex System – Semi-Gloss Finish:
 - a. Block Filler: Block filler, latex, interior/exterior, MPI #4.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, semi-gloss, (Gloss Level 5), MPI #54.

E. Steel Substrates, Hollow Metal Doors & Frames – MPI INT 5.1B:

1. Water-Based Light Industrial Coating System – Eggshell Finish:
 - a. Prime Coat: Primer, rust-inhibitive, water based MPI #107.
 - b. Intermediate Coat: Light industrial coating, interior, water based, matching topcoat.
 - c. Topcoat: Light industrial coating, interior, water based (Gloss Level 3), MPI #151.
2. Water-Based Light Industrial Coating System – Semi-Gloss Finish:
 - a. Prime Coat: Primer, rust-inhibitive, water based MPI #107.
 - b. Intermediate Coat: Light industrial coating, interior, water based, matching topcoat.
 - c. Topcoat: Light industrial coating, interior, water based, semi-gloss (Gloss Level 5), MPI #153.

F. Steel Substrates, Metal Deck Above – MPI INT 5.1C:

1. Water-Based Dry-Fall System:
 - a. Prime Coat: Shop primer specified in Section where substrate is specified.
 - b. Topcoat for Shop Primed or Previously Painted Steel: Dry fall, latex, flat, MPI #118.

G. Steel Substrates, Other – MPI INT 5.1R:

1. High-Performance Architectural Latex System – Semi-Gloss:
 - a. Prime Coat: Shop primer specified in Section where substrate is specified.
 - b. Intermediate Coat: Latex, interior, high performance architectural, matching topcoat.
 - c. Topcoat: Latex, interior, high performance architectural, (Gloss Level 4), MPI #140.

2. High-Performance Architectural Latex System – Gloss:
 - a. Prime Coat: Shop primer specified in Section where substrate is specified.
 - b. Intermediate Coat: Latex, interior, high performance architectural, matching topcoat.
 - c. Topcoat: Latex, interior, high performance architectural, semi-gloss (Gloss Level 5), MPI #141.
- H. Galvanized-Metal Substrates, Hollow Metal Doors & Frames – MPI INT 5.3K:
 1. Water-Based Light Industrial Coating Over Waterborne Primer System – Eggshell Finish:
 - a. Prime Coat: Primer, galvanized, water based, MPI #134.
 - b. Intermediate Coat: Light industrial coating, interior, water based, matching topcoat.
 - c. Topcoat: Light industrial coating, interior, water based (Gloss Level 3), MPI #151.
 2. Water-Based Light Industrial Coating Over Waterborne Primer System – Semi-Gloss Finish:
 - a. Prime Coat: Primer, galvanized, water based, MPI #134.
 - b. Intermediate Coat: Light industrial coating, interior, water based, matching topcoat.
 - c. Topcoat: Light industrial coating, interior, water based, semi-gloss (Gloss Level 5), MPI #153.
- I. Galvanized-Metal Substrates, Metal Deck Above – MPI 5.3H:
 1. Water-Based Dry-Fall System:
 - a. Prime Coat: Dry fall, water based, for galvanized steel, flat (Gloss Level 1), MPI #133.
 - b. Topcoat: Dry fall, water based, for galvanized steel, flat (Gloss Level 1), MPI #133.
- J. Galvanized-Metal Substrates, Other – MPI INT 5.3M:
 1. High-Performance Architectural Latex System – Semi-Gloss:
 - a. Prime Coat: Primer, galvanized, water based, MPI #134.
 - b. Intermediate Coat: Latex, interior, high performance architectural, matching topcoat.
 - c. Topcoat: Latex, interior, high performance architectural, (Gloss Level 4), MPI #140.
 2. High-Performance Architectural Latex System – Gloss:
 - a. Prime Coat: Primer, galvanized, water based, MPI #134.

- b. Intermediate Coat: Latex, interior, high performance architectural, matching topcoat.
 - c. Topcoat: Latex, interior, high performance architectural, semi-gloss (Gloss Level 5), MPI #141.
- K. Aluminum (Not Anodized or Otherwise Coated) Substrates – MPI INT 5.4F:
 - 1. High-Performance Architectural Latex System – Semi-Gloss:
 - a. Prime Coat: Primer, quick dry, for aluminum, MPI #95.
 - b. Intermediate Coat: Latex, interior, high performance architectural, matching topcoat.
 - c. Topcoat: Latex, interior, high performance architectural, (Gloss Level 4), MPI #140.
 - 2. High-Performance Architectural Latex System – Gloss:
 - a. Prime Coat: Primer, quick dry, for aluminum, MPI #95.
 - b. Intermediate Coat: Latex, interior, high performance architectural, matching topcoat.
 - c. Topcoat: Latex, interior, high performance architectural, semi-gloss (Gloss Level 5), MPI #141.
- L. Wood Substrates: Including wood trim, architectural woodwork, doors, and wood-based panel products – MPI INT 6.3T:
 - 1. Latex System – Satin Finish:
 - a. Prime Coat: Primer, latex, for interior wood, MPI #39.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, (Gloss Level 4), MPI #43.
 - 2. Latex System – Semi-Gloss Finish:
 - a. Prime Coat: Primer, latex, for interior wood, MPI #39.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, semi-gloss, (Gloss Level 5), MPI #54.
 - 3. Latex System – Gloss Finish:
 - a. Prime Coat: Primer, latex, for interior wood, MPI #39.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, gloss, (Gloss Level 6, except minimum gloss of 65 units at 60 degrees), MPI #114.
- M. Gypsum Board Substrates, Walls & Ceilings – MPI INT 9.2M:
 - 1. Institutional Low-Odor/VOC Latex System – Flat Finish:
 - a. Prime Coat: Primer sealer, interior, institutional low odor/VOC, MPI #149.

- b. Intermediate Coat: Latex, interior, institutional low odor/VOC, matching topcoat.
 - c. Topcoat: Latex, interior, institutional low odor/VOC, flat (Gloss Level 1), MPI #143.
- 2. Institutional Low-Odor/VOC Latex System – Eggshell Finish:
 - a. Prime Coat: Primer sealer, interior, institutional low odor/VOC, MPI #149.
 - b. Intermediate Coat: Latex, interior, institutional low odor/VOC, matching topcoat.
 - c. Topcoat: Latex, interior, institutional low odor/VOC, (Gloss Level 3), MPI #145.
- 3. Institutional Low-Odor/VOC Latex System – Semi-Gloss Finish:
 - a. Prime Coat: Primer sealer, interior, institutional low odor/VOC, MPI #149.
 - b. Intermediate Coat: Latex, interior, institutional low odor/VOC, matching topcoat.
 - c. Topcoat: Latex, interior, institutional low odor/VOC, semi-gloss (Gloss Level 5), MPI #147.
- N. Cotton or Canvas Insulation-Covering Substrates: Including pipe and duct coverings – MPI INT 10.1A:
 - 1. Latex System – Flat:
 - a. Prime Coat: Primer sealer, latex, interior, MPI #50.
 - b. Intermediate Coat: Latex, interior, matching topcoat.
 - c. Topcoat: Latex, interior, flat (MPI Gloss Level 1), MPI #53.

END OF SECTION

SECTION 09 96 53
ELASTOMERIC COATINGS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Surface preparation and application of elastomeric coatings to the following exterior substrates:
1. Concrete.
 2. Concrete unit masonry.

1.2 ACTION SUBMITTALS

- A. Product Data: Technical data and product information for block fillers, primers, paints, and coatings, including label analysis and instructions for handling, storing, surface preparation, and application for each paint and coating system.
1. For field painting of factory primed metal products and fabrications, submit technical data for each type of paint product, surface preparation requirements, and application instructions.
 2. Indicate manufacturer's instructions for special surface preparation procedures and substrate conditions requiring special attention.
 3. Product List: Provide inclusive list of required coating systems and materials. Indicate each material and cross reference specific coatings, finish system, and application. Identify each material by manufacturer's catalog number, series, and general classification. Use same designations indicated in Finish Schedules.
- B. Samples: Submit for each type of elastomeric coating indicated and each color and gloss.
1. Submit Samples on same type of substrate as that to receive application, 8 inches (200 mm) square.
 2. Apply coats on Samples in steps to show each separate coat, including primers and block fillers as applicable.
 3. Label each coat of each Sample.
 4. Label each Sample for location and application area.
- C. Product List: Cross reference to paint and coating system and locations of application areas. Use same designations indicated on Drawings and in schedules. Include color designations.

1.3 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with Federal and local toxicity and air quality regulations and with Federal requirements on content of for heavy metals including but not limited to: lead and mercury. Do not use solvents in paint products that contribute to air pollution.
- B. Applicator Qualifications: Entity having minimum 5 years documented experience in applying paints and coatings similar in material, design, and extent to those indicated.
- C. Source Limitations: Obtain cementitious coating materials from single source and from single manufacturer.
- D. The Contractor shall coordinate the Project Schedule to allow compliance with the required application temperatures for the specific products specified, without substitution. Unspecified products that do not meet all criteria of this Section are subject to rejection.
 - 1. Lower Temperature Application Option: Refer to Part 2 for acceptable products for elastomeric coatings for application in low temperature conditions.
- E. Mockups: Apply mockups of each coating system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Architect will select one surface to represent surfaces and conditions for application of each coating system.
 - a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft (9 sq. m).
 - b. Other Items: Architect will designate items or areas required.
 - 2. Final approval of color selections will be based on mockups.
 - a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Architect at no added cost to Owner.
 - 3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
 - 4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.4 COORDINATION

- A. Coordination of Work: Coordinate field finishing of shop primed metals are provided to ensure compatibility of total systems for various substrates.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 degrees F (7 degrees C).
 - 1. Protect elastomeric coating materials from freezing or becoming hot.
 - 2. Maintain containers in clean condition, free of foreign materials and residue.

1.6 FIELD CONDITIONS

- A. Do not thin or add water to waterbased coatings, including waterbased alkyds.
- B. Weather Conditions:
 - 1. Apply coatings when temperature of surfaces to be treated and ambient air temperatures are between 50 degrees F and 95 degrees F (10 degrees C and 35 degrees C).
 - 2. Do not apply coatings when relative humidity exceeds 85 percent; at temperatures less than 5 degrees F (3 degrees C) above the dew point; or to damp or wet surfaces.
 - 3. Minimum Application Temperatures for Latex Coatings: 45 degrees F (7 degrees C) for interiors; 50 degrees F (35 degrees C) for exterior, unless otherwise indicated by manufacturer's Product Data Sheet.
- C. Lower Temperature Option: Refer to Part 2 for an option for lower temperature application. Installation shall be by a single product for the entire Project.
- D. Coating application may continue during inclement weather if surfaces and areas to receive paint and coatings are enclosed and heated within temperature limits specified by the manufacturer during application and drying periods.
- E. Provide lighting level of 80 foot candles (860 lx) measured midheight at substrate surface.
- F. Labels: Do not apply coatings over Underwriters Laboratories, Factory Mutual, other code required labels, or equipment name, identification, performance rating, or nomenclature plates.

1.7 WARRANTY

- A. Written warranty signed by manufacturer in which manufacturer agrees to repair or replace elastomeric coatings that fail within specified warranty period.
 - 1. Failures include, but are not limited to, the following:
 - a. Water penetration through the coating.
 - b. Deterioration of coating beyond normal weathering.
 - 2. Warranty Period: 10 years from date of Substantial Completion.

1.8 EXTRA MATERIALS

- A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Quantity: Furnish an additional 5 percent but not less than 1 gal. (3.8 L) of each material, color, and texture applied.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Performance and Durability:
 - 1. ASTM D 16 Standard Test Method for Load Testing Refractory Shapes at High Temperatures.
 - 2. ASTM D 412 Rubber Properties in Tension.
 - 3. ASTM D 16 53 Test Methods for Water Vapor Transmission of Organic Coating Films.
 - 4. ASTM D 2486 Standard Test Method for Scrub Resistance of Interior Wall Paint.
 - 5. ASTM D 2805 Standard Test Method for Hiding Power of Paints by Reflectometry.
 - 6. ASTM D 4828 Standard Test Method for Practical Washability of Organic Coatings.
- B. Material Compatibility:
 - 1. Materials for use within each coating system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
 - 2. For each coat in a coating system, products shall be recommended in writing by manufacturers of topcoat for use in paint system and on substrate indicated.
- C. Chemical Components of Interior Paints and Coatings: Provide products complying with limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24). These requirements do not apply to paints and coatings applied in a fabrication or finishing shop.
 - 1. Aromatic Compounds: Paints and coatings shall not contain more than 1.0 percent by weight of total aromatic compounds (hydrocarbon compounds containing one or more benzene rings).
 - 2. Restricted Components: Paints and coatings shall not contain components restricted by the EPA.

2.2 MATERIALS

- A. Basis of Design: Conflex ConFlex XL Smooth Elastomeric High Build Coating A5-800 Series by Sherwin Williams; Subject to compliance with requirements, products by one of the following may also be acceptable:
 - 1. BASF Building Systems: MasterProtect EL 750.
 - 2. Kelly-Moore: Kelly-Moore Premium; Kel-Seal Urethane Modified Acrylic Elastomeric Coating DU0080920.
 - 3. PPG: Perma-Crete Pitt-Flex Elastomeric Coating 4-110.
 - 4. Pratt & Lambert: Pro-Hide Gold; Exterior Elastomeric Coating Z6000.
 - 5. Sto: StoSilco Lastic; StoSilco Lastic 80222.
- B. Substitutions: Not Permitted.
 - 1. Refer to the Lower Application Temperature Option article of this Section for the sole solution for applications in lower temperatures. Installation shall be by a single product for the entire Project.
- C. Elastomeric Coating: Factory formulated, 100 percent acrylic or silicone emulsion finish coating materials, compatible with substrate, filler, and primer required.
 - 1. Physical Properties:
 - a. Elongation at Break: Not less than 240 percent when tested according to ASTM D 412 or ASTM D 2370.
 - b. Low Temperature Flexibility: Passes 1/8 inch, 180 degree mandrel bend at minus 10 degrees F at 20 mil dry film thickness when tested according to ASTM D 552.
 - c. Tensile Strength: Not less than 220 psi when tested according to ASTM D 412 or ASTM D 2370.
 - d. Water Vapor Permeability: Not less than 10 perms when tested according to ASTM D 1653 or ASTM E 96.
 - e. Wind Driven Rain Test: No water penetration when tested according to ASTM D 6904.
 - f. Minimum Dry Film Thickness per Coat: Not less than 10 mils.
 - g. Minimum Solids Content by Volume: Not less than 45 percent.
- D. Moisture Vapor Transmission: Minimum 34.4 perms, based on testing according to ASTM D1653.
- E. Texture: Medium.
- F. Colors: Selected by Architect from manufacturer's standard range.
- G. Crack Fillers: Elastomeric coating manufacturer's recommended, factory formulated crack fillers or sealants, including crack filler primers, compatible with substrate and other materials indicated.
- H. Primer: Elastomeric coating manufacturer's recommended, factory formulated, alkali resistant primer compatible with substrate and other materials indicated.

- I. Concrete Unit Masonry Block Filler: Elastomeric coating manufacturer's recommended, factory formulated, high performance latex block filler compatible with substrate and other materials indicated.

2.3 LOWER APPLICATION TEMPERATURE OPTION

- A. The Basis of Design and other listed products are susceptible to freezing and shall not be applied outside the indicated temperature conditions. The only acceptable option for lower application temperatures is as included in this Article. Installation shall be by a single product for the entire Project; exercising this option requires the same material to be applied for all elastomeric coating areas of the Project.
 1. Silicone Elastomeric Coating: Single component pigmented silicone elastomer.
 2. Coating properties meeting the following:
 - a. Solids Content: 58.6 percent by weight; 50.1 percent by volume when tested in accordance with ASTM D 2369.
 - b. Specific Gravity: 9.64 lb / gallon when tested in accordance with ASTM D 1475.
 - c. Durometer Hardness, Shore A: 38 points when tested in accordance with ASTM D 2240.
 - d. Tensile Strength: >145 psi when tested in accordance with ASTM D 412.
 - e. Elongation: 600 Percent when tested in accordance with ASTM D 412.
 - f. Permeance: 43.2 English Perms per ASTM D 1653.
 - g. Fungus Resistance: No growth when tested in accordance with ASTM D 3274.
 - h. Wind Driven Rain: Pass when tested in accordance with ASTM D 6904.
 - i. Freeze/Thaw Resistance: No change when tested in accordance with ASTM D 2243.
 3. Basis of Design: DOWSIL AllGuard Silicone Elastomeric Coating. May be applied only within the temperature limitations indicated in the manufacturer's published literature.

2.4 SOURCE QUALITY CONTROL

- A. Testing of Paints and Coating Materials: Owner reserves the right to invoke the following procedure:
 1. Owner will engage the services of a qualified testing agency to sample coating materials. Contractor will be notified in advance and may be present when samples are taken. If coating materials have already been delivered to project site, samples may be taken at project site. Samples will be identified, sealed, and certified by testing agency.
 2. Testing agency will perform tests for compliance with product requirements.
 3. Owner may direct Contractor to stop applying coatings if test results show materials being used do not comply with product requirements. Contractor shall remove noncomplying paint and coating materials from project site, pay for testing, and recoat surfaces painted with rejected materials. Contractor will be

required to remove rejected materials from previously painted or coated surfaces if, on repainting and recoating with complying materials, the two paint and coating products are incompatible.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions for compliance with manufacturer's requirements for maximum moisture content, alkalinity, and conditions affecting performance of work.
- B. Test substrates for moisture and alkalinity after repairing and cleaning substrates but prior to application of paint and coatings.
 - 1. Commence coating when moisture content of substrate is 12 percent or less when measured with an electronic moisture meter.
- C. Begin coating no sooner than 28 days after substrate is constructed and is visually dry on both sides.
- D. Verify substrate is within the range of alkalinity recommended by manufacturer.
- E. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.
- F. Verify suitability of substrates, including surface conditions and compatibility, with existing finishes and primers.
 - 1. Verify previously painted surfaces can be stripped to bare substrate, repaired if necessary, and prepared to receive new paint or coating system consisting of primer and two top coats at a minimum.
 - a. Where previously painted surfaces failed to accept new coating systems, determine cause of failure and take corrective measures to ensure each surface accepts new paint or coating system. Failure of new paint or coating systems is not permitted.
- G. Commence coating application after correcting unsatisfactory conditions and surfaces are dry.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions applicable to substrates and coating systems indicated.
- B. Remove hardware and hardware accessories, plates, machined surfaces, light fixtures, and similar items already installed that are not to be coated. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and coating.

1. After completing coating operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface applied protection if any.
- C. Clean substrates of substances that could impair bond of coatings, including dirt, oil, grease, and incompatible paints, coatings and encapsulants. Do not coat surfaces if moisture content or alkalinity of surfaces to be coated exceeds that permitted in manufacturer's written instructions.
1. Remove incompatible primers and reprime substrate with compatible primers as required to produce coating systems indicated.
 2. Perform cleaning and coating application so dust and other contaminants from cleaning process will not fall on wet, newly coated surfaces.
 3. Cementitious Surfaces: Prepare surfaces of concrete, concrete masonry, stucco, and similar surfaces to receive elastomeric coatings by removing efflorescence, chalk, dust, dirt, release agents, grease, oils, and similar conditions by water blasting followed by a clear water rinse.
 4. Remove mildew and neutralize surfaces according to manufacturer's recommendations before materials are applied.
 5. Determine alkalinity and moisture content of surfaces to be coated by testing. Do not apply coatings over surfaces where moisture content exceeds that permitted in manufacturer's printed directions.
- D. Crack Repair: Fill cracks according to manufacturer's written instructions before coating surfaces.

3.3 APPLICATION

- A. Apply elastomeric coatings according to manufacturer's written instructions.
1. Use equipment and techniques best suited for substrate and type of material being applied.
 2. Coat surfaces behind movable items the same as similar exposed surfaces.
 3. Apply each coat separately according to manufacturer's written instructions.
- B. Primers: Apply at a rate to ensure complete coverage.
- C. Block Fillers: Apply at a rate to ensure complete coverage with pores filled.
- D. Elastomeric Finish Coat(s): Minimum two coats with a total dry film thickness of 16 to 18 mils (0.41 to 0.45 mm).
- E. Tint each undercoat a lighter shade to facilitate identification of each coat if multiple coats of same material are to be applied. Tint undercoats similar to color of topcoat but provide sufficient difference in shade of undercoats to distinguish each separate coat.
- F. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform finish, color, and appearance.

- G. Apply coatings to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.
- H. Apply coatings to prepared surfaces as soon as practicable after preparation and before subsequent surface soiling or deterioration.
- I. Spray Application: Use spray equipment for application when permitted by authorities having jurisdiction. Wherever spray application is used, do not double back with spray equipment to build up film thickness of two coats in one pass.

3.4 FIELD QUALITY CONTROL

- A. Testing of Paint and Coating Materials: Owner reserves the right to invoke the following testing procedures:
 - 1. Owner will engage the services of a qualified testing agency to sample materials being used. Samples of material delivered to Project site will be taken, identified, sealed, and certified in presence of Contractor.
 - 2. Testing agency will perform the following tests for compliance of materials with product requirements.
 - a. Quantitative materials analysis.
 - b. Elongation at break.
 - c. Tensile strength.
 - d. Percent of recovery.
 - e. Resistance to wind driven rain.
 - f. Water vapor transmission.
 - 3. Owner may direct Contractor to stop coating application if test results show materials being used do not comply with requirements. Remove noncomplying materials from Project site, pay for testing, and recoat surfaces that were coated with rejected materials. Remove rejected materials from previously coated surfaces if, on recoating with complying materials, the two coatings are incompatible.
- B. Field Testing and Inspection: Owner reserves the right to engage the services of a qualified testing agency to verify installed thickness of elastomeric coatings.

3.5 CLEANING AND PROTECTION

- A. At end of each day, remove rubbish, empty cans, rags, and discarded materials from project site.
- B. After completing coating application, clean spattered surfaces. Remove spattered coatings by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

- C. Protect work of other trades against damage from coating application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.
- D. At completion of construction activities, touch up and restore damaged or defaced coated surfaces.

3.6 COATING SCHEDULE

- A. Concrete: Two finish coats over primer with total dry film thickness not less than 20 mils.
 - 1. Primer: Manufacturer's recommended primer.
 - 2. First Coat: Elastomeric coating.
 - 3. Second Coat: Elastomeric coating.
- B. Concrete Masonry (CMU): Two finish coats over block filler with a total dry film thickness not less than 20 mils, excluding block filler.
 - 1. Block Filler: Manufacturer's recommended masonry block filler.
 - 2. First Coat: Elastomeric coating.
 - 3. Second Coat: Elastomeric coating.

END OF SECTION

SECTION 102113.13
METAL TOILET COMPARTMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes painted steel toilet compartments configured as toilet enclosures and urinal screens.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings: For toilet compartments. Include plans, elevations, sections, and attachment details.
- C. Samples for each type of toilet compartment material indicated.

1.3 CLOSEOUT SUBMITTALS

- A. Maintenance data.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Regulatory Requirements: Comply with applicable provisions in the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines for Buildings and Facilities and ICC A117.1 for toilet compartments designated as accessible.

2.2 PAINTED STEEL TOILET COMPARTMENTS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Accurate Partitions Corporation.
 - 2. Bradley Corporation; Mills Partitions.
 - 3. Flush Metal Partition Corp.
 - 4. General Partitions Mfg. Corp.
 - 5. Global Steel Products Corp.
 - 6. Hadrian Manufacturing Inc.

7. Knickerbocker Partition Corporation.
 8. Metpar Corp.
- B. Toilet-Enclosure Style: From the following options, as indicated on Drawings:
1. Overhead braced.
 2. Floor anchored.
 3. Ceiling hung.
 4. Floor and ceiling anchored.
- C. Urinal-Screen Style: Wall hung, flat panel.
- D. Door, Panel, and Pilaster Construction: Seamless, metal facing sheets pressure laminated to core material; with continuous, interlocking molding strip or lapped-and-formed edge closures; corners secured by welding or clips and exposed welds ground smooth. Exposed surfaces shall be free of pitting, seam marks, roller marks, stains, discolorations, telegraphing of core material, or other imperfections.
1. Core Material: Manufacturer's standard sound-deadening honeycomb of resin-impregnated kraft paper in thickness required to provide finished thickness of 1 inch for doors and panels and 1-1/4 inches for pilasters.
 2. Grab-Bar Reinforcement: Provide concealed internal reinforcement for grab bars mounted on units of size and material adequate for panel to withstand applied downward load on grab bar of at least 250 lbf, when tested according to ASTM F 446, without deformation of panel.
 3. Tapping Reinforcement: Provide concealed reinforcement for tapping (threading) at locations where machine screws are used for attaching items to units.
- E. Urinal-Screen Construction:
1. Flat-Panel Urinal Screen: Matching panel construction.
- F. Facing Sheets and Closures: Electrolytically coated steel or hot-dip galvanized-steel sheet with nominal base-metal (uncoated) thicknesses as follows:
1. Pilasters, Braced at Both Ends: Manufacturer's standard thickness, but not less than 0.036 inch.
 2. Pilasters, Unbraced at One End: Manufacturer's standard thickness, but not less than 0.048 inch.
 3. Panels: Manufacturer's standard thickness, but not less than 0.030 inch.
 4. Doors: Manufacturer's standard thickness, but not less than 0.030 inch.
 5. Flat-Panel Urinal Screens: Thickness matching the panels.
- G. Pilaster Shoes and Sleeves (Caps): Stainless-steel sheet, not less than 0.031-inch nominal thickness and 3 inches high, finished to match hardware.
- H. Brackets (Fittings):
1. Stirrup Type: Ear or U-brackets; stainless steel or chrome-plated brass.
- I. Steel Sheet Finish: Manufacturer's standard baked-on finish.

1. Color: As selected by Architect from manufacturer's full range.

2.3 HARDWARE AND ACCESSORIES

- A. Hardware and Accessories: Manufacturer's standard operating hardware and accessories.
 1. Material: Chrome-plated brass.
 2. Provide units that comply with regulatory requirements for accessibility at compartments designated as accessible.
- B. Overhead Bracing: Manufacturer's standard continuous, extruded-aluminum head rail with antigrip profile and in manufacturer's standard finish.
- C. Anchorages and Fasteners: Manufacturer's standard exposed fasteners of stainless steel, finished to match the items they are securing, with theft-resistant-type heads. Provide sex-type bolts for through-bolt applications. For concealed anchors, use stainless steel, hot-dip galvanized steel, or other rust-resistant, protective-coated steel compatible with related materials.

2.4 FABRICATION

- A. Fabrication, General: Fabricate toilet compartment components to sizes indicated. Coordinate requirements and provide cutouts for through-partition toilet accessories, and solid blocking within panel where required for attachment of toilet accessories.
- B. Overhead-Braced Units: Provide manufacturer's standard corrosion-resistant supports, leveling mechanism, and anchors at pilasters to suit floor conditions. Provide shoes at pilasters to conceal supports and leveling mechanism.
- C. Floor-Anchored Units: Provide manufacturer's standard corrosion-resistant anchoring assemblies with leveling adjustment nuts at pilasters for structural connection to floor. Provide shoes at pilasters to conceal anchorage.
- D. Ceiling-Hung Units: Provide manufacturer's standard corrosion-resistant anchoring assemblies with leveling adjustment nuts at pilasters for connection to structural support above finished ceiling. Provide assemblies that support pilasters from structure without transmitting load to finished ceiling. Provide sleeves (caps) at tops of pilasters to conceal anchorage.
- E. Floor-and-Ceiling-Anchored Units: Provide manufacturer's standard corrosion-resistant anchoring assemblies with leveling adjustment at tops and bottoms of pilasters. Provide shoes and sleeves (caps) at pilasters to conceal anchorage.
- F. Door Size and Swings: Unless otherwise indicated, provide 24-inch-wide, in-swinging doors for standard toilet compartments and 36-inch-wide, out-swinging doors with a minimum 32-inch-wide, clear opening for compartments designated as accessible.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General: Comply with manufacturer's written installation instructions. Install units rigid, straight, level, and plumb. Secure units in position indicated with manufacturer's recommended anchoring devices.
 - 1. Maximum Clearances:
 - a. Pilasters and Panels: 1/2 inch.
 - b. Panels and Walls: 1 inch.
 - 2. Stirrup Brackets: Secure panels to walls and to pilasters with no fewer than three brackets attached at midpoint and near top and bottom of panel.
 - a. Locate wall brackets so holes for wall anchors occur in masonry or tile joints.
 - b. Align brackets at pilasters with brackets at walls.

3.2 ADJUSTING

- A. Hardware Adjustment: Adjust and lubricate hardware according to hardware manufacturer's written instructions for proper operation. Set hinges on in-swinging doors to hold doors open approximately 30 degrees from closed position when unlatched. Set hinges on out-swinging doors to return doors to fully closed position.

END OF SECTION

SECTION 10 26 00
WALL PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Corner guard.
2. Wall protection.

1.2 ACTION SUBMITTALS

A. Product Data: Technical data including material descriptions, impact strength, dimensions of individual components and profiles, and finishes for each impact resistant wall protection unit.

1. Include fire ratings of units recessed in fire rated walls and listings for door protection items attached to fire rated doors.

B. Shop Drawings: Submit plans, elevations, sections, and attachment details. Show handrail design and support spacing required to withstand structural loads.

1.3 CLOSEOUT SUBMITTALS

A. Maintenance Data: Submit data for each type of wall and door protection product to include in maintenance manuals.

1. Include recommended methods and frequency of maintenance for maintaining best condition of plastic covers under anticipated traffic and use conditions. Include precautions against using cleaning materials and methods that may be detrimental to finishes and performance.

1.4 QUALITY ASSURANCE

A. Source Limitations: Obtain wall protection products from single source from single manufacturer.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Store wall protection in original undamaged packages and containers inside well ventilated area protected from weather, moisture, soiling, extreme temperatures, and humidity.

1. Maintain room temperature within storage area at not less than 70 degrees F (21 degrees C) during the period plastic materials are stored.
2. Keep plastic materials out of direct sunlight.
3. Store plastic wall and door protection components for a minimum of 72 hours, or until plastic material attains a minimum room temperature of 70 degrees F (21 degrees C).
 - a. Store corner guard covers in a vertical position.

1.6 WARRANTY

- A. Written warranty signed by manufacturer in which manufacturer agrees to repair or replace components of wall and door protection units that fail in materials or workmanship within specified warranty period.
 1. Failures include, but are not limited to, the following:
 - a. Structural failures including detachment of components from each other or from the substrates, delamination, and permanent deformation beyond normal use.
 - b. Deterioration of metals, metal finishes, plastics, and other materials beyond normal use.
 2. Warranty Period: Five years from date of Substantial Completion.

1.7 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 1. Wall Guard Covers: Full size plastic covers of maximum length equal to 2 percent of each type, color, and texture of cover installed, but no fewer than two, 96 inch (2400 mm) long units.
 2. Mounting and Accessory Components: Amounts proportional to the quantities of extra materials. Package mounting and accessory components with each extra material.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Surface Burning Characteristics: Comply with ASTM E 84 or UL 723; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
 1. Flame Spread Index: 25 or less.
 2. Smoke Developed Index: 450 or less.

2.2 WALL PROTECTION

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Construction Specialties, Inc.
 - 2. InPro Corporation (IPC).
 - 3. Korogard Wall Protection Systems.
 - 4. Pawling Corporation.
- B. Corner Guard: Assembly consisting of snap on, resilient plastic cover installed over retainer; including mounting hardware; fabricated with 90 degree or 135 degree turn to match wall condition.
 - 1. Cover: Extruded rigid plastic, minimum 0.078 inch (2.0 mm) wall thickness; in dimensions and profiles indicated on Drawings.
 - a. Profile: Nominal 2 inch (50 mm) long leg and 1/4 inch (6 mm) corner radius.
 - b. Height: Full height.
 - c. Color and Texture: Selected by Architect.
 - 2. Continuous Retainer: Minimum 0.060 inch (1.5 mm) thick, one piece, extruded aluminum.
 - 3. Retainer Clips: Impact absorbing clips.
 - 4. Top and Bottom Caps: Prefabricated, injection molded plastic; color matching cover; field adjustable for close alignment with snap on cover.
- A. Abuse Resistant Sheet Wall Covering: Fabricated from semirigid, plastic sheet wall covering material.
 - 1. Size: As indicated.
 - 2. Sheet Thickness: 0.125 inch (3.0 mm).
 - 3. Color and Texture: Selected by Architect.
 - 4. Height: As indicated.
 - 5. Trim and Joint Moldings: Extruded rigid plastic matching wall covering color.
 - a. Provide top cap, inside corners, divider bars, and outside corners.
 - 6. Mounting: Adhesive.

2.3 MATERIALS

- A. Plastic Materials: Chemical and stain resistant, high impact resistant plastic with integral color throughout; extruded and sheet material as required, thickness as indicated.
- B. Fasteners: Aluminum, nonmagnetic stainless steel, or noncorrosive metal screws, bolts, and fasteners compatible with items being fastened. Use security type fasteners where exposed to view.

- C. Adhesive: Recommended by protection product manufacturer.

2.4 FABRICATION

- A. Fabricate wall protection according to requirements indicated for design, performance, dimensions, and member sizes, including thicknesses of components.
- B. Factory Assembly: Assemble components in factory to greatest extent possible to minimize field assembly. Disassemble only as necessary for shipping and handling.
- C. Quality: Fabricate components with uniformly tight seams and joints and with exposed edges rolled. Provide surfaces free of wrinkles, chips, dents, uneven coloration, and other imperfections. Fabricate members and fittings to produce flush, smooth, and rigid hairline joints.

2.5 FINISHES

- A. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
- B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and wall areas for compliance with requirements for installation tolerances and conditions affecting performance of the work.
- B. Examine walls to which wall protection will be attached for blocking, grounds, and other solid backing that have been installed in the locations required for secure attachment of support fasteners.
 - 1. For wall and door protection attached with adhesive, verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
- C. Proceed with installation after correcting unsatisfactory conditions.

3.2 PREPARATION

- A. Complete finishing operations, including painting, before installing wall and door protection.
- B. Before installation, clean substrate to remove dust, debris, and loose particles.

3.3 INSTALLATION

- A. Installation Quality: Install wall and door protection according to manufacturer's written instructions, level, plumb, and true to line without distortions. Do not use materials with chips, cracks, voids, stains, or other defects that might be visible in the finished Work.
- B. Mounting Heights: Install wall and door protection in locations and at mounting heights indicated on Drawings and compliant with TAS.
- C. Accessories: Provide splices, mounting hardware, anchors, trim, joint moldings, and other accessories required for a complete installation.

3.4 CLEANING

- A. Immediately after completion of installation, clean plastic covers and accessories using a standard household cleaning agent.
- B. Remove excess adhesive using methods and materials recommended in writing by manufacturer.

END OF SECTION

SECTION 104413
FIRE PROTECTION CABINETS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes fire-protection cabinets for portable fire extinguishers.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings: For fire-protection cabinets.

1.3 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.4 COORDINATION

- A. Coordinate size of fire-protection cabinets to ensure that type and capacity of fire extinguishers indicated are accommodated.
- B. Coordinate sizes and locations of fire-protection cabinets with wall depths.
- C. Coordinate fire-rating of cabinets with fire-rating of wall in which cabinet is located.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Fire-Rated Fire-Protection Cabinets: Listed and labeled to comply with requirements in ASTM E 814 for fire-resistance rating of walls where they are installed.

2.2 FIRE-PROTECTION CABINETS

- A. Cabinet Type: Suitable for size of fire extinguisher specified.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. JL Industries, Inc.; a division of the Activar Construction Products Group.
 - b. Larsens Manufacturing Company.
 - c. Nystrom, Inc.
 - d. Potter Roemer LLC.
- B. Cabinet Construction: Nonrated or fire rated as required.
 - 1. Fire-Rated Cabinets: Construct fire-rated cabinets with double walls fabricated from 0.043-inch-thick cold-rolled steel sheet lined with minimum 5/8-inch-thick fire-barrier material. Provide factory-drilled mounting holes.
- C. Cabinet Material: As indicated on the Drawings from the following options:
 - 1. Cold-rolled steel sheet.
 - 2. Stainless-steel sheet.
 - 3. Copper-alloy brass sheet.
 - 4. Copper-alloy bronze sheet.
- D. Fire Extinguisher Mounting Style: As indicated on the Drawings from the following options:
 - 1. Recessed Cabinets: Exposed Flat Trim: One-piece combination trim and perimeter door frame overlapping surrounding wall surface with exposed trim face and wall return at outer edge (backbend).
 - 2. Semirecessed Cabinets: One-piece combination trim and perimeter door frame overlapping surrounding wall surface with exposed trim face and wall return at outer edge (backbend).
 - a. Rolled-Edge Trim: 2-1/2-inch backbend depth.
 - 3. Surface-Mounted Cabinet: Cabinet box fully exposed and mounted directly on wall with no trim.
- E. Cabinet and Door Trim Material: As indicated on the Drawings from the following options:
 - 1. Cold-rolled steel sheet.
 - 2. Stainless-steel sheet.
 - 3. Copper-alloy brass sheet.
 - 4. Copper-alloy bronze sheet.
- F. Door Style: Vertical duo panel with frame.

- G. Door Glazing: Acrylic sheet.
 - 1. Acrylic Sheet Color: Clear transparent acrylic sheet.
- H. Door Hardware: Manufacturer's standard door-operating hardware of proper type for cabinet type, trim style, and door material and style indicated.
- I. Accessories:
 - 1. Door Lock: Cam lock that allows door to be opened during emergency by pulling sharply on door handle.
 - 2. Identification: Lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated.
 - a. Identify fire extinguisher in fire-protection cabinet with the words "FIRE EXTINGUISHER."
 - 1) Location: Applied to cabinet door.
 - 2) Application Process: Pressure-sensitive vinyl letters.
 - 3) Lettering Color: Red.
 - 4) Orientation: Vertical.
- J. Materials:
 - 1. Cold-Rolled Steel: ASTM A 1008/A 1008M, Commercial Steel (CS), Type B.
 - a. Finish: Baked enamel or powder coat.
 - b. Color: White.
 - 2. Stainless Steel: ASTM A 666, Type 304.
 - a. Finish: No. 4 directional satin finish.
 - 3. Copper Alloy, Brass: ASTM B 36/B 36M, alloy as standard with manufacturer.
 - a. Finish: Satin.
 - 4. Copper Alloy, Bronze: ASTM B 36/B 36M, alloy as standard with manufacturer.
 - a. Finish: Satin oxidized oil rub.
 - 5. Transparent Acrylic Sheet: ASTM D 4802, Category A-1 (cell-cast sheet), 3 mm thick, with Finish 1 (smooth or polished).

2.3 FABRICATION

- A. Fire-Protection Cabinets: Provide manufacturer's standard box (tub) with trim, frame, door, and hardware to suit cabinet type, trim style, and door style indicated.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Prepare recesses for recessed and semi-recessed fire-protection cabinets as required by type and size of cabinet and trim style.
- B. Install fire-protection cabinets in locations indicated or, if not indicated, at a minimum rate of one fire protection cabinet for each 6,000 sq. ft. of floor area or fraction thereof.
 - 1. Mounting Height: 54 inches above finished floor to top of cabinet.
- C. Fire-Protection Cabinets: Fasten cabinets to structure, square and plumb.
- D. Identification: Apply vinyl lettering at locations indicated.
- E. Adjust fire-protection cabinet doors to operate easily without binding. Verify that latching devices operate properly.

END OF SECTION

SECTION 104416

FIRE EXTINGUISHERS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes portable, hand-carried fire extinguishers and mounting brackets for fire extinguishers.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1.3 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.4 COORDINATION

- A. Coordinate type and capacity of fire extinguishers with fire-protection cabinets to ensure fit and function.

1.5 WARRANTY

- A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace fire extinguishers that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: Six years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. NFPA Compliance: Fabricate and label fire extinguishers to comply with NFPA 10, "Portable Fire Extinguishers."
- B. Fire Extinguishers: Listed and labeled for type, rating, and classification by an independent testing agency acceptable to authorities having jurisdiction.

2.2 PORTABLE, HAND-CARRIED FIRE EXTINGUISHERS

- A. Fire Extinguishers: Type, size, and capacity for each fire-protection cabinet or mounting bracket indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Ansul Incorporated.
 - b. Badger Fire Protection.
 - c. JL Industries, Inc.; a division of the Activar Construction Products Group.
 - d. Larsens Manufacturing Company.
 - e. Nystrom Building Products.
 - f. Potter Roemer LLC.
 - 2. Instruction Labels: Include pictorial marking system complying with NFPA 10, Appendix B.
- B. Multipurpose Dry-Chemical Type: UL-rated 4-A:80-B:C, 10-lb nominal capacity, with monoammonium phosphate-based dry chemical in manufacturer's standard enameled container.

2.3 MOUNTING BRACKETS

- A. Mounting Brackets: Manufacturer's standard galvanized steel, designed to secure fire extinguisher to wall or structure, of sizes required for types and capacities of fire extinguishers indicated, with plated or black baked-enamel finish.
- B. Identification: Lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated by Architect.
 - 1. Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Examine fire extinguishers for proper charging and tagging.
 - 1. Remove and replace damaged, defective, or undercharged fire extinguishers.
- B. Install fire extinguishers in locations indicated or, if not indicated, at a minimum rate of one fire extinguisher for each 6,000 sq. ft. of floor area or fraction thereof.

- C. Mounting Brackets: Fasten mounting brackets to surfaces, square and plumb, at locations indicated.
 - 1. Mounting Height: 54 inches above finished floor to top of fire extinguisher.

END OF SECTION

SECTION 105123
PLASTIC-LAMINATE-CLAD LOCKERS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes:
 - 1. Plastic-laminate-clad wood lockers
 - 2. Locker benches.

1.2 ACTION SUBMITTALS

- A. Product data.
- B. Shop Drawings: For plastic-laminate-clad wood lockers. Include plans, elevations, sections, details, attachments to other work, and locker identification system and numbering sequence.
- C. Samples: For each exposed finish.

1.3 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.4 FIELD CONDITIONS

- A. Environmental Limitations: Do not deliver or install lockers until building is enclosed, wet work is complete, and HVAC system is operating and maintaining temperature at occupancy levels during the remainder of the construction period.

1.5 COORDINATION

- A. Coordinate sizes and locations of concealed wood support bases.
 - 1. Requirements are specified in Section 06 10 53 "Miscellaneous Rough Carpentry."

1.6 WARRANTY

- A. Special Warranty: Manufacturer agrees to repair or replace components of lockers that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Three years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Accessibility Requirements: For lockers indicated to be accessible, comply with applicable provisions in the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines and ICC A117.1.

2.2 PLASTIC-LAMINATE-CLAD WOOD LOCKERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 1. Classic Woodworking, LLC.
 2. Club Resource Group.
 3. Famous Lockers.
 4. Hollman, Inc.
 5. Ideal Products, Inc.
 6. Legacy Lockers.
 7. List Industries Inc.
 8. Treeforms.
- B. Construction Style: Flush overlay.
- C. Locker Body: Fabricated from particleboard-core panels covered on both sides with thermoset decorative overlay.
 1. Side, Top, and Bottom Panels: Manufacturer's standard 3/4 or 5/8 inch thick.
 2. Back Panel: Manufacturer's standard 1/2 or 3/8 inch thick.
 3. Exposed Panel Edges: 1-mm-thick PVC.
- D. Plastic-Laminate-Clad Wood Doors: High-pressure decorative laminate, Grade VGS, over both sides of medium-density-fiberboard core.
 1. Thickness: Manufacturer's standard 3/4 or 5/8 inch thick.
 2. Panel Edges: High-pressure decorative laminate, Grade VGS, to match panels.
- E. End Panels: Match style, material, construction, and finish of plastic-laminate-clad wood doors.
- F. Shelves: Fabricated from particleboard-core panels covered on both sides with thermoset decorative overlay; fixed unless otherwise indicated.
 1. Thickness: 3/4 inch.
 2. Exposed Edges: Thermoset decorative overlay to match panels.

- G. Corners and Filler Panels: 3/4-inch-thick panels. Match style, material, construction, and finish of plastic-laminate-clad wood doors.
- H. Continuous Finish Base: Plastic-laminate-clad, 3/4-inch-thick panel that matches door faces; fabricated in lengths as long as practical to enclose base and base ends of lockers.
- I. Plastic-Laminate Colors, Patterns, and Finishes:
 - 1. As selected by Architect from plastic-laminate manufacturer's full range of solid colors, wood grains, and patterns.

2.3 MATERIALS

- A. Composite Wood: Provide materials that comply with requirements of referenced quality standard for each type of woodwork and quality grade specified unless otherwise indicated.
 - 1. Medium-Density Fiberboard: ANSI A208.2, Grade 130, made with binder containing no urea formaldehyde.
 - 2. Softwood Plywood: DOC PS 1, medium-density overlay.
 - 3. Thermoset Decorative Panels: Medium-density fiberboard finished with thermally fused, melamine-impregnated decorative paper and complying with requirements of NEMA LD 3, Grade VGL, for Test Methods 3.3, 3.4, 3.6, 3.8, and 3.10.
- B. High-Pressure Decorative Laminate: NEMA LD 3, grades as follows:
 - 1. Horizontal Surfaces: Grade HGS.
 - 2. Postformed Surfaces: Grade HGP.
 - 3. Vertical Surfaces: Grade VGS.
- C. Furring, Blocking, Shims, and Hanging Strips: Softwood or hardwood lumber, kiln dried to less than 15 percent moisture content.
- D. Anchors: Select material, type, size, and finish required for each substrate for secure anchorage. Provide metal expansion sleeves or expansion bolts for post-installed anchors. Use nonferrous-metal or hot-dip galvanized anchors and inserts at inside face of exterior walls and at floors.
- E. Wood Support Base: 2-by-4-inch nominal-size lumber treated with manufacturer's standard preservative-treatment, pressure process.

2.4 HARDWARE

- A. General: Provide manufacturer's standard locker hardware complying with the requirements in this Section.
- B. Digital Keypad Locks: Battery-powered electronic keypad with reprogrammable manager and owner codes that override access. Three consecutive incorrect code entries shall disable lock for three minutes.

1. Designed for permanently assigned access via entry of user's four-digit code.
 2. Designed for shared or temporary access by multiple users, with user-defined code to lock and unlock. Provide LED indicator to show when lock is in use.
- C. Frameless Hinges (European Type): Fully concealed, nickel-plated steel, with not less than 125 degrees of opening.
1. Provide two hinges for doors 36 inches high and less.
 2. Provide three hinges for doors more than 36 inches high.
- D. Wire Pulls: Back mounted; 5 inches long, 2-1/2 inches deep, and 5/16 inch in diameter.
- E. Accessible Handle: Metal, fixed, graspable lever handle and rose trim; surface mounted.
- F. Shelf Rests: BHMA A156.9, B04013.
- G. Hooks: Manufacturer's standard, ball-pointed aluminum or steel; finished to match other locker hardware. Attach hooks with at least two fasteners.
1. Provide two single-prong wall hooks for each compartment of single-tier and double-tier lockers.
- H. Coat Rods: 1-inch-diameter steel; finished to match other locker hardware.
1. Provide coat rod for each compartment of single-tier and double-tier triple-tier lockers.
- I. Exposed Hardware Finishes: Satin chrome unless otherwise indicated.

2.5 ACCESSORIES

- A. Number Plates: 1-1/2-inch-diameter, etched, embossed, or stamped, stainless-steel plates with black numbers and letters at least 1/2 inch high. Identify lockers in sequence indicated on Drawings.

2.6 LOCKER BENCHES

- A. Provide bench units with overall assembly height of 17-1/2 inches.
- B. Bench Tops: Manufacturer's standard one-piece units, with rounded corners and edges.
1. Size: Minimum 15 inches wide by 1-1/4 inches thick except provide minimum 20-inch-wide tops where accessible benches are indicated.
 2. Laminated clear hardwood with one coat of clear sealer on all surfaces and one coat of clear lacquer on top and sides.
- C. Back Supports: Where freestanding accessible benches are indicated, provide manufacturer's standard one-piece units, with rounded corners and edges.

1. Size: 16 inches high by 1-1/4 inches thick, mounted 2 inches off bench top surface.
 2. Laminated clear hardwood with one coat of clear sealer on all surfaces and one coat of clear lacquer on top and sides
- D. Fixed Pedestals: Manufacturer's standard stainless steel supports, with predrilled fastener holes for attaching bench top and anchoring to floor, complete with fasteners and anchors.
1. Materials: Stainless Steel: ASTM A 666, Type 304.

2.7 FABRICATION

- A. Fabricate components square, rigid, without warp, and with finished faces flat and free of scratches and chips. Accurately factory machine components for attachments. Make joints tight and true.
- B. Accessible Lockers: Fabricate as follows:
1. Locate bottom shelf no lower than 15 inches above the floor.
 2. Where hooks, coat rods, or additional shelves are provided, locate no higher than 48 inches above the floor.
- C. Venting: Fabricate lockers with space between doors and locker assembly of not less than 1/4 inch.
- D. Complete fabrication, including assembly, finishing, and hardware application, to maximum extent possible, before shipment to Project site. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.
- E. Shop cut openings, to maximum extent possible, to receive hardware, electrical work, and similar items. Locate openings accurately and use templates or roughing-in diagrams to produce accurately sized and shaped openings. Sand edges of cutouts to remove splinters and burrs.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install wood support base with 1/2-inch-thick plywood top.
- B. Install lockers level, plumb, and true; use concealed shims.
- C. Connect groups of lockers together with manufacturer's standard fasteners, through predrilled holes, with no exposed fasteners on face frames. Fit lockers accurately together to form flush, tight, hairline joints.

- D. Install lockers without distortion so doors and drawers fit openings properly and are accurately aligned. Adjust hardware to center doors and drawers in openings, providing unencumbered operation. Complete installation of hardware and accessory items as indicated.
 - 1. Installation Tolerance: No more than 1/8 inch in 96-inch sag, bow, or other variation from a straight line. Shim as required with concealed shims.
- E. Locker Anchorage: Fasten wood lockers through back, near top and bottom, at ends with No. 8 flush-head wood screws sized for 1-inch penetration into wood framing, blocking, or furring and spaced not more than 16 inches o.c.
- F. Scribe and cut corner and filler panels to fit adjoining work using fasteners concealed where practical. Repair damaged finish at cuts.
- G. Install number plates after lockers are in place.
- H. Anchor locker benches to floors.

END OF SECTION

SECTION 122413.13
MANUAL ROLLER WINDOW SHADES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes manually operated roller shades.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
 - 1. Include styles, material descriptions, construction details, dimensions of individual components and profiles, features, finishes, and operating instructions for roller shades.
- B. Shop Drawings: Show fabrication and installation details for roller shades, including shadeband materials, their orientation to rollers, and their seam and batten locations.
- C. Samples: For each exposed product and for each color and texture specified.

1.3 INFORMATIONAL SUBMITTALS

- A. Product test reports.

1.4 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Fabricator of products.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. DFB Sales Inc.

2. Draper Inc.
3. Hunter Douglas Contract.
4. Lutron Electronics Co., Inc.
5. MechoShade Systems, Inc.
6. Nysan Solar Control Inc.; a Hunter Douglas company.
7. Springs Window Fashions; SWFcontract.

2.2 ROLLER SHADES

- A. Chain-and-Clutch Operating Mechanisms: With continuous-loop bead chain and clutch that stops shade movement when bead chain is released; permanently adjusted and lubricated.
 1. Bead Chains: Stainless steel.
 - a. Loop Length: Full length of roller shade.
 - b. Limit Stops: Provide upper and lower ball stops.
 - c. Chain-Retainer Type: Chain tensioner, sill mounted.
 2. Spring Lift-Assist Mechanisms: Manufacturer's standard for balancing roller-shade weight and lifting heavy roller shades.
 - a. Provide for shadebands that weigh more than 10 lb or for shades as recommended by manufacturer, whichever criteria are more stringent.
- B. Spring Operating Mechanisms: Roller contains spring sized to accommodate shade size indicated. Provide with positive locking mechanism that can stop shade movement at each half-turn of roller and with manufacturer's standard pull.
 1. Pole: Manufacturer's standard type in length required to make operation convenient from floor level and with hook for engaging pull.
- C. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.
 1. Roller Mounting Configuration: Single roller.
 2. Direction of Shadeband Roll: Regular, from back of roller.
 3. Shadeband-to-Roller Attachment: Manufacturer's standard method.
- D. Mounting Hardware: Brackets or endcaps, corrosion resistant and compatible with roller assembly, operating mechanism, installation accessories, and mounting location and conditions indicated.
- E. Roller-Coupling Assemblies: Coordinated with operating mechanism and designed to join up to three inline rollers into a multiband shade that is operated by one roller drive-end assembly.

F. Shadebands:

1. Shadeband Material: Light-filtering or light-blocking fabric as indicated on Drawings.
2. Shadeband Bottom (Hem) Bar: Steel or extruded aluminum.
 - a. Type: Enclosed in sealed pocket of shadeband material.

G. Installation Accessories:

1. Front Fascia: Aluminum extrusion that conceals front and underside of roller and operating mechanism and attaches to roller endcaps without exposed fasteners.
 - a. Shape: L-shaped.
 - b. Height: Manufacturer's standard height required to conceal roller and shadeband when shade is fully open, but not less than 4 inches.
2. Exposed Headbox: Rectangular, extruded-aluminum enclosure including front fascia, top and back covers, endcaps, and removable bottom closure.
 - a. Height: Manufacturer's standard height required to enclose roller and shadeband when shade is fully open, but not less than 4 inches.
3. Endcap Covers: To cover exposed endcaps.
4. Recessed Shade Pocket: Where indicated on Drawings, provide rectangular, extruded-aluminum enclosure designed for recessed ceiling installation; with front, top, and back formed as one piece, end plates, and removable bottom closure panel.
 - a. Height: Manufacturer's standard height required to enclose roller and shadeband when shade is fully open, but not less than 5 inches.
 - b. Provide pocket with lip at lower edge to support acoustical ceiling panel.
5. Closure Panel and Wall Clip: Removable aluminum panel designed for installation at bottom of site-constructed ceiling recess or pocket and for snap-in attachment to wall clip without fasteners.
 - a. Closure-Panel Width: As indicated on Drawings.
6. Side Channels: When light-blocking shades are indicated, provide shades with light seals designed to eliminate light gaps at sides of shades as shades are drawn down. Provide side channels with shadeband guides or other means of aligning shadebands with channels at tops.
7. Bottom (Sill) Channel or Angle: When light-blocking shades are indicated, provide shades with light seals and designed to eliminate light gaps at bottoms of shades when shades are closed.

2.3 SHADEBAND MATERIALS

- A. Shadeband Material Flame-Resistance Rating: Comply with NFPA 701. Testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
- B. Light-Filtering Fabric: Woven fabric, stain and fade resistant.
 - 1. Source: Roller-shade manufacturer.
 - 2. Type: As indicated on Drawings.
 - 3. Weave: As indicated by product designation.
 - 4. Weight: As indicated by product designation.
 - 5. Orientation on Shadeband: Up the bolt.
 - 6. Openness Factor: No greater than 3 percent.
 - 7. Color: As indicated on Drawings.
- C. Light-Blocking Fabric: Opaque fabric, stain and fade resistant.
 - 1. Source: Roller-shade manufacturer.
 - 2. Type: As indicated on Drawings.
 - 3. Weight: As indicated by product designation.
 - 4. Orientation on Shadeband: Up the bolt.
 - 5. Features: Antistatic treatment.
 - 6. Color: As indicated on Drawings.

2.4 ROLLER-SHADE FABRICATION

- A. Product Safety Standard: Fabricate roller shades to comply with WCMA A 100.1, including requirements for flexible, chain-loop devices; lead content of components; and warning labels.
- B. Unit Sizes: Fabricate units in sizes to fill window and other openings as follows, measured at 74 deg F:
 - 1. Between (Inside) Jamb Installation: Width equal to jamb-to-jamb dimension of opening in which shade is installed less 1/4 inch per side or 1/2-inch total, plus or minus 1/8 inch. Length equal to head-to-sill or -floor dimension of opening in which shade is installed less 1/4 inch, plus or minus 1/8 inch.
 - 2. Outside of Jamb Installation: Width and length as indicated, with terminations between shades of end-to-end installations at centerlines of mullion or other defined vertical separations between openings.
- C. Shadeband Fabrication: Fabricate shadebands without battens or seams to extent possible except as follows:
 - 1. Vertical Shades: Where width-to-length ratio of shadeband is equal to or greater than 1:4, provide battens and seams at uniform spacings along shadeband length to ensure shadeband tracking and alignment through its full range of movement without distortion of the material.

PART 3 - EXECUTION

3.1 ROLLER-SHADE INSTALLATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, operational clearances, and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
- C. Install roller shades level, plumb, and aligned with adjacent units, according to manufacturer's written instructions.
 - 1. Opaque Shadebands: Located so shadeband is not closer than 2 inches to interior face of glass. Allow clearances for window operation hardware.
- D. Adjust and balance roller shades to operate smoothly, easily, safely, and free from binding or malfunction throughout entire operational range.
- E. Clean roller-shade surfaces after installation, according to manufacturer's written instructions.

END OF SECTION

SECTION 123661.19
QUARTZ AGGLOMERATE COUNTERTOPS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes quartz agglomerate countertops.

1.2 ACTION SUBMITTALS

- A. Product Data: For countertop materials.
- B. Shop Drawings: For countertops. Show materials, finishes, edge and backsplash profiles, methods of joining, and cutouts for plumbing fixtures.
- C. Samples: For each type of material exposed to view.

PART 2 - PRODUCTS

2.1 QUARTZ AGGLOMERATE COUNTERTOP MATERIALS

- A. Quartz Agglomerate: Solid sheets consisting of quartz aggregates bound together with a matrix of filled plastic resin and complying with ICPA SS-1, except for composition.
 - 1. Products: Subject to compliance with requirements, provide products indicated on Drawings.
 - a. Colors and Patterns: As indicated on Drawings.
- B. Composite Wood Products: Products shall be made without urea formaldehyde.
 - 1. Plywood: Exterior softwood plywood complying with DOC PS 1, Grade C-C Plugged, touch sanded.
 - 2. Medium-Density Fiberboard: ANSI A208.2, Grade 130.

2.2 COUNTERTOP FABRICATION

- A. Fabricate countertops according to quartz agglomerate manufacturer's written instructions and the AWI/AWMAC/WI's "Architectural Woodwork Standards."
 - 1. Grade: Custom.

- B. Configuration:
 - 1. Front Edge: As indicated on Drawings.
 - 2. Exposed End and Side Edge: As indicated on Drawings.
 - 3. Backsplash: Straight, slightly eased at corner.
 - 4. End Splash: Matching backsplash.
- C. Countertops: 3/4-inch-thick, quartz agglomerate with front edge built up with same material.
- D. Backsplashes: 3/4-inch-thick, quartz agglomerate.
- E. Joints: Fabricate countertops without joints wherever possible. Fabricate oversized countertops in sections for joining in field, with joints at locations indicated on approved Shop Drawings.
- F. Cutouts and Holes:
 - 1. Undercounter Plumbing Fixtures: Make cutouts for fixtures using template or pattern furnished by fixture manufacturer. Form cutouts to smooth, even curves.

2.3 INSTALLATION MATERIALS

- A. Adhesive: Product recommended by quartz agglomerate manufacturer.
 - 1. Adhesives shall have a VOC content of 70 g/L or less.
- B. Sealant for Countertops: Comply with applicable requirements in Section 07 92 00 "Joint Sealants."

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Fasten subtops to cabinets by screwing through subtops into cornerblocks of base cabinets. Shim as needed to align subtops in a level plane.
- B. Secure countertops to subtops with adhesive according to quartz agglomerate manufacturer's written instructions.
- C. Bond joints with adhesive and draw tight as countertops are set. Mask areas of countertops adjacent to joints to prevent adhesive smears.
- D. Install backsplashes and end splashes by adhering to wall and countertops with adhesive.
- E. Install aprons to backing and countertops with adhesive.

- F. Complete cutouts not finished in shop. Mask areas of countertops adjacent to cutouts to prevent damage while cutting. Make cutouts to accurately fit items to be installed, and at right angles to finished surfaces unless beveling is required for clearance. Ease edges slightly to prevent snipping.
- G. Apply sealant to gaps at walls; comply with Section 07 92 00 "Joint Sealants."

END OF SECTION

SECTION 220000 GENERAL PLUMBING REQUIREMENTS

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 DESCRIPTION

- A. Intent of drawings and Specifications is to obtain complete systems, tested, adjusted, and ready for operation.
- B. The terms "provide", "furnish" and "install" as used in Contract Documents shall have the following meanings:
 - 1. "Provide" or "provided" shall mean "furnish and install".
 - 2. "Furnish" or "furnished" does not include installation.
 - 3. "Install" or "installed" does not include furnishing.
- C. Include incidental details not usually shown or specified, but necessary for proper installation and operation.
- D. Check, verify and coordinate work with drawings and specifications prepared for other trades. Include modifications, relocations or adjustments necessary to complete work or to avoid interference with other trades.
- E. Information given herein and on drawings is as exact as could be secured but is not guaranteed. Do not scale drawings for exact dimensions.
- F. Where Architectural features govern location of work, refer to architectural drawings.
- G. Contractor may install additional piping, fittings and valves, not shown on drawings, for testing purposes or for convenience of installation. Where such materials are installed, they shall comply with specifications and shall be sized to be compatible with system design. Remove such installed materials when they interfere with design conditions or as directed by Architect.

1.3 RELATED WORK

- A. Utility Services:
 - 1. Determine utility connection requirements and include in Base Bid all costs to Owner for utility service.

2. Include costs for temporary service, temporary routing of piping or any other requirements of a temporary nature associated with utility service.
- B. Temporary Services:
 1. Division 01 - Temporary Facilities and Controls.
- C. Continuity of Service:
 1. No service shall be interrupted or changed without permission from Architect and Owner. Obtain written permission before any work is started.
- D. When interruption of services is required, Architect, Owner, and other concerned parties shall be notified and shall determine a time.
- E. Demolition:
 1. Division 02 - Selective Demolition.
 2. Perform demolition as required to accomplish new work.
 3. Accomplish work in neat workmanlike manner to minimize interference, annoyance or inconvenience such work might impose on Owner or other Contractors.
 4. Unless otherwise noted, remove from premises materials and equipment removed in demolition work.
 5. Equipment noted to be removed and turned over to Owner, shall be delivered to Owner at place and time Owner designates.
 6. Where materials are to be turned over to Owner or reused and installed by Contractor, it shall be Contractor's responsibility to maintain condition of materials and equipment equal to that existing before work began. Repair or replace damaged materials or equipment at no additional cost to Owner.
 7. Where demolition work interferes with Owner's use of premises, schedule work through Architect, Owner and with other Contractor to minimize inconvenience to Owner. Architect must approve schedule before Contractor begins such Work.

1.4 REQUIREMENTS OF REGULATORY AGENCIES

- A. Rules and regulations of Federal, State and Local Authorities and utility companies, in force at time of execution of Contract shall become part of this specification.

1.5 REFERENCE STANDARDS

- A. Agencies or publications referenced herein refer to the following:
 1. AGA - American Gas Association
 2. AMCA - Air Movement and Control Association
 3. ANSI - American National Standards Institute
 4. AHRI - Air-Conditioning, Heating and Refrigeration Institute
 5. ASHRAE - American Society of Heating Refrigerating and Air Conditioning Engineers
 6. ASPE - American Society of Plumbing Engineers
 7. ASSE - American Society of Sanitary Engineering
 8. AWS - American Welding Society
 9. AWWA - American Water Works Association
 10. ASME - American Society of Mechanical Engineers

11. ASTM - American Society for Testing and Materials
12. CDA - Copper Development Association
13. CISPI - Cast Iron Soil Pipe Institute
14. FMG - FM Global
15. FS - Federal Specifications
16. IEEE - Institute of Electrical and Electronics Engineers
17. MCA - Mechanical Contractors Association
18. MSS - Manufacturers Standardization Society
19. NEC - National Electrical Code
20. NEMA - National Electrical Manufacturers Association
21. NFPA - National Fire Protection Association
22. NIST - National Institute of Standards & Technology
23. NSF - National Sanitation Foundation
24. NSPI - National Spa and Pool Institute
25. OSHA - Occupational Safety and Health Administration
26. PDI - Plumbing and Drainage Institute
27. SMACNA - Sheet Metal and Air Conditioning Contractors National Association
28. UL - Underwriters Laboratories, Inc.
29. WQA - Water Quality Association

- B. Work shall be in accordance with latest edition of codes, standards or specifications unless noted otherwise.

1.6 SUBMITTALS

A. Product Data:

1. Refer to Division 01 - Submittal Procedures.
2. Submit for equipment and systems as requested in the respective specification sections.
3. Unless otherwise indicated submittals shall be in electronic Portable Document Format (pdf) and bookmarks shall be included to facilitate navigation through submittal documentation.
4. Specifically mark general catalog sheets and drawings to indicate specific items submitted and its correlation to specific designation for product in drawings.
5. Specifically indicate proper identification of equipment by name and/or number, as indicated in specification and shown on drawings.
6. When manufacturer's reference numbers are different from those specified, provide correct cross-reference numbers for each item. Clearly mark and note submittals accordingly.
7. Submit complete record of required components when fixtures, equipment and items specified include accessories, parts and additional items under one designation.
8. Include composite wiring diagrams for electrically powered equipment and devices.
9. Where submittals cover products containing non-metallic materials, include "Material Safety Data Sheet" (MSDS) from manufacturer stating physical and chemical properties of components and precautionary considerations required.
10. Submit shop drawings or product data as soon as practicable after signing contracts. Submittals must be approved before installation of materials and equipment.

11. Submittals that are not complete, not permanent or not properly checked by Contractor will be returned without review.
- B. Certificates and Inspections:
1. Obtain and pay for inspections required by authorities having jurisdiction and deliver certificates approving installations to Owner unless otherwise directed.
- C. Operation and Maintenance Manuals:
1. Refer to Division 01 - Operation and Maintenance Data.
 2. Upon completion of Work but before final acceptance of system, submit to Architect for approval, electronic version of operation and maintenance manuals in pdf format. 2" Electronic file shall contain bookmarks for each submittal contained in manual.
 3. Organize manuals by specification section number and furnish table of contents and tabs for each piece of equipment or system.
 4. Fire protection system shall be separately bound.
 5. Manuals shall include the following:
 - a. Copies of Shop Drawings and Product Data
 - b. Manufacturer's operating and maintenance instructions. Include parts lists of items or equipment, with component exploded views and part numbers. Where manufacturer's data includes several types or models, designate applicable type or model.
 - c. Electronic versions of O&M data with exploded parts lists where available
 - d. Phone numbers and addresses of local parts suppliers and service companies
 - e. Internet/WEB page addresses where applicable
 - f. Wiring diagrams
 - g. Startup and shutdown procedures
 - h. Composite electrical diagrams
 - i. Flow diagrams
 - j. Lubrication instructions
 - k. Factory and field test records (Refer to Test and Balancing in Part 3 of this section.)
 - l. Air and water balance reports
 - m. Valve identification charts as specified in Section 220553 - Plumbing Systems Identification
 - n. Access panel identification charts as specified in Section 220553 - Plumbing Systems Identification
 - o. Additional information, diagrams or explanations as designated under respective equipment or systems specification sections.
 6. Instruct Owner's Representative in operation and maintenance of equipment. Instruction shall include complete operating cycle on all apparatus.
 7. Furnish O&M Manuals and instructions to Owner prior to request for final payment.
- D. Record Documents:
1. Refer to General Conditions of Contract, and Division 01 - Project Record Documents. Prepare complete set of record drawings in accordance with Division 01.

2. Use designated set of prints of Contract Documents as prepared by Architect to mark-up for record drawing purposes.

1.7 JOB CONDITIONS

A. Building Access:

1. Arrange for necessary openings in building to allow for admittance of all apparatus.

B. Electrical Coordination:

1. Contractor shall provide the following items as specified under their respective Division(s):
 - a. Motors
 - b. Electrically powered equipment
 - c. Electrically controlled equipment
 - d. Starters, where specified
 - e. Variable frequency drives, where specified
 - f. Control devices, where specified
 - g. Temperature Control wiring
 - h. Wiring diagrams to Electrical Contractor for apparatus indicating external connection and internal controls.
 - i. Disconnect devices furnished with units (VFDs, chillers, prepackaged control devices, etc.)
 - 1) Devices shall have an interrupting rating not less than that of the upstream overcurrent device as shown on electrical drawings.
 - 2) Equipment electrical connection points shall be labeled with listed electrical short circuit current rating (SCCR). SCCR shall not be less than interrupting rating of upstream overcurrent device as shown on electrical drawings. SCCR shall be marked on equipment control enclosure in accordance with UL 508, or other acceptable, accredited third-party testing agency standards.
2. Electrical Contractor will provide the following devices required for control of motors or electrical equipment, unless noted otherwise.
 - a. Starters
 - b. Disconnect devices
 - c. Control devices:
 - 1) Pushbuttons
 - 2) Pilot lights
 - 3) Contacts
 - d. Conduit, boxes and wiring for power wiring.
 - e. Conduit, boxes and wiring for control wiring, except temperature control wiring.
3. Electrical Contractor will make connections, from power source to starter or variable frequency drive and from starter or variable frequency drive, where specified, to motor.
4. Where starters or other similar control devices are furnished by this Contractor, they shall be installed by this Contractor and wired by Electrical Contractor.

5. Should any change in size, hp rating, voltage, or means of control be made to any motor or other electrical equipment after Contracts are awarded, this Contractor shall immediately notify Electrical Contractor of change. Additional costs due to these changes shall be responsibility of this Contractor.

C. Cutting and Patching:

1. Refer to General Conditions of the Contract, and Division 01 - Cutting and Patching.
2. Perform cutting and patching required for complete installation of systems, unless otherwise noted. Patch and restore work cut or damaged to original condition. This includes openings remaining from removal or relocation of existing system components.
3. Provide materials required for patching unless otherwise noted.
4. Do not pierce beams or columns without permission of Architect and then only as directed. If openings are required through walls or floors where no sleeve has been provided, hole shall be core drilled to avoid unnecessary damage and structural weakening.
5. Where alterations disturb lawns, paving, walks, etc., replace, repair or refinish surfaces to condition existing prior to commencement of work. This may include areas beyond construction limits.

D. Housekeeping and Cleanup:

1. Refer to Division 01 - Closeout Procedures.
2. As work progresses and/or as directed by Architect, periodically remove waste materials from building and leave area of work broom clean. Upon completion of Work, remove tools, scaffolding, broken and waste materials, etc., from site.

1.8 WARRANTY

- A. Refer to Division 01 for general warranty requirements.
- B. Refer to technical sections for warranty requirement for each system.
 1. Where no warranty requirements are called out, warrant equipment, materials, and workmanship to be free from defect as called out in Division 01.
- C. Warrant that systems will operate without objectionable noise, vibration and uncontrolled expansion.
- D. Repair, replace or alter systems or parts of systems found defective at no extra cost to Owner.
- E. In any case, wherein fulfilling requirements of any warranty, if this Contractor disturbs any work warranted under another contract, this Contractor shall restore such disturbed work to condition satisfactory to Architect and warranty such restored work to same extent as it was warranted under such other contract.
- F. Warranty shall include labor, materials, and travel time.

PART 2 - PRODUCTS

2.1 PRODUCT SUBSTITUTIONS

- A. Refer to Division 01 - Product Requirements.

PART 3 - EXECUTION

3.1 GENERAL

- A. Verify elevations and dimensions prior to installation of materials.
- B. Size of pipe segments shall be equal to largest pipe segment to which it is connected where size for pipes are not indicated. Transition to smaller size shall occur on side of fitting where smaller size is indicated.

3.2 DELIVERY, STORAGE, HANDLING, AND PROTECTION

- A. Deliver products to the site under provisions of Division 01.
- B. Store and protect products under provisions of Division 01.
- C. Store in clean, dry space.
- D. Maintain factory wrapping or provide cover to protect units from dirt, water, construction debris, and traffic.
- E. Handle in accordance with manufacturer's written instructions.
- F. Handle carefully to avoid damage to components, enclosure, and finish. Lift only with lugs provided for the purpose.
- G. Provide supplemental heat if required to prevent equipment from moisture contamination.
- H. Protect openings in equipment until connected to system to prevent entry of foreign materials.

3.3 EXCAVATION AND BACKFILL

- A. Refer to Division 31 - Earthwork.
- B. Provide excavation and backfill for underground work unless otherwise indicated. Blasting is not allowed on this project without written permission of Architect and Owner.

- C. Backfill trenches beneath concrete floor and stair slabs within building and beneath concrete slabs, walks, stairs and drives at exterior of building with gravel fill and compact to same density as surrounding area.

3.4 FLOOR, WALL, ROOF AND CEILING OPENINGS

- A. Coordinate location of openings, chases, furred spaces, etc., with appropriate Contractors. Provide sleeves and inserts that are to be built into structure during progress of construction.
- B. Remove temporary sleeves, if used to form openings, prior to installation of permanent materials. Utilize minimum 24 ga galvanized sheet metal for permanent sleeves unless otherwise noted.
- C. Provide Schedule 40 carbon steel pipe with integral water stop and 2" extension above floor slab for steel sleeves required in interior floor slabs.
- D. Submit to Structural Engineer for review and approval size and location of core-drilled holes prior to execution.
- E. Submit product data and installation details for penetrations of building structure. Include schedule indicating penetrating materials (metal pipe, plastic pipe, conduit, etc.), sizes of each, opening sizes and sealant products intended for use.
- F. Where penetrations of fire-rated assemblies are involved, seal penetrations with appropriate firestopping systems as specified in Division 07 Thermal and Moisture Protection.
- G. Submit complete penetration layout drawings showing openings in building structural members including floor slabs, bearing walls, shear walls, etc. Indicate and locate, by dimension, all required openings, including those sleeved, formed or core drilled. Drawings shall be approved prior to preparing openings in structural member.
- H. Provide minimum 1" clearance around penetration openings intended for pipe. Where fire resistant penetrations are required, size openings in accordance with written recommendations of firestopping systems manufacturer.
- I. Openings for underground pipes passing through foundations or under footings shall have minimum clearance of 1-1/2" to concrete. Do not disturb footing bearing soil.
- J. Openings for underground pipe passing through on grade concrete slabs shall have minimum 1/4" clearance to concrete. Seal openings with urethane caulk.
- K. Openings for insulated piping shall be sized based on outside diameter of insulation when it is specified or detailed to be continuous through opening.

- L. Seal non fire-rated floor penetrations with non-shrink grout equal to Embeco by Master Builders, or urethane caulk, as appropriate.
- M. Seal non fire-rated wall openings with urethane caulk.
- N. Where penetrations occur through exterior walls into building spaces, use sleeves with integral water stop. For piping having outer surface temperature less than 150°F, use plastic (HDPE) sleeves, similar to PSI Link-Seal Model CS, rated to 150°F. For piping having outer surface temperature 150°F or higher, or where steel sleeves are shown or walls are fire rated, use steel sleeves with hot dip galvanizing, similar to PSI Link-Seal Model WS. Seal annular space between sleeves and pipe with Thunderline "Link-Seal" modular wall and casing seals, or sealing system by another manufacturer approved as equal by Architect. Where "Link-Seals" are used with insulated pipe, insulation shall be butted against seals on both sides. Sealing system shall utilize Type 316 stainless steel bolts, washers and nuts.
- O. In lieu of openings as specified herein penetration systems as manufactured by Pro Set may be used, including sleeve couplings and plug.
- P. If total ProSet system with Water Guard "CR" is used, opening shall not need additional water proofing or riser clamps.
- Q. Finish and trim penetrations as shown on details and as specified.
- R. Provide chrome or nickel plated escutcheons where piping passes through walls, floors or ceilings and is exposed in finished areas. Size escutcheons to fit pipe and pipe covering for finished appearance. Finished areas shall not include mechanical/electrical rooms, janitors' closets, storage rooms, etc., unless suspended ceilings are specified.

3.5 EQUIPMENT SHUTOFF VALVES

- A. Provide shutoff valves at equipment connected to piping system. Refer to valve section or system section for requirements of valve type.

3.6 EQUIPMENT ACCESS

- A. Install piping, conduit and accessories to permit access to equipment for maintenance. Relocate piping, equipment or accessories to provide access at no additional cost to Owner.
- B. Install equipment with sufficient maintenance space for removal, repair or changes to equipment. Provide ready accessibility to equipment without moving other future or installed equipment (including light fixtures) or system components.

- C. Access doors in walls, chases, or inaccessible ceilings will be provided under Division 08 - Access Doors and Frames, unless otherwise indicated. Access doors for valves, shock stops or other equipment shall provide access for servicing, repairs, and/or maintenance.
- D. Provide necessary coordination and information to the Trade Contractor under Division 08 - Access Doors and Frames. This information shall include required locations, sizes, and rough-in dimensions.
- E. Provide access doors in walls, chases or above inaccessible ceilings for valves, shock stops, unions or equipment/devices requiring access for servicing, repairs or maintenance, unless otherwise noted. Access frames and doors shall be as manufactured by Milcor, Incorporated, or similar, of style applicable to surface. Provide access doors used in fire rated construction with UL (FRD) Label. Provide steel, prime coated access doors unless otherwise specified. Provide stainless steel doors in ceramic tile walls, toilet rooms, locker rooms and in areas subject to excessive moisture. Provide access doors of sufficient size to allow complete maintenance. Coordinate location of access doors with General Contractor and rough-in equipment accordingly.

3.7 EQUIPMENT SUPPORTS

- A. Provide supporting steel not indicated on drawings as required for installation of equipment and materials including angles, channels, beams, hangers, etc.

3.8 EQUIPMENT GUARDS

- A. Provide equipment guards over belt driven assemblies, pump shafts, exposed fans, and elsewhere as indicated in this Specification or required by Code.
- B. Paint equipment guards bright yellow.
- C. Equipment guards shall comply with OSHA requirements.

3.9 SUPPORT PROTECTION

- A. In occupied areas, mechanical rooms and areas requiring normal maintenance access, guard certain equipment to protect personnel from injury.
- B. Provide minimum 1/2" thick Armstrong Armaflex insulation or similar product applied with Armstrong 520 adhesive on lower edges of equipment and mechanical supporting devices suspended less than 7 ft above floors, platforms or catwalks in these areas.
- C. Protect threaded rod or bolts at supporting elements as described above. Trim threaded rod or bolts such that they do not extend beyond supporting element and devices.

3.10 MECHANICAL SYSTEMS IDENTIFICATION

- A. Refer to Section 220553 - Plumbing Systems Identification.

3.11 TEST AND BALANCING

- A. Tests for equipment, ductwork and piping systems shall be performed as specified in their respective specification sections in accordance with technical requirements noted.
- B. Provide equipment required for testing, including fittings for additional openings required for test apparatus.
- C. All ductwork and piping inspections and testing shall be successfully completed and approved before application of covering materials.
- D. When equipment or systems fail to meet minimum test requirements, replace or repair defective work or material as necessary and repeat inspection and test until equipment or systems meet test requirements. Make repairs with new materials. Caulking of holes or threaded joints is not allowed.
- E. Contractor is responsible for certifying in writing equipment and system test results. Certification shall include identification of portion of system tested, date, time, test criteria, test medium and pressure used, duration of test and name and title of person signing test certification document.
- F. Maintain copies of certified test results, including those for any failed tests, at project site. At completion of project, include copies of test records and certifications in O&M Manuals.
- G. Balancing of various systems shall be in accordance with associated specification sections in addition to requirements noted herein.
- H. If exterior domestic water supply also serves as source for fire protection systems, either exterior or interior or both, it shall be tested according to fire protection system requirements as specified in applicable Specification Section.

3.12 START-UP

- A. Systems and equipment shall be started, tested, adjusted and turned over to Owner ready for operation. This includes "Owner-Furnished, Contractor-Installed" (OFICI) and "Contractor-Furnished, Contractor-Installed" (CFCI) systems and equipment.
- B. Follow manufacturer's pre-start-up check-out, start-up, trouble shooting and adjustment procedures.
- C. Contractor shall provide services of technician/mechanic knowledgeable in start-up and check-out of types of systems and equipment on project.

- D. Provide start-up services by manufacturer's representative where specified or where Contractor does not have qualified personnel.
- E. Coordinate start-up with all trades.

3.13 LUBRICATION

- A. Upon completion of work and before turning over to Owner, clean and lubricate bearings except sealed and permanently lubricated bearings. Use only lubricant recommended by manufacturer.
- B. Contractor is responsible for maintaining lubrication of mechanical equipment under this Contract until Work is accepted by Owner.

3.14 CLEANING

- A. Clean systems after installation is complete.
- B. Clean piping and ductwork both internally and externally to remove dirt, plaster dust or other foreign materials. When external surfaces of piping are rusted, clean and restore surface to original condition.
- C. Clean pipeline strainers to restore them to original condition or replace with new strainer elements.
- D. Clean equipment and plumbing fixtures as recommended by manufacturers.
- E. Replace throwaway or replaceable media air filters used during construction period with new filters or new filter media after construction has been completed and before building is turned over to Owner. Filter replacement shall be as hereinafter specified.
- F. Blow and clean dirt, plaster dust and other foreign matter from coils, terminal devices, diffusers, registers and grilles.
- G. Thoroughly clean equipment of stains, paint spots, dirt and dust. Remove temporary labels not used for instruction or operation.
- H. Provide additional cleaning of individual piping systems and apparatus as hereinafter specified.

END OF SECTION

SECTION 220529
PLUMBING PIPING AND EQUIPMENT SUPPORTING DEVICES

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 220700 - Plumbing Systems Insulation

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Provide all supporting devices not provided as part of building structure or indicated on structural drawings or structural details, as specified and as required for proper supporting, anchoring, and guiding of piping, equipment, materials and systems.
- B. Support for all conditions of operation, including variations in installed and operating weight of equipment, piping and ductwork, to prevent excess stress and allow for proper expansion and contraction.

1.4 SUBMITTALS

- A. Product Data for each piping system for all pipe sizes and all applicable equipment including the following:
 - 1. Manufacturer's name
 - 2. Model numbers
 - 3. Materials of construction and load ratings lbs.
 - 4. Schedule of hangers and support devices with pipe support spacing
 - 5. Insulated pipe supports along with application chart or table including pipe support spacing.
 - 6. Insulation protection saddles and weight bearing insulation table
 - 7. Details and calculations for sizing supplementary steel utilized for trapeze or specially designed supports
 - 8. Structural attachments, inserts, and concrete anchors. Submit ICC-ES Evaluation Report for each type of anchor.
 - 9. Calculations and drawings for concrete inserts and anchors for each application
 - 10. Drawings showing specific locations of any weld attachments to structure, including weight supported by such attachments

11. Drawings showing specific locations of any suspended loads which exceed 100 lbs within joist chord panel to be attached to open web steel joist structural members. Include weight supported by such attachments. (Panel is length of chord between two adjacent diagonal web members at point of connection to chord.)
12. Equipment mounting devices
13. Pipe guides and anchors
14. All other appropriate data

- B. Submittals in PDF format shall be organized using PDF bookmarks per detailed instructions on assembling submittal packages identified under Submittal Requirements section of 220000 - General Plumbing Requirements.

1.5 DESIGN CRITERIA

- A. Materials and application of pipe hangers and supports shall conform to the latest requirements of ANSI B31/ASME B31.1 Code for Pressure Piping and MSS Standard Practice SP-58-2018 (Pipe Hangers and Supports Materials, Design, Manufacture, Selection, Application, and Installation), except as supplemented or modified herein.
- B. Support materials shall be steel or stainless steel unless specifically indicated.
- C. Support devices shall be factory fabricated and have published load ratings.
- D. Unless otherwise indicated, design supports, anchors, and related components with safety factor in accordance with AISC Manual of Steel Construction, but not less than 2.0.
- E. Determine maximum deflection using the following equation.
- F. Maximum deflection of support members, including trapeze supports, shall be in accordance with the following equation, but shall not exceed 0.3":

$$D = L/240$$

Where D = Max deflection in "

L = Support member length in "

- G. Materials in contact with pipe shall be galvanically compatible with piping material to eliminate conductive path for galvanic corrosion. Provide galvanic separation, such as nonmetallic coating or inserts between piping and metallic supports where piping and support materials have galvanic potential. Pipe insulation is acceptable galvanic separation. Galvanic potential shall be determined by table below:

	Galvanized Steel	Carbon Steel	Stainless Steel (Type 304 or 316)	Copper Brass Bronze
Copper, Brass, Bronze	Yes	Yes	No	NA
Stainless Steel (Type 304 or 316)	Yes Note (1)	Yes Note (1)	NA	
Carbon Steel	No	NA		
Galvanized Steel	NA			

(1) Required where stainless steel surface area near interface is equal or greater than steel surface area

- H. Unless otherwise indicated, steel support devices exposed to ventilation air stream shall be stainless steel or steel with either galvanized finish or paint finish. Paint type shall be approved by Architect/Engineer.
- I. This Contractor is responsible for proper placement and sizing of supporting devices to accommodate insulation thickness and pitching of pipe. Coordinate with Contractor performing work specified in Section 220700 - Plumbing Systems Insulation.
- J. Where piping can be conveniently grouped to allow trapeze type supports, supporting steel shall be by means of standard structural shapes.
- K. Hangers and rods shall be plumb when pipelines are at their normal operating temperatures.
- L. Unless otherwise indicated, continuous insert channels are not allowed.
- M. Punching, drilling, or welding of building structural steel is not allowed unless approved by Structural Engineer.
- N. Refer to Structural Documents and ICC-ES Evaluation Report for application of concrete inserts and concrete anchors.
- O. Lateral braces shall be designed and detailed to apply loads as directly as possible to structural floor slabs, roof decks, or other building lateral elements. Braces shall not be applied to bottom flanges of steel beams or bottom chords of open web steel joists.

- P. Coordinate with Contractor for any proposed weld attachments to building structure. This may result in use of welding codes or standards, which may apply to "structural work". and may necessitate repair of fireproofing and/or extension of fireproofing to support members. Execution of this work may be assigned to General Trades responsible for building structural steel. Cost for this work, however, will remain the responsibility of this Contractor.
- Q. Top or bottom chords of open web steel joists may be used to support loads, provided total load within panel does not exceed 100 lbs and load is placed concentric to joist. (Panel is length of chord between two adjacent diagonal web members at point of connection to chord).
- R. Where fire rated fiberglass products are used for channel and support devices, the following properties shall apply:
1. Flame Spread Properties
 - a. Polyester Fiberglass (PF) Class 1 ASTM E84
 - b. Vinylester Fiberglass (VF) Class 1 ASTM E84
- S. Fasteners including concrete anchors for seismic application shall have ICC Evaluation Service Report (ESR) and meet requirements of local authorities.

PART 2 - PRODUCTS

2.1 STRUCTURAL SUPPORTS

- A. Unless specifically indicated on structural drawings, design and provide all supporting devices including miscellaneous steel (angles, channels, beams, etc.), required for proper support of piping, equipment and materials.

2.2 PIPE HANGERS AND SUPPORTS (METALLIC)

- A. Manufacturers: Anvil, Erico, Tolco, PHD, National Pipe Hanger Corporation, or B-Line, equal to Anvil figures listed. Corresponding MSS Type is indicated where applicable.

- B. Clevis and Roller Type Hangers:

<u>System</u>	<u>Pipe Size</u>	<u>Clevis</u>	<u>Roller</u>
Hot Pipes with Insulation (105°F and above)	and 2" and smaller	65 (MSS Type-1), 260 (MSS Type-1)	---
	2-1/2" to 6"	---	171 (MSS Type-41), 181 (MSS Type-43)
	8" and larger	---	171 (MSS Type-41)

Ambient Bare Pipes (61°F to 104°F)	2" and smaller	65 (MSS Type-1), 260 (MSS Type-1)	---
	2-1/2" and larger	260 (MSS Type-1), 216 (MSS Type-4)	---

C. Flat Surfaces (Trapeze, Rack Type):

1. Use structural steel members such as struts, angles, channels and beams to support pipes as required. Select members properly for pipe support types and loading conditions. Refer to Part 1 for design criteria. Submit support details with type of members selected and load calculations. Provide straps, clamps, rollers or slides indicated below at each support point.

<u>System</u>	<u>Pipe Size</u>	<u>Straps or Clamps</u>	<u>Rollers</u>	<u>Slides</u>
Hot Pipes with Insulation (105°F and above)	2" and smaller	Anvil Klo-Shure	---	
	2-1/2" and larger	---	171 or 177(MSS Type-41), 271 (MSS Type-45), 274 (MSS Type-46)	257 or 436 with 212 or 432 clamps, Type 1, 2 or 3 for longitudinal movement only. Type 4, 5 or 6 for both longitudinal and transverse movement
Ambient Bare Steel Pipes (61°F to 104°F)	6" and smaller 8" and larger	B-Line BVT Unistrut Cush-a-Clamp	---	---
Ambient Bare (Copper) pipes (61°F to 104°F)	all sizes	B-Line BVT Unistrut Cush-a-Clump		

D. Vertical Pipe within Wall Cavities

1. Use clamps, straps, inserts or channels to support pipes concealed in wall cavity. Select members for pipe support types and loading conditions. Refer to Part 1 for design criteria.

2. Provide clamps, strut channels, insulated supports, or brackets and inserts equal to manufacturer indicated below:

System	Pipe Size	Supports	Clamps, Brackets/Inserts	Pre-insulated Inserts
Hot and Cold Pipes with Insulation	All sizes	Anvil 137	HoldRite 260 series with SBIS bracket, Anvil 262	Anvil Klo-Sure, Pipe Shields A2000
Ambient Bare Pipes (61°F to 104°F)	All sizes	Anvil 137, 138-R	Anvil 262	---
Ambient Bare Copper Pipes (61°F to 104°F)	All sizes	Anvil CT-138R	HoldRite 260 series with SBIS bracket	---

2.3 INSULATION PROTECTION SHIELDS

- A. Anvil Fig. 167 (MSS Type-40) constructed of galvanized carbon steel. Per the latest edition of Standard MSS SP-58, select shield to accommodate outer diameter of insulation. Shield length and gauge for insulation compression strength not less than 15 psi, shall be as follows:

Pipe Size	Length	Gauge
1/4" thru 3"	12"	18
4"	12"	16
5" and 6"	18"	16
8" thru 14"	24"	14
16" thru 24"	24"	12

2.4 INSULATED PIPE SUPPORTS

- A. Description:
- Products designed specifically for weight-bearing support of insulated pipes. Apply products in accordance with manufacturer's recommendations and requirements indicated below:
 - Refer to PART 3 – EXECUTION for application of Type A, Type B, and Type C Insulated Pipe Supports specified below.
- B. General:
- Supports shall be designed and rated for applied load, including weight of pipe, fluid, insulation, and any other imposed loads, with minimum 1.5 safety factor. Ratings shall be published by manufacturer and included in submittals.
 - High compressive strength inserts utilized to support loads shall encircle circumference of pipe. Block-style inserts are not allowed.
 - Supports shall be suitable for hot or cold pipe service as applicable.
 - Submit chart or table indicating selected model along with pipe sizes, rated loads, support device types and support spacing for each piping system.

5. Pipe support spacing shall be in accordance with manufacturer's recommendations but shall not exceed maximum spacing indicated under Hanger and Support Spacing in Part 3 of this Section.
6. Testing of insulation for compressive strength properties shall comply with ASTM D1621.
7. Insulation thickness shall match adjacent pipe insulation thickness.
8. Integrity of vapor barrier jacket shall be maintained continuously through support assembly.
9. Insulated pipe support style shall be specifically selected for the application and shall consider the following criteria at minimum:
 - a. Vertical, lateral and axial support design load limits.
 - b. Vertical, lateral, and axial support design travel limits
 - c. Test or pre-operational loads that may exceed normal operating conditions
 - d. Material for any items that will be welded directly to the pipe
 - e. Loading and displacements caused by seismic, hydraulic surge, or other forces
 - f. Temperature at support steel
10. All steel components shall have corrosion protection coating consisting of hot-dip galvanizing or zinc-rich primer coating.

C. Type A Insulated Pipe Supports (Light Duty):

1. Description:
 - a. Pipe insulation specified in Section 220700 - Plumbing Systems Insulation with insulation protection shields specified in this Section. Weight-bearing inserts are not required.
 - b. Type B or Type C supports may be utilized in lieu of Type A supports.

D. Type B Insulated Pipe Supports (Standard Duty):

1. Manufacturers:
 - a. SNAPP ITZ insulation inserts by KB Enterprise, Tru-Balance Insulated Saddles by Buckaroos, Inc., Value Engineered Products, or approved equal.
 - b. Klo-Shure insulation couplings may be used for cold pipes insulated with elastomeric insulation. Mount shall be 7 Series Strup Mount with metal clamps or Clevis System for clevis hangers.
 - c. Type C supports may be utilized in lieu of Type B supports
 - d. Contractor may propose to utilize contractor-fabricated insulated pipe supports in lieu of manufactured Type B Supports. Use of contractor-fabricated assemblies is subject to approval of appropriate submittal data. Submit detail drawings of assemblies and product data showing equivalency to specified manufactured products for approval.
2. Description:
 - a. Load-rated assembly consisting of high compressive strength insulation material completely encompassing circumference of pipe, vapor barrier jacket, and insulation protection shield.
 - b. Insulation protection shield shall conform to ANSI/MSS SP-58. Shield shall be G90 galvanized steel and shall span full circumference of pipe insulation. Half-shields spanning lower 180° degrees arc of insulation outer circumference will be acceptable when used with clevis hangers.

- c. Axial length of insulation material shall be not less than 9" or 2" longer than insulation protection shields (1" minimum on each end), whichever is longer.
 - 3. Insulation Materials:
 - a. Hot Pipes 105°F to 250°F:
 - 1) Rigid closed cell, polyisocyanurate or phenolic insulation by ITW, Resolco, or Kingspan. Minimum compressive strength shall be 100 psi.
 - b. Hot Pipes 251°F to 1200°F:
 - 1) High-density calcium silicate insulation similar to Johns Manville Thermo-12 Gold. Minimum compressive strength shall be 100 psi.
 - c. Cold Pipes 60°F and below:
 - 1) Rigid closed cell, polyisocyanurate, phenolic insulation similar to ITW, Resolco, Kingspan, or cellular glass insulation similar to Pittsburgh Corning Foamglas.
- E. Type C Insulated Pipe Supports (Heavy Duty):
 - 1. Manufacturers:
 - a. Pipe Shields, Inc., Bergen Pipe Supports, or Rilco equal to Pipe Shields models listed.
 - b. Unless otherwise indicated, pre-insulated pipe supports shall be as indicated in the following schedule. Model numbers are based on Pipe Shields, Inc.
 - 1) Pipe supported on hangers: Model "A" Series and Model "D" Series
 - 2) Pipe supported on flat surfaces and pipe rollers: Models "A" Series
 - 3) Pipe supported on slides: Model "B" Series with lateral guide or restraint
 - 4) Pipe anchors: Model "C" Series
 - 5) Riser clamps: Model "E" Series with thrust plates. Select proper model for restraint for downward load or upward load.
 - c. Contractor may propose to utilize contractor-fabricated insulated pipe supports in lieu of manufactured Type C Supports. Use of contractor-fabricated assemblies is subject to approval of appropriate submittal data. Submit detail drawings of assemblies and product data showing equivalency to specified manufactured products for approval.
 - 2. Description:
 - a. Load-rated assembly consisting of high compressive strength insulation material completely encircling circumference of pipe, vapor barrier jacket system incorporating structural inserts and insulation protection shield/casing where applicable.
 - 3. Insulation Material:
 - a. Jacket:
 - 1) Jacket shall consist of G90 galvanized steel conforming to ASTM A653/A653M and shall provide complete vapor barrier around insulation and bearing surface for protection of insulation.
 - 2) When recommended by manufacturer, use reinforced insulation protection shield at support bearing surface. Insulation shall extend 1" beyond insulation protection shield to maintain vapor barrier integrity.

2.5 HANGER RODS (METALLIC)

- A. Rods shall conform to the latest MSS Standards except as modified herein. Furnish rods complete with adjusting and lock nuts.
- B. Rods shall have electroplated zinc or hot dip galvanized finish.
- C. Unless otherwise indicated, size rods for individual hangers and trapeze support as indicated in the following schedule. Rod size may be reduced one size for double rod hangers. Total weight of equipment, including valves, fittings, pipe, pipe content and insulation, shall not exceed limits indicated.

Max. Pipe Size with Single Rigid Rod	Rod Diameter inches	Max Load lbs of Hanger Rod (Not exceeding 650 °F Service Temp.)
2"	3/8	730
3"	1/2	1350
5"	5/8	2160
8"	3/4	3230
12"	7/8	4480
18"	1	5900
30"	1-1/4	9500

- D. Threaded rods are not allowed in clean rooms.

2.6 BOLTS, NUTS, STUDS AND WASHERS

- A. ASTM A307, electroplated zinc finish

2.7 ROD ATTACHMENTS

- A. Anvil Fig. 290 (MSS Type-17), galvanized finish

2.8 U-BOLTS

- A. Anvil Fig. 137 (MSS Type-24), galvanized finish

2.9 BEAM CLAMPS

- A. Beam Clamps: Anvil Fig. 133/134 (MSS Type-21), 218 (MSS Type-30), 228 (MSS Type-28 or 29) and 292 (MSS Type-28 or 29)
- B. Top Beam Clamps: Anvil Fig. 227 (MSS Type-25)
- C. C-Clamps: Anvil Fig. 86, 92 or 93 (MSS Type-19 or 23) with set screw and lock nut

2.10 ADJUSTABLE PIPE SADDLE SUPPORTS

- A. Anvil Fig. 264 (MSS Type-38), galvanized finish. Provide Anvil Fig. 63 Type T stanchion with base, galvanized finish, where applicable.

2.11 RISER CLAMPS (BARE PIPE)

- A. Anvil Fig. 261 (MSS Type-8), galvanized finish
- B. B-Line B3373C, PVC coated carbon steel, in area at pipe contact, for bare copper tubing
- C. Proset system, proseal plug and fire-fill for sleeved and cored holes.

2.12 RISER CLAMPS (INSULATED PIPE)

- A. Unless otherwise indicated, insulated pipe riser clamps shall be Type C insulated pipe supports. Refer to Insulated Pipe Supports in Part 2.
- B. Contractor may propose to utilize contractor-fabricated riser supports. Use of contractor-fabricated riser supports is subject to approval of appropriate submittal data. Submit support detail drawings, bearing stamp of Structural Engineer registered in project jurisdiction, for approval. Supports shall be engineered to withstand static and dynamic forces with minimum safety factor of 2.0. Submit insulation details addressing thermal break from building structure and vapor barriers.

2.13 CONCRETE INSERTS (WOODEN FORMED CONCRETE)

- A. Anvil Fig. 281 or 282, or Hilti HCI-WF (MSS Type-18), suitable for rod diameter and weight supported.

2.14 CONCRETE INSERTS (METAL DECK FORMED CONCRETE)

- A. Anvil Fig. 284, Tolco No. 109DD, B-Line Fig. B3019, DeWalt/Powers "Bang-It+", Hilti HCI-MD, or MSCO No. MX34.

2.15 CONCRETE ANCHORS

- A. Manufacturers: Hilti, DeWalt/Powers or Red Head
- B. Anchors shall be selected, sized, and detailed by Contractor's structural engineer registered in project's jurisdiction, based on project conditions and in accordance with project building code. Calculations and drawings shall be submitted.
- C. Anchors shall meet ICC Acceptance Criteria, and ICC-ES Evaluation Reports (ESRs) shall specifically list the current applicable codes.

- D. Anchors installed in hardened concrete for purpose of transmitting structural loads from one connected element to another, or for safety related elements such as sprinkler pipes, heavy suspended pipes, and barrier rails shall have ICC-ES report demonstrating anchors have met requirements of AC 193 for mechanical anchors in concrete elements.
- E. Post-installed expansion anchors and undercut anchors installed in hardened concrete shall be qualified for strength design and tested according to ACI 355.2. Designs shall be per the requirements of ACI 318, Appendix D.
- F. Anchors shall be zinc plated in accordance with ASTM B633.
- G. Select anchors with load ratings based on cracked concrete conditions.

2.16 CONTINUOUS INSERT CHANNELS

- A. Manufacturers: Unistrut, Anvil, B-Line, Michigan, Halfen, Hilti or Kindorf. Brackets inserts and accessories shall be suitable for channel insert selected.
- B. Length and support capabilities to be suitable for supporting plumbing piping in vicinity.

2.17 METAL FRAMING SUPPORT SYSTEM (STRUT SYSTEM)

- A. Manufacturers: Unistrut, B-Line Strut Systems, Anvil-Strut, Power-Strut, Erico, Superstrut, Kindorf, Hilti, and Hydra-Zorb
- B. Channels shall have epoxy paint or electroplated zinc finish.
- C. Channels shall not be lighter than 12 ga.

2.18 PIPE GUIDES

- A. Unless otherwise indicated, guides shall be equal to Pipe Shields "B" Series, selected by load and movement. Refer to Insulated Pipe Supports in Part 2.

2.19 PIPE ANCHORS

- A. Unless otherwise indicated, anchors shall be Type C Insulated Pipe Supports. Refer to Insulated Pipe Supports in Part 2.
- B. Contractor may propose to utilize contractor-fabricated anchors. Use of contractor-fabricated anchors is subject to approval of appropriate submittal data. Submit anchor system detail drawings, bearing stamp of Structural Engineer registered in project jurisdiction, for approval. Anchors shall be engineered to withstand static and dynamic forces with minimum safety factor of 3.0. Submit insulation details addressing thermal break from building structure and vapor barriers.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install supports to allow for free expansion of piping. Support piping from building structural members using concrete inserts, beam clamps, ceiling plates, wall brackets, or floor stands. At no time shall hangers and supports overload building structural members. Fasten ceiling plates and wall brackets securely to structure and test to demonstrate adequacy of fastening.
- B. Select and size building attachments properly in accordance with MSS Standards and manufacturer's published load rating information.
- C. Coordinate hanger and support installation to properly group piping of all trades.
- D. Suspend piping hangers by means of hanger rods. Perforated band iron and flat wire (strap iron) are not allowed.
- E. Piping and ductwork shall be supported independently from other piping or ductwork.
- F. Pipe hangers and supports shall not penetrate vapor barrier of pipe insulation.
- G. Do not support equipment, or piping from metal roof decking or ceiling grid.
- H. Install adequate supports so as not to over stress either piping or equipment to which piping is connected.
- I. Refer to Section 220000 - General Plumbing Requirements for requirements of personnel injury protection guards for supporting devices.

3.2 HANGER AND SUPPORT SPACING

- A. Space pipe hangers and supports for horizontal pipe accordance with the following schedule, with exceptions as indicated herein:

- B. Copper Tube (Domestic Water, Laboratory Water, Non-potable Water):

Pipe Size	Max Spacing
1-1/4" and smaller	6 ft-0"
1-1/2" and larger	10 ft-0"

- C. Plastic Pipe

- 1. PVC Pipe:

Pipe Size	Max Spacing
All sizes	4 ft-0"

- 2. Support plastic pipe at all changes of direction. Adequate consideration shall be given to piping expansion.

- D. Cast Iron Pipe:
 - 1. Maximum hanger and support spacing shall be 10 ft for all pipe sizes. Provide minimum of one hanger per pipe section close to joint on barrel, at each pipe fitting, at change of direction and branch connections.
 - 2. Support Cast Iron No-Hub pipe as recommended in CISPI Publication "Cast Iron Soil Pipe and Fittings Handbook, Chapter IV - Installation of Cast Iron Soil Pipe and Fittings."
 - E. Maximum spacing shown above may be restricted by strength of attachment to building structure. Submit data with calculations with published load ratings showing attachment to be utilized and maximum spacing allowable for that type of attachment and pipe size.
 - F. Spacing less than indicated above may be required to conform to building structure design or loading limitations.
 - G. Spacing less than indicated may be required depending on compressive strength of pipe insulation and insulated pipe supports.
 - H. If pipe size changes between support points, maximum spacing shall be based on the smaller pipe size.
 - I. If trapeze hangers are used to support multiple services, spacing shall be based on the most restrictive pipe size and material on trapeze hanger.
 - J. For non-metallic pipe, follow manufacturer's installation recommendations in addition to requirements noted herein.
 - K. Install supports for vertical piping and anchors as recommended by pipe manufacturer.
 - L. Place hangers and supports to meet requirements of Section 232116 - Pipe and Pipe Fittings or specific pipe system sections, with regard to pitch for drainage and venting and clearance between services.
 - M. Hangers and supports shall bear on outside of insulation when pipes are to be insulated.
 - N. Place hangers and supports within 1 ft of each fitting, such as elbows and tees, and at each valve, strainer, and other piping specialty for piping 4" and larger.
 - O. Place hanger or support at first elbow upstream of pump inlet and first elbow downstream of pump outlet.
- 3.3 INSULATED PIPE SUPPORTS APPLICATION
- A. Install insulated pipe support at each support point of insulated pipe.

- B. Pipe Size 1-1/2" and Smaller:
 - 1. Use Type A insulated pipe support. Pipe insulation specified in Section 220700 - Plumbing Systems Insulation shall be continuous through support points.
 - 2. Use one shield (bottom) for clevis hanger.
 - 3. Use 2 shields (top and bottom) for roller hanger/support or strap/clamp support. Apply 2 metal straps to hold top and bottom shields onto insulation jacket.
 - 4. Type B or Type C insulated pipe supports may be used in lieu of Type A support.
- C. Pipe Size 2" through 5":
 - 1. Use Type B insulated pipe supports. Refer to Part 2 for acceptable products.
 - 2. Type C insulated pipe supports may be used in lieu of Type B supports.
- D. Pipe Size 6" and Larger:
 - 1. Use Type C insulated pipe supports. Refer to Part 2 for applicable products.

3.4 PIPE FLOOR SUPPORTS

- A. Unless specifically shown otherwise, use adjustable pipe saddle supports with associated stanchion similar to Anvil Fig. 264/63. Select supports properly for weight and height of pipe stand.

3.5 CONCRETE INSERTS

- A. Concrete insert application, size, loading, and placement shall be this Contractor's responsibility.
- B. Coordinate with General Contractor for placement of inserts before concrete pour. Minimize use of inserts and anchors after concrete pour.

3.6 CONTINUOUS INSERT CHANNELS

- A. Mount continuous insert channels when used for pipe support on 8 ft-0" maximum centers and 8 ft-0" from corners.

3.7 BEAM CLAMPS

- A. Provide locknut for hanging rod at clamp.
- B. C-clamps are allowed for rod size 3/8" or smaller and only for static loading such as air piping, cold water piping, fire protection piping and, other similar piping. C-clamps are not allowed for hot water piping and steam and steam condensate piping, except hot water runouts to terminal heating devices.
- C. C-clamps are not allowed for open web steel joist application.
- D. C-clamps are not allowed for seismic application.

3.8 TRAPEZE SUPPORTS

- A. Construct trapeze supports with struts, angles, or channels and hang them by inserts or welded beam attachments and rods.
- B. Determine trapeze supports spacing by the smallest pipe on trapeze.
- C. If trapeze supports are used to support multiple services, support spacing shall be based on the most restrictive pipe size and material on trapeze supports.
- D. Refer to Part 1, Design Criteria for maximum deflection allowed for trapeze supports.

3.9 CONCRETE ANCHORS

- A. Anchor application, size, and placement shall be this Contractor's responsibility.

3.10 PIPE ROOF PENETRATION PROTECTIONS

- A. Install at points where pipes are penetrating roof. Install as shown and according to manufacturer's installation instructions.

3.11 PIPE GUIDES

- A. Install where shown on drawings.
- B. For manufactured expansion devices, install minimum of 2 pipe guides at each side of manufactured pipe expansion device. Locate first guide no more than 4 pipe diameters from expansion device and second guide at 14 pipe diameters from first guide. Install intermediate guides in accordance with guide spacing data recommended by manufacturer or the following table, whichever is more stringent.

MAXIMUM DISTANCE BETWEEN INTERMEDIATE GUIDES (ft)				
<u>Pipe Operating Pressure</u>				
Pipe Size (inches)	0-50 psig	51-100 psig	101-150 psig	151-200 psig
3	21	19	17	16
4	35	29	25	22
6	57	44	37	32
8	66	52	45	40
10	91	69	58	51
12	107	79	66	58
14	115	85	71	62
16	127	94	78	68

- C. If anchor is located within 4 pipe diameters from expansion joints, guides need not be installed on anchor side.

3.12 PIPE ANCHORS

- A. Install anchors where shown on drawings or in conjunction with expansion joints, loops and swing joints as required to allow proper expansion and contraction of piping without damage to structure, equipment or piping.
- B. Do not anchor piping to concrete block walls, wood, or partition walls.

END OF SECTION

SECTION 220553 PLUMBING SYSTEMS IDENTIFICATION

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 SUBMITTALS

- A. Product Data: For identification materials and devices
- B. Valve Schedules: For each piping system
- C. Samples: Of color, lettering style, and graphic representation required for each identification material and device.

PART 2 - PRODUCTS

2.1 IDENTIFYING DEVICES

- A. Stencil Paint:
 - 1. Oil-based, alkyd enamel, black color
- B. Marker System:
 - 1. Manufacturers: Brady USA, Marking Services Inc. (MSI), Kolbi, or Seton
 - 2. Manufacturer's standard, preprinted with color coding, lettering size, and length of color field according to ASME A13.1.
 - 3. Use pressure-sensitive type or "snap-on" type.
 - 4. "Strap-on" type may be used for piping over 6" size, including insulation.
- C. Valve Tags:
 - 1. Minimum 1-1/2" diameter, 0.032" thick, polished brass or 316 stainless steel.
- D. Laminated Plastic Nameplates:
 - 1. Nameplates shall be approximately 1-1/2" x 4", 1/16" thick, and have 1/2" high lettering. Face of plastic nameplates shall be black with white letters.
 - 2. Fasteners shall be self-tapping, stainless steel screws or contact type with permanent adhesive.
- E. Underground Warning Tape:
 - 1. Manufacturers: Brady USA, Marking Services Inc. (MSI), Kolbi, or Seton

2. Underground warning tape, 5.0 mil overall thickness, 2" width minimum, aluminum foil core where required to be detectable, bonded polyethylene jacket. Brady "Identoline" or equal.
3. Following services shall be provided with warning tape with colors and trace wire as indicated:
 - a. Sanitary Sewer: Green with black letters, (non) detectable

PART 3 - EXECUTION

3.1 GENERAL

- A. After painting and/or covering is completed, identify equipment and piping as indicated. Locate identification as conspicuously as possible except where such would distract from finished area.
- B. Where markers are used in high heat applications or exposed to harsh chemical or acid environments, specifically select marker materials for those applications.
- C. Coordinate, obtain and confirm mechanical systems identification criteria and requirements from Owner.

3.2 PIPING SYSTEM IDENTIFICATION

- A. Install pipe identification on each system.
- B. Place flow directional arrows at each pipe identification location.
- C. Identify all piping (except medical gas) not less than once every 25 ft, not less than once in each room, at each branch, adjacent to each access door or panel, at each valve and where exposed piping passes through walls and floors.
- D. Identify piping by stenciling. Height of lettering shall be same as pipe diameter up to maximum of 1" in height. When finished color of piping is dark, stenciling shall be on white background.
- E. Identify piping with marker system.
 1. For "strap-on" type, ensure marker is fitted snugly to pipe or pipe insulation surface with sufficient straps.

3.3 VALVE IDENTIFICATION

- A. Identify valves with brass tags bearing system identification and valve sequence number in 1/2" black characters. Attach tag to valve body with brass jack chain and "S" hook for brass tag and SS jack chain or SS braided wires with swag sleeves and "S" hook for stainless steel tag. Non-metallic fasteners are not allowed.

- B. Valve numbers shall be prefixed with corresponding piping system identification in 1/4" black letters.
- C. Furnish typewritten valve schedule indicating valve number, fixtures, equipment or areas served by each numbered valve and incorporate in O&M Manuals.

3.4 EQUIPMENT IDENTIFICATION

- A. Identify major equipment, including heat exchangers, pumps, water heaters, tanks, compressors, etc.
- B. Identify equipment by stenciling equipment number and service in 2" high letters.
- C. Identify equipment with marker system.
- D. Identify equipment with laminated plastic nameplates.
- E. Identify control equipment and panels with laminated plastic nameplates.
- F. Nameplate Markings:
 - 1. Identify model number, size, capacity, electrical characteristics, serial number, along with other items scheduled for equipment on drawings.
 - 2. Indicate motor horsepower, voltage, phase, cycles, RPM, full load amps, locked motor amps, frame size, manufacturer's name and model number, Service Factor, Power Factor, efficiency, minimum circuit amps, minimum feeder conductor size, disconnect or fuse size, refrigerant, and other pertinent information.
- G. Locate motor nameplates for easy reading. Relocate or provide new nameplates on motors if original nameplates are not located for easy reading.

3.5 ACCESS PANEL IDENTIFICATION

- A. Identify each service opening or access opening with minimum 1/2" high letters indicating type of damper.
- B. Furnish typewritten charts with identification and location of all access panels serving equipment and valves and incorporate in O&M Manuals.

3.6 UNDERGROUND WARNING TAPE

- A. Install detectable warning tape 4" to 6" below grade to allow for accurate above surface detection.
- B. Install non-detectable warning tape 6" to 12" below grade.
- C. Repair and replace existing underground warning tape where disturbed by excavation.

END OF SECTION

**SECTION 220594
DOMESTIC WATER SYSTEMS BALANCE**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 221118 - Water Distribution System
- B. Section 222114 - Plumbing Specialties

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Plumbing Contractor shall be responsible for providing complete testing and balancing work of liquid fluid handling systems, such as domestic hot water return, laboratory hot water return, water mixing valves, and other processes included in this Project.
- B. Work required shall consist of setting volume flow rates and adjusting speed controls, recording data, making tests, and preparing reports as specified herein.
- C. Scope of work includes new work specified herein and includes all equipment, distribution systems, and terminal units connected.
- D. Scope of work also includes existing liquid fluid handling systems as defined by drawings and/or schedules.
- E. Work is limited to new areas within construction boundaries and does not include central pumping equipment or other areas. Adjust and balance flows to values indicated or scheduled. If flow is abnormal, attempt to proportional balance flows to the same percentage below design and contact Engineer for additional instruction.
- F. Procedures shall be in accordance with the latest edition of AABC or NEBB and as per detailed herein.
- G. TAB work shall be performed by persons trained in TAB work and certified by Associated Air Balance Council (AABC) or National Environmental Balancing Bureau (NEBB). Contractors who are members of AABC or NEBB and who have qualified personnel available to perform work may submit Quality Assurance Submittal for approval.

- H. Contractor who are members of AABC or NEBB and who have qualified personnel available to perform Work may submit Quality Assurance Submittal for approval. Contractors who cannot meet these requirements shall subcontract with independent TAB Contractor. TAB subcontractor shall prepare Quality Assurance Submittal for Contractor who will submit it for approval.
- I. Owner will separately contract with an independent TAB Contractor to perform all testing, adjusting and balancing of HVAC hydronic systems required for this Project. Work related to testing, adjusting, and balancing that must be performed by Mechanical Contractor is specified in other sections of these Specifications.
- J. Upon direction of Architect/Engineer or TAB subcontractor, Contractor shall provide (at no additional cost to Owner) any additional work and/or devices necessary to properly balance the system, including calibrated balancing valves, gauge tapings, flow sensors, and thermometer wells. Contractor shall be responsible for trimming and balancing pump impellers as necessary to obtain design pump flow rates at minimum pressure differential.
- K. TAB work shall not proceed until all assigned personnel have been approved by, Engineer via Quality Assurance Submittal. Coordinate each phase of TAB work with overall project schedule. Each phase of TAB work shall be done in timely manner as detailed herein. Fieldwork must be complete before occupancy. Certificate of Substantial Completion shall not be issued until after Final Report is accepted by Engineer.

1.4 SUBMITTALS

- A. General:
 - 1. Make submittals in accordance with Section 013300 – Submittals. Submit minimum of 5 copies of all submittals unless otherwise directed.
 - 2. Reports shall be assembled using a 3-ring hard cover binder with Project Name and location on the cover and the side panel. Information sheets shall be 8-1/2" x 11" white bond paper. Use pre-printed forms of NEBB or AABC wherever possible. Assemble report in the following order.
 - a. Transmittal letter
 - b. Cover sheet with Project title, location, submittal date, and names and addresses of Owner, Contractor, TAB subcontractor, Architect, and Engineer
 - c. Index of numbered tabs listing major systems
 - d. Data organized by system in the following order:
 - 1) Equipment data and measurement summary
 - 2) Equipment measurement data
 - 3) Branch main measurement data
 - 4) Terminal device measurement data
 - e. Provide numbered tabs for each system.

B. Quality Assurance Submittal:

1. Within 30 days of signing Contract, Contractor shall submit the following information:
 - a. Firm resume
 - 1) AABC or NEBB active membership required
 - 2) Names of 3 recent relevant completed projects along with the project address, Owner's contact person, supervising design professional.
 - b. Supervisor resume
 - c. Balance technician(s) resume
2. Architect/Engineer and Owner reserve the right to contact previous project representatives and to reject persons whom Architect/Engineer and/or Owner feel are not qualified for this Project due to lack of relevant experience or problems on previous projects.

C. Planning Report:

1. Submit Planning Report as detailed in Part 3-EXECUTION of this Section to demonstrate to Engineer and Owner that proper procedures are being followed. Planning Report shall be submitted after Quality Assurance submittal and 30 days before fieldwork starts.

D. Initial Test Report:

1. Prior to starting Final Balance Phase, submit Initial Test Report as detailed in Part 3 of this Section to indicate to A/E and Contractor incomplete work or problem areas to be resolved before final balance is completed.

E. Final Report:

1. Within 30 days after fieldwork is completed, submit Final Report as detailed in Part 3 of this Section to assure design objectives are met and to assist Owner in future maintenance.

1.5 REFERENCE STANDARDS

- A. Refer to the latest publications of the NEBB, the American Society of Plumbing Engineers (ASPE) and the American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE) for establishing required procedures.

PART 2 - PRODUCTS

2.1 INSTRUMENTATION

- A. Provide required instrumentation to obtain proper measurements. Application of instruments and accuracy of instruments and measurements shall be in accordance with requirements of NEBB or AABC Standards and instrument manufacturer's specifications.

- B. Instruments used for measurements shall be accurate, and calibration histories for each instrument shall be available for examination by Architect/Engineer upon request. Calibration and maintenance of all instruments to be in accordance with requirements of NEBB or AABC Standards.

PART 3 - EXECUTION

3.1 GENERAL

- A. TAB work shall be done in separate phases as outlined herein. Project schedule shall allow ample time to complete TAB work before occupancy. Follow procedures outlined herein and as described in Planning Phase narratives.

3.2 PLANNING PHASE

- A. Procedure:
 - 1. Obtain latest contract documents including addenda and change orders. Obtain shop drawings and performance curves from Contractor for pumps, flow measuring devices, and terminal devices. Prepare Planning Report as detailed herein. Make adjustments in Planning Report and/or measuring instrument calibration.
- B. Planning Report:
 - 1. Planning Report shall contain the following minimum requirements.
 - a. Narratives:
 - 1) Provide written narratives of procedures used. Provide separate narratives for each pump and liquid fluid handling system.
 - 2) Identify flow-measuring devices to be used at each pump and terminal device. Provide different narratives for constant and variable flow systems.
 - 3) For non-standard water systems, include narratives on how to measure and adjust for different viscosities.
 - 4) Narratives shall include references to published standards of NEBB or AABC. Narratives shall include measuring instruments to be used and ranges required for each procedure. Narratives shall include specified adjustment tolerances. For this Project, minimum acceptable is $\pm 10\%$ of design flow.
 - b. Prebalance Checklist: include:
 - 1) Check for completeness of work
 - 2) System cleaning
 - 3) System fill and air venting
 - 4) Place system into operation
 - 5) Check expansion tanks and fill pressures
 - 6) Pump bearings, alignment, starters, vibration isolators, rotation
 - 7) Setting valves to proper position including shutoff and bypass valves
 - 8) Set up of controls and control devices

- c. Measuring Instrument List: list measuring instruments to be used for each procedure. Indicate ranges required for each procedure. Provide data on each measuring instrument to be used. This data shall include:
 - 1) Manufacturer name and model number
 - 2) Measurement range
 - 3) Pressure/temperature limits
 - 4) Date put into service
 - 5) Date of last calibration
 - 6) Include certificate from calibration firm
- 2. Architect/Engineer reserves the right to request adjustments in any procedure and/or ask for recalibration of any measuring instrument, which has not been recalibrated within the past year.
- 3. Samples: Submit copies of TAB forms to be used.
- 4. Branch circuit and terminal measurements: indicate on pre-printed forms of AABC or NEBB measurements to be taken in the field. Include branch circuit or terminal identification, system, space served, location, design flows (include zone and system summaries), and flow measuring device size, type, Cv, and manufacturer. Indicate initial setpoint on forms.

3.3 SET-UP PHASE

- A. Procedure:
 - 1. Perform prebalance checkout as per Planning Phase narrative.
- B. Initial Test:
 - 1. Measure pump data and flows in "as found" condition after initial valve settings are made.
- C. Initial Test Report:
 - 1. Submit report to Architect/Engineer and Contractor indicating measurements made and make notes of items, which are not complete or are not within design tolerance.

3.4 FINAL BALANCE PHASE

- A. Procedure:
 - 1. Perform procedures as per Planning Phase narrative. Correct deficiencies and redo procedures as required before submitting Final Report.
- B. Final Report:
 - 1. Submit report to Engineer and to Contractor indicating data and measurements as per requirements herein and per Planning Phase narrative. Do not submit partial or incomplete reports.
- C. Final Report Adjustments:
 - 1. Architect/Engineer reserves the right to check any measurement made and to reject any portion of work not within the design tolerance of $\pm 10\%$ of design flow. Contractor shall resubmit all or portions of Final Report as directed by Engineer.

END OF SECTION

**SECTION 220700
PLUMBING SYSTEMS INSULATION**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 220529 - Plumbing Piping and Equipment Supporting Devices

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Provide insulating materials and accessories as required for mechanical systems as specified below.
- B. Insulating products delivered to construction site shall be labeled with manufacturer's name and description of materials.

1.4 DEFINITIONS

- A. Concealed areas, where indicated in this Section, shall apply to shafts, furred spaces and space above finished ceilings, inaccessible tunnels and crawl spaces. All other areas, including walk-through tunnels, shall be considered as exposed.
- B. Unless otherwise indicated, unit of thermal conductivity is Btu in/(h ft² °F).
- C. Interstitial spaces are considered as concealed areas.

1.5 SUBMITTALS

- A. Product Data for each piping system for all pipe sizes and all equipment, including but not limited to, the following:
 - 1. Manufacturer's name
 - 2. Schedule of insulating materials
 - 3. Insulation material and thickness
 - 4. Jacket
 - 5. Adhesives
 - 6. Fastening methods
 - 7. Fitting materials
 - 8. Intended use of each material

9. Manufacturer's data sheets indicating density, thermal characteristics, temperature ratings
10. Insulation installation details (manufacturer's installation instruction/details, Contractor's installation details, MICA plates where applicable)
11. Literature data sheet from sealants and adhesive manufacturers stating VOC compliance with USGBC LEED IEQ 4.1.
12. Literature data sheet from coatings and mastics (including lagging adhesives) manufacturers stating VOC compliance with USGBC LEED IEQ 4.2.
13. All other appropriate data

1.6 DELIVERY, STORAGE AND HANDLING

- A. Insulation material shall be delivered to project site in original, unbroken factory packaging labeled with product designation and thickness. Shipment of materials from manufacturer to installation location shall be in weather-tight transportation. Protect insulation materials from moisture and weather during storage and installation. Protect insulation material against long exposure to UV light from sun.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Insulation:
 1. Owens Corning, Johns Manville, Manson, Knauf or CertainTeed similar to product indicated except where product of manufacturers not listed above is specifically identified for special type of insulation.
- B. Coatings, Mastics, Sealants and Adhesives:
 1. Foster, Childers, Design Polymeric, Vimasco, Miracle or Pittsburgh Corning

2.2 MATERIALS

- A. Products used for or related to air conditioning and ventilating systems shall conform to NFPA 90A possessing flame spread index of not over 25 and smoke developed index no higher than 50.
- B. Unless otherwise indicated, all products, material itself or on composite basis, shall have flame spread index not more than 25 and smoke developed index not more than 50, when tested in accordance with ASTM E84 or UL 723.
- C. Pipe insulation installed outdoors may have smoke developed index not exceeding 450.
- D. Insulation shall not contain formaldehyde.
- E. Foam insulation materials shall not use CFC or HCFC blowing agents in manufacturing process.

- F. Insulation applied on stainless steel shall meet requirements of ASTM C795 and NRC 1.36.

2.3 INSULATION

- A. Insulation materials shall be fire retardant, moisture and mildew resistant, vermin proof, and suitable to receive jackets, adhesives and coatings as indicated.
- B. Glass fiber insulation shall be of inert inorganic material, non-corrosive to mechanical surfaces.
- C. Insulating cement shall be Quick-Cote by PK Insulation MFG Co., or Ryder GP, with dry density of no more than 38 lb/ft³ thermal conductivity of 0.96 at 400°F mean temperature, and service temperature to 1200°F.
- D. Filling and finishing cement shall be Super-Stik by PK Insulation MFG Co., or Ryder MW, with dry density of no more than 24 lb/ft³, thermal conductivity of 0.74 at 500°F mean temperature, and service temperature to 1900°F.
- E. Type R Insulation (Rigid Mineral Fiber Pipe Insulation):
 - 1. Mineral fiber (fiberglass) preformed pipe insulation faced with specified jacket.
 - a. ASTM Standard: ASTM C547, Type I
 - b. Density: Minimum nominal 3 lb/ft³
 - 1) Thermal Conductivity: Maximum 0.23 at 75°F and 0.29 at 200°F
 - 2) 0.29 at 200°F mean temperature
 - c. Service Temperature: To 850°F
 - 2. Manufacturers/Products:
 - a. Johns Manville, Micro-Lok HP
 - b. Owens Corning, ASJ Max Fiberglas Pipe Insulation
 - c. Knauf, Earthwool 1000° Pipe Insulation

2.4 JACKETS

- A. Jacket puncture resistances shall be based on ASTM D781 test methods. Vapor barrier permeance ratings shall be based on ASTM E96/E96M Procedure A.
- B. Type P-1 Jackets:
 - 1. Heavy-duty, fire retardant material with glass fiber reinforcing and self-sealing lap. Jacket shall have neat, white Kraft finish suitable for painting, with burst strength of 1.5 Joules(50 beach units) minimum and tensile strength 45 lbs/in minimum. Vapor barrier shall be adhered to inner surface of jacket. Permeance shall not exceed 0.02 perm.
 - 2. Manufacturer/Products:
 - a. Owens Corning: ASJ-SSL
 - b. Johns Manville, ASJ
 - c. Knauf, ASJ+

2.5 ADHESIVES, MASTIC, COATINGS, SEALANTS, AND REINFORCING MATERIALS

- A. Adhesives and sealants shall comply with the South Coast Air Quality Management District SCAQMD 1168; VOC limits shall comply with Indoor Environmental Quality Section, Credit IEQ-4.1.
- B. Coatings and mastics shall comply with VOC limits set forth by Green Seal BS-11 and comply with the South Coast Air Quality Management District SCAQMD 1113; VOC limits shall comply with Indoor Environmental Quality Section, Credit IEQ-4.2.
- C. Products shall be compatible with surfaces and materials on which they are applied and shall be suitable for use at operating temperatures of systems to which they are applied.
- D. Products shall be fire retardant, moisture resistant and mildew resistant and vermin proof.
- E. Vapor Barrier Mastic: Below ambient insulation. Water vapor permeance shall be less than 0.08 perms at 45 mils dry film thickness per ASTM F1249.
 - 1. Foster 30-33
 - 2. Childers CP-33
 - 3. Vimasco 749
- F. Weather Barrier Breather Mastic: Above ambient insulation. Permeance shall be greater than 1.0 perms at 1/16" dry film thickness per ASTM E96/E96M.
 - 1. Foster 46-50 Weatherite
 - 2. Childers CP-10/CP-11 Vi Cryl
 - 3. Vimasco WC-5
- G. Lagging Adhesive/Coatings: Indoors applications used in conjunction with canvas/glass cloth.
 - 1. Foster 30-36
 - 2. Childers CP-50 AMV1
 - 3. Vimasco 713
- H. Metal jacketing sealant for aluminum jacketing:
 - 1. Foster 95-44 Elastolar
 - 2. Childers CP-76 Chil Byl
 - 3. Pittsburgh Corning 727
- I. Insulation joint sealant for Type P, Type PP, and Type G insulation:
 - 1. Foster 95-50 Flextra
 - 2. Childers CP-76 Chil Byl
 - 3. Pittsburgh Corning CW Sealant
- J. Glass fiber fabric reinforcing shall be 10 x 10 mesh similar to Childers Chil Glas #10 or Foster Mast A Fab.

- K. Wire mesh reinforcing shall be 22 ga, 1" galvanized.
- L. Insulation cement shall be ANSI/ASTM C195, hydraulic setting mineral wool.
- M. Finishing cement shall be ASTM C449.
- N. Butt joint and longitudinal joint adhesive for Type A insulation shall be Armstrong 520, Rubatex 373, Childers CP-82 or Foster 85-75.
- O. Weather-resistant protective finish for Type A insulation shall be equal to Armstrong WB Armaflex finish or Foster 30-64 elastomeric coating.

2.6 METAL BANDS AND WIRES

- A. Aluminum bands shall be 0.5 x 0.20" up to 48" diameter and 0.75 x 0.020" over 48" diameter.
- B. Stainless steel bands shall be 0.5 x 0.015" or 0.75 x 0.015".
- C. Stainless steel wires shall be 16 ga.

2.7 ACOUSTICAL BARRIER MATERIALS

- A. Acoustical barrier material shall be combination of absorber and barrier, similar to Sound Seal Model B-10 LAG/QFA-3. Absorber/Barrier material shall have acoustic ratings of STC-29.
- B. Minimum sound transmission loss at each octave band shall be as follows:

Sound Transmission Loss (dB)					
Octave Band Center Frequency (Hz)					
125	250	500	1000	2000	4000
18	18	23	30	39	46

PART 3 - EXECUTION

3.1 APPLICATION

- A. Provide insulation and jackets with minimum insulation thickness as indicated in the following schedule. The schedule applies to both exposed and concealed applications unless noted otherwise:

Piping System							
Service	Jacket Type	Minimum Insulation Thickness According to Pipe Size					
		Insulation Type	3/4" and Smaller	1" - 1-1/4"	1-1/2" - 3"	4" - 6"	8" and Larger
Domestic Hot Water and Hot Water Return 105-140°F	P-1	R	1"	1"	1-1/2"	1-1/2"	1-1/2"

3.2 INSTALLATION - GENERAL

- A. All insulation installation methods shall be performed in accordance with the latest edition of National Commercial and Industrial Insulation Standards published by MICA (Midwest Insulation Contractors Association) and manufacturer's installation instructions, except as modified in this Section of specifications.
- B. Install products with good workmanship, with smooth and even surfaces. Use full-length factory-furnished material where possible. Do not use scrap pieces.
- C. Apply insulation only on clean, dry surfaces, after all rust and scale have been removed and testing of systems has been completed. Do not insulate any section of system that must be pressure tested until after it has been successfully tested. Any removal and reinstallation to correct system defects prior to end of guarantee period shall be accomplished at no expense to Owner.
- D. Install insulating materials with necessary joints and terminations, to permit easy access and removal of equipment sections where inspection, service or repair is required, and to allow for expansion.
- E. Where possible longitudinal joints in jackets shall face toward wall or ceiling.
- F. Apply insulation to each pipe individually. Common insulation applied to adjacent pipes will not be accepted.
- G. Unless otherwise indicated, pipe insulation shall be continuous through walls and floors.
- H. Where multiple layers of insulation are used, stagger and secure each layer with metal bands.
- I. Where penetrations occur through fire-rated walls, partitions, or floors, provide fire seal as specified in Section 220000 - General Plumbing Requirements and Section 078400 - Firestopping.

- J. Insulate water piping within casework up to penetration of casework pipe chase at fixture stop. Insulate water piping within walls up to pipe penetration through the wall at fixture stop when serving wall-mounted fixtures. Termination of insulation shall be in neat and workman like manner with insulation jacket cap.
- K. Apply Type A insulation for insulation and jackets requiring vapor barrier protection where specified insulations are cut for mounting sensors, control devices, parts of valves, devices or components which extend out from specified insulation to prevent condensation.

3.3 PIPING, VALVE AND FITTING INSULATION

- A. Apply insulation to pipe, unions, flanges, fittings, valves and piping specialties with butt joints and longitudinal seams closed tightly. Valve insulation shall cover entire valve body including bonnets and packing nuts.
- B. Laps on factory-applied jackets shall be 2" minimum width firmly cemented with lap adhesive, or shall be pressure sealing type lap.
- C. Cover joints with factory furnished tape 3" minimum width to match jacket. Cement firmly with lap adhesive. On systems requiring a vapor barrier (ASJ), vaporseal all longitudinal and butt joints ASJ/Saran seams with 4" wide coat of vapor barrier mastic or 3" minimum tape.
- D. Where staples are used, they shall be on 6" maximum centers. When used for systems requiring vapor barrier, cover lap and staples with finish coat of vapor barrier mastic or 3" minimum tape.
- E. For finishing of insulated pipe fittings and valves where surface temperature of insulation is not higher than 125°F, use one piece PVC fitting covers, minimum thickness of 20 mil, Fitting cover shall be Johns Manville Zeston 2000 PVC, PROTO Fitting Covers, or similar by other manufacturers listed. Where fitting and valve insulation requires vapor barrier, seal joints of PVC covers with vapor barrier adhesives. Insulation type, R-value and density of insulation used at fittings shall match insulation of adjacent piping. Install insulation at pipe fittings and valves completely prior to applying PVC covers.
 - 1. For Type R (Rigid glass fiber) pipe insulation, PVC fitting covers with flexible mineral fiber blanket insulation inserts are acceptable, except those located in mechanical rooms within 6' above floor. For fitting covers located in mechanical room within 6' above floor, insulation inserts shall be pre-molded rigid fiber glass type wrapped around elbows.
- F. Stove pipe style insulation on elbows (Detail A on Plate 2-200 of MICA 8th Edition) is not allowed. It may be used for closed cell elastomeric insulation.

- G. Where terminations of pipe insulation are required, insulation shall have tapered ends, built up and finished as specified for fittings.
- H. For pipes 1" and smaller, install specified pipe insulation and jacket continuous through hanger or support locations. Install insulation protection shields to protect insulation from compressing.
- I. For pipes 1-1/2" and larger, where manufactured pre-insulated pipe supports are used at hanger or support locations, extend insulation to insulated pipe supports. Where vapor barrier is required, this Contractor shall be responsible for continuity of vapor barrier at insulated pipe supports. 3" wide vapor barrier tape on hot and cold systems at pipe supports.
- J. For pre-insulated pipe supports and insulation protection shields, refer to Section 220529 - Plumbing Piping and Equipment Supporting Devices.
- K. For Contractor-fabricated anchors, secure insulation directly to pipe surface and extend insulation up anchor for distance of 4 times insulation thickness. For pre-insulated anchors, cover entire surface of anchors with Type A insulation. Where applicable, take special care to assure vapor seal at anchor.
- L. Where mechanical grooved pipe connections are used in piping system, insulate couplings as specified for pipe.
- M. Piping, fittings and valves not to be insulated:
 - 1. Valves furnished with removable insulation/jacket

END OF SECTION

SECTION 221118 WATER DISTRIBUTION SYSTEM

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This Section covers interior domestic cold water, domestic hot water, domestic hot water return lines to a point as shown on the drawings.
- B. All components shall comply with NSF 61 and NSF 372 to be compliant with requirement for lead content of $\leq 0.25\%$ maximum weighted average.

1.2 RELATED WORK

- A. Section 220529 - Plumbing Piping and Equipment Supporting Devices
- B. Section 220553 - Plumbing Systems Identification
- C. Section 220594 - Domestic Water Systems Balance
- D. Section 220700 - Plumbing Systems Insulation
- E. Section 222114 - Plumbing Specialties
- F. Section 262913 - Enclosed Controllers

1.3 QUALITY ASSURANCE

- A. Order pipe with each length marked with manufacturer's name or trademark and type of pipe; with each shipping unit marked with purchase order number, metal or alloy designation, temper, size, and supplier's name.
- B. Installed material not meeting specification requirements must be replaced with material that meets these Specifications without additional cost to Owner.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Promptly inspect shipments to ensure material is undamaged and complies with specifications.
- B. Cover pipe to prevent corrosion or deterioration while allowing sufficient ventilation to avoid condensation. Do not store materials directly on grade. Protect pipe, tube, and fitting ends from damage. End caps shall remain in place. Protect fittings, flanges, and unions by storage inside or by durable, waterproof, above ground packaging.

- C. Offsite storage agreements will not relieve Contractor from using proper storage techniques.
- D. Storage and protection methods must allow inspection to verify products.
- E. Before shipping, piping shall be cleaned, free of rust and scale, and chemically treated to protect inside of pipe from rusting and furnished with end caps.

1.5 SUBMITTALS

- A. Manufacturer's technical data for the following:
 - 1. Pipe
 - 2. Fittings
 - 3. Joints
 - 4. Valves
 - 5. Unions and Flanges
 - 6. Dielectric fittings
 - 7. Water hammer arrestors
 - 8. In-line Centrifugal Pumps
- B. Shop Drawings on items specified herein.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials as specified shall be new unless otherwise noted.
- B. Materials shall be provided from list of approved manufacturers. Home Market, Generic Broker, or Wholesaler's house brands are not acceptable.

2.2 PIPE, FITTINGS, AND JOINTS

- A. Above Ground:
 - 1. Copper (2 1/2" and Smaller):
 - a. Pipe: Copper tube, Type L, hard drawn, ASTM B88
 - b. Fittings:
 - 1) Cast copper alloy, solder joint, pressure rated, ANSI/ASME B16.18
 - 2) Wrought copper, solder joint, pressure rated, ANSI/ASME B16.22
 - 3) Copper, press fit joint, EPDM O-ring, ASME B16.51, 0 degrees - 250°F, maximum 200 psi. Propress by Viega, ApolloPress by Apollo Flow Controls, Copper Press by Merit Brass or Presssystem by Nibco.
 - c. Joints:
 - 1) Lead free (<0.2%) solder, ASTM B32, flux, ASTM B813
 - 2) Press fit joint, EPDM O-ring, made with electro-hydraulic crimping tool and jaw correct for pipe size.
 - d. Nipples: Red brass pipe, threaded

- e. Exposed tubing and fittings in kitchen and areas subject to chemical cleaning shall have chrome plated finish.
- 2. Copper 3" thru 8":
 - a. Pipe: Copper tube, Type L, hard drawn, ASTM B88
 - b. Fittings:
 - 1) Cast copper alloy, solder joint, pressure rated, ANSI/ASME B16.18
 - 2) Wrought copper, solder joint, pressure rated, ANSI/ASME B16.22
 - c. Joints: Brazed, BCuP-3 or BCuP-5 type, AWS A5.8M/A5.8, 1250°F minimum melting point

2.3 UNIONS AND FLANGES

- A. General:
 - 1. Unions, flanges and gasket materials to have pressure rating of not less than 150 psi at 180°F.
- B. Copper (3" and Smaller):
 - 1. Wrought copper union, Nibco Figure 633-W. Mueller Brass equal.

2.4 VALVES

- A. Shutoff Valves:
 - 1. Ball Valves (2" and smaller):
 - a. Acceptable manufacturers: Apollo, Hammond, Milwaukee, Nibco, Stockham and Watts with indicated features and equal to model listed. Note that not all manufacturers make all sizes. Basis of design valves have threaded or soldered ends. Equivalent valves with press ends are acceptable when press fit joints and fittings are allowed.
 - b. Full Port, 2 Piece: Bronze body, ASTM B584, stainless steel ball and stem, teflon seats, stem extension with length according to installed system insulation thickness, 600 psi CWP pressure rating, Nibco 585-66-LF.
 - c. Full Port, 3 Piece: Bronze body, ASTM B584, stainless steel ball and stem, teflon seats, stem extension with length according to installed system insulation thickness, 600 psi CWP pressure rating, Nibco 595-Y-LF.
 - d. Insulated Handle: For insulated systems to prevent condensation on valve body with thermal and vapor seal, equal to Nibco Nib Seal.
- B. Swing Check Valves:
 - 1. Size 2" and Smaller:
 - a. Bronze body, ASTM B62, Y pattern, PTFE resilient disc, horizontal swing, 200 psi CWP rating, Nibco 413-Y-LF.
- C. Balancing Valves:
 - 1. Circuit Setter:
 - a. Acceptable Manufacturers: Bell and Gossett, Victaulic, Watts or approved equal

- b. 2" and Smaller: Shall be of lead-free brass or bronze construction with glass and carbon-filled TFE seat rings and have differential pressure read-out ports across valve seat area. Read-out ports to be filled with internal EPT insert and better connection with check valve. Valve bodies to have 1/4" NPT tapped drain/purge port. Valves to have memory stop feature and calibrated nameplate to assure specific valve setting. Valve to be leak-tight at full-rated working pressure and temperature 300 psi/250°F. B&G Circuit Setter Model CB, or approved equal.

D. Mixing Valves

- 1. Thermostatic Master Mixing Valve
 - a. Acceptable manufacturers: Armstrong, Lawler, Leonard, Powers or approved equal
 - b. Master thermostatic mixing valve, capable of maintaining mixed water temperature within 5°F of setpoint at zero demand and peak flow with recirculated flow for temperature maintenance. Valve assembly shall be capable of sensing temperature in recirculated water temperature and returning water to main distribution if temperature is adequate or diverting partial flow to heater when temperature falls below setpoint.
 - c. Mixing valve shall be bronze construction with dual thermostatic elements, integral thermometer, check valves, strainer, visual flow indicator and integral thermostatic return limiter. Valve shall be rated for 150 psi operating pressure and be certified under ASSE 1017.
- 2. Thermostatic Point of Use Tempered Water Mixing Valve
 - a. Acceptable manufacturers: Bradley, Lawler, Leonard, Powers or approved equal
 - b. Master emergency fixture thermostatic mixing valve, capable of maintaining mixed water temperature within 5°F of setpoint. Valve shall fail to cold water supply only on loss of hot water supply. Valve shall fail closed on loss of cold water supply.
 - c. Mixing valve shall be bronze construction with dual thermostatic elements, high temperature limit stop, locked temperature regulator and integral thermometers on inlet, outlet and mixed water lines. Valve shall be rated for 125 psi operating pressure and be certified per ANSI Z358.1 requirements.

2.5 DIELECTRIC FITTINGS

- A. Insulating nipple, metal casing, inert thermoplastic lining; Anvil Figure 7090, Clearflow dielectric fitting by Perfection Corporation or Victaulic Style 47.
- B. Dielectric unions 2"and smaller; dielectric flanges 2-1/2"and larger; with iron female pipe thread to copper solder joint or brass female pipe thread end connections, non-asbestos gaskets and pressure rating of not less than 175 psi at 180°F. Watts Regulator Company, Lochinvar, Wilkins or Epco Sales, Inc.
- C. Copper-silicon casting, UNS C87850, threaded or grooved end. UL classified in accordance with NSF 61 for potable water service. Victaulic Style 647

2.6 WATER HAMMER ARRESTORS

- A. Mechanical Water Hammer Arrestors:
 - 1. Mechanically engineered water hammer arrestor.
 - 2. Manufacturers: JR Smith or Zurn.

2.7 IN-LINE CENTRIFUGAL PUMPS FOR TEMPERATURE MAINTENANCE OF POTABLE HOT WATER

- A. Manufacturers: Armstrong, Aurora, Bell and Gossett, Deming, Ingersoll-Rand, Taco, Weinman, or Worthington
- B. Pumps shall be pipeline mounted, single suction type with cast iron casing, bronze fitted with working pressure of 175 psi and operating temperature of 225°F continuous, 250°F intermittent.
- C. Casings shall have tapped and plugged openings for vent, drain, and suction and discharge gauge connections.
- D. Impellers to be single suction enclosed type made of bronze, hydraulically and dynamically balanced, keyed and locked to carbon steel pump shafts and protected by replaceable bronze shaft sleeves.
- E. Impellers shall be directly hung from motor shafts without using flexible couplings.
- F. Pump shafts shall be high strength carbon steel, sealed and gasketed from pumped fluid.
- G. Pumps shall be furnished with mechanical seals of single unbalanced type with rotating element and ceramic stationary seat. Provide spare pump seals for each pump.
- H. Bearing assemblies shall have oil lubricated sleeve bearings or regreasable ball bearings.
- I. Pump shall be controlled by aquastat located in return pipe at pump with 100°F to 240°F operating temperature with 5°F to 30°F adjustable differential, remote bulb, UL listed similar to Honeywell L6006A1012.
- J. Refer to Section 262913 - Enclosed Controllers.
- K. Refer to schedule on Drawings for capacity requirements of pumps.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install pipe and fittings in accordance with reference standards, manufacturer's recommendations and recognized industry practices.
- B. Maintain piping system in clean condition during installation. Remove dirt and debris from assembly of piping as work progresses. Cap open pipe ends where left unattended or subject to contamination.
- C. Include connections to plumbing fixtures, to equipment by others, and to equipment requiring water. Provide proper backflow and back siphonage protection to safeguard potable water system from contamination.
- D. Lay out water system so as to conform to intent of drawings. Coordinate piping with building features and work of other trades. Install water piping plumb and square with building. Plans indicate, general routing, provide additional offsets as required. Install piping with necessary swing joints and offsets to allow for expansion.
- E. Install shut-off valves on branch lines near mains to avoid long dead-leg branches when valves are closed.
- F. Install shut-off valves where indicated and at base of risers to allow isolation of portions of system for repair.
- G. Do not install water piping within exterior walls.
- H. Provide drain valves at base of risers and at low points of trapped piping 2" and larger where trapped water volume exceeds 5 gallons.
- I. Provide protective sleeve covering of elastomeric pipe insulation where copper or steel piping is embedded in masonry or concrete.
- J. Provide dielectric fittings between dissimilar piping materials.
- K. Do not route piping through transformer vaults or above transformers, panelboards, or switchboards, including required service space for this equipment, unless piping is serving this equipment.
- L. Install valves and piping specialties, including items furnished by others, as specified and/or detailed. Provide access to valves and specialties for maintenance. Make connections to equipment, fixtures and systems installed by others where same requires piping services indicated in this Section.
- M. Install water pipe using proper pipe and fittings. Use reducing fittings for changes in pipe size.

3.2 COPPER TUBING

- A. Copper tubing shall be installed per Copper Development Association guidelines in addition to methods specified herein.
- B. Soldered Copper Joints:
 - 1. Use non-acidic and lead free flux on cleaned pipe and fittings for soldered joints.
 - 2. Cut tube square, remove burrs from exterior of tube and ream interior of tube before assembly.
 - 3. Fill joints with solder by capillary action. Solder shall cover joint periphery. Wipe joint clean.
 - 4. Apply heat carefully to prevent damage to pipe, fittings and valves.
 - 5. Follow manufacturer`s recommendations when heating valves and equipment for soldered connections.
- C. Brazed Copper Joints:
 - 1. Cut tube square, remove burrs from exterior of tube and ream interior of tube before assembly.
 - 2. Joints shall be cleaned and polished before brazing.
 - 3. Flux of any type shall not be used.
 - 4. Apply heat carefully to prevent damage to pipe, fittings and valves. Disassemble valves where possible to prevent damage to seats during brazing.
- D. Press Fit Copper Joint:
 - 1. Cut tube square, remove burrs from exterior of tube and ream interior of tube before assembly.
 - 2. Tubing shall be clean and dry before inserting into fittings.
 - 3. Insert pipe fully into fitting and mark on pipe at shoulder of fitting.
 - 4. Check fitting alignment against mark on pipe to ensure pipe is fully engaged.
 - 5. Crimp joint with pressing tool approved by fitting manufacturer. Mark crimped joints with permanent marker after each joint has been pressed.
 - 6. Provide initial joint leak test at 45 psig prior to performing final pressure test as defined later in this Section.

3.3 WATER HAMMER ARRESTORS

- A. Use water hammer arrestors to control water hammer. Installed devices shall be sized and located according to manufacturer`s recommendations, PDI Standards, or as shown on drawings.
- B. Use water hammer arrestors with flush valves and quick-closing valves. Provide access panels when water hammer arrestors are installed in non-accessible concealed locations.

3.4 DIELECTRIC UNIONS AND FLANGES

- A. Install dielectric unions or flanges at points where copper-to-steel pipe connection is required in domestic water systems.

- B. Install unions on equipment side of shutoff valves for items such as: water heaters, water softeners, pumps, filters, and similar equipment requiring periodic replacement.

3.5 CLEANING

- A. Flush and clean piping prior to testing. Remove corrosion by mechanical or chemical means. Use chemicals that are non-toxic.

3.6 TESTING

- A. Refer to Testing paragraph of Section 220000 - General Plumbing Requirements.
- B. Water test system may be applied to system in its entirety or in sections. Test piping with water to pressure of 100 psi for 2 h. No decrease in pressure allowed. Provide pressure gauge with shutoff and bleeder valve at highest point of system tested. Inspect joints in system under test.
- C. Defective work or material shall be replaced or repaired as necessary and inspection and test repeated. Repairs shall be made with new materials. No caulking of threaded joints or holes will be allowed.
- D. Do not conceal pipe until satisfactorily tested.
- E. Testing with air will not be allowed.

3.7 BALANCING

- A. Balance water distribution system. Adjust control valves for proper operation. Set balancing valves to maintain hot water in hot water system.
- B. Balance flush valves, flow control valves and mixing valves for adequate flow and temperature to plumbing fixtures and equipment.

3.8 DISINFECTION

- A. Disinfect water piping in the following manner:
 - 1. Clean and flush water pipe with water until water at remote tap is clear.
 - 2. Fill water systems with solution containing 50 ppm of chlorine (minimum concentration). Allow solution to stay in water system for 24 h. Alternately use solution of 200 ppm of chlorine (minimum concentration) for 3 h.
 - 3. Flush water system of chlorine solution.
 - 4. Allow clean water to stand in system for 24 h. Take sample from remote tap for bacteriological test.
- B. Do not use water system for potable water supply until safe bacteriological test is obtained. Repeat steps 1 through 4 until safe water system is obtained.

3.9 BACTERIOLOGICAL TESTS

- A. Take representative water samples and test to ensure bacteriologically safe water supply system. Include HPC (Heterotrophic Plate Count) test and test for presence of *Pseudomonas aeruginosa* as well as regular coliform bacteria test. HPC test maximum containment level of 500 organisms/ml. Perform bacteriological tests shortly before Owner's acceptance of building. If tests fail, make corrections and retest.
- B. When connecting to existing water supply of unknown quality, sample for analysis and comparison with finished water system analysis shall be taken prior to making new connection. This will allow isolating source of contamination from within scope of work or pre-existing water supply. Final conditions shall meet criteria specified above for areas within scope of work.

END OF SECTION

SECTION 221314
SANITARY WASTE AND STORM DRAINAGE SYSTEMS

PART 1 - GENERAL

1.1 DESCRIPTION

This Section includes materials and methods for sanitary waste and vent piping systems to a point as shown on the drawings.

1.2 RELATED WORK

- A. Section 220529 - Plumbing Piping and Equipment Supporting Devices
- B. Section 220553 - Plumbing Systems Identification
- C. Section 221118 - Water Distribution System
- D. Section 222114 - Plumbing Specialties
- E. Section 224000 - Plumbing Fixtures
- F. Section 262913 - Enclosed Controllers

1.3 QUALITY ASSURANCE

- A. Order piping with each length marked with manufacturer's name or trademark and type of pipe; with each shipping unit marked with purchase order number, metal or alloy designation, temper, size, and supplier's name.
- B. Installed material not meeting specification requirements must be replaced with material that meets these specifications without additional cost to Owner.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Promptly inspect shipments to ensure material is undamaged and complies with Specifications.
- B. Cover pipe to prevent corrosion or deterioration while allowing sufficient ventilation to avoid condensation. Do not store materials directly on grade. Protect pipe, tube, and fitting ends from damage. End caps shall remain in place. Protect fittings by storage inside or by durable, waterproof, above ground packaging.
- C. Offsite storage agreements will not relieve Contractor from using proper storage techniques.

- D. Storage and protection methods must allow inspection to verify products.

1.5 SUBMITTALS

- A. Manufacturer's technical data for the following:
 - 1. Pipe and fittings
 - 2. Joints
 - 3. Cleanouts
 - 4. Floor drains and floor sinks
 - 5. Air gap fittings
 - 6. Traps

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials herein specified shall be new, unless otherwise noted.

2.2 PIPE, FITTINGS, AND JOINTS

- A. Interior Underground 15" and Smaller:
 - 1. Cast Iron:
 - a. Pipe: Hubless cast iron pipe, ASTM A888, CISPI 301, NSF certified.
 - b. Fittings: Hubless cast iron fittings, ASTM A888, CISPI 301, NSF.
 - c. Joints: Heavyweight no-hub couplings with stainless steel clamps, FM 1680 Class 1, ASTM C-1540, Mission Heavyweight, Husky Series 4000, Ideal Tridon "HD", or Clamp-All Hi-Torq 125
- B. Interior Above Ground:
 - 1. Cast Iron:
 - a. Pipe: Hubless cast iron pipe, ASTM A888, CISPI 301, NSF certified with material test reports from marked with collective trademark of Cast Iron Soil Pipe Institute or receive prior approval by Engineer.
 - b. Fittings: Hubless cast iron fittings, ASTM A888, CISPI 301, NSF certified with material test reports from marked with collective trademark of Cast Iron Soil Pipe Institute or receive prior approval by Engineer.
 - c. Joints: Heavyweight no-hub couplings with stainless steel clamps, FM 1680 Class 1, ASTM C-1540, Mission Heavyweight, Husky Series 4000, Ideal Tridon "HD", or Clamp-All Hi-Torq 125

2.3 CLEANOUTS

- A. Josam, Mifab, Smith, Wade, Watts or Zurn, equal to number listed on the drawings.
- B. Provide recessed, solid brass, cleanout plugs where fittings are used as cleanouts. Provide taper-thread plug with Teflon tape thread wrap.

2.4 FLOOR DRAINS

- A. Josam, Mifab, Smith, Wade, Watts or Zurn, equal to number listed on the drawings.
- B. Floor drains shall be in accordance with ANSI/ASME A112.21.1M. Provide with caulked or no-hub connection. Floor drains shall have internal seepage collar for embedding in floor construction and weep holes to provide adequate drainage to drain pipe. Include trap seal insert where indicated on drawings.

2.5 AIR GAP FITTINGS

- A. Air gap fittings constructed of cast iron with integral air gap having free area of at least twice the inlet area. Josam, Mifab, Smith, Wade, Watts or Zurn, equal to J.R. Smith 3950 or 3951.

2.6 TRAPS

- A. Same material as pipe or fittings unless specified with fixtures. Refer to Section 224000 - Plumbing Fixtures. Provide 17 ga brass, chrome plated traps for exposed traps.

PART 3 - EXECUTION

3.1 INSTALLATION - GENERAL

- A. Install pipe and fittings in accordance with reference standards, manufacturer`s recommendations and recognized industry practices.
- B. Connect piping to fixtures, each piece of equipment, and drains. Install required piping as shown on drawings.
- C. Grade horizontal lines with minimum of 1/4" per ft, except piping 4" diameter or larger which may be run at 1/8" per ft slope with approval of local authority.
- D. Install piping parallel with building lines and at heights, which do not obstruct any portion of window, doorway, stairway, or passageway, except, as may be shown on plans. Install overhead piping as high as possible.
- E. Grade vent pipe for complete drainage by gravity to soil or waste pipes. Vent terminations shall be set true and level. Locate vent piping at least 10 ft away from window, door or intake openings. Coordinate closely with roofing Contractor to prevent damage to roofing membrane. Flashing shall be in accordance with requirements of roofing manufacturer.

- F. Where interferences develop, offset or reroute piping as required to clear interferences. Coordinate locations of plumbing piping with piping, ductwork, conduit and equipment of other trades to allow sufficient clearances. Consult drawings for exact location of pipe spaces, ceiling heights, door and window openings, or other architectural details before installing piping.
- G. Provide protective sleeve covering of elastomeric pipe insulation, where piping and/or fittings are embedded in masonry or concrete.
- H. Maintain piping in clean condition internally during construction.
- I. Mitered ells, notched tees, and orange peel reducers are not allowed. Bushings are not allowed on threaded piping.
- J. Do not route piping through transformer vaults or above transformers, panelboards, or switchboards, including required service space for this equipment, unless piping is serving this equipment.
- K. Set cleanouts true and level and protect properly throughout construction.
- L. Set floor drains true and level and protect properly throughout construction. Weep holes shall be filled with removable material and kept free from concrete and other debris during construction. Weep holes shall be cleaned out for final working order. Provide safing for floor drains installed in elevated slabs.
- M. Trap each fixture and piece of equipment requiring sanitary drainage connections. Trap seals shall be standard depth, except when deep seals are required by code. Traps shall be set true and level and located within limits of code requirements. Traps shall not be used as separator, interceptor or other type of device to retain solids. Traps shall be provided with thread type approved cleanout plugs when specified. Protect traps during construction and seal off to prevent stones, debris and other foreign matter from entering before use. Locate running traps for full accessibility with double cleanout.
- N. Provide plugs or caps for pipe openings during construction to prevent debris from entering pipe. Temporary plug shall be plastic cap or equivalent.

3.2 CAST IRON PIPE

- A. No-hub Piping: Place gasket on end of one pipe or fitting and clamp assembly on end of other pipe or fitting. Firmly seat pipe or fittings ends against integrally molded shoulder inside neoprene gasket. Slide clamp assembly into position over gasket. Tighten fasteners to manufacturer's recommended torque.
- B. Install cast iron pipe and fittings as recommended by CISPI in their publication "Installation of Cast Iron Soil Pipe and Fittings".

- C. Support piping at every coupling. Locate hanger within 18" of coupling.
- D. Installations with multiple joints within a 4 ft developed length shall be supported at every second joint.
- E. Secure base of risers with thrust restraints to prevent joint separation. Restraint shall be in accordance with CISPI recommendations.

3.3 TESTING

- A. Refer to Testing paragraph of Section 220000 - General Plumbing Requirements.
- B. Gravity Systems:
 - 1. Water test may be applied to system either in its entirety or in sections. Piping shall be tightly plugged and submitted to 10 ft head of water located at highest point. Provide separate standpipe above highest point being tested or extend system to obtain required 10 ft head of water. Head shall be maintained for at least 30 minutes before inspection starts.
- C. Defective work or material shall be replaced or repaired as necessary and inspection and test repeated. Repairs shall be made with new materials. No caulking of threaded joints or holes will be allowed.
- D. Do not backfill pipe until successfully tested.
- E. Testing with air will not be allowed.

END OF SECTION

SECTION 221600 NATURAL GAS PIPING

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This Section specifies natural gas piping and accessories to 5 ft outside the building wall.

1.2 RELATED WORK

- A. Section 220529 - Plumbing Piping and Equipment Supporting Devices
- B. Section 220553 - Plumbing Systems Identification
- C. Division 26 - Electrical

1.3 QUALITY ASSURANCE

- A. Order piping with each length marked with manufacturer's name or trademark and type of pipe; with each shipping unit marked with purchase order number, metal or alloy designation, temper, size, and supplier's name.
- B. Installed material not meeting specification requirements must be replaced with material that meets these Specifications without additional cost to Owner.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Promptly inspect shipments to ensure material is undamaged and complies with specifications.
- B. Cover pipe to prevent corrosion or deterioration while allowing sufficient ventilation to avoid condensation. Do not store materials directly on grade. Protect pipe, tube, and fitting ends from damage. End caps shall remain in place. Protect fittings, flanges, and unions by storage inside or by durable, waterproof, above ground packaging.
- C. Off-site storage agreements will not relieve Contractor from using proper storage techniques.
- D. Storage and protection methods must allow inspection to verify products.

1.5 SUBMITTALS

- A. Manufacturer's technical data for the following:
 - 1. Pipe and fittings

2. Joints
3. Valves
4. Regulators

B. Shop Drawings and Product Data on items specified herein.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Materials herein specified shall be new, unless otherwise noted.

2.2 ABOVE GROUND PIPE, FITTINGS AND JOINTS (UNDER 1 PSIG)

A. 2" and Smaller:

1. Pipe: ASTM A53/A53M, Grade A or B, Type E, or ASTM A106/A106M, Grade B, standard weight, (Schedule 40), carbon steel
2. Fittings: ASTM A197/A197M/ASME B16.3 Class 150, black malleable iron, threaded
3. Joints: Threaded

B. 2-1/2" and Larger:

1. Pipe: ASTM A53/A53M, Grade B, Type E or S, standard weight, (Schedule 40), carbon steel
2. Fittings: ASTM A234/A234M Grade WPB/ASME B16.9, standard weight, (Schedule 40), seamless, carbon steel, welded
3. Joints: Welded

2.3 UNIONS

A. Steel Pipe, 2" and Smaller:

1. Malleable iron, ground brass seat, 150 psi steam working pressure; Anvil 2125 or equivalent
2. Forged steel, spiral wound gasket seats, ASTM A105/A105M, ASME B16.5

2.4 FLANGES

A. Steel Pipe, 2-1/2" and Larger:

1. ANSI 150 lb class forged steel flanges, ASTM A105/A105M/ASME B16.5. Standard bolt pattern, ANSI 150 lb class 1/8" thick gasket, Type 304 stainless steel, spiral wound metal with graphite filler.

2.5 VALVES

A. Ball Valves:

1. Acceptable Manufacturers: Neles-Jamesbury, Apollo, Kerotest, Nibco and Watts equal to manufacturer's Figure number listed

2. 2" and Smaller:
 - a. Bronze body, threaded, quarter turn, chrome plated brass ball, large port, reinforced TFE seat and stem packing, blowout-proof stem, 250 psig LP-Gas, UL Listed, CSA certified. Apollo 50 GB series.
 - b. Carbon steel body, threaded, quarter turn, 2 piece design, 316 stainless steel ball and stem, full port, spiral wound 316 stainless steel and teflon seats and seals, blowout proof stem, 800 psi CWP rated, lever handle, CSA certified. Jamesbury Fire-Tite, Series 2000 Style 23-22
 - c. Carbon steel body, quarter turn, 3 piece design, chrome plated steel ball, full port, TFE seats and seals, blowout proof stem, 1000 psi CWP, lever handle, CSA certified. Jamesbury Fire-Tite, Series 4000.
 3. 2-1/2" through 8":
 - a. Carbon steel body, ASTM A216/A216M-WCB Class 150, quarter turn, carbon steel flanged ends, Type 316 stainless steel ball, stainless steel stem, PTFE seats and seals, API-6D test requirement. Apollo 88A-140-35-59-TD
- B. Plug Valves:
1. Acceptable Manufacturers: DeZurik, Homestead, Key Port, Milliken and Resun equal to manufacturer's Figure number listed
 2. 2" and Smaller:
 - a. Cast iron body, threaded, permanently lubricated bearings, bronze plug, corrosion resistant Hycar plug seal, Buna stem seal packing, lever actuator, 175 psi CWP, UL Listed. Key Port Figure 425S.
 3. 2-1/2" through 4":
 - a. Cast iron body, flanged, permanently lubricated bearings, electroless nickel plated cast iron plug, corrosion resistant Hycar plug seal, Buna stem seal packing, lever actuator, 175 psi CWP, UL Listed. Key Port Figure 425F.
- C. Vented Gas Pressure Regulators:
1. Acceptable Manufacturers: Fisher, Rockwell, Sensus, or American meeting capacity and performance as scheduled
 2. 2" and Smaller Venting: Cast iron body, aluminum spring case, plated steel spring, Nitrile diaphragm and disc, threaded, vent to exterior of building, 150 psi CWP, -20°F to 160°F, listed in compliance with ANSI Z21.80/CSA 6.22, Fisher S200 series. .
- D. Non-Vented Gas Pressure Regulators:
1. Acceptable Manufacturers: Maxitrol 325 with VLimit, Pietro Fiorentine "Gas Governor", or approved equal.
 2. 2" and Smaller: Cast iron or cast steel body, plated steel spring, threaded, external vent limiter, 2 psi maximum inlet pressure, listed in compliance with ANSI Z21.80/CSA 6.22.

PART 3 - EXECUTION

3.1 GENERAL

- A. Install gas piping according to requirements of this Section, local gas utility, NFPA 54 National Fuel Gas Code, AGA pamphlets and as shown on drawings.
- B. Piping through roof to be run through approved roof penetration with flashing and counter flashing.
- C. Manufacturer's representative shall instruct workmen in proper techniques for installation of underground polyethylene gas piping and provide certification to Owner that instructions have been given and proficiency been demonstrated by Contractor for installation of that piping system. Joints must be made by "Qualified" personnel proficient in joining methods of ASTM D2513 thermoplastic gas pressure pipe.
- D. Grounding to gas piping is prohibited.
- E. Gas piping shall be installed with dirt legs adjacent to equipment. The dirt legs shall have drain tees and plugs at low points.
- F. Gas piping in plenum ceilings shall have welded joints.
- G. Install gas piping above ground in buildings. Gas piping shall not be installed below building floor or footings.
- H. Pitch horizontal piping downward at 1" per 60 ft in direction of flow toward risers or appliances. Install minimum of 4" deep dirt leg at bottom of each vertical run and at each appliance. When installing mains and branches, cap gastight each tee or pipe end, which will not be immediately extended. Take branch connections to main from top or side of main.
- I. Make threaded joints by cutting pipe square and reaming inside. Threads shall be cut so exposed threads do not exceed 3 in number. Protect exposed threads against corrosion. Use only joint compounds approved for gas piping.
- J. Do not exceed maximum torque (inch-pounds) specified by valve Manufacturer during installation of seismic shut-off valve.
- K. Do not route piping through transformer vaults or above transformers, panelboards, or switchboards, including the required service space for this equipment, unless piping is serving this equipment.

3.2 PRESSURE REGULATORS

- A. Install regulators in accordance with manufacturer's instructions.

- B. Regulator shall be accessible for maintenance and protected from fire and mechanical damage. Regulator shall be supported from structure by brackets and supports.
- C. Vent(s) from vented pressure regulators or pressure safety valves shall be piped to outside. Terminate vent with protection screen and return bend. If above ground vent terminates in area subject to snow accumulation, terminate line at least 5 ft above grade. Coordinate vent routing with other trades to point of termination. Size vents in accordance with regulator manufacturer's requirements for regulator flow rate and length of run.
- D. Provide unions on both sides of regulators for removal and maintenance.
- E. Provide gas cock for pressure verification.

3.3 CONNECTIONS

- A. Provide appliance connectors at ranges, ovens and other appliances, which require that they be relocated before accessing gas connection. Do not use appliance connectors for water heaters or for other equipment or appliances, which do not require that they be relocated before accessing gas connection.
- B. Install shut off valve at each appliance. Provide valved connection at main for equipment and appliances furnished by others.
- C. Shutoff valves shall be accessible in case of emergency; installed minimum of 5 ft from equipment. Provide shutoff valves at each piece of equipment.

3.4 WELDER QUALIFICATIONS

- A. Welding procedures, welders, and welding operators for building service piping to be in accordance with certified welding procedures of National Certified Pipe Welding Bureau and Section 927.5 of ASME B31.9 Building Services Piping or AWS 10.9 Qualification of Welding Procedures and Welders for Piping and Tubing.
- B. Before metallic welding is performed, submit Welding Procedure Specification together with Procedure Qualification Record as required by Section 927.6 of ASME B31.9 Building Services Piping.
- C. Before polyethylene fusion welding is performed, submit certification that welders to be used on this project have successfully demonstrated proper welding procedures in accordance with Code of Federal Regulations, Title 49, Part 192, Section 192.285.
- D. Before welding is performed, submit Standard Welding Procedure Specification together with Procedure Qualification Record as required by Section IX of ASME Boiler and Pressure Vessel Code.

- E. Welded joints shall be made in conformance with latest provision of Code for Pressure Piping, ANSI Standard B31-8 - Gas Transmission and Distribution Systems. Welds to be made by qualified welders experienced in piping work. Welding, piping fabrication, etc. shall be in accordance with ASME Code, State Codes, and Welding Manual of Mechanical Contractors Association of America.
- F. Architect or Engineer reserves right to test work of welder employed on Project, at Contractor's/Owner's expense. If work of welder is found to be unsatisfactory, welder shall be prevented from doing further welding on project and defective welds replaced at no additional cost to Owner.

3.5 TESTING

- A. Test above ground steel gas piping with dry compressed air at 50 psi for 2 h. Soap test of each joint shall be done to detect leaks during 2 h period. No loss of pressure allowed during test period. No piping shall be concealed until successfully tested.
- B. Test underground polyethylene gas piping at 50 psi with dry air for 2 h. No loss in pressure allowed. Defective joints shall be cut out, pipe repaired, and retested. No piping may be backfilled until successfully tested.
- C. Types and extent of non-destructive examinations required for pipe welds are as shown in Table 136.4 of ASME Code for Pressure Piping, ANSI/ASME B31.1 - Power Piping. If requirements for non-destructive examination are to be other than that stated above, degree of examination, and basis for rejection shall be matter of prior written agreement between fabricator, or Contractor and purchaser.
- D. Local regulating and governing agencies may require periodic testing of seismic valve's ability to shut-off gas flow during seismic event. Follow procedures specified by local governing agency for test.

3.6 CLEANING

- A. Before actuation of gas system, flush system with dry nitrogen to ensure clean system free of oil and construction debris.

END OF SECTION

**SECTION 222114
PLUMBING SPECIALTIES**

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This Section covers material specialties for piping systems.
- B. All components installed on water systems defined in Section 221118 - Water Distribution System shall comply with NSF 372 to be compliant with requirement for lead content of <0.25% maximum weighted average.

1.2 RELATED WORK

- A. Section 220594 - Domestic Water Systems Balance
- B. Section 221118 - Water Distribution System
- C. Section 221314 - Sanitary Waste and Storm Drainage Systems

1.3 SUBMITTALS

- A. Manufacturer's technical data for the following:
 - 1. Thermometers
 - 2. Pressure gauges
 - 3. Pressure relief valves
 - 4. Strainers
 - 5. Backflow preventers
 - 6. Trap guards
 - 7. In-line check valves
 - 8. Safings
- B. Shop drawings and product data on items specified herein.
- C. Certificates: Submit performance testing certificates for reduced pressure backflow preventers and double check backflow preventers.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials herein specified shall be new unless otherwise noted.

2.2 THERMOMETERS

- A. Manufacturers: Miljoco, Taylor, Trerice, Weksler, Winters and Weiss equal to Trerice number listed
- B. Thermometers shall be 9" die cast aluminum case and frame, double strength glass window, adjustable angle stem, permanently stabilized glass tube with mercury free indicating fluid, readable scale with gradations from 30°F to 240°F. Provide brass extension neck sockets of appropriate length. Trerice Series No. A400 (old catalog number BX91400).
- C. Thermometers shall be 5" round bi-metal type, stainless steel case, readable scale and gradations from 30°F to 240°F, external calibrator adjustment, back or bottom connection as appropriate. Provide brass extension neck sockets of appropriate length. Trerice Series No. B85200. Provide with minimum or maximum registering pointers.

2.3 THERMOMETER SOCKETS AND TEST WELLS

- A. Brass construction with threaded connections suitable for thermometer stems or bulbs and control sensing devices, well length suitable for pipe diameter with extended neck as required to suit pipe insulation. Furnish with brass cap and short chain to secure cap to body for test well.
- B. Test wells for stainless steel piping shall be same material as piping.

2.4 PRESSURE GAUGES

- A. Manufacturers: Ashcroft, Marsh, Marshalltown, Miljoco, Taylor, Trerice, U.S. Gauge, Weiss, and Winters, equal to Trerice number listed.
- B. Pressure gauge shall be 4-1/2" die cast aluminum case, double strength glass window, readable dial scale with gradations from 0 to 200 psi, phosphor bronze bourdon tube, brass socket. Provide shutoff valve with pressure gauge, Trerice Series No. 600.
- C. Gauge accuracy shall meet ASME B40.1 Grade 1A (1% full scale).
- D. Pressure gauges shall be calibrated for the following pressure ranges:
 - 1. Domestic Water: 0 to 160 psi at 2 psi increments
- E. Pressure Snubbers:
 - 1. 1/4" or 1/2" size, matching gauge size, 1000 psig WP. Brass for copper or carbon steel pipe, stainless steel for stainless steel pipe.

2.5 PRESSURE RELIEF VALVE

- A. Manufacturers: Apollo, Cash-Acme, Consolidated, Kunkle, Lonergan, and Watts

- B. Bronze body, resilient seat/seal, ASME Section VIII, stainless steel spring
- C. Refer to Schedules on drawings for performance requirements.

2.6 STRAINERS

- A. Manufacturers: Apollo, Hoffman, Keckley, Metraflex, Mueller, Watts, Wheatley or Zurn
- B. Strainers shall be comparable to regulator or control valve specified. Strainers shall be "Y" type for liquid service to 300 lbs WOG at 210°F, with 20 mesh stainless steel screen. Body material shall be compatible with installed piping, stainless steel, or FDA approved, heat fused, epoxy coated interior.

2.7 BACKFLOW PREVENTER

- A. Reduced Pressure Zone Backflow Preventers:
 - 1. Manufacturers: Apollo, Cla-Val, Febco, Watts or Zurn, equal to model listed
 - 2. 3/4" through 2": Bronze body, resilient check valve seats, shut-off valves, Y-pattern strainer with bronze body and stainless steel screen, drain line air-gap fitting, bronze test cocks, certified in accordance with ASSE 1013 and AWWA C511.
- B. Double Check Valve Backflow Preventers:
 - 1. Manufacturers: Cla-Val, Febco, Apollo or Watts, equal to model listed
 - 2. 2-1/2" through 10": Cast iron body, bronze trimmed check valves, OS&Y gate valves, bronze test cocks, certified in accordance with ASSE 1015 and AWWA C506, UL listed or FM approved.

2.8 IN-LINE TRAP SEALER

- A. Manufacturers: Mi-Gard by Mifab, Quad Close Trap Seal by Jay R. Smith, Sure Seal by Rector Seal, Trap Guard by Proset, Green Drain by Green Drains Inc., or ZShield by Zurn.
- B. Inline trap sealer shall be ABS plastic housing and neoprene rubber diaphragm, silicone, or Elastomeric PVC material with self-closing bottom. In-line trap sealer shall conform to ASSE 1072.

2.9 SAFINGS

- A. 4 lb/ft² sheet lead, to 18" beyond edge of drain on all sides.
- B. Chlorinated polyethylene (CPE) as manufactured by Noble Company under trade name Chloraloy 240.
- C. Polyvinyl Chloride (PVC) shower pan line, 40 mil thickness, ASTM D4551.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Provide thermometers where indicated on drawings. Thermometers shall be easily read from floor or maintenance platforms. Calibrate thermometers to ensure accuracy.
- B. Install pressure gauges where indicated on drawings. Gauges shall be easily read from floor or maintenance platforms. Provide extensions as required to make gauges easily readable. Calibrate gauges to ensure accuracy.
- C. Install backflow preventers as indicated on drawings. Flush debris from strainers. Certified tester shall test reduced pressure zone backflow preventers to verify that functions are operational. Route vent line to adjacent hub drain.
- D. Install strainers for equipment including pumps, meters, backflow preventers, reducers and regulators, and as shown on drawings.
- E. Install in-line trap sealer as recommended by manufacturer.
- F. Install in-line check valves where specified or as indicated on drawings.
- G. Safing:
 - 1. Install safing for floor drains. Extend safing to 18" from edge of drain. Safing shall be clamped to floor drain body and pitched to drain to weep holes. Floor drains installed in unexcavated areas do not require safing.
 - 2. Where core drilled floor drain installation into existing floor slab has been approved by A/E, drain strainer inlet shall be grouted in place with non-shrink epoxy concrete approved by Structural Engineer.

3.2 TESTING

- A. Safings shall be subject to standing water test to detect leaks and proper drainage to weep holes of floor drain.

END OF SECTION

SECTION 223314 WATER HEATING EQUIPMENT

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions of Contract, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 QUALITY ASSURANCE

- A. Water heating equipment shall conform to State and Local Codes, meet national standards, and be certified by respective organization and bear its stamp.

1.3 SUBMITTALS

- A. Shop drawings and product data on items specified herein.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials specified herein shall be new unless otherwise noted.

2.2 STORAGE TYPE ELECTRIC WATER HEATER

- A. Manufacturers: A. O. Smith, Bradford White, Rheem
- B. Water heater shall be electric storage type heater. Water heater shall be UL listed and have 5 yr tank warranty and 1 yr parts warranty.
- C. Heating element(s) shall be zinc plated copper sheaths. Temperature control shall be adjustable from 110° to 170°F for single heating element.
- D. Water heater design shall include electronic controls for element sensing low water cut off and economy mode with programmable operating set points.
- E. Tank shall be glass lined steel ASME rated for 150 psi. Tank shall have removable magnesium anode rod, pressure and temperature relief valve, drain valve, polyurethane insulation and painted steel jacket.
- F. Provide thermometers on inlet and outlet to heater.

2.3 POINT OF USE ELECTRIC WATER HEATER

- A. Manufacturers: Chronomite, Eemax
- B. Water heater shall be point of use, tankless, under counter electric water heater.
- C. Replaceable heating element shall be stainless steel or iron-free nickel-chrome wire. Water heater to operate by electric flow switch and to be tested at 150 psig.

2.4 THERMAL EXPANSION TANK

- A. Manufacturers: Amtrol, Goulds, Wessels
- B. Expansion tank shall be precharged, diaphragm-type tank designed for hot water systems. Tank shall be ASME rated for 125 psig and come with fixed rubber bladder, air charge fitting, steel base ring stand (floor mounted tanks only), factory primed and enamel painted exterior, and ASME relief valve. Materials exposed to water to be NSF or FDA approved for potable water service.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install water heaters as recommended by manufacturer. Provide final connections as required. Coordinate water heater location with other Contractors.
- B. Initial start up and balancing service shall be provided by representative of manufacturer.

END OF SECTION

SECTION 224000 PLUMBING FIXTURES

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This Section lists plumbing fixtures and accessories including method of installation.

1.2 RELATED WORK

- A. Section 221118 - Water Distribution System
- B. Section 221314 - Sanitary Waste and Storm Drainage Systems

1.3 SUBMITTALS

- A. One package of manufacturer's technical data for all items. Submittal shall be assembled brochure, showing cuts and full detailed descriptions for each item.
- B. Shop drawings and product data on items specified herein.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials as specified shall be new unless otherwise noted.
- B. Vitreous china fixtures shall be of highest quality, non-absorbent, hard-burned, and vitrified throughout.
- C. Enameled ware shall be quality cast iron of uniform thickness and density, glazed to uniform depth and high gloss rubbed smooth, without chips or flaws, craze, or cracks, and completely acid resisting.
- D. Stainless steel fixtures shall be 302/304 types of non-corrosive steel, 18 ga self-rim for cabinet sinks, 14 ga for free standing compartment type sinks. Sink material shall have satin finish and coved corners, with faucet holes punched to match specified faucet fitting.
- E. Precast receptors and shower basins shall be "terrazzo" concrete. Receptor and basin colors shall be standard colors unless otherwise noted. Assembly of drain to waste piping shall be made from floor level on which basin or receptor is installed.

- F. Insulation for traps and supplies shall be molded closed cell vinyl insulation and shall meet ASTM E84 for flame and smoke spread. Insulation shall be vandal resistant and be color as listed.

2.2 MANUFACTURERS

- A. Plumbing fixtures shall be provided from list of approved manufacturers. Home Market, Generic Broker, or Wholesaler's house brands are not acceptable.
- B. Stainless Steel Sinks: As listed on the drawings.
- C. Manual Faucets: As listed on the drawings.
- D. Sensor Activated Faucets: As listed on the drawings.
- E. Fixture Traps: As listed on the drawings.
- F. Insulated Traps and Supplies: As listed on the drawings.
- G. Supplies and Stops: As listed on the drawings.

2.3 DRAINS

- A. Stainless Steel Sinks: Stainless steel, 1-1/2" diameter tailpiece with grid strainer

2.4 TRAPS

- A. Sinks: P-traps shall be chrome plated brass body with cleanout plug, 17 ga seamless tubular wall bend, cast brass slip nuts and wall escutcheon. Trap size to match fixture connections.

2.5 STOPS AND SUPPLIES

- A. Sinks: Angle pattern, lock shield cap, loose key handle, copper alloy control valve body, stem and gland nut, 1/2" NPT inlet x 1/2" compression outlet, McGuire LF2167LK

2.6 PLUMBING FIXTURES

- A. Refer to schedule on drawings for detailed fixture selection criteria not contained herein.
- B. Submittal sheets for water closets, urinals, public lavatories and shower heads shall contain EPA WaterSense Label.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install plumbing fixtures as recommended by manufacturer. Caulk around fixtures mounted on irregular surfaces such as tile or stone with silicone sealant, same color as fixture.
- B. Install each fixture with trap easily removable for servicing and cleaning. Install fixture stops in readily accessible location for servicing.
- C. Install barrier free fixtures in compliance with local code and Federal ADA Accessibility Guidelines. Install barrier free lavatory traps parallel and adjacent to wall and supplies and stops elevated to 27" above finished floor to avoid contact by wheelchair users.
- D. Return fixture waste and supply piping into wall as high as practical under fixture. Provide accessible shutoff in fixture supply. Protect "barrier free" supply and drain piping with white colored wrap neatly trimmed to prevent contact with hot or sharp surfaces by user.
- E. Provide individual supplies to fixtures and rough-in fixture piping with adequate support to prevent movement fore, aft and laterally. Provide additional blocking as required.

3.2 PROTECTION

- A. Protect finished surfaces of fixtures from accidental damage or discoloration by use of protective covering.

3.3 CLEANING

- A. Prior to Owner acceptance, clean fixtures with compounds recommended by manufacturer and remove stains and marks from surrounding walls and countertops.

END OF SECTION

SECTION 230000 GENERAL MECHANICAL REQUIREMENTS

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 DESCRIPTION

- A. Intent of drawings and Specifications is to obtain complete systems, tested, adjusted, and ready for operation.
- B. The terms "provide", "furnish" and "install" as used in Contract Documents shall have the following meanings:
 - 1. "Provide" or "provided" shall mean "furnish and install".
 - 2. "Furnish" or "furnished" does not include installation.
 - 3. "Install" or "installed" does not include furnishing.
- C. Include incidental details not usually shown or specified, but necessary for proper installation and operation.
- D. Check, verify and coordinate work with drawings and specifications prepared for other trades. Include modifications, relocations or adjustments necessary to complete work or to avoid interference with other trades.
- E. Information given herein and on drawings is as exact as could be secured but is not guaranteed. Do not scale drawings for exact dimensions.
- F. Where Architectural features govern location of work, refer to architectural drawings.
- G. Contractor may install additional piping, fittings and valves, not shown on drawings, for testing purposes or convenience of installation. Where such materials are installed, they shall comply with specifications and shall be sized to be compatible with system design. Remove such installed materials when they interfere with design conditions or as directed by Architect.

1.3 RELATED WORK

- A. When interruption of services is required, Architect, Owner, and other concerned parties shall be notified and shall determine a time.

B. Demolition:

1. Division 02 - Selective Demolition.
2. Perform demolition as required to accomplish new work.
3. Accomplish work in neat workmanlike manner to minimize interference, annoyance or inconvenience such work might impose on Owner or other Contractors.
4. Unless otherwise noted, remove from premises materials and equipment removed in demolition work.
5. Equipment noted to be removed and turned over to Owner, shall be delivered to Owner at place and time Owner designates.
6. Where materials are to be turned over to Owner or reused and installed by Contractor, it shall be Contractor's responsibility to maintain condition of materials and equipment equal to that existing before work began. Repair or replace damaged materials or equipment at no additional cost to Owner.
7. Where demolition work interferes with Owner's use of premises, schedule work through Architect, Owner and with other Contractor to minimize inconvenience to Owner. Architect must approve schedule before Contractor begins such Work.

C. Painting:

1. Painting of mechanical equipment will be done under Division 09 unless specified otherwise or unless equipment is to be furnished with factory applied finish coats.
2. Equipment
 - a. Furnish equipment with factory applied prime finish unless otherwise specified.
 - b. If factory finish on equipment furnished by Contractor is damaged in shipment or during construction, refinish equipment to satisfaction of Architect.
3. Piping:
 - a. Uninsulated Piping
 - 1) Paint cast iron, carbon steel, and copper piping located outside building and inside building within unfinished spaces without ceilings.
 - 2) Galvanized steel and stainless steel piping shall not be painted.
 - b. Insulated Piping
 - 1) Paint exposed insulated piping.
 - 2) Aluminum, stainless steel, PVC, and pre-colored insulation jackets shall not be painted.

1.4 REQUIREMENTS OF REGULATORY AGENCIES

- A. Rules and regulations of Federal, State and Local Authorities and utility companies, in force at time of execution of Contract shall become part of this specification.

1.5 REFERENCE STANDARDS

- A. Agencies or publications referenced herein refer to the following:
1. AGA - American Gas Association
 2. AMCA - Air Movement and Control Association
 3. ANSI - American National Standards Institute
 4. AHRI - Air-Conditioning, Heating and Refrigeration Institute

5. ASHRAE - American Society of Heating Refrigerating and Air Conditioning Engineers
6. ASPE - American Society of Plumbing Engineers
7. ASSE - American Society of Sanitary Engineering
8. AWS - American Welding Society
9. AWWA - American Water Works Association
10. ASME - American Society of Mechanical Engineers
11. ASTM - American Society for Testing and Materials
12. CDA - Copper Development Association
13. CISPI - Cast Iron Soil Pipe Institute
14. FMG - FM Global
15. FS - Federal Specifications
16. IEEE - Institute of Electrical and Electronics Engineers
17. MCA - Mechanical Contractors Association
18. MSS - Manufacturers Standardization Society
19. NEC - National Electrical Code
20. NEMA - National Electrical Manufacturers Association
21. NFPA - National Fire Protection Association
22. NIST - National Institute of Standards & Technology
23. NSF - National Sanitation Foundation
24. NSPI - National Spa and Pool Institute
25. OSHA - Occupational Safety and Health Administration
26. PDI - Plumbing and Drainage Institute
27. SMACNA - Sheet Metal and Air Conditioning Contractors National Association
28. UL - Underwriters Laboratories, Inc.
29. WQA - Water Quality Association

- B. Work shall be in accordance with latest edition of codes, standards or specifications unless noted otherwise.

1.6 SUBMITTALS

- A. Product Data:
1. Refer to Division 01 - Submittal Procedures.
 2. Unless otherwise indicated submittals shall be in electronic Portable Document Format (pdf) and bookmarks shall be included to facilitate navigation through submittal documentation.
 3. Submit for equipment and systems as requested in the respective specification sections.
 4. Specifically mark general catalog sheets and drawings to indicate specific items submitted and its correlation to specific designation for product in drawings.
 5. Specifically indicate proper identification of equipment by name and/or number, as indicated in specification and shown on drawings.
 6. When manufacturer's reference numbers are different from those specified, provide correct cross-reference numbers for each item. Clearly mark and note submittals accordingly.
 7. Submit complete record of required components when fixtures, equipment and items specified include accessories, parts and additional items under one designation.

8. Include composite wiring diagrams for electrically powered equipment and devices.
 9. Where submittals cover products containing non-metallic materials, include "Material Safety Data Sheet" (MSDS) from manufacturer stating physical and chemical properties of components and precautionary considerations required.
 10. Submit shop drawings or product data as soon as practicable after signing contracts. Submittals must be approved before installation of materials and equipment.
 11. Submittals that are not complete, not permanent or not properly checked by Contractor will be returned without review.
- B. Certificates and Inspections:
1. Obtain and pay for inspections required by authorities having jurisdiction and deliver certificates approving installations to Owner unless otherwise directed.
- C. Operation and Maintenance Manuals:
1. Refer to Division 01 - Operation and Maintenance Data.
 2. Upon completion of Work but before final acceptance of system, submit to Architect for approval, operation and maintenance manuals. 2" Submit set initially for review. After securing approval, submit to Owner.
 3. Organize manuals by specification section number and furnish table of contents and tabs for each piece of equipment or system.
 4. Fire protection system shall be separately bound.
 5. Manuals shall include the following:
 - a. Copies of Shop Drawings and Product Data
 - b. Manufacturer's operating and maintenance instructions. Include parts lists of items or equipment, with component exploded views and part numbers. Where manufacturer's data includes several types or models, designate applicable type or model.
 - c. Electronic versions O&M data with exploded parts lists where available
 - d. Phone numbers and addresses of local parts suppliers and service companies
 - e. Internet/WEB page addresses where applicable
 - f. Wiring diagrams
 - g. Startup and shutdown procedures
 - h. Composite electrical diagrams
 - i. Flow diagrams
 - j. Lubrication instructions
 - k. Factory and field test records (Refer to Test and Balancing in Part 3 of this section.)
 - l. Air and water balance reports
 - m. Valve identification charts as specified in Section 220553 - Plumbing Systems Identification
 - n. Access panel identification charts as specified in Section 220553 - Plumbing Systems Identification
 - o. Additional information, diagrams or explanations as designated under respective equipment or systems specification sections.
 6. Instruct Owner's representative in operation and maintenance of equipment. Instruction shall include complete operating cycle on all apparatus.
 7. Furnish O&M Manuals and instructions to Owner prior to request for final payment.

D. Record Documents:

1. Refer to General Conditions of Contract, and Division 01 - Project Record Documents. Prepare complete set of record drawings in accordance with Division 01.
2. Use designated set of prints of Contract Documents as prepared by Architect to mark-up for record drawing purposes.

1.7 JOB CONDITIONS

A. Building Access:

1. Arrange for necessary openings in building to allow for admittance of all apparatus.

B. Electrical Coordination:

1. Contractor shall provide the following items as specified under their respective Division(s):
 - a. Motors
 - b. Electrically powered equipment
 - c. Electrically controlled equipment
 - d. Starters, where specified
 - e. Variable frequency drives, where specified
 - f. Control devices, where specified
 - g. Temperature Control wiring
 - h. Wiring diagrams to Electrical Contractor for apparatus indicating external connection and internal controls.
 - i. Disconnect devices furnished with units (VFDs, chillers, prepackaged control devices, etc.)
 - 1) Devices shall have an interrupting rating not less than that of the upstream overcurrent device as shown on electrical drawings.
 - 2) Equipment electrical connection points shall be labeled with listed electrical short circuit current rating (SCCR). SCCR shall not be less than interrupting rating of upstream overcurrent device as shown on electrical drawings. SCCR shall be marked on equipment control enclosure in accordance with UL 508, or other acceptable, accredited third-party testing agency standards.
2. Electrical Contractor will provide the following devices required for control of motors or electrical equipment, unless noted otherwise.
 - a. Starters
 - b. Disconnect devices
 - c. Control devices:
 - 1) Pushbuttons
 - 2) Pilot lights
 - 3) Contacts
 - d. Conduit, boxes and wiring for power wiring.
 - e. Conduit, boxes and wiring for control wiring, except temperature control wiring.
3. Electrical Contractor will make connections, from power source to starter or variable frequency drive and from starter or variable frequency drive, where specified, to motor.

4. Where starters or other similar control devices are furnished by this Contractor, they shall be installed by this Contractor and wired by Electrical Contractor.
5. Should any change in size, hp rating, voltage, or means of control be made to any motor or other electrical equipment after Contracts are awarded, this Contractor shall immediately notify Electrical Contractor of change. Additional costs due to these changes shall be responsibility of this Contractor.

C. Cutting and Patching:

1. Refer to General Conditions of the Contract, and Division 01 - Cutting and Patching.
2. Perform cutting and patching required for complete installation of systems, unless otherwise noted. Patch and restore work cut or damaged to original condition. This includes openings remaining from removal or relocation of existing system components.
3. Provide materials required for patching unless otherwise noted.
4. Do not pierce beams or columns without permission of Architect and then only as directed. If openings are required through walls or floors where no sleeve has been provided, hole shall be core drilled to avoid unnecessary damage and structural weakening.
5. Where alterations disturb lawns, paving, walks, etc., replace, repair or refinish surfaces to condition existing prior to commencement of work. This may include areas beyond construction limits.

D. Housekeeping and Cleanup:

1. Refer to Division 01 - Closeout Procedures.
2. As work progresses and/or as directed by Architect, periodically remove waste materials from building and leave area of work broom clean. Upon completion of Work, remove tools, scaffolding, broken and waste materials, etc., from site.

1.8 WARRANTY

- A. Refer to Division 01 for general warranty requirements.
- B. Refer to technical sections for warranty requirement for each system.
 1. Where no warranty requirements are called out, warrant equipment, materials, and workmanship to be free from defect as called out in Division 01.
- C. Warrant that systems will operate without objectionable noise, vibration and uncontrolled expansion.
- D. Repair, replace or alter systems or parts of systems found defective at no extra cost to Owner.
- E. In any case, wherein fulfilling requirements of any warranty, if this Contractor disturbs any work warranted under another contract, this Contractor shall restore such disturbed work to condition satisfactory to Architect and warrant such restored work to same extent as it was warranted under such other contract.
- F. Warranty shall include labor, materials, and travel time.

PART 2 - PRODUCTS

2.1 PRODUCT SUBSTITUTIONS

- A. Refer to Division 01 - Product Requirements.

PART 3 - EXECUTION

3.1 GENERAL

- A. Verify elevations and dimensions prior to installation of materials.

3.2 DELIVERY, STORAGE, HANDLING, AND PROTECTION

- A. Deliver products to the site under provisions of Division 01.
- B. Store and protect products under provisions of Division 01.
- C. Store in clean, dry space.
- D. Maintain factory wrapping or provide cover to protect units from dirt, water, construction debris, and traffic.
- E. Handle in accordance with manufacturer's written instructions.
- F. Handle carefully to avoid damage to components, enclosure, and finish. Lift only with lugs provided for the purpose.
- G. Provide supplemental heat if required to prevent equipment from moisture contamination.
- H. Protect openings in equipment until connected to system to prevent entry of foreign materials.

3.3 FLOOR, WALL, ROOF AND CEILING OPENINGS

- A. Coordinate location of openings, chases, furred spaces, etc., with appropriate Contractors. Provide sleeves and inserts that are to be built into structure during progress of construction.
- B. Remove temporary sleeves, if used to form openings, prior to installation of permanent materials. Utilize minimum 24 ga galvanized sheet metal for permanent sleeves unless otherwise noted.
- C. Provide Schedule 40 carbon steel pipe with integral water stop for steel sleeves required in interior floor slabs.

- D. Submit to Structural Engineer for review and approval size and location of core-drilled holes prior to execution.
- E. Submit product data and installation details for penetrations of building structure. Include schedule indicating penetrating materials (metal pipe, plastic pipe, conduit, etc.), sizes of each, opening sizes and sealant products intended for use.
- F. Where penetrations of fire-rated assemblies are involved, seal penetrations with appropriate firestopping systems as specified in Division 07.
- G. Submit complete penetration layout drawings showing openings in building structural members including floor slabs, bearing walls, shear walls, etc. Indicate and locate, by dimension, all required openings, including those sleeved, formed or core drilled. Drawings shall be approved prior to preparing openings in structural member.
- H. Provide minimum 1" clearance around penetration openings intended for pipe. Where fire resistant penetrations are required, size openings in accordance with written recommendations of firestopping systems manufacturer.
- I. Openings for underground pipes passing through foundations or under footings shall have minimum clearance of 1-1/2" to concrete. Do not disturb footing bearing soil.
- J. Openings for underground pipe passing through on grade concrete slabs shall have minimum 1/4" clearance to concrete. Seal openings with urethane caulk.
- K. Openings for insulated piping shall be sized based on outside diameter of insulation when it is specified or detailed to be continuous through opening.
- L. Openings for duct penetrations shall be no more than 1/2" larger on all sides than size of duct or duct including duct insulation, if applicable. Where firestopping systems are required at penetrations, size in accordance with recommendations of firestopping systems manufacturer, but opening shall not exceed 1" average clearance on all sides. Openings for ducts with fire dampers shall be in accordance with fire damper installation requirements.
- M. Duct penetrations through concrete floors in mechanical rooms containing liquid heat exchangers and/or pumps shall have 2" high water stopped curbs surrounding openings. This applies to mechanical rooms above the lowest floor level.
- N. Seal non fire-rated floor penetrations with non-shrink grout equal to Embeco by Master Builders, or urethane caulk, as appropriate.
- O. Seal non fire-rated wall openings with urethane caulk.

- P. Where penetrations occur through exterior walls into building spaces, use sleeves with integral water stop. For piping having outer surface temperature less than 150°F, use plastic (HDPE) sleeves, similar to PSI Link-Seal Model CS, rated to 150°F. For piping having outer surface temperature 150°F or higher, or where steel sleeves are shown or walls are fire rated, use steel sleeves with hot dip galvanizing, similar to PSI Link-Seal Model WS. Seal annular space between sleeves and pipe with Thunderline "Link-Seal" modular wall and casing seals, or sealing system by another manufacturer approved as equal by Architect. Where "Link-Seals" are used with insulated pipe, insulation shall be butted against seals on both sides. Sealing system shall utilize Type 316 stainless steel bolts, washers and nuts.
- Q. In lieu of openings as specified herein penetration systems as manufactured by Pro Set may be used, including sleeve couplings and plug.
- R. If total ProSet system with Water Guard "CR" is used, opening shall not need additional water proofing or riser clamps.
- S. Finish and trim penetrations as shown on details and as specified.
- T. Provide chrome or nickel plated escutcheons where piping passes through walls, floors or ceilings and is exposed in finished areas. Size escutcheons to fit pipe and pipe covering for finished appearance. Finished areas shall not include mechanical/electrical rooms, janitors' closets, storage rooms, etc., unless suspended ceilings are specified.
- U. Trim duct penetrations exposed in finished areas with 2" wide galvanized or aluminum trim collars properly sized to fit duct. Collars shall be same gauge as duct, prime finish unless noted otherwise. Finished areas shall not include mechanical rooms, janitors' closets, storage rooms, etc., unless suspended ceilings are specified.

3.4 EQUIPMENT SHUTOFF VALVES

- A. Provide shutoff valves at equipment connected to piping system. Refer to valve section or system section for requirements of valve type.

3.5 EQUIPMENT ACCESS

- A. Install piping, conduit and accessories to permit access to equipment for maintenance. Relocate piping, equipment or accessories to provide access at no additional cost to Owner.
- B. Install equipment with sufficient maintenance space for removal, repair or changes to equipment. Provide ready accessibility to equipment without moving other future or installed equipment (including light fixtures) or system components.

- C. Access doors in walls, chases, or inaccessible ceilings will be provided under Division 08 - Access Doors and Frames, unless otherwise indicated. Access doors for valves, shock stops or other equipment shall provide access for servicing, repairs, and/or maintenance.
- D. Provide necessary coordination and information to the Trade Contractor under Division 08 - Access Doors and Frames. This information shall include required locations, sizes, and rough-in dimensions.
- E. Provide access doors in walls, chases or above inaccessible ceilings for valves, shock stops, unions or equipment/devices requiring access for servicing, repairs or maintenance, unless otherwise noted. Access frames and doors shall be as manufactured by Milcor, Incorporated, or similar, of style applicable to surface. Provide access doors used in fire rated construction with UL (FRD) Label. Provide steel, prime coated access doors unless otherwise specified. Provide stainless steel doors in ceramic tile walls, toilet rooms, locker rooms and in areas subject to excessive moisture. Provide access doors of sufficient size to allow complete maintenance. Coordinate location of access doors with General Contractor and rough-in equipment accordingly.

3.6 EQUIPMENT SUPPORTS

- A. Provide supporting steel not indicated on drawings as required for installation of equipment and materials including angles, channels, beams, hangers, etc.

3.7 EQUIPMENT GUARDS

- A. Provide equipment guards over belt driven assemblies, pump shafts, exposed fans, and elsewhere as indicated in this Specification or required by Code.
- B. Paint equipment guards bright yellow.
- C. Equipment guards shall comply with OSHA requirements.

3.8 SUPPORT PROTECTION

- A. In occupied areas, mechanical rooms and areas requiring normal maintenance access, guard certain equipment to protect personnel from injury.
- B. Provide minimum 1/2" thick Armstrong Armaflex insulation or similar product applied with Armstrong 520 adhesive on lower edges of equipment and mechanical supporting devices suspended less than 7 ft above floors, platforms or catwalks in these areas.
- C. Protect threaded rod or bolts at supporting elements as described above. Trim threaded rod or bolts such that they do not extend beyond supporting element and devices.

3.9 LEAD SHIELDING

- A. Wherever installation of this Contractor's equipment destroys radiologic integrity of wall, floor, or ceiling, this Contractor shall be responsible to provide suitable lead shielding to restore that integrity. Coordinate these requirements with Contractor.

3.10 MECHANICAL SYSTEMS IDENTIFICATION

- A. Refer to Section 230553 - Mechanical Systems Identification

3.11 TEST AND BALANCING

- A. Tests for equipment, ductwork and piping systems shall be performed as specified in their respective specification sections in accordance with technical requirements noted.
- B. Provide equipment required for testing, including fittings for additional openings required for test apparatus.
- C. All ductwork and piping inspections and testing shall be successfully completed and approved before application of covering materials.
- D. When equipment or systems fail to meet minimum test requirements, replace or repair defective work or material as necessary and repeat inspection and test until equipment or systems meet test requirements. Make repairs with new materials. Caulking of holes or threaded joints is not allowed.
- E. Contractor is responsible for certifying in writing equipment and system test results. Certification shall include identification of portion of system tested, date, time, test criteria, test medium and pressure used, duration of test and name and title of person signing test certification document.
- F. Maintain copies of certified test results, including those for any failed tests, at project site. At completion of project, include copies of test records and certifications in O&M Manuals.
- G. Balancing of various systems shall be in accordance with associated specification sections in addition to requirements noted herein.

3.12 START-UP

- A. Systems and equipment shall be started, tested, adjusted and turned over to Owner ready for operation. This includes "Owner-Furnished, Contractor-Installed" (OFICI) and "Contractor-Furnished, Contractor-Installed" (CFICI) systems and equipment.
- B. Follow manufacturer's pre-start-up check-out, start-up, trouble shooting and adjustment procedures.

- C. Contractor shall provide services of technician/mechanic knowledgeable in start-up and check-out of types of systems and equipment on project.
- D. Provide start-up services by manufacturer's representative where specified or where Contractor does not have qualified personnel.
- E. Coordinate start-up with all trades.

3.13 LUBRICATION

- A. Upon completion of work and before turning over to Owner, clean and lubricate bearings except sealed and permanently lubricated bearings. Use only lubricant recommended by manufacturer.
- B. Contractor is responsible for maintaining lubrication of mechanical equipment under this Contract until Work is accepted by Owner.

3.14 CLEANING

- A. Clean systems after installation is complete.
- B. Clean piping and ductwork both internally and externally to remove dirt, plaster dust or other foreign materials. When external surfaces of piping are rusted, clean and restore surface to original condition.
- C. Clean pipeline strainers to restore them to original condition or replace with new strainer elements.
- D. Clean equipment and plumbing fixtures as recommended by manufacturers.
- E. Replace throwaway or replaceable media air filters used during construction period with new filters or new filter media after construction has been completed and before building is turned over to Owner. Filter replacement shall be as hereinafter specified.
- F. Blow and clean dirt, plaster dust and other foreign matter from coils, terminal devices, diffusers, registers and grilles.
- G. Thoroughly clean equipment of stains, paint spots, dirt and dust. Remove temporary labels not used for instruction or operation.
- H. Provide additional cleaning of individual piping systems and apparatus as hereinafter specified.

END OF SECTION

SECTION 230513
MOTORS FOR MECHANICAL EQUIPMENT

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 262913 - Enclosed Controllers

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 SUBMITTALS

- A. Product Data including the following:
 - 1. Manufacturer
 - 2. hp, voltage, phase, hertz, rpm
 - 3. Motor type
 - 4. Enclosure type
 - 5. Frame type
 - 6. Insulation class
 - 7. NEMA design designation
 - 8. Service factor
 - 9. Nominal efficiency at full load
 - 10. Power factor at full load
 - 11. Full load amperes
 - 12. Bearings
 - 13. Mountings
 - 14. Dimensions
 - 15. Weight
 - 16. Shaft grounding brush for motors driven by Variable Frequency Drives (VFD)

1.4 PRODUCT CRITERIA

- A. Motors covered by this Specification shall conform to applicable requirements of NEMA, IEEE, ANSI, and NEC Standards and shall be UL Listed where applicable for service specified.
- B. Motors shall be designed for conditions in which they will be required to perform; i.e., general purpose, splash proof, explosion proof, standard duty, high torque or other special type as required by equipment manufacturers.
- C. Select motors so they do not exceed nameplate rating nor operate into service factor to meet specified duty.

- D. Motors located inside air handling units or exposed located in outdoor or wash down environments shall have totally enclosed fan cooled (TEFC) enclosures.
- E. Motors shall be furnished for starting in accordance with utility requirements and be compatible with starters specified hereinafter or under Electrical sections of Specifications.
 - 1. Refer to Section 262913 - Enclosed Controllers for reduced voltage starting requirements.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials shall be new and guaranteed for service intended.

2.2 ELECTRICALLY COMMUTATED MOTORS (ECM)

- A. Motor assembly shall be designed for use on direct drive fans or pumps and shall be electrically commutated, DC, brushless type, specifically designed for use with 120 V or 208-230/277 V, 60 Hz single-phase electrical input or three-phase, 60 Hz electrical input as scheduled.
- B. Motor speed shall be variable via compatible input signal and shall be complete with and operated by single-phase integrated controller/inverter that operates wound stator and senses rotor position to electronically commutate stator. Motor shall maintain minimum of 70% efficiency over entire operating range.
- C. Motors shall:
 - 1. have permanently lubricated ball bearings
 - 2. be designed for synchronous rotation.
 - 3. be permanent magnet type with near-zero rotor losses.
 - 4. be able to be mounted with shaft in horizontal or vertical orientation.
 - 5. be controllable from local and remote speed-control adjustments
 - 6. be designed to overcome reverse rotation without affecting life expectancy
- D. Provide Power Factor Correction / Harmonic Correction module to limit harmonic distortion to less than 10% THDi at input of the filter.
- E. Speed Control (Drive):
 - 1. Motor manufacturer shall provide factory installed PWM speed control drive for local (manual) and remote speed adjustment. Local PWM drive shall be field adjustable with standard screwdriver. Drive shall also be capable of receiving 0-10 vdc signal from DDC controller to control fan or pump speed.
 - 2. Drive shall provide soft start and adjustable speed change ramp rate.
 - 3. Drive shall provide motor protection consisting of locked rotor, over-current, and thermal overload protection at minimum.

2.3 INDUCTION MOTORS

- A. Voltage Ratings
 - 1. Refer to equipment schedules and specification sections for voltages required.
 - 2. Unless otherwise indicated, motors 1/3 hp and smaller shall be rated 115 V for operation on 120 V, 1 Ph, 60 Hz service.
 - 3. Unless otherwise indicated, motors 1/2 hp and larger shall be rated:
 - a. 460 V for operation on 480 V, 3 Ph, 60 Hz service.
 - b. 200 V for operation on 208 V, 3 Ph, 60 Hz service.
 - c. 230 V for operation on 240 V, 3 Ph, 60 Hz service.
- B. Motors shall be 4 pole (approximately 1750 rpm) unless otherwise noted.
- C. Single-phase motors shall be furnished with built-in thermal overload protection.
- D. Use NEMA Design B motors, normal starting torque with regreasable ball bearings, and Class B insulation unless specified otherwise or unless manufacturer of equipment on which motor is being used has more stringent requirements.
 - 1. Bearings shall be rated for minimum AFBMA 9, L-10 life of 26,280 hours (belted) and 200,000 hours (direct-coupled) at full-load.
- E. Motors shall be rated continuous duty and have 1.15 service factor unless otherwise noted.
- F. Motors Driven by Variable Frequency Drives (VFD)
 - 1. Motors shall comply with the latest NEMA MG 00001, Section IV, Part 31.
 - 2. Motors shall have service factor not less than 1.0 at rated load.
 - 3. Insulation shall be Class F or H.
 - 4. Furnish each motor with shaft grounding ring utilizing conductive microfiber similar to AEGIS SGR to protect motor bearings from electrical damage.
- G. Vibration shall not exceed 0.15" per second, unfiltered peak unless otherwise noted.
- H. Motors (180 frames and larger) shall have provisions for lifting eyes or lugs capable of safety factor of 5.
- I. Full load nominal efficiency of motors 1 hp and larger, except special-purpose motors including 2-speed or multi-speed motors, and rewind motors, shall meet or exceed listed values when tested in accordance with IEEE Standard 112 Method B as defined by NEMA MG 00001-12.6C. Efficiency values listed are based on NEMA Premium Efficiency Electric Motors of NEMA MG 00001-2011, Table 12-12.

	<u>Open Drip-Proof Motors</u>			<u>Totally Enclosed Fan-Cooled Motors</u>		
	1200 rpm	1800 rpm	3600 rpm	1200 rpm	1800 rpm	3600 rpm
(hp)	(6 pole)	(4 pole)	(2 pole)	(6 pole)	(4 pole)	(2 pole)
1 hp	82.5	85.5	77.0	82.5	85.5	77.0
1.5 hp	86.5	86.5	84.0	87.5	86.5	84.0
2 hp	87.5	86.5	85.5	88.5	86.5	85.5
3 hp	88.5	89.5	85.5	89.5	89.5	86.
5 hp	89.5	89.5	86.5	89.5	89.5	88.5
7.5 hp	90.2	91.0	88.5	91.0	91.7	89.5
10 hp	91.7	91.7	89.5	91.0	91.7	90.2
15 hp	91.7	93.0	90.2	91.7	92.4	91.0
20 hp	92.4	93.0	91.0	91.7	93.0	91.0
25 hp	93.0	93.6	91.7	93.0	93.6	91.7
30 hp	93.6	94.1	91.7	93.0	93.6	91.7
40 hp	94.1	94.1	92.4	94.1	94.1	92.4
50 hp	94.1	94.5	93.0	94.1	94.5	93.0
60 hp	94.5	95.0	93.6	94.5	95.0	93.6
75 hp	94.5	95.0	93.6	94.5	95.4	93.6
100 hp	95.0	95.4	93.6	95.0	95.4	94.1
125 hp	95.0	95.4	94.1	95.0	95.4	95.0
150 hp	95.4	95.8	94.1	95.8	95.8	95.0
200 hp	95.4	95.8	95.0	95.8	96.2	95.4
250 hp	95.4	95.8	95.0	95.8	96.2	95.8

- J. Single-phase motors for hard starting applications including outdoor applications shall be capacitor start type. Motors for fans and pumps located indoors may be split phase or permanent split-capacitor. Motors shall be equipped with permanently lubricated and sealed ball bearings and shall be selected for quiet operation. Motors 1/8 hp and below may be shaded pole type.
 - 1. Refer to individual equipment section for additional requirements or specific type of motors.
- K. 3 Ph, 2-speed motors shall be one winding, consequent pole, variable torque type and 1 Ph, 2-speed motors shall be capacitor start capacitor run type.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install materials in accordance with drawings, approved Shop Drawings and manufacturer's recommendations.

END OF SECTION

SECTION 230529
MECHANICAL PIPING AND EQUIPMENT SUPPORTING DEVICES

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 230700 - Mechanical Systems Insulation
- B. Section 230550 - Vibration Isolation
- C. Section 233114 - Ductwork

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Provide all supporting devices not provided as part of building structure or indicated on structural drawings or structural details, as specified and as required for proper supporting, anchoring, and guiding of piping, equipment, materials and systems.
- B. Support for all conditions of operation, including variations in installed and operating weight of equipment, piping and ductwork, to prevent excess stress and allow for proper expansion and contraction.

1.4 SUBMITTALS

- A. Product Data for each piping system for all pipe sizes and all applicable equipment including the following:
 - 1. Manufacturer's name
 - 2. Model numbers
 - 3. Materials of construction and load ratings lbs.
 - 4. Schedule of hangers and support devices with pipe support spacing
 - 5. Insulated pipe supports along with application chart or table including pipe support spacing.
 - 6. Insulation protection saddles and weight bearing insulation table
 - 7. Details and calculations for sizing supplementary steel utilized for trapeze or specially designed supports
 - 8. Structural attachments, concrete inserts, and concrete anchors. Submit ICC-ES Evaluation Report for each type of concrete insert and anchor.
 - 9. Calculations and drawings for concrete inserts and anchors for each application shall be submitted when requested by Owner or Engineer.

10. Drawings showing specific locations of any weld attachments to structure, including weight supported by such attachments.
 11. Drawings showing specific locations of any suspended loads which exceed 100 lbs within joist chord panel to be attached to open web steel joist structural members. Include weight supported by such attachments. (Panel is length of chord between two adjacent diagonal web members at point of connection to chord.)
 12. Equipment mounting devices
 13. Pipe guides and anchors
 14. All other appropriate data
- B. Submittals in PDF format shall be organized using PDF bookmarks per detailed instructions on assembling submittal packages identified under Submittal Requirements section of 230000 - General Mechanical Requirements.
- C. Calculations and drawings prepared by the Contractor's structural engineer shall be submitted bearing the stamp of a licensed Structural Engineer registered in project jurisdiction, for approval.

1.5 DESIGN CRITERIA

- A. Materials and application of pipe hangers and supports shall conform to the latest requirements of ANSI B31/ASME B31.1 Code for Pressure Piping and MSS Standard Practice SP-58-2018 (Pipe Hangers and Supports Materials, Design, Manufacture, Selection, Application, and Installation), except as supplemented or modified herein.
- B. Support materials shall be steel or stainless steel unless specifically indicated.
- C. Support devices shall be factory fabricated and have published load ratings.
- D. Unless otherwise indicated, design supports, anchors, and related components with safety factor in accordance with AISC Manual of Steel Construction, but not less than 2.0.
- E. Determine maximum deflection using the following equation.
- F. Maximum deflection of support members, including trapeze supports, shall be in accordance with the following equation, but shall not exceed 0.3":
1. $D = L/240$
D = Max deflection in inches
L = Support member length in inches
- G. Unless otherwise indicated, hangers, support devices and hardware shall be steel and shall have factory standard black, primed, galvanized or electroplated finish for indoor application, and hot-dipped galvanized finish for outdoor application and corrosive atmospheres. Coat cut edges, welds or any damaged finish with galvanized paint.
1. Corrosive atmospheres include the following locations:
 - a. Exterior locations

b. Chemical storage and hazardous waste storage rooms

- H. Materials in contact with pipe shall be galvanically compatible with piping material to eliminate conductive path for galvanic corrosion. Where piping and support materials have galvanic potential, provide galvanic separation, such as nonmetallic coating or inserts between piping and metallic supports. Pipe insulation is acceptable galvanic separation. Galvanic potential shall be determined by table below:

	Galvanized Steel	Carbon Steel	Stainless Steel (Type 304 or 316)	Copper Brass Bronze
Copper, Brass, Bronze	Yes	Yes	No	NA
Stainless Steel (Type 304 or 316)	Yes Note (1)	Yes Note (1)	NA	
Carbon Steel	No	NA		
Galvanized Steel	NA			

(1) Required where stainless steel surface area near interface is equal or greater than steel surface area

- I. Unless otherwise indicated, steel support devices exposed to ventilation air stream shall be stainless steel or steel with either galvanized finish or paint finish. Paint type shall be approved by Architect.
- J. This Contractor is responsible for proper placement and sizing of supporting devices to accommodate insulation thickness and pitching of pipe. Coordinate with Contractor performing work specified in Section 220700 - Plumbing Systems Insulation.
- K. Piping connected to coils, which are in assembly mounted on vibration isolators, shall have vibration isolation hangers or supports as indicated above. Piping connected to coils, which are in equipment where fan assembly is separately isolated by vibration isolators and flexible connections, does not require additional vibration isolation hangers or supports. Refer to Section 230550 - Vibration Isolation for flexible connections, vibration isolators and additional requirements.
- L. Where piping can be conveniently grouped to allow trapeze type supports, supporting steel shall be by means of standard structural shapes.
- M. Hangers and rods shall be plumb when pipelines are at their normal operating temperatures.

- N. Unless otherwise indicated, continuous insert channels are not allowed.
- O. Punching, drilling, or welding of building structural steel is not allowed unless approved by Structural Engineer.
- P. Lateral braces shall be designed and detailed to apply loads as directly as possible to structural floor slabs, roof decks, or other building lateral elements. Braces shall not be applied to bottom flanges of steel beams or bottom chords of open web steel joists.
- Q. Coordinate with Contractor for any proposed weld attachments to building structure. This may result in use of welding codes or standards, which may apply to "structural work". and may necessitate repair of fireproofing and/or extension of fireproofing to support members. Execution of this work may be assigned to General Trades responsible for building structural steel. Cost for this work, however, will remain the responsibility of this Contractor.
- R. Top or bottom chords of open web steel joists may be used to support loads, provided total load within panel does not exceed 100 lbs and load is placed concentric to joist. (Panel is length of chord between two adjacent diagonal web members at point of connection to chord).
- S. Where fire rated fiberglass products are used for channel and support devices, the following properties shall apply:
 - 1. Flame Spread Properties
 - a. Polyester Fiberglass (PF) Class 1 ASTM E84
 - b. Vinylester Fiberglass (VF) Class 1 ASTM E84
- T. Fasteners including concrete anchors for seismic application shall have ICC Evaluation Service Report (ESR) and meet requirements of local authorities.

PART 2 - PRODUCTS

2.1 STRUCTURAL SUPPORTS

- A. Unless specifically indicated on structural drawings, design and provide all supporting devices including miscellaneous steel (angles, channels, beams, etc.), required for proper support of piping, equipment and materials.

2.2 PIPE HANGERS AND SUPPORTS (METALLIC)

- A. Manufacturers: Anvil, Erico, Tolco, PHD, National Pipe Hanger Corporation, or B-Line, equal to Anvil figures listed. Corresponding MSS Type is indicated where applicable.

B. Clevis and Roller Type Hangers:

System	Pipe Size	Clevis	Roller
Hot Pipes with Insulation (105°F and above)	2" and smaller	65 (MSS Type-1), 260 (MSS Type-1)	---
	2-1/2" to 6"	---	171 (MSS Type-41), 181 (MSS Type-43)
	8" and larger	---	171 (MSS Type-41)
Ambient Bare Pipes (61°F to 104°F)	2" and smaller	65 (MSS Type-1), 260 (MSS Type-1)	---
	2-1/2" and larger	260 (MSS Type-1), 216 (MSS Type-4)	---
Cold Pipes with Insulation (33°F to 60°F)	2" and smaller	65 (MSS Type-1), 260 (MSS Type-1)	---
	2-1/2" and larger	260 (MSS Type-1), 295 (MSS Type-1)	---

C. Flat Surfaces (Trapeze, Rack Type):

1. Use structural steel members such as struts, angles, channels and beams to support pipes as required. Select members properly for pipe support types and loading conditions. Refer to Part 1 for design criteria. Submit support details with type of members selected and load calculations. Provide straps, clamps, rollers or slides indicated below at each support point.

System	Pipe Size	Straps or Clamps	Rollers	Slides
Hot Pipes with Insulation (105°F and above)	2" and smaller	Anvil Klo- Shure	---	---
	2-1/2" and larger	---	171 or 177(MSS Type-41), 271 (MSS Type-45), 274 (MSS Type-46)	257 or 436 with 212 or 432 clamps, Type 1, 2 or 3 for longitudinal movement only. Type 4, 5 or 6 for both longitudinal and transverse movement

Ambient Bare Steel Pipes (61°F to 104°F)	6" and smaller 8" and larger	B-Line BVT Unistrut Cush-a-Clamp	---	---
Ambient Bare (Copper) pipes (61°F to 104°F)	all sizes	B-Line BVT Unistrut Cush-a- — Clump		

D. Vertical Pipe within Wall Cavities

1. Use clamps, straps, inserts or channels to support pipes concealed in wall cavity. Select members for pipe support types and loading conditions. Refer to Part 1 for design criteria.
2. Provide clamps, strut channels, insulated supports, or brackets and inserts equal to manufacturer indicated below:

System	Pipe Size	Supports	Clamps, Brackets/Inserts	Pre-insulated Inserts
Hot and Cold Pipes with Insulation	All sizes	Anvil 137	HoldRite 260 series with SBIS bracket, Anvil 262	Anvil Klosure, Pipe Shields A2000
Ambient Bare Steel Pipes (61°F to 104°F)	All sizes	Anvil 137, 138-R	Anvil 262	---
Ambient Bare Copper Pipes (61°F to 104°F)	All sizes	Anvil CT-138R	HoldRite 260 series with SBIS bracket	---

2.3 INSULATION PROTECTION SHIELDS

- A. Anvil Fig. 167 (MSS Type-40) constructed of galvanized carbon steel. Per the latest edition of Standard MSS SP-58, select shield to accommodate outer diameter of insulation. Shield length and gauge for insulation compression strength not less than 15 psi, shall be as follows:

Pipe Size	Length	Gauge
1/4" thru 3"	12"	18
4"	12"	16
5" thru 6"	18"	16
8" thru 14"	24"	14
16" thru 24"	24"	12

2.4 INSULATED PIPE SUPPORTS

A. Description:

1. Products designed specifically for weight-bearing support of insulated pipes. Apply products in accordance with manufacturer's recommendations and requirements indicated below.
2. Refer to PART 3 – EXECUTION for application of Type A, Type B, and Type C Insulated Pipe Supports specified below.

B. General:

1. High compressive strength inserts utilized to support loads shall encircle circumference of pipe. Block-style inserts are not allowed.
2. Supports shall be suitable for hot or cold pipe service as applicable.
3. Submit chart or table indicating selected model along with pipe sizes, rated loads, support device types and support spacing for each piping system.
4. Pipe support spacing shall be in accordance with manufacturer's recommendations but shall not exceed maximum spacing indicated under Hanger and Support Spacing in Part 3 of this Section.
5. Testing of insulation for compressive strength properties shall comply with ASTM D1621.
6. Insulation thickness shall match adjacent pipe insulation thickness.
7. Integrity of vapor barrier jacket shall be maintained continuously through support assembly.
8. Insulated pipe support style shall be specifically selected for the application and shall consider the following criteria at minimum:
 - a. Vertical, lateral and axial support design load limits.
 - b. Vertical, lateral, and axial support design travel limits
 - c. Temperature of support, at pipe surface, and ambient conditions
 - d. Test or pre-operational loads that may exceed normal operating conditions
 - e. Material for any items that will be welded directly to the pipe
 - f. Loading and displacements caused by seismic, hydraulic surge, or other forces
 - g. Temperature at support steel

C. Type A Insulated Pipe Supports (Light Duty)

1. Description:
 - a. Pipe insulation specified in Section 230700 - Mechanical Systems Insulation with insulation protection shields specified in this Section. Weight-bearing inserts are not required.
 - b. Type B or Type C supports may be utilized in lieu of Type A supports.

D. Type B Insulated Pipe Supports (Standard Duty):

1. Manufacturers:
 - a. SNAPP ITZ insulation inserts by KB Enterprise, Tru-Balance Insulated Saddles by Buckaroos, Inc., Value Engineered Products, or approved equal.
 - b. Klo-Shure 7-series or 9-Series insulation couplings by Hydra-Zorb 7-series for pipes with elasometric insulation, or approved equal.
 - c. Type C supports may be utilized in lieu of Type B supports.

- d. Contractor may propose to utilize contractor-fabricated insulated pipe supports in lieu of manufactured Type B Supports. Use of contractor-fabricated assemblies is subject to approval of appropriate submittal data. Submit detail drawings of assemblies and product data showing equivalency to specified manufactured products for approval.
 - 2. Description:
 - a. Load-rated assembly consisting of high compressive strength insulation material completely encompassing circumference of pipe, vapor barrier jacket, and insulation protection shield.
 - b. Insulation protection shield shall conform to ANSI/MSS SP-58. Shield shall be G90 galvanized steel and shall span full circumference of pipe insulation. Half-shields spanning lower 180° degrees arc of insulation outer circumference will be acceptable when used with clevis hangers.
 - c. Axial length of insulation material shall be not less than 9" or 2" longer than insulation protection shields (1" minimum on each end), whichever is longer.
 - 3. Insulation Materials:
 - a. Hot Pipes 105°F to 250°F:
 - 1) Rigid closed cell, polyisocyanurate or phenolic insulation by ITW, Resolco, or Kingspan. Minimum compressive strength shall be 100 psi.
 - b. Cold Pipes 60°F and below:
 - 1) Rigid closed cell, polyisocyanurate, phenolic insulation similar to ITW, Resolco, Kingspan, or cellular glass insulation similar to Pittsburgh Corning Foamglas.
- E. Type C Insulated Pipe Supports (Heavy Duty):
 - 1. Manufacturers:
 - a. Pipe Shields, Inc., Bergen Pipe Supports, or Rilco equal to Pipe Shields models listed.
 - b. Unless otherwise indicated, pre-insulated pipe supports shall be as indicated in the following schedule. Model numbers are based on Pipe Shields, Inc.
 - 1) Pipe supported on hangers: Model "A" Series and Model "D" Series
 - 2) Pipe supported on flat surfaces and pipe rollers: Models "A" Series
 - 3) Pipe supported on slides: Model "B" Series with lateral guide or restraint
 - 4) Pipe anchors: Model "C" Series
 - 5) Riser clamps: Model "E" Series with thrust plates. Select proper model for restraint for downward load or upward load.
 - c. Contractor may propose to utilize contractor-fabricated insulated pipe supports in lieu of manufactured Type C Supports. Use of contractor-fabricated assemblies is subject to approval of appropriate submittal data. Submit detail drawings of assemblies and product data showing equivalency to specified manufactured products for approval.
 - 2. Description:
 - a. Load-rated assembly consisting of high compressive strength insulation material completely encircling circumference of pipe, vapor barrier jacket system incorporating structural inserts and insulation protection shield/casing where applicable.

3. Insulation Material:
 - a. Water-resistant high density calcium silicate with minimum density of 13 lb/ft³ and compressive strength not less than 100 psi. Thermal conductivity shall be not more than 0.38 Btu in/(hr sq ft °F at 75°F.
 - b. Structural Inserts:
 - 1) Structural inserts used by manufacturer to reinforce between pipe and insulation jacket for clamping devices shall be water-resistant high compressive strength inorganic materials selected by manufacturer for desired combination of structural strength and insulating properties. Structural inserts shall have minimum compressive strength of 600 psi. Thermal conductivity shall be not more than 0.58 Btu in/(hr sq ft °F at 75°F.
 - c. Jacket:
 - 1) Jacket shall consist of G90 galvanized steel conforming to ASTM A653/A653M and shall provide complete vapor barrier around insulation and bearing surface for protection of insulation.
 - 2) When recommended by manufacturer, use reinforced insulation protection shield at support bearing surface. Insulation shall extend 1" beyond insulation protection shield to maintain vapor barrier integrity.

2.5 HANGER RODS (METALLIC)

- A. Rods shall conform to the latest MSS Standards except as modified herein. Furnish rods complete with adjusting and lock nuts.
- B. Rods shall have electroplated zinc or hot dip galvanized finish.
- C. Unless otherwise indicated, size rods for individual hangers and trapeze support as indicated in the following schedule. Rod size may be reduced one size for double rod hangers. Total weight of equipment, including valves, fittings, pipe, pipe content and insulation, shall not exceed limits indicated.

Max. Pipe Size with Single Rigid Rod	Rod Diameter	Max Load of Hanger Rod (Not exceeding 650°F Service Temp.)
2"	3/8"	730 lbs
3"	1/2"	1350 lbs
5"	5/8"	2160 lbs
8"	3/4"	3230 lbs
12"	7/8"	4480 lbs
18"	1"	5900 lbs
30"	1-1/4"	9500 lbs

2.6 BOLTS, NUTS, STUDS AND WASHERS

- A. ASTM A307, electroplated zinc finish

2.7 ROD ATTACHMENTS

- A. Anvil Fig. 290 (MSS Type-17), galvanized finish

2.8 U-BOLTS

- A. Anvil Fig. 137 (MSS Type-24), galvanized finish

2.9 BEAM CLAMPS

- A. Beam Clamps: Anvil Fig. 133/134 (MSS Type-21), 218 (MSS Type-30), 228 (MSS Type-28 or 29) and 292 (MSS Type-28 or 29)
- B. Top Beam Clamps: Anvil Fig. 227 (MSS Type-25)
- C. C-Clamps: Anvil Fig. 86, 92 or 93 (MSS Type-19 or 23) with set screw and lock nut

2.10 ADJUSTABLE PIPE SADDLE SUPPORTS

- A. Anvil Fig. 264 (MSS Type-38), galvanized finish. Provide Anvil Fig. 63 Type T stanchion with base, galvanized finish, where applicable.

2.11 RISER CLAMPS (BARE PIPE)

- A. Anvil Fig. 261 (MSS Type-8), galvanized finish
- B. B-Line B3373C, PVC coated carbon steel, in area at pipe contact, for bare copper tubing
- C. Proset system, proseal plug and fire-fill for sleeved and cored holes.

2.12 RISER CLAMPS (INSULATED PIPE)

- A. Unless otherwise indicated, insulated pipe riser clamps shall be Type C insulated pipe supports. Refer to Insulated Pipe Supports in Part 2.
- B. Contractor may propose to utilize contractor-fabricated riser supports. Use of contractor-fabricated riser supports is subject to approval of appropriate submittal data. Submit support detail drawings, bearing stamp of Structural Engineer registered in project jurisdiction, for approval. Supports shall be engineered to withstand static and dynamic forces with minimum safety factor of 2.0. Submit insulation details addressing thermal break from building structure and vapor barriers.

2.13 CONCRETE INSERTS (WOODEN FORMED CONCRETE)

- A. Anvil Fig. 281 or 282, or Hilti KCM-WF or DeWalt Wood-Knocker II+, suitable for rod diameter and weight supported.

2.14 CONCRETE INSERTS (METAL DECK FORMED CONCRETE)

- A. Tolco No. 109DD, DeWalt Pan-Knocker II+, "Bang-It+", Hilti KCM-MD or KCM-PD, or MSCO No. MX34.

2.15 CONCRETE ANCHORS

- A. Manufacturers: Hilti, DeWalt or Red Head
- B. Anchors shall be selected, sized, and detailed by Contractor's structural engineer registered in project's jurisdiction, based on project conditions and in accordance with project building code.
- C. Anchors shall meet ICC Acceptance Criteria, and ICC-ES Evaluation Reports (ESRs) shall specifically list the current applicable codes.
- D. Anchors installed in hardened concrete for purpose of transmitting structural loads from one connected element to another, or for safety related elements such as sprinkler pipes, heavy suspended pipes, and barrier rails shall have ICC-ES report demonstrating anchors have met requirements of AC 193 for mechanical anchors in concrete elements.
- E. Post-installed expansion anchors and undercut anchors installed in hardened concrete shall be qualified for strength design and tested according to ACI 355.2. Designs shall be per the requirements of ACI 318, Appendix D.
- F. Anchors shall be zinc plated in accordance with ASTM B633.
- G. Select anchors with load ratings based on cracked concrete conditions.

2.16 CONTINUOUS INSERT CHANNELS

- A. Manufacturers: Unistrut, Anvil, B-Line, Michigan, Halfen, Hilti or Kindorf. Brackets inserts and accessories shall be suitable for channel insert selected.
- B. Length and support capabilities to be suitable for supporting plumbing piping in vicinity.

2.17 METAL FRAMING SUPPORT SYSTEM (STRUT SYSTEM)

- A. Manufacturers: Unistrut, B-Line Strut Systems, Anvil-Strut, Power-Strut, Erico, Superstrut, Kindorf, Hilti, and Hydra-Zorb
- B. Channels shall have epoxy paint or electroplated zinc finish.
- C. Channels shall not be lighter than 12 ga.

2.18 PIPE MOUNTING PEDESTALS

- A. Equal to Roof Products & System Corporation consisting of equipment rail, "U" shaped mounting brackets, galvanized threaded rod and cast iron pipe rollers. Rail shall have built-in raised cant to match roof deck insulation.

2.19 EQUIPMENT RAILS

- A. Manufacturers: Roof Products & Systems, ThyCurb, Custom Curb, Inc. or Vent Products equal to Roof Products & Systems Model ER-4 with raised cant style for built-up roofing system. Mounting rails shall be galvanized steel with integral base plate, continuous welded corner seams, factory installed 2" x 4" wood nailer and 18 ga galvanized steel counter flashing.
- B. Manufacturers: Roof Products & Systems, ThyCurb, Custom Curb, Inc. or Vent Products equal to Roof Products & Systems Model ER-2 for elastic sheet roofing system. Mounting rails shall be galvanized steel with integral base plate, continuous welded corner seams, factory installed 2" x 4" wood nailer and 18 ga galvanized steel counter flashing.
- C. Mounting rail gauge shall be selected to support equipment adequately but shall be not less than 18 ga.
- D. Height shall be as detailed, but not less than 8" above finished roof.
- E. Equipment rails shall span minimum of 2 joists and not cantilever more than 6" where joists are used. Rails shall be level at top with pitch built in when deck slopes 1/4" per ft or greater.

2.20 PIPE ROOF PENETRATION PROTECTIONS

- A. Manufacturers: Roof Products & Systems, ThyCurb or Vent Products equal to Roof Products & Systems "RPS-Pipe Portals" consisting of 12" OD prefabricated roof curb, laminated acrylic coated ABS plastic curb cover with EPDM protective rubber cap and stainless steel clamp.

2.21 PIPE ANCHORS

- A. Unless otherwise indicated, anchors shall be Type C Insulated Pipe Supports. Refer to Insulated Pipe Supports in Part 2.

- B. Contractor may propose to utilize contractor-fabricated anchors. Use of contractor-fabricated anchors is subject to approval of appropriate submittal data. Submit anchor system detail drawings, bearing stamp of Structural Engineer registered in project jurisdiction, for approval. Anchors shall be engineered to withstand static and dynamic forces with minimum safety factor of 3.0. Submit insulation details addressing thermal break from building structure and vapor barriers.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install supports to allow for free expansion of piping. Support piping from building structural members using concrete inserts, beam clamps, ceiling plates, wall brackets, or floor stands. At no time shall hangers and supports overload building structural members. Fasten ceiling plates and wall brackets securely to structure and test to demonstrate adequacy of fastening.
- B. Select and size building attachments properly in accordance with MSS Standards and manufacturer's published load rating information.
- C. Coordinate hanger and support installation to properly group piping of all trades.
- D. Suspend piping hangers by means of hanger rods. Perforated band iron and flat wire (strap iron) are not allowed.
- E. Piping and ductwork shall be supported independently from other piping or ductwork.
- F. Pipe hangers and supports shall not penetrate vapor barrier of pipe insulation.
- G. Do not support equipment, or piping from metal roof decking or ceiling grid.
- H. Install adequate supports so as not to over stress either piping or equipment to which piping is connected.
- I. Refer to Section 230000 - General Mechanical Requirements for requirements of personnel injury protection guards for supporting devices.

3.2 HANGER AND SUPPORT SPACING

- A. Space pipe hangers and supports for horizontal pipe accordance with the following schedule, with exceptions as indicated herein:

B. Steel Pipe (Standard Weight and Extra Strong):

Pipe Size	Max Spacing
1-1/4" and smaller	7 ft
1-1/2"	9 ft
2"	10 ft
2-1/2"	11 ft
3"	12 ft
4"	14 ft
6"	17 ft
8"	19 ft
10" and larger	20 ft

C. Copper Tube (Unless Otherwise Noted):

Pipe Size	Max Spacing
3/4" and smaller	5 ft
1" 1-1/4"	6 ft
1-1/2" 2-1/2"	8 ft
3" and larger	10 ft

D. Plastic Pipe

1. PP-R Pipe:

Pipe Size	Max Spacing
2" and smaller	4 ft
3"	6 ft

2. PEX-a Pipe:

Pipe Size	Max Spacing
3" and smaller	2 ft-8"

- a. Pipe can be supported with PEX-a Pipe Channel as an alternate. Spacing for supports with pipe channel shall be 6 ft for pipes 3/4" and smaller and 8 ft for pipes 1" and larger.
 - b. Provide copper tube size riser clamps at base of each floor and at top of every other floor with mid-floor guides for hot water systems.
 - c. Provide copper tube size riser clamps at base of each floor and top of every 4 floors with mid-floor guides for cold water systems.
3. Support plastic pipe at all changes of direction. Adequate consideration shall be given to piping expansion.

E. Maximum spacing shown above may be restricted by strength of attachment to building structure. Submit data with calculations with published load ratings showing attachment to be utilized and maximum spacing allowable for that type of attachment and pipe size.

F. Spacing less than indicated above may be required to conform to building structure design or loading limitations.

- G. Spacing less than indicated may be required depending on compressive strength of pipe insulation and insulated pipe supports.
- H. If pipe size changes between support points, maximum spacing shall be based on the smaller pipe size.
- I. If trapeze hangers are used to support multiple services, spacing shall be based on the most restrictive pipe size and material on trapeze hanger.
- J. For non-metallic pipe, follow manufacturer's installation recommendations in addition to requirements noted herein.
- K. Install supports for vertical piping and anchors as recommended by pipe manufacturer.
- L. Place hangers and supports to meet requirements of Section 232116 - Pipe and Pipe Fittings or specific pipe system sections, with regard to pitch for drainage and venting and clearance between services.
- M. Hangers and supports shall bear on outside of insulation when pipes are to be insulated.
- N. Place hangers and supports within 1 ft of each fitting, such as elbows and tees, and at each valve, strainer, and other piping specialty for piping 4" and larger.
- O. Place hanger or support at first elbow upstream of pump inlet and first elbow downstream of pump outlet.

3.3 RISER SUPPORTS

- A. Insulated Piping:
 - 1. Unless otherwise indicated, support vertical piping as indicated below:
 - 2. Support vertical piping at bottom of riser, secured and anchored to building structure. Provide guides on vertical piping. Use spring hangers at top of riser and at take offs from riser at each floor. Use spring hangers for minimum 3 hangers away from top and bottom elbows and from each take off at riser.
 - 3. Guide vertical piping 2-1/2" and smaller at every floor. Guide 2-1/2" and larger at every other floor. Spring hangers (Type 6) and guides (Type VSG) are specified in Section 230550 - Vibration Isolation.
- B. Bare Piping:
 - 1. Unless otherwise indicated, maximum vertical support spacing for ambient bare steel and cast iron pipes shall be 15 ft.
 - 2. Maximum vertical support spacing for other piping including copper tubing and plastic piping shall be 10 ft.
 - 3. Install riser clamps and intermediate supports as required.
 - 4. Rest riser clamps on floor or on pipe sleeve.
 - 5. Bare piping above 120°F such as steam vents shall be supported per insulated piping requirements.

3.4 INSULATED PIPE SUPPORTS APPLICATION

- A. Install insulated pipe support at each support point of insulated pipe.
- B. Pipe Size 1-1/2" and Smaller:
 - 1. Use Type A insulated pipe support. Pipe insulation specified in Section 220700 - Plumbing Systems Insulation shall be continuous through support points.
 - 2. Use one shield (bottom) for clevis hanger.
 - 3. Use 2 shields (top and bottom) for roller hanger/support or strap/clamp support. Apply 2 metal straps to hold top and bottom shields onto insulation jacket.
 - 4. Type B or Type C insulated pipe supports may be used in lieu of Type A support.
- C. Pipe Size 2" through 5":
 - 1. Use Type B insulated pipe supports. Refer to Part 2 for acceptable products.
 - 2. Type C insulated pipe supports may be used in lieu of Type B supports.
- D. Pipe Size 6" and Larger:
 - 1. Use Type C insulated pipe supports. Refer to Part 2 for applicable products.

3.5 PIPE FLOOR SUPPORTS

- A. Unless specifically shown otherwise, use adjustable pipe saddle supports with associated stanchion similar to Anvil Fig. 264/63. Select supports properly for weight and height of pipe stand.

3.6 CONCRETE INSERTS

- A. Concrete insert application, size, loading, and placement shall be this Contractor's responsibility.
- B. Coordinate placement of inserts before concrete pour. Minimize use of inserts and anchors after concrete pour.

3.7 CONTINUOUS INSERT CHANNELS

- A. Mount continuous insert channels when used for pipe support on 8 ft-0" maximum centers and 8 ft-0" from corners.

3.8 BEAM CLAMPS

- A. Provide locknut for hanging rod at clamp.
- B. C-clamps are allowed for rod size 3/8" or smaller and only for static loading such as air piping, cold water piping, fire protection piping and, other similar piping. C-clamps are not allowed for hot water piping and steam and steam condensate piping, except hot water runouts to terminal heating devices.

- C. C-clamps are not allowed for open web steel joist application.
- D. C-clamps are not allowed for seismic application.

3.9 TRAPEZE SUPPORTS

- A. Construct trapeze supports with struts, angles, or channels and hang them by inserts or welded beam attachments and rods.
- B. If trapeze supports are used to support multiple services, support spacing shall be based on the most restrictive pipe size and material on trapeze supports.
- C. Refer to Part 1, Design Criteria for maximum deflection allowed for trapeze supports.

3.10 PIPE MOUNTING PEDESTALS

- A. Use for all piping on roof. Install bottom of pedestal flat on roof deck, insulate exterior of pedestal, flash and counter flash.

3.11 EQUIPMENT RAILS

- A. Use for all roof-mounted equipment, which is not curb mounted. Install bottom of equipment rail flat on roof deck. Insulate exterior of equipment rail.
- B. Flashing will be by General Contractor. Provide counter flashing as specified and secure to wood nailer with stainless steel truss head screws.

3.12 CONCRETE ANCHORS

- A. Anchor application, size, and placement shall be this Contractor's responsibility.

3.13 PIPE ROOF PENETRATION PROTECTIONS

- A. Install at points where pipes are penetrating roof. Install as shown and according to manufacturer's installation instructions.

3.14 PIPE ANCHORS

- A. Install anchors where shown on drawings or in conjunction with expansion joints, loops and swing joints as required to allow proper expansion and contraction of piping without damage to structure, equipment or piping.
- B. Do not anchor piping to concrete block walls, wood, or partition walls.

END OF SECTION

**SECTION 230550
VIBRATION ISOLATION**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 230529 - Mechanical Piping and Equipment Supporting Devices
- B. Section 233314 - Ductwork Specialties (Duct Flexible Connections)

1.2 DESIGN CRITERIA

- A. Isolate motor driven mechanical equipment, unless otherwise noted, from building structure, and from systems that they serve, to prevent equipment vibrations from being transmitted to structure. Unless specifically indicated, follow the latest edition of ASHRAE Handbook HVAC Applications - Sound and Vibration Control, or manufacturer's recommendations for isolator selection whichever is more stringent.
- B. Select and locate isolators to produce uniform loading and deflection. Use minimum of 4 isolators to support each piece of equipment.
- C. Select vibration isolation devices based on the lowest operating speed of equipment.
- D. Vibration Criteria:
 - 1. All rotating equipment shall operate at speeds less than 80% of their true critical speed. Unless otherwise required, equipment shall be balanced according to recommendations given in the following schedules.
 - 2. Vertical vibration of rotating equipment shall not be greater than levels indicated. Vibration shall be measured on equipment. If equipment has inertia base, allowable vibration level is reduced by ratio of equipment weight alone to equipment weight plus inertia base weight.

Equipment Speed rpm	Maximum Allowable Vibration Displacement Peak -to-Peak (mil)
100 to 200	10
200 to 300	6
300 to 600	4
600 to 1000	3
1000 or 2000	2
over 2000	1

1.3 SUBMITTALS

- A. Submit Shop Drawings and product data including the following:
 - 1. Manufacturer's name
 - 2. Isolator type and model number
 - 3. Materials of construction and finish
 - 4. Dimensional data
 - 5. Load ratings lbs
 - 6. Isolator free and operating heights
 - 7. Static deflections
 - 8. Isolation efficiency based on lowest operating speed
 - 9. All other appropriate data
- B. Provide seismic restraints for all vibration-isolated equipment. Restraints shall not be in contact with the equipment during its normal operation but shall be capable of withstanding loads imposed by seismic acceleration of the equipment in any direction during seismic event.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials used shall retain their isolation characteristics for life of equipment served. Elastomeric materials shall comply with ASTM D2240 and shall be oil-resistant industrial grade neoprene.
- B. Isolators shall be treated to resist corrosion.
- C. Isolation devices subject to weather shall have either hot-dip or cold-dip galvanized, cadmium plated, or neoprene coated finish after fabrication and be furnished with limit stops to resist wind.
- D. Vibration isolator springs shall have minimum additional travel to solid equal to 50% of rated deflection.
- E. Ratio of lateral to vertical stiffness of vibration isolators shall not be less than 0.8 or greater than 2.0.
- F. Coordinate selection of devices with isolator and equipment manufacturer.

2.2 MANUFACTURERS

- A. Mason Industries, Amber/Booth Co., Aeroflex-VMC-Korfund. Vibration Eliminator, Vibro-Acoustics, Kinetics, or equal to manufacturer's model listed, except flexible pipe connections.

- B. Mason, Metraflex, Proco, Twin City Hose, Engineered Flexible Products (EFP) or Flex-Weld/Keflex for flexible pipe connections.

2.3 TYPE 1 MOUNTS (NEOPRENE PAD)

- A. Mason Type Super W, neoprene waffle pads, 50 durometer. Select number and size of pads as required to accept equipment operating weight evenly.

2.4 TYPE 2 MOUNTS (NEOPRENE PAD)

- A. Mason Type ND or rails Type RND, double deflection neoprene mounts with cast-in metal inserts for bolting to equipment.
- B. Both surfaces shall be rib molded for skid resistance. On equipment such as small vent sets and close coupled pumps, steel rails shall be used above mountings to compensate for overhang.

2.5 TYPE 3 MOUNTS (UNHOUSED SPRING WITH NEOPRENE)

- A. Mason Type SLF, combination spring and neoprene with rib molded base. Spring type isolators shall be free standing and laterally stable without any housing and complete with 1/4" neoprene acoustical friction pads between baseplate and support.
- B. Mountings shall have leveling bolts rigidly bolted to the equipment. Spring diameters shall be no less than 0.8 of compressed height of spring at rated load.

2.6 TYPE 4 MOUNTS (RESTRAINED SPRING WITH NEOPRENE)

- A. Mason Type SLR, combination spring and neoprene with rib molded base similar to Type 3 above, but shall have rigid housing with hold-down bolts to limit vertical as well as horizontal movement.
- B. Installed and operating heights shall be the same. Maintain minimum clearance of 1/2" around restraining bolts and between housing and spring so as not to interfere with spring action. Limit stops shall be out of contact during normal operations. Use height saving brackets.

2.7 TYPE S BASES (STEEL BASE)

- A. Mason Type WF, structural steel bases, rectangular in shape for all equipment other than centrifugal refrigeration machines and pump bases which may be "T" or "L" shaped. Pump bases for split case pumps shall include supports for suction and discharge base ells. Perimeter members shall be beams with minimum depth equal to 1/10 of longest base span between isolators, but not less than 4". Beam depth need not exceed 14" provided that deflection and misalignment is kept within acceptable limits as determined by manufacturer. Employ height saving brackets in all mounting locations to provide base clearance of at least 1" above floor or housekeeping pad.

2.8 TYPE I BASES (INERTIA BASE)

- A. Mason Type K, or BMK rectangular or T shaped structural beam or channel concrete forms for floating foundations. Bases for split case pumps shall be large enough to provide support for suction and discharge base ells. Base depth need not exceed 12" unless specifically recommended by base manufacturer for mass, rigidity or component alignment. Base depth shall be a minimum of 1/10 of longest base span between isolators, but not less than 6". Forms shall include concrete reinforcement bars welded in place running both ways. Furnish forms with drilled steel members with sleeves welded below holes to receive equipment anchor bolts where anchor bolts fall in concrete locations. Employ height saving brackets in all mounting locations to maintain base clearance of at least 1" above floor or housekeeping pad.

2.9 TYPE 5 HANGERS (SPRING HANGER WITH NEOPRENE)

- A. Mason Type 30N, vibration hangers with steel spring and neoprene element in series. Neoprene element shall be molded with rod isolation bushing that passes through hanger box. Spring diameters and hanger box lower hole sizes shall be large enough to permit hanger rod to swing through 30° arc before contacting hole and short circuiting spring.
- B. Mason Type DNHS may be used where load rating and specified deflection cannot be accommodated by Type 30N.

2.10 TYPE 6 HANGERS (PRECOMPRESSED SPRING HANGER WITH NEOPRENE)

- A. Mason Type PC30N, vibration hangers similar to Type 5, but precompressed to rated deflection so as to keep piping or equipment at fixed elevation during installation. Design hangers with release mechanism to free spring after installation complete and hanger is subjected to its full load.

2.11 TYPE 7 HANGERS (SPRING HANGER WITH DEFLECTION INDICATOR)

- A. Mason Type HES, steel spring in steel housing including deflection indicator scale. Hangers shall be preset at factory for required load. Select hangers so that actual loads do not exceed rated capacities lbs.
- B. Submittals shall include isolator rated deflection, required deflection and supporting calculation. Calculations shall be made by registered mechanical or civil engineer demonstrating structural adequacy of hanger and that hanger connections to building and pipe are adequate for live and dead loads encountered.

2.12 TYPE 8 HANGERS (SPRING HANGER WITH NEOPRENE)

- A. Mason Type 30, W30, or PC30 steel spring located in neoprene cup manufactured with grommet to prevent short-circuiting of hanger rod. Neoprene cup to contain steel washer designed to properly distribute load on neoprene and prevent its extrusion. Spring diameters and hanger box lower hole size to be large enough to permit hanger rod to swing through 30° arc before contacting hole and short-circuiting spring. Provide hangers with rod attachments or eyebolts on spring end.

2.13 TYPE AG PIPE ANCHORS/GUIDES

- A. Mason Type ADA all-directional acoustical pipe anchors and Type VSG guides for vertical piping consisting of telescopic arrangement of 2 sizes of steel tubing separated by minimum 1/2" thickness of heavy duty neoprene and neoprene isolation material. Vertical restraints shall prevent vertical travel in either direction. Allowable loads on isolation materials shall not exceed 500 psi and design shall be balanced for equal resistance in any direction.
- B. Submittals shall include supporting calculations by registered mechanical or civil engineer indicating anchor/guide loads and isolator selection.

2.14 TYPE T THRUST RESTRAINTS

- A. Mason Type WB, horizontal thrust restraint consisting of spring element in series with neoprene pad as described for Type 3 mounts with the same deflection as specified for mountings or hangers. Spring element shall be contained within steel frame and designed so it can be preset for thrust at factory and adjusted in field for maximum of 1/4" movement at start and stop. Furnish thrust restraints complete with rods and angle brackets for attachment to both equipment and ductwork or equipment and structure. Attach horizontal restraints at centerline of thrust and symmetrically on either side of unit.

2.15 FLEXIBLE PIPING CONNECTORS

- A. Flexible connectors shall be suitable for pressure, temperature and fluid involved, but not less than 215 psig working pressure at 250°F for 14" and smaller and 150 psi working pressure at 250°F for 16" and larger.
- B. Flexible connectors shall be straight pipe configuration and shall not be used to replace pipe fittings such as elbows.
- C. Refrigerant System:
1. Seamless corrugated bronze or stainless steel corrugated liner with bronze or stainless steel wire braided cover and standard copper tube ends suitable for brazing for copper piping. Seamless corrugated stainless steel flexible connector with stainless steel wire braided cover and carbon steel ends for steel piping. Minimum pressure-temperature rating 400 psig at 200°F.
 2. Seamless corrugated stainless steel liner with stainless steel braided cover and carbon steel butt weld ends for steel piping. Minimum pressure-temperature rating 400 psig at 200°F.

Nominal Pipe Diameter (in)	Minimum Live Length (in)
2-1/2" and smaller	12"
3" and 4"	18"
5" and larger	24"

- D. Do not provide flexible piping connectors for gas piping.

2.16 PERFORMANCE

- A. Unless otherwise noted, select vibration isolation devices to achieve either minimum 95% isolation efficiency or minimum static deflection and mounting requirements listed in Table below, whichever is greater. For floor spans between increments indicated, use longer floor span indicated in table. Minimum static deflections listed below are not nominal but certifiable minimums with actual installed load. Unless otherwise indicated, apply requirements listed for floor mount for roof-mounted equipment.

Type of Equipment	Ground Supported Slab	Floor Span							
		Up to 20 ft				Up to 30 ft		Up to 40 ft	
	Type	Min Defl in	Type	Min Defl in	Type	Min Defl in	Type	Min Defl in	
Condensing Units									
Air Cooled Condensers	-	-	4	0.75	4	1.5	4	1.5	
Air Cooled Condensing Units	-	-	4	0.75	4	1.5	4	2.5	
Packaged Air Handling Units (Floor Mounted)									
Thru 5 hp	3	0.75	3	0.75	3	0.75	3	0.75	
7-1/2 hp and over	3	0.75	3	3.5	3	3.5	3	3.5	
Thru 400 rpm									
7.5 to 40 hp	3	0.75	3	0.75	3	1.75	3	1.75	
401 rpm and over									
Emergency Generator									
Muffler (Suspended)	-	-	5	1.5	5	1.5	5	2.0	
Utility Set Fans									
Roof Mounted	Use Type 4 mount with deflection from blower minimum deflection guide up to 0.75" deflection. Over 0.75" deflection, use Type 4-S mount.								
Vertical Pipe Risers									
	Use Type 6 hangers, Type AG anchors and guides. Refer to Section 230529 - Mechanical Piping and Equipment Supporting Devices for additional riser support requirements.								

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install and adjust vibration isolation devices as specified, as shown on drawings and according to manufacturer's recommendations.
 1. Adjust isolators after piping system is at operating weight.
 2. Adjust limit stops on restrained spring isolators to mount equipment at normal operating height. After equipment installation is complete, adjust limit stops so they are out of contact during normal operation.
 3. Adjust active height of spring isolators.
 4. Adjust restraints to permit free movement of equipment within normal mode of operation.
 5. Adjust air-spring leveling mechanism.
- B. In no case shall installation short circuit isolation devices.

3.2 PACKAGED AIR HANDLING UNITS

- A. Install Type T thrust restraints at fan sections when fan section is isolated from remainder of air handling unit.

3.3 FLEXIBLE PIPING CONNECTIONS

- A. Provide flexible connections for piping connected to rotating or reciprocating equipment, equipment such as coils mounted on vibration isolators, and as indicated on plans and details.
- B. Piping connected to coil which is in assembly where fan is separately isolated by vibration isolators and duct flexible connections does not require flexible piping connectors or piping vibration hangers.
- C. Install flexible connections on equipment side of shut off valves and horizontal and parallel to equipment shafts where applicable.

END OF SECTION

SECTION 230553 MECHANICAL SYSTEMS IDENTIFICATION

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 RELATED WORK

- A. Section 230700 - Mechanical Systems Insulation

1.3 SUBMITTALS

- A. Product Data: For identification materials and devices
- B. Valve Schedules: For each piping system
- C. Samples: Of color, lettering style, and graphic representation required for each identification material and device.

PART 2 - PRODUCTS

2.1 IDENTIFYING DEVICES

- A. Stencil Paint:
 - 1. Oil-based, alkyd enamel, black color
- B. Marker System:
 - 1. Manufacturers: Brady USA, Marking Services Inc. (MSI), Kolbi, or Seton
 - 2. Manufacturer's standard, preprinted with color coding, lettering size and length of color field according to ASME A13.1.
 - 3. Use pressure-sensitive type or "snap-on" type.
 - 4. "Strap-on" type may be used for piping over 6" size including insulation.
- C. Valve Tags:
 - 1. Minimum 1-1/2" diameter, 0.032" thick, polished brass or 316 stainless steel.
- D. Laminated Plastic Nameplates:
 - 1. Nameplates shall be approximately 1-1/2" x 4", 1/16" thick, and have 1/2" high lettering. Face of plastic nameplates shall be black with white letters.
 - 2. Fasteners shall be self-tapping, stainless steel screws or contact type with permanent adhesive.

PART 3 - EXECUTION

3.1 GENERAL

- A. After painting and/or covering is completed, identify equipment and piping as indicated. Locate identification as conspicuously as possible except where such would distract from finished area.
- B. Where markers are used in high heat applications or exposed to harsh chemical or acid environments, specifically select marker materials for those applications.
- C. Coordinate, obtain and confirm mechanical systems identification criteria and requirements from Owner.

3.2 PIPING SYSTEM IDENTIFICATION

- A. Install pipe identification on each system.
- B. Place flow directional arrows at each pipe identification location.
- C. Identify all piping not less than once every 25 ft, not less than once in each room, at each branch, adjacent to each access door or panel, at each valve and where exposed piping passes through walls and floors.
- D. Identify piping with marker system.
 - 1. For "strap-on" type, ensure marker is fitted snugly to pipe or pipe insulation surface with sufficient straps.
- E. In addition to marker system, piping systems shall be identified by fully painted jacket according to the color chart below. Painting shall be under applicable sections of Division 09 - Finishes. Colored PVC jackets are to be provided under Section 230700 - Mechanical Systems Insulation.

Piping System	System Abbreviation	Pipe Jacket Color
Cooling Coil Condensate	CCC	White
Non-Potable Cold Water	NPW	Green
Refrigerant Suction	RS	N/A
Refrigerant Liquid	RL	N/A
Refrigerant Hot Gas	RHG	N/A

- F. Piping specified Section 230700 - Mechanical Systems Insulation to have aluminum or stainless steel jacket shall not be painted.

3.3 VALVE IDENTIFICATION

- A. Identify valves with brass tags bearing system identification and valve sequence number in 1/2" black characters. Attach tag to valve body with brass jack chain and "S" hook for brass tag and SS jack chain or SS braided wires with swag sleeves and "S" hook for stainless steel tag. Non-metallic fasteners are not allowed.
- B. Valve numbers shall be prefixed with corresponding piping system identification in 1/4" black letters.
- C. Valve tags are not required at terminal devices unless valves are greater than 10 ft from device or located in another room not visible from terminal unit.
- D. Furnish typewritten valve schedule indicating valve number, fixtures, equipment or areas served by each numbered valve and incorporate in O&M Manuals.

3.4 DUCT SYSTEM IDENTIFICATION

- A. Install duct identification for each supply, return and exhaust air system.
- B. Identify all ductwork not less than once every 25 ft and not less than once in each room.
- C. Identify duct system by stenciling exterior of duct or insulation jacket by name as either "Supply Air (AHU-x)", "Return Air (RF-x)", or "Exhaust Air (EF-x)". "-x" shall indicate system number (e.g., AHU-1).
- D. Stencils shall be 2" (min) lettering, shall include direction arrow and shall be on bottom of duct or insulation jacket such that it is visible from floor below.
- E. Do not identify systems exposed in architecturally "finished" spaces.
- F. Hazardous ductwork shall have appropriate warning signs posted to protect personnel from exposure.
- G. Where fire dampers or combination fire-smoke dampers are identified by individual tag/number on drawings or Life-Safety Damper Schedule, apply identification to damper with stencil or tag.

3.5 EQUIPMENT IDENTIFICATION

- A. Identify major equipment, including air handling units, fans, boilers, chillers, heat exchangers, air terminal devices, pumps, tanks, etc.
- B. Identify equipment by stenciling equipment number and service in 2" high letters.
- C. Identify control equipment and panels with laminated plastic nameplates.

- D. Nameplate Markings:
 - 1. Identify model number, size, capacity, electrical characteristics, serial number, along with other items scheduled for equipment on drawings.
 - 2. Indicate motor horsepower, voltage, phase, cycles, RPM, full load amps, locked motor amps, frame size, manufacturer's name and model number, Service Factor, Power Factor, efficiency, minimum circuit amps, minimum feeder conductor size, disconnect or fuse size, refrigerant, and other pertinent information.
- E. Locate motor nameplates for easy reading. Relocate or provide new nameplates on motors if original nameplates are not located for easy reading.

3.6 ACCESS PANEL IDENTIFICATION

- A. Identify each service opening or access opening for fire, smoke, and fire/smoke damper with minimum 1/2" high letters indicating type of damper.
- B. Furnish typewritten charts with identification and location of all access panels serving equipment and valves and incorporate in O&M Manuals.

END OF SECTION

SECTION 230595A
AIR SYSTEMS TEST ADJUST BALANCE

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 DESCRIPTION

- A. This Contractor shall be responsible for providing complete testing, adjusting and balancing (TAB) work of all air systems including distribution systems and the equipment and apparatus connected.
- B. Work required shall consist of setting volume (flow) and speed adjusting facilities provided or specified for the systems, recording data, making tests and preparing reports, all as hereinafter specified.
- C. TAB work shall be performed by separate, Independent Contractor who is certified by either National Environmental Balancing Bureau (NEBB), Associated Air Balance Council (AABC) or Testing, Adjusting and Balancing Bureau (TABB).
- D. TAB work, may be performed by Mechanical Contractor with prior written approval by Engineer.
- E. Upon direction of Architect/Engineer or TAB Contractor, Mechanical Contractor shall provide at no additional cost to Owner, any additional work and/or devices necessary to properly balance system, including fan sheave, motor sheave and/or drive belts.

1.3 PROCEDURES

- A. Unless otherwise specified, test, adjust and balance air systems including all equipment, apparatus and distribution systems in accordance with the latest edition of NEBB, AABC, or TABB Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems.

1.4 INSTRUMENTS

- A. Instruments used for measurements shall be accurate, and calibration histories for each instrument shall be available for examination. Calibration and maintenance of instruments shall be in accordance with requirements of NEBB, AABC, or TABB Standards.

- B. Application of instruments and accuracy of measurements shall be in accordance with Standards.

1.5 REPORTS

- A. Submit Final Reports on applicable NEBB, AABC or TABB Reporting forms. Each Final Report form shall bear name of person who recorded data and seal of supervisor of TAB Contractor.
- B. Submit Final Reports in pdf format.
 - 1. PDF files shall include bookmarks to facilitate navigation.
- C. Include identification of all types of instruments used and their last dates of calibration with Final Reports.
- D. Note any and all discrepancies in design flows on report forms.
- E. Incomplete report forms will not be approved.

1.6 GUARANTEE

- A. Guarantee that all test-adjust and balance work, be performed in accordance with NEBB Standards and that all air systems operate within plus or minus 10% of design flow rates as shown on plans and/or as scheduled.

PART 2 - PRODUCTS

2.1 NOT APPLICABLE TO THIS SECTION.

PART 3 - EXECUTION

3.1 GENERAL

- A. Test, adjust and balance all air systems and the associated components in accordance with procedures outlined in the Standards.
- B. Upon completion of TAB work, mark equipment settings, including damper control positions, fan speed control levers, and similar devices to indicate final setting in approved manner.
- C. Plug holes in insulation, ductwork and housings with acceptable test plugs.

END OF SECTION

**SECTION 230700
MECHANICAL SYSTEMS INSULATION**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 230529 - Mechanical Piping and Equipment Supporting Devices
- B. Section 230553 - Mechanical Systems Identification (pipe jacket color chart)
- C. Section 233114 - Ductwork

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Provide insulating materials and accessories as required for mechanical systems as specified below.
- B. Insulating products delivered to construction site shall be labeled with manufacturer's name and description of materials.

1.4 DEFINITIONS

- A. Concealed areas, where indicated in this Section, shall apply to shafts, furred spaces and space above finished ceilings, inaccessible tunnels and crawl spaces. All other areas, including walk-through tunnels, shall be considered as exposed.
- B. Unless otherwise indicated, unit of thermal conductivity is BTU in(h ft²F).
- C. Interstitial spaces are considered as concealed areas.

1.5 SUBMITTALS

- A. Product Data for each piping system for all pipe sizes, each ductwork system but not limited to, the following:
 - 1. Manufacturer's name
 - 2. Schedule of insulating materials
 - 3. Insulation material and thickness
 - 4. Jacket
 - 5. Adhesives

6. Fastening methods
7. Fitting materials
8. Intended use of each material
9. Manufacturer's data sheets indicating density, thermal characteristics, temperature ratings
10. Insulation installation details (manufacturer's installation instruction/details, Contractor's installation details, MICA plates where applicable)
11. Literature data sheet from sealants and adhesive manufacturers stating VOC compliance with USGBC LEED IEQ 4.1.
12. Literature data sheet from coatings and mastics (including lagging adhesives) manufacturers stating VOC compliance with USGBC LEED IEQ 4.2.
13. All other appropriate data

1.6 DELIVERY, STORAGE AND HANDLING

- A. Insulation material shall be delivered to project site in original, unbroken factory packaging labeled with product designation and thickness. Shipment of materials from manufacturer to installation location shall be in weather-tight transportation. Protect insulation materials from moisture and weather during storage and installation. Protect insulation material against long exposure to UV light from sun.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Insulation:
 1. Owens Corning, Johns Manville, Manson, Knauf or CertainTeed similar to product indicated except where product of manufacturers not listed above is specifically identified for special type of insulation.
- B. Coatings, Mastics, Sealants and Adhesives:
 1. Foster, Childers, Design Polymerics, Vimasco, Miracle or Pittsburgh Corning

2.2 MATERIALS

- A. Products used for or related to air conditioning and ventilating systems shall conform to NFPA 90A possessing flame spread index of not over 25 and smoke developed index no higher than 50.
- B. Unless otherwise indicated, all products, material itself or on composite basis, shall have flame spread index not more than 25 and smoke developed index not more than 50, when tested in accordance with ASTM E84 or UL 723.
- C. Pipe insulation installed outdoors may have smoke developed index not exceeding 450.
- D. Insulation shall not contain formaldehyde.

- E. Foam insulation materials shall not use CFC or HCFC blowing agents in manufacturing process.
- F. Insulation applied on stainless steel shall meet requirements of ASTM C795 and NRC 1.36.

2.3 INSULATION

- A. Insulation materials shall be fire retardant, moisture and mildew resistant, vermin proof, and suitable to receive jackets, adhesives and coatings as indicated.
- B. Glass fiber insulation shall be of inert inorganic material, non-corrosive to mechanical surfaces.
- C. Insulating cement shall be Quick-Cote by PK Insulation MFG Co. or Ryder GP, with dry density of no more than 38 lb/ft³ thermal conductivity of 0.96 at 400°F mean temperature, and service temperature to 1200°F.
- D. Filling and finishing cement shall be Super-Stik by PK Insulation MFG Co., or Ryder MW, with dry density of no more than 24 lb/ft³, thermal conductivity of 0.74 at 500°F mean temperature, and service temperature to 1900°F.
- E. Type A Insulation (Flexible Elastomeric):
 - 1. Flexible, closed-cell, elastomeric cellular insulation.
 - a. ASTM Standard: ASTM C534/C534M, Type I (tube form) or Type II (sheet form), Grade I
 - b. Thermal Conductivity: Maximum 0.28 at 75°F mean temperature
 - c. Water Vapor Permeability: Maximum 0.08 perm-inch
 - d. Water Absorption: Maximum 0.2 % by volume
 - e. Service Temperature: -297 to 220°F
 - 2. Manufacturers/Products:
 - a. Armacell, AP Armaflex or AP Armaflex FS
 - b. Aeroflex, Aerocel
 - c. K-Flex, Insul-Tube
- F. Type F Insulation (Flexible Fiberglass Blanket):
 - 1. Flexible fiberglass blanket insulation, faced with specified jacket.
 - a. ASTM Standard: ASTM C1290
 - b. Density: Minimum 1.0 lb/ft³
 - c. Thermal Conductivity: Out-of-package (uncompressed) conductivity shall not exceed Maximum 0.27 btu-in/(hr-ft²-°F) at 75°F mean temperature
 - d. Service Temperature: Maximum 250°F.
 - 2. Manufacturers/Products:
 - a. Johns Manville, Microlite EQ FSK
 - b. Owens Corning, SOFTR Duct Wrap FRK
 - c. Knauf, Performance Plus Duct Wrap

- G. Type R Insulation (Rigid Mineral Fiber Board Insulation):
 - 1. Mineral fiber (fiberglass) duct and equipment insulation faced with specified jacket.
 - a. ASTM Standard: ASTM C612, Type IA and IB
 - b. Density: Minimum nominal 3 lb/ft³
 - c. Thermal Conductivity: Maximum 0.23 at 75°F mean temperature
 - d. Service Temperature: To 450°F
 - 2. Manufacturers/Products:
 - a. Johns Manville, 800 Series Spin-Glas, Type 814
 - b. Owens Corning, Fiberglas Fiberglass 700 Series, Type 703
 - c. Knauf, Earthwool Insulation Board

2.4 JACKETS

- A. Jacket puncture resistances shall be based on ASTM D781 test methods. Vapor barrier permeance ratings shall be based on ASTM E96/E96M Procedure A.
- B. Type D-1 Jacket:
 - 1. Heavy-duty, fire retardant material with glass fiber reinforcing. Jackets shall have neat, white Kraft finish suitable for painting, with beach puncture resistance of 50 units minimum. Vapor barrier shall be adhered to inner surface of jacket. Permeance shall not exceed 0.02 perm.
 - 2. Manufacturers/Products:
 - a. Owens Corning, ASJ Max
 - b. Johns Manville, ASJ-4535
 - c. Knauf, ASJ+
- C. Type D-2 Jacket:
 - 1. Glass fiber reinforced foil Kraft laminate with permeance not exceeding 0.02 perm and beach puncture resistance 25 units minimum.
 - 2. Manufacturers/Products:
 - a. Owens Corning, FRK
 - b. Johns Manville, FSK
 - c. Knauf, FSK
- D. Type P-1 Jackets:
 - 1. Heavy-duty, fire retardant material with glass fiber reinforcing and self-sealing lap. Jacket shall have neat, white Kraft finish suitable for painting, with burst strength of 1.5 Joules(50 beach units) minimum and tensile strength 45 lbs/in minimum. Vapor barrier shall be adhered to inner surface of jacket. Permeance shall not exceed 0.02 perm.
 - 2. Manufacturers/Products:
 - a. Owens Corning, ASJ-SSL
 - b. Johns Manville, ASJ
 - c. Knauf, ASJ+

2.5 ADHESIVES, MASTIC, COATINGS, SEALANTS, AND REINFORCING MATERIALS

- A. Adhesives and sealants shall comply with the South Coast Air Quality Management District SCAQMD 1168; VOC limits shall comply with Indoor Environmental Quality Section, Credit IEQ-4.1.
- B. Coatings and mastics shall comply with VOC limits set forth by Green Seal BS-11 and comply with the South Coast Air Quality Management District SCAQMD 1113; VOC limits shall comply with Indoor Environmental Quality Section, Credit IEQ-4.2.
- C. Products shall be compatible with surfaces and materials on which they are applied and shall be suitable for use at operating temperatures of systems to which they are applied.
- D. Products shall be fire retardant, moisture resistant and mildew resistant and vermin proof.
- E. Vapor Barrier Mastic: Below ambient insulation. Water vapor permeance shall be less than 0.08 perms at 45 mils dry film thickness per ASTM F1249.
 - 1. Foster 30-33
 - 2. Childers CP-33
 - 3. Vimasco 749
- F. Weather Barrier Breather Mastic: Above ambient insulation. Permeance shall be greater than 1.0 perms at 1/16" dry film thickness per ASTM E96/E96M.
 - 1. Foster 46-50 Weatherite
 - 2. Childers CP-10/CP-11 Vi Cryl
 - 3. Vimasco WC-5
- G. Lagging Adhesive/Coatings: Indoors applications used in conjunction with canvas/glass cloth.
 - 1. Foster 30-36
 - 2. Childers CP-50 AMV1
 - 3. Vimasco 713
- H. Metal jacketing sealant for aluminum jacketing:
 - 1. Foster 95-44 Elastolar
 - 2. Childers CP-76 Chil Byl
 - 3. Pittsburgh Corning 727
- I. Insulation joint sealant for Type P, Type PP, and Type G insulation:
 - 1. Foster 95-50 Flextra
 - 2. Childers CP-76 Chil Byl
 - 3. Pittsburgh Corning CW Sealant
- J. Glass fiber fabric reinforcing shall be 10 x 10 mesh similar to Childers Chil Glas #10 or Foster Mast A Fab.

- K. Wire mesh reinforcing shall be 22 ga, 1" galvanized.
- L. Insulation cement shall be ANSI/ASTM C195, hydraulic setting mineral wool.
- M. Finishing cement shall be ASTM C449.
- N. Butt joint and longitudinal joint adhesive for Type A insulation shall be Armstrong 520, Rubatex 373, Childers CP-82 or Foster 85-75.
- O. Weather-resistant protective finish for Type A insulation shall be equal to Armstrong WB Armaflex finish or Foster 30-64 elastomeric coating.

2.6 METAL BANDS AND WIRES

- A. Aluminum bands shall be 0.5 x 0.20" up to 48" diameter and 0.75 x 0.020" over 48" diameter.
- B. Stainless steel bands shall be 0.5 x 0.015" or 0.75 x 0.015".
- C. Stainless steel wires shall be 16 ga.

2.7 INSULATION FASTENERS

- A. Insulation fasteners shall be cup head weld pins, galvanized low carbon steel, minimum 12 ga (0.105") pins.
- B. Washer edge shall be beveled.
- C. Fasteners shall be stainless steel for stainless steel ductwork application.
- D. Insulation fasteners using adhesive are not allowed.

2.8 REMOVABLE INSULATING BLANKETS

- A. Custom designed removable, reusable, flexible, blanket thermal insulation system.
- B. Acceptable Manufacturers: Thermal Energy Products, Inc., Advanced Thermal Corp., Temptec, Remco Technology, Inc., or approved equal.
- C. Removable insulation system shall be custom designed for each individual item to provide close contour fit. Overlapping seams and gaps are not acceptable.
- D. Removable insulation shall be designed to overlap adjoining pipe insulation by minimum of 2".
- E. Insulation: Minimum 2" thick, 2.4 lb/ft³ density, 1000°F continuous service temperature thermal insulating wool; Owens Corning Fiberglass or equal.

- F. Interior and Exterior Fabric: Minimum weight 0.59 kg/m²silicone rubber coated fiberglass cloth.
- G. Securement: Blanket seams shall be closed with buckle and strap assembly (D ring closure).
- H. Identification/Tagging: Label each removable insulation device with plastic or 304 stainless steel tag with raised letters. Tag as directed by Owner.

2.9 ACOUSTICAL BARRIER MATERIALS

- A. Acoustical barrier material shall be combination of absorber and barrier, similar to Sound Seal Model B-10 LAG/QFA-3. Absorber/Barrier material shall have acoustic ratings of STC-29.
- B. Minimum sound transmission loss at each octave band shall be as follows:

Sound Transmission Loss (dB)					
Octave Band Center Frequency (Hz)					
125	250	500	1000	2000	4000
18	18	23	30	39	46

PART 3 - EXECUTION

3.1 APPLICATION

- A. Provide insulation and jackets with minimum insulation thickness as indicated in the following schedule. The schedule applies to both exposed and concealed applications unless noted otherwise:

Piping System							
Service	Jacket Type	Minimum Insulation Thickness According to Pipe Size					
		Insulation Type	3/4" and Smaller	1" - 1-1/4"	1-1/2" - 3"	4" - 6"	8" and Larger
Refrigerant Suction Lines and Hot Gas By-pass Line if Used (Interior)	---	A	1"	1"	1"	1"	1-1/2"
Refrigerant Suction Lines and Hot Gas By-pass Line if Used (Exterior)	P-2	A	1"	1"	1"	1"	1-1/2"
Cooling Coil Condensate Drain	P-1	R	1"	1"	1"	1"	1"
Cooling Coil Condensate Drain	---	A	3/4"	3/4"	3/4"	3/4"	3/4"

(Two layers of insulation shall be provided) (A-2 jacket shall be provided over P-1 jacket for exposed piping)							
Non-Potable Cold Water	P-1	R	1"	1"	1"	1"	1"
Non-Potable ColdWater	---	A	3/4"	3/4"	3/4"	3/4"	3/4"
Piping Provided with Heat Tracing	P-1	R	1"	1-1/2"	2"	2"	2"
Insulation thickness shall be the greater thickness specified for piping system or thickness specified above.							
Insulated Exterior Piping	Unless otherwise indicated, provide protective insulation shield (Type A-1 jacket) in addition to pipe insulation and jacket specified in this schedule.						
Ductwork/Equipment & Systems							
Service		Jacket Type		Insulation Type		Insulation Thickness	
OA Ducts Exposed		D-1 or D-2		R		2"	
OA Ducts Concealed		D-2		R		2"	
Mixed Air (Outside Air & Return Air Ducts & Plenum)		D-1		R		1-1/2"	
Supply Ducts Exposed		D-1		R		1-1/2"	
Supply Ducts Concealed		D-2		F		2"	
Supply Ducts Exterior		D-3		S		2"	
Return Ducts Exterior		D-3		S		2"	
Exhaust Ducts Exterior		D-1		S		3"	

3.2 INSTALLATION - GENERAL

- A. All insulation installation methods shall be performed in accordance with the latest edition of National Commercial and Industrial Insulation Standards published by MICA (Midwest Insulation Contractors Association) and manufacturer's installation instructions, except as modified in this Section of specifications.
- B. Install products with good workmanship, with smooth and even surfaces. Use full-length factory-furnished material where possible. Do not use scrap pieces.
- C. Apply insulation only on clean, dry surfaces, after all rust and scale have been removed and testing of systems has been completed. Do not insulate any section of system that must be pressure tested until after it has been successfully tested. Any removal and reinstallation to correct system defects prior to end of guarantee period shall be accomplished at no expense to Owner.

- D. Install insulating materials with necessary joints and terminations, to permit easy access and removal of equipment sections where inspection, service or repair is required, and to allow for expansion.
- E. Where possible longitudinal joints in jackets shall face toward wall or ceiling.
- F. Apply insulation to each pipe or duct individually. Common insulation applied to adjacent pipes or ducts will not be accepted.
- G. Unless otherwise indicated, pipe and duct insulation shall be continuous through walls and floors.
- H. Where multiple layers of insulation are used, stagger and secure each layer with metal bands.
- I. Insulate the following systems for complete vapor barrier protection:
 - 1. Refrigerant
 - 2. Cooling coil condensate drain
 - 3. Non-Potable Cold Water
 - 4. Insulated ductwork
 - 5. Piping and/or equipment with surface temperature below 60°F
- J. Apply Type A insulation for insulation and jackets requiring vapor barrier protection where specified insulations are cut for mounting sensors, control devices, parts of valves, devices or components which extend out from specified insulation to prevent condensation.

3.3 PIPING, VALVE AND FITTING INSULATION

- A. Apply insulation to pipe, unions, flanges, fittings, valves and piping specialties with butt joints and longitudinal seams closed tightly. Valve insulation shall cover entire valve body including bonnets and packing nuts.
- B. Laps on factory-applied jackets shall be 2" minimum width firmly cemented with lap adhesive, or shall be pressure sealing type lap.
- C. Cover joints with factory furnished tape 3" minimum width to match jacket. Cement firmly with lap adhesive. On systems requiring a vapor barrier (ASJ), vaporseal all longitudinal and butt joints ASJ/Saran seams with 4"wide coat of vapor barrier mastic or 3" minimum tape.
- D. Where staples are used, they shall be on 6" maximum centers. When used for systems requiring vapor barrier, cover lap and staples with finish coat of vapor barrier mastic or 3" minimum tape.

- E. For finishing of insulated pipe fittings and valves where surface temperature of insulation is not higher than 125°F, use one piece PVC fitting covers, minimum thickness of 20 mil, Fitting cover shall be Johns Manville Zeston 2000 PVC, PROTO Fitting Covers, or similar by other manufacturers listed. Where fitting and valve insulation requires vapor barrier, seal joints of PVC covers with vapor barrier adhesives. Insulation type, R-value and density of insulation used at fittings shall match insulation of adjacent piping. Install insulation at pipe fittings and valves completely prior to applying PVC covers.
 - 1. For Type R (Rigid glass fiber) pipe insulation, PVC fitting covers with flexible mineral fiber blanket insulation inserts are acceptable, except those located in mechanical rooms within 6 ft above floor. For fitting covers located in mechanical room within 6 ft above floor, insulation inserts shall be pre-molded rigid fiber glass type wrapped around elbows.
- F. Stove pipe style insulation on elbows (Detail A on Plate 2-200 of MICA 8th Edition) is not allowed. It may be used for closed cell elastomeric insulation.
- G. Where terminations of pipe insulation are required, insulation shall have tapered ends, built up and finished as specified for fittings.
- H. For pipes 1" and smaller, install specified pipe insulation and jacket continuous through hanger or support locations. Install insulation protection shields to protect insulation from compressing.
- I. For pipes 1-1/2" and larger, where manufactured pre-insulated pipe supports are used at hanger or support locations, extend insulation to insulated pipe supports. Where vapor barrier is required, this Contractor shall be responsible for continuity of vapor barrier at insulated pipe supports 3" wide vapor barrier tape on hot and cold systems at pipe supports.
- J. For Contractor-fabricated anchors, secure insulation directly to pipe surface and extend insulation up anchor for distance of 4 times insulation thickness. For pre-insulated anchors, cover entire surface of anchors with Type A insulation. Where applicable, take special care to assure vapor seal at anchor.
- K. Where mechanical grooved pipe connections are used in piping system, insulate couplings as specified for pipe.
- L. Piping, fittings, and valves not to be insulated:
 - 1. Heating hot water piping inside fin tube radiation enclosures
 - 2. Control valves and balancing valves for heating terminal devices
 - 3. Valves and steam traps furnished with removable insulation/jacket

3.4 EQUIPMENT INSULATION

- A. Install removable insulation where access is required for cleaning, repair and inspection. Construct removable insulation with Type A insulation for cold equipment and Type R insulation with V-1 jacket for hot equipment. Do not apply bonding adhesive to equipment surface.
- B. Provide removable insulating blankets on expansion joints.
- C. Do not insulate over equipment nameplates or ASME stamps. Bevel and seal insulation at these locations.
- D. Equipment not to be insulated:
 - 1. Equipment furnished with factory insulation.

3.5 DUCTWORK AND COMPONENTS

- A. Apply duct insulation evenly over duct surface. Unless otherwise indicated, insulation and jacket shall run continuously between duct and duct supports. Maintain insulation thickness specified over duct reinforcing members.
- B. For support points of ducts supported by trapeze hangers, place weight-supporting insulation at bottom of duct over trapeze. Weight supporting insulation inserts shall be minimum 6" wide and cover entire bottom of duct. Insulation insert shall have same thickness as insulation specified and shall be Type G, H, P or PP insulation. Size inserts based on compression strength and weight being supported.
- C. Insulation may be installed outside of support for rectangular or oval ducts having width 60" or smaller or round ducts 24" or smaller, and supported with strap hangers, provided that vapor barrier integrity is maintained at strap penetration.
- D. For support points of round ducts 16" diameter and larger, place weight-supporting insulation between duct and strap or trapeze. Weight-supporting insulation shall be minimum 6" long with same thickness as insulation specified and shall be Type G, H, P or PP insulation. Size inserts based on compression strength and weight being supported.
- E. Securing glass fiber insulation (Type F, R, and RR) for rectangular or oval ductwork.
 - 1. Horizontal ductwork:
 - a. Secure to bottom of duct where duct width is 24" diameter or greater.
 - b. Secure to sides of duct where duct side is 24" diameter or greater.
 - 2. Vertical ductwork:
 - a. Secure to all sides where duct width is 18" diameter or greater.
 - 3. Install fasteners as required to secure, but not over 18" on center and within 3" of butt joint or edge.
 - 4. Fastener shall be weld pin mechanical type.

- F. Fastening insulation anchors to ductwork with adhesives is not allowed. Where weld pin fasteners are used, install them without damage to interior galvanized surface. Where weld pin fasteners cannot be used, use other type of fasteners such as metal bands.
- G. Where ductwork is specified to be insulated, entire system including components with external surfaces equivalent to duct exterior (filter housings, mixing air chambers, air measuring stations, reheat coils, fire dampers, fire-smoke dampers, sound attenuators, etc.) shall be insulated and integrity of vapor barrier shall be continuous. For fire dampers, smoke dampers and combination F/S dampers in ductwork requiring insulation, install insulation and jacket to wall and apply vapor barrier sealant to prevent condensation.
- H. Where ductwork is specified to be internally lined, duct system components which are not lined (filter housings, mixing chambers, air measuring devices, reheat coils, sound attenuators, etc.), shall be externally insulated and vapor barrier jacket shall be sealed to duct. External insulation shall extend a minimum of 6 inches in each direction from component.
- I. Provide insulation over back pan or plenum of supply air diffusers, registers, or grilles primary air plenums of active chilled beams, and unlined boots after termination point of flexible ducts or rigid duct insulation to prevent sweating.
- J. Where vapor barrier jackets are specified, pins and staples if used shall be jacketed over with matching material using 4" tape. Vaporseal insulation seams, punctures, and tears with two 4" wide tape.
- K. Insulation without factory jacket shall be cut and mitered to suit surface. Build up voids, seams and joints with insulating cement, cover with glass fabric as specified herein and finish to smooth surface.
- L. D-1 jackets:
 - 1. Butt together joints and seams firmly, cover with glass fiber fabric 4" minimum width.
- M. D-2 jackets:
 - 1. Butt together joints and seams firmly and cover with 4" wide FSK tape.
- N. D-3 jackets:
 - 1. Install jacket in strict accordance with manufacturer's installation requirements.
 - 2. For rectangular ducts, overlap bottom half jacket and top half jacket at each side of duct at least 4" (bottom inside top outside at overlap).
 - 3. For round ducts, wrap duct insulation continuously with spiral shaped overlapping. Overlap at least 4".

- O. Acoustical Barrier Materials:
 - 1. Provide 2" thick Type R insulation with Type D-1 jacket as inner layer. Over insulation install acoustical barrier materials as outer layer in accordance with manufacturer's installation instruction.
 - 2. For ductwork specified with Type F insulation, replace it with insulation specified above.
 - 3. Refer to plans for ductwork requiring acoustical barrier materials.
- P. Ductwork not to be insulated:
 - 1. Internally lined ductwork
 - 2. Ductwork components with factory installed insulation
 - 3. Exposed supply and return ductwork in air conditioned spaces
- 3.6 PROTECTIVE INSULATION SHIELD (A-1 JACKET) FOR PIPE JACKETS EXTERIOR TO BUILDING
 - A. Unless otherwise indicated, install shields (A-1 jacket) around insulated pipe and fittings exterior to building. Seal water and vapor tight at terminations.
 - B. Longitudinal overlap shall be at least 2" wide with vapor barrier sealant.
 - C. Secure jacketing with 3/4" wide .015" stainless steel or 3/4" wide .020" aluminum bands and wing seals on maximum 18" centers.
- 3.7 TYPE S INSULATION (POLYSTYRENE)
 - A. Pipe:
 - 1. Use sectional insulation (semi-circular form) for pipe sizes 10" and smaller. Use segmental or sectional insulation for pipe sizes above 10".
 - 2. Plain sectional insulation shall be applied so that end joints are broken by making one-half of first section 18" long and leaving other half 36" in length. Longitudinal joints shall be on top and bottom of pipe. Apply insulation with sealer such as Foster 30-45 Foam-seal or Childers CP-70 Chil Joint on joints of single layer and outer layer insulation with band placed approximately 80 mm or 3 or 4-1/2" (see band schedule) back from end joints. Do not cement insulation to pipe. Omit joint sealer from inner layer of double layer insulation.
 - 3. Plain insulation lagging (blocks) for segmental insulation may be factory fabricated or beveled lagging assembled on job. Lags shall fit pipe snugly and maximum width of each lag shall be such as to leave not more than 1/8" void between pipe and joints of segments. Joints shall be broken by, starting with alternating 18" and 36" lags. Apply sealer and bands same as specified for plain sectional insulation.
 - 4. Apply pipe insulation in double layer construction with joints staggered (2 layers at 1" thick each).

B. Fittings, Valves and Flanges:

1. Insulation at fittings shall not be of less thickness than insulation on adjacent piping. Fitting insulation (covers) shall be held together and applied with sealer. Insulation on welded fittings shall fit snugly to fitting contour and shall be applied in same manner and with same materials as specified for pipe.
2. Apply fitting insulation with sealer on joints and band insulation in place using not fewer than 2 bands on threaded fittings and 4 bands on flanged fitting.
3. Insulate threaded fittings before straight pipe is covered. Insulate flanged fittings after straight pipe is covered.

C. Band Material:

1. Secure single layer or outer layer insulation with stainless steel bands, 6" spacing for insulated outside diameter under 12" and 9" spacing for 12" and over.
2. Secure inner layer insulation with stainless steel bands 9" spacing.
3. Tighten bands with mechanical tightening tool and secure with 304 stainless steel wing type seals.

D. Vapor Barrier and Jacket:

1. Finish plain pipe insulation, fittings, valves and flanges with vapor barrier mastic.
2. After thoroughly dry, apply service jacket (Type P-1) and insulation shield jacket (Type A-1).

END OF SECTION

SECTION 230901A CONTROL SYSTEMS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 230550 - Vibration Isolation
- B. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- C. Section 260533 - Raceway and Boxes for Electrical Systems
- D. Control Sequences: Refer to Drawings

1.2 DEFINITIONS

- A. The following abbreviations, acronyms, and definitions are used in addition to those details found elsewhere in contract documents.
 - 1. Actuator: Control device to provide motion of valve or damper in response to control signal
 - 2. AI: Analog Input
 - 3. AO: Analog Output
 - 4. Analog: Continuously variable state over stated range of values
 - 5. Auto-Tune: Software routine used to adjust tuning parameters based on historical data
 - 6. BAS: Building Automation System
 - 7. BMS Building Management System
 - 8. DDC: Direct Digital Control
 - 9. DDCP: Direct Digital Control Panel
 - 10. Discrete: Binary or digital state
 - 11. DI: Discrete Input (Sometimes referred to as Binary Input BI)
 - 12. DO: Discrete Output (Sometimes referred to as Binary Output BO)
 - 13. EMCS: Energy Management and Control System (Typically interchangeable with BAS or BMS)
 - 14. E/P: Voltage to pneumatic transducer (Often a solenoid valve is referred to as an E/P transducer)
 - 15. FA Field Adjustable
 - 16. FC: Fail Closed position of control device or actuator. Device moves to closed position on loss of control signal or energy source
 - 17. FMS: Facility Management System linking two or more BAS
 - 18. FO: Fail Open position of control device or actuator. Device moves to open position on loss of control signal or energy source
 - 19. I/P: Current to pneumatic transducer
 - 20. Instrument: Device used for sensing input parameters or used for actuation
 - 21. Modulating: Movement of a control device through an entire range of values proportional to an infinitely variable input value
 - 22. Motorized: Control device with actuator

- 23. NC: Normally Closed position of switch after control signal is removed
- 24. NO: Normally Open position of switch after control signal is removed
- 25. Node: DDCP, operator workstation, or other control device connected to communications network
- 26. Operator: Same as actuator for motorized devices. Also refers to an individual who physically "operates" the facility
- 27. PC: Personal Computer
- 28. Peer-to-Peer: Mode of communication between controllers in which each device connected to network has equal status and each shares its database values with all other devices connected to network.
- 29. P: Proportional control, control mode with continuous linear relationship between observed input signal and final controlled output element.
- 30. PI: Proportional - Integral control, control mode with continuous proportional output plus additional change in output based on both amount and duration of change in controlled variable (Reset control).
- 31. PID: Proportional - Integral - Derivative control, control mode with continuous correction of final controlled output element versus input signal based on proportional error, its time history (reset), and rate at which its changing (derivative).
- 32. Point: Analog or discrete instrument with addressable database value
- 33. Self-Tune: Same as Auto-Tune
- 34. Solenoid: Electric two position actuator (See E/P)
- 35. TCC: Temperature Control Contractor (Same as Control Contractor).
- 36. TCP: Temperature Control Panel

1.3 ACCEPTABLE CONTROL CONTRACTORS

- A. Control Contractor shall have full service office within 100 miles of Project site. Full service office is defined as having complete parts inventory, having all required test and diagnostic equipment, and to be home office of Applications Engineers, Supervisors, and Field Technicians. Control Contractors shall be factory authorized agent or dealer of controllers and control hardware as manufactured by:
 - 1. Alerton Technologies
 - 2. Automated Logic Corporation
 - 3. Ceylon Auto-Matrix
 - 4. Delta Controls Inc
 - 5. Distech
 - 6. Honeywell, Inc
 - 7. Johnson Controls, Inc
 - 8. KMC Controls
 - 9. Schneider Electric
 - 10. Siemens Building Technologies
 - 11. Trane Company

1.4 SYSTEMS DESCRIPTION

- A. System shall be electric and/or electronic.
- B. Control system shall be Direct Digital Control (DDC).

- C. Damper and valve actuators shall be electric type, unless otherwise noted.
- D. Provide BAS architecture consisting of communication network, operator workstations, modular designed DDCP with all points addressable and modifiable from operator workstations or from master DDCP using laptop computer. BAS shall be fully expandable with addition of hardware and/or software. Expansion shall not require removal of existing DDCP, sensors, actuators, or communication networks.
- E. System shall support operator workstations as specified and shall be capable of additional workstations, limited only by systems maximum node capacity.
- F. System intelligence shall be such that operator workstation(s) can be used for programming controls, performing analysis on filed data, generating maintenance and operation reports and providing permanent storage for programs and data.
- G. Workstation PCs and printer will be furnished by Owner. Provide hardware interface card to communicate with BAS network and all required software for each workstation, as defined in this Section, to make each PC full function workstation.
- H. Provide dial-up modem for communication between BAS and remote workstation stations via dial-up voice grade phone lines. Owner will provide a dedicated phone line for modem connection.
- I. All safety devices shall function in both auto and hand modes on starter.

1.5 SCOPE OF WORK

- A. Provide all labor and materials for complete and fully functioning control systems in accordance with Contract Documents including this Section.
- B. Engineering services shall be performed by factory-trained engineers. System shall be installed either by trained mechanics directly employed by Control Contractor or by subcontractors who are under direct supervision of Control Contractor's field representative.
- C. Control Contractor shall be responsible for complete installation of all control devices, (except as noted), wiring, and pneumatic terminations at DDCP locations to accomplish control sequences specified in this project manual or on drawings. Power for field mounted devices that require 24 VAC, 60 Hz shall be powered from 120 to 24 VAC transformer panels provided by the Control Contractor. Control Contractor shall also be responsible for any additional instrumentation described in point schedules found in Contract Documents, which may not be directly related to any specified control sequences.

- D. Mechanical Contractor shall provide all wells, taps, and other mechanical interfaces required for control equipment mounting into piping systems. Mechanical Contractor shall install all in-line mounted devices, such as valves, dampers, flow meters, static pressure probes, etc., furnished by Control Contractor. Control Contractor is responsible for installation of all other control devices, such as actuators, linkages, sensors, air terminal controllers, flow transducers, remote mounted control devices, control panels, control transformers, etc.
- E. All electrical work required, as integral part of control work is responsibility of Control Contractor. Control Contractor is responsible for providing final control power connections including conduit, wire, and/or disconnect switches to all control devices from appropriate electrical distribution panels.
 - 1. Electrical Contractor will provide circuit breakers required to provide electrical power to DDCP's panels.
 - 2. 120 to 24 VAC transformer panels shall be mounted adjacent to DDCP panels and powered from dedicated electrical circuit.
 - 3. Should any change in number of DDCPs or addition of other electrical equipment after contracts are awarded, Control Contractor shall immediately notify Electrical Contractor of change. Additional costs due to these changes shall be responsibility of this Contractor.
 - 4. Coordinate with Electrical Contractor for additional power requirements.
- F. Materials shall be as specified unless approved through procedures for product substitution specified in Division 01. Control Contractor shall provide components not specifically indicated or specified, but necessary to make system function within the intent of specification.
- G. All electrical products to be listed and labeled by UL and comply with NEMA Standards.
- H. Provide weather protection cover or weatherproof control devices where required for control devices located outdoors.

1.6 SUBMITTALS

- A. Shop Drawings:
 - 1. Refer to Division 01 General Requirements.
 - 2. Submit manufacturer's printed product data sheets for all control devices and all materials listed in bill of material in Control Contractor's control drawings. Datasheets shall be submitted electronically in pdf format with bookmarks provided for each individual device and table of contents listing each device manufacturer and full model number with links to device pages. Organize sheets in order of model number, alphabetically, then numerically.
 - 3. Submit data concerning type of signal wiring and installation methods including raceway types and grounding methods.
 - a. Submit voltage drop calculations for all low voltage DDC circuits. Voltage drop to include number of devices and wiring run lengths, calculated voltage available at each device.

4. Submit engineering calculations for sizing air compressor assembly if pneumatic control system is specified.
 5. Submit control drawings including the following:
 - a. Front sheet index for projects with more than 10 control drawing sheets.
 - b. Overall system/network architecture drawings: Provide block diagram showing relationship of each controller, control panel, or other network devices relative to each other, label room location of each device, number and indicate model number of each device, indicate network types, and general cabling routing.
 - c. Control Drawings: Provide graphic representation of systems with all major inline components to properly locate all control devices. Identify controlled devices with their software designation on drawings, including unique valve and damper tag numbers.
 - d. Detailed wiring and piping diagrams show point-to-point hookup details of all transducers, relays, outputs, inputs and subsystem components. Label all pneumatic lines and control wires with field ID numbers/colors.
 - e. Bill of Materials: Identify actual product model number used for each control device for each schematic control drawing.
 - 1) Bill of material shall be included on flow diagrams for each system and on panel layouts showing panel components.
 - f. Sequence of operation: Provide written narrative describing each control sequence indicating method of control. Identify sensors, controllers, and actuators used with references to tag number of the controlled device. Include setpoints and offsets of each control loop.
 6. Instrumentation submittals can be submitted as a separate submittal from control shop drawings but must be submitted at the same time as control shop drawings.
 7. Submit valve schedules with shop drawings that indicate unique tag numbers for each device, equipment or system served, device model numbers, sizes, shutoff head required, actuator air pressure or force required to meet shutoff head, torque requirements for rotary valves, flow coefficients (Cv) for 10% and 100% valve stem travel, actual flow requirements based on equipment shop drawings, calculation of actual pressure drops, actuator model number, actuator torque capacities and pilot positioner locations.
 8. Valve and damper shop drawing submittals will not be processed unless all supporting data and sizing calculations are included.
 9. Submit damper schedules with shop drawings that indicate unique tag numbers for each device, equipment or system served, device model numbers, duct sizes, damper sizes, flow rates, pressure differentials, calculation of actual damper pressure drops, approach velocities, leakage rates, torque requirements, actuator model number, actuator torque capacities and pilot positioner locations.
 10. Select dampers to meet their intended service with respect to maximum approach velocities and maximum pressure differential. Damper materials shall match duct construction materials in which they are installed (galvanized steel, aluminum, 304 or 316 stainless steel, etc.).
 11. Aluminum dampers may be used in galvanized ductwork.
- B. Thermostat/Room Sensor Schedules:
1. Submit thermostat/room sensor schedule with shop drawings. Thermostat/room sensor schedule shall have detailed listing of which type is used for each room including data concerning service and model numbers, sizes, cover types, and engineering data sheets for each control device.

C. Operation and Maintenance Manuals:

1. Refer to Division 01 - General Requirements.
2. Operation and maintenance manuals shall provide descriptions of maintenance on all system components, including sensors and controlled devices. These shall include inspection requirements, periodic preventative maintenance, fault diagnosis, and repair or replacement of defective components, calibration instructions, parts lists, and name, address, and phone number of manufacturers' representative.

D. Software Manual:

1. As part of operation and maintenance manuals, submit one software manual per workstation plus one extra copy for archive use. Software manuals shall be divided into separate parts with tabs for each part.
2. Software Manual parts shall include:
 - a. Complete description of operating system including all commands, configuration programs, printouts, logs, database functions and passwords. Describe general operating procedures, starting with system overview and proceeding to detailed description of each software command feature with sample printed displays and system function description for each option. Include instructions on verifying errors, status, changing passwords and initiating or disabling control programs.
 - b. Complete description of programming language including all commands, configuration programs, control loop functions and testing. Describe general programming procedures, starting with system overview and proceeding to detailed description of each software command feature. Include instructions on creating or modifying any control algorithm or parameter, debugging, etc. This shall include all control functions, algorithms, mathematic equations, variables, setpoints, time periods, messages, and other information necessary to load, alter, test and execute custom or pre-written programs.
 - c. Software Backup: Upon successful completion of acceptance testing, submit to Owner 2 archive copies of all accepted versions of source code and compiled code for all application programs and data files, on CD ROM backup disks. All control software must be readily accessible by Owner using BAS workstation hardware and software.

E. Record Drawings:

1. Refer to Division 01 - General Requirements.
2. Submit revised Shop Drawings indicating all changes made during project.

1.7 DAMPER SELECTION AND SIZING

- A. Submit Engineering calculations for sizing modulating control dampers including outside, return, and relief air dampers of air handling units unless dampers are scheduled.

- B. Calculations for sizing dampers shall be based on actual characteristics of ductwork system being installed. Opposed blade dampers shall be sized for minimum of 10% of duct system pressure drop. Parallel blade dampers shall be sized for minimum of 30% of duct system pressure drop. Duct section is defined as ductwork containing flow control damper starting with inlet or branch tee and ending with outlet or branch tee. Calculate actual duct pressure drops for each duct section containing modulating damper using latest version of ASHRAE Handbook of Fundamentals. If control systems fixes pressure drop, use those pressure setpoints. Use balance damper to provide additional pressure drop as required to obtain linear damper response.
- C. Control Contractor is responsible for obtaining adequate system information necessary for sizing.
- D. Provide dampers as shown on drawings or as scheduled.
- E. Two position dampers shall be sized as close as possible to duct size, but in no case is damper size to be less than duct area.
- F. Submit leakage and flow characteristics data for all control dampers along with shop drawings. Leakage ratings to be based on AMCA Standard 500 and dampers to bear AMCA Air Leakage Seals.

1.8 FCC COMPLIANCE

- A. All digital equipment furnished under this contract shall have been tested and made to comply with limits for Class A computing device pursuant to Subpart J of Part 15 of FCC Rules, which are designed to provide reasonable protection against interference when operated in commercial environment. All literature shall so note and all equipment shall be so labeled to show this compliance.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Pressure and temperature ratings of devices indicated in Part 2 are minimum required. All devices shall be designed to withstand maximum pressure and temperature encountered in the system.

2.2 CONTROL WIRING

- A. Control wiring shall be in accordance with National Electric Code and local electrical codes. Final connection points at devices and panels shall be made at terminal blocks either integral to device or separate terminal blocks mounted inside of control panel enclosures. Use of wire nuts and crimped connections are not allowed for terminating control wiring unless approved by Engineer.

- B. Refer to Division 26 for specification requirements for conduits and conductors, except as noted.
- C. Signal and Power Conductors (24 V and Under):
 - 1. No wire smaller than #18 AWG shall be used, except for manufacturer supplied instrument specific wire, or unless otherwise specified. Use 2 wire stranded twisted/shielded pair 24 VDC for analog and/or discrete input and 24 VAC/VDC output devices. For RTD signal wiring, use #18 AWG stranded, tinned copper twisted/shielded three lead conductor.
 - 2. Conductors not concealed in raceway shall have UL listed plenum rated teflon insulation.
 - 3. Communication Cable: Use control system manufacturer's standard communications cable or #22 to #24 AWG twisted, shielded pairs, coaxial cable, or fiber optics for communications between remote control panels/devices. Provide 250 ohm, 5 watt, 0.1% tolerance dropping resistors in 4 – 20 mA circuits as required to generate 1 to 5 V signals in 24 VDC powered instrument loops. Provide isolated instrument grounding system per manufacturer's recommendations.
 - 4. 24 VAC Power Conductors shall be #18 AWG, 2-wire twisted pair or larger. Provide Metal Oxide Varistors (MOVs) on 24 VAC/VDC discrete outputs connected to inductive loads to reduce noise levels (i.e., solenoid valves, motor contactors, relays, damper/valve electric actuators, etc.).
 - 5. Stranded twisted/shielded control conductors are required with the shields to be terminated within variable frequency Drive enclosures to reduce the effects of noise from VFD. Follow VFD manufacturer's installation instructions for wiring control conductors to VFD.

2.3 LOCAL CONTROL PANELS

- A. Local control panels shall be constructed of steel or extruded aluminum with hinged door and keyed lock, with baked enamel finish of manufacturer's standard color. Construction shall comply with NEMA 1 Standards for interior panels, NEMA 4 for exterior panels.
- B. Controlling instruments, temperature indicators, relays, switches and gauges shall be factory installed and permanently labeled. Devices shall be located inside or mounted on face of panel.
- C. Unless otherwise indicated, mount control and adjusting switches, temperature indicators and other indicating or manually operated devices on front face of panel with black phenolic engraved nameplates.

2.4 DIRECT DIGITAL CONTROL PANEL (DDCP)

A. General:

1. Direct Digital Control Panels (DDCP) shall be microprocessor based, field programmable controllers, capable of performing control and energy management functions, and shall be UL Listed as Signaling Systems. Each DDCP shall include its own microprocessor, power supply, input/output modules, and termination modules as required to perform its intended function.
2. DDCP shall receive discrete electrical or analog electronic field input signals, convert signals for use by controller, perform control sequences, convert controller information into output signals, and provide control output signals to actuators and field control device. All inputs and outputs, including communication connections, shall be electrically or optically isolated from controller.
3. DDCP with analog input modules shall be capable of accepting any form of linear or non-linear voltage (0-5 VDC or 0-10 VDC), current (4-20 mA) or resistive input (0-1000 ohm).
4. DDCP with discrete input modules shall be capable of accepting discrete inputs from any device with isolated, dry-type contacts (no grounds or no voltage) of either normally open (NO) or normally closed (NC) configuration. Provide visible status lights (LEDs) to indicate input point status.
5. Provide input modules capable of interfacing with pulsed output type sensors as required.
6. DDCP with discrete output modules shall have isolated, dry-type contacts (no grounds or no voltage) of either normally open (NO) or normally closed (N.C.) configuration.
7. DDCP shall have capability to scale, offset, and display proper analog value without field hardware modification. DDCP shall convert analog input signals to digital values (A/D conversion) and convert digital values to analog outputs (D/A conversion) for modulating control purposes.
8. Failsafe hardware shall be provided such that BAS failures result in immediate return to local control. If DDCP uses database values from other DDCP and communication network fails or malfunctions, control loop outputs shall continue to function using last value received from BAS.

B. DDCP Operator's Interface:

1. Provide communications port to connect display device for operator access to all information and all operating system functions except database reports and graphics functions in each mechanical room.
2. Display keyboards to be alphanumeric with special dedicated keys for functions such as manual-auto, test, function, value, and enter.
3. Keyboard/display units shall be able to display analog variables, binary conditions, point values and condition, and other information required for analysis and adjustment.
4. Provide minimum of one permanent panel mounted keyboard and display for each mechanical equipment room. If manufacturer does not offer dedicated panel mounted keyboard and display unit, provide portable hand held keyboard and display unit. If manufacturer does not offer dedicated hand held keyboard and display unit, provide battery powered portable PC unit with proper software and hardware to interface with local DDCP.

2.5 DIRECT DIGITAL CONTROL SOFTWARE

- A. DDCP control strategies shall be Owner definable from operator terminals or workstations.
- B. Software functions and algorithms shall be sufficient to enable implementation of control sequences as specified and able to maintain continuous control as intended.
- C. Control functions shall include both mathematical and logical operators. Control algorithms shall include proportional, integral and derivative control (PID). Adaptive (self-tuning) PID loop parameters, if offered by DDCP manufacturer, shall not be used unless adaptive limits are used to adjust limit values based on system status; or written request is submitted and approved by Engineer.
- D. Allow operator to assign unique identifiers of their choice to each connected point. Identifiers must have at least 8 alpha/numeric characters. All reference to these points in programs, reports and command messages shall be by these identifiers.
- E. Provide access control (user defined passwords) for system operation. There shall be minimum of 3 access levels. First level shall allow system monitoring only. Second level shall allow monitoring and setpoint, and scheduling revision. Third level shall allow modification of control algorithms. System shall return to secured (monitoring only) mode after 5 minutes of inactive operation.
- F. Each DDCP shall contain self-diagnostics that continuously monitor proper operation of panel.
- G. If microprocessor malfunctions, control loop outputs shall continue to function using last value received from microprocessor.

2.6 OPERATOR'S WORKSTATION HARDWARE

- A. Workstations shall communicate with BAS on as-needed basis such that other executable programs may be processed without affecting control functions of BAS.
- B. Provide 1 operator's workstations for operator interface to BAS for monitoring, control, and database management. Each workstation shall consist of the following as a minimum:
 - 1. Case and 250 watt power supply
 - 2. Motherboard with Pentium 4 microprocessor running at 3 GHZ
 - 3. BIOS in Flash Memory
 - 4. Sufficient expansion slots to be properly configured for its intended use
 - 5. 1 GB SDRAM expandable to 2 GB
 - 6. 101 key, keyboard with 12 function keys
 - 7. 17" color SVGA monitor with SVGA interface card with 128 MB video SDRAM
 - 8. 3 interface ports (2 serial, 1 parallel)
 - 9. one 1.44 MB 3.5 disk drive

10. 48X max. variable speed CD-Rom drive
11. 160 GB hard disk drive
12. Network interface card and associated hardware
13. All required cables for connecting to network and other remote devices
14. Mouse and mouse pad
15. Surge Protector
16. 8 page per minute letter size laser printer with direct network interface port.

2.7 MODEMS

- A. Provide one modem at one of DDCCP or direct network access type modem for remote monitoring and control from remote workstation(s) provided by Owner.
- B. Modems shall have auto-dial/auto-answer features and be compatible with Hayes command and transmission protocol. Unit shall be equal to Hayes ACCURA V.92 using V.44 standard and MNP protocols with V.42bis and MNP5 data compression. Baud rates shall be up to 56K baud.

2.8 NETWORK HARDWARE

- A. Provide network interface hardware for each device connected to network. Each device shall have sufficient performance as not to degrade processing speed specified.
- B. Provide network cabling with sufficient performance as not to degrade communication speed specified. Cabling shall be compatible with proposed system.
- C. Provide other network support devices that are required for proper operation of network, such as file servers, signal repeaters, network hubs, etc.

2.9 CONTROL DAMPERS

- A. General:
 1. If control damper sizes are not shown or scheduled, refer to Part 1 of this Section for sizing criteria.
 2. Unless otherwise indicated, modulating control dampers shall be opposed blade or parallel blade type and two position (open/close) dampers shall be parallel blade type.
 3. All blade linkage hardware shall have corrosion-resistant finish and be readily accessible for maintenance.
- B. Standard Modulating and Two Position Dampers:
 1. Damper frames shall be minimum of 16 ga galvanized steel or 14 ga extruded aluminum. Blades shall be minimum of 16 ga galvanized steel or 14 ga aluminum. Blades shall have maximum blade width of 8" with steel trunnions mounted in bronze sleeve, nylon or ball bearings.
 2. Furnish dampers with blade seal and side seals. Dampers and seals shall be suitable for maximum temperature, pressure differential and approach velocity to be encountered in the system, but not less than temperature range of -40 to 200°F, pressure differential of 4" WG, and approach velocity of 2500 fpm.

3. Dampers when closed shall be guaranteed by control manufacturer not to leak air in excess of 176 cfm at 4" WG static pressure for 48" x 48" damper size (11 cfm per ft²).

2.10 DAMPER AND VALVE ACTUATORS

A. Analog Electronic:

1. Actuators shall be hydraulic or electric motor/gear drives that respond proportionally to analog voltage or current input. Stroke time for major equipment shall be 90 seconds or less for 90° rotation. Stroke time for terminal equipment shall be compatible with its associated local controller, but no more than 6 minutes.
2. Provide spring return feature for fail open or closed positions as required by control sequence for critical applications such as outside, return, or exhaust dampers, heating and cooling coils on major air handling units, humidifiers, heat exchangers, and flow control for major equipment items such as chillers, cooling towers, boilers, etc.
3. Provide position feedback potentiometers connected to controller for closed loop control on major equipment analog control loops.
4. Actuators for terminal heating/cooling equipment do not require spring return feature.

B. Discrete Two-Position Electric:

1. Actuators shall be hydraulic or electric motor/gear drives for two-position control. Stroke time shall be 90 seconds or less for 90° rotation.
2. Provide spring return feature for fail open or closed positions as required by control sequence.

2.11 GENERAL INSTRUMENTATION

A. General:

1. No devices containing mercury will be allowed under this specification.

2.12 DISCRETE ELECTRIC INSTRUMENTATION

A. General:

1. Electrical devices, switches, and relays shall be UL listed and of type meeting current and voltage characteristics of the project. Terminal connections shall be made at terminal blocks inside of NEMA 1 enclosures unless otherwise specified. Outdoor units shall be NEMA 4 with concealed adjustment.
2. Ratings of normally open and closed contacts shall be adequate for applied load (Minimum 5 amps at 240 V).
3. Accuracy of devices shall be $\pm 1\%$ of scale with adjustable offset unless otherwise specified.

B. Temperature Switches (Electric Thermostats):

1. Line voltage or low voltage type suitable for application with adjustable setpoint and setpoint indication.
2. Low voltage type to have heat anticipation.

3. Thermostats with remote sensing bulb shall have liquid filled sensing element and exposed setpoint adjustment.
 4. Wall mounted space thermostat enclosure shall have concealed sensing element and exposed setpoint adjustment.
 5. Unless otherwise stated, space thermostat covers shall be brushed aluminum or brushed nickel.
 6. Manufacturer's standard plastic covers may be used in lieu of metal covers.
- C. Temperature Low Limit Switches (Freezestats):
1. Electric 2-position type with temperature sensing element and manual reset. Controls shall be capable of opening circuit if any one-foot length of sensing element is subject to temperature below setting.
 2. Sensing element shall not be less than one lineal foot per square foot of coil surface areas. Unless otherwise indicated, calibrate temperature switch setpoint to 38°F.
- D. Relays:
1. Equal to IDEC type RH2B-U, miniature 8 blade pilot relay with DPDT silver cadmium oxide contacts rated at 15A, 30 VDC, or 120 VAC. Coil shall match control circuit characteristics. DDC outputs shall be 24 VDC with maximum current burden of 50 milliamps. Rectangular base socket mount with blade type plug-in terminals and polycarbonate dust cover.
 2. Provide DIN rail mountable (Snap type) mounting sockets equal to IDEC SH2B-05.
- E. Time Switches (Time Clocks):
1. Programmable electronic clock type consisting of electronic clock, LED or LCD display, operator interface keypad, and multiple normally open/fail close contacts. Time clocks shall be programmable for up to 8 start/stops per day for each 7-day period.
 2. Provide each time clock with battery that will maintain programming schedule up to 8 hrs upon electric power failure and shall return to its programmed position after re-start.
- F. Current Switches - Constant Load, Constant Speed:
1. Manufacturers: Veris Industries, N-K Technologies, Absolute Process Instruments, Kele & Associates, R-K Electronics or approved equal
 2. These shall be induction type sensors clamped over single-phase conductor of AC electrical power and shall be solid-state sensors with adjustable threshold and normally open contacts. Each current switch shall be selected for proper operating range of current.
 - a. Output: Solid state relay or relay contacts
 - b. Trip Setpoint: Adjustable by multi-turn potentiometer
 - c. Operating Temperature: 32 to 131°F
 - d. Response Time: < 0.5 second
- G. Current Switches - Variable Load, Variable Speed
1. Manufacturers: Veris Industries, N-K Technologies or approved equal

2. These shall be induction type sensors clamped over single-phase conductor of AC electrical power and shall consist of solid-state sensors with self-calibrating threshold and normally open contacts. Each current switch shall be selected for proper operating range of current.
 - a. Output: Solid state relay or relay contacts
 - b. Trip Setpoint: Self-calibrating through microprocessor
 - c. Operating Temperature: 32 to 131°F
 - d. Response Time: < 0.5 seconds

H. Indicator Lights:

1. 1/4" minimum size. Use green for normal, yellow for warning (low/high values), and red for alarm or fail (low-low or high-high conditions). AC or DC type with voltage matched to control circuit without transformers.

2.13 ANALOG ELECTRONIC INSTRUMENTATION

A. Space Temperature Sensors:

1. Sensors shall be nickel or platinum RTD type, with the following minimum performance:
 - a. Temperature Coefficient of Resistivity (TCR): .00385 ohm/ohm/°C
 - b. Accuracy: $\pm (.54^{\circ}\text{F} + (0.005 \times T))$ (Class B)
 - c. Accuracy: $\pm (.27^{\circ}\text{F} + (0.005 \times T))$ (Class A)
 - d. Conformance: T = Temperature of interest
DIN-IEC 751
 - e. Operating Range: -50 to 500°F
0 to 99% rh
2. Thermistors or nickel RTD's will be acceptable in lieu of platinum RTD provided thermistor carries 5 year guarantee that device will maintain its accuracy within tolerance of $\pm 0.36^{\circ}\text{F}$ between 32°F and 150°F, and 0.5°F between -20°F and 212°F.
3. Unless otherwise stated, space sensor cover shall be brushed aluminum or brushed nickel.
4. Manufacturer's standard plastic covers may be used in lieu of metal covers.
5. Provide visible setpoint, setpoint adjustment, and space temperature indication.

B. Electronic Controllers:

1. Provide dedicated function type controllers with electronic analog and/or discrete electric type inputs and electronic analog and/or discrete electric type outputs, capable of performing sequences specified. Analog loop controllers shall have PID programs. Units shall have face-plate with adjustable setpoints, calibration, offset, gain factors, and visual display of all parameters.

PART 3 - EXECUTION

3.1 GENERAL

- A. Install control equipment, wiring and air piping in neat and workmanlike manner to satisfaction of Architect and/or Engineer, and in accordance with manufacturer's recommendations. Maintain clearances, straight length distances, etc. required for proper operation of each device. Mark and detail exact location of inline devices, wells, and taps to be installed by Mechanical Contractor on Coordination Drawings.
- B. Coordinate timely delivery of materials and supervise activities of other trade contractors to install inline devices such as immersion wells, pressure tappings, any associated shut-off valves, flow switches, level switches, flow meters, air flow stations, and other such items furnished by Control Contractor that are to be installed by Mechanical Contractor.
- C. Install control devices in accessible location.
- D. Mount motor control devices within 5 ft of disconnect switch, or starting device furnished by Electrical Contractor. Maintain required NEC clearances.

3.2 CONTROL WIRING

- A. Provide all electrical wiring required for complete functional control systems, including power circuit to control panels, both line and low voltage, in accordance with all applicable local codes, and the latest version of National Electric Code and NFPA when applicable.
 - 1. Voltage drops for all low voltage circuits shall be calculated prior to installing low voltage circuits. Voltage drop calculations shall be made available to Engineer on demand.
- B. Control panels serving equipment fed by emergency power shall also be served by emergency power. Equipment fed by emergency power is so indicated on mechanical equipment schedules and electrical motor schedules.
- C. Control panels shall be powered by local UPS (Uninterruptible Power Supply) to ensure continued control of equipment powered by site standby power sources when primary power source is lost. Devices such as Operator Workstations, Floor Level and Building Level Controllers and other critical devices shall be provided with UPS power.
- D. Power wiring to control compressors and dryers will be provided by Electrical Contractor. Provide field mounted starters to Electrical Contractor for installation and supervise installation.
- E. Install control wiring in raceway system per Division 26 - Electrical, unless otherwise noted.

F. BAS Network Communication Cable:

1. Run communication cable in separate raceways or in cable trays with proper clearances.
2. Install special cable connectors in accordance with manufacturer's recommendations.
3. BAS network communication cable shall not be spliced.

3.3 LOCAL CONTROL PANELS

- A. Provide local control panel for each system where more than one control device requires field mounting (air handling units, exhaust fans, miscellaneous control systems including pump controls, heat exchanger controls, etc.). Single devices may be exposed mounted on piping, wall or ductwork. Install local control panel where indicated on drawings or suitable location adjacent to system served.
- B. Mount panel on wall with suitable brackets or on self-supporting stand. Mount top of panel no higher than 6 ft above floor. Install panels so front cover door can swing full open without interference.
- C. Label all local control panels with its respective unique ID number per Section 220553 - Plumbing Systems Identification.

3.4 ADJUSTMENT AND COMPLETION CHECK LIST

- A. After completion of installation, follow check standard list procedures defined by instrument vendors to adjust and calibrate all thermostats, control valves, control actuators, controllers, sensors, and other equipment provided in this Contract.

3.5 OWNER TRAINING

- A. Provide minimum of 8 hrs of on-site training to Owner's representatives. Conduct training sessions during normal business hours after system start-up and acceptance by Owner. Scheduling of training session(s) will be established by Owner.

3.6 OVERALL BAS ARCHITECTURE

- A. Provide hardware/software to update database in less than 1 second for fast-acting control loops such as pressure control, air or water volume control, and air handling unit temperature control, or 10 seconds or less for all other control loops.
- B. Control loop software algorithm for each analog control loop shall reside on the same controller as all inputs and outputs required for that specific control loop.

- C. Networks that operate via polled response or other types of protocols that rely on central processors, file servers, or other such devices to maintain or manage peer-to-peer communications shall have redundant components to maintain network in the event of failure at central device. Provide automatic changeover to redundant device upon failure of any central type processor without operator intervention.
- D. Network shall be multi-drop digital transmission network. Network shall provide communication link between operator's workstation and all remote DDCP and field panels.
- E. Each multi-drop trunk shall be within manufacturer's allowable line lengths without signal degradation. All multi-drop trunks shall be interfaced to system via standard EIA or other industry recognized interfaces so that single failure does not disrupt or halt network.
- F. Communications between DDCP's and operator's workstations shall allow multiple users to access and use system simultaneously with no loss of system performance.
- G. Provide levels of connected networks to connect all DDCP, including terminal DDCP. Communications to terminal devices shall be similar to capabilities and functions of other DDCP and shall be transparent to operator.
- H. Number of nodes (devices connected) on any one network shall not exceed 50 % of maximum node capacity published by equipment manufacturer. Provide additional hardware, DDCP, network controllers, etc.) to meet this requirement.
- I. Alarm reports from DDCP shall not be impeded by use of either remote or local monitor, nor control stations on network either in access mode or programming mode.
- J. Provide telephone modem for remote access to system from remote operator's workstation. Coordinate telephone communications with Owner for modem connection point.

3.7 DIRECT DIGITAL CONTROL PANELS

- A. DDC Panel Usage:
 - 1. Select DDCP to provide speed of response required for each control loop type.
 - 2. Each DDCP shall have sufficient I/O capacity to perform specified control sequences and/or included points listed in any point schedules. If DDCP does not have sufficient capacity, provide additional panels to achieve required point count.
 - 3. Analog and critical safety discrete control loops shall have inputs and outputs into/from the same DDCP. Analog control loops for major equipment (chilled water, hot water, convertors, air handling units, etc.) shall have PID control.

4. For valves and dampers within 100 ft of its DDCP, mount current to pneumatic (I/P's) convertors within DDCP or in adjacent panel. Otherwise mount I/P convertors at valve or damper. Provide pressure gauges on main air, and all control output signals.

B. Cabinets:

1. Provide local control cabinets for all DDCP. DDCP cabinets may be used directly if enclosures are rated for NEMA 1. All cabinets shall utilize one masterkey. Provide 2 spare key sets to Owner.

3.8 OPERATOR'S WORKSTATIONS

- A. Furniture is provided by others. Provide all necessary connections and extensions to integrate hardware with furniture system.

3.9 SOFTWARE

- A. Software from panels shall be permanently stored on CD ROM.
- B. Provide the latest version of all standard software, including, operating system and control software. Include any software updates for period of 1 yr coinciding with warranty period. No beta released software shall be used.

3.10 INITIAL PROGRAMMING

- A. Control Contractor shall provide initial programming of all controllers to accomplish sequences specified.
- B. Provide back-up documentation per software manual submittals for all programs in both written and magnetic media formats.
- C. Provide programming of menus to assist new users in accessing screen displays of each point group. Point groups (user definable) shall be initially arranged by DDCP for major equipment and by floor and area for terminal devices. Terminal devices shall also be grouped by air handling system where applicable.
- D. Program historical file for run-times and number of start/stops of all motor driven equipment and filters.
- E. Program maintenance alarms based on run-times and number of start/stops for all motor driven equipment.
- F. Program alarms using the following levels:
 1. Level 1 - Maintenance alarm, requiring attention within one to 2 days. (Examples; 2-3°F temperature variance from setpoint, 15-25% relative humidity variance, etc.)

2. Level 2 - Low Level Alarm, requiring attention within 8 hours, preferably during the same shift. (Examples; 4°F or more variance from setpoint, 30% relative humidity or more variance from setpoints, etc., excess start/stops per day, etc.)
3. Level 3 - Critical Alarm, requiring immediate attention. (Examples; non-operation of primary equipment, H-O-A overrides.)
4. Level 1 and 2 alarms shall not interrupt current user operation, but shall be logged into alarm summary file indicating status, acknowledgment, and by whom. Level 3 alarms shall interrupt user via audible and/or flashing warning until acknowledged without losing any work in progress. When alarms are acknowledged, the program shall display point group or appropriate graphic display. Level 3 alarms shall also be logged into alarm summary file similar to Level 1 and 2 alarms.

3.11 POINT LIST

- A. Provide all points required to implement control sequences specified, whether or not they are listed in schedules. In addition to control points, provide additional monitoring points listed in point schedules or defined in Control Sequences.
- B. All outputs, whether sequenced or not, shall have separate programmable hardware outputs. For air handling units, minimum outside air, maximum (economizer) outside air, return, relief air, smoke dampers, heating coil valves, cooling coil valves, humidifier valves, etc., shall each have separate output.

3.12 CONTROL DAMPERS

- A. Furnish control dampers as shown on drawings and/or as required to perform control sequence specified except those furnished with other equipment.
- B. Control dampers furnished by Control Contractor will be installed by Mechanical Contractor under coordinating control and supervision of Control Contractor.
- C. Blank-off plates or transitions required to facilitate dampers will be provided by Mechanical Contractor.

3.13 GENERAL INSTRUMENTATION

- A. Thermometers (Temperature Indicators):
 1. Install thermometers at each point of temperature transmission and control except those indicated at local control panels. Install thermometers to permit easy reading from floor or operating platform (within 3 ft of line of sight). Provide remote bulb thermometers with readout indicators mounted within 3 ft of line of sight whenever sensing point is more than 3 ft from line of sight.
 2. Thermometer wells in piping will be installed by Mechanical Contractor.

3.14 LOCAL CONTROL PANELS

- A. Install remote mounted devices, controllers, I/O terminal blocks, power supplies, etc. inside of local control panels.
- B. Locate panels as shown on drawings.
- C. Locate panels adjacent to equipment served with a minimum of 3 ft clearance in front of the door. Provide sufficient clearances to allow full door swing and full access to all internal components. Submit proposed panel locations with shop drawings.
- D. Mount top of panel between 5 and 6 ft above floor so gauges and indicators are at eye level.

3.15 DISCRETE AND ANALOG INSTRUMENTATION

- A. Wall Mounted Space Sensors:
 - 1. Install space thermostats/sensors where indicated, as required to perform specified controls, or directed to meet job site conditions.
 - 2. Mount thermostats/sensors 5 ft above floor unless otherwise indicated.
 - 3. Mount space thermostats/sensors with accessible setpoint adjustment or temperature reading (thermometer or digital temperature readout) at 4 ft above floor meeting ADA requirements.
 - 4. Any room thermostats/sensors mounted on exterior walls shall be mounted on thermally insulated sub-base.
 - 5. Relocate room thermostats/sensors if required due to draft, interferences with cabinets, chalkboards, etc., or improper sensing.
 - 6. Mount room thermostats/sensors in corridors, stairways and public toilets 7 ft above floor.
 - 7. Room thermostats/sensors in gymnasium, locker rooms, corridor, stairways, vestibules and toilets shall be aspirating type.
- B. Low Limit Temperature Switches (Freeze Stats):
 - 1. Install low limit controls where indicated on drawings or as specified. Unless otherwise indicated, install sensing element at downstream side of heating coils.
 - 2. Distribute sensing element across entire area of medium being sensed. Install controls at accessible location with suitable mounting brackets and element duct collars where required.

END OF SECTION

SECTION 232116 PIPE AND PIPE FITTINGS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 078400 - Firestopping
- B. Section 230000 - General Mechanical Requirements (submittal requirements)
- C. Section 232314 - Refrigeration Systems

1.2 DESCRIPTION

- A. Specification of an item in this or any other sections shall not relieve Contractor from providing all items, articles, materials, operations, methods, labor, equipment and incidentals necessary for a complete and functional system.
- B. Use only new material, free of defects, rust and scale, and guarantee for services intended.
- C. Use material meeting the latest revision of ASTM specifications as listed in this specification.
- D. Follow local codes if they require other types of pipe or joints.
- E. Where weld fittings or mechanical grooved fittings are used, use only long radius elbows having centerline radius of 1.5 pipe diameters unless otherwise indicated.
- F. Manufacturer, pressure class, size and heat code of each fitting and flange shall be permanently identified on its body in accordance with MSS SP-25.
- G. Where size for a pipe segment is not indicated, the pipe segment size shall be equal to the largest pipe segment to which it is connected. Transition to smaller size shall occur on the side of fitting where smaller size is indicated.
- H. Unless otherwise indicated, fittings and accessories connected to pipe shall be of the same material and wall thickness as the pipe.

1.3 SUBMITTALS

- A. Submittals shall be in PDF format and shall include bookmarks to each product to enhance navigation and facilitate review process.

- B. Shop Drawings for each piping system for all pipe sizes including the following:
 - 1. Name of system
 - 2. Pipe; ASTM number, grade if known, type, wall thickness, material
 - 3. Fittings; ASME number, grade if known, class, type, wall thickness, material
 - 4. Joint type
 - 5. Flanges; ASTM number, grade, class, type, material
 - 6. Bolts and nuts; material
 - 7. Thread joint sealants; material
 - 8. Flange gaskets; material, rating
 - 9. Unions; ASTM number, type, material, rating
 - 10. Test pressure and media
 - 11. All other appropriate data
- C. Submit pipe certification as specified under Pipe Certification in this Section.

1.4 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Furnish pipe with plastic end-caps/plugs on each end of pipe. Maintain end-caps/plugs through shipping, storage and handling to prevent pipe-end damage and eliminate dirt and construction debris from accumulating inside of pipe.
- B. Where possible, store materials inside and protect from weather. Where necessary to store outside, elevate well above grade and enclose with durable, waterproof wrapping.
- C. Before shipping, all carbon steel piping shall be free of rust and scale, and furnished with plastic end caps/plugs on each end of pipe.

PART 2 - PRODUCTS

2.1 VENTS AND RELIEF VALVES

- A. Unless otherwise indicated, use pipe and pipe fittings as indicated for the system to which relief valve or vent is connected.
- B. ASTM A53/A53M, Grade B, Type E or S, carbon steel pipe with standard weight, carbon steel fittings may be used for steam vents smaller than 4".
- C. Use ASTM A53/A53M, Grade B, Type E or S carbon steel pipe with ASTM A234/A234M Grade WPB/ASME B16.9, standard weight, seamless carbon steel weld fittings for refrigerant vent piping.

2.2 PRESSURE GAUGES AND TAPPINGS

- A. Use pipe and pipe fittings as indicated for the system to which pressure gauge or tapping is connected. Use "Threadolets", "Sokolets" or tee fittings for tappings. Refer to Part 3 under General for use of "Threadolets" and "Sokolets".

- B. Gauge pipe shall be 1/4" unless otherwise indicated.
- C. Gauge pipe shall be 1/2" for high pressure steam (101 psig and over) systems.

2.3 COOLING COIL CONDENSATE DRAIN

- A. Piping shall be one of the following, unless otherwise indicated on drawings:
 - 1. Galvanized Steel
 - a. Pipe: ASTM A53/A53M Grade A or B, Type F, standard weight, galvanized steel
 - b. Fittings: ASTM A126/ASME B16.4, cast iron, threaded, ASTM A123/A123M galvanized
 - 2. Copper
 - a. Pipe: ASTM B88 Type M, hard temper copper tubing
 - b. Fittings: ASME B16.22 wrought copper fittings
 - c. Joint: ASTM B32, 95-5 tin-antimony solder, Bridgit or Silvabrite
 - 3. PVC
 - a. Pipe: ASTM D1785, Schedule 40, polyvinyl chloride (PVC)
 - b. Fittings: ASTM D2665, solvent weld PVC fittings

2.4 FUEL SYSTEM PIPING – CARBON STEEL

- A. 2" and Smaller:
 - 1. Pipe: ASTM A106/A106M Grade B, Schedule 40, carbon steel
 - 2. Fittings: ASTM A105/A105M Grade II/ASME B16.11, Class 3000 forged steel, socket weld
 - 3. Unions: ASTM A105/A105M Grade II, Class 3000 socket weld.
- B. 2-1/2" and Larger:
 - 1. Pipe: ASTM A106/A106M Grade B, Schedule 40, carbon steel
 - 2. Fittings: ASTM A234/A234M Grade WPB/ASME B16.9, Schedule 40, seamless, carbon steel weld
 - 3. Flanges:
 - a. Class 150, ASTM A105/A105M Grade II, ASME B16.5, welding neck
 - 4. Bolts and Nuts:
 - a. ASTM A193/A193M Grade B7 or B16 with nuts conforming to ASTM A194/A194M Grade 2H.
 - 5. Gaskets:
 - a. ASME B16.21 and ASTM F104 flat ring type, fire-safe, asbestos-free, compressed inorganic fiber with nitrile binder.
 - b. 2-1/2" thru 4"
 - 1) Gasket Thickness: 1/16"
 - 2) Maximum Seating Stress (y): 3050 psi
 - 3) Gasket Factor (m): 3

2.5 POLYVINYL CHLORIDE (PVC) PIPING

- A. All Sizes:
 - 1. Pipe: Schedule 80, ASTM D1785, and ASTM D2467
 - 2. Fittings: ASTM D3467 socket type, 125 psi rating fittings

3. Joint: ASTM D2564 PVC solvent cement

2.6 DIELECTRIC UNIONS, FLANGES AND FITTINGS (STEEL PIPE TO COPPER PIPE)

A. 2" and Smaller:

1. Dielectric fittings similar to Victaulic Style 647 or Clearflow Dielectric Waterway fittings may be used in lieu of dielectric unions for pipe sizes 2" and smaller.
 - a. Clearflow fittings shall be ASTM A53/A53M electro zinc-plated steel pipe with high temperature polyolefin polymer liner, suitable for continuous use at temperatures up to 230°F and pressures up to 300 psig.
2. ASTM A197/A197M/ASME B16, equal to Stockham Figure 693-1/2, Watts Series 3000 or Wilkins (Zurn) Model DU series dielectric unions with EPDM dielectric gasket, 250 psi at 180°F.

2.7 DIELECTRIC UNIONS, FLANGES AND FITTINGS (STEEL TO STEEL PIPE)

- A. 1" and Smaller: Similar to Epco model HA-B with dielectric gasket, 250 psi at 210°F.
- B. 1-1/2" and Larger: Similar to Epco model W with bolt insulators, dielectric gasket, bolts and nuts, 175 psi at 210°F. Pikotek model VSC dielectric gasket with viton sealing element, G-10 sleeve and double washers, suitable to 350°F, may be used with specified flanges.

2.8 THREADED JOINT SEALANTS

- A. Paste type for brush application or cord type. Products shall be non-toxic, chemically inert, non-hardening, rated for -50 to 400°F and up to 10,000 psi (liquids) and 2000 psi (gases), certified by UL, CSA, and NSF.
- B. Use sealant similar to Loctite Model 54531 for piping handling oil or petroleum products.

2.9 REFRIGERANT PIPING

- A. ASTM B88 Type L hard drawn copper tube, cleaned and capped in accordance with ASTM B280, and marked "ACR" with ANSI B16.22 wrought copper or forged brass solder-type fittings.

PART 3 - EXECUTION

3.1 GENERAL

- A. Remove foreign materials before erection. Ream ends of piping to remove burrs.

- B. Install piping parallel to building walls and ceilings and at such heights so as not to obstruct any portion of window, doorway, stairway, or passageway. Install piping to allow adequate service space for equipment. Refer to drawings and/or manufacturer's recommendations. Install vertical piping plumb. Where interferences develop in field, offset or reroute piping as required to clear such interferences. In all cases, consult drawings for exact location of pipe spaces, ceiling heights, door and window openings or other architectural details before installing piping.
- C. Provide anchors, expansion joints, swing joints and expansion loops so that piping may expand and contract without damage to itself, equipment or building.
- D. Mitered elbows, welded branch connections, notched tees, and "orange peel" reducers are not allowed. Unless specifically indicated, reducing flanges and reducing bushings are not allowed. Reducing bushings may be used for air vents and instrumentation connections.
- E. Bullhead tees (use of "branch" connection as only inlet or only outlet of tee) are not allowed.
- F. Unless otherwise indicated, use fittings as specified in Part 2 of this Section for elbows, tees, reducers, etc.
- G. Diverging tees (one inlet, two outlets) shall have inlet on "run" of tee. Converging tees (two inlets, one outlet) shall have outlet on "run" of tee. Bullhead tees will not be allowed unless specifically indicated or with approval of engineer.
- H. "Weldolets" with outlet size 2-1/2" and larger and "Threadolets" or "Sockolets" with outlet size 2" and smaller may be used for branch connections up to one pipe size smaller than main. Use "Threadolets" where threaded fittings are specified and use "Sockolets" where socket weld fittings are specified. Install in accordance with PFI (Pipe Fabrication Institute) Standard ES49.
- I. Install drains throughout systems to permit complete drainage of entire system.
- J. Do not install piping over electrical panelboards, switchgear, switchboards or motor control centers.
- K. Install valves, control valves and piping specialties, including items furnished by others, as specified and/or detailed.
- L. Make connections to equipment installed by others where that equipment requires piping services indicated in this Section.

3.2 THREADED PIPE JOINTS

- A. Threads of pipe and fittings shall conform to ASME B1.20.1.

- B. Ream pipe ends after cutting and clean before erection. Apply thread sealants to cleaned male threads. Assemble joint to appropriate depth and remove any excess pipe joint compound from tightened joint.

3.3 FLANGED JOINTS

- A. Clean flange surfaces and align them parallel. Bolt holes of gaskets shall be cut slightly larger than bolt diameter. Gasket ID shall be slightly larger than flange ID.
- B. Position gasket concentrically so compression is equally distributed over entire gasket surface.
- C. Lubricate bolts and run nuts down by hand.
- D. By using torque wrench, tighten nuts in the proper sequence so gasket is compressed evenly, and to the appropriate torque specified by bolt manufacturer.
- E. Re-torque bolts 12 to 24 h after start up.

3.4 COPPER PIPE JOINTS

- A. Cutting of tubing shall not make tubing out of round. Ream cut tube ends to full inside diameter.
- B. Remove slivers and burrs remaining from tube cut by reaming and filing both pipe surfaces. Clean fitting and tube with emery or sand cloth. Remove residue from cleaning operation, apply flux and assemble joint. Use solder or brazing to secure joint as specified for specific piping service.
- C. Press Joint Option:
 - 1. Cut pipe square and ream before assembly
 - 2. Insert pipe fully into fitting and mark on pipe at shoulder of fitting
 - 3. Check fitting alignment against mark on pipe to ensure pipe is fully engaged
 - 4. Press joint with press tool approved by fitting manufacturer

3.5 PLASTIC PIPE JOINTS

- A. Use cleaning procedure and solvent cement as recommended by pipe and fitting manufacturers for particular material being used.

3.6 POLYVINYL CHLORIDE (PVC) PIPING

- A. Install PVC piping in strict accordance with pipe and fitting manufacturer's recommendations, including support spacing, compensation for thermal expansion and contraction and solvent cementing.

3.7 WATER SYSTEMS

- A. Unless otherwise indicated, install horizontal piping level. Install manual air vents at all high points where air may collect. If vent is not in accessible location, extend air vent piping to nearest code acceptable drain location with vent valve located at nearest accessible location to pipe.
- B. Main branches and runouts to terminal equipment may be made at top, top 45°, side or bottom 45° of main provided that there are drain valves suitably located for complete system drainage and manual air vents are located as described above.
- C. Unless otherwise indicated, use top or top 45° connection to main for upfeed risers, and use side or bottom 45° connection to main for downfeed risers. Bottom connection is not allowed.
- D. Use minimum of 3 elbows in each pipeline to terminal equipment to provide flexibility for expansion and contraction of piping systems. Offset pipe connections at equipment to allow for service, such as removal of terminal device.
- E. Unless otherwise indicated, use concentric fittings for changes in pipe sizes and for valves smaller than pipe sizes.
- F. Notch and dimple branch tubes. Braze joints. Apply heat properly so that pipe and tee do not distort. Remove distorted connections.
- G. Where mechanically formed tee fittings are allowed, form mechanically extracted collars in continuous operation, consisting of drilling pilot hole and drawing out tube surface to form collar having height of not less than 3 times thickness of tube wall. Collaring device to be adjustable.

3.8 VENTS AND RELIEF VALVES

- A. Install vent and relief valve discharge lines as indicated on drawings, as detailed, and as specified for each specific valve or piping specialty item.
- B. Steam safety valve vent pipes:
 - 1. Horizontal steam safety valve vent pipes not exceeding 10 feet in horizontal run shall pitch back toward safety valve drip pan elbow or condensate drain as detailed.
 - 2. Horizontal steam safety valve vent pipes exceeding 10 feet in horizontal run shall pitch in direction of steam flow and shall be drained through drip trap with thermostatic trap at each rise where condensate may collect.

3.9 COOLING COIL CONDENSATE DRAIN

- A. Trap each cooling coil drain pan connection with trap seal of sufficient depth to prevent conditioned air from moving through piping. Extend drain piping to nearest code approved drain location. Construct trap with plugged tee for cleanout purposes.
- B. Pitch pipe down at not less than 1/8" per ft for proper drainage.
- C. Where copper piping is allowed, joints and fittings may be secured with 95-5 tin-antimony solder or brazing alloys.

3.10 DIELECTRIC UNIONS AND FITTINGS

- A. Install dielectric unions, flanges or fittings in main and branch piping of water systems at each point where copper to steel pipe connection occurs. Dielectric unions or fittings shall not be used at terminal device connections.
- B. Concealed dielectric unions and fittings are not allowed.

3.11 UNIONS AND FLANGES

- A. Install union or flange at each automatic control valve and at each piping specialty or piece of equipment that requires tube pull or removal for maintenance, repair or replacement. If required, provide additional unions or flanges in order to facilitate removal of piping sections that interfere with tube pulls or equipment removal. Where valve is located at piece of equipment, provide flange or union connection on equipment side of valve.
- B. Concealed unions or flanges are not allowed.

3.12 FLUSHING AND CLEANING PIPING SYSTEMS

- A. Flushing and cleaning piping systems is not required.

3.13 INITIAL SYSTEM FILL AND VENT

- A. Fill and vent systems with proper working fluids.

3.14 PIPE PAINTING

- A. Exposed exterior carbon steel, black iron or other ferrous pipe and fittings shall be prepared and painted by qualified painters using corrosion inhibitive paints. Pipe shall be prepared in accordance with paint manufacturer's instructions and primed (2 coats) and finish painted (2 coats). Paint type shall be approved by Architect/Engineer.
- B. Protect piping from weather and paint promptly to prevent corrosion.

END OF SECTION

SECTION 232314 REFRIGERATION SYSTEMS

PART 1 - GENERAL

1.1 SUBMITTALS

- A. Submit Shop Drawings for all items, including data concerning dimensions, capacities, materials of construction, ratings, and appropriate identification.

1.2 MATERIAL

- A. All equipment and material shall be new and guaranteed for service intended.
- B. All tubing and specialties shall be delivered to jobsite with original mill end caps in place.
- C. Refrigeration equipment and specialties shall be of same make and capacity specified or scheduled, or as required for proper system operation.

1.3 MINIMUM CODE AND STANDARD COMPLIANCE

- A. All equipment, material and specialties for this project shall be tested, rated and approved in accordance with the latest versions of the following standards and codes:
 - 1. ANSI B16.22 - Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
 - 2. ANSI B31.5 - Refrigeration Piping and Heat Transfer Components
 - 3. ASHRAE Std 15 - Safety Standard for Refrigeration Systems
 - 4. ASHRAE Std 34 – Designation and Safety Classification of Refrigerants
 - 5. AHRI 750 (I-P) - Performance Rating of Thermostatic Refrigerant Expansion Valves
 - 6. UL 207 - Refrigerant-Containing Components and Accessories
 - 7. All governing Local, State and Federal Codes

1.4 DESCRIPTION

- A. These Specifications are intended to cover installation of compressors, condensers, coils, condensing units, fixtures, and other fittings, devices, and accessories required to complete refrigeration systems as shown or called for on Drawings and schedules. Omission from these specifications or from drawings and schedules of express reference to any parts necessary for complete installation is not to be construed as releasing Contractor from responsibility for providing such parts.

- B. For details of installation, refer to drawings, manufacturer's installation instructions, and to applicable codes and ordinances. With exception of governing regulatory codes and ordinances, whenever these Specifications are in conflict with manufacturer's installation instructions, manufacturer's recommendations shall govern. Provide necessary refrigerant piping, fittings, specialties, and accessories necessary to make a complete installation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Superior HVACR Products, Mueller Refrigeration, Anaconda Universal, Henry Technologies, Sporlan, Emerson Climate Technologies, AC & R Components, Refrigerating Specialties, Refrigeration Research

2.2 REFRIGERANTS

- A. R-410A

2.3 REFRIGERANT PIPING

- A. ASTM B280 ACR hard-temper copper tubing, cleaned and capped.
- B. Alternate proposals may be submitted for ASTM B88, Type L refrigeration grade annealed temper (soft) copper tubing for long underfloor runs only.

2.4 FITTINGS

- A. ANSI B16.22 wrought copper or forged brass solder-type fittings.
- B. Copper pressure-seal refrigeration fittings manufactured by Rapid Locking Systems (RLS™) or ZoomLock™ Flame-Free Refrigerant Fittings by Sporlan Division of Parker Hannifin Corporation will be accepted within manufacturer's published application limits for temperature, pressure, and refrigerant use.
- C. Pressure-seal fittings for refrigeration piping shall be installed by personnel trained and certified under manufacturer's training and certification program.

2.5 REFRIGERATION VALVES

- A. Manufacturers: Refrigerating Specialties (Parker), Superior HVACR Products, Henry Technologies, Mueller Refrigeration, Emerson Climate Technologies, or Danfoss, except as indicated in specific paragraphs.

- B. Straight shutoff:
 - 1. 2-1/8" OD and smaller: Henry 907 Series or equal, forged brass body, triple sealed, chrome plated ball valve with teflon seals, ball equalizing orifice, and stem seal cap. 300°F, 500 psig.
 - 2. 2-5/8" OD and larger: Mueller or Henry globe valve equal to Henry type No. 203 or No. 216 back seating seal packed globe valve, with brass body, steel stem, self aligning disc, forged brass bolted bonnet, 275°F, 450 psig.
- C. Angle shutoff 1-5/8" OD and smaller: Mueller or Henry equal to Henry type No. 647 and No. 642 standard diaphragm packless globe valve, with brass body, nondirectional, with heat stabilized nylon disc, 275°F, 500 psig.
- D. Charging and purging valves: Henry or Mueller equal to Henry type No. 927, No. 916, No. 926 packed globe type valves with forged brass body, 300°F, 500 psig.
- E. Receiver shutoff valves: Mueller or Henry valves equal to Henry type No. 783, No. 779B packed globe valves, 300°F, 500 psig.
- F. Check valves 7/8" OD and smaller: Henry, Mueller, Superior equal to Henry type No. 120 spring-loaded piston check with accessible internal parts. Valve shall be provided with soft teflon seat, piston guide, stainless steel spring, 250°F, 500 psig.
- G. Check valve 1-1/8" OD and larger: Henry, Mueller, Superior equal to Henry type No. 205, No. 215 bronze body spring loaded piston type check valves. Valve shall be provided with soft teflon seat, stainless steel spring and shall have accessible parts, 300°F, 450 psig.
- H. 3-way drier bypass and dual relief valve shutoff valves 7/8" OD and smaller: Henry type No. 92, brass bodied 2-way shutoff valve, meeting ANSI B9.1, 300°F, 450 psig.
- I. Pressure relief valves: Henry type Nos. 522, 52, 524, 523, 541, 542, 545, ASME Stamped pressure relief valves.
- J. Expansion valves: Alco, Sporlan, AP, or equal, externally equalized thermostatic expansion valves with superheat adjustment. Standard ratings to be in accordance with ARI Standard 750-76.
- K. Liquid line solenoid valves: Alco, Sporlan, or approved equal, UL Listed coil voltage compatible with unit controls.
- L. Head pressure regulating valves (air cooled condensing unit): Alco, Sporlan, Flo-Con or equal head pressure regulating system. Main head pressure regulating valve and associated accessories to be selected for specific system operating conditions. Each valve system shall maintain the minimum condensing pressures 200 psig. As a minimum, regulating valve system shall maintain refrigerant receiver 20 psig below condensing pressure. All valves shall be UL Listed and shall have maximum working pressure of 400 psig.

- M. Hot gas bypass valves: Sporlan, Alco, Flo-Con or equal, electrically pilot operated. Unless specified otherwise, each valve shall be selected to maintain each system's minimum evaporating pressure down to 15% of full load for air conditioning systems, and to zero load for refrigeration systems. Multiple evaporator hot gas systems shall include pilot operating DE super heating expansion valve, mixing tee and any additional accessories required for proper system performance. All valves shall be UL Listed.
- N. Evaporator pressure regulating valve: Sporlan, Alco, Flo-Con or equal. Unless specified otherwise, evaporator regulating valves shall have maximum pressure drop of 2 psig. All valves shall be UL Listed.
- O. Crankcase pressure regulating valves: Sporlan, Alco, Flo-Con or equal. Each valve shall maintain compressor crankcase pressure at or below maximum pressure recommended for specific refrigeration system application. Unless specified otherwise, maximum open valve pressure drop shall not exceed 2 psig. Each valve shall be selected for specific system operating condition.

PART 3 - EXECUTION

3.1 GENERAL

- A. Install refrigeration piping system in accordance with the latest ASHRAE and AHRI Practices.
- B. Install piping parallel to building walls and ceilings and at such heights not to obstruct any portion of window, doorway, stairway, or passageway. Where interferences develop in the field, offset or reroute piping to clear such interferences. Install piping straight and free from kinks and bends. In all cases, consult drawings for location of pipe spaces, ceiling heights, door and window openings or other Architectural details before installing piping.

3.2 REFRIGERANT

- A. Use only that refrigerant in any equipment for which equipment was designed by manufacturer. Use only one refrigerant in a system or circuit.

3.3 REFRIGERANT PIPING INSTALLATION

- A. Horizontal hot gas and suction piping runs shall be sloped away from and toward the compressor respectively at a rate of 2-1/2" per 20 feet. Support lines at intervals of not more than 8 ft and anchor appropriately. Use rubber grommets on uninsulated piping between pipe and hanger to prevent line chafing.

- B. Where oil is lifted in vertical hot gas and suction risers of more than 5 ft, provide trap at bottom of riser. Provide additional traps for each additional 12 ft of riser length.
- C. Where branch suction line enters main suction line, connect branch line at top of main. Arrange piping so refrigerant or oil cannot drain from suction line into coil.
- D. Individual fixture or unit suction and liquid lines shall be of size shown on drawings or as recommended by manufacturer in applicable installation and service instruction. Install liquid and hot gas refrigerant lines as shown on drawings and/or in accordance with good industry practice to avoid excess pressure drops. Install branch and main suction line to maintain adequate velocities to properly return oil to compressor under minimum load conditions at lowest saturated suction pressure to be expected.
- E. Compressor discharge piping and hot gas piping brazed joints shall be ASTM Grade 4 or 5, having melting point of approximately 1250°F. Solder impurities shall not exceed 0.15%. Use only suitable silver solder alloy on copper-to-copper connections in suction line and liquid line. At any copper brass joint, except on compressor discharge line, use 95/5 solder. Use solder with at least 35% silver content on copper to steel, brass to steel, or steel-to-steel joints. During brazing operation, dry nitrogen must be bled through piping to prevent oxidation and scaling. Clean and polish joints before soldering. Avoid prolonged heating and burning during soldering. To avoid damage to internal Silfos joint in vibration eliminators, line connections to vibration eliminators shall be made with silver solder alloy equal to Easy-Flo having melting temperature of 900°F to 1,200°F, (well below 1300°F melting point of Silfos). To prevent contamination of line internally, limit soldering paste or flux to minimum required. Flux only male portion of connection, never female portion.
- F. Take extreme care to keep refrigerant piping clean and dry. It shall be kept sealed except when cutting or fabricating. Each length shall be inspected and swabbed with cloth soaked in refrigeration oil if any dirt, filings, or visible moisture are present.
- G. Protect refrigerant piping adequately. Install permanent guards as required to protect piping and fittings from damage.
- H. Arrange piping so normal inspection and servicing of compressor and other equipment is not hindered. Do not obstruct view of crankcase oil sight glass, or run piping so it interferes with removal of compressor or other components.
- I. Tubing installed in trenches or conduit under floor shall be pitched 2-1/2" per 20 ft to prevent oil trapping. Guard against deformation or damage from trucks carrying heavy loads, or cement being poured.

3.4 FITTINGS/JOINTS

- A. Manufactured soldered ACR type fittings are preferred and shall be used wherever practical. Use of sliding block or lever type tube bending tools or flaring block or punch type swagging tools to field fabricate elbow or coupling type fittings will be allowed on piping 1-1/8" OD and smaller as long as fittings are of long radius type and are free from buckling or kinks. Field fabricated mitered elbows, tees, and oil traps will not be allowed.
- B. Use tools and procedures recommended by fitting manufacturer for pressure-seal fittings.

3.5 GENERAL INSTALLATION OF ACCESSORIES

- A. Install vibration eliminators in suction and discharge lines of compressors. Vibration eliminator must be applied according to manufacturer's recommendation. Mount vibration eliminator for compressors parallel to crankshaft, as close to compressor as possible. Installation in horizontal plane at right angles to crankshaft is not acceptable. If installed in vertical position, eliminator joints must be sealed against dripping from condensation to protect from freezing.
- B. Provide minimum of two evacuation fittings, one in suction line at inlet side of suction line filter and the other in liquid line at outlet side of filter-drier. If properly valved, connection in liquid line may serve as charging valve. After evacuation and charging, cap or remove fittings. Connections to be at least 3/8" and preferably 1/2" in size.

3.6 VALVES IN GENERAL

- A. Provide isolating, charging, purging check and relief valves where indicated or as required to allow for directional control, isolation, charging, excessive pressure protection and pressure monitoring of all major refrigeration equipment and specialties.
- B. Install valves per manufacturer's instructions.
- C. When brazing any valve to connecting piping, protect valve properly against damage from excessive heating. At minimum, wrap damp cloth around valve body or remove internal valve parts before brazing and direct flame away from valve body.

3.7 RELIEF VALVES AND RUPTURE DISCS

- A. Provide at least one pressure relief device in each system according to ASME Boiler and Pressure Vessel Code and/or UL Standard 207.

- B. Provide additional relief devices for portion of system, specialty device and pressure vessel that contains more than 3 ft³ of internal gross volume and may contain liquid refrigerant, and may be isolated from any other system excessive pressure protection device.
- C. Provide one relief device for internal gross volumes of less than 10 ft³ and 2 relief devices for internal gross volumes larger than 10 ft³.
- D. Pipe all relief devices to atmosphere.

3.8 THERMOSTATIC EXPANSION VALVES

- A. Install expansion valve as close to evaporator as possible. Clamp valve's remote bulb to cleaned straight section of evaporator's horizontal suction line as close to evaporator as possible. Protect bulb from ambient influence by wrapping bulb and suction line contact point with "Presstite" or equal, 40% virgin cork, moisture proof, insulating tape.
- B. On suction lines 7/8" OD and smaller, install bulb on top of suction line.
- C. On suction lines 1-1/8" OD to 2-1/8" OD, install bulb at about 4 or 8 o'clock, in relation to suction line pipe diameter.
- D. On suction lines 2-5/8" OD and larger, install bulb in remote bulb well. Install bulb well in evaporator's horizontal suction line, vertical and perpendicular to line, or in run of tee with opposite run connected to evaporator suction line and tee branch drop connected to either trap or evaporator manifold.
- E. Do not, under any circumstances, locate remote bulb in location where suction line is trapped.
- F. Use externally equalized expansion valves whenever combined pressure drop of evaporator and distributor exceed 2.5 psig for air conditioning, 1.5 psig for commercial and 0.5 for low temperature systems. Connect equalizer to top of evaporator suction line on compressor side of remote bulb location and before any trap, multiple evaporator manifold or evaporator pressure regulating valve. On multiple evaporator system, locate each equalizer line so as to not be influenced by pressure changes occurring from other evaporators being fed by other expansion valves.
- G. Provide charging valve and thermometer well as close as possible to external equalizer connection on all evaporator suction lines larger than 2-1/8" OD.
- H. Adjust valve superheat as required for system installation.

3.9 LIQUID LINE SOLENOID VALVES

- A. Install liquid control solenoid valves on refrigeration system incorporating multiple evaporators or pumpdown cycle.
- B. Valve coil voltage shall be compatible with electric control system and shall never be higher than 120 V.
- C. Install in-line valves on 1-1/8" OD and smaller liquid lines ahead of and close to each evaporator sight glass.
- D. Install pilot control solenoid valves on systems with 1-/8" OD and larger liquid lines or systems where in-line valve pressure drops would exceed 2 psig at design conditions. Install pilot type solenoid so that common port connects to expansion valve external equalizer port, normally open port connects to refrigerant liquid line ahead of sight glass, and normally closed port connects to evaporator suction line on compressor side of expansion valve's remote bulb.
- E. Install liquid line drier ahead of any liquid control solenoid valve.

3.10 CHARGING AND PURGING VALVES

- A. Install one valve in suction line on inlet side of suction line filter. Install second valve on outlet side of liquid line filter drier. Close valve and cap flare connection after evacuation and charging.

3.11 HEAD PRESSURE REGULATING VALVES (AIR COOLED CONDENSING UNIT)

- A. Provide complete head pressure regulating valve system as required by each individual refrigeration system. Install according to manufacturer's recommendations.

3.12 HOT GAS BYPASS VALVES (AIR CONDITIONING SERVICE)

- A. Each valve coil voltage shall be compatible with electric control system and must never be higher than 120 V. Connect valve as close to compressor to condenser gas discharger line as possible. Hot gas piping to evaporator or suction line shall be pitched and trapped so as to promote recirculation of refrigerant oil air to compressor through system suction line.
- B. Provide hot gas bypass to evaporator inlet or expansion valve liquid refrigerant distributor on all single evaporator systems. Connect valve refrigerant suction pressure sensing line to top evaporator suction line. Adjust valve to maintain minimum saturated suction temperature of 35°F.

- C. Provide hot gas bypass to suction line on multiple evaporator systems. Hot gas to suction line connection shall be made after last evaporator and as far from compressor as possible, through one run of bull head mixing tee. Connect the other run of bull head tee to electrical pilot operated desuperheating expansion valve and connect branch of mixing tee to system suction line through lateral connection, opposite direction of suction gas flow. Adjust desuperheating expansion valve to maintain compressor's suction gas temperature below 75°F. Expansion valve shall have minimum superheat adjustment range of 40°F. Locate expansion valve's remote bulb between compressor and suction bypass connection, as close to suction connection as possible.
- D. Install suction line accumulation on multiple evaporator systems when bypass connection must be made within 10 feet of compressor suction port.

3.13 EVAPORATOR PRESSURE REGULATING VALVES

- A. Provide evaporator pressure regulating valves for each evaporator on multi-temperature refrigeration system. Install one valve on each evaporator suction line after all expansion valve equalizer and bulb connection, and before any trap or multiple evaporator manifold connection. Install valve so as to promote return of refrigerant oil to compressor.
- B. Adjust each valve to maintain each evaporator's minimum saturated suction pressure.

3.14 CRANKCASE PRESSURE REGULATING VALVES

- A. Provide compressor crankcase pressure regulating valves on medium and low temperature refrigeration systems. Install valve between compressor suction port of just ahead of system suction line accumulator and suction piping. Install valve so as to promote return of refrigerant oil to compressor.
- B. Adjust valve to maintain compressor maximum crankcase pressure.

3.15 LIQUID LINE FILTER DRIERS

- A. Provide at least one liquid line filter drier in each refrigeration system.
- B. Provide at least one liquid line drier in each refrigeration system.
- C. Provide at least one liquid line drier in each evaporator's liquid line.
- D. Provide Schrader valves at inlet and outlet of each drier.
- E. Liquid line driers shall be of filter drier type and of size recommended by manufacturer. Do not install drier cartridges until second evacuation has been completed.

3.16 SUCTION LINE FILTER DRIERS

- A. Provide at least one suction line filter drier for each compressor. Install drier as close to compressor or suction line accumulator as possible.
- B. Install valved bypass around drier, shut-off valves upstream and downstream, and pressure taps upstream and downstream of drier if not installed in drier shell. Position drier so it does not trap an excessive amount of oil.
- C. Install permanent suction line filter in each compressor suction line. Provide pressure fitting ahead and after filter, preferably in shell, to facilitate checking pressure drop. If pressure drop across filter is in excess of 1 psig after initial 24 h of operation, replace suction line filter cartridge, or if filter is of sealed permanent type, replace filter.

3.17 COMBINATION MOISTURE AND LIQUID INDICATORS (SIGHT GLASS)

- A. Provide minimum of 2 indicators for each refrigeration system. Install one indicator on condenser or receiver liquid line, and install second between inlet of each evaporator expansion valve and last line specialty device (solenoid or drier).
- B. Provide in-line indicators for 2-1/8" OD and smaller lines. On liquid line sizes larger than 2-1/8" OD, install 3/8" indicators in bypass circuit connected in parallel with primary liquid line.
- C. Install at least 2 solder-type combination liquid sight glass and moisture indicators in each system and locate for easy visibility.

3.18 TESTING EVACUATION AND CHARGING

- A. Contractor shall notify Engineer 24 h in advance of any test so Owner and/or manufacturer's representative may be present for the test if desired.
- B. When refrigeration connections have been completed, test system at minimum of 300 psig with compressor suction and discharge valves closed, and other valves in system open. (If local codes require higher test pressures, comply with such codes.) Charge sufficient liquid refrigerant into system to raise pressure to 35 psig, and add dry nitrogen to obtain desired test pressure. Rap all joints with rubber or rawhide mallet and check for leaks with electric leak detector having its sensitivity certified within the past year.
- C. Refrigeration piping will not be acceptable unless it is gas tight. If leaks are found, isolate defective area, discharge gas and repair leaks, and then repeat test. When testing has been completed, release all pressure freely to atmosphere.

- D. Evacuate system with vacuum pump specifically manufactured for refrigeration vacuum duty, having capability of pulling vacuum of at least 50 micron. Evacuation of system shall not be performed with refrigeration compressor. Connect pump to both low and high side evacuation valves with copper tube or high vacuum hoses 1/4" ID minimum. Compressor service valves shall remain closed. Attach a high vacuum gauge capable of registering pressure in microns to system for pressure reading. Hermetic or accessible-hermetic motor compressors must not be operated during evacuation because of reduced dielectric strength of atmosphere within motor chamber. To check system pressure, provide hand valve between pressure gauge and vacuum pump, which can be closed to isolate system and check pressure.
- E. Evacuate each system to absolute pressure not to exceed 1,500 micron. Break vacuum to 2 psig with refrigerant to be used in system. Repeat evacuation process, again breaking vacuum with refrigerant. Install drier of required size in liquid line, open compressor suction and discharge valves, and evacuate to absolute pressure not to exceed 500 micron. Leave vacuum pump running for not less than 2 h without interruption. Raise system pressure to 2 psig with refrigerant, and remove vacuum pump.
- F. Charge refrigerant directly from original drums through combination filter-drier. Each drier may be used for maximum of 3 cylinders of refrigerant, and then must be replaced with fresh drier. Charge system by means of charging fitting in liquid line. Weigh refrigerant drum before charging so that accurate record can be kept of weight of refrigerant put in system. If refrigerant is added to system through suction side of compressor, charge in vapor form only.

3.19 START-UP

- A. Check compressors for proper oil level, and if necessary add sufficient oil to bring level to center of crankcase sight glass. Use only refrigeration oil recommended by compressor manufacturer. Oil must be delivered to job site in factory sealed containers.
- B. Before operating any motor or other moving parts, they are to be lubricated with proper oil or grease as necessary.
- C. Remove or loosen shipping retainers under motor compressor assembly. Confirm hold-down nuts on spring-mounted compressors are not touching compressor feet, and are not more than 1/16" above mounting foot.
- D. Check high and low pressure control cut-in and cut-out points. Check water valve settings and adjust if necessary.

- E. After compressor is started, continue charging until system has sufficient refrigerant for proper operation. Do not overcharge. During start-up, no compressor shall be left operating unattended and unwatched until system is properly charged with refrigerant and oil.
- F. Do not add refrigeration oil while system is short of refrigerant unless oil level is dangerously low. If oil has been added during charging, carefully check compressor crankcase sight glass after reaching normal operating condition to be sure system does not contain excessive amount of oil, which can cause slugging or loss of refrigerating capacity.
- G. During first 24 h of system operation, use high efficiency acid removing cores in suction line filter. Replace cores if pressure drop exceeds 1 psi. After system is clean, install new standard duty cores in suction and liquid line filter assemblies.

3.20 OPERATION AND CHECK-OUT

- A. Contractor shall coordinate work with Control Contractor and shall be responsible for proper adjustment of all refrigeration controls in system, including controls on each refrigeration circuit, air temperature controls in machine room, remote condenser or cooling tower controls, water regulating valves, or other controls required.
- B. Check compressor overload protectors with manufacturer's specifications, and inform Engineer if they are incorrect.
- C. Furnish competent refrigeration service mechanic to check and make any necessary adjustments to controls during first hours of system operation. Mechanic shall remain at site for at least 8 h during first day of full operation.

END OF SECTION

SECTION 233114 DUCTWORK

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 230000 - General Mechanical Requirements (submittal requirements)
- B. Section 230700 - Mechanical Systems Insulation
- C. Section 230595A - Air Systems Test Adjust Balance
- D. Section 233314 - Ductwork Specialties

1.2 SUBMITTALS

- A. For each duct system, submit schedule utilizing reinforcement tables from SMACNA HVAC Duct Construction Standards Metal and Flexible where applicable. Each duct system schedule shall include, but not be limited to, the following:
 - 1. Name of Contractor/manufacturer fabricating each duct system
 - 2. Material and gauge
 - 3. Pressure class
 - 4. Transverse joint type and length and reinforcement rigidity class with designated joint T number or proprietary duct connection if utilized for each system
 - 5. Certified test results of proprietary joint products, if used, tested in accordance with SMACNA procedures
 - 6. Intermediate reinforcement spacing and rigidity class with metal angle dimensions and gauge
 - 7. Type of longitudinal seam
 - 8. Fitting construction details
 - 9. Support methods including spacing, upper attachments, and lower attachments
 - 10. Sealant and gasket
 - 11. Sealing class
- B. Duct leakage testing methods, apparatus and apparatus certification signifying meter is in conformance with ASME Requirements for testing meters.
- C. Duct liner including data on thermal conductivity, air friction correction factor, and temperature and velocity limitation.
- D. Submit shop drawings for manufactured duct system products

1.3 DELIVERY, STORAGE, AND HANDLING

- A. Protect duct and fittings from damage due to normal handling during shipment and storage. Protection shall be applied to ends of duct to prevent dirt and moisture from entering ducts and fittings.

1.4 DESCRIPTION

- A. Furnish and erect ductwork free of objectionable vibration, chatter, and pulsations. Verify dimensions at site, making field measurements and drawings necessary for fabrication and erection.
- B. Duct sizes indicated are net inside dimensions.
- C. Where size for a duct segment is not indicated, the duct segment size shall be equal to the largest duct segment to which it is connected. Transition to smaller size shall occur on side of fitting where smaller size is indicated.

1.5 DESIGN CRITERIA

- A. All products shall conform to NFPA 90A, and shall possess flame spread rating of not over 25 and smoke developed rating no higher than 50.
- B. Unless otherwise indicated, construct all ductwork of galvanized sheet metal for pressure class not less than 2" WG for positive pressure ductwork and not greater than -2" WG for negative pressure ductwork.
- C. Ductwork shall comply with Local, State and Federal requirements.
- D. Use material, weight, thickness, gauge, construction, and installation methods as outlined in the following SMACNA publications, unless noted otherwise:
 - 1. HVAC Duct Construction Standards Metal and Flexible, 3rd Edition, 2005
 - 2. HVAC Air Duct Leakage Test Manual, 2nd Edition, 2012
 - 3. Round Industrial Duct Construction Standards, 2nd Edition, 1999
 - 4. Rectangular Industrial Duct Construction Standards, 2nd Printing 2004
 - 5. Accepted Industry Practice for Industrial Duct Construction, 2nd Edition, 2008
 - 6. Thermoset FRP Duct Construction Manual" 1st Edition (1997)
 - 7. Phenolic Duct Construction Standards ANSI/SMACNA 022 (2015)
- E. Unless otherwise indicated, pressure class for constant air volume system ductwork shall be equal to external static pressure (fan entrance or discharge pressure minus associated unit internal component pressure drop), but not less than ± 2 " WG.
- F. Duct transverse joints and reinforcement material, including angle ring flanges and stiffeners, shall be of same material as duct.

- G. Adhesives and sealants used on installation of ducts shall comply with South Coast Air Quality Management District (SCAQMD) Rule #1168; Indoor Environmental Quality Section, Credit EQ-4.

PART 2 - PRODUCTS

2.1 ALUMINUM SHEET

- A. Aluminum alloy, ASTM B209/B209M, Type 3003H-14 capable of double seaming without fracture.

2.2 STAINLESS STEEL SHEET

- A. First quality, cold rolled annealed, pickled, ASTM A167 and A480, Finish No. 2B for concealed work and Finish No. 4 for exposed work. Unless otherwise indicated, use Type 304L where welded duct construction is specified and Type 304 where non-welded duct construction is allowed.

2.3 FLEXIBLE DUCT

- A. Manufacturers: Thermaflex, Casco, or Flexmaster
- B. Factory fabricated, UL listed under UL 181 as Class 1 duct, meeting requirements of NFPA 90A with flame spread of 25 or less and smoke developed rating of 50 or under.
- C. Flexible duct shall have minimum ratings as follows:
 - 1. Operating Temperature: -20 to 250°F
 - 2. Internal Working Pressure:
 - Positive: 6" WG
 - Negative: 1" WG
 - 3. Burst Pressure: 2-1/2 times working pressure
 - 4. Velocity: 5000 fpm
- D. Unless otherwise indicated, duct shall be nonmetallic insulated type composed of polyester film, polyethylene film, nylon film, CPE film, or coated woven fiberglass liner bonded permanently to corrosion resistant coated steel wire helix without adhesive.
- E. Insulation shall be flexible fiberglass insulation with minimum R-value of 6 at mean temperature of 75°F. Vapor barrier jacket shall be aluminum foil reinforced, polyethylene, or metalized polyester film with maximum perm rating of 2.87 ng/(s·m²·Pa)(0.05 perm) per ASTM.
- F. Insulation material shall not be exposed to air stream.

- G. Lined flexible duct shall have the following minimum acoustical performance in accordance with ARI Standard 885. Dynamic Insertion Loss in each octave band of 5 ft or 10 ft straight duct shall not be less than the following:

Duct Diameter (in)	Dynamic Insertion Loss (dB)					
	Octave Band Center Frequency (Hz)					
	(Based on 5 ft length)					
	125	250	500	1000	2000	4000
6"	6	9	18	22	24	15
8"	6	10	18	20	21	12
10"	5	11	18	18	18	9
Duct Diameter (in)	(Based on 10 ft 3 m(10 foot) length)					
	125	250	500	1000	2000	4000
	125	250	500	1000	2000	4000
6"	10	15	28	33	35	22
8"	10	18	29	32	32	20
10"	9	19	28	31	29	18

- H. Stainless Steel Flexible Duct:

- Duct shall be 304 stainless steel construction, spiral wound into tube and spiral corrugated. Construction to be triple mechanical lock to form continuous secure airtight joint. Duct shall be suitable for temperature -40 to 2100°F, pressure -1" WG to 12" WG.
- Duct shall be Flexmaster Triple Lock T/L-SS.

2.4 DUCT FLEXIBLE CONNECTORS

- A. Refer to Specification Section 233314 - Ductwork Specialties

2.5 DUCT SEALANTS AND GASKETS

- A. Sealant:

- Flexible, water or solvent based, adhesive sealant compounded specifically for sealing joints and seams in ductwork. Hardcast, McGill AirSeal, Ductmate PROseal, Mon-Eco Industries, Childers, Foster, or Design Polymerics.
- Sealants shall be UL 723 (ASTM E84) classified, and meet NFPA 90A and 90B.
- Sealants shall comply with requirements of SCAQMD Rule 1168 and LEED v4 EQ-4.
- Select sealants as recommended by manufacturer for specific application.
- Submit sealant manufacturer's data sheets including performance data, pressure ratings, surface burning characteristics data, VOC compliance with LEED v4, detailed installation instructions.
 - Sealants for exterior ductwork shall include explicit statement by manufacturer indicating suitability for application.
 - Sealants for exposed (uninsulated) exterior ductwork shall be UV resistant and include explicit statement by manufacturer indicating suitability for exposed exterior application.
- Duct tapes are not allowed.

- B. Gaskets:
 - 1. Butyl, copolymer or neoprene based tape similar to Ductmate 440 Gasket Tape or Neoprene Gasket Tape for flanged joints.
- C. Duct sealant for high temperature ductwork:
 - 1. Sealant shall be 100% silicone and rated to 500°F continuous operation, similar to Surebond SB-188 or ASI 600.

2.6 CABLE SUSPENSION SYSTEM

- A. Suspension system shall be similar to Gripple Hang-Fast as manufactured and supplied by Gripple Incorporated or Ductmate Industries "Clutcher" cable hanging system.
- B. Suspension system shall be load rated and verified by SMACNA Testing and Research Institute to be in compliance with SMACNA HVAC Duct Construction Standards, 2005, Chapter 5.

PART 3 - EXECUTION

3.1 GENERAL

- A. Unless otherwise indicated, install ductwork level, parallel, and/or perpendicular to building structure, walls, and ceilings and at such heights not to obstruct any portion of ceiling, window, doorway, stairway, or passageway. Install ductwork to allow adequate access and service space for equipment. Refer to drawings and/or manufacturer's recommendations. Install vertical ductwork plumb. Where interferences develop in field, offset or reroute ductwork as required to clear such interferences. In all cases, consult drawings for exact location of duct spaces, ceiling heights, door and window openings or other architectural details before installing ductwork.
- B. Make allowances for beams, pipes or other obstructions in building construction and for work of other contractors. Check plans showing work of other trades and consult with Engineer in event of interference. Transform, divide, or offset ducts as required, in such a manner as to maintain same cross sectional area of duct as indicated on drawings. Where it is necessary to install pipes or similar obstructions through ducts, consult with Engineer and obtain written approval from Engineer and Owner. If approved, provide streamlined encasement or collar designed in accordance with SMACNA Standards and seal to prevent air leakage.
- C. Ductwork shall be free of kinks and dents.
- D. Fabricate and install duct, fittings, joints, seams, reinforcement, supports, sealing, liner, etc., in sizes indicated on drawings and in accordance with manufacturer's published data and SMACNA Standards except as modified in this section of specifications or on drawings.

- E. Provide transitions where different size or different shape ductwork segments are connected. Use concentric transitions unless otherwise shown. Unless otherwise indicated, make diverging transitions with maximum angle of 15° per side (30° total diverging) and converging transitions with maximum angle of 25° per side (50° total converging).
- F. Provide transitions at ductwork system components and connections to equipment. Refer to Specification Section 233713 - Diffusers, Registers and Grilles, for additional information regarding diffuser/register/grille connections.
- G. Refer to ductwork symbols list on drawings for additional and dimensional requirements for fittings.
- H. Seal duct seams and joints to meet SMACNA Seal Class A as minimum for all ductwork including low-pressure ductwork.
- I. Construct ductwork so that interior surfaces are smooth. Internal duct hangers and internal bracing are not allowed. Refer to Part 1, Design Criteria for internal tie rods.
- J. Support coils, filters, air terminals, dampers, sound attenuator devices or other devices installed in duct systems with angles or channels and make all connections to such equipment including equipment furnished by others. Secure frames with gaskets, nuts, bolts and washers.
- K. Air terminal devices may be supported by strap hangers if air terminal manufacturer approves. Strap hangers are not allowed for fan powered devices, double wall type and Titus Steri-Loc type devices.
- L. Install outside air intake duct to pitch down at minimum 1" per 20 ft toward intake louver or plenum and to drain to outside of building. Solder or seal seams to form watertight joints.
- M. Install exhaust air duct to pitch down at minimum 1" per 20 ft toward exhaust louver.
- N. Blank off unused portion of outside air intake or exhaust louvers.
- O. Where 2 different metal ducts meet, install joint in such a manner that metal ducts do not contact each other by using proper gasket seal or compound.
- P. Do not install ductwork over electrical panelboards, switchgear, switchboards or motor control centers.
- Q. When original galvanized finish is altered or damaged, apply field galvanizing paint as follows:
 - 1. Prepare surface with use of power sanders or wire brushes to remove rust, paint, etc.
 - 2. Apply cold galvanizing material equal to ZRC Products, Inc.

3.2 ELBOWS

A. Rectangular Duct:

1. Use radius elbows with centerline radius to width ratio of 1.5 (SMACNA Type RE 1).
2. Where 1.5 centerline radius elbows do not fit due to limited space, use radius elbows with centerline radius to width ratio of 1.0 (SMACNA Type RE1 or RE-3).
3. Where 1.0 centerline radius elbows do not fit, use radius elbows with centerline radius to width ratio of 0.75 (SMACNA Type RE-1 or RE-3) or 45° throat with radius heel elbows (SMACNA Type RE 8).
4. Use single splitter vanes for 1.0 radius elbows, 0.75 radius elbows and 45° throat with radius heel elbows for elbows width 18" and over.
5. Fabricate splitter vanes in accordance with SMACNA HVAC Duct Construction Standards, Chart 4-1, (page. 4.11) and Figure 4-9 (page. 4.13).
6. Square throat elbows with or without turning vanes are not allowed unless specifically indicated. Square throat elbows without turning vanes may be used for transfer air ducts.

B. Round and Oval Duct:

1. Unless specific type is indicated, use radius elbows with centerline radius to diameter ratio of 1.5 regardless of duct velocity. Where 1.5 radius elbows do not fit due to limited space, use 1.0 radius elbows.

3.3 LONGITUDINAL SEAM

A. Rectangular Duct:

1. Unless otherwise indicated, use Pittsburgh lock seam.
2. Seal longitudinal seams with approved sealant or pre-sealed with encapsulated mastic.
3. Button punch snap lock construction is not allowed for ductwork in chases and areas above inaccessible ceiling.
4. Button punch snap lock construction is not allowed on aluminum ductwork.
5. Button punch snap lock construction (SMACNA L-2) may be used for ductwork that is both 2" WG (+ or -) and lower, and 36" and smaller in width or height. For ductwork over 24" in width or height, add screw 4" from each end.

B. Round and Oval Duct:

1. Unless otherwise indicated, longitudinal seams shall be in accordance with SMACNA HVAC Duct Construction Standards with the following exceptions.
 - a. SMACNA Snaplock Seam Types 6A, 6B, 7 and 8 shown in Figure 3-2 are not allowed.

3.4 TRANSVERSE JOINT

A. Rectangular Duct:

1. Transverse joints shall be in accordance with SMACNA HVAC Duct Construction Standards.

2. Ductmate 25/35/45 connection systems with corner clips or optional nuts and bolts may be used. Incorporate use of all Ductmate accessories to ensure integrity of transverse connection. Install joints in strict accordance with the latest edition of Ductmate 25/35/45 Assembly and Installation Instruction Manual and Duct Construction Standards. Nexus or WDCI will be acceptable.
3. Lockformers TDC or Engles TDF may be used in accordance with T-25 flanges of SMACNA HVAC Duct Construction Standards Metal and Flexible, 2005, provided that corner pieces with bolts are used. If TDF/TDC flanges are damaged, replace the damaged joint(s) by straightening and reinforcing with minimum 1-1/2" x 1-1/2" x 1/4" angle at each side of transverse joint.
4. Where any type of flanged transverse joint is used for exterior uninsulated ductwork, top flange shall have continuous sheetmetal cap or continuous cleat to protect joint.

B. Round and Flat Oval Duct:

1. Unless otherwise indicated, use beaded sleeve joints (SMACNA RT-1) with minimum 2" insertion length or flange joints (SMACNA RT-2 or RT-2A).
2. Connection systems manufactured by Ductmate Industries (Spiralmate and Ovalmate) or McGill AirFlow (Uni-flange) may be used for supply air ductwork.
3. AccuFlange connected systems may be used with gaskets specified in Part 2 of this Section.

3.5 DUCT SUPPORTS

- A.** Unless otherwise indicated, use straps or Z bar hangers with 3/8" rods to support rectangular ducts 60" wide and smaller and trapeze hangers with rods or angles to support rectangular ducts over 60" wide.
1. Use trapeze hangers to support externally insulated ductwork with weight bearing inserts. Refer to Section 230700 - Mechanical Systems Insulation and details.
- B.** For round ducts 24" diameter or smaller, use single hanger.
1. Cable Suspension System may be used up to 24" diameter at spaces higher than 8 ft above floor or platform.
 2. Round Duct Strap Bracket by Ductmate Industries may be used up to 24" diameter.
- C.** For round ducts 25 inches diameter or larger, use 2 minimum 3/8" rods, with trapeze in accordance with the following schedule:

Duct Size	Trapeze (Half Round)
25" to 36"	1-1/2" x 1-1/2" x 1/8"
37" to 48"	or 2" x 2" x 1/8"
49" to 60"	2" x 2" x 1/4"
61" to 84"	2-1/2" x 2-1/2" x 1/4"

- D.** Refer to Section 230700 - Mechanical Systems Insulation for ductwork insulation, weight bearing inserts and insulation protection shield requirements.

- E. Riser Supports:
 - 1. Unless otherwise noted, support risers at every floor in accordance with SMACNA Standard Chapter 5, 5.2.10, Figure 5-8 and Figure 5-10.
 - 2. If floor to floor exceeds 16 ft, support riser by its reinforcing as shown on FIG. B on Figure 5-8.
- F. The following upper attachments, upper attachment devices, lower hanger attachments, hanger devices, and/or hanger attachments are not allowed except where specifically indicated:
 - 1. Hook or loop
 - 2. Nailed pin fasteners
 - 3. Expansion nails without washers
 - 4. Powder actuated fasteners (forced entry anchors).
 - a. Exception: Forced entry anchors may be used for upper attachments of flexible ductwork supports.
 - 5. Beam or "C" clamps without retaining clips or friction clamps (provide retaining clips for "C" clamps)
 - 6. Non-factory manufactured upper attachments for metal pan deck including rods, straps, or wedged studs directly attached to metal deck (Items 16 through 21 shown in Fig 5-4 of SMACNA HVAC Duct Construction Standards)
 - 7. Wire hanger shown in Fig. 5-5 of SMACNA HVAC Duct Construction Standards
 - 8. Trapeze hangers supported by wires or straps
 - 9. Drilled hole with attachment to structural steel
 - 10. Lag screw expansion anchor
 - 11. Rivets
 - 12. Non-metallic hangers or straps
- G. Supporting devices shall be standard products of manufacturers having published load ratings.
- H. For welded ducts, soldered ducts or ducts with watertight joints, do not use supports utilizing screws or other penetrations into ductwork.
- I. Unless Architectural Documents indicate the required framing, provide angle iron framing around roof opening where duct penetrates through roof decking, to maintain roof decking structural integrity in accordance with roof decking manufacturer's recommendations. This is not required for concrete decking. For concrete decking, consult with the project structural engineer for location and size of opening prior to execution of Work.

3.6 PROTECTION OF DUCTWORK

- A. Protect ductwork during construction against entry of foreign matter and construction dirt.
- B. Keep ductwork capped when work is complete for the day or when duct is not being worked on or added to. Use of polyvinyl (VISQUEEN) with duct tape wrap is an adequate measure as long as it is secure with no openings or tears in product.

- C. If ductwork is not protected, remove dirt and foreign matter from the duct system and obtain inspection and approval from Engineer upon completion of cleaning before operating fans.

3.7 DUCT LEAKAGE TEST

- A. Refer to Test and Balancing portion of Section 230000 - General Mechanical Requirements.
- B. Owner and/or Owner's representative may elect to witness leakage tests. Notify Owner and/or Owner's representative at least 3 days in advance.
- C. Test supply, return, and exhaust ductwork as follows, unless otherwise indicated in this section or in schedules.
 - 1. Test 25% of total installed duct area for duct pressure positive or negative Class 3" wg and higher.
 - 2. Test 25% of total installed duct area for duct pressure positive or negative Class 2" wg.
 - 3. Test 100% of total installed duct area for ductwork located outdoors.
 - 4. Test 100% of total installed duct area for ductwork located in inaccessible spaces including shafts.
 - 5. Engineer will select sections of ductwork to be tested.
 - 6. If test results are acceptable to Engineer, remainder of ductwork is permitted to proceed without further testing. If ductwork fails test, repair all ductwork including ductwork not tested. Then repeat leakage tests for new sections of ductwork as described above.
- D. Leakage test procedures shall be in accordance with test method described in Section 3 of SMACNA HVAC Air Duct Leakage Test Manual, except as modified in this Section. Test apparatus shall be in accordance with Section 5 of SMACNA HVAC Air Duct Leakage Test Manual.
- E. Test pressure shall be equal to duct pressure class. Negative pressure ductwork shall be tested with negative test pressure.
- F. Air leakage shall not exceed limits specified or scheduled. If leakage exceeds allowable limits, identify leaked areas, repair, seal and retest.
- G. Provide filter system on duct inlet to test blower. Filter system shall be equal to final filtering efficiency of AHU supply air duct system. Filters are not required for negative pressure testing.
- H. Do not insulate ductwork until it has been successfully tested.

3.8 LOW PRESSURE DUCT CONSTRUCTION (PRESSURE CLASS 2" WG AND UNDER)

- A. Use welds, rivets or nuts, and bolts for fabricating ductwork. Fully threaded sheet metal screws may be used on duct hangers, transverse joints and other SMACNA approved locations if screw does not extend more than 1/2" into duct. Sheet metal "TEK" screws 3/4" in length may be used as fasteners in conjunction with factory made transverse joints.
- B. Unless otherwise indicated, construct branch take-off fittings as follows:
 - 1. For branch take-offs including branch ducts serving more than one diffuser or grille, use 45° entry fittings. For supply air ducts, conical taps may be used.
 - 2. For take-offs serving single diffuser, register or grille, use straight spin-in collars with manual balancing dampers.
- C. Splitter dampers and/or extractors are not allowed.

3.9 LOW PRESSURE DUCT LININGS

- A. All portions of duct designated to receive duct liner shall be completely covered with duct liner. Transverse joints to be neatly butted and there shall be no interruptions or gaps.
- B. Duct liner shall be adhered to sheet metal with 90% coverage of adhesive, and all exposed leading edges and all transverse joints shall be coated with adhesive. Adhesives shall meet ASTM C916 Type II. Foster 85-60/85-00, Childers CP-127, Miracle-Kingco PF-101/PF-102.
- C. Duct liner shall be additionally secured with mechanical fasteners, which shall compress duct liner sufficiently to hold it firmly in place. Use weld pins only. Adhesive bonded pins or impact pins are not allowed. Spacing of mechanical fasteners with respect to duct liner interior width shall be in accordance with SMACNA Standard, "HVAC Duct Construction Standards, Metal and Flexible."

3.10 AIR HANDLING UNIT SYSTEM DUCTWORK PRESSURE CLASS

- A. Air Handling Unit
 - 1. Duct pressure class to be as follows:
 - a. Outside air intake plenum: 2" WG
 - b. From air handling unit discharge to inlet of air terminal device (where not in shafts): +4 " WG
 - c. Ductwork located in shafts: +6 " WG
- B. Return Fan
 - 1. Duct pressure class to be as follows:
 - a. From return grille to return fan: -2 " WG

- b. From return fan to outside air intake and relief from return fan to relief air louver: +2 " WG

3.11 FLEXIBLE DUCT

- A. Install flexible ducts in accordance with manufacturer's installation instructions and SMACNA Standards, except as modified in this Section of Specifications.
- B. In supply air systems without air terminal devices, flexible ducts may be used for final connections to diffusers, grilles, and registers. Flexible ducts shall be of minimum length required to make connections, but shall not be greater than 6 ft in length, unless otherwise noted.
- C. In return and general exhaust air systems without air terminal devices, flexible ducts may be used for final connections to return grilles and registers and general exhaust grille and registers. Flexible ducts shall be of minimum length required to make connections, but no greater than 6 ft in length, unless otherwise noted.
- D. In return and general exhaust air systems with air terminal devices, flexible ducts shall be used for duct connections to grilles and registers for sound attenuation purposed, except above non-accessible ceilings. Flexible ducts shall be minimum 6 ft long and maximum 8 ft long.
- E. Centerline radius of bends shall not be less than one duct diameters. FlexFlow Elbow supports by Thermaflex or similar products shall be used at diffuser/grille connection to assure full radius elbow.
- F. Support flexible ductwork as recommended by manufacturer and with minimum 0.5" wide saddles with maximum sag of 0.5" per ft between supports. Spacing of supports shall be a maximum of 5 ft on center, with no portion lying on ceiling supporting system.
- G. Individual sections of flexible ductwork shall be of one-piece construction. Splicing of short sections is not allowed.
- H. Connect flexible duct liner to collars and rigid duct in accordance with manufacturer's installation instructions. If manufacturer's installation instructions do not include stainless steel draw-bands, provide stainless steel draw-bands in addition to manufacturer's installation instructions. One draw-band shall secure flexible duct to rigid duct and another shall secure flexible duct outer jacket to rigid duct. If collars have beads, position draw bands behind beads.
- I. Pull insulation and vapor barrier jacket over liner connection and secure with draw band. For terminations at externally insulated ductwork, fittings, grilles, diffusers, etc., secure flexible duct jacket to ductwork insulation jacket with compatible vapor barrier tape.

- J. Flexible ducts are not allowed in the following locations or applications:
 - 1. Above non-accessible ceilings
 - 2. In high pressure ductwork
 - 3. In exposed duct applications
 - 4. For special exhaust systems, such as fume hood exhaust, BSC exhaust, animal room exhaust, BSL-3 exhaust, cagewash exhaust, shower room exhaust, kitchen hood exhaust, dishwasher exhaust, etc.
 - 5. For supply air systems serving BSL-3, BSL-3Ag, or BSL-4 spaces.
 - 6. Through any partition, wall, floor or ceiling.
- K. Use non-insulated high temperature type for ductwork related to solar collector system.

3.12 BATH OR SHOWER EXHAUST DUCT CONSTRUCTION

- A. Use aluminum sheet where indicated on drawings or as specified below, with joints and seams sealed with duct sealant or welded for watertightness.
- B. Seal or weld aluminum exhaust duct for watertightness.

END OF SECTION

SECTION 233314
DUCTWORK SPECIALTIES

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 230595A - Air Systems Test Adjust Balance

1.2 SUBMITTALS

- A. Shop Drawings including the following:
 - 1. Manufacturer's name and model number
 - 2. Capacities
 - 3. Temperature/pressure ratings
 - 4. Materials of construction
 - 5. Dimensions
 - 6. Manufacturer's installation instructions and/or detailed drawings
 - 7. All other appropriate data

1.3 DESIGN CRITERIA

- A. Products and materials shall conform to NFPA 90A, possessing flame spread rating of not over 25 and smoke developed rating no higher than 50.
- B. Ductwork specialties exposed to air stream, such as dampers, turning vanes, and access doors, shall be of same material as duct or unit within which specialties are mounted. When complying with this material requirement performance requirements (pressure, temperature, velocity limits, etc.) shall be as specified for same product constructed of the standard specified materials found in Part 2 of this Section
- C. Unless otherwise noted, ductwork specialties shall be designed and constructed for pressure class of ductwork in which they are installed.

PART 2 - PRODUCTS

2.1 MANUAL BALANCING DAMPERS

- A. Manufacturers: Ruskin, Greenheck, Vent Products, Pottorff or Air Balance, constructed in accordance with SMACNA (HVAC DUCT CONST) Standards, except as modified below.
- B. Rectangular Dampers:
 - 1. For low pressure ductwork, for damper blade height up to 12", use single blade type with minimum 22 ga blade with minimum 3/8" rod for blade width up to 18",

and with minimum 18 ga blade with minimum 1/2" continuous rod for blade width from 19-48 inch. For damper blade height more than 12", use multiple blade type with minimum 16 ga channel frames, opposed blade linkage operation, with blades minimum 16 ga and 6" to" maximum blade width, minimum 1/2" continuous rod and 1/2" x 1/2" angle blade stops. Bearings shall be nylon or molded synthetic. Construct dampers over 48" in width or height in multiple sections with mullions.

2. For high pressure ductwork, dampers shall be constructed to withstand maximum pressure of 5" WG (1250 Pa) at closed position and maximum velocity of 2000 fpm at open position. Frame and blades shall be constructed of minimum 16 ga with minimum 1/2" diameter or square rod.

C. Single Blade Round Dampers:

1. For low pressure ductwork, damper shall have blade 24 ga, but not less than two gauges heavier than duct gauge. Rod shall be minimum 3/8" diameter or square continuous. Bearings shall be nylon or molded synthetic.
2. For high pressure ductwork, damper blade shall be minimum 16 ga. Rod shall be minimum 1/2" square continuous and tack welded to blade. Provide sealed end bearing similar to Ventlok #609 and acorn nut type dial regulator similar to Ventlok #635 or 641.

D. Provide damper operators with locking devices and damper position indicators. Sheet metal screws are not allowed in construction or installation of dampers. Use rivets or tack welds.

E. Dampers shall be properly stiffened and fabricated to prevent vibration, flutter or other noise.

F. Extend damper shafts through duct insulation or use elevated regulators for externally insulated ducts to accommodate specified insulation thickness.

G. Segmented Blade (Iris) Dampers:

1. Damper shall be segmented blade (iris) type with 22 ga galvanized steel frame and blades. Casing leakage shall not exceed 6 cfm.
2. Iris blade segments shall be internally linked and driven by manual adjustment knob. Linkage parts shall be fully encapsulated and out of air stream. Manual adjustment knob shall be factory calibrated to the exact aperture position and aligned with the K factor set point to provide linear response flow control.
3. Accuracy of flow measurement shall be $\pm 5\%$. Assembled units shall be furnished with specific charts designed for exact size and blade aperture configuration. Air pressure taps shall be integral to damper frame and positioned on either side of blade segments.
4. Dampers shall be similar to Ruskin Model VFBD35.

2.2 SPLITTER VANES AND TURNING VANES

A. Radius Elbow Splitter Vanes (SMACNA Type RE-3):

1. Splitter vanes for radius elbows shall be constructed in accordance with SMACNA (HVAC DUCT CONST) Standards Chart 4-1, (p. 4.11) and Figure 4-9 (p. 4.13).

B. Turning Vanes (SMACNA Type RE-2):

1. Turning vanes are not allowed unless specifically indicated.

2.3 BACKDRAFT DAMPERS

- A. Manufacturers: Air Balance, Inc., American Warming and Ventilating, Pottorff, Louvers and Dampers, Inc., Advanced Air, Vent Products, Greenheck or Ruskin
- B. Dampers shall be multi-blade, weighted type with counter-balanced blades, equal to Ruskin Type CBD-4. Blade edges to have vinyl gasket or polyurethane foam seals with ball bearings. Dampers to be suitable for flange and gasket connection to ductwork.

2.4 BACKDRAFT DAMPERS

- A. Manufacturers: Ruskin or Greenheck
- B. Dampers shall be multi-blade, weighted type with counter-balanced blades and with 12 ga frame and extruded airfoil-shaped blades equal to Ruskin Type CBS 92. Blade edges shall have silicon rubber seals with ball bearings. Dampers shall be suitable for flange and gasket connection to ductwork or fan outlet.
- C. Dampers shall be rated to maximum velocity of 4000 fpm, maximum temperature of 250°F and maximum system pressure of 5" WG (1250 Pa) for damper width of 60" and 14" WG for damper width of 12".
- D. Maximum damper leakage shall be 13.5 cfm/sf based on pressure differential of 1" WG.

2.5 FIRE DAMPERS

- A. Manufacturers: Air Balance, NCA, Greenheck, Nailor, Cesco, Pottorff Louvers and Dampers, or Ruskin
- B. Fire damper assemblies shall be listed by UL 555 with 165°F fusible link and shall meet construction standards as set forth in NFPA 90A.
- C. Fire resistance rating of fire dampers shall be as shown on drawings.
- D. Dampers shall be dynamic type dampers suitable for maximum air velocity and pressure to which they are subjected, but not less than 2000 fpm and 4" WG.

- E. Dampers shall be curtain type with blades out of air stream when in open position. Where curtain type dampers are not available because of size, use multiple blade type dampers.
- F. For round ducts, dampers similar to Ruskin Model FDR25 may be used where products are suitable for duct size, velocity and static pressure.
- G. Damper fire rating shall be compatible with rating of building surface in which damper is used.
- H. Submit UL installation details showing mounting method and duct connection method.
- I. Where ceiling fire dampers are used, they shall be similar to Ruskin CFD(R) 2 or 3, UL Classified for installation in fire rated floor or roof/ceiling assemblies.

2.6 ACCESS DOORS

- A. Access doors shall be rectangular, minimum 22 ga frame and minimum 24 ga (0.709 mm) door, fit airtight with neoprene gasket and shall be suitable for duct pressure class. When access doors are installed in insulated ductwork or equipment provide insulated doors with insulation equivalent to what is provided for adjacent ductwork or equipment. Access doors constructed with sheet metal screw fasteners are not acceptable.
- B. Low Pressure Ducts (Pressure Class 2" and Under):
 - 1. Doors shall be hinged type with sash lock for exposed application and non-hinged type with cam latches for concealed application.
 - 2. Access doors constructed in accordance with SMACNA HVAC Duct Construction Standard (Figure 7-2) or similar to Ruskin Model ADC or ADH will be acceptable.
 - 3. Sandwich style access doors made by Ductmate, Ward Industries, Greenheck, or Flexmaster are acceptable, provided that they meet insulation requirements.
- C. High Pressure Ducts (Pressure Class 3" and Over):
 - 1. Use access doors factory fabricated and rated by manufacturer's published literature for installation in systems with pressures to positive or negative 10" WG.
 - 2. Sandwich access doors made by Ductmate, Ward Industries, Pottorf, Greenheck, or Flexmaster are acceptable, provided that they meet insulation requirements.
- D. Kitchen Grease Exhaust Ducts:
 - 1. Access doors shall be grease and airtight, passing ASTM E2336 test standard, shall have ICC ESR. Access doors shall be furnished by duct wrap manufacturer as part of duct wrap system.

2.7 FLASHINGS

- A. Construct counterflashings of 16 ga galvanized Armco Zinc-Grip. Flashings are by General Contractor unless otherwise indicated.

2.8 DUCT FLEXIBLE CONNECTIONS

- A. Manufacturers: Unless specifically indicated, Ventfabrics, Inc., Ductmate or Duro Dyne. Material shall be glass fabric, fire retardant, waterproof, airtight and comply with NFPA 90A and 701 (formally UL 214).
- B. General Supply, Return and Exhaust Ductwork:
 - 1. Material for indoor use to be 30 oz per sq yd, double coated with neoprene, tensile strength (warp/fill) of 475/375 lbs, tear strength of 13x13 lbs, suitable for -40-200°F continuous operation.
 - 2. Material for outdoor use shall be combination of inner layer of Duro Dyne Neoprene or similar and outer layer of 24 oz per sq yd, coated with Hypalon, UV resistant, suitable for -40-200°F, similar to Duro Dyne Durolon.
- C. Special Exhaust Ductwork:
 - 1. Material for indoor use shall be 17 oz per sq yd, teflon or silicon coated, tensile strength of 200x250 lbs, tear strength of 50x40 lbs, suitable for -65-500°F, similar to Duro Dyne Thermafab.
 - 2. Material for outdoor use shall be combination of inner layer of Duro Dyne Thermafab and outer layer of Durolon.
- D. High Temperature Exhaust Fans:
 - 1. 17 oz per sq yd, teflon or silicon coated, suitable for 500°F, similar to Duro Dyne Thermafab.
 - 2. Swartwout multi-layer fabric, Type M, Style 2, suitable for low acid content air, up to 2000°F and 120" WG.
 - 3. General Rubber Corporation, Non-metallic expansion joint Style 1094, 1000°F continuous service, ±3 psi pressure rating, 1/2" thick laminated fiberglass insulation layer, laminated fluoroplastic PTFE corrosion barrier and integral flanged design.

2.9 INTAKE AND RELIEF HOODS

- A. Manufacturers: Carnes, Greenheck, Ammerman, Acme, Cook, Louvers and Dampers, Vent Products, Jenn-Air, or Penn
- B. Hoods shall be low silhouette type.
- C. Hoods shall be louvered penthouse type with stormproof louvers.
- D. Hoods shall be aluminum, all welded construction with mitered corner.
- E. Hoods shall be removable, cross broken, undercoated with insulating mastic and fabricated of aluminum alloy 3003-H14. Louver blades and extruded members shall be aluminum alloy 6063-T5.
- F. Finish shall be baked enamel; color selected by Architect. Submit color selection charts with shop drawings.

- G. Intake hoods shall be furnished with 1/2", 19 ga, galvanized steel bird screen and motor operated damper except those furnished by Control Contractor.
- H. Relief hoods shall be furnished with 1/2", 19 ga galvanized steel bird screen and motor operated damper except those furnished by Control Contractor.
- I. Hoods shall be furnished with factory fabricated curbs and extended bases, as required for mounting with inlet/outlet minimum of 24" above roof.
- J. Curbs shall be galvanized steel or extruded aluminum with continuous welded corner seams, treated wood nailer, minimum 1-1/2" thick 3 lb/ft³ density rigid mineral fiberboard insulation with metal liner.

2.10 BIRD AND INSECT SCREENS

- A. Bird screens shall be 1/2" square mesh formed with 0.063" diameter aluminum wire. Insect screen to be 18 x 16 with 0.017" diameter aluminum mesh.
- B. Frame shall be removable type of minimum 12 ga, extruded aluminum.

2.11 INSTRUMENT TEST HOLES

- A. Manufacturers: Ventlok 699 (up to 1" insulation thickness) or Ventlok 699-2 (over 1" insulation thickness).
- B. Use concave gaskets for round ductwork.

PART 3 - EXECUTION

3.1 MANUAL BALANCING DAMPERS

- A. Install manual balancing dampers in supply, return and exhaust branch ducts, as shown on drawings and as required to regulate airflow to meet air balance requirements.
- B. Install manual balancing damper in branch duct to each diffuser and grille. Install dampers as close as possible to take-offs.
- C. Install balancing dampers so as not to flutter or vibrate and as far as possible upstream from the air outlet.
- D. Do not install balancing dampers in supply ductwork upstream of air terminal devices.
- E. Balancing dampers are not required for return ductwork at outlet side of air terminal devices.

3.2 SPLITTER VANES AND TURNING VANES

- A. Install splitter vanes (SMACNA Type RE-3) as shown on drawings and as specified in Section 233114 - Ductwork, for rectangular radius elbows. Install splitter vanes in accordance with SMACNA Standards and/or manufacturer's recommendations.
- B. Turning vanes (SMACNA Type RE-2) are not allowed unless specifically indicated.

3.3 BACKDRAFT DAMPERS

- A. Install backdraft dampers where indicated on drawings.
- B. Where motorized dampers are shown in exhaust fan discharge duct, or in duct connecting to relief or exhaust louver, backdraft dampers are not required unless specifically indicated. Where motorized dampers are not shown, provide backdraft dampers in these locations.

3.4 FIRE DAMPERS AND COMBINATION FIRE/SMOKE DAMPERS

- A. Install dampers where shown on drawings in accordance with manufacturer's installation instructions and requirements of NFPA 90A. Install dampers complete with mounting collars, retaining angles, connections to adjoining ductwork and duct access doors. Install duct access door at each damper with door size large enough to permit replacement of fusible links and resetting of dampers.
- B. Test and demonstrate proper operation of each damper after system is installed and ready for operation.
 - 1. Manually test each damper for proper operation by removing fusible link or actuating EFL or PFL. Repair or replace any damper that does not close completely. Replace fusible link and certify in writing that each damper was installed according to manufacturer's installation instructions and that each damper can be expected to close completely when fusible link melts.
 - 2. Notify Owner and/or Owner's representative at least 48 h prior to testing to allow for witnessing.
- C. Contractor shall provide letter from manufacturer's representative indicating that dampers are installed per manufacturer's installation instructions.

3.5 ACCESS DOORS

- A. Install access doors where specified, indicated on drawings, and in locations where maintenance, service, cleaning or inspection is required, including automatic dampers, fire dampers, smoke dampers, smoke detectors, fan bearings, heating and cooling coils, reheat coils, humidifiers, filters, bird/insect screens, valves and control devices within duct or casing, at outside air intake duct and at inlet side of turning vanes in return ductwork.

- B. Locate access doors for greatest ease of access.
- C. Size and quantity of duct access doors shall be sufficient to perform intended service, but not less than the following:

Rectangular Duct Size	Minimum Access Door Quantity and Size
10" and smaller	(1) 8"x 8"
12" to 15" and smaller	(1) 10" x 10"
16" to 21"	(1) 14" x 14"
22" x 27"	(1) 18" x 18"
28" to 51"	(1) 24" x 24"
52" to 96"	(2) 24" x 24"
Round Duct Size	Minimum Access Door Size
10" and smaller	8" x 4"
15" and smaller	12" x 8"
29" and smaller	16" x 12"
30" and over	24" x 18"

- D. Increase duct size to accommodate access door size indicated above where required.
- E. Access doors for life safety dampers shall be adequate size for inspection and maintenance, but not less than 12" x 12". If adjacent duct or damper is smaller than 12" x 12", provide larger size duct or damper to accommodate minimum access door size without transition.

3.6 FLASHINGS

- A. Install counterflashings where shown on drawings. Install in accordance with SMACNA recommendations.

3.7 DUCT FLEXIBLE CONNECTIONS

- A. Connect ductwork to fans or casings containing rotating equipment or mounted on vibration isolators with duct flexible connections. Installed width shall be suitable for specific application but shall not be less than 4". Install flexible connections in accordance with SMACNA Standards with double lock or "Grip Loc" connection.
- B. Duct flexible connections are not allowed for fan connection serving kitchen hood, or perchloric acid hood.

3.8 INTAKE AND RELIEF HOODS

- A. Install hoods as shown on the drawings and/or as scheduled.
- B. General Contractor will install curbs furnished with hoods unless otherwise indicated.

3.9 BIRD AND INSECT SCREENS

- A. Provide bird screens or insect screens as indicated on drawings at louvers and at intake/exhaust openings.
- B. Unless otherwise indicated, provide bird screens where filters are specified and insect screens where filters are not specified.

3.10 CONTROL DAMPERS

- A. Install dampers in locations indicated on drawings, as detailed and according to manufacturer's instructions.
- B. Install blank-off plates or transitions as specified in Control Sections.
- C. Provide adequate operating clearance and access to operators.
- D. For dampers located outdoor, provide weather protection enclosure for parts of damper such as linkage and actuator located outside of duct. Enclosure shall be removable and made of material matching associated duct or housing.

3.11 SMOKE DAMPERS AND DETECTORS

- A. Install smoke dampers in locations indicated on drawings. Allow room for operator installation and access to operator. Install access door in ductwork for access to damper. When damper is installed in location other than in smoke or fire rated wall, use SMACNA seal class A for ductwork between smoke damper and smoke rated wall for affected zone.
- B. Smoke detectors will be provided by Electrical Contractor in locations indicated on drawings. Install access door in ductwork for access to detector sampling device.

3.12 INSTRUMENT TEST HOLES

- A. Provide instrument test holes at air entering and air leaving side of all internal air handling system components for static pressure differential (Delta P) or temperature differential (Delta T) measurements.
- B. Provide ductwork instrument test holes as shown on drawings, or as directed by TAB personnel, or Engineer.

END OF SECTION

SECTION 233400
FANS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 262816 - Enclosed Switches and Circuit Breakers

1.2 SUBMITTALS

- A. Shop Drawings including the following:
1. Manufacturer's name and model number
 2. Identification as referenced in the documents
 3. Capacities/ratings
 4. Fan curves
 5. Materials of construction
 6. Sound power levels
 7. Fan type, size, class, drive arrangement, discharge/rotation, bearings, drives
 8. Wheel type, diameter, rpm, tip speed
 9. Required fan hp including drive losses
 10. Vibration isolators furnished with fans
 11. Dimensions and weights
 12. Special coatings where applicable
 - a. Product descriptive data
 - b. Complete chemical resistance charts
 - c. Coating manufacturer's base metal preparation requirements (cleaning/pretreatment)
 - d. Facility where special coatings will be applied
 - e. Recommended method of repairing damaged coating
 13. Color selection charts where applicable
 14. Manufacturer's installation instructions
 15. All other appropriate data
- B. Fan curves shall include series of curves indicating relationship of flow rate (cfm) to static or total pressure for various fan speeds, brake hp (kW) curves, and selection range (surge curves, maximum rpm, etc.).
- C. Indicate performance data, based on both design air quantity and 110% of design air quantity.
- D. For variable air volume application, indicate operating points at 100, 80, 60 and 40% of design capacity on fan curves including data to indicate effect of variable frequency drives on flow, pressure and hp.

- E. Complete equipment data sheet attached at end of this Section for each piece of equipment and submit with shop drawings. Shop Drawings will be returned without review if data sheets are not provided for each piece of equipment and if data sheet is not filled out completely.

1.3 DESIGN CRITERIA

- A. Fan ratings shall be tested and certified in accordance with AMCA Standards 211 and 311 and fans shall bear AMCA Seal.
- B. Fans shall be furnished complete with motors, wheels, drive assemblies, bearings and accessories as hereinafter specified. Motors for V-belt drives shall be furnished with adjustable rails or bases.
- C. Where backward Inclined or backward curved (BI) centrifugal fans are indicated, fans shall have solid (single-thickness) blades.
- D. Each fan wheel shall be statically and dynamically balanced to Balance Quality Grade G6.3 per ANSI S2.19 and AMCA 204-05. Complete fan assembly shall be factory balanced statically and dynamically in accordance with Standard AMCA 204-05 for Balance Quality and Vibration Levels for Fans and meet or exceed guidelines in Application Category BV-3.
- E. For fans furnished with 5 hp or larger hp (kW) motors, each fan assembly shall have factory run test including vibration signatures taken on each bearing in horizontal, vertical and axial direction. Filter-in reading as measured at fan's scheduled rpm shall not exceed the following values when fan is rigidly mounted.
 - 1. 0.15 in/sec peak velocity
 - 2. 0.08 in/sec peak velocity
 - 3. 0.08 in/sec peak velocity
 - 4. Written records of run test and vibration test shall be available upon request.
- F. Unless otherwise indicated, vibration level of installed fan assembly flexibly mounted shall not exceed 0.35 in/sec.
- G. Unless otherwise indicated, belt drives shall be V-belt drives selected for 150% of motor nameplate hp.
- H. Furnish fans specified with V-belt drives with either variable-pitch or fixed-pitch sheaves for drives 3 hp and smaller and fixed-pitch sheaves for drives 5 hp and larger. Select variable pitch sheaves to drive fan at such speed as to produce specified capacity at approximate midpoint of sheave adjustment.

- I. When fixed-pitch sheaves are furnished, system air balancing shall be accomplished by either trial of different fixed-pitch sheaves or use of temporary adjustable-pitch sheaves. This Contractor shall provide necessary trial and final sheaves and drive belts as required by TAB Contractor.
- J. Select each fan to operate at single stable operating point as predicted by fan curve. Fans having 2 potential operating points on fan curves are not acceptable.
- K. Provide OSHA Compliant belt and shaft guards for belt driven fans. Provide speed test openings at shaft locations. Paint guards bright yellow.
- L. Sound power levels shall be based on tests performed in accordance with AMCA Standards 300 and 301.
- M. Each fan and motor combination shall be capable of delivering 110% of air quantity scheduled at scheduled static pressure. Motor furnished with fan shall not operate into motor service factor in any of these cases.
- N. Consider drive efficiency in motor selection according to manufacturer's published recommendation, or according to AMCA Publication 203, Appendix L.
- O. Where inlet and outlet ductwork at any fan is changed from that shown on drawings, submit scaled layout of change and system effect factor calculations indicating increased static pressure requirements as described in AMCA 201. This Contractor shall be responsible for costs associated with any motor, drive, and/or wiring changes required as a result of duct configuration changes at fan.

PART 2 - PRODUCTS

2.1 CENTRIFUGAL FANS GENERAL APPLICATION

- A. Manufacturers: Twin City, Greenheck, Cook, Chicago Blower, or PennBarry
- B. Housings: Heavy gauge steel, continuously welded throughout, braced and supported by structural channels or angle irons to prevent vibration or pulsation, flanged outlet, fully streamlined inlet.
- C. Wheels: Non-overloading airfoil blades welded to spun wheel cones unless otherwise indicated.
- D. Shaft: Fan shaft shall be turned and polished steel and sized so first critical speed is at least 25% over maximum operating speed for each pressure class.

- E. Bearings: Air handling quality, heavy duty, grease lubricated, ball or roller, self-aligning, pillow block type. Bearings shall be selected for minimum life (ABMA L-10) of not less than 80,000 h (equivalent to L-50 average life of 400,000 h) at maximum cataloged operating speed based on ABMA STD 9 and ABMA STD 11. Furnish bearings with pressure relief type external grease fittings. Bearings shall be fixed to the fan shaft using concentric mounting locking collars, which reduce vibration, increase service life, and improve serviceability. Bearings that use set screws are not allowed.
- F. Painting: All metal parts shall be painted with prime coat after metal cleaning and surface preparation. In addition, apply second coat of paint to all exterior surfaces.
- G. Mounting Rails: Furnish common mounting rails for fan and motor as appropriate for fan arrangement indicated.
- H. Discharge Dampers: Provide gravity backdraft dampers, for fans not provided with motorized dampers, for mounting at fan discharge outlet. Backdraft dampers shall be similar to Ruskin Model CBS 92, having aluminum airfoil-shaped blades, ball bearings and be rated to 4000 fpm. Dampers shall have blade and jamb seals with leakage rating no higher than 13, 5 cfm/ft² for damper width 36" and larger based on pressure differential of 1 WG.
- I. Inlet Screens: Minimum 16 ga galvanized, 1" wire mesh for fans without inlet ductwork.
- J. Access Doors: Bolted and gasketed type in fan housing for inspection of interior and wheel.
- K. Drain Connection: 3/4" NPT external threads located at the lowest point of housing.
- L. Air Flow Meters:
 - 1. Provide factory mounted piezometer ring at inlet cone of each fan. Piezometer ring shall be completely outside of airstream and shall impose no airflow restriction.
 - 2. Terminate pressure tubing at exterior of fan housing.
 - 3. Display and Transmitter:
 - a. Provide electronics package, for remote mounting, for each fan flow meter consisting of digital display of volumetric flow and BACNet interface or analog outputs (4-20 mA, 0-10 VDC, etc.) for use by Controls Contractor. Refer to Instrumentation and Control Diagrams or System Flow Diagrams for requirements. Volumetric flow shall be calculated with empirically derived formulas based on testing by fan manufacturer. Provide necessary information to Controls Contractor for interpreting output signals or scaling output signal(s) to volumetric air flow.
 - b. Unit shall operate with 120 VAC power.

2.2 UTILITY SETS

- A. Manufacturers: Twin City, Greenheck, Cook, Chicago Blower, PennBarry, or ACME

- B. Housings: Heavy gauge steel with welded or lockseam fabrication and adequate reinforcing to prevent housing distortion. Furnish access doors.
- C. Bearings: Grease lubricated, self-aligning ball bearing type with grease seal. Bearing shall be selected for a minimum L 10 life in excess of 40,000 h (equivalent to L-50 average life of 200,000 h), at maximum cataloged operating speed.
- D. Painting: All fan parts shall be painted with prime coat after metal cleaning and surface preparation. In addition, apply second coat of paint to all exterior surfaces.
- E. Motor Supports: Motor supports shall be designed to adequately hold motor in place and provide adjustment on V-belt drive models.
- F. Backdraft Dampers: Provide gravity backdraft dampers, for fans not provided with motorized dampers, for mounting in discharge ductwork.
- G. Inlet Screens: 1" mesh for fans without inlet ductwork.
- H. Drain Connection: Drain connection to be located at the lowest point of fan housing.
- I. Weather Covers: Where fans are indicated for outdoor installation, they shall be equipped with factory fabricated weather covers and weather-resistant enamel finish.

2.3 POWER ROOF EXHAUSTERS (UPBLAST TYPE)

- A. Manufacturers: Twin City, Greenheck, PennBarry, Cook, Jenn-Air, ACME, or Carnes
- B. Construct housing of heavy gauge seamless spun aluminum.
- C. Housing shall be finished with baked enamel; color selected by Architect.
- D. Fan wheels shall be centrifugal, solid (single-thickness) backward inclined (BI), non-overloading type, and be spark proof aluminum construction.
- E. Motor, bearings and drives to be completely enclosed and sealed from exhaust air stream and be positively ventilated with outside air by forced air cooling tubes.
- F. Entire drive assembly or entire unit to be mounted on factory installed vibration isolators.
- G. Units shall be furnished with factory mounted and wired NEMA weatherproof disconnect switches.
- H. Units shall be furnished with hinged access arrangement, hold-open retainer chain, and flexible weatherproof electric cable.
- I. Units shall be furnished with minimum 18" high factory fabricated curb.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install units as shown on drawings, and according to manufacturer's installation instructions. On units provided with drain connection, install drain valve and cap discharge of drain.
- B. Verify lubrication of motor and fan bearings and lubricate properly in accordance with manufacturer's recommendation and Section 230000 - General Mechanical Requirements, Part 3 under LUBRICATION.
- C. General Contractor will install curbs furnished with fans unless otherwise indicated.
- D. Perform field mechanical balancing, if necessary, to meet vibration tolerance specified in Part 1 of this Section.

END OF SECTION

**SECTION 233713
DIFFUSERS, REGISTERS AND GRILLES**

PART 1 - GENERAL

1.1 SUBMITTALS

- A. Shop Drawings including the following:
 - 1. Manufacturer's name and model number
 - 2. Identification as referenced in the Documents
 - 3. Capacities/ratings
 - 4. Materials of construction
 - 5. Sound ratings
 - 6. Dimensions
 - 7. Finish
 - 8. Color selection charts where applicable
 - 9. Manufacturer's installation instructions
 - 10. All other appropriate data

1.2 DESIGN CRITERIA

- A. Performance data shall be based on tests conducted in accordance with ASHRAE Std 70-2006.
- B. Screw holes on surface shall be counter sunk to accept recessed type screws.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Titus, Price, Carnes, Nailor, Anemostat, Greenheck (formerly Metalaire), or Krueger
- B. Acceptable manufacturers for specialty products are listed under each item.

2.2 CEILING DIFFUSERS

- A. Diffusers shall be aluminum or steel as scheduled, unless otherwise indicated, and furnished with frame type appropriate to installation. Furnish diffusers with equalizing grids where it is not possible to maintain minimum 2 times duct diameter straight duct into diffuser. Equalizing grids shall consist of individually adjustable vanes designed for equalizing airflow into diffuser neck and providing directional control of airflow.
- B. Diffuser models, sizes and finishes shall be as shown on drawings and/or as scheduled. Unless otherwise noted, diffusers shall have baked enamel or powder coat finish with color selected by Architect.

- C. Perforated face ceiling diffusers shall have minimum 51% free face area and pattern controllers accessible through removable or hinged faceplate. Unless otherwise indicated, pattern controllers shall be curved vane type mounted in neck of diffuser. Unless otherwise indicated, furnish diffusers with round neck inlets with minimum 1" depth.

2.3 REGISTERS AND GRILLES

- A. Registers and grilles shall be aluminum or steel as scheduled unless otherwise indicated, and furnished with frame type appropriate to installation.
- B. Supply registers shall have double deflection type blades to provide for air deflection adjustment in all directions. Registers shall be furnished complete with opposed blade volume control dampers, operable from face.
- C. Return and exhaust registers and grilles shall have fixed blade core.
- D. Register and grille models, sizes and finishes shall be as shown on drawings and/or as scheduled. Unless otherwise noted, registers and grilles shall have baked enamel finish with color selected by Architect.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install grilles, registers and diffusers as shown on drawings and according to manufacturer's instructions.
- B. Unless otherwise indicated, size ductwork drops to diffusers or grilles to match unit collar sizes.
- C. Seal connections between ductwork drops and diffusers/registers/grilles airtight.
- D. Support independently diffusers and grilles designed for T-bar mounting that exceed weight limit of ceiling suspension system in which they are to be installed.
- E. Blank off unused portion of linear diffusers and grilles.
- F. Where diffusers, registers and grilles cannot be installed to avoid seeing inside duct, paint inside of duct with flat black paint to reduce visibility.
- G. Protect diffusers, registers and grilles from construction dirt. Clean or replace those soiled or stained prior to turnover to Client.

END OF SECTION

**SECTION 237400
PACKAGED ROOFTOP UNITS**

PART 1 - GENERAL

1.1 SUBMITTALS

- A. Shop Drawings including, but not limited to, the following:
 - 1. Manufacturer's name and model number
 - 2. Identification as referenced in the documents
 - 3. Capacities/ratings
 - 4. Materials of construction
 - 5. Filters; size, efficiency
 - 6. Fans; type, drives
 - 7. Power and control wiring diagrams
 - 8. Vibration isolation furnished with units
 - 9. Dimensions and weights
 - 10. Manufacturer's installation instructions
 - 11. Other appropriate data
- B. Fan curves shall include a series of curves indicating relationship of airflow cfm and static pressure for various rpm and fan brake power curves. Indicate design operating point clearly on fan curves.

1.2 DESIGN CRITERIA

- A. Design units specifically for outdoor rooftop application.
- B. Units to be completely factory assembled and run tested, piped, internally wired, fully charged with refrigerant and compressor oil when specified, and shipped in one piece.
- C. Units shall be factory run tested to check cooling operation, fan and blower rotation and control sequence.
- D. Furnish units with components, accessories, and operating and safety controls to provide the intended performance as specified in this section, as shown on Drawings and/or as scheduled.
- E. Insulation and liners to meet NFPA 90A requirements.
- F. Unit shall be UL Listed and carry UL Label.
- G. V-belt drives shall be designed for 150% of motor rating.
- H. Cooling and/or heating coils shall be rated in accordance with AHRI 410.

- I. Heating system on gas-fired units shall be certified by AGA.
- J. Each fan and motor combination shall be capable of meeting both of the following conditions while maintaining stable fan performance:
 - 1. Deliver 110% of air quantity scheduled at scheduled static pressure.
 - 2. Unit static pressure shall take into consideration actual static pressure loss of components furnished within unit.
- K. Fan motor shall not operate into motor service factor in any case. Drive efficiency shall be considered in motor selection according to manufacturer's published recommendation, or according to AMCA Publication 203, Appendix L.
- L. Where inlet and outlet ductwork at any fan is changed from that shown on drawings, submit scaled layout of the change and system effect factor calculations, indicating increased static pressure requirement as described in AMCA Publication 201. Contractor shall be responsible for any motor, drive and/or wiring changes required as result of duct configuration changes at fan.

1.3 TEST REPORTS

- A. Manufacturer or manufacturer's representative shall field check, test and start units after they have been properly installed. Component systems to be run and adjusted to perform quietly and efficiently at capacities scheduled.
- B. Manufacturer or its representative shall provide services to field instruct and check unit operation and control of complete system. Instruction shall not take place until HVAC system has been field checked, tested, started and balanced. Manufacturer shall establish instruction dates, and give Engineer and Owner 10 days' written notice of the time. Instruction shall be deemed completed when affidavit of instruction has been signed by representative of manufacturer and Owner.

1.4 OPERATION AND MAINTENANCE DATA

- A. Refer to Section 230000 - General Mechanical Requirements.

1.5 GUARANTEE

- A. Manufacturer and Contractor shall provide warranty on rooftop units for a period of 1 yr after acceptance by Owner.
- B. Warranty shall include cost of parts and labor for work necessary to repair malfunctioning unit or piece of equipment furnished by unit manufacturer. Warranty need not cover normal maintenance such as changing of filters, cleaning of coils, replacement of belts and lubrication of bearings.
- C. Provide additional 4 yr warranty for refrigeration compressors.

- D. Provide additional 4 yr warranty for stainless steel heat exchangers.

PART 2 - PRODUCTS

2.1 ROOFTOP UNITS (5 TONS AND SMALLER)

- A. Manufacturers:
1. Equal to manufacturer's model with capacity and operating characteristics as scheduled
- B. Casings:
1. Weatherproof and constructed of heavy gauge galvanized steel phosphatized with baked enamel finish. Insulate interior surfaces of cabinet with minimum of 1/2" glass fiber insulation.
 2. Access doors shall be heavy gauge galvanized or phosphatized zinc coated steel complete with hinges or fasteners to secure door. Access doors and removable panels shall have neoprene gaskets.
 3. Cabinet base shall be constructed so as to mate and seal with roof mounting curb. Furnish sealing gasket for positive, waterproof seal when unit is installed on curb.
- C. Supply Air Fans:
1. Double width, double inlet centrifugal type with adjustable V-belt drives, statically and dynamically balanced and tested in factory.
 2. Entire fan and drive assemblies to be mounted on common base and be isolated from unit by factory mounted vibration isolators.
- D. Return-Exhaust Air Section:
1. Furnish exhaust hood with rain gutter to protect exhaust air openings.
 2. Provide baffle in return air stream entering filter section to balance pressure drop between return air and outside air entering supply fan.
- E. Filters:
1. Filter racks shall be integral part within unit and easily accessible from both sides of unit via hinged access doors.
 2. Furnish units with 1" glass fiber throwaway filters in commercially available sizes with low velocity V bank type filter casing.
 3. In addition, furnish a set of 1" glass fiber throwaway filters to be used during construction and initial start-up periods.
- F. Electric Heating Section:
1. Furnish unit with factory assembled and wired electric resistance type heating coils. Heating element shall be heavy-duty nickel chromium. Provide distribution baffle to ensure even air distribution across coil.
 2. Heater circuit current shall be 48 amps or less, each individually fused in accordance with National Electrical Code. Capacity control shall use SCR control for electric heating.
 3. Provide main disconnect switch in main control panel for electric heating.

4. Provide safety and operating controls including manual reset thermal cutouts on each bank of electric heating element and flow switch.
- G. Heat Pump Section:
1. Provide reversing type heat pump system with capacities indicated on schedules.
 2. Provide outdoor coil defrost system to remove frost accumulation during heating cycle.
 3. Defrost system shall activate reverse cycle defrost time and temperature initiated, temperature termination.
 4. Defrost system shall have integral time override to limit defrost cycle to maximum of 5 minutes.
- H. Cooling System:
1. Direct Expansion Cooling:
 - a. Factory charged with refrigerant and fully tested. System shall include refrigeration compressor(s), evaporator coil, air cooled condenser assembly, thermal expansion valves, liquid line valves, sight glass, filter drier, liquid and suction line service valves, and insulated interconnecting refrigerant piping.
 - b. Evaporator coils to be of non-ferrous construction with 5/8" OD seamless copper tubes mechanically bonded to configured aluminum fins.
 - c. Provide drain pan fabricated from zinc-coated steel with additional coat of insulating sealer.
 - d. Compressor shall be hermetic type with spring isolators.
 - e. Provide compressor with internal thermal overloads, crankcase sight glass and crankcase heater. Provide capacitor start kit for single phase units.
 - f. Condenser coils shall be seamless copper tubing mechanically bonded to heavy duty configured aluminum fins. Condenser coils of aluminum tubes and aluminum fins constructed with Microchannel coil technology will be acceptable.
 - g. Refrigeration safety controls shall include high and low pressure cutout, oil pressure cutout, compressor overload protections, magnetic contactors and low voltage control transformer.
 - h. Provide system with necessary controls to maintain head pressure at ambient to 40°F.
 - i. System shall be fully charged with refrigerant as scheduled on Drawings, ready for operation.
- I. Condenser Fan:
1. Condenser fans to shall be direct drive propeller type with exposed fan and shaft surfaces suitably weatherproofed. Fan motors to shall be furnished with normal and current overload protection and permanently lubricated motor bearings.
- J. Electrical:
1. Furnish each unit with factory mounted disconnect device with short circuit and over current protection, motor starter and contactor.
 2. Each unit shall include low voltage control transformer.
 3. Motor power and voltages shall be as scheduled.

K. Unit Controls:

1. Contractor shall provide temperature control and interlocking devices necessary to perform the specified control sequence.
2. Provide low voltage space thermostat to control stages of heating, cooling, automatic changeover and fan control. Provide matching heat-cool/fan on-off-auto sub-base.

2.2 ROOFTOP UNITS (7-1/2 THROUGH 17-1/2 TONS)

A. Manufacturers:

1. Unit shall be manufacturer's model with capacity and operating characteristics as scheduled.

B. Casings:

1. Casings to be completely weatherproof and constructed of heavy gauge galvanized with baked enamel finish. Insulate interior surfaces of cabinet with minimum of 1" foil-faced glass fiber insulation.
2. Access doors shall be heavy gauge galvanized steel complete with hinges or fasteners to secure door. Access doors and removable panels to have neoprene gaskets.
3. Cabinet base to be constructed so as to mate and seal with the roof mounting curb. Provide sealing gasket for positive, waterproof seal when unit is installed on curb.

C. Direct Expansion Cooling:

1. Factory charged with refrigerant and fully tested. System to include refrigeration compressors, evaporator coil, air cooled condenser assembly, thermal expansion valves, liquid line solenoid valves, sight glass, filter drier, liquid and suction line service valves, and insulated interconnecting refrigerant piping. Unit shall be capable of operation down to 0°F ambient temperature.
2. Evaporator coils shall be non-ferrous construction with seamless copper tubes mechanically bonded to configured aluminum fins. Furnish adequate refrigeration circuits, each with thermostatic expansion valve and solenoid valve for proper system part load operation.
3. Furnish unit with fully insulated drain pan fabricated from zinc-coated steel with an additional coat of insulating sealer. Extend drain pan into fan section.
4. Compressor shall be hermetic scroll, designed to operate at 1750 rpm.
5. Furnish compressor lubrication system of forced feed type with positive displacement reversible oil pump, filter, oil pump magnets, crankcase site glass and crankcase heater.
6. Condenser coils shall be seamless copper tubing mechanically bonded to heavy-duty configured aluminum fins. Condenser coils of aluminum tubes and aluminum fins constructed with Microchannel coil technology will be acceptable.
7. Refrigeration safety controls shall include high and low pressure cutout, oil pressure cutout, non-recycling pumpdown; manual lockout reset relay, compressor overload protections, magnetic contractors and low voltage control transformer.
8. Furnish each unit with fully automatic compressor cylinder unloading capacity control system.

9. Provide system with necessary controls to maintain head pressure to 40°F ambient.
 10. For multizone unit, design of refrigeration system to be such that during mechanical cooling cycle, return and outside air will pass through evaporator coil and be fully dehumidified and reheated where necessary before being delivered to conditioned space.
 11. System shall be fully charged with refrigerant as scheduled on Drawings, ready for operation.
- D. Condenser Fan:
1. Condenser fans shall be propeller type with exposed fan and shaft surfaces suitably weatherproofed. Fan motors shall be furnished with disconnect and starter and permanently lubricated motor bearings.
- E. Supply Air Fan:
1. Double width, double inlet forward curved centrifugal type with adjustable motor sheaves, statically and dynamically balanced and factory tested.
 2. Entire fan and drive assemblies shall be mounted on a common base and isolated from unit by factory mounted vibration isolators.
 3. Mount fan wheels on solid steel shafts having self-aligning 200,000 h average life permanently lubricated bearings.
- F. Return-Exhaust Air Section:
1. Double-inlet, forward-curved fans mounted on common shaft with fixed sheave drive. Fans shall be dynamically balanced and tested in factory. Exhaust fan test shall be run in unit as part of unit test. Unit shall reach rated rpm before fan shaft passes through first critical speed. Fan shaft shall be mounted on two grease lubricated ball bearings designed for 20,000-h average life. Provide extended grease lines to allow greasing of ball bearings from unit filter section.
 2. Fan and motor assembly shall be mounted on common base to allow consistent belt tension with no relative motion between fan and motor shafts.
 3. Entire assembly isolated from unit and fan board by spring isolators. Provide discharge dampers at unit outlet to modulate exhaust airflow in response to building pressure.
 4. Furnish exhaust hood with rain gutter to protect exhaust air openings.
 5. Provide baffle in return air stream entering filter section to balance pressure drop between return air and outside air entering supply fan.
- G. Electric Heaters:
1. Furnish units with factory assembled and wired electric resistance type heating coil speed to eliminate temperature stratification.
 2. Provide distribution baffle to ensure even airflow across coil.
 3. Heating element shall be constructed of heavy-duty nickel chromium.
 4. Heater circuit currents shall be maximum 48 amps or less, each individually fused in accordance with National Electrical Code.
 5. Heaters to have branch circuit protection in accordance with National Electrical Code. Capacity control to use SCR control for control.
 6. Provide safety and operating controls including manual reset thermal cutouts on each bank of electric heating element and flow switch.

7. Provide main disconnect switch in main control panel for electric heating of scheduled voltage.

H. Filters:

1. Filter racks shall be integral within unit and accessible via hinged access doors.
2. Filters shall be Farr 30/30, Cambridge Series APII or equal.
3. Media shall be non-woven, fine fibered material laminated to rigid backing to hold pleat formation, having average efficiency of 25 to 30% on ASHRAE Standard 52-76.
4. Filter housing shall consist of rooftop unit manufacturers low velocity filter section, or holding frame, as scheduled.
5. Filters to be 2" thick of size and capacity scheduled. Clean pressure drop shall not exceed 0.38" WG based on 500 fpm (2.54 m/s) face velocity.
6. In addition, furnish a set of 2" glass fiber throwaway filters to be used during construction and initial startup periods.

I. Electrical:

1. Furnish each unit with factory mounted disconnect device with short circuit and over current protection, motor starter, and contactor.
2. Each unit to include low voltage control transformer and convenience outlet.
3. Motor power hp and voltages shall be as scheduled.

J. Controls:

1. Space thermostat shall control staging of heating and cooling and shall provide automatic changeover and fan control. Provide appropriate thermostat sub-base.

K. Economizer Controls:

1. Provide dry-bulb type economizer control. Provide devices required for control including: fresh air damper, damper drive motor, relief exhaust fans, switchover thermostat, minimum OA damper position potentiometer, etc.

2.3 ROOF MOUNTING CURBS:

- A. Construct roof mounting curb to mate and seal with unit casing base. Curb to have minimum height of 12".
- B. Construct curb to support perimeter of entire unit. Form cross section of curb members to accept wood nailing strips and insulation. Form top curb members to provide counter flashing.

- C. Curbs Serving Units 7-1/2 tons and larger:
1. Curb mounted roof top equipment shall be flexibly ducted. Units shall be supported by spring isolation curb, lower member of which is rigid steel tube or specially formed steel section containing adjustable and removable steel springs that support upper floating section. Upper frame must provide continuous support for equipment and must remain captive when resiliently resisting wind and seismic forces. Directional neoprene snubber bushings must be minimum of 1/4" thick. Steel springs shall rest on 1/4" neoprene acoustical pads. Hardware must be cadmium or zinc electroplated and springs similarly plated or provided with approved rust resistant finish.
 2. Curb waterproofing shall consist of continuous galvanized flexible counterflashing nailed over lower curb's waterproofing and joined at corners by EPDM bellows. Spring locations shall have access ports with removable waterproof covers. Lower curbs shall have provision for 2" insulation.
 3. Floating member of roof curb shall have perimeter angle and cross members to support 2 layers of 5/8" waterproof sheetrock laid on with staggered joints. Sheetrock must surround ducts to provide continuous sound break. This acoustical barrier shall be caulked to minimize sound transmission. Where mechanical arrangement makes attachment to the floating member unfeasible, barrier shall be attached at highest practical elevation of fixed curb with provision for 1" thick closed cell neoprene flexible seals around the ductwork. 4" layer of 1.5 lb/cubic foot density fiberglass shall cover entire solid roof surface under the unit. Complete instructions shall be provided by spring isolation curb manufacturer.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install units as indicated on Drawings and according to unit manufacturer's instructions.
- B. Provide roof mounting curbs and set and install units on curbs.
- C. Install ductwork, electrical and piping connections to unit as shown on Drawings for complete installation.
- D. Coordinate with controls contractor for wiring between rooftop units, remote panels, time clocks and other temperature control devices.
- E. Manufacturer and/or manufacturer's representative shall calibrate control devices and adjust unit automatic dampers to ensure proper operation of system.
- F. Install space thermostats approximately 5 ft-0" above floor and as shown on drawings.
- G. Install remote control panels as indicated on Drawings.

END OF SECTION

SECTION 238144 HEAT PUMPS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 230513 - Motors for Mechanical Equipment
- B. Section 230550 - Vibration Isolation
- C. Section 230901A - Control Systems
- D. Section 230993 - Control Sequences

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 SUBMITTALS

- A. Shop Drawings including the following:
 - 1. Manufacturer's name and model number
 - 2. Identification as referenced in the documents
 - 3. Capacities/ratings/efficiencies
 - 4. Materials of construction
 - 5. Filters; size, efficiency
 - 6. Fans, type, drives
 - 7. Motor data (refer to Section 230513 - Motors for Mechanical Equipment)
 - 8. Power and control wiring diagrams
 - 9. Vibration isolation furnished with units
 - 10. Dimensions and weights
 - 11. Manufacturer's installation instructions
 - 12. All other appropriate data
- B. Fan curves to include a series of curves indicating relationship of cfm and static pressure for various RPM and brake hp (kW) curves. Indicate design-operating point clearly on fan curves.

1.4 DESIGN CRITERIA

- A. Design unit specifically for outdoor, horizontal airflow, rooftop application.
- B. Unit shall be completely factory assembled and run tested, piped, internally wired, fully charged with refrigerant and compressor oil when specified, and shipped in one piece.

- C. Unit shall be factory run tested to check heating and cooling operation, fan and blower rotation and control sequence.
- D. Furnish unit complete with all components, accessories, and operating and safety controls to provide the intended performance as specified in this Section, as shown on drawings and/or as scheduled.
- E. Insulation and liners shall meet NFPA 90A Requirements.
- F. Unit shall be UL Listed and carry UL Label.
- G. Coils shall be rated in accordance with AHRI Standards.
- H. Each fan and motor combination shall be capable of meeting both of the following conditions while maintaining stable fan performance:
 - 1. Deliver 110% of air quantity scheduled at scheduled static pressure.
 - 2. Unit static pressure shall take into consideration actual static pressure loss of components furnished within unit.
- I. Fan motor shall not operate into motor service factor in any of 3 cases. Drive efficiency shall be considered in motor selection according to manufacturer's published recommendation, or according to AMCA Publication 203, Appendix L.
- J. Where inlet and outlet ductwork at any fan is changed from that shown on drawings, submit scaled layout of the change and system effect factor calculations, indicating increased static pressure requirement as described in AMCA Publication 201. This Contractor shall be responsible for any motor, drive and/or wiring changes required as result of duct configuration changes at fan.

1.5 TEST REPORTS

- A. Manufacturer or manufacturer's representative shall field check, test and start all units after they have been properly installed. Component systems to be run and adjusted to perform quietly and efficiently at capacities scheduled.
- B. Manufacturer or his representative shall provide services to field instruct and check unit operation and control of complete system. Instruction shall not take place until HVAC system has been field checked, tested, started and balanced. Manufacturer shall establish instruction dates, and give Architect and Owner 10 days written notice of the time. Instructions shall be deemed completed when affidavit of instruction has been signed by representative of manufacturer and Owner.

1.6 OPERATION AND MAINTENANCE DATA

- A. Manuals shall indicate unit model numbers, serial numbers, sizes, type, and indicate name and address of service company.

- B. Description of maintenance required weekly, monthly, quarterly, semi-annually and annually shall be included along with recommended lubrication type.

1.7 GUARANTEE

- A. Manufacturer and Contractor shall provide warranty on heat pump for a period of 1 yr after acceptance by Owner.
- B. Warranty shall include all cost of parts and labor for any and all work unit manufacturer and/or contractor has to accomplish to repair malfunctioning unit or piece of equipment furnished by unit manufacturer. Warranty shall not cover normal maintenance such as changing of filters, cleaning of coils, replacement of belts and lubrication of bearings.
- C. Manufacturer shall provide additional 4 yr warranty for refrigeration compressors.

PART 2 - PRODUCTS

2.1 HEAT PUMPS

- A. Manufacturer:
 - 1. Trane, York, FHP or Carrier
 - 2. Based on Trane Voyager, Model No. WCH090C3
 - 3. Equal to manufacturer's model with capacity and operating characteristics as scheduled.
- B. Casings:
 - 1. Completely weatherproof and constructed of heavy gauge galvanized steel phosphatized with baked enamel finish. Insulate interior surfaces of cabinet with minimum of 1/2" glass fiber insulation.
 - 2. Access doors to be heavy gauge galvanized or phosphatized zinc-coated steel complete with hinges or fasteners to secure door. Access doors and removable panels to have neoprene gaskets.
 - 3. Rooftop unit shall have an engraved metal nameplate permanently attached to the housing. Nameplates shall provide at least the following information: manufacturer, model number, rated capacity, and rated efficiency.
 - 4. Base of the unit shall have provisions for crane lifting.
- C. Supply Air Fans:
 - 1. Statically and dynamically balanced and tested in factory.
 - 2. Entire fan and drive assemblies shall be mounted on common base and be completely isolated from unit by factory mounted vibration isolators.
- D. Economizer Air Section:
 - 1. Provide economizer hood with rain gutter to protect exhaust and supply air openings.
 - 2. Section shall be capable of introducing 0 to 100% outdoor air.
 - 3. Provide all hardware and controls to provide cooling with outdoor air.

- E. Filters:
 - 1. Filters shall be integral to unit and to accessible.
 - 2. Filters shall be 2" fiberglass throwaway type.
 - 3. In addition, furnish one set of 1" glass fiber throwaway filters to be used during construction and initial start-up periods.
- F. Refrigerant Circuits:
 - 1. System shall include refrigeration compressors, evaporator coil, air cooled condenser assembly, thermal expansion valves, liquid line valves, sight glass, filter drier, liquid and suction line service valves, and insulated interconnecting refrigerant piping. Provide all other components required to modulate between heating and cooling modes.
- G. Evaporator and Condenser Coils:
 - 1. Direct expansion type, factory charged with R-410A refrigerant and fully tested.
 - 2. Evaporator coils to be of non-ferrous construction with 3/8" OD seamless copper tubes mechanically bonded to configured aluminum fins.
 - 3. Provide fully insulated drain pan fabricated from zinc-coated steel with additional coat of insulating sealer.
 - 4. Condenser coils shall be 3/8" OD seamless copper tubing mechanically bonded to heavy-duty configured aluminum fins. Condenser fans to be direct drive type with exposed fan and shaft surfaces suitably weatherproofed. Fan motors to be furnished with normal and current overload protection and permanently lubricated motor bearings.
 - 5. Coils shall be leak tested at factory to ensure pressure integrity.
- H. Compressor:
 - 1. Compressor shall be hermetic type with spring isolators.
 - 2. Provide compressor with internal thermal overloads, crankcase sight glass and crankcase heater.
 - 3. Provide capacitor start kit for single-phase units.
- I. Electrical:
 - 1. Furnish each unit with short circuit and over current protection motor starter, and contactors.
 - 2. Each unit shall include low voltage control transformer.
 - 3. Motor hp and voltages shall be as scheduled.
- J. Unit Controls:
 - 1. Unit shall be completely factory wired for 24V controls.
 - 2. Microprocessor based unit controls shall contain all required algorithms to meet requirements of Mechanical Drawing Controls Sequences.
 - 3. Provide heating/cooling/deadband thermostat.
 - 4. Provide dry-bulb economizer control. Provide all devices required for control including: fresh air damper, damper drive motor, barometric relief damper with hood, minimum OA damper position potentiometer, etc.
 - 5. Refrigeration safety controls to include high and low pressure cutout, oil pressure cutout, compressor overload protections, magnetic contractors and low voltage control transformer.

6. Provide system with necessary controls to maintain head pressure at ambient to 0°F.
7. Provide defrost controls for use during heating cycle.
8. Furnish differential pressure switches for fan failure and dirty filter indication.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install all units as indicated on plans and according to unit manufacturer's instructions.
- B. Contractor shall install units on roof equipment support stands.
- C. Provide all necessary connections to unit as shown on drawings and as required to complete installation.
- D. Manufacturer and/or manufacturer's representative shall calibrate all of the unit's control devices and adjust unit automatic dampers to ensure proper operation of system. Manufacturer shall coordinate with Controls Contractor to ensure proper operation of system.

END OF SECTION

SECTION 238214
HEATING AND COOLING TERMINAL DEVICES

PART 1 - GENERAL

1.1 SUBMITTALS

- A. Shop Drawings for all items in this Section including the following:
 - 1. Manufacturer's name and model number
 - 2. Identification as referenced in the documents
 - 3. Capacities/ratings
 - 4. Materials of construction
 - 5. Dimensions and weights
 - 6. Color selection chart where applicable
 - 7. Wiring diagrams
 - 8. For chilled beams, submit performance data from independent testing agency. Performance data to include cooling capacity at supply water temperatures scheduled, air pressure drop, sound data, and water pressure drop for each beam length.
 - 9. All other appropriate data

1.2 REFERENCE STANDARDS AND DESIGN CRITERIA

- A. Duct Reheat Coils: Ratings shall be certified in accordance with AHRI 410 Standard for Forced-Circulation Air-Cooling and Air-Heating Coils for reheat coils, unit heaters, cabinet unit heaters and air curtain door heaters.
- B. AHRI 1410 Standard for Performance Rating of Commercial Finned-Tube Radiation.
- C. AHRI 440 Standard for Performance Rating of Fan-coil Units.
- D. AHRI 1240 Standard for Performance Rating of Active Chilled Beams.
- E. AHRI 210/240 Standard for Performance of Unitary Air-conditioning and Air-source Heat Pump Equipment.
- F. Units including electrical work shall be UL Listed.
- G. Electric Heaters: Shall be listed by UL, bear appropriate UL Label, contain the latest devices for protection of installation, and shall be installed in strict accordance with the latest revision of National Electric Code and other applicable State and Local Codes. Provide grounding lugs on all apparatus.

PART 2 - PRODUCTS

2.1 MINI-SPLIT SYSTEM

- A. Manufacturers: Mitsubishi, Toshiba, Carrier, Daikin or LG. Manufacturer shall be responsible for entire system consisting of DX cooling fan coil unit and condensing unit.
- B. Unit data shall be certified in accordance with AHRI Standard 210/240.
- C. Provide factory assembled and tested, outdoor mounted, air-cooled condensing unit suitable for installation on grade. Include compressor, air cooled condenser, refrigerant, lubrication system, interconnecting wiring, safety and operating controls, motor starting components and additional features as specified herein or required for safe, automatic operation. Capacity and operating characteristics as indicated in the equipment schedule. Coil performance data shall be in accordance with AHRI 410. Refrigerant shall be R-410A.
- D. Condensing unit cabinet is to be constructed of heavy gauge, galvanized steel coated with weather resistant paint. Provide removable access panels to facilitate full access to the compressor, fan and control components.
- E. Furnish head pressure control to operate compressor satisfactorily at ambient temperature to -20°F at minimum compressor load.
- F. Provide factory assembled indoor fan coil unit. Cabinet shall be impact resistant and fully insulated. Supply fan shall be direct drive tangential blower. Motor shall be multi-speed, ODP with permanently lubricated bearings. Cooling coil shall be copper tube with aluminum fin and galvanized steel tube sheets. Drip pan under coil shall have drain connection with internal trap. Unit shall have filter rack with factory supplied cleanable filters. Unit shall be furnished with integral wall mounting bracket and mounting hardware.
- G. Indoor fan coil unit shall be powered by outdoor condensing unit.
- H. Provide with complete electronic microprocessor control system with shall control space temperature, determine optimum fan speed, and run self-diagnostics. The temperature shall be controlled by a programmable wired remote thermostat with integral temperature sensor and wall mounting bracket.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install units as indicated on drawings, and according to manufacturer's installation instructions.

- B. Install branch piping to each unit with minimum of 3 elbows to allow for expansion and contraction.
- C. Protect units during construction against entry of foreign matter and construction dirt.
- D. Unless otherwise shown on drawings, mount vertical type wall mounted heating units 12" above finished floor.

END OF SECTION

SECTION 260000
GENERAL ELECTRICAL REQUIREMENTS

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 DESCRIPTION

- A. Intent of drawings and Specifications is to obtain complete systems tested, adjusted, and ready for operation.
- B. Except as otherwise defined in greater detail, the terms "provide", "furnish" and "install" as used in Division 26 Contract Documents shall have the following meanings:
 - 1. "Provide" or "provided" shall mean "furnish and install".
 - 2. "Furnish" or "furnished" does not include installation.
 - 3. "Install" or "installed" does not include furnishing.
- C. Include incidental details not usually shown or specified, but necessary for proper installation and operation.
- D. Check, verify and coordinate work with drawings and specifications prepared for other trades. Include modifications, relocations or adjustments necessary to complete work or to avoid interference with other trades.
- E. Included in this Contract are electrical connections to equipment provided by others. Refer to Architectural, Mechanical, Plumbing, and final shop drawings for equipment being furnished under other sections for exact locations of electrical outlets and various connections required.
- F. Information given herein and on drawings is as exact as could be secured but is not guaranteed. Do not scale drawings for dimensions.
- G. Where architectural features govern location of work, refer to Architectural Drawings.
- H. Perform work in "neat and workmanlike" manner as defined in ANSI/NECA 1, Standard Practices for Good Workmanship in Electrical Contracting.

1.3 RELATED WORK

A. Temporary Services:

1. Division 01 - Temporary Facilities and Controls.

B. Continuity of Service:

1. No service shall be interrupted or changed without permission from Architect and Owner. Obtain written permission before work is started.
2. When interruption of services is required, Architect, Owner and other concerned parties shall be notified and shall determine a time.

C. Demolition:

1. Division 02 - Selective Demolition
2. Division 02 - Building Demolition
3. Perform required demolition to accomplish new work.
 - a. Remove abandoned wiring to source of supply.
 - b. Remove exposed abandoned conduit, including abandoned conduit above accessible ceiling finishes. Cut conduit flush with walls and floors, and patch surfaces.
 - c. Disconnect abandoned outlets and remove devices.
 - d. Remove abandoned outlets if conduit servicing them is abandoned and removed.
 - e. Provide blank cover for abandoned outlets that are not removed.
 - f. Disconnect and remove electrical devices and equipment serving utilization equipment that has been removed.
 - g. Disconnect and remove abandoned luminaires. Remove brackets, stems, hangers, and other accessories.
 - h. Disconnect electrical systems in walls, floors, and ceilings scheduled for removal.
4. Accomplish work in neat workmanlike manner to minimize interference; annoyance or inconvenience such work might impose on Owner or other Contractors.
5. Unless otherwise noted, remove from premises materials and equipment removed in demolition work.
6. Equipment noted to be removed and turned over to Owner, shall be delivered to Owner at place and time Owner designates.
7. Where materials are to be turned over to Owner or reused and installed by Contractor, it shall be Contractor's responsibility to maintain condition of materials and equipment equal to that existing before work began. Repair or replace damaged materials or equipment at no additional cost to Owner.
8. Where demolition work interferes with Owner's use of premises, schedule work through Architect, Owner and with other Contractors to minimize inconvenience to Owner. Architect must approve schedule before Contractor begins such work.

D. Cleaning and Repair

1. Clean and repair existing materials and equipment that remain or are to be reused.
2. Panelboards.
 - a. Clean exposed surfaces and check tightness of electrical connections.

- b. Replace damaged circuit breakers and provide closure plates for vacant positions.
 - c. Provide typed circuit directory showing revised circuiting arrangement.
- 3. Luminaires:
 - a. Remove existing luminaires for cleaning.
 - b. Use mild detergent to clean exterior and interior surfaces; rinse with clean water and wipe dry.
 - c. Replace lamps ballasts, and broken electrical parts.

E. Concrete Work:

- 1. Provide cast-in-place concrete as required by Contract Documents unless otherwise noted.
- 2. Concrete shall comply with Division 03 - Concrete.
- 3. Provide anchor bolts, metal shapes and templates to be cast in concrete or used to form concrete as required for anchoring and supporting electrical equipment.

F. Painting:

- 1. Furnish equipment with factory-applied finish coats or paint equipment per Division 09 – Finishes unless specified otherwise.
- 2. Furnish equipment with factory applied prime finish unless otherwise specified.
- 3. If factory finish on equipment furnished by Contractor is damaged in shipment or during construction, refinish equipment to satisfaction of Architect.
- 4. Furnish one can of touch up paint for each final factory-applied finish coat of product.

1.4 REQUIREMENTS OF REGULATORY AGENCIES

- A. Rules and regulations of Federal, State and local authorities and utility companies, in force at time of execution of Contract shall become part of this specification.

1.5 REFERENCE STANDARDS

- A. Agencies or publications referenced herein refer to the following:

- 1. Association of Edison Illuminating Companies
- 2. American National Standards Institute
- 3. American Society of Mechanical Engineers
- 4. American Society for Testing and Materials
- 5. Building Industry Consulting Services International
- 6. Electronic Industries Association
- 7. Federal Information Processing Standards
- 8. Federal Communications Commission
- 9. Insulated Cable Engineers Association
- 10. Institute of Electrical & Electronics Engineers
- 11. Illuminating Engineering Society of North America
- 12. National Electrical Code
- 13. National Electrical Contractors Association
- 14. National Electrical Manufacturers Association
- 15. National Electrical Safety Code
- 16. National Electrical Testing Association
- 17. National Fire Protection Association

18. National Institute of Standards & Technology
19. Occupational Safety and Health Administration
20. Telecommunications Industries Association
21. Underwriters Laboratories, Inc.

- B. Work shall be in accordance with latest edition of codes, standards or specifications unless otherwise noted.

1.6 LISTING

- A. Install materials bearing UL label or UL listing, unless UL label or listing is not available for that type of material.
- B. Other nationally recognized testing agencies, acceptable to AHJ, are approved.

1.7 ENCLOSURES

- A. Typical NEMA Enclosures and Usage
1. NEMA 1 - Indoors. Falling dirt.
 2. NEMA 2 - Indoors. Falling dirt. Falling liquids. Light splashing.
 3. NEMA 3 - Outdoors. Sleet, snow, rain. Windblown dust.
 4. NEMA 3X - Same as NEMA 3 plus corrosion resistant.
 5. NEMA 3S - Same as NEMA 3 plus mechanism operable when ice covered.
 6. NEMA 3SX - Same as NEMA 3S plus corrosion resistant.
 7. NEMA 3R - Outdoors. Rain, snow, sleet.
 8. NEMA 3RX - Same as NEMA 3R plus corrosion resistant.
 9. NEMA 4 - Indoors: Falling dirt. Falling and light splashing liquids. Flying dust, lint and fibers. Hose down. Outdoors: Rain, sleet, snow. Wind blown dust. Hose down.
 10. NEMA 4X - Same as NEMA 4 - Indoors / Outdoor plus corrosion resistant.
 11. NEMA 5 - Indoors. Falling Dirt. Falling Liquids. Settling dust, lint and fibers.
 12. NEMA 6 - Indoors: Falling dirt. Falling and light splashing liquids. Flying dust, lint and fibers. Hose down. Temporary submersion. Outdoors: Rain, snow, sleet. Windblown dust. Hose down. Temporary
 13. NEMA 6P - Same as NEMA 6 - Indoors plus corrosion resistant. Prolonged submersion.
 14. NEMA 6P - Same as NEMA 6 - Outdoors plus corrosion resistant. Prolonged Submersion.
 15. NEMA 7 - Indoors. Class I, Division 1 or 2, Groups A, B, C or D. (Flammable gas).
 16. NEMA 9 - Indoors. Class II, Division 1 or 2. Groups E, R, or G. (Combustible dust).
 17. NEMA 12 - Indoors. Falling Dirt. Falling liquids. Flying dust, lint and fibers. Oil or coolant seepage.
 18. NEMA 13 - Same as NEMA 12 plus oil or coolant spraying or splashing.

1.8 SUBMITTALS

- A. Shop Drawings (Product Data):
1. Refer to Division 01 - Submittal Procedures.

2. Note that for satisfying submittal requirements for Division 26, "Product Data" is usually more appropriate than true "Shop Drawings" as defined in Division 01. However, the expression "Shop Drawings" is generally used throughout Specification.
3. Submit shop drawings for equipment and systems as requested in respective specification sections. Submittals which are not requested may not be reviewed.
4. Specifically mark general catalog sheets and drawings to indicate specific items submitted and its correlation to specific designation for product in drawings.
5. Specifically indicate proper identification of equipment by name and/or number, as indicated in specification and shown on drawings.
6. When manufacturer's reference numbers are different from those specified, provide correct cross-reference number for each item. Clearly mark and note submittal accordingly.
7. Submit complete record of required components when luminaires, equipment and items specified include accessories, parts and additional items under one designation.
8. Include wiring diagrams for electrically powered or controlled equipment.
9. Submit electrical equipment room layouts drawn to scale, including equipment, raceways, accessories and required working clearances. Submit electrical equipment room layouts concurrently with electrical distribution equipment submittals.
10. Where submittals cover products containing non-metallic materials, include "Material Safety Data Sheet" (MSDS) from manufacturer stating physical and chemical properties of components and precautionary considerations required.
11. Submit shop drawings or product data as soon as practicable after signing contracts. Submittals must be approved before installation of materials and equipment.
12. Refer to Division 01 - Coordination Drawings.
13. Submittals that are not complete, not permanent, or not properly checked by Contractor, will be returned without review.
14. Bidders shall provide a full compliance review of specifications. Compliance review shall accompany submittals. Compliance review shall include paragraph-by-paragraph review of specifications with the following information "C", "D", "E" marked in the margin of the specification section. Unless a deviation or exception is specifically noted in the compliance review, it is assumed that Bidder is in complete compliance with plans and specifications. Lack of these requirements in the submittal shall result in rejection of submittal. Text shall be provided accompanying compliance review as follows:
 - a. "C" – Comply with no exceptions
 - b. "D" – Comply with deviations. For each of the deviations, provide numbered footnote with reasons for proposed deviation
 - c. "E" – Exceptions; do not comply

B. Bookmarks:

1. Submitted documents greater than 50 pages in length shall include electronic bookmarks setup to quickly navigate and easily locate submitted information. Each major series of equipment shall have a bookmark.
2. Hyperlinks can be used to enhance bookmark tools but are not an acceptable substitute for electronic bookmarks. If hyperlinks are used, they should be clearly identifiable as a hyperlink by using a different color text similar to how MS Office identifies hyperlinks.

- C. Certificates and Inspections:
 - 1. Obtain and pay for inspections required by authorities having jurisdiction and deliver certificates approving installations to Owner unless otherwise directed.
- D. Operation and Maintenance Manuals:
 - 1. Refer to Division 01 - Operation and Maintenance Data.
 - 2. Upon completion of work but before final acceptance of system, submit electronic copy to Architect for approval. After securing approval, submit 3 copies to Owner.
 - 3. Organize manuals by specification section number and furnish table of contents and tabs for each piece of equipment or system.
 - 4. Manuals shall include the following:
 - a. Copies of shop drawings
 - b. Manufacturer's operating and maintenance instructions. Include parts lists of items or equipment, with component exploded views and part numbers. Where manufacturer's data includes several types or models, designate applicable type or model.
 - c. CD ROM's or flash/thumb drives of O&M data with exploded parts lists where available
 - d. Phone numbers and addresses of local parts suppliers and service companies
 - e. Internet/WEB page addresses where applicable
 - f. Wiring diagrams
 - g. Start up and shut down procedure
 - h. Factory and field test records
 - i. Additional information, diagrams or explanations as designated under respective equipment or systems specification section
 - 5. Instruct Owner's representative in operation and maintenance of equipment. Instruction shall include complete operating cycle on all apparatus.
 - 6. Furnish O&M manuals and instructions to Owner prior to request for final payment.
 - 7. Include bookmarks as indicated above.
- E. Record Documents:
 - 1. Refer to General Conditions of Contract and Division 01 - Project Record Documents. Prepare complete set of record drawings in accordance with Division 01.
 - 2. Use designated set of prints of Contract Documents as prepared by Architect to mark-up for record drawing purposes.

1.9 JOB CONDITIONS

- A. Building Access:
 - 1. Arrange for necessary openings in building to allow for admittance of all apparatus.
- B. Coordination:
 - 1. Equipment provided under other Divisions of these specifications.
 - a. Motors
 - b. Electrically powered equipment
 - c. Electrically controlled equipment

- d. Starters, where specified
 - e. Variable frequency drives, where specified
 - f. Control devices, where specified
 2. Provide the following devices required for control of motors or electrical equipment, unless otherwise noted:
 - a. Starters
 - b. Disconnect devices
 - c. Control devices:
 - 1) Pushbuttons
 - 2) Pilot lights
 - 3) Contacts
 - d. Conduit, boxes and wiring for Power wiring
 - e. Conduit, boxes and wiring for Control wiring.
 3. Connect and wire equipment complete and ready to operate according to wiring diagrams furnished by various trades.
 4. Wire starters or other similar control devices furnished by others.
 5. This Contractor's drawings and/or specifications show number and hp rating of motors furnished by others, together with their actuating devices. Should any change in size, hp rating, voltage, or means of control be made to any motor or other electrical equipment after Contracts are awarded, Contractor responsible for change shall immediately notify this Contractor. Additional costs due to these changes shall be responsibility of Contractor initiating change.
 6. Equipment and wiring shall be selected and installed for conditions in which it will be required to perform. (i.e., general purpose, weatherproof, rain tight, explosion proof, dust tight, or any other special type as required.)
 7. Comply with local utility motor starting requirements and provide starters for motors furnished by others as specified herein or under various trade sections of those specifications.
- C. Cutting and Patching:
 1. Refer to General Conditions of the Contract and Division 01 - Cutting and Patching.
 2. Perform cutting and patching required for complete installation of systems, unless otherwise noted. Patch and restore work cut or damaged to original condition. This includes openings remaining from removal or relocation of existing system components.
 3. Provide materials required for patching unless otherwise noted.
 4. Do not pierce beams or columns without permission of Architect and then only as directed. If openings are required through walls or floors where no sleeve has been provided, hole shall be core drilled to avoid unnecessary damage and structural weakening.
 5. Where alterations disturb lawns, paving, walks, etc., replace, repair or refinish surfaces to condition existing prior to commencement of work. This may include areas beyond construction limits.
- D. Housekeeping and Cleanup:
 1. Refer to Division 01 - Closeout Procedures.
 2. As work progresses or as directed by Architect, periodically remove waste materials from building and leave area of work broom clean. Upon completion of work, remove tools, scaffolding, broken and waste materials, etc. from site.

1.10 WARRANTY

- A. Refer to Division 01 for general warranty requirements.
- B. Refer to technical sections for warranty requirement for each system.
 - 1. Where no warranty requirements are called out, warrant for 1 year after acceptance by Owner equipment, materials, and workmanship to be free from defect.
- C. Repair, replace, or alter systems or parts of systems found defective at no extra cost to Owner.
- D. In any case, wherein fulfilling requirements of any guarantee, if this Contractor disturbs any work guaranteed under another contract, this Contractor shall restore such disturbed work to condition satisfactory to Architect and guarantee such restored work to same extent as it was guaranteed under such other contract.
- E. Warranty shall include labor, material, and travel time.

PART 2 - PRODUCTS

2.1 PRODUCT SUBSTITUTIONS

- A. Refer to Division 01 - Product Requirements.

PART 3 - EXECUTION

3.1 GENERAL

- A. Verify elevations and dimensions prior to installation of materials.
- B. Where conduits are shown on plans, they are shown for reference only. Conduit routes shown on plans may not show all required junction boxes, pull boxes, control conduits, or coordination with other trades. Contractor is responsible for coordination of all conduit routes with other trades, and providing junction boxes and pull boxes as required by code.

3.2 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to the site under provisions of Division 01.
- B. Store and protect products under provisions of Division 01.
- C. Store in clean, dry space.

- D. Maintain factory wrapping or provide cover to protect units from dirt, water, construction debris, and traffic.
- E. Handle in accordance with manufacturer's written instructions.
- F. Handle carefully to avoid damage to components, enclosure, and finish. Lift only with lugs provided for the purpose.
- G. Provide supplemental heat if required to prevent moisture contamination. Provide temporary circuits as required.

3.3 FLOOR, WALL, ROOF AND CEILING OPENINGS

- A. Coordinate location of openings, chases, furred spaces, etc. with appropriate Contractors. Provide sleeves and inserts that are to be built into structure during progress of construction.
- B. Remove temporary sleeves, if used to form openings, prior to installation of permanent materials. Utilize minimum 24 ga galvanized sheet metal for permanent sleeves unless otherwise noted.
- C. Provide Schedule 40 carbon steel pipe with integral water stop for steel sleeves required below grade or to exterior.
- D. Submit to Structural Engineer for review and approval size and location of core-drilled holes prior to execution.
- E. Submit product data and installation details for penetrations of building structure. Include schedule indicating penetrating materials, (steel conduit, PVC conduit, cables, cable tray, etc.), sizes of each, opening sizes and sealant products intended for use.
- F. Where penetrations of fire-rated assemblies are involved, seal penetrations with appropriate firestopping systems as specified in 260593 - Electrical Systems Firestopping.
- G. Provide 2" clearance around penetration openings intended for raceways and cables. Where fire resistant penetrations are required, size openings in accordance with written recommendations of firestopping systems manufacturer.
- H. Seal non fire-rated floor penetrations with non-shrink grout equal to Embeco by Master Builders, or urethane caulk, as appropriate.
- I. Seal non-rated wall openings with urethane caulk.

- J. Where penetrations occur through exterior walls into building spaces, use steel sleeves with integral water stop, similar to type "WS" wall sleeves by Thunderline Corporation. Seal annular space between sleeves and pipe with "Link-Seal" modular wall and casing seals by Thunderline Corporation, or sealing system by another manufacturer approved as equal by Engineer. Sealing system shall utilize Type 316 stainless steel bolts, washers and nuts.
- K. Finish and trim penetrations as shown on details and as specified.
- L. Provide chrome or nickel plated escutcheons where raceways pass through walls, floors or ceilings and are exposed in finished areas. Size escutcheons to fit raceways for finished appearance. Finished areas shall not include mechanical/electrical rooms, janitor's closets, storage rooms, etc., unless suspended ceilings are specified.

3.4 EQUIPMENT ACCESS

- A. Install raceways, cable tray, junction and pull boxes, and accessories to permit access to equipment for maintenance. Relocate raceways or accessories to provide maintenance access at no additional cost to Owner.
- B. Install equipment with sufficient maintenance space for removal, repair or changes to equipment. Provide ready accessibility to equipment and wiring without moving other future or installed equipment.
- C. Access doors in walls, chases, or inaccessible ceilings will be provided under Division 08 - Access Doors and Frames, unless otherwise indicated. Access doors for equipment shall provide access for servicing, repairs and/or maintenance.
- D. Provide necessary coordination and information to the Trade Contractor under Division 08 - Access Doors and Frames. This information shall include required locations, sizes and rough-in dimensions.
- E. Provide access doors in walls, chases or inaccessible ceilings for equipment requiring access for servicing, repairs and maintenance, unless otherwise noted. Access frames and doors shall be as manufactured by Milcor, Incorporated, or similar, of style applicable to surface. Provide access doors used in fire-rated construction with UL label. Provide steel, prime-coated access doors in dry locations. Provide stainless steel access doors for use in ceramic tile walls, toilet rooms, locker rooms, and in areas subject to excessive moisture. Provide access doors of sufficient size to allow complete maintenance. Coordinate location of access doors with General Contractor and rough-in equipment accordingly.
- F. Locate electrical outlets and equipment to fit details, panels, decorating or finish at space. Architect reserves right to make minor position changes of outlet locations before work has been installed.

- G. Verify door swings before installing room light switch boxes. Install boxes on latch side of door unless otherwise noted

3.5 EQUIPMENT SUPPORTS

- A. Provide supporting steel not indicated on drawings as required for installation of equipment and materials including angles, channels, beams, hangers, etc.
- B. Provide steel shell with plug type concrete anchors for attaching equipment to concrete. Plastic, rawhide or anchors using lead are not allowed.
- C. Do not support equipment or luminaires from metal roof decking.

3.6 SUPPORT PROTECTION

- A. In occupied areas, mechanical and electrical rooms and areas requiring normal maintenance access, guard certain equipment to protect personnel from injury.
- B. Provide minimum 1/2" thick Armstrong Armaflex insulation or similar product applied with Armstrong 520 adhesive on lower edges of equipment, including bus duct, cable tray, pull boxes and electrical supporting devices suspended less than 7 ft above floors, platforms or catwalks in these areas.
- C. Protect threaded rods or bolts at supporting elements as described above. Trim threaded rods or bolts such that they do not extend beyond supporting element.

3.7 ELECTRICAL SYSTEMS IDENTIFICATION

- A. Refer to Section 260553 - Identification for Electrical Systems.

3.8 ACCEPTANCE TESTING

- A. Contractor shall engage testing and inspection agency to perform acceptance tests. Equipment to be tested is noted as "Testing by Testing Agency" in technical specification sections. Perform in accordance with Section 260812 - Power Distribution Acceptance Tests and Section 260813 - Power Distribution Acceptance Test Tables.
- B. Contractor shall perform acceptance testing. Equipment to be tested is noted as "Testing by Electrical Contractor" in technical specification sections. Perform in accordance with Section 260812 - Power Distribution Acceptance Tests and Section 260813 - Power Distribution Acceptance Test Tables.
- C. When testing is to be witnessed by Architect/Engineer or Inspector, notify them at least 10 days prior to testing date.

- D. When equipment or systems fail to meet minimum test requirements, replace or repair defective work or materials as necessary and repeat inspection and test until equipment or systems meet test requirements. Make repairs with new materials.
- E. Contractor is responsible for certifying in writing equipment and system test results. Certification shall include identification of portion of system tested, date, time, test criteria and name and title of person signing test certification documents.
- F. Maintain copies of certified test results, including those for any failed tests, at project site. At completion of project, include copies of test records and certifications in O&M Manuals.

3.9 START-UP

- A. Systems and equipment shall be started, tested, adjusted, and turned over to Owner ready for operation. This includes "Owner-Furnished, Contractor-Installed" (OFICI) and "Contractor-Furnished, Contractor-Installed" (CFCI) systems and equipment.
- B. Follow manufacturer's pre-start-up checkout, start-up, trouble shooting and adjustment procedures.
- C. Contractor shall provide services of technician/mechanic knowledgeable in start-up and checkout of types of systems and equipment on project.
- D. Provide start-up services by manufacturer's representative where specified or where Contractor does not have qualified personnel.
- E. Coordinate start-up with all trades.

3.10 CLEANING

- A. Clean systems after installation is complete.
- B. Vacuum debris from panelboards, switchboards, motor starter and disconnect switch enclosures, junction boxes and pull boxes two weeks before energization and again prior to completion.
- C. Where louvers are provided in switchgear or transformer enclosures, vacuum louvers free of dust and dirt.
- D. Clean luminaire lenses and lamps at time of installation and clean lens exteriors just prior to final inspection.
- E. Thoroughly clean equipment of stains, paint spots, dirt and dust. Remove temporary labels not used for instruction or operation.

3.11 CONSTRUCTION WASTE MANAGEMENT

- A. Construction waste management shall be managed in accordance with provisions of Section 017400 - Cleaning and Waste Management. Documentation shall be submitted to satisfy the requirements of that section.

END OF SECTION

SECTION 260516 OWNER-FURNISHED EQUIPMENT

PART 1 - GENERAL

1.1 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.2 DESCRIPTION

- A. Section includes electrical connection(s) to Owner-purchased pieces of equipment, which are required in construction.
- B. Owner-furnished equipment requiring work by Contractor is specified in the following Sections:
- C. Contractor shall be responsible for receipt of equipment from Owner, storage after receipt, installation, and electrical connection.
- D. Owner-furnished equipment requiring work by the Contractor is shown on the drawings and schedules.
- E. Owner-furnished, Contractor-installed equipment is labeled OFCI.

1.3 SUBMITTALS

- A. Shop Drawings: Owner-supplied shop drawings of equipment furnished by Owner.
- B. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, installation, and starting of product.

1.4 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Comply with NFPA 70 for components and installation.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory unopened packaging until ready for installation.

1.6 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr written warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 (NOT APPLICABLE TO THIS SECTION)

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Provide labor, materials and electrical connections for Owner-furnished equipment in accordance with contract drawings.
- B. Install and connect Owner-furnished equipment as though it had been purchased by Contractor.
 - 1. This shall include:
 - a. Receiving equipment at jobsite
 - b. Rigging and setting equipment in place
 - c. Making electrical connections
 - d. Starting
 - e. Testing
- C. Coordinate with owner equipment provider and start-up technician as required for a complete and working installation.
- D. Install equipment in accordance with manufacturer's installation instructions.
- E. Maintain equipment until facility is accepted by Owner.
- F. Review Owner-supplied shop drawings of Owner-furnished equipment to ascertain that necessary labor and materials have been provided to install equipment and complete the system it serves.

END OF SECTION

SECTION 260519
LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260519.16 - Manufactured Wiring System
- B. Section 260529 - Hangers and Supports for Electrical Systems
- C. Section 260533 - Raceway and Boxes for Electrical Systems
- D. Section 260553 - Electrical Systems Identification
- E. Section 260593 - Electrical Systems Firestopping
- F. Section 260812 - Power Distribution Acceptance Tests
- G. Section 260813 - Power Distribution Acceptance Test Tables

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes conductors and cables rated 600 V and less, connectors, splices, and terminations rated 600 V and less, sleeves and sleeve seals for cables.
- B. Conductor and conduit sizes in these contract documents are based on copper wire, and only copper wire shall be used.

1.4 REFERENCE STANDARDS

- A. ASTM B1 – Standard Specification for Hand-Drawn Copper Wire.
- B. ASTM B8 – Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.
- C. NEMA WC 70 – Non-Shielded Power Cable 2000 V or less for the Distribution of Electrical Energy (ICEA S-95-658).
- D. NFPA 70 – National Electrical Code.

- E. UL 44 – Thermoset-Insulated Wires and Cables.
- F. UL 83 – Thermoplastic-Insulated Wires and Cables.
- G. UL 486A-486B – Wire Connectors.
- H. UL 486C – Splicing Wire Connectors.
- I. UL 486D – Standard for Insulated Wire Connector Systems for Underground Use or in Damp or Wet Locations.
- J. UL 486E – Standard for Equipment Wiring Terminals for Use with Aluminum and/or Copper Conductors.
- K. UL 1569 – Standard for Metal-Clad Cables.

1.5 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Comply with NFPA 70 for components and installation.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.
- B. Wire and cable boxes and reels shall bear the date of manufacture.
 - 1. Date of manufacture shall not precede contract date by more than one year.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Protect from dirt, fumes, water, corrosive substances, and construction debris.

1.7 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. General Cable Corporation
- B. Cerrowire

- C. Southwire Company
- D. Encore Wire Corporation
- E. VFD Cable: Aetna Insulated Wire, Americable, General Cable, Southwire [Approved Equal]
- F. Approved equal

2.2 DESCRIPTION

- A. NEMA WC 70; single copper conductor insulated wire; 600 V rated insulation; 90°C maximum operating temperature for dry and wet or damp locations.
 - 1. Thermoplastic-insulated wires and cables: NEMA WC 70, UL 83; Type THWN-2.
 - 2. Thermoset-insulated wires and cables: NEMA WC 70, UL 44; Type XHHW-2.
- B. Multi-conductor Cable: NEMA WC 70; for metal-clad cable, Type MC with ground wire.
- C. Metal-clad cable, Type MC; UL 1569:
 - 1. Impervious, corrugated, continuous, seam welded metal sheath.
 - 2. Single grounding conductor.
 - 3. Three grounding conductors for use as VFD to motor feeder.
- D. VFD Cable:
 - 1. Cable
 - a. 600V/2000V rated, high stranded tinned copper conductors, shielded, engineered for use with Variable Frequency Drives.
 - b. Insulation shall be rated for 90 degrees Celsius Wet/Dry operating temperature.
 - 2. Conductors
 - a. Conductor shall be annealed fine wire flexible high strand count tinned copper or standard Class B stranded bare copper.
 - b. Three (3) phase conductors, three (3) ground conductors. The total cross sectional area of the three ground conductors shall be the same size as the single ground conductor shown on the drawings.
 - 3. Insulation
 - a. Flame-Retardant Cross-Linked Polyethylene.
 - b. Conductors shall be cabled together. Ground conductors shall be symmetrical. Fillers shall be included as necessary to make the cable round.
 - 4. Shielding
 - a. The following are acceptable:
 - 1) Overall tinned copper braid plus aluminum/polyester tape foil, 100% coverage.
 - 2) 5 mil helically applied copper tape.
 - 3) Impervious corrugated welded continuous armor.
 - 5. Jacket
 - a. Flame-retardant Thermoplastic, suitable for 90°C use.
 - 6. Termination Kit

- a. Pre-sized and pre-formed specifically for VFD cable constructions. Obtain from VFD cable manufacturer.

2.3 REMOTE CONTROL AND SIGNAL CIRCUITS

A. Class 1

- 1. Copper conductor, single insulated wire.
- 2. Insulation type THHN rated 90°C, 600 V insulation class.
- 3. Type XHHW for ambient temperature less than 32°F.
- 4. UL 83 listed, ASTM B1 for solid conductors; ASTM B 8 for stranded conductors.

B. Classes 2 and 3

- 1. Copper conductor, multiple twisted conductors covered with an overall non-metallic jacket unless otherwise noted.
- 2. Insulation type XLPE, rated 105°C, 300 V insulation class.
- 3. UL listed for use in space in which circuits will be installed.

2.4 CONNECTORS, SPLICES, AND TERMINALS

A. Manufacturers:

- 1. AFC Cable Systems, Inc.
- 2. Hubbell Power Systems, Inc.
- 3. O-Z/Gedney; EGS Electrical Group LLC.
- 4. 3M; Electrical Products Division
- 5. Tyco Electronics Corp.
- 6. Approved equal

- B. Description: UL 486A-486B, UL 486C, UL 486D, UL 486E; factory-fabricated connectors, splices, and terminals of size, ampacity rating, material, type, and class for application and service indicated.

2.5 TERMINATIONS

- A. Compression set, bolted or screw type lug, or direct to bolted or screw type terminal.

2.6 PLASTIC CABLE TIES

- A. Nylon or approved; locking type; metallic ties not permitted.

PART 3 - EXECUTION

3.1 INSTALLATION OF CONDUCTORS AND CABLES

- A. Install conductors in a raceway system, unless otherwise specified or indicated.
- B. Install conductors only after:
 - 1. Building interior is enclosed and weather tight
 - 2. Mechanical work likely to damage conductors has been completed

3. Raceway installation is complete and supported

- C. Pull conductors into raceway at same time.
- D. Neatly train and lace conductors inside boxes, equipment, and panelboards.
- E. Conceal cables in finished walls, ceilings, and floors, unless otherwise indicated.
- F. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- G. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.
- H. Install exposed cables parallel and perpendicular to surfaces of exposed structural members and follow surface contours where possible. Protect exposed cables from damage.
- I. Support cables above accessible ceiling using plastic cable ties to support cables from structure. Do not rest cable on ceiling panels.
- J. Support cables and conductors in vertical raceways per requirements in Section 260529 - Hangers and Supports for Electrical Systems.
- K. Identify and color-code conductors and cables according to Section 260553 - Electrical Systems Identification.
- L. Wiring at Outlets: Install conductor at each outlet, with minimum 6" of slack.
- M. Limit conduit fill to a maximum of 9 current-carrying conductors.
- N. Install stranded conductors where conductors terminate in crimp type lugs. Do not place bare stranded conductors directly under screws.
- O. Install VFD input wiring, output wiring and control wiring in their own separate conduit systems.
- P. Provide dedicated neutrals for branch circuits unless otherwise noted on drawings.

3.2 CONDUCTOR MATERIAL APPLICATIONS

- A. Feeders: Copper. Solid or stranded for #10 AWG and smaller; stranded for #8 AWG and larger.
- B. Branch Circuits: Copper. Solid or stranded for #10 AWG and smaller; stranded for #8 AWG and larger.

- C. Minimum conductor sizes shall be as follows:
 - 1. #12 AWG – Branch circuits of any kind.
 - 2. #14 AWG – Fire alarm system.
 - 3. #16 AWG – Remote control and signal systems.
- D. Branch wiring length limitations:
 - 1. Where wire sizes are shown on project drawings and do not indicate they have been adjusted for voltage drop based on circuit length, they shall be increased as noted below depending on Contractor routing.
 - a. 208Y/120 V circuits over 75 ft in length: Increase wire size one size for each 75 ft in length. Increase conduit size as required.
 - b. 480Y/277 V circuits over 150 ft in length: Increase wire size one size for each 150 ft of length. Increase conduit size as required.

3.3 CONDUCTOR INSULATIONS AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

- A. Feeders Concealed in Ceilings, Walls, Partitions, and Crawlspace: Type THHN/THWN-2, rated 90°C for wet locations, single conductors in raceway.
- B. Feeders Concealed in Concrete, below Slabs-on-Grade, and Underground: Type XHHW-2, rated 90°C for wet locations, single conductors in raceway.
- C. Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THWN-2, rated 90°C for dry and wet or damp locations, single conductors in raceway.
- D. Branch Circuits Concealed in Concrete, below Slabs-on-Grade, and Underground: Type XHHW-2, rated 90°C for dry and wet or damp locations, single conductors in raceway.
- E. Motor Circuit Branch Wiring and Associated Control Wiring: Type THHN, rated 90°C for dry and wet or damp locations, single conductors in raceway, stranded.
- F. Motor Circuit Branch Wiring Between Motor and VFD: VFD Cable
 - 1. Terminate VFD cable using pre-sized and pre-formed termination kits supplied by cable manufacturer. Install per manufacturer's recommendations.
 - 2. Requirements for raceway installation when VFD cable is not used:
 - a. Tighten all raceway fittings and connections per manufacturer's requirements.
 - b. Bond and secure raceways to junction boxes.
 - c. Set screw connectors are not permitted.
 - d. Install equipment grounding conductor to be continuous between supply and load.
 - e. Install fittings to provide proper 360° connection between raceway and enclosure at concentric knockouts.
 - f. Remove paint from any surfaces between bond of enclosure, fittings and raceway.
 - g. Provide grounding bushings at enclosures.
 - h. Re-torque all fittings and terminations prior to building turnover.

- G. Wiring in Light Fixture Channels: Type THHN/THWN-2, rated 90°C for dry and damp locations, single conductors.
- H. Branch Circuits Single Conductors in Raceway: 90°C rated conductors sized at 75°C rating for connection to equipment and devices.
- I. Circuits 100A and less, utilize 60C for ampacity (unless both sides of the circuit are listed for 75C). Circuits 100A and over use 75C for ampacity. If the wire insulation ratings are higher, they may be utilized for de-rating purposes only and wire size shall be increased if the de-rated insulation rating is less than the required load current.
- J. Metal-clad cable, Type MC, use for the following:
 - 1. Vertical branch circuits from the homerun junction box in suspended ceiling to device outlet (receptacles, switches)
 - 2. Recessed and semi-recessed lighting fixtures
- K. Cord Drops and Portable Appliance Connections: Type SO, hard service cord with stainless-steel, wire-mesh, strain relief device at terminations to suit application.

3.4 REMOTE CONTROL AND SIGNAL CIRCUITS

- A. Sizing – #16 AWG minimum.
- B. Installation:
 - 1. Install cables in cable tray and cable rings.
 - 2. Provide protection for exposed cables where subject to damage.
 - 3. Support cables above accessible ceilings; do not rest on ceiling tiles.
 - 4. Use suitable cable fittings and connectors.

3.5 CONNECTORS, SPLICES, AND TERMINALS

- A. Connectors:
 - 1. Except where equipment is furnished with bolted or screw type lug, use compression set pressure connectors with insulating covers. Use compression tools and die compatible with connectors being installed.
 - 2. Use bolt or compression-set type with application of insulating tape, pre-stretched or heat-shrinkable insulating tubing for splices and taps of #8 AWG conductors and larger. Install with hydraulic compression tool.
 - 3. Use pre-insulated “twist-on” connectors with integral spring for splices and taps of #10 AWG conductors and smaller and in quantities not to exceed manufacturer's recommendations.
 - 4. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- B. Splices:
 - 1. Splice wires and cable only in accessible locations such as within junction boxes.
 - 2. Make splices to carry full capacity of conductors with no perceptible temperature rise.

3. Make below-grade splices in manholes and handholes watertight with pre-stretched or heat-shrinkable insulating tubing, or resin-filled insulator.
4. Use electrical tape to build up insulation level equivalent to cable insulation and cover with not less than two half-lapped layers of plastic electrical tape, for joints, taps, and splices of #1 AWG conductors and larger.
5. Plastic snap-on splice insulators are not allowed.
6. Make splices and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.
7. Use oxide inhibitor in each splice and tap conductor for aluminum conductors.

C. Terminals:

1. Insulate ends of spare conductors with electrical tape and identify spare circuit number where appropriate.
2. Eye type crimped terminal for removable screw type terminal. Forked torque terminal when screw terminal cannot be removed.
3. Train wires to eliminate fanning of strands, crimp with proper tool and die.
4. Torque screw termination per manufacturer's recommended values.

3.6 CABLE TIES

- A. Neatly bundle conductors and cables together for support. Size cable ties sufficiently to accommodate the multiple cables being supported.

3.7 FIELD QUALITY CONTROL

- A. Test 600 volt conductors and cables per requirements in Sections 260812 - Power Distribution Acceptance Tests and 260813 - Power Distribution Acceptance Test Tables.
- B. Interpret test results in writing and submit to Engineer.
- C. Replace conductors and cables that are found defective, at no expense to Owner.

END OF SECTION

**SECTION 260519.16
MANUFACTURED WIRING SYSTEM**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260000 - General Electrical Requirements
- B. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- C. Section 260533 - Raceway and Boxes for Electrical Systems
- D. Section 260553 - Electrical Systems Identification
- E. Section 265000 - Lighting

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION OF SYSTEM

- A. At Contractor's option, provide Manufactured Wiring System for connection to Luminaires instead of wire and conduit system.

1.4 REFERENCE STANDARDS

- A. UL 183 - Manufactured Wiring Systems.

1.5 SUBMITTALS

- A. Submit shop drawings for equipment provided under this Section.
- B. Submit as-built floor plans of fixture interconnections.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Manufacturers: Day-brite, Electro/Connect Walker Flexible Systems, Reloc, and Hubbell.

2.2 PRODUCTS

- A. Provide luminaires and/or other equipment with power-in and power-out connector. Flush mount all connectors.
- B. Provide factory prewired luminaires with system fixture receptacle assemblies and ship to job site ready for connection.
- C. Provide coordination between system used and luminaire manufacturer.
- D. Provide system components that are prewired, labeled as to function, voltage, number of circuits, capacity of each circuit, catalog number, and manufacturer's name and address.
- E. Identifications can be embossed, painted, or stick-on label type.
- F. Provide proper interfacing between system and out-of-system components.
- G. Provide whips that have female power-in receptacle on one end and power-out receptacle or pre-stripped leads on other end.
- H. Provide leads that are a minimum of 6" long.
- I. Provide plugs that couple, lock, and complete circuit only with receptacles of like configuration.
- J. Provide whips that are of ample length to make proper connection without bending damage to cable.
- K. Provide polarized connectors.
- L. Wiring shall be #12 AWG, 90°C insulated wire and rated at 20 A.
- M. Carry an equipment grounding conductor throughout all system components, except those that are serving only low-energy power circuits, signaling circuits, or other remote control circuits.
- N. Provide conversion units to convert from permanent wiring to flexible wiring, cable sets, junction units, fixture receptacles. Provide switch and receptacle whips to feed switches and receptacles from flexible system. Provide cable couplers and splitters as required.
- O. Provide connectors that are held in place to prevent accidental disconnection.
- P. Route cable in accordance with manufacturer's instructions.
- Q. Color code plugs and receptacles.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install components and materials in accordance with code requirements, manufacturer's instructions, and approved shop drawings.
- B. Install switch whips and wiring adapters with nonsystem components in approved manner, ready for interface with flexible wiring system.
- C. When conversion units, luminaires, and switching units have been installed, complete manufactured wiring system by connecting system components with cable sets of lengths, types, and color coding as indicated or required.
- D. Support cable sets with nylon tie wraps or metal clips designed for purpose at 4 ft on center.
- E. Cap unused receptacles and connectors with dust cover.

END OF SECTION

SECTION 260526
GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- B. Section 260519.16 - Manufactured Wiring System
- C. Section 260533 - Raceway and Boxes for Electrical Systems
- D. Section 260543 - Underground Ducts and Raceways for Electrical Systems
- E. Section 260543.19 - Manholes and Hardware
- F. Section 260812 - Power Distribution Acceptance Tests
- G. Section 260813 - Power Distribution Acceptance Test Tables
- H. Section 260923 - Lighting Control Devices
- I. Section 262416.13 - Lighting and Appliance Panelboards
- J. Section 262726 - Wiring Devices
- K. Section 262816 - Enclosed Switches and Circuit Breakers
- L. Section 262913 - Enclosed Controllers
- M. Section 263213 - Engine Generators
- N. Section 263623 - Automatic Transfer Switches
- O. Section 264300 - Surge Protective Devices (SPD)
- P. Section 265000 - Lighting

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes methods and materials for grounding systems and equipment, as required by State Codes, NFPA 70, applicable portions of other NFPA codes, as indicated herein, plus the following special applications:
 - 1. Underground distribution grounding.
- B. Maximum resistance to ground shall be less than 25 ohms.

1.4 REFERENCE STANDARDS

- A. TIA-607 -B – Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
- B. ASTM B3 – Specification for Soft or Annealed Copper Wire
- C. ASTM B8 – Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard or Soft
- D. ASTM B33 – Specification for Tinned Soft or Annealed Copper Wire for Electrical Purposes
- E. IACS - International Annealed Copper Standard
- F. IEEE C2 – National Electrical Safety Code (ANSI)
- G. IEEE 142 - Recommended Practice for Grounding of Industrial and Commercial Power Systems
- H. IEEE 837 - Standard for Qualifying Permanent Connections used on Substation Grounding
- I. IEEE 857 – Standard for Qualifying Permanent Connections Used in Substation Grounding
- J. NETA MTS – Maintenance Testing Specifications
- K. NFPA 70 – National Electrical Code
- L. NFPA 70B – Recommended Practice for Electrical Equipment Maintenance
- M. NFPA 780 – Lightning Protection Systems
- N. UL 96 – Lightning Protection Components
- O. UL 467 – Grounding and Bonding Equipment

1.5 TELECOMMUNICATIONS GROUNDING SYSTEM DEFINITIONS

- A. Grounding Equalizer (GE): Conductor that interconnects elements of telecommunications grounding infrastructure.
- B. Telecommunications Bonding Backbone (TBB): Conductor that interconnects telecommunications main grounding busbar (TMGB) to telecommunications grounding busbar (TGB).
- C. Telecommunications Bonding Conductor: Conductor that interconnects telecommunications bonding infrastructure to building's service equipment (power) ground.
- D. Telecommunications Grounding Busbar (TGB): Interface to building telecommunications grounding system, common point of connection for telecommunications system and equipment to ground, and located in telecommunications room or equipment room.
- E. Telecommunications Main Grounding Busbar (TMGB): Busbar placed in convenient and accessible location and bonded by means of bonding conductor for telecommunications to building service equipment (power) ground.

1.6 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70.
 - 2. Comply with UL 467 for grounding and bonding materials and equipment.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store products in clean, dry space. Protect from dirt, fumes, water, corrosive substances, and construction debris.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr written warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 CONDUCTORS

- A. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction, insulation color: green.
 - 1. Grounding Conductor: Solid #12 & #10, stranded #8 and larger
 - 2. Bonding Cable: 28 kcmil, 14 strands of #17 AWG conductor, 1/4" in diameter.
 - 3. Bonding Conductor: #4 AWG or #6 AWG, stranded conductor.
- B. Bare Copper Conductors:
 - 1. Solid Conductors: ASTM B3.
 - 2. Stranded Conductors: ASTM B8.
 - 3. Tinned Conductors: ASTM B33.
 - 4. Bonding Jumper: Copper tape, braided conductors, terminated with copper ferrules; 1-5/8" wide and 1/16" thick.
 - 5. Tinned Bonding Jumper: Tinned-copper tape, braided conductors, terminated with copper ferrules; 1-5/8" wide and 1/16" thick.

2.2 WALL MOUNTED GROUND BUS

- A. Grounding Bus: Horizontal rectangular bars of annealed copper, 1/4" by 2" in cross section; with insulators.

2.3 CONNECTORS

- A. Listed and labeled by a nationally recognized testing laboratory acceptable to authorities having jurisdiction for applications in which used, and for specific types, sizes, and combinations of conductors and other items connected.
- B. Bolted Connectors for Conductors and Pipes: Electro-tin plated copper or copper alloy, bolted pressure-type, with at least two bolts.
 - 1. Pipe connectors: Clamp type, sized for pipe.
- C. Welded Connectors: Exothermic welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.
- D. Compression Connectors: Irreversible type.

2.4 GROUNDING ELECTRODES

- A. Ground Rods: Copper-clad steel; 3/4" in diameter by 10' in length.

2.5 TELECOMMUNICATIONS BUSBARS

- A. Material: Copper
 - 1. 1/4" thick

- B. Pre-drilled
 - 1. 7/16" diameter
 - 2. Hole spacing per ANSI Joint Standard TIA-607-B
 - 3. Hole pattern shall accommodate two-hole lugs
- C. Insulators and stand-off brackets shall electrically isolate busbar from wall or other mounting surface.
- D. Provide busbars listed by nationally recognized testing laboratory.
- E. Size:
 - 1. Telecommunications Main Ground Busbar (TMGB) – 20" x 4" (minimum)
 - 2. Telecommunications Grounding Busbar (TGB) – 12" x 2" (minimum)
- F. Size:
 - 1. Telecommunications Primary Bonding Busbar (PBB) – 20 x 4" (minimum)

2.6 TELECOMMUNICATIONS GROUNDING CONDUCTORS

- A. Material: Stranded copper
- B. Provide insulated bonding conductors.
 - 1. Green Jacket or Black Jacket marked with Green Tape or Green adhesive labels per NEC.
- C. Size:
 - 1. Bonding Conductor for Telecommunications (BCT; TMGB to Grounding Electrode), Telecommunications Bonding Backbone (TBB; TMGB to TGB), Grounding Equalizer (GE; TBB to TBB): #4/0 AWG
 - 2. Telecommunications Bonding Backbone (TBB; TMGB to TGB) #4/0 AWG
- D. Size:
 - 1. Telecommunications Bonding Conductor (TBC; PBB to Electrical Service Entrance main ground bar), Backbone Bonding Conductor (BBC; PBB to local steel structure): #4/0 AWG
 - 2. Telecommunications Bonding Backbone (TBB; PBB to SBB), Backbone Bonding Conductor (BBC; SBB to local steel structure): #4/0 AWG

2.7 TELECOMMUNICATIONS GROUNDING CONNECTIONS

- A. Mechanical Connectors:
 - 1. Connector Body:
 - a. High-strength, high-conductivity cast copper alloy
 - b. 2-bolt type
 - 2. Bolts, nuts, washers, and lock-washers: 300 series stainless steel
 - a. Supplied as part of connector body
 - b. Split-bolt connector types are not allowed.
 - 3. Connector:
 - a. Meet or exceed UL 467

- b. Clearly marked with catalog number, conductor size, and manufacturer
- B. Irrever Compression Connectors:
 - 1. Connector Body: Pure wrought copper.
 - 2. Conductivity shall be no less than 99%.
 - 3. Connector:
 - a. Meet or exceed requirements of IEEE 837, latest revision.
 - b. Filled with an oxide-inhibiting compound.
 - c. Clearly marked with manufacturer, catalog number, conductor size, and required compression tool settings.

PART 3 - EXECUTION

3.1 APPLICATIONS

- A. Conductors: Install solid conductor for #8 AWG and smaller, and stranded conductors for #6 AWG and larger, unless otherwise indicated.
 - 1. Size conductors and jumpers as indicated on drawings, unless specifically noted otherwise.
- B. Underground Grounding Conductors: Install bare copper conductor. No. #2/0 AWG
 - 1. Bury at least 24" below grade.
 - 2. Ductbank Grounding Conductor: Bury 12" above ductbank when indicated as part of ductbank installation.
- C. Wall Mounted Grounding Bus: Install in electrical and communications rooms, in rooms housing service equipment.
 - 1. Install bus on insulated spacers, 1", minimum, from wall; 6" above finished floor.
 - 2. Where indicated on both sides of doorways, route bus up to top of door frame, across top of doorway, and down to specified height above floor, and connect to horizontal bus.
- D. Conductor Terminations and Connections:
 - 1. Pipe Terminations: Mechanical lugs
 - 2. Equipment Grounding Conductor Terminations: Mechanical lugs
 - 3. Underground Connections: Welded connectors
 - 4. Connections to Ground Rods at Test Wells: Mechanical connectors
 - 5. Connections to Structural Steel: Mechanical connections
 - a. Use mechanical connectors with theft resistant hardware where connections are exposed in public areas.

3.2 GROUNDING UNDERGROUND DISTRIBUTION SYSTEM COMPONENTS

- A. Grounding Manholes and Handholes: Install a driven ground rod through manhole or handhole floor, close to wall, and set rod depth so 4" will extend above finished floor. If necessary, install ground rod before manhole is placed and provide #1/0 AWG bare, tinned-copper conductor from ground rod into manhole through a waterproof PVC sleeve in manhole wall. Protect ground rods passing through concrete floor with a double wrapping of pressure-sensitive insulating tape or heat-shrunk insulating sleeve from 2" above to 6" below concrete. Seal floor opening with waterproof, nonshrink grout.
- B. Grounding Connections to Manhole Components: Bond exposed-metal parts such as inserts, cable racks, pulling irons, ladders, catch basins, metallic cover frame and cable shields within each manhole or handhole, to ground rod or grounding conductor. Make connections with #4 AWG minimum, stranded, copper bonding conductor. Train conductors level or plumb around corners and fasten to manhole walls. Connect to cable armor and cable shields as recommended by manufacturer of splicing and termination kits.
- C. Pad-Mounted Transformers, Unit Substations, and Switches: Install two ground rods and ground ring around the pad. Ground pad-mounted equipment and noncurrent-carrying metal items associated with transformer and switches by connecting them to underground cable and grounding electrodes. Install copper conductor not less than #2 AWG for ground ring and for taps to equipment grounding terminals. Bury ground ring not less than 24" from the foundation. Install as indicated on drawings.

3.3 EQUIPMENT GROUNDING

- A. Install insulated equipment grounding conductors with the following items, in addition to those required by NFPA 70:
 - 1. Feeders and branch circuits
 - a. Install a single insulated equipment ground conductor for each branch circuit conduit originating from panelboards.
 - b. Install a single insulated equipment ground conductor for each branch circuit neutral originating from panelboards, including lighting circuits.
 - 2. Flexible raceway runs
 - 3. Armored and metal-clad cable runs
 - 4. Computer and Rack-Mounted Electronic Equipment Circuits: Install insulated equipment grounding conductor in branch-circuit runs from equipment-area power panels and power-distribution units.

- B. Air-Duct Equipment Circuits: Install a separate insulated equipment grounding conductor to duct-mounted electrical devices operating at 120 V and more, including air cleaners, heaters, dampers, humidifiers, and other duct electrical equipment. Bond conductor to each unit and to air duct and connected metallic piping. Ground ductwork of fans serving flammable liquid storage rooms or fume hoods. Install continuous ground around any flexible connections in this ductwork system. Bond lower end of exhaust ducts, vent stacks, etc., which pass through roof.
- C. Water Heater, Heat-Tracing, and Antifrost Heating Cables: Install a separate insulated equipment grounding conductor to each electric water heater and heat-tracing cable. Bond conductor to heater units, piping, connected equipment, and components.
- D. Wiring devices, receptacles and light switches of any amperage: Install separate jumper between grounding terminal on device and metallic box.
- E. Isolated Equipment Enclosure Circuits: For designated equipment supplied by a branch circuit or feeder, isolate equipment enclosure from supply circuit raceway with a nonmetallic raceway fitting listed for the purpose. Install fitting where raceway enters enclosure, and install a separate insulated equipment grounding conductor. Isolate conductor from raceway and from panelboard grounding terminals. Terminate at equipment grounding conductor terminal of the applicable derived system or service, unless otherwise indicated.
- F. Size of equipment grounding conductors for branch circuits: As indicated in NEC-70, except minimum size shall be #12 AWG.
- G. Size of branch panel feeder originating at switchboards/switchgear: As indicated in NEC-70, except in no instance smaller than #8 AWG.
- H. Signal and Communication Equipment: For alarm and other communication equipment (see Telecommunications Grounding System Installation section below for voice and data systems), install #4 AWG minimum grounding conductor in raceway from grounding electrode system to each service location, terminal cabinet, wiring closet, and central equipment location.
 - 1. Service and Central Equipment Locations and Wiring Closets: Terminate grounding conductor on a 1/4 x 2 x 12" grounding bus.
 - 2. Terminal Cabinets: Terminate grounding conductor on cabinet grounding terminal.
- I. Install grounding conductor from each standby-emergency generator to grounding electrode system. Provide flexible jumper between base and isolated generator.
- J. Install grounding electrode conductor from secondary side of each transformer to grounding electrode system as required for separately derived system.

- K. Install supply side bonding jumper from secondary side of each transformer to downstream distribution equipment, as required for separately derived system.
- L. Install grounding conductor to luminaires hanging from conduit swivel hangers.
- M. Install grounding conductor across all flexible or isolated support systems to maintain effective ground fault path.

3.4 GROUNDING ELECTRODE SYSTEM

- A. Install grounding for equipment room or generator enclosure consisting of ground bus, ground conductors, and grounding rods arranged as indicated on drawings.
 - 1. Ground bus shall be horizontal 1/4" x 12", copper bar. Bolt to wall at 3 ft intervals with 1 " stand-offs at each bus support.
 - 2. Install ground bus per details on drawings.
- B. Install No. #4/0 copper grounding conductor from ground bus to each ground rod.
- C. Install No. #4/0 copper grounding conductor from ground bus to buried ground loop.
- D. Install No. #4/0 copper grounding conductor from ground bus to structural steel.
- E. Install No. #4/0 copper grounding conductor from ground bus to cold water pipe where pipe enters the building.
- F. Install No. #4/0 copper grounding conductor from ground bus to concrete encased electrode.
- G. Install No. #4/0 copper grounding conductor from standby engine/generator to grounding electrode system.

3.5 SEQUENCING, SCHEDULING

- A. Permanently attach equipment grounds prior to energizing equipment.

3.6 INSTALLATION

- A. Contractor shall take photographs of all underground grounding and bonding connections prior to covering. Furthermore, Contractor shall coordinate with the Authority Having Jurisdiction to confirm approval of installation prior to covering.
- B. Connections: Exposed and visible for inspection at all times. Do not install insulation over ground connections.
- C. Identify all grounding conductors by system and room number of termination at building grounding electrode point.

- D. Grounding Conductors: Route along shortest and straightest paths possible, unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- E. Ground Rods: Install ground rods with the top 1' minimum below finished grade
 - 1. Interconnect ground rods with grounding electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating, if any.
 - 2. For grounding electrode system, install at least 3 rods spaced at least one rod length from each other and located at least the same distance from other grounding electrodes, and connect to the service grounding electrode conductor.
- F. Test Wells: Ground rod driven through drilled hole in bottom of handhole. Handholes shall be at least 12" deep, with cover.
 - 1. Test Wells: Install at least one test well for each service, unless otherwise indicated. Install at the ground rod electrically closest to service entrance. Set top of test well flush with finished grade or floor.
- G. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance, except where routed through short lengths of conduit.
 - 1. Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate any adjacent parts.
 - 2. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install so vibration is not transmitted to rigidly mounted equipment.
 - 3. Use exothermic-welded connectors for outdoor locations, but if a disconnect-type connection is required, use a bolted clamp.
- H. Grounding and Bonding for Piping:
 - 1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes, using a bolted clamp connector or by bolting a lug-type connector to a pipe flange, using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect grounding conductor on street side of fitting. Water pipe, by itself, is not an adequate grounding electrode and must be supplemented by another electrode system. Bond system together.
 - 2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
 - 3. Bond each aboveground portion of gas piping system downstream from equipment shutoff valve.
 - a. Confirm application with local gas and electric utility ordinances and notify the engineer if restrictions do not allow bonding as required by the project documents.
- I. Grounding Electrode Conductor installed in metal conduit or sleeves, shall be bonded at each end using grounding hubs. The bonding conductor shall be the same size as the grounding electrode conductor. Conduits terminated to grounding enclosures, shall be secured to enclosure using grounding locknuts.

- J. Bonding Interior Metal Ducts: Bond metal air ducts to equipment grounding conductors of associated fans, blowers, electric heaters, and air cleaners. Install tinned copper bonding jumper to bond across flexible duct connections to achieve continuity.
 - K. Bond steel columns at bases using copper conductor to column footing rebar.
 - L. Make grounding connections on surface that has been cleaned of paint, dirt, oil, etc., so that connections are bare metal to bare metal contact.
 - M. Make grounding connections tight with UL listed grounding devices, fittings, bushings, etc.
 - N. Concrete-Encased Grounding Electrode: Fabricate according to NFPA 70, using a minimum of 20 ft of bare copper conductor not smaller than #4 AWG.
 - 1. If concrete foundation is less than 20 ft long, coil excess conductor within base of foundation.
 - 2. Bond grounding conductor to reinforcing steel in at least four locations and to anchor bolts. Extend grounding conductor below grade and connect to building grounding grid or to grounding electrode external to concrete.
 - O. Equipment Grounding Conductor: Terminate in panelboard at equipment ground bus.
 - P. Multiple Conductors on Single Lug: Not permitted. Terminate each grounding and grounded conductor on its own terminal lug.
 - Q. Flexible Metallic Conduit, Non-Metallic Rigid Conduit, or Liquid Tight Flexible Conduit: Install equipment grounding conductor with phase conductors in conduit.
- 3.7 TELECOMMUNICATIONS BONDING AND GROUNDING SYSTEM INSTALLATION
- A. Provide required elements and miscellaneous hardware necessary to establish Telecommunication Bonding and Grounding infrastructure as specified.
 - B. Install products in accordance with manufacturer's instructions. Install Compression Connectors with compression, tool-and-die system, as recommended by manufacturer of connectors.
 - C. Telecommunications Bonding Conductor, Telecommunications Bonding Backbone (TBB), and Grounding Equalizer (GE): Compression or Exothermic type connections.
 - D. Locate TGBs and TMGB per drawings.
 - E. Telecommunications Bonding Backbone (TBB) shall be continuous and not interrupted by Telecommunications Grounding Busbars (TGB).
 - 1. TGBs shall be bonded to TBB via tap off of TBB. Exception: "last" TGB on TBB (e.g., furthest from TMGB).
 - 2. Grounding Equalizer(s) (GE) shall connect to TGBs to be interconnected.

- F. Insulate busbars from their support.

3.8 FIELD QUALITY CONTROL

- A. After installing grounding system but before permanent electrical circuits have been energized, test grounding system.
 - 1. Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal at ground test wells.
 - 2. Test grounding systems per requirements in Section 260812 - Power Distribution Acceptance Tests and 260813 - Power Distribution Acceptance Test Tables.
- B. Interpret test results in writing and submit to Engineer.

END OF SECTION

**SECTION 260529
HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260533 - Raceway and Boxes for Electrical Systems
- B. Section 262416.13 - Lighting and Appliance Panelboards
- C. Section 262816 - Enclosed Switches and Circuit Breakers
- D. Section 262913 - Enclosed Controllers
- E. Section 263213 - Engine Generators
- F. Section 263623 - Automatic Transfer Switches
- G. Section 265000 - Lighting

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes the following:
 - 1. Manufactured hangers and supports for individual raceways and cables, slotted channel and angle systems for multiple conduit runs, and most electrical equipment that is not floor mounted.
 - 2. Construction requirements for concrete housekeeping pads for floor-mounted electrical equipment.

1.4 REFERENCE STANDARDS

- A. AWS D1.1/D1.1M – Structural Welding Code-Steel.
- B. ASTM A36/A36M – Carbon Structural Steel.
- C. ASTM F3125/F3125M – Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength.

- D. ASTM A780/A780M – Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings.
 - E. MSS SP-58 – Pipe Hangers and Supports - Materials, Design and Manufacture.
 - F. MSS SP-69 – Pipe Hangers and Supports - Selection and Application.
 - G. MFMA-4 – Metal Framing Standards Publication.
 - H. NECA 1 – Standard Practices for Good Workmanship in Electrical Construction.
 - I. NECA 101 – Standard for Installing Steel Conduits (Rigid, IMC, EMT).
 - J. NFPA 70 – National Electrical Code.
 - K. SSPC-PA 1 – Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel.
 - L. ETL PVC-001 – PVC Coated Conduit
- 1.5 QUALITY ASSURANCE
- A. Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M, “Structural Welding Code – Steel.”
 - B. Comply with NFPA 70.
 - C. Certification:
 - 1. Installer of PVC-coated hangers and supports shall be certified by a PVC conduit manufacturer.

PART 2 - PRODUCTS

2.1 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

- A. Rated Strength: Adequate in tension, shear, and pullout force to resist maximum loads calculated or imposed for this Project, with a minimum structural safety factor of 5 times the applied force.
- B. Steel Slotted Support Systems: Comply with MFMA-4, factory-fabricated components for field assembly.
 - 1. Finishes
 - a. Metallic Coatings:
 - 1) Factory standard primed, galvanized or electroplated finish and applied according to MFMA-4, for indoor applications.
 - 2) Hot-dip galvanized after fabrication and applied according to MFMA-4, for outdoor applications.

- b. Nonmetallic Coatings: Manufacturer's standard PVC, polyurethane, or polyester coating applied according to MFMA-4, for corrosive environments.
 - c. Painted Coatings: Manufacturer's standard painted coating applied according to MFMA-4.
 - 2. Channel Dimensions: Selected for applicable load criteria.
 - 3. Manufacturers:
 - a. Allied Support Systems; Power-Strut Unit.
 - b. Cooper B-Line, Inc.; A division of Cooper Industries.
 - c. ERICO International Corporation.
 - d. GS Metals Corporation.
 - e. Thomas & Betts Corporation.
 - f. Unistrut; Tyco International, Ltd.
 - g. Wesanco, Inc.
 - h. National Pipe Hanger Corporation.
 - i. Michigan Hanger Co., Inc.; O-Strut Division.
 - j. Approved equal.
- C. Nonmetallic Slotted Support Systems: Structural-grade, factory-formed, glass-fiber-resin channels and angles with 9/16" diameter holes at a maximum of 8" o.c., in at least one surface.
 - 1. Fittings and Accessories: Products of channel and angle manufacturer and designed for use with those items.
 - 2. Fitting and Accessory Materials: Same as channels and angles, except metal items may be stainless steel.
 - 3. Rated Strength: Selected to suit applicable load criteria.
 - 4. Manufacturers:
 - a. Allied Support Systems; Power-Strut Unit
 - b. Cooper B-Line, Inc.; A division of Cooper Industries
 - c. Fabco Plastics Wholesale Limited
 - d. Seasafe, Inc.
 - e. Approved equal
- D. Raceway and Cable Supports: As described in NECA 1 and NECA 101.
- E. Raceway and Cable Support Devices: Steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.
- F. PVC Raceway Support Devices: ANSI C80.1, UL 6, ETL PVC-001.
- G. Support for 600V and under Conductors in Vertical Raceway: Factory-fabricated assembly consisting of threaded body and insulating wedging plug or plugs for non-armored electrical conductors or cables in riser conduits. Plugs shall have number, size, and shape of conductor gripping pieces as required to suit individual conductors or cables supported. Body shall be malleable iron.
 - 1. Manufacturers:
 - a. O-Z/Gedney
 - b. Approved equal

- H. Support for Conductors 601V and greater in Vertical Raceway: Heavy duty strain relief grip consisting of solder protector, galvanized steel mesh, endless multi-weave style.
 - 1. Manufacturers:
 - a. Hubbell-Kellems
 - b. Approved equal
- I. Structural Steel for Fabricated Supports and Restraints: ASTM A36/A36M, steel plates, shapes, and bars; black and galvanized.
- J. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:
 - 1. Concrete Anchors
 - a. Anchors shall be selected, sized, and detailed by Contractor's structural engineer registered in project's jurisdiction, based on project conditions and in accordance with project building code. Calculations and drawings shall be submitted.
 - b. Anchors shall meet ICC Acceptance Criteria, and ICC-ES Evaluation Reports (ESRs) shall specifically list the current applicable codes.
 - c. Anchors installed in hardened concrete for purpose of transmitting structural loads from one connected element to another, or for safety related elements such as sprinkler pipes, heavy suspended pipes, and barrier rails shall have ICC-ES report demonstrating anchors have met requirements of AC 193 for mechanical anchors in concrete elements.
 - d. Post-installed expansion anchors and undercut anchors installed in hardened concrete shall be qualified for strength design and tested according to ACI 355.2. Designs shall be per the requirements of ACI 318, Appendix D.
 - e. Anchors shall be zinc plated in accordance with ASTM B633.
 - f. Select anchors with load ratings based on cracked concrete conditions.
 - g. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
 - 1) Manufacturers:
 - a) Hilti Inc.
 - b) ITW Ramset/Red Head; A division of Illinois Tool Works, Inc.
 - c) MKT Fastening, LLC.
 - d) Simpson Strong-Tie Co., Inc.; Masterset Fastening Systems Unit
 - e) Approved equal
 - h. Mechanical-Expansion Anchors: Insert-wedge-type, stainless steel, for use in hardened portland cement concrete with tension, shear, and pullout capacities appropriate for supported loads and building materials in which used.
 - 1) Manufacturers:
 - a) Cooper B-Line, Inc.; A division of Cooper Industries
 - b) Empire Tool and Manufacturing Co., Inc.
 - c) Hilti Inc.
 - d) ITW Ramset/Red Head; A division of Illinois Tool Works, Inc.
 - e) MKT Fastening, LLC.

- f) Approved equal
- 2. Concrete Inserts: Steel or malleable-iron, slotted support system units similar to MSS Type 18; complying with MFMA-4 or MSS SP-58.
- 3. Clamps for Attachment to Steel Structural Elements: MSS SP-58, type suitable for attached structural element.
- K. Beam Clamps: C-clamps are allowed 3/8" or smaller and only for static loading such conduits. Provide locknut for hanging rod at clamp. C-clamps are not allowed for open web steel joist applications nor seismic applications.
- L. Through Bolts: Structural type, hex head, and high strength. Comply with ASTM A 325.
- M. Toggle Bolts: All-steel springhead type.
- N. Hanger Rods:
 - 1. MSS SP-58; threaded steel, with adjusting and lock nuts; electroplated zinc finish.
 - 2. MSS SP-58; nonmetallic, with adjusting and lock nuts.
- 2.2 FABRICATED METAL FRAMING EQUIPMENT SUPPORT ASSEMBLIES
 - A. Description: Welded or bolted, structural steel shapes, shop or field fabricated to fit dimensions of supported equipment.
 - B. Materials: Comply with requirements in Division 05 Section "Metal Fabrications" for steel shapes and plates; not be lighter than 12 ga.
 - C. Finish: Epoxy paint
 - D. Manufacturers: Same as in paragraph 2.1.B.3 above.
- 2.3 CONTINUOUS INSERT CHANNELS
 - A. Length and support capabilities to be suitable for application.
 - B. Brackets, inserts and accessories suitable for channel insert selected.
 - C. Manufacturers:
 - 1. Unistrut; Tyco International, Ltd.
 - 2. Cooper B-Line, Inc.; A division of Cooper Industries
 - 3. Michigan Hanger Co., O-Strut Division
 - 4. Anvil International, Inc.
 - 5. Approved equal

PART 3 - EXECUTION

3.1 APPLICATION

- A. Comply with NECA 1 and NECA 101 for application of hangers and supports for electrical equipment and systems except if requirements in this Section are stricter.
- B. Maximum Support Spacing and Minimum Hanger Rod Size for Raceway: Space supports for EMT, IMC, and RMC as required by NFPA 70.
 - 1. Size steel hanger rods for individual hangers and trapeze supports as indicated in the following schedule. Total weight of equipment shall not exceed limits indicated.

Maximum Loads	Rod Diameter	Maximum Pipe Size With Single Rod
730lbs	3/8"	2"
1130lbs	1/2"	3"
1818lbs	5/8"	5"

- C. Multiple Raceways or Cables: Install trapeze-type supports fabricated with steel slotted or other support system, sized so capacity can be increased by at least 25% in future without exceeding specified design load limits.
 - 1. Secure raceways and cables to these supports with 2-bolt conduit clamps.
- D. Spring-steel clamps designed for supporting single conduits without bolts may be used for 3/4" and smaller raceways serving branch circuits and communication systems above suspended ceilings and for fastening raceways to trapeze supports.
- E. Install PVC-coated hangers and supports in areas with corrosive atmosphere as noted on plans.

3.2 SUPPORT INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements, except as specified in paragraphs below.
- B. Raceway Support Methods: In addition to methods described in NECA 1, EMT may be supported by openings through structure members, as permitted in NFPA 70.
- C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.
- D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor application size and placement shall be reviewed and approved by Structural Engineer prior to installation. Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:

1. To Wood: Fasten with lag screws or through bolts.
 2. To New Concrete: Bolt to concrete inserts.
 3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
 4. To Existing Concrete: Expansion anchor fasteners.
 5. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4" thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4" thick.
 6. To Steel: Welded threaded studs complying with AWS D1.1/D1.1M, with lock washers and nuts.
 7. To Light Steel: Sheet metal screws.
 8. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate by means that meet seismic-restraint strength and anchorage requirements.
- E. Drill holes for expansion anchors in concrete at locations and to depths that avoid reinforcing bars.
- F. Do not support raceway by other raceway.
- G. Do not support equipment or raceway from metal roof decking or floor decking.
- H. Do not impose weight of electrical equipment, raceways, or lighting fixtures on support provided for other trades or systems.
- I. Top or bottom chords of open web steel joists may be used to support loads provided total load within panel does not exceed 100 lbs and load is placed concentric to joist (panel is length of chord between two adjacent diagonal web members at point of connection to chord).
1. C-clamps are not permitted for use in open web steel joist applications.
- J. Suspend hangers by means of hanger rods. Perforated band iron and flat wire (strap iron) are not allowed.
- K. Use conduit-mounting pedestals for piping on roof. Install bottom of pedestal flat on roof deck and insulate exterior of pedestal, flashing and counter flashing.
- L. Minimize use of concrete anchors and inserts after concrete pour.
- M. Punching, drilling, welding of building structural steel or welding attachment to building structural steel is not allowed, unless approved by structural engineer.
- N. Use tools approved for use with PVC-coated conduits and fittings.

3.3 INSTALLATION OF FABRICATED METAL SUPPORTS

- A. Comply with installation requirements in Division 05 Section "Metal Fabrications" for site-fabricated metal supports.
- B. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.
- C. Field Welding: Comply with AWS D1.1/D1.1M.

3.4 CONCRETE HOUSEKEEPING PADS

- A. Construct concrete housekeeping pads for all floor-mounted electrical equipment.
- B. Dimensions: 3.5" high and not less than 2" larger in both directions than supported equipment, so anchors will be a minimum of 10 bolt diameters from edge of the base.
- C. Use 3000psi, 28-day compressive-strength concrete. Concrete materials, reinforcement, and placement requirements are specified in Division 03 Section "Cast-in-Place Concrete."
- D. Anchor equipment to concrete housekeeping pad.
 - 1. Place and secure anchorage devices. Use supported equipment manufacturer's setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 - 2. Install anchor bolts to elevations required for proper attachment to supported equipment.
 - 3. Install anchor bolts according to anchor-bolt manufacturer's written instructions.
- E. Coordinate with Architect installation of housekeeping pads on roof.

3.5 PAINTING

- A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.
 - 1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils.
- B. Touchup: Comply with requirements in Division 09 Section "Painting" for cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.
- C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A780/A780M.

END OF SECTION

**SECTION 260533
RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 260529 - Hangers and Supports for Electrical Systems
- D. Section 260553 - Electrical Systems Identification
- E. Section 260593 - Electrical Systems Firestopping
- F. Section 262726 - Wiring Devices

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes raceways, fittings, wireways, wall ducts, indoor service poles, outlet boxes, pull and junction boxes, floor boxes, tap boxes and raceway seals.

1.4 REFERENCE STANDARDS

- A. ANSI/NECA 1 – Standard Practices for Good Workmanship in Electrical Contracting
- B. ANSI C80.1 – Rigid Steel Conduit-Zinc Coated (GRS)
- C. ANSI C80.3 – Electrical Metallic Tubing-Zinc Coated (EMT)
- D. ANSI C80.5 – Aluminum Rigid Conduit-Zinc Coated (ARC)
- E. ANSI C80.6 – Intermediate Metal Conduit-Zinc Coated (IMC)
- F. ASTM A53/A53M – Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
- G. BICSI TDMM – Telecommunications Distribution Methods Manual, Latest Edition

- H. ETL PVC-001 – PVC-Coated Conduit
- I. NEMA EN 10250 – Enclosures for Electrical Equipment (1000 V Maximum)
- J. NEMA FB 1 – Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing and Cable
- K. NEMA OS 1 – Sheet-Steel Outlet Boxes, Device Boxes, Covers, and Box Supports
- L. NEMA OS 2 – Nonmetallic Outlet Boxes, Device Boxes, Covers, and Box Supports
- M. NEMA RN 1 – Polyvinyl Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
- N. NEMA TC 2 – Electrical Polyvinyl Chloride (PVC) Conduit
- O. NEMA TC 3 – PVC Fittings for Use with Rigid PVC Conduit and Tubing
- P. NEMA BI 50058 – Electrical Nonmetallic Tubing (ENT)
- Q. NFPA 70 – National Electrical Code
- R. TIA-569-B – Commercial Building Standard for Telecommunications Pathways and Spaces
- S. UL 1 – Flexible Metal Conduit
- T. UL 6 – Electrical Rigid Metallic Conduit-Steel
- U. UL 6A – Electrical Rigid Metallic Conduit-Aluminum and Stainless Steel
- V. UL 360 – Liquid-Tight Flexible Steel Conduit
- W. UL 514A – Metallic Outlet Boxes
- X. UL 514B – Conduit, Tubing, and Cable Fittings
- Y. UL 514C – Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
- Z. UL 651 – Schedule 40 and 80 Rigid PVC Conduit and Fittings
- AA. UL 797 – Electrical Metallic Tubing-Steel
- BB. UL 870 – Wireways, Auxiliary Gutters, and Associated Fittings
- CC. UL 1242 – Electrical Intermediate Metal Conduit-Steel
- DD. UL 1660 – Liquid-Tight Flexible Nonmetallic Conduit

EE. UL 2024 – Optical Fiber and Communication Cable Raceway

1.5 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Comply with NFPA 70.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.
- B. Certification:
 - 1. Installer of PVC-coated conduits and fitting shall be certified by a PVC conduit manufacturer.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory wrapping or provide additional canvas or plastic cover to protect from dirt, water, construction debris, and traffic.
- B. Protect PVC conduit from sunlight.
- C. Comply with manufacturer's written instructions.

1.7 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr written warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 RIGID METAL CONDUIT (RMC)

- A. Rigid Steel Conduit (RSC): ANSI C80.1, UL 6; heavy wall galvanized steel
- B. Intermediate Metal Conduit (IMC): ANSI C80.6, UL 1242; thinner wall, galvanized steel
- C. Rigid Aluminum Conduit (RAC): ANSI C80.5; heavy wall aluminum
- D. PVC coated rigid steel conduit: NEMA RN 1, ANSI C80.1, UL 6, ETL PVC-001; plastic cap protector caps

- E. Fittings (couplings, connectors and bushings): NEMA FB 1, UL 514B; steel and aluminum alloy (concrete-tight where applicable); threaded; connectors with double locknuts and steel insulating bushings, thermoplastic insulating bushings for conduits 2" and smaller.
- F. Fittings (conduit bodies): NEMA FB 1, UL 514B; stainless-steel; cover: steel, with stainless steel screws and neoprene gaskets; PVC coated to match conduit.
- G. Fittings Manufacturers: Cooper Crouse-Hinds; Carlon Electric Products/Prime Conduit Inc.; O-Z/Gedney; Appleton; Hubbell; Approved equal

2.2 ELECTRICAL METALLIC TUBING (EMT)

- A. ANSI C80.3, UL 797; galvanized steel tubing
- B. Fittings (couplings and connectors): NEMA FB 1, UL 514B; steel, watertight gland compression type connectors with double locknuts and insulated throat. Indentor, drive-on, zinc die-cast or pressure cast not permitted.
- C. Fittings (conduit bodies): NEMA FB 1, UL 514B: steel, with stainless steel screws and neoprene gaskets.
- D. Fittings Manufacturers: Same as manufacturers listed in 2.1.G.

2.3 FLEXIBLE METAL CONDUIT (FMC)

- A. UL 1; interlocked steel
- B. Fittings: NEMA FB 1, UL 514B; steel, die-cast fittings not permitted

2.4 LIQUIDTIGHT FLEXIBLE METAL CONDUIT (LFMC)

- A. UL 360; interlocked steel or, with PVC jacket
- B. Fittings: NEMA FB 1, UL 514B; steel

2.5 RIGID NONMETALLIC CONDUIT (RNC)

- A. NEMA TC 2, UL 651; Schedule 40 PVC
- B. Fittings: NEMA TC 3, UL 651

2.6 ELECTRICAL NONMETALLIC TUBING (ENT)

- A. NEMA BI 50058; hand-bendable, corrugated PVC conduit
- B. Fittings: NEMA TC 3

2.7 LIQUIDTIGHT FLEXIBLE NONMETALLIC CONDUIT (LFNC)

- A. UL 1660; flexible conduit with a plastic sheath
- B. Fittings: UL 514B

2.8 NONMETALLIC WIREWAYS

- A. NEMA EN 10250, UL 870; flame-retardant, nonmetallic troughs with removable cover
- B. Material: PVC plastic, extruded and fabricated to size and shape indicated, with snap-on cover and mechanically coupled connections with plastic fasteners.
- C. Fittings and Accessories: Couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps, and other fittings to match and mated with wireways as required for complete system.
- D. Manufacturers: Hoffman Engineering Co.; Lamson & Sessions; Carlon Electrical Products; Square D Co.; Approved equal

2.9 INDOOR SERVICE POLES

- A. Description: Freestanding, two compartments utility columns with receptacles for power branch circuits and provisions for communication outlets.
- B. Materials: Main Body: Steel; cover plates: cover for each compartment.
- C. Components:
 - 1. Wiring devices factory installed, wired, and covers labeled with panel number and circuit number, voltage, phase, and amperes, as identified on drawings, per requirements in Sections 260519 - Low-Voltage Electrical Power Conductors and Cables and 262726 - Wiring Devices.
 - 2. 12" pigtails at feed locations customized to match length shown on drawings.
 - 3. Equivalent distance between receptacles; number of receptacles per length of pole as shown on drawings.
 - 4. Factory installed, NFPA 70 sized grounding conductors, per requirements in Section 260526 - Grounding and Bonding for Electrical Systems.
 - 5. Pole cover with hole-cut provisions for communication outlets.
- D. Accessories: Trim plates for closing ceiling opening; fittings available as standard accessories; foot: suitable for floor finish as indicated on drawings; top clamp: concealed, designed to fasten pole to inverted "T" grid ceiling suspension member.
- E. Communications Outlets and Accessories:
 - 1. Factory-fabricated provisions for connectivity outlets, modular inserts, and other outlets for indicated configurations and service requirements.
 - a. Fasteners: Types specified in manufacturer's installation instructions.

- F. Finish: Manufacturer's standard gray color.
- G. Manufacturers: Wiremold; Approved equal

2.10 OUTLET BOXES

- A. Sheet Metal Outlet Boxes: NEMA OS 1, UL 514A galvanized steel with stamped knockouts.
 - 1. Luminaire and Equipment Supporting Boxes: Rated for weight of equipment supported; 1/2" male fixture studs, where required.
 - 2. Concrete Ceiling Boxes: Concrete type
- B. Cast-Metal Outlet Boxes: NEMA FB 1, ferrous alloy, Type FD, with gasketed cover and threaded hubs
 - 1. For applications requiring more than 2 gang boxes, provide stainless steel custom fabricated welded boxes with threaded hubs and coverplate. For applications including terminations and splicing of power conductors, a standard UL Listed box shall be used inside of the custom fabricated box.
- C. Nonmetallic Outlet Boxes: NEMA OS 2
- D. Gangable type boxes are not allowed.
- E. Manufacturers: O-Z/Gedney; Raco; Cooper Crouse-Hinds; Approved equal

2.11 OUTLET BOXES FOR COMMUNICATIONS

- A. Minimum outlet box size: 4-11/16" square by 2-1/8" deep minimum, with single-gang trim ring, unless otherwise noted on drawings.
 - 1. Total depth of the assembly including the trim ring shall not be less than 2-1/2".

2.12 PULL AND JUNCTION BOXES

- A. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1; galvanized steel
- B. Cast-Metal, Pull, and Junction Boxes: NEMA FB 1; cast aluminum with ground flange, gasketed cover and stainless steel cover screws
- C. Minimum size: 4" square by 2-1/8" deep for use with 1" conduit and smaller; 4-11/16" square by 2-1/8" deep for use with 1-1/4" conduit and larger
- D. Sheet Metal Boxes Larger Than 12" in any direction: Hinged cover or a chain installed between box and cover
- E. Field-fabricated boxes not allowed without prior approval of local authority having jurisdiction.
- F. Manufacturers: O-Z/Gedney; Raco; Cooper Crouse-Hinds; Approved equal

2.13 PULL AND JUNCTION BOXES FOR COMMUNICATIONS

- A. Size: Per TIA-569-B, unless otherwise noted on drawings.
- B. Minimum pull box size: 4-11/16" square by 2-1/8" deep, where pull box is used with raceway(s) smaller than 1-1/4" trade size, unless otherwise noted on drawings.
- C. Minimum pull box size, where pull box is used with raceway(s) 1-1/4" trade size or larger:
 - 1. For straight pull through: Length of at least 8 times trade-size diameter of largest raceway.
 - 2. For angle and U pulls:
 - a. Have distance between each raceway entry inside box and opposite wall of box of at least 6 times trade-size diameter of largest raceway, this distance being increased by sum of trade-size diameters of other raceways on same wall of box; and
 - b. Have distance between nearest edges of each raceway entry enclosing same conductor of at least:
 - 1) Six times trade-size diameter of raceway; or
 - 2) Six times trade-size diameter of larger raceway if raceways are of different sizes.
 - c. For raceway entering wall of pull box opposite to removable cover, have distance from wall to cover of not less than trade-size diameter of largest raceway plus 6 times diameter of largest conductor.

2.14 POKE-THROUGH FITTINGS

- A. Poke-Through Fittings: Assembly comprising service fitting, poke-through component, fire stops and smoke barriers, and junction box for conduit termination.
 - 1. Fire Rating: 3 h
 - 2. Service Fitting Type: Flush
 - 3. Housing: Satin aluminum
 - 4. Configuration: One receptacle and one communication outlet or as indicated on drawings.
 - 5. Manufacturers: Hubbell, Legrand/Wiremold, FSR; Approved equal

2.15 FLOOR BOXES

- A. Metal Floor Boxes: NEMA OS 1; cast metal; semi-adjustable; rectangular; Moisture-proof, with forged brass blank cover with each box and close up covers and/or carpet flanges as required for finished floor.
- B. Nonmetallic Floor Boxes: nonadjustable, round

2.16 MULTISERVICE FLOOR BOXES

- A. Above Grade: Stamped steel, watertight design approved for use on above-grade concrete floor applications, with four independent wiring compartments and capacity for up to four duplex receptacles and/or communication services. The box: fully adjustable providing pre-pour and after-pour adjustment, tunnel compartment, and two receptacle brackets. Conduit knockouts per drawing requirements. Comply with UL 514A and UL 514C scrub water exclusion test for tile, terrazzo, carpet and wood floors.
- B. On Grade: Cast iron or steel pour box, watertight design approved for use in on-grade and above-grade concrete floor applications, with four independent wiring compartments and capacity for up to four duplex receptacles and/or communication devices. The box: fully adjustable providing pre-pour and after-pour adjustment, tunnel compartment, and two receptacle brackets. Conduit knockouts per drawing requirements. Comply with UL 514A and UL 514C scrub water exclusion test for tile, terrazzo, carpet and wood floors.
- C. Covers: Activation Covers – Die-cast aluminum with textured aluminum finish, and black or brass powder-coated paint finishes as selected by the Architect. Cover: flanged or flangeless, as required, with options for tile or carpet inserts, blank covers, or covers with one or two 1" liquid tight conduit openings for furniture feed applications.
- D. Manufacturers:
 - 1. Hubbell - HBLCFB Series
 - 2. Spider - AFB/CFB Series
 - 3. Legrand/Wiremold - Evolution Series

2.17 TAP BOXES

- A. Multi-tap connectors as indicated on drawings.
- B. Manufacturers: IlSCO; Approved equal

2.18 EXPANSION FITTINGS

- A. Malleable iron, hot dip galvanized allowing 4" raceway movement.
- B. Manufacturers: OZ/Gedney AX Series; or equivalent by manufacturer listed in 2.1.G.

2.19 RACEWAY PENETRATION SEALS

- A. Thruwall and Floor Seals.
- B. Manufacturers: New construction – OZ/Gedney FSK Series; existing construction – OZ/Gedney CSM Series; or equivalent by manufacturer listed in 2.1.G.

2.20 RACEWAY SEALING FITTINGS

- A. For one through four conductors: Manufacturers: OZ/Gedney CSB Series; Approved equal
- B. For greater than four conductors: Manufacturers: OZ/Gedney EYA Series with sealing compound; Approved equal
- C. Low-temperature or hazardous locations: Manufacturers: OZ/Gedney EYA Series with sealing compound; Approved equal

2.21 CABLE SUPPORTS

- A. Manufacturers: OZ/Gedney Type S; or equivalent by manufacturer listed in 2.1.G.

2.22 SLEEVES FOR RACEWAYS

- A. Steel Pipe Sleeves: ASTM A53/A53M, Type E, Grade B, Schedule 40, galvanized steel, plain ends, with integral water stop.
- B. Sleeves for Rectangular Openings: Galvanized sheet steel with minimum 0.052" or 0.138" thickness and of length to suit application.
- C. Integral Water Stop: Manufacturer: Thunderline Corporation; Approved equal
 - 1. High density polyethylene (HDPE). Type Century-Line engineered sleeve with end caps.
 - 2. Steel. Type WS engineered sleeve.

2.23 SLEEVE SEALS

- A. Description: Modular sealing device, designed for field assembly, to fill annular space between sleeve and cable.
 - 1. Sealing Elements: EPDM interlocking links shaped to fit surface of cable or conduit. Include type and number required for material and size of raceway or cable.
 - 2. Pressure Plates: Stainless steel. Include two for each sealing element.
 - 3. Connecting Bolts and Nuts: Stainless steel of length required to secure pressure plates to sealing elements. Include one for each sealing element.

PART 3 - EXECUTION

3.1 COORDINATION

- A. Coordinate with Architect/Engineer size and location of required built-in openings in building structure, including those sleeved, formed or core drilled.

- B. Coordinate with Architect/Engineer cutting, removing, or piercing general or mechanical insulation, fire-rated walls, ceilings or steelwork.
- C. Verify with Architect/Engineer all surface raceway installations except in mechanical, electrical, and communications rooms.
- D. Coordinate with Architect/Engineer exact locations of floor boxes, where shown on drawings, prior to rough-in.
- E. Coordinate routing of through-roof conduits.
- F. Coordinate sleeve selection and application with selection and application of firestopping specified in Section 260593 - Electrical Systems Firestopping.
- G. Verify that exterior wall or wet location boxes are gasketed type cast boxes with matching cover.
- H. Verify with manufacturer that "touch-up" paint kit and PVC-coating kit are available for use.

3.2 EXAMINATION

- A. Examine surfaces to receive raceways and boxes for compliance with installation tolerances and other conditions affecting performance of raceway's installation. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.3 INSTALLATION

- A. Raceways:
 - 1. Comply with ANSI/NECA 1 and NFPA 70 for installation requirements applicable to products specified in Part 2 except where requirements on drawings or in this Section are stricter.
 - 2. Arrange raceways to maintain headroom and present neat appearance.
 - 3. Raceway routing is shown in approximate locations, unless dimensioned. Route to complete raceway installation before starting conductor installation.
 - 4. Keep raceways at least 6 " away from parallel runs of flues, steam, hot-water pipes or ductwork. Install horizontal raceway runs above water and steam piping. Install raceways level and square and at proper elevations: 6'-6" minimum headroom, except in exit pathways 7'0" minimum headroom. Do not block access to junction boxes, mechanical equipment or prevent removal of ceiling panels, etc.
 - 5. Run raceways concealed in construction to avoid adverse conditions such as heat and moisture, to permit drainage, and to avoid materials and equipment of other trades, except where noted otherwise.

6. Avoid exposed raceway runs. Run raceways exposed where impractical or impossible to conceal or where specific approval is obtained. Run exposed raceways grouped and parallel or perpendicular to construction. Do not route exposed raceways over boilers or other high-temperature machinery or in contact with such equipment. Offset exposed raceways at boxes.
7. Route raceways installed above accessible ceilings parallel or perpendicular to construction.
8. Do not install raceways in structural or topping floor slabs, except where noted on the plans. Install raceway in structural or topping floor slabs, where noted on plans, as follows:
 - a. Center raceways in structural slabs clear of reinforcing steel, except where crossing same, and spaced on centers equal or exceeding 3 times the raceway diameter. Secure raceways to reinforcing rods to prevent sagging or shifting during concrete placement. Space raceways laterally to prevent voids in concrete.
 - b. Outside diameter of raceway shall not exceed 1/3 the structural slab thickness.
 - c. Obtain approval from Engineer for each run of raceway 1" or larger.
 - d. Do not install raceways in topping slabs of 2" or less.
 - e. Locate raceways to avoid conflict with equipment, door bucks, partitions and other equipment bolted to floor.
 - f. Arrange stub-ups so curved portions of bends are not visible above finished slab. Install with an adjustable top or coupling threaded inside for plugs set flush with finished floor. Extend conductors to equipment with rigid steel conduit; use flexible metal conduit 6" above the floor. Install threaded plugs flush with floor for future equipment connections.
 - g. Change from nonmetallic raceway to RMC or IMC before rising above floor.
9. Cut raceways square using saw or pipecutter.
10. Use hydraulic one-shot raceway bender or factory elbows for bends in raceway larger than 2", unless sweep elbows required. Bend raceways according to manufacturer's recommendations. Do not use torches or open flame to aid in bend of PVC conduit.
11. Use raceway fittings compatible with raceways and suitable for use and environment.
12. Provide bushings on all raceways stubbed above accessible ceilings.
13. Raceways minimum sizes:
 - a. Minimum raceway size 3/4", except as noted on drawings.
 - b. Minimum home run size: 1", except as noted on drawings.
 - c. Minimum size for flexible metal conduit is 1/2" except 3/8" for luminaires.
 - d. Minimum size for liquidtight flexible metal conduit is 1/2"
14. Install empty raceways 2-1/2" and larger with 10 ga galvanized fishwire; install 200 lb nylon pull cord in raceways smaller than 2-1/2"; leave at least 12" of slack at each end of pull wire. Cap raceways at both ends.
15. Feed devices on same wall vertically from above or junction box in suspended ceiling.
 - a. Do not install horizontal bends in conduit around corners.
 - b. Feed devices in exterior or load-bearing walls by horizontal conduit runs.

16. Raceways Supports:
 - a. Independently support or attach raceway system to structural parts of construction. Suspended ceiling systems shall not be considered as structural parts of construction for raceway support. Do not attach raceways to piping system.
 - b. Raceway supports for horizontal or vertical single runs:
 - 1) Hot dipped galvanized heavy-duty sheet steel straps, mineralac clamps or steel slotted support channel system with appropriate components.
 - 2) Spring steel type pressure clamps for raceways 3/4" and smaller.
 - c. Raceway supports for horizontal and vertical multiple runs:
 - 1) Trapeze-type supports fabricated with steel slotted channel systems with appropriate components.
 - 2) Support horizontal runs with appropriately sized rods.
 - 3) Anchor vertical runs to structure.
 - 4) Spring-steel type pressure clamps for raceways 3/4" and smaller.
 - d. Vertical raceway runs 1-1/4" and larger passing through floors: Support at each floor with pipe riser clamps.
 - e. Do not support raceways with wire, perforated pipe straps or plastic tie-wrap. Remove wires used for temporary support.
 - f. Secure raceways in metal stud walls to prevent rattling.
 - g. Arrange raceway supports to prevent misalignment during wiring installation.
 - h. Do not fasten raceways to corrugated metal roof deck.
 - i. For fasteners and supports, including steel slotted support systems, support devices, support spacing, support of conductors in vertical raceways, and hanger rod size, refer to Section 260529 - Hangers and Supports for Electrical Systems and NFPA 70.
17. Raceways Seismic Restraints:
 - a. Avoid raceway runs crossing building seismic joints. Use flexible connections where crossings cannot be avoided.
18. Identify raceways per requirements in Section 260553 - Electrical Systems Identification.
19. Ground raceways per requirements in Section 260526 - Grounding and Bonding for Electrical Systems.
20. Flexible Conduit Connections: Use maximum of 72" of flexible conduit for recessed and semi-recessed lighting fixtures, equipment subject to vibration, noise transmission, or movement; and for motors.
 - a. Use LFMC or LFNC in damp or wet locations not subject to severe physical damage.
21. Install PVC-coated raceways in areas with corrosive atmosphere as noted on plans.
22. Use tools approved for use with PVC coated conduits and fittings.
23. Install stainless steel raceway clamps, mounting hardware, supports, hangers, etc., when located in "wet" or "wash-down" areas.
24. Communications Raceways:
 - a. Minimum communications raceway size: 1", unless otherwise noted on drawings.
 - b. Install one raceway from each communications outlet box. Horizontal raceway runs between wall outlet boxes are not allowed.
 - c. Terminate raceway above closest accessible ceiling space.

- d. Install insulated bushings on end of each raceway.
- e. Use UL listed metallic grounding clamps, when terminating raceway on cable tray.
- f. Install flush two-gang box with single-gang trim ring for each communications outlet or as noted on drawings.
- g. Install with no more than 180 degrees of bends between pull or junction boxes or terminations at distribution frames or cabinets where necessary to comply with these requirements.
- h. Conduit bend radii (minimum) shall be:
 - 1) Six (6) times internal conduit diameter for conduit 2" or less internal diameter.
 - 2) Ten (10) times internal conduit diameter for conduit greater than 2" internal diameter.
- i. Conduit bends shall be smooth, even, and free of kinks or other discontinuities that may have detrimental effects on pulling tension or cable integrity during or after installation.
- j. Do not install 90-degree condulets. Install continuous radius sweeps of 45° minimum for 90-degree bends.
- k. Do not install continuous sections longer than 100 ft.
- l. Reference Legend for nurse call outlet box sizes.
- m. Install nylon pull cord in empty raceways. Leave at least 12" of slack at each end of pull wire. Cap raceways at both ends.

B. Indoor Service Poles:

- 1. Install wiring devices and communications outlets of type, quantity and spacing as indicated on drawings.
- 2. Maintain ground continuity throughout entire pole length per requirements in Section 260526 - Grounding and Bonding for Electrical Systems.
- 3. Raceway receptacle faceplates shall be labeled with adhesive labels with 1/4" high lettering per requirements in Section 260553 - Electrical Systems Identification, indicating receptacle's voltage, phase, and amperage (i.e., 120V, 1-phase, 20A) at top of receptacle, and panel and circuit designation (i.e., NLP-D2-2/12) at bottom of receptacle.
- 4. Neatly cut openings in ceiling panels. Install trim plate.

C. Boxes:

- 1. Install boxes to accommodate device indicated by symbol, in conformance with code requirements, number and size of conductors and splices and consistent with type of construction.
- 2. Install the appropriate cover on surface-mounted boxes:
 - a. Raised device covers on 4" square and 4-11/16" boxes and handy box covers on handy boxes, etc.
 - b. Device covers that are square drawn or square cut on boxes in block.
 - c. Tile covers on boxes in tile.
 - d. Round drawn device covers on boxes in lath and plaster walls or dry wall only.
 - e. Set front edge of device boxes flush with finished wall surfaces except on walls of non-combustible materials where boxes may have maximum set back of 1/4". Secure flush-mounted box to interior wall and partition studs. Accurately position to allow for surface finish thickness.
- 3. Set outlet boxes parallel to construction and independently attached to same.

4. Do not install back-to-back and through-the-wall boxes. Install with minimum 6" horizontal separation between closest edges of the boxes. Install with minimum 24" separation in acoustic-rated walls and fire-rated walls.
 5. Install multi-ganged boxes where 2 or more devices are in same location, unless otherwise noted.
 6. Box Support:
 - a. Mount boxes straight.
 - b. Install horizontal bracing at top or bottom of box for 3 or more gang device boxes in stud walls.
 - c. Install stud support one side, with short piece of stud, for up to 2 gang device boxes.
 - d. Do not support boxes with tie-wire.
 - e. For one and two gang box support, manufactured bracket supports shall be accepted alternate.
 - f. Support boxes independently of raceways.
 - g. Install adjustable steel channel fasteners for hung ceiling outlet box.
 - h. Install stamped steel bridges to fasten flush-mounted outlet box between studs.
 - i. Do not install boxes to ceiling support wires or piping systems.
 7. Install partitions in multi-ganged boxes where different types of devices are installed, or devices installed operate at different voltages.
 8. Mount boxes in block walls at block joint nearest to indicated height.
 9. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block, and install box flush with surface of wall.
 10. When boxes are installed in fire-resistive walls and partitions, provide 24" horizontal separation between boxes on opposite sides of a wall. In addition, limit penetrations to 16 sq in per penetration and not to exceed a total of 100 sq in per 100 sq ft of wall area. Apply fire stop putty pads acceptable to the fire marshal.
 11. Pull and junction boxes: Install as shown, or as necessary to facilitate pulling of wire and to limit number of bends within code requirements. Install above accessible ceilings and in unfinished areas.
 12. Install boxes to be permanently accessible.
 13. Do not intermix conductors from more than one system in same junction box or pull box, unless shown or specifically authorized otherwise.
 14. Adjust box location up to 10' prior to rough-in to accommodate intended purpose. Coordinate exact location with Architectural drawings prior to rough-in.
 15. Orient boxes to accommodate wiring devices oriented as specified in Section 262726 - Wiring Devices.
 16. Inaccessible Ceiling Areas: Install outlet and junction boxes no more than 6" from ceiling access panel or from removable recessed luminaire.
 17. The drawings do not necessarily show every outlet, pull or junction box required. Add all required boxes as necessary.
- D. Outlet Boxes for Communications:
1. Install communications outlet boxes for each communications outlet, or as noted on drawings.
 2. Coordinate with other trades to maintain 8" clear space (minimum, measured from box centerline) on all sides of wall-mounted telephone outlet box.

- E. Pull and Junction Boxes for Communications:
 - 1. Position Communications Pull and Junction Boxes:
 - a. In any section of conduit longer than 100 ft
 - b. Where there are bends totaling more than 180 degrees between pull points or pull boxes
 - c. Wherever there is a reverse bend in run
 - 2. Do not use pull boxes in place of bends on straight section of raceway, unless otherwise shown on drawings.
- F. Floor Boxes:
 - 1. Set metal floor boxes level and flush with finished floor surface.
 - 2. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.
 - 3. Use cast floor boxes for installations in slab on grade.
 - 4. Install floor boxes and fittings to preserve fire-resistant rating of slabs and other elements, using materials and methods specified in Section 260593 - Electrical Systems Firestopping.
 - 5. Power and IT or AV conduits require a minimum 12" separation where routed parallel including entry into floor boxes.
- G. Expansion Fittings:
 - 1. Install raceway expansion and deflection fittings in all raceway runs embedded in or penetrating concrete where movement perpendicular to axis of the raceway may be encountered.
 - 2. Install raceway expansion fittings complete with bonding jumpers in raceway runs that cross expansion joints in structure and raceway runs mechanically attached to 2 separate structures.
 - 3. Use couplings and flexible connection made up of 24" length of flexible metal conduit, where EMT runs across expansion joints in ceiling spaces.
 - 4. Install fitting(s) that provide expansion and contraction for at least 0.0004" per ft of length of straight run per 1°F of temperature change.
 - 5. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at time of installation.
- H. Raceway Penetration Seals:
 - 1. Seal space outside of sleeves with grout for penetrations of concrete and masonry and with approved joint compound for gypsum board assemblies.
 - 2. Interior Penetrations of Non-Fire-Rated Walls and Floors: Seal annular space between sleeve and raceway, using joint sealant appropriate for size, depth, and location of joint. Refer to Division 07 Section "Maintenance of Joint Protection" for materials and installation.
 - 3. Fire-Rated-Assembly Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at raceway penetrations. Install sleeves and seal with firestop materials. Comply with Section 260593 - Electrical Systems Firestopping.
 - 4. Roof-Penetration Sleeves: Seal penetration of individual raceways with flexible, boot-type flashing units applied in coordination with roofing work.

5. Aboveground, Exterior-Wall Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Select sleeve size to allow for 1" annual clear space between pipe and sleeve for installing mechanical sleeve seals.
 6. Underground, Exterior-Wall Penetrations: Install cast-iron "wall pipes" for sleeves. Size sleeves to allow for 1" annual clear space between raceway and sleeve for installing mechanical sleeve seals.
 7. Sleeve-Seal Installation: Use type and number of sealing elements recommended by manufacturer for raceway material and size. Position raceway in center of sleeve. Assemble mechanical sleeve seals and install in annular space between raceway and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.
 8. Provide chrome- or nickel-plated escutcheons where raceways pass through walls, floors or ceilings and are exposed in finished areas. Size escutcheons to fit raceways for finished appearance. Finished areas shall not include mechanical/electrical rooms, janitor's closets, storage rooms, etc., unless suspended ceilings are specified.
 9. Remove temporary sleeves, if used for form wall openings, prior to installation of permanent materials.
- I. Raceway Sealing Fittings:
1. Install listed watertight seals to prevent the passage of moisture and water vapor through raceway, where raceway passes from interior to exterior of the building, where raceway passes between areas of different temperatures such as into or out of cold rooms or freezers, where raceway enters room which at any time is subject to low or high temperatures and where raceway enters a room which at any time is subject to internal air pressures above or below normal.
 2. Install watertight seals in interior of all raceways passing through building roof, ground floor slab (when the raceway does not extend beyond building footprint), or through outside walls of building above or below grade. Seal on the end inside building, using raceway sealing fittings manufactured for the purpose. Locate fittings at suitable accessible locations. For concealed raceways install each fitting in flush steel box with blank coverplate to match finish of adjacent plates or surfaces.
 3. Seal raceways entering or passing through "hazardous (classified) areas" as defined in NFPA 70.
- J. Raceway and Outlet Boxes Sealing in Bio-sensitive Areas:
1. Where outlet boxes and raceways are recessed mounted, seal box to adjacent wall, ceiling, or floor surface with silicone caulk.
 2. Where outlet boxes and raceways are surface mounted:
 - a. Seal box to adjacent wall, ceiling, or floor surface with continuous bead of silicone caulk.
 - b. Seal both sides of surface-mounted raceway to adjacent surfaces with silicone caulk. Where raceways are threaded rigid steel on minimum 3/4" standoffs, sealing of raceway sides is not required.
 3. Install gasketed device cover plates with additional continuous bead of silicone caulk between device plate and adjacent wall, ceiling, or floor surface.
 4. After wiring is installed, surround wiring with 1" barrier of silicone caulk around conductors within device box hub.
 5. Silicone Caulk: Resistant to microbiological growth.

6. No escutcheons are allowed where raceways pass through walls, floors or ceilings and are exposed in finished areas. Cut and patch holes to within 1/4" of raceway and seal opening with sprayable vinyl, flexible PVC coating equal to Cocoon material.

K. Sleeve Installation for Electrical Penetrations:

1. Coordinate sleeve selection and application with selection and application of firestopping specified in Section 260593 - Electrical Systems Firestopping.
2. Concrete Slabs and Walls: Install sleeves for penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of slabs and walls.
3. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
4. Rectangular Sleeve Minimum Metal Thickness:
 - a. For sleeve cross-section rectangle perimeter less than 50" and no side greater than 16", thickness shall be 0.052".
 - b. For sleeve cross-section rectangle perimeter equal to, or greater than, 50" and 1 or more sides equal to, or greater than, 16", thickness shall be 0.138".
5. Fire-Rated Assemblies: Install sleeves for penetrations of fire-rated floor and wall assemblies, unless openings compatible with firestop system used are fabricated during construction of floor or wall.
6. Cut sleeves to length for mounting flush with both surfaces of walls.
7. Extend sleeves installed in floors 2" above finished floor level.
8. Size pipe sleeves to provide 1/4" annular clear space between sleeve and raceway, unless sleeve seal is to be installed or unless seismic criteria require different clearance.

3.4 APPLICATION

A. Provide raceways and boxes in accordance with the following table:

Application	Approved Raceways	Approved Boxes	Application Notes
Underground More than 5' outside Foundation Walls	Rigid steel conduit, Schedule 80 PVC or Schedule 40 PVC.	Cast Metal Boxes or Nonmetallic Handholes	
Underground More than 5' from Foundation Walls	Rigid steel conduit, Schedule 80 PVC or Schedule 40 PVC.	Cast Metal Boxes or Nonmetallic Handholes	
In or Under Slab on Grade	Rigid steel conduit, Schedule 80 PVC or Schedule 40 PVC	Cast or Nonmetallic Boxes	Use steel elbows with RNC when encased in concrete
Outdoor Locations, Above Grade	Rigid steel conduit	Cast Metal or Nonmetallic	
In Slab Above Grade	Rigid steel conduit, Schedule 80 PVC	Cast, or Nonmetallic	Paint conduits and boxes in accordance with Division 09 -

Application	Approved Raceways	Approved Boxes	Application Notes
			Finishes, Painting Section
Wet and Damp Locations	Rigid steel conduit,	Cast Metal or Nonmetallic. Install flush mounting outlet boxes in finished areas.	
Concealed Dry Locations	electrical metallic tubing	Sheet Metal Boxes; Install flush mounting outlet boxes in finished areas. Install hinged enclosure for large pull boxes.	
Exposed Dry Locations	Rigid steel conduit, Schedule 80 PVC	Sheet Metal boxes; Install flush mounting outlet boxes in finished areas; Install hinged enclosure for large pull boxes.	
Vibrating equipment (including transformers & hydraulic, pneumatic, electric solenoid or motor-driven equipment)	Flexible Metal Conduits (FMC) – Dry Locations Only Liquid Tight Flexible Metal Conduits (LFMC) – Wet Locations		Lengths for FMC & LFMC may range between 2 ft to 4 ft
Instruments & control devices	Liquid Tight Flexible Metal Conduits		Lengths for LFMC may range between 2 ft to 4 ft

3.5 FIELD QUALITY CONTROL

- A. Inspect raceway, boxes, indoor service poles, and wireways for physical damage, proper alignment, supports and seismic restraints, where applicable.
- B. Replace any damaged component of the raceway system or install new raceway system.
- C. Inspect components, wiring, connections and grounding.

3.6 REPAINTING

- A. Repair damage to galvanized finishes with manufacturer-supplied zinc-rich paint kit. Leave remaining paint with Owner.

- B. Repair damage to PVC or paint finishes with manufacturer-supplied touch-up coating. Leave remaining coating with Owner.
- C. Wireways, indoor service poles: Remove paint splatters and other marks from surface; touch-up chips, scratches, or marred finished to match original finish using manufacturer-supplied paint kit. Leave remaining paint with Owner.

3.7 ADJUSTING

- A. Adjust flush-mounted boxes pre-pour and after-pour to be flush with finished materials.
- B. Install knockout closures in unused openings in boxes.
- C. Align adjacent wall-mounted outlet boxes for switches and similar devices.
- D. Adjust outlet boxes to allow luminaires to be positioned as indicated on reflected ceiling plan.

3.8 CLEANING

- A. Clean interior and exterior of boxes, wireways, and indoor poles to remove dust, debris and other material.

END OF SECTION

SECTION 260543
UNDERGROUND DUCTS AND RACEWAYS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 033000 - Cast-In-Place Concrete
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 260533 - Raceway and Boxes for Electrical Systems
- D. Section 260543.13 - Excavation and Backfill
- E. Section 260543.19 - Manholes and Hardware
- F. Section 260553 - Electrical Systems Identification

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes conduits, ducts, and duct accessories for direct buried and concrete encased for underground distribution for electrical power and communications.
- B. The terms duct and duct bank, as used in this Section, are defined as follows:
 - 1. Duct: A single underground conduit, encased in concrete or direct buried.
 - 2. Duct Bank: Two or more ducts run together.

1.4 REFERENCE STANDARDS

- A. ANSI C2 – National Electrical Safety Code
- B. ANSI C80.1 – Rigid Steel Conduit-Zinc Coated (GRC)
- C. ASTM F512 – Specification for Smooth-Wall Poly (Vinyl Chloride) (PVC) Conduits and Fittings for Underground Installation
- D. ETL PVC-001 – PVC Coated Conduit

- E. NEMA RN 1 – Polyvinyl Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
- F. NEMA TC 2 – Electrical Polyvinylchloride (PVC) Conduit
- G. NEMA TC 3 – PVC Fittings for Use with Rigid PVC Conduit and Tubing
- H. NEMA BI 50063 – PVC Plastic Utilities Duct for Underground Installation
- I. NEMA TC 9 – Fittings for PVC Plastic Utility Duct for Underground Installation
- J. NFPA 70 – National Electrical Code
- K. UL 651 – Schedule 40 and 80 Rigid PVC Conduit
- L. UL 651A – Type EB and A Rigid PVC Conduit and HDPE Conduit
- M. UL E53373 – Underground Fiber Reinforced Epoxy Conduit (FRE)
- N. UL 6 – Electrical Rigid Metallic Conduit-Steel

1.5 SUBMITTALS

- A. Product data for the following:
 - 1. Duct bank materials, including spacers and miscellaneous components
 - 2. Ducts and conduits and their accessories, including elbows, end bells, bushings, seals, bends, fittings, plugs, pull tape, and solvent cement
 - 3. Warning tape
 - 4. Warning plank
- B. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, and installation of product.
- C. Closeout Submittals:
 - 1. Project Record Documents:
 - a. Record actual routing of conduits and duct banks.
 - 2. Operation and Maintenance Data:
 - a. Include manufacturer's recommended operating instructions, maintenance procedures and intervals, and preventive maintenance instructions.
 - b. Include spare parts data listing, source, and current prices of replacement parts and supplies.

1.6 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Comply with NFPA 70
 - 2. Comply with ANSI C2

3. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

B. Certification:

1. Installer of PVC coated conduits and fitting shall be certified by a PVC conduit manufacturer.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory wrapping or provide additional canvas or plastic cover to protect from dirt, water, construction debris, and traffic.
- B. Deliver ducts to project site with end capped. Store nonmetallic ducts with supports to prevent bending, warping, and deforming.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr written warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 CONDUITS

- A. Rigid Steel Conduit (RSC): ANSI C80.1, UL 6, heavy wall, hot dipped, galvanized steel with ETL PVC-001 certified PVC coating.
- B. Rigid Nonmetallic Conduit (RNC): NEMA TC 2 Type EPC-40-PVC, UL 651, with matching fittings by same manufacturer, complying with NEMA TC 3 and UL 651, listed for underground use, direct buried, concrete encased.
- C. Size:
 1. 4" nominal for voltages below 600 V and communications

2.2 DUCT ACCESSORIES

- A. Duct Spacers:
 1. Rigid PVC interlocking spacers.
 2. Factory-fabricated, sized for type and sizes of ducts with which used, and selected to provide minimum duct spacings indicated while supporting ducts during concreting or backfilling. Horizontal and vertical locking separation of 3" between ducts.

- B. Elbows: Material to match conduit; minimum bend radius of 48".
- C. Bell Ends: Manufactured bell ends of appropriate sizes at each end of conduit; pre-manufactured system for PVC with conduit seals, provisions for roughing into the concrete pour and waste stops, when entering a new building or a new manhole.
- D. Bushings: Groundable steel bushings of appropriate sizes on all metal conduits where bell ends are not used; pre-manufactured system for PVC with conduit seals, provisions for roughing into concrete pour and water stops, when entering a new or existing building or a new or existing handhole.
- E. Seals: Mechanical interlocking assembly seal of modular synthetic rubber links properly sized to fit the pipe and tightened in place, in accordance with manufacturer's instructions, when entering an existing building or handhole below grade and concrete shall be core drilled for the appropriate size conduit and seal.
- F. Plugs: Closure plugs or caps of same material as conduit at ends of unused sections.
- G. Pull Tape: Nylon pull tape with measurement markings in uniform lengths in each empty duct.
- H. Grounding:
 - 1. Steel grounding bushings, where bell ends are not used.
 - 2. Bonding fitting with bonding strap on steel conduit with end bells.
- I. Warning Tape: Underground line warning tape specified in Section 260553 - Electrical Systems Identification.
- J. Solvent Cement: Recommended by conduit manufacturer.

PART 3 - EXECUTION

3.1 COORDINATION

- A. Coordinate layout and installation of ducts with final arrangement of other utilities, site grading, and surface features as determined in the field.
- B. Coordinate elevations of ducts and duct bank entrances into manholes, handholes, and pull boxes with final locations and profiles of ducts and duct banks as determined by coordination with other utilities, underground obstructions, and surface features. Revise locations and elevations from those indicated as required to suit field conditions and to ensure that duct runs drain to manholes and handholes, and as approved by Architect. For manholes and handholes construction, refer to Section 260543.19 - Manholes and Hardware.

- C. Adjust the depth of electrical utilities to avoid existing utilities with no change to contract price.
- D. Utility Coordination: When duct lines are being constructed for use by a utility serving the project, consult with them for duct size and quantity, minimum bending radii, maximum distance between pulling points, grounding details, termination arrangement, and other criteria.
- E. Duct Bank Coordination Drawings: Show duct profiles and coordination with other utilities and underground structures.
 - 1. Include plans and sections, drawn to scale, and show bends and locations of expansion fittings.
 - 2. Drawings shall be signed and sealed by a qualified professional engineer.

3.2 EXISTING UTILITIES

- A. The existing utilities shown on contract drawings have been plotted from available records. No guarantee is made as to accuracy of locations indicated, and is shown for the benefit of Contractor.
- B. Contact all serving utility companies and have them locate their lines prior to commencing work. Telephone "Call Before You Dig" at 1-800-424-5555, 48 hours prior to commencing work. Coordinate with Owner all existing utility lines prior to commencing work.
- C. Protect shown, visible and located utilities from damage. Promptly repair all active shown, visible and located utilities damaged by construction. This repair shall be made solely at the expense of the Contractor.
- D. Demolish and completely remove from site existing underground utilities indicated to be removed. Coordinate with utility companies to shut off services if lines are active.

3.3 PROJECT CONDITIONS

- A. Interruption of Existing Electrical Service: Do not interrupt electrical service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary electrical service according to requirements indicated:
 - 1. Notify Owner no fewer than 2 days in advance of proposed interruption of electrical service.
 - 2. Do not proceed with interruption of electrical service without written permission from the Owner.

3.4 DUCT INSTALLATION

- A. Slope: Pitch ducts a minimum slope of 1:300 down toward manholes and handholes and away from buildings and equipment. Slope ducts from a high point in runs between two manholes to drain in both directions.
- B. Curves and Bends: Use 5-degree angle couplings for small changes in direction. Use manufactured long sweep bends, both horizontally and vertically, at other locations, unless otherwise indicated. All 90-degree sweeps with radius 10 ft or less shall be rigid steel conduit.
- C. Joints: Use solvent-cemented joints in ducts and fittings and make watertight according to manufacturer's written instructions. Stagger couplings so those of adjacent ducts do not lie in same plane. Do not use conduit that requires the use of couplings for straight runs. Use acceptable PVC terminal adapters when joining PVC conduit to metallic fittings or rigid metal conduit.
- D. Duct Entrances to Manholes and Concrete and Polymer Concrete Handholes: Use end bells, spaced approximately 10" o.c. for 5" ducts, and vary proportionately for other duct sizes.
 - 1. Begin change from regular spacing to end-bell spacing 10 ft from the end bell without reducing duct line slope and without forming a trap in the line.
 - 2. Direct Buried Duct Banks: Install an expansion and deflection fitting in each conduit in area of disturbed earth adjacent to manhole or handhole.
 - 3. Concrete Enclosed Duct Banks: Install watertight expansion fitting in each conduit, with internal bonding jumper to allow for 3/4" movement in any direction.
 - 4. Grout end bells into structure walls from both sides to provide watertight entrances.
- E. Building Wall Penetrations: Make a transition from underground duct to rigid steel conduit at least 5 ft outside the building wall without reducing duct line slope away from the building, and without forming a trap in the line. Use fittings manufactured for duct-to-conduit transition. Install conduit penetrations of building walls as specified in Section 260533 - Raceway and Boxes for Electrical Systems.
- F. Expansion Fittings: Provide suitable expansion fittings or other suitable means to compensate for expansion and contraction for raceways crossing expansion joints in structures or concrete slabs between two adjacent structures and between a duct bank and structure. Provide for the high rate of thermal expansion and contraction of PVC conduit by providing PVC expansion joints as recommended by manufacturer and as required. Refer to structural drawings for location of expansion joints in structures.

- G. Sealing: Provide temporary closure at termination of ducts that have cables pulled. Seal spare ducts at terminations. Use sealing compound and plugs to withstand minimum of 15 psig hydrostatic pressure. Provide watertight entrance sealing device where an underground conduit enters a structure through a concrete roof or membrane waterproofed wall or floor.
- H. Fire Stops: Provide fire stop openings around electrical penetrations to maintain fire-resistance rating, where underground raceways penetrate fire-rated walls or floors.
- I. Pulling Cord: Install 100 lb test nylon cord in ducts, including spares. Identify with tags at each end and at any intermediate pull point the origin and destination of each spare duct. Provide a removable permanent cap over each end of each spare duct.
- J. Concrete Encased Ducts: Support ducts on duct spacers.
 - 1. Spacer Installation:
 - a. Provide spacers close enough to prevent sagging and deforming of ducts, with not less than 4 spacers per 20 ft of duct. Secure spacers to earth and to ducts to prevent floating during concreting. Stagger spacers approximately 6" between tiers. Tie entire assembly together using tie wires and reinforcing steel. Install base and intermediate spacers at every coupling point of each duct line for a separation horizontally and vertically per NEC.
 - 2. Concreting Sequence: Pour each run of envelope between manholes or other terminations in one continuous operation.
 - a. Start at one end and finish at the other, allowing for expansion and contraction of ducts as their temperature changes during and after the pour. Use expansion fittings installed according to manufacturer's written recommendations, or use other specific measures to prevent expansion-contraction damage.
 - b. Terminate each pour in a vertical plane if more than one pour is necessary, and install 3/4" reinforcing rod dowels extending 18" into concrete on both sides of joint near corners of envelope. Obtain Architect's approval for the number and location of dowels.
 - 3. Pouring Concrete: Space concrete carefully during pours to prevent voids under and between conduits and at exterior surface of envelope. Do not allow a heavy mass of concrete to fall directly onto ducts. Use a plank to direct concrete down sides of bank assembly to trench bottom. Allow concrete to flow to center of bank and rise up in middle, uniformly filling all open spaces. Do not use power-driven agitating equipment unless specifically designed for duct bank application.
 - 4. Reinforcement: Reinforce concrete-encased duct banks where they cross disturbed earth and where indicated. Arrange reinforcing bars and ties without forming conductive or magnetic loops around ducts or duct groups. Size reinforcing bars and wire ties as indicated on drawings. Provide rebars with minimum of 2" of concrete on sides, top and bottom. Reinforcing bars shown in sections are required throughout.
 - 5. Forms: Use walls of trench to form side walls of duct bank where soil is self-supporting and concrete envelope can be poured without soil inclusions; otherwise, use forms of materials and in a manner acceptable to Architect.

6. Minimum Space between Ducts: 3" between ducts and exterior envelope wall, 2" between ducts for like services, and 4" between power and signal ducts.
7. Depth: Install top of duct bank at least 24" below finished grade in areas not subject to deliberate traffic, and at least 30" below finished grade in deliberate traffic paths for vehicles, unless otherwise indicated.
8. Maintain a grade of at least 4" per 100 ft, either from one manhole or pull box to the next, or from a high point between them, depending on surface contour.
9. Warning Tape: Bury warning tape approximately 12" above all concrete-encased ducts and duct banks. Align tape parallel to and within 3" of the centerline of duct bank. Provide an additional warning tape for each 12" increment of duct bank width over a nominal 18". Space additional tapes 12" apart, horizontally.
10. Place duct banks on an undisturbed soil base if possible. Where concrete encased duct bank is installed over an extensive area of disturbed earth such that within the periphery of a building, provide a separate concrete base under the duct bank to ensure stability of raceways during installation. Allow this base to set before duct bank is installed.

K. Direct Buried Duct Banks:

1. Support ducts on duct spacers coordinated with duct size, duct spacing, and outdoor temperature.
2. Space spacers close enough to prevent sagging and deforming of ducts, with not less than 4 spacers per 20 ft of duct. Secure spacers to earth and to ducts to prevent displacement during backfill and yet permit linear duct movement due to expansion and contraction as temperature changes. Stagger spacers approximately 6" between tiers.
3. Excavate trench bottom to provide firm and uniform support for duct bank. Prepare trench bottoms as specified in Section 260543.13 - Excavation and Backfill.
4. Install backfill as specified in Section 260543.13 - Excavation and Backfill.
5. After installing first tier of ducts, backfill and compact. Start at tie-in point and work toward end of duct run, leaving ducts at end of run free to move with expansion and contraction as temperature changes during this process. Repeat procedure after placing each tier. After placing last tier, hand place backfill to 4" over ducts and hand tamp. Firmly tamp backfill around ducts to provide maximum supporting strength. Use hand tamper only. After placing controlled backfill over final tier, make final duct connections at end of run and complete backfilling with normal compaction as specified in Section 260543.13 - Excavation and Backfill.
6. Install ducts with a minimum of 3" between ducts for like services and 6" between power and signal ducts.
7. Depth: Install top of duct bank at least 36" below finished grade, unless otherwise indicated.
8. Set elevation of bottom of duct bank below the frost line.
9. Warning Plank: Bury warning plank approximately 12" above direct buried ducts and duct banks, placing them 24" o.c. Align plank along the width and along the centerline of duct bank. Provide an additional plank for each 12" increment of duct bank width over a nominal 18". Space additional planks 12" apart, horizontally.

L. Stub-Ups:

1. Use manufactured PVC duct elbows for stub-ups at poles and equipment and at building entrances through floor, unless otherwise indicated. Extend concrete encasement throughout length of the elbow. Concrete encasement applies to concreted encased ducts.
2. Use manufactured rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through floor.
 - a. Couple steel conduits to ducts with adapters designed for this purpose, and encase couplings with 3" of concrete. Concrete encasement applies to concrete encased ducts.
 - b. Stub-Ups to Equipment: For equipment mounted on outdoor concrete pads, extend steel conduit horizontally a minimum of 5 ft from edge of equipment pad or foundation. Encase in concrete for concrete encased ducts. Install insulated grounding bushings on terminations at equipment.

M. Arrangement and Routing:

1. Arrange multiple duct runs substantially in accordance with details shown on drawings. Locate underground ducts where indicated on drawings and grade to the elevations shown on civil drawings.
2. Make minor changes in location or cross-section as necessary to avoid obstructions or conflicts. Where duct runs cannot be installed substantially as shown because of conditions not discoverable prior to digging of trenches, refer the condition to the Construction Manager for written instructions before further work is done.
3. Maintain a 12" minimum vertical separation between ducts and other systems at crossings where other utility piping systems are encountered or being installed along a raceway route. Maintain a 12" minimum separation between ducts and other systems in parallel runs. Do not place ducts over valves or couplings in other piping systems. Refer conflicts with these requirements to the Construction Manager for written instructions before further work is done.
4. Provide markers at grade to indicate direction of underground conduits provided under this contract. Provide markers consisting of double-ended arrows, straight for straight runs and bent at locations where runs change direction. Provide markers at all bends and at intervals not exceeding 100 ft in straight runs. Use markers made of sheet bronze not less than 1/4" thick embedded in and secured to the top of concrete posts. User markers not less than 10" long and 3/4" wide and marked ELECTRIC CABLES in letters 1/4" high incised into the bronze to a depth of 3/32".
5. Enter manholes and structures with ducts at right angles.

- N. PVC Coated Conduits: Use tools approved for use with PVC coated conduits and fittings.

3.5 UNDERGROUND DUCT APPLICATION

- A. Ducts for Electrical Feeders 600V and Less: RNC, NEMA Type EPC-40-PVC, in concrete encased duct bank, unless otherwise indicated.
- B. Ducts for Electrical Feeders 600V and Less: RNC, NEMA Type EPC-40-PVC, in direct buried duct bank, unless otherwise indicated.

- C. Ducts for Electrical Branch Circuits: RNC, NEMA Type EPC-40-PVC, in direct buried duct bank, unless otherwise indicated.

3.6 EARTHWORK

- A. Excavation and Backfill: Comply with Section 260543.13 - Excavation and Backfill, do not use heavy-duty, hydraulic-operated compaction equipment.

3.7 CONCRETE

- A. Concrete: 3000 psi, 28-day strength, complying with Division 03 – Concrete, where concrete encased.

3.8 GROUNDING

- A. Ground underground ducts according to Section 260526 - Grounding and Bonding for Electrical Systems.

3.9 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections:
 - 1. Demonstrate capability and compliance with requirements on completion of installation of underground ducts.
 - 2. Pull aluminum or wood test mandrel through duct to prove joint integrity and test for out-of-round duct. Provide mandrel equal to 80% fill of duct. If obstructions are indicated, remove obstructions and retest.
- B. Preparation for pulling in conductors:
 - 1. Do not install crushed or deformed raceways. Avoid traps in raceways where possible. Take care to prevent the lodging of plaster, concrete, dirt, or trash in raceways, boxes, fittings, and equipment during the course of construction. Make raceways entirely free of obstructions or replace them. Ream all raceways, remove burrs, and clean raceway interior before introducing conductors or pull wires.
 - 2. Immediately after installation, plug or cap all raceway ends with watertight and dust-tight seals until the time for pulling in conductors.
- C. Do not backfill underground direct buried and concrete encased ducts until the Architect has inspected them. Notify Architect 24 h in advance of duct concrete pour, or backfill of direct buried ducts.

3.10 CLEANING

- A. Pull leather-washer-type duct cleaner, with graduated washer sizes, through full length of ducts. Follow with rubber duct swab for final cleaning and to assist in spreading lubricant throughout ducts.

END OF SECTION

SECTION 260543.13 EXCAVATION AND BACKFILL

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 312000 - Earth Moving
- B. Section 260543 - Underground Ducts and Raceways for Electrical Systems

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section lists methods and materials for trench excavation and backfill for electrical and communications conduits in duct banks. Refer to Section 260543 - Underground Ducts and Raceways for Electrical Systems.

1.4 DEFINITIONS

- A. Backfill: Soil material or controlled low-strength material used to fill an excavation.
- B. Excavation: Removal of material encountered above sub-grade elevations and to lines and dimensions indicated.
- C. Duct: A single underground conduit encased in concrete or direct buried.
- D. Duct Bank: Two or more ducts run together.
- E. Fill: Soil materials used to raise existing grades.
- F. Sub-grade: Surface or elevation remaining after completing excavation, or top surface of fill or backfill immediately below subbase, drainage fill, or topsoil materials.
- G. Utilities: On-site underground ducts and duct banks as well as underground services within buildings.

1.5 QUALITY ASSURANCE

- A. Pre-excavation Conference: Conduct conference at project site to comply with requirements in Division 01 Section "Project Coordination."

PART 2 - PRODUCTS

2.1 FILL MATERIAL

- A. Type 1 Fill:
 - 1. Material from excavation separated from materials, which do not compact by tamping and rolling. No stones larger than 3" and no building, organic, or corrosive or frozen materials and no lumps larger than 6".
- B. Type 2 Fill:
 - 1. Sand or gravel materials with none larger than 2" and of that portion passing the #4 sieve less the 5% to pass #200 sieve.
- C. Type 3 Fill:
 - 1. Gravel of rounded to sub-angular shape, screened, which will pass 3/4" sieve and retained on #4 sieve.
- D. Type 4 Fill:
 - 1. Pit run rock or gravel with maximum stone size of 1".
- E. Type 5 Fill:
 - 1. Pea gravel, screened, which will pass 3/8" sieve and retained on #4 (4.75 mm) sieve.

2.2 CRUSHED ROCK

- A. Crushed Rock: 1-1/4" minus, unless smaller is required for bedding material.

2.3 SAND

- A. Sand: Clean and washed building sand.

2.4 TOPSOIL

- A. Topsoil: Equal in quality to that removed.

2.5 SOD

- A. New Sod: Mature, densely rooted grass free of weeds and objectionable grasses.

2.6 PLANTS

- A. Plants: Obtained from a commercial nursery and be similar to those replaced.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Establish grade lines and locations of roadways and sidewalks, grade beams, and pill caps. Provide necessary stakes and batter boards.
- B. Verify elevations of existing utilities prior to excavation for new ducts.
- C. Verify locations of vaults and manholes with civil drawings
- D. Coordinate excavation and backfill with Section 312000 – Earth Moving.
- E. Verify footings and foundations with structural drawings
- F. Verify underground sanitary and sewer with plumbing and civil drawings.

3.2 EXCAVATION

- A. Provide excavation for underground work, including ducts, vaults, manholes, handholes, pull boxes, unless otherwise shown or specified. Lay duct in open trench, except when Architect gives written permission for tunneling.
- B. Excavate trench to 24" wider than duct or duct bank dimensions and minimum of 3" below bottom of duct.
- C. Include clearing, tree removal, grubbing, pavement removal, substructure removal such as walls, footings, and piers, and all incidental work such as tunneling, sheet piling, shoring, underpinning, pumping, bailing, and transportation. Coordinate excavation extending beyond construction limits with Construction Manager and Owner.
- D. Do not provide blasting on this project without written permission of Architect and Owner.
- E. Dispose of excess excavation material on site in location designated by Construction Manager.
- F. Over excavate 3" and fill with 3" of sand, where trench bottom is rock, or rocky, or contains debris larger than 1", or material with sharp edges.
- G. Perform all crossings of concrete or asphalt after surface material has been saw cut to required width and removed.
- H. Conform to utility company requirements for excavation and vault installation in addition to contract document requirements where excavations are for installing utility company's ducts and vaults.

3.3 ROCK EXCAVATION

- A. Use mechanical methods to remove rock in trenches for underground ducts.
- B. Refer to Geotechnical Report available from Architect/Engineer for data on rock.
- C. Include rock excavation in the Bid, unless otherwise indicated.

3.4 INSTALLATION

- A. Keep underground ducts to proper line and grade and sealed to prevent entrance of animals or foreign matter.
- B. Provide bracing and sheet piling as necessary to support trenches. Comply with Local Regulations, applicable provisions of OSHA Regulations on trenching, or with provisions of "Manual of Accident Prevention in Construction" published by Associated General Contractors of America.
- C. Do not lay duct in water.
- D. Review project drawings for underground duct bank details.
- E. Keep trench free from water until duct joint material has hardened and concrete encasement is in place.
- F. Do not increase the contract cost due to presence of ground water in soil or necessity of sheet piling or bracing trenches. Adjust contract cost when sheet piling is left in place, on written order of Owner.
- G. Do not remove sheet piling until trench is substantially backfilled. Cut off sheet piling left in place not less than 2 ft below new, finished grade.
- H. Place underground ducts on 3" compacted bedding of sand. Shape bedding for clearance for joints and fittings, tamped in place and graded evenly to ensure uniform bearing for the full length of duct. Do not support duct by blocking, planking or mounding of bedding material.
- I. Install lines passing under foundations with minimum of 3" clearance to concrete and ensure there is no disturbance of bearing soil.

3.5 BACKFILL

- A. Backfill around ducts by hand to depth of 12" above top of duct with Type 3 or 5 fill in 6" layers. Compact backfill thoroughly with compactor of suitable weight or with approved mechanical tamper. Do not use flooding or jetting with water.

- B. Place backfill from 12" above duct to elevation of subgrade in layers not exceeding 8" in depth with Type 3, 4 or 5 fill.
- C. Backfill from 12" above duct to sub-grade with Type 2, 3, or 4 fill, when excavating through areas which are to become walks, roads, driveways or parking areas of concrete, bituminous or exposed gravel surfacing or such areas are existing to remain. Backfill in 12" layers and compact with mechanical means to density 95% modified proctor.
- D. Conform excavation, duct laying, backfilling, grading and surfacing, as herein specified, when excavation occurs on public property or areas beyond the property line. Comply with additional requirements for public utility or other authorities. Check with each utility and incorporate cost of any additional requirements in base bid.
- E. Backfill around vaults and handholes to be free of debris larger than 1-3/4" in all directions to 1 ft from vault.
- F. Provide 6" of pea gravel or sand bedding for vaults and handholes larger than 3x3 FT. For handholes smaller than 3x3 ft, provide 3" pea gravel or sand.
- G. Other backfill shall be free of debris larger than 6" in diameter.
- H. Place backfill material so as to obtain a minimum degree of compaction of 95% of maximum density at optimum moisture content. Moisten backfill material as required to obtain proper compaction.
- I. Broken pavement, concrete, sod, roots, and debris shall not be used for backfill.

3.6 DEWATERING

- A. Provide, operate, and maintain all pumps or other dewatering equipment required for control of water in trenches and excavations for electrical and communications site work during the entire construction period.

3.7 SHORING

- A. Provide as required by trenching and excavating to secure site work. Comply with applicable safety regulations.

3.8 FINISHING

- A. On completion of trenching and backfilling operations, restore grades to original elevation or to new sub-grade elevation.
- B. Replace surfaces to existing conditions when trenching is through existing areas or beyond construction limits.

- C. Use 6" of topsoil and sod to match existing elevations in landscaped areas or as otherwise approved by Landscape Architect.

3.9 SURFACE FINISHING

- A. Refinish every disturbed surface to its original condition.
- B. Replace planted materials not surviving 90 days after contract acceptance at Contractor's own expense.
- C. Return after 1 year and re-fill, compact and refinish settled areas to grade.

3.10 CARE OF PLANTS AND TREES

- A. Remove and safely store plants and trees with trunks smaller than 6" diameter prior to commencing site work. Avoid trees larger than 6" diameter when so indicated on drawings. Replace plants and trees upon completion of site work.

END OF SECTION

**SECTION 260543.19
MANHOLES AND HARDWARE**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 033000 - Cast-in-Place Concrete
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 260543 - Underground Ducts and Raceways for Electrical Systems
- D. Section 260543.13 - Excavation and Backfill

1.2 DESCRIPTION

- A. Section includes underground utility structures: manholes and handholes for power and communication cable systems.

1.3 REFERENCED STANDARDS

- A. AASHTO HB 17 – Standard Specifications for Highway Bridges
- B. ANSI C2 – National Electrical Safety Code
- C. ASTM A48/A48M – Specification for Gray Iron Castings
- D. ASTM C270 – Specification for Mortar for Unit Masonry
- E. ASTM C387/C387M – Specification for Packaged, Dry, Combined Materials for Mortar and Concrete
- F. ASTM C858 – Specification for Underground Precast Concrete Utility Structures
- G. ASTM C891 – Standard Practice for Installation of Underground Precast Concrete Utility Structures
- H. ASTM C1037 – Practice for Inspection of Underground Precast Concrete Utility Structures
- I. ASTM E329 – Standard Specification for Agencies Engaged in Construction Inspection and/or Testing
- J. ISO 9000 – Quality Management

- K. ISO 10012 – Measurement Management Systems
- L. NFPA 70 – National Electrical Code
- M. SCTE 77 – Specification for Underground Enclosure Integrity

1.4 SUBMITTALS

- A. Product Data:
 - 1. Accessories for underground utility structures.
- B. Shop Drawings for Precast or Factory-Fabricated Underground Utility Structures: Include plans, elevations, sections, details, attachments to other work, and accessories, including the following:
 - 1. Duct entry provisions, including locations and duct sizes
 - 2. Reinforcement details
 - 3. Frame and cover design and manhole frame support rings
 - 4. Ladder details
 - 5. Grounding details
 - 6. Dimensioned locations of cable rack inserts, pulling-in and lifting irons, and sumps
 - 7. Joint details
- C. Shop Drawings for Factory-Fabricated Handholes and Pull Boxes Other Than Precast Concrete: Include dimensioned plans, sections, and elevations, and fabrication and installation details, including the following:
 - 1. Duct entry provisions, including locations and duct sizes
 - 2. Cover design
 - 3. Grounding details
 - 4. Dimensioned locations of cable rack inserts, and pulling-in and lifting irons
- D. Product Certificates: For concrete and steel used in precast concrete manholes and handholes, as required by ASTM C858.
- E. Qualification Data: For professional engineer and testing agency.
- F. Source quality-control test reports.
- G. Field quality-control test reports.
 - 1. Indicate field test and inspection procedures and interpret test results and corrective action taken for compliance with specification requirements.
- H. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, and installation of product.
- I. Closeout Submittals:
 - 1. Project Record Documents:

- a. Record actual location of underground utility structures.
- 2. Operation and Maintenance Data:
 - a. Include manufacturer's recommended operating instructions, maintenance procedures and intervals, and preventive maintenance instructions.
 - b. Include spare parts data listing, source, and current prices of replacement parts and supplies.

1.5 QUALITY ASSURANCE

- A. Testing Agency Qualifications: Qualified according to ASTM E329 for testing indicated
- B. Comply with ANSI C2
- C. Comply with NFPA 70

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Store manholes and handholes underground utility structures at project site as recommended by manufacturer to prevent physical damage. Arrange so identification markings are visible.
- B. Lift and support precast concrete units only at designated lifting or supporting points.

1.7 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr written warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

1.8 MAINTENANCE

- A. Extra Materials:
 - 1. Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 2. Furnish cable-support stanchions, arms, insulators, and associated fasteners in quantities equal to 5% of quantity of each item installed.

PART 2 - PRODUCTS

2.1 PRECAST CONCRETE AND MANHOLES

- A. Description: Factory-fabricated, reinforced-concrete, monolithically poured walls and bottom. Frame and cover shall form top of enclosure and shall have load rating consistent with that of manhole.
- B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Carder Concrete Products
 - 2. Christy Concrete Products
 - 3. Elmhurst-Chicago Stone Co.
 - 4. Oldcastle Precast Group
 - 5. Riverton Concrete Products; a division of Cretex Companies, Inc.
 - 6. Utility Concrete Products, LLC.
 - 7. Utility Vault Co.
 - 8. Wausau Tile, Inc.
 - 9. Jensen Precast
- C. Comply with ASTM C858 and with interlocking mating sections, complete with accessories, hardware, and features.
 - 1. Windows: Precast openings in walls, arranged to match dimensions and elevations of approaching ducts and duct banks plus an additional 12" vertically and horizontally to accommodate alignment variations.
 - a. Windows: Located not less than 6" from interior surfaces of walls, floors, or roofs of manholes, but close enough to corners to facilitate racking of cables on walls.
 - b. Window opening: Cast-in-place, welded wire fabric reinforcement for field cutting and bending to tie in to concrete envelopes of duct banks.
 - c. Window openings: Framed with at least two additional No. 4 steel reinforcing bars in concrete around each opening.
 - 2. Duct Entrances in Manhole Walls: Cast end-bell or duct-terminating fitting in wall for each entering duct.
 - a. Type and size: Fittings matched to duct or conduit to be terminated.
 - b. Fittings: Aligned with elevations of approaching ducts and located near interior corners of manholes to facilitate racking of cable.
- D. Concrete Knockout Panels: 1-1/2" to 2" thick, for future conduit entrance and sleeve for ground rod.
- E. Joint Sealant: Asphaltic-butyl material with adhesion, cohesion, flexibility, and durability properties necessary to withstand maximum hydrostatic pressures at the installation location with the groundwater level at grade.

2.2 CAST-IN-PLACE MANHOLES

- A. Description: Underground utility structures, constructed in place, complete with accessories, hardware, and features, with concrete knockout panels for conduit entrance and sleeve for ground rod.
- B. Materials: Comply with ASTM C858 and with Section 033000 - Cast-in-Place Concrete.
- C. Structural Design Loading: As specified in Part 3 "Underground Enclosure Application" Article.

2.3 PRECAST CONCRETE HANDHOLES AND PULL BOXES

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Carder Concrete Products
 - 2. Christy Concrete Products
 - 3. Elmhurst-Chicago Stone Co.
 - 4. Oldcastle Precast Group
 - 5. Riverton Concrete Products; a division of Cretex Companies, Inc.
 - 6. Utility Concrete Products, LLC.
 - 7. Utility Vault Co.
 - 8. Wausau Tile, Inc.
 - 9. Jensen Precast
- B. Comply with ASTM C858 for design and manufacturing processes.
- C. Structural Load Ratings: As specified in Part 3 "Underground Enclosure Application."
- D. Description: Factory-fabricated, reinforced-concrete, monolithically poured walls and bottom unless open-bottom enclosures are indicated. Frame and cover shall form top of enclosure and shall have load rating consistent with that of handhole or pull box.
 - 1. Frame and Cover: Weatherproof cast-iron frame, with cast-iron cover with recessed cover hook eyes and tamper-resistant, captive, cover-securing bolts.
 - 2. Frame and Cover: Weatherproof steel frame, with steel cover with recessed cover hook eyes and tamper-resistant, captive, cover-securing bolts.
 - 3. Frame and Cover: Weatherproof steel frame, with hinged steel access door assembly with tamper-resistant, captive, cover-securing bolts.
 - a. Cover Hinges: Concealed, with hold-open ratchet assembly
 - b. Cover Handle: Recessed
 - 4. Frame and Cover: Weatherproof aluminum frame with hinged aluminum access door assembly with tamper-resistant, captive, cover-securing bolts.
 - a. Cover Hinges: Concealed, with hold-open ratchet assembly
 - b. Cover Handle: Recessed
 - 5. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
 - 6. Cover Legend: Molded lettering, As indicated for each service.
 - 7. Configuration: Units shall be designed for flush burial and have integral closed bottom, unless otherwise indicated.

8. Extensions and Slabs: Designed to mate with bottom of enclosure. Same material as enclosure.
 - a. Extension shall provide increased depth of 12".
 - b. Slab: Same dimensions as bottom of enclosure, and arranged to provide closure.
9. Windows: Precast openings in walls, arranged to match dimensions and elevations of approaching ducts and duct banks plus an additional 12" vertically and horizontally to accommodate alignment variations.
 - a. Windows: Located no less than 6" from interior surfaces of walls, floors, or frames and covers of handholes, but close enough to corners to facilitate racking of cables on walls.
 - b. Window opening: Cast-in-place, welded wire fabric reinforcement for field cutting and bending to tie-in to concrete envelopes of ductbanks.
 - c. Window openings: Framed with at least two additional NO. 4 steel reinforcing bars in concrete around each opening.
10. Duct Entrances in Handhole Walls:
 - a. Type and size: Fittings matched to duct or conduit to be terminated.
 - b. Fittings: Aligned with elevations of approaching ducts and located near interior corners of handholes to facilitate racking of cable.
11. Handholes 12" wide by 24" long and larger: Inserts for cable racks and pulling-in irons installed before concrete is poured.

2.4 HANDHOLES AND PULL BOXES OTHER THAN PRECAST CONCRETE

- A. Description: Comply with SCTE 77.
 1. Color: Gray
 2. Configuration: Designed for flush burial and have integral closed bottom, unless otherwise indicated.
 3. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure.
 4. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
 5. Cover Legend: Molded lettering, As indicated for each service.
 6. Direct Buried Wiring Entrance Provisions: Knockouts equipped with insulated bushings or end-bell fittings, selected to suit box material, sized for wiring indicated, and arranged for secure, fixed installation in enclosure wall.
 7. Duct Entrance Provisions: Duct-terminating fittings mated with entering ducts for secure, fixed installations in enclosure wall.
 8. Handholes 12" wide by 24" long and larger: Factory-installed inserts for cable racks and pulling-in irons.
- B. Polymer Concrete Handholes and Pull Boxes with Polymer Concrete Cover: Molded of sand and aggregate, bound together with a polymer resin, and reinforced with steel or fiberglass or a combination of the two.
 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Armorcast Products Company
 - b. Carson Industries LLC.
 - c. CDR Systems Corporation
 - d. NewBasis

- C. Fiberglass Handholes and Pull Boxes with Polymer Concrete Frame and Cover: Sheet-molded, fiberglass-reinforced, polyester resin enclosure jointed to polymer concrete top ring or frame.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Armorcast Products Company
 - b. Carson Industries LLC.
 - c. Christy Concrete Products
 - d. Synertech Moulded Products, Inc.; a division of Oldcastle Precast
- D. Fiberglass Handholes and Pull Boxes: Molded of fiberglass-reinforced polyester resin, with covers of polymer concrete.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Carson Industries LLC.
 - b. Christy Concrete Products
 - c. Nordic Fiberglass, Inc.
- E. High-Density Plastic Boxes: Injection molded of high-density polyethylene or copolymer-polypropylene. Cover: polymer concrete.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Carson Industries LLC.
 - b. Nordic Fiberglass, Inc.
 - c. PenCell Plastics

2.5 SOURCE QUALITY CONTROL

- A. Test and inspect precast concrete utility structures according to ASTM C1037.
- B. Nonconcrete Handhole and Pull Box Prototype Test: Test prototypes of manholes and pull boxes for compliance with SCTE 77. Strength tests shall be for specified tier ratings of products supplied.
 - 1. Tests of materials shall be performed by an independent testing agency.
 - 2. Strength of tests of complete pull boxes and covers shall be by either an independent testing agency or the manufacturer. A qualified registered professional engineer shall certify tests by manufacturer.
 - 3. Testing machine pressure gages shall have current calibration certification complying with ISO 9000 and ISO 10012, and traceable to NIST standards.

PART 3 - EXECUTION

3.1 UNDERGROUND ENCLOSURE APPLICATION

- A. Manholes: Precast concrete:
 - 1. Manholes Located in Roadways and Other Deliberate Traffic Paths by Heavy or Medium Vehicles: H-20 structural load rating according to AASHTO HB 17.

2. Manholes Not Located in Deliberate Traffic Paths by Heavy or Medium Vehicles: H-10 load rating according to AASHTO HB 17.
- B. Handholes and Pull Boxes for 600 V and Less, Including Telephone, Communications, and Data Wiring:
1. Handholes and Pull Boxes in Roadways and Other Deliberate Traffic Paths: Precast concrete AASHTO HB 17, H-10 structural load rating.
 2. Handholes and Pull Boxes in Driveway, Parking Lot, and Off-Roadway Locations, Subject to Occasional, Nondeliberate Loading by Heavy Vehicles: Precast concrete, AASHTO HB 17, H-20 structural load rating.
 3. Handholes and Pull Boxes in Sidewalk and Similar Applications with a Safety Factor for Nondeliberate Loading by Vehicles: Precast concrete, AASHTO HB 17, H-10 structural load rating.
 4. Handholes and Pull Boxes Subject to Light-Duty Pedestrian Traffic Only: Fiberglass-reinforced polyester resin, structurally tested according to SCTE 77 with 3000 lbf vertical loading.
- 3.2 INSTALLATION OF CONCRETE MANHOLES, HANDHOLES, AND PULL BOXES
- A. Cast-in-Place Manhole Installation:
1. Finish interior surfaces with a smooth-troweled finish.
 2. Windows for Future Duct Connections: Form and pour concrete knockout panels 1-1/2" to 20" thick, arranged as indicated.
 3. Cast-in-place concrete, formwork, and reinforcement are specified in Section 033000 - Cast-in-Place Concrete.
- B. Precast Concrete Manhole and Handhole Installation
1. Comply with ASTM C891, unless otherwise indicated.
 2. Install units level and plumb and with orientation and depth coordinated with connecting ducts to minimize bends and deflections required for proper entrances.
 3. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1" sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.
 4. Concrete: 5000 psi 28-day strength, complying with Division 03 – Concrete.
- C. Elevations:
1. Manhole Roof: Install with rooftop at least 15" below finished grade.
 2. Manhole Frame: In paved areas and traffic-ways, set frames flush with finished grade. Set other manhole frames 1" above finished grade.
 3. Install handholes with bottom below the frost line, below grade.
 4. Handhole Covers: In paved areas and traffic-ways set surface flush with finished grade. Set covers of other handholes 1" above finished grade.
 5. Where indicated, cast handhole cover frame integrally with handhole structure.
- D. Drainage: Install drains in bottom of manholes via drain lines where indicated. Coordinate with drainage provisions indicated.

- E. Manhole Access: Circular opening in manhole roof; sized to match cover size.
 - 1. Manhole with Fixed Ladders: Offset access opening from manhole centerlines to align with ladder.
 - 2. Install chimney, constructed of precast concrete collars and rings, to support frame and cover and to connect cover with manhole roof opening. Provide moisture-tight masonry joints and waterproof grouting for cast-iron frame to chimney.
 - F. Waterproofing: Apply waterproofing to exterior surfaces of manholes and and handholes after concrete has cured at least 3 days. Waterproofing materials and installation are specified in Division 07 Section Thermoplastic Sheet Waterproofing." After ducts have been connected and grouted, and before backfilling, waterproof joints and connections and touch up abrasions and scars. Waterproof exterior of manhole chimneys after mortar has cured at least 3 days.
 - G. Dampproofing: Apply dampproofing to exterior surfaces of manholes and and handholes after concrete has cured at least 3 days. Dampproofing materials and installation are specified in Division 07 Section "Bituminous Dampproofing." After ducts have been connected and grouted, and before backfilling, dampproof joints and connections and touch up abrasions and scars. Dampproof exterior of manhole chimneys after mortar has cured at least 3 days.
 - H. Hardware: Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated.
 - I. Fixed Manhole Ladders: Arrange to provide for safe entry with maximum clearance from cables and other items in manholes.
 - J. Field-Installed Bolting Anchors in Manholes and Concrete Handholes: Do not drill deeper than 3-7/8" for manholes and 20" for handholes, for anchor bolts installed in the field. Use a minimum of two anchors for each cable stanchion.
 - K. Warning Sign: Install "Confined Space Hazard" warning sign on the inside surface of each manhole cover.
 - L. Pack and smooth non-shrink grout at all rough edges around duct entrances at each manhole.
- 3.3 INSTALLATION OF HANDHOLES AND PULL BOXES OTHER THAN PRECAST CONCRETE
- A. Install handholes and pull boxes level and plumb and with orientation and depth coordinated with connecting ducts to minimize bends and deflections required for proper entrances. Use pull box extension if required to match depths of ducts, and seal joint between box and extension as recommended by the manufacturer.

- B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2" sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.
- C. Elevation: In paved areas and traffic-ways, set so cover surface will be flush with finished grade. Set covers of other handholes 1" above finished grade.
- D. Install handholes and pull boxes with bottom below the frost line. Insert depth of frost line below grade at project site below grade.
- E. Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated. Select arm lengths to be long enough to provide spare space for future cables, but short enough to preserve adequate working clearances in the enclosure.
- F. Field-cut openings for ducts and conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetration after fittings are sealed.
- G. For enclosures installed in asphalt paving and subject to occasional, non-deliberate, heavy-vehicle loading, form and pour a concrete ring encircling, and in contact with, enclosure and with top surface screeded to top of box cover frame. Bottom of ring shall rest on compacted earth.
 - 1. Concrete: 3000 psi, 28-day strength, complying with Section 033000 - Cast-in-Place Concrete, with a troweled finish.
 - 2. Dimensions: 10" wide by 12" deep.
- H. Install drainage through the side of each handhole and pull box via drain lines where indicated.

3.4 GROUNDING

- A. Ground utility structures according to Section 260526 - Grounding and Bonding for Electrical Systems.

3.5 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections and prepare test reports:
 - 1. Demonstrate capability and compliance with requirements on completion of installation of underground utility structures.
 - 2. Test manholes and handholes grounding to ensure electrical continuity of grounding and bonding connections. Measure and report ground resistance as specified in Section 260526 - Grounding and Bonding for Electrical Systems.
- B. Correct deficiencies and retest as specified above to demonstrate compliance.

- C. Interpret test results in writing and submit to Engineer.

3.6 CLEANING

- A. Clean internal surfaces of manholes, including sump and remove foreign material, after completing the installation of all devices, equipment, cables and terminations.
- B. Remove water from manholes. If manholes continue to fill up with water, Contractor shall pump them regularly until the source of water has been detected and corrected.

END OF SECTION

**SECTION 260553
ELECTRICAL SYSTEMS IDENTIFICATION**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260516 - Owner-Furnished Equipment
- B. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- C. Section 260533 - Raceway and Boxes for Electrical Systems
- D. Section 260543 - Underground Ducts and Raceways for Electrical Systems
- E. Section 260573 - Power System Studies
- F. Section 260923 - Lighting Control Devices
- G. Section 262416.13 - Lighting and Appliance Panelboards
- H. Section 262726 - Wiring Devices
- I. Section 262816 - Enclosed Switches and Circuit Breakers
- J. Section 262913 - Enclosed Controllers
- K. Section 263213 - Engine Generators
- L. Section 263623 - Automatic Transfer Switches
- M. Section 264300 - Surge Protective Devices

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 - General Requirements.

1.3 DESCRIPTION

- A. Section includes the following:
 - 1. Identification for raceway and metal-clad cable
 - 2. Identification for conductors and communication and control cable
 - 3. Underground-line warning tape
 - 4. Warning labels and signs

5. Instruction signs and posted drawings
6. Equipment identification nameplates
7. Wiring devices identification
8. Miscellaneous identification products

- B. Refer to the respective Division 26 Sections, and Sections in other Divisions that specify electrical components, for additional electrical identification requirements.

1.4 REFERENCE STANDARDS

- A. ANSI A13.1 – Scheme for the Identification of Piping Systems
- B. ANSI C2 – National Electrical Safety Code
- C. ANSI Z535.4 – National Standards for Product Safety Signs and Labels
- D. 29 CFR 1910 – Labor, Part 1910 – Occupational Safety and Health Standards, Section 1910.145 – Specifications for Accident Prevention Signs and Tags
- E. NFPA 70 – National Electrical Code

1.5 SUBMITTALS

- A. Product Data: For each electrical identification product indicated.
- B. Torque log for all terminations 100A and over. Log shall identify target torque values, as they are found via manufacturer documentation and individual locations.
- C. Nameplate Schedule: Prior to making nameplates, submit a complete schedule to Architect for approval indicating nameplate size, lettering size, color and actual nameplate information.
- D. Samples: For each type of label and sign to illustrate size, colors, lettering style, mounting provisions, and graphic features of identification products.

1.6 QUALITY ASSURANCE

- A. Comply with ANSI A13.1 and ANSI C2.
- B. Comply with NFPA 70.
- C. Comply with 29 CFR 1910.145.

1.7 COORDINATION

- A. Coordinate identification names, abbreviations, colors, and other features with requirements in Contract Documents, Shop Drawings, manufacturer's wiring diagrams, and Operation and Maintenance Manual, and with those required by codes, standards, and 29 CFR 1910.145. Use consistent designations throughout project.
- B. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.
- C. Coordinate installation of identifying devices with location of access panels and doors.
- D. Install identifying devices before installing acoustical ceilings and similar concealment.

PART 2 - PRODUCTS

2.1 RACEWAY AND METAL-CLAD CABLE IDENTIFICATION MATERIALS

- A. Comply with ANSI A13.1 for minimum size of letters for legend and for minimum length of color field for each raceway and cable size.
- B. Manufacturers: Brady USA, Ideal, Marking Services, Inc. (MSI), Seton, or approved equal.
- C. Color for Printed Legend:
 - 1. Power Circuits: Black letters on an orange field.
 - 2. Legend: Indicate system or service and voltage, if applicable.
- D. Self-Adhesive Vinyl Labels: Preprinted, flexible label laminated with a clear, weather- and chemical-resistant coating and matching wraparound adhesive tape for securing ends of legend label.
- E. Snap-Around Labels: Slit, pretensioned, flexible, preprinted, color-coded acrylic sleeves, with diameter sized to suit diameter of raceway or cable it identifies and to stay in place by gripping action when placed in position.
- F. Snap-Around, Color-Coding Bands: Slit, pretensioned, flexible, solid-colored acrylic sleeves, 2" long, with diameter sized to suit diameter of raceway or cable it identifies and to stay in place by gripping action when placed in position.
- G. Self-Adhesive Vinyl Tape: Colored, heavy duty, waterproof, fade resistant; 2" wide; compounded for outdoor use.

2.2 CONDUCTOR AND COMMUNICATION- AND CONTROL-CABLE IDENTIFICATION MATERIALS

- A. Comply with ANSI A13.1 for minimum size of letters for legend.
- B. Manufacturers: Brady USA, Ideal, Marking Services, Inc. (MRI), Seton, or approved equal.
- C. Color-Coding Conductor Tape: Colored, self-adhesive vinyl tape not less than 3 mils thick by 1 to 2" wide.
- D. Marker Tapes: Vinyl or vinyl-cloth, self-adhesive wraparound type, with circuit identification legend machine printed by thermal transfer or equivalent process.
- E. Aluminum Wraparound Marker Labels: Cut from 0.014" thick aluminum sheet, with stamped, embossed, or scribed legend, and fitted with tabs and matching slots for permanently securing around wire or cable jacket or around groups of conductors.
- F. Metal Tags: Brass or aluminum, 2" x 2" x 0.05", with stamped legend, punched for use with self-locking nylon tie fastener.
- G. Write-On Tags: Polyester tag, 0.010" thick, with corrosion-resistant grommet and polyester or nylon tie for attachment to conductor or cable.
 - 1. Marker for Tags: Permanent, waterproof, black ink marker recommended by tag manufacturer.
- H. Plasticized Card-Stock Tags: Vinyl cloth with preprinted and field-printed legends. Orange background, unless otherwise indicated, with eyelet for fastener.

2.3 UNDERGROUND-LINE WARNING TAPE

- A. Manufacturers: Ideal, Marking Services, Inc. (MRI), Seton, or approved equal.
- B. Description: Permanent, bright-colored, continuous-printed, polyethylene tape.
 - 1. Not less than 6" wide by 4 mils thick.
 - 2. Compounded for permanent direct-burial service.
 - 3. Embedded continuous metallic strip.
 - 4. Printed legend shall indicate type of underground line.
 - 5. Red tape for electrical and orange tape for communications / controls installations.

2.4 WARNING LABELS AND SIGNS

- A. Comply with NFPA 70 and 29 CFR 1910.145.
- B. Self-Adhesive Warning Labels: Factory printed, multicolor, pressure-sensitive adhesive labels, configured for display on front cover, door, or other access to equipment, unless otherwise indicated.

- C. Self-Adhesive Arc Flash Warning Labels: Industrial grade, made of durable polyester with over-laminate to withstand harsh environments (UV rays, scratches and most chemicals).
 - 1. Manufacturer: Seton or approved equal
- D. Engraved Plastic Signs: Engraving stock, melamine plastic laminate, minimum 1/16" thick for signs up to 20 sq inch and 1/8" thick for larger sizes.
 - 1. Engraved legend with black letters on white face.
 - 2. Punched or drilled for mechanical fasteners.
- E. Baked-Enamel Warning Signs for Interior Use: Preprinted aluminum signs, punched or drilled for fasteners, with colors, legend, and size required for application. 1/4" grommets in corners for mounting. Nominal size, 7x10".
- F. Metal-Backed, Butyrate Warning Signs for Exterior Use: Weather-resistant, nonfading, preprinted, cellulose-acetate butyrate signs with 0.0396" galvanized-steel backing; and with colors, legend, and size required for application. 1/4" grommets in corners for mounting. Nominal size, 10x14".
- G. Warning label and sign shall include, but are not limited to, the following legends:
 - 1. Multiple Power Source Warning: "DANGER – ELECTRICAL SHOCK HAZARD – EQUIPMENT HAS MULTIPLE POWER SOURCES."
 - 2. Workspace Clearance Warning: "WARNING – OSHA REGULATION – AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36"."
 - 3. Arc Flash Labels: Per ANSI Z535.4, the signal word WARNING appearing in black letters on an orange background, with second line below (Arc Flash and Shock Hazard) in black letters on white background and third line below (Appropriate PPE Required) in black letters on white background. Include the following information on the label:
 - a. Equipment name
 - b. Available bolted current
 - c. Flash protection boundary distance
 - d. Incident energy level at 18" expressed in cal/cm2
 - e. Personnel protective equipment (PPE) class
 - f. Voltage shock hazard
 - g. Limited shock approach boundary
 - h. Restricted shock approach boundary

2.5 TORQUE MARKING

- A. Contractor shall keep a log of torque values used for project. This record shall incorporate name of installer for each specific location with date/time.
- B. Torque marks shall be made on nut side and extend to a non-rotating surface. Torque marks shall be made at time of torqueing – placing torque marks after the fact is not permitted, but instead must be re-torqued and marked.

- C. Permanent marker: Contractor shall confirm torque requirements with manufacturer and use tools calibrated within past 6 months. Terminations over 100A shall have black permanent marker utilized for visual confirmation.
- D. Torque seal: (Blue) Cross-Check or similar manufacturer shall be used for terminations 100A and over. Use of permanent black marker is not acceptable.

2.6 INSTRUCTION SIGNS AND POSTED DRAWINGS

- A. Instruction Signs: Engraved, laminated acrylic or melamine plastic, minimum 1/16" thick for signs up to 20 sq inch and 1/8" thick for larger sizes.
 - 1. Engraved legend with black letters on white face.
 - 2. Punched or drilled for mechanical fasteners.
 - 3. Mounting Frames: Extruded aluminum, 4-point screw mount with 1/8" clear plexiglass cover.
- B. Posted Drawings: Print electrical riser diagrams on 20 lb bond paper. (Blueprint paper is not acceptable.) Reduce drawings to approximately 1/2 size using Xerox reduction process. Contact Engineer to obtain updated original plans for printing.

2.7 EQUIPMENT IDENTIFICATION NAMEPLATES

- A. Engraved, Three-layer, Laminated Acrylic or Melamine Nameplate: Punched or drilled for screw mounting. White letters on a black background, except emergency power equipment nameplates are to have white letters on a red background. Minimum letter height shall be 3/8" unless otherwise noted.

2.8 WIRING DEVICES IDENTIFICATION

- A. Refer to Section 262726 - Wiring Devices for requirements.

2.9 MISCELLANEOUS IDENTIFICATION PRODUCTS

- A. Cable Ties: Fungus-inert, self-extinguishing, 1-piece, self-locking, type 6/6 nylon cable ties.
 - 1. Minimum Width: 3/16"
 - 2. Tensile Strength: 50 lb minimum
 - 3. Temperature Range: -40°F to 185°F
 - 4. Color: Black, except where used for color-coding
- B. Paint: Paint materials and application requirements are specified in Division 09 – Finishes painting Sections.
- C. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Raceway and Ductbanks More Than 600 V Concealed within Buildings: 4" wide black stripes on 10" centers over orange background that extends full length of raceway or duct and is 12" wide. Stencil legend "DANGER CONCEALED HIGH-VOLTAGE WIRING" with 3" high black letters on 20" centers. Stop stripes at legends. Apply to the following finished surfaces:
1. Floor surface directly above conduits running beneath and within 12" of a floor that is in contact with earth or is framed above unexcavated space.
 2. Wall surfaces directly external to raceways concealed within wall.
 3. Accessible surfaces of concrete envelope around raceways in vertical shafts, exposed in the building, or concealed above suspended ceilings.
- B. Accessible Raceways and Metal-Clad Cables, 600 V or Less, for Feeder, and Branch Circuits More Than 30A: Identify with orange self-adhesive vinyl labels.
1. Identify 4" round, 4" square and 4-11/16" junction boxes concealed above ceiling or exposed with neat lettering on cover with permanent black marking pen. Identify source, circuit number, phase, and control circuit number.
- C. Accessible Raceways and Cables of Auxiliary Electrical Systems: Identify the following systems with color-coded, self-adhesive vinyl tape applied in bands:
1. Fire Alarm System (including covers of pull and junction boxes): Red
 2. Fire-Suppression Supervisory and Control System: Red and yellow
 3. Combined Fire Alarm and Security System: Red and blue
 4. Security System: Blue and yellow
 5. Mechanical and Electrical Supervisory System: Green and blue
 6. Telecommunication System: Green and yellow
- D. Power-Circuit Conductor Identification: For primary conductors No. 1/0 AWG and larger in vaults, pull and junction boxes, manholes, and handholes use color-coding conductor tape. Identify source and circuit number of each set of conductors or other appropriate number or letter to expedite future tracing and troubleshooting. For single conductor cables, identify phase in addition to the above. Phase identification shall be consistent throughout the system.
- E. Branch-Circuit Conductor Identification: Where there are conductors for more than three branch circuits in same junction or pull box, use color-coding conductor tape. Identify each ungrounded conductor according to source and circuit number.
- F. Conductors to Be Extended in the Future and Spare Conductors: Attach write-on tags to conductors and list source and circuit number.
- G. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, signal, sound, intercommunications, voice, and data connections.
1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.

2. Use system of marker tape designations that is uniform and consistent with system used by manufacturer for factory-installed connections.
 3. Coordinate identification with project drawings, manufacturer's wiring diagrams, and Operation and Maintenance Manual.
- H. Locations of Underground Lines: Identify with underground-line warning tape for power, lighting, communication, and control wiring and optical fiber cable. Install underground-line warning tape for both direct-buried cables and cables in raceway.
- I. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Comply with 29 CFR 1910.145 and apply self-adhesive warning labels. Identify system voltage with black letters on an orange background. Apply to exterior of door, cover, or other access to equipment.
1. Equipment with Multiple Power or Control Sources: Apply to door or cover of equipment including, the following:
 - a. Power transfer switches
 - b. Controls with external control power connections
 2. Equipment Requiring Workspace Clearance According to NFPA 70: Unless otherwise indicated, apply to door or cover of equipment but not on flush panelboards and similar equipment in finished spaces.
 3. Arc Flash Warning Labels: install per NFPA 70 for each switchgear, switchboard, panelboard, motor control center, industrial control panel (every enclosure that may contain energized conductors or components). Locate labels so they are visible to the personnel before examination, adjustment, servicing, or maintenance of the equipment.
 4. Available Fault Current Labels: install per NFPA 70 for each piece of service entrance equipment. Locate labels so they are visible to the personnel before examination, adjustment, servicing or maintenance of the equipment.
- J. Instruction Signs and Posted Drawings:
1. Operating Instructions: Install instruction signs to facilitate proper operation and maintenance of electrical systems and items to which they connect. Install instruction signs with approved legend printed in all capital letters of 12 pt size minimum where instructions are needed for system or equipment operation.
 2. Emergency Operating Instructions: Install instruction signs with white legend on a red background with minimum 3/8" high letters for emergency instructions at equipment used for power transfer.
- K. Equipment Identification Nameplates: On each unit of equipment, install unique designation nameplate that is consistent with wiring diagrams, schedules, and Operation and Maintenance Manual. Apply nameplates to disconnect switches and protection equipment, central or master units, control panels, control stations, terminal cabinets, and racks of each system. Systems include power, lighting, control, communication, signal, monitoring, and alarm systems unless equipment is provided with its own identification.

1. Nameplate Instructions:
 - a. Indoor Equipment: Engraved, laminated acrylic or melamine nameplate. Unless otherwise indicated, provide a single line of text with 1/2" high letters 1/4" where space is limited) on 1-12" high nameplate; where 2 lines of text are required, use nameplates sized 2" high.
 - b. Outdoor Equipment: Engraved, laminated acrylic or melamine nameplates sized similar to indoor equipment nameplates.
 - c. Elevated Components: Increase sizes of labels and letters to those appropriate for viewing from the floor.
2. Install nameplates for equipment including, the following:
 - a. Panelboards, electrical cabinets, and enclosures
 - b. Access doors and panels for concealed electrical items
 - c. Transformers
 - d. Motor-control centers, including each device
 - e. Disconnect switches
 - f. Enclosed circuit breakers
 - g. Motor controllers
 - h. Pushbutton stations
 - i. Power transfer equipment
 - j. Contactors
 - k. Remote-controlled switches, dimmer modules, and control devices
 - l. Battery inverter units
 - m. Power-generating units
 - n. Non-concealed junction box covers of auxiliary electrical systems
3. Provide the following information on each nameplate:
 - a. Equipment name/tag:
 - 1) Matching the designation from the contract documents, or identifying the load controlled or function of the equipment where no specific tag is shown on the contract documents.
 - 2) For disconnect switches, use the prefix "SW-" followed by the name of the equipment served, example: "SW-PMP-201."
 - b. Equipment operating voltage, phase, wiring configuration, and ampacity:
 - 1) Example: "208V/3PH/4W/225A"
 - c. Source of power supply, including circuit number:
 - 1) Example: "FED FROM LP-2/45"

- L. Access Panel Identification: Furnish typewritten charts with identification and location of access panels serving equipment and incorporate in O&M Manuals.

3.2 INSTALLATION

- A. Verify identity of each item before installing identification products.
- B. Location: Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment.
- C. Apply identification devices to surfaces that require finish after completing finish work.
- D. Self-Adhesive Identification Products: Clean surfaces before application, using materials and methods recommended by manufacturer of identification device.

- E. Install non-adhesive signs and plastic nameplates parallel to equipment lines; attach with screws and auxiliary hardware appropriate to the location and substrate. Secure to inside surface of door or panelboard that is recessed in finished locations.
- F. Posted Drawings and Operating Instructions: Mount drawings and operating procedures on the wall immediately adjacent to the piece of equipment for which the instructions apply. If sufficient wall space is available, mount directly to one of the sheet metal panels of the equipment.
- G. Warning Signs: Install warning signs where there is hazardous exposure or danger associated with access to or operation of electrical facilities. Provide text of sufficient clarity and lettering of sufficient size to convey adequate information at each location; mount permanently in an appropriate and effective location. Comply with ANSI A13.1 standard color and design.
 - 1. Operational Tags: Where needed for proper and adequate information on operation and maintenance of electrical systems, provide tags of plasticized card stock, either preprinted or hand printed to convey the message; example: "DO NOT OPEN THIS SWITCH WHEN BREAKER IS CLOSED."
- H. System Identification Color Banding for Raceways and Cables: Each color band shall completely encircle cable or conduit. Place adjacent bands of two-color markings in contact, side by side. Locate bands at changes in direction, at penetrations of walls and floors, at 50 ft maximum intervals in straight runs, and at 25 ft maximum intervals in congested areas.
- I. Color-Coding for Phase and Voltage Level Identification, 600 V or Less: Use colors listed below for ungrounded feeder and branch circuit conductors.
 - 1. Color shall be factory applied.
 - 2. Colors for 208/120 V Circuits:
 - a. Phase A (left bus in panelboard): Black
 - b. Phase B (center bus in panelboard): Red
 - c. Phase C (right bus in panelboard): Blue
 - d. Neutral: White
 - 1) Dedicated neutral, Phase A; white with black tracer
 - 2) Dedicated neutral, Phase B: white with red tracer
 - 3) Dedicated neutral, Phase C: white with blue tracer
 - e. Equipment Ground: Green
 - 3. Colors for 480/277 V Circuits:
 - a. Phase A (left bus in panelboard): Brown
 - b. Phase B (center bus in panelboard): Orange
 - c. Phase C (right bus in panelboard): Yellow
 - d. Neutral: Gray
 - 1) Dedicated neutral, Phase A; gray with brown tracer
 - 2) Dedicated neutral, Phase B: gray with orange tracer
 - 3) Dedicated neutral, Phase C: gray with yellow tracer
 - e. Equipment Ground: Green

4. Where dedicated neutrals are required or are shown on the drawings, neutral insulation color shall be either white or gray with colored tracer to match color of phase conductor to which neutral is dedicated.
 5. Field-applied, Color-Coding Conductor Tape: Apply in half-lapped turns for a minimum distance of 6" from terminal points and in boxes where splices or taps are made. Apply last two runs of tape with no tension to prevent possible unwinding. Locate bands to avoid obscuring factory cable markings.
- J. Aluminum Wraparound Marker Nameplates and Metal Tags: Secure tight to surface of conductor or cable at a location with high visibility and accessibility.
- K. Underground-Line Warning Tape: During backfilling of trenches install continuous underground-line warning tape directly above line at 6" to 8" below finished grade. Use multiple tapes where width of multiple lines installed in a common trench or concrete envelope exceeds 16" overall.
- L. Painted Identification: Prepare surface and apply paint according to Division 09 – Finishes painting Sections.

END OF SECTION

**SECTION 260573
POWER SYSTEM STUDIES**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260553 - Electrical Systems Identification
- B. Section 260812 - Power Distribution Acceptance Tests
- C. Section 260813 - Power Distribution Acceptance Test Tables
- D. Section 262416.13 - Lighting and Appliance Panelboards
- E. Section 262813 - Fuses
- F. Section 262816 - Enclosed Switches and Circuit Breakers
- G. Section 262913 - Enclosed Controllers
- H. Section 263623 - Automatic Transfer Switches

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes computer based, fault current, arc flash, and overcurrent protective device coordination studies for an electrical distribution system, based on actual equipment supplied. Set protective devices based on results of the protective device coordination study.
 - 1. Coordination of series-rated devices is permitted only where indicated on drawings.
- B. Furnish field information and data needed for the studies.
- C. Available fault current and electrical equipment interrupting capacity indicated on drawings are based on the short circuit study performed during design as part of the construction documents.
- D. Provide studies and reports prior to manufacture of the electrical distribution equipment.

- E. Equipment submittal will not be approved until the coordination study is complete and the equipment submittals indicate compliance with the study recommendations.

1.4 REFERENCE STANDARDS

- A. IEEE C57.12.10 – American National Standard for Transformers-230 kV and Below 833/958-8333/10 417 kVA, Single-Phase, and 750/862-60 000/80 000/100 000 kVA, Three-Phase, w/o Load Tap Changing; and 3750/4687-60 000/80 000 kVA with Load Tap Changing-Safety Requirements
- B. ANSI C57.12.22 – American National Standard for Transformers-Pad-Mounted, Compartmental-Type, Self-Cooled, Three-Phase Distribution Transformers with High-Voltage Bushings, 2500 kVA & Smaller: High-Voltage, 34 500 GrdY/19 920 V & Below; Low Voltage, 480 V & Below-Requirements
- C. ANSI C57.12.40 – American National Standard for Secondary Network Transformers-Subway and Vault Types (Liquid Immersed)-Requirements
- D. ANSI C57.12.90 – General Requirements for Liquid-Immersed Distribution Power and Regulating Transformers
- E. IEEE C57.96 – Distribution and Power Transformers, Guide for Loading Dry-Type (Appendix to ANSI C57.12 Standards)
- F. IEEE 141 – Recommended Practice for Electric Power Distribution for Industrial Plants
- G. IEEE 241 – Recommended Practice for Electric Power Systems in Commercial Buildings
- H. IEEE 242 – Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems
- I. IEEE 399 – Recommended Practice for Power System Analysis
- J. IEEE 620 – Guide for the Presentation of Thermal Limit Curves for Squirrel Cage Induction Machines
- K. IEEE 1015 – Recommended Practice for Applying Low-Voltage Circuit Breakers Used in Industrial and Commercial Power Systems
- L. IEEE 1584 – Guide for Performing Arc-Flash Hazard Calculations
- M. IEEE C37.010 – Application Guide for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis
- N. IEEE C37.20.1 – IEEE Standard for Metal-Enclosed, Low-Voltage Power Circuit Breaker Switchgear

- O. ANSI C37.46 – American National Standard Specifications for Power Fuses and Fuse-Disconnecting Switches
- P. IEEE C57.12.00 – General Requirements for Liquid-Immersed Distribution, Power and Regulating Transformers
- Q. IEEE C57.96 – IEEE Guide for Loading Dry-Type Distribution and Power Transformers
- R. ICEA P-32-382 – Short-Circuit Characteristics of Insulated Cable
- S. ICEA P-45-482 – Short-Circuit Performance of Metallic Shielding and Sheaths of Insulated Cable
- T. NEMA MG 00001 – Motors and Generators
- U. NFPA 70 – National Electrical Code (NEC)
- V. NFPA 70B – Recommended Practice for Electrical Equipment Maintenance
- W. NFPA 70C – Hazardous Locations Classification
- X. NFPA 70E – Standard for Electrical Safety in Workplace

1.5 SUBMITTALS

- A. Product Data: Computer software program to be used for studies. Include specific software version for owner record.
- B. Product Certificates:
 - 1. Coordination-study and fault-current-study computer software programs, certifying compliance with IEEE 3002.3.
 - 2. Arc flash calculations computer software programs, certifying compliance with IEEE 1584.
- C. Qualification Data: For coordination study specialist.
 - 1. Submit qualifications of the organization proposed for performing the study. Include description of the equipment and computer-based computation methods or programs used and the names and experience histories of the personnel who will perform the study.
- D. Other Action Submittals: Subsequent to having approval for system protective devices submit the following in digital format:
 - 1. Electrical one-line drawing drafted in computer software program with component names.
 - a. Component names shall be easily traceable to the names used in the Construction Documents.
 - b. Drawing minimum text height of 3/32". Maximum paper size 30"x42". Provide multiple drawing sheets as required.

2. Fault current study report
3. Equipment evaluation report
4. Coordination study input data, including completed computer program input data sheets
5. Coordination Study Report
6. Arc Flash Study and Report
7. Arc Flash labels
8. Serving utility information with utility letterhead, including:
 - a. Protective device part numbers/settings
 - b. Maximum available 1P and 3P fault
 - c. Line conductor sizes/lengths
 - d. Transformer impedance
 - e. Serving voltage
9. Provide copy of Owner's electrical safety program if site specific standards vary from instructions noted in this project specification. All software files, including report documents and system study native files (including relevant library files), to allow review and future use of files
10. Sample energized work permit form

1.6 QUALITY ASSURANCE

- A. Perform studies using computer programs that are distributed nationally and are in wide use. Software algorithms shall comply with requirements of standards and guides specified in this Section. Manual calculations are not acceptable.
- B. Coordination Study Specialist Qualifications: An organization experienced in the application of computer software used for studies, having performed successful studies of similar magnitude on electrical distribution systems using similar devices.
 1. Perform study under the direct supervision and control of a Registered Professional Electrical Engineer licensed in the State of Illinois, with a minimum of 5 yrs recent experience in performing protective device coordination studies, arc flash calculations, and electrical system analysis. Final report shall be signed and sealed by said engineer.
- C. Comply with IEEE 3002.3 for general study procedures.
- D. Comply with IEEE 1584 for arc flash calculations.

PART 2 - PRODUCTS

2.1 COMPUTER SOFTWARE DEVELOPERS

- A. Computer Software Developers: Subject to compliance with requirements, provide products by one of the following:
- B. Basis of Design Product:
 1. CYME International, Inc.
 2. EDSA Micro Corporation

3. SKM Systems Analysis, Inc.
4. ETAP

2.2 COMPUTER SOFTWARE PROGRAM REQUIREMENTS

- A. Comply with IEEE 3002.3 and IEEE 1584.
- B. Analytical features of fault current study computer software program shall include "mandatory," "very desirable," and "desirable" features as listed in IEEE 399 Table 7-4.
- C. Computer software program shall be capable of plotting and diagramming time-current characteristic curves as part of its output. Computer software program reports device settings and ratings of all overcurrent protective devices and demonstrates selective coordination by computer-generated, time-current coordination plots.
- D. Arc Flash Calculations: Software program capable of calculating Arc Flash Incident Energy (AFIE) levels and flash protection boundary distances.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine project overcurrent protective device submittals for compliance with electrical distribution system coordination requirements and other conditions affecting performance. Devices for coordination are indicated on drawings.
- B. Proceed with coordination study only after relevant equipment submittals have been assembled. Overcurrent protective devices that have not been submitted and approved prior to coordination study may not be used in study.
- C. Provide the study based on the actual electrical equipment supplied for the project.

3.2 POWER SYSTEM DATA

- A. Gather and tabulate the following input data to support coordination study:
 1. Product Data for overcurrent protective devices specified in other Division 26 Sections and involved in overcurrent protective device coordination studies. Use equipment designation tags that are consistent with power riser diagrams, overcurrent protective device submittals, input and output data, and recommended device settings.
 2. Impedance of utility service entrance(s).
 3. Power Riser Diagrams: In hard copy and electronic copy formats, showing the following:
 - a. Circuit breaker and fuse-current ratings and types
 - b. Relays and associated power and current transformer ratings and ratios
 - c. Transformer kilovolt amperes, primary and secondary voltages, connection type, impedance, and X/R ratios

- d. Generator kilovolt amperes, size, voltage, and source impedance
- e. Cables: Indicate conduit material, sizes of conductors, conductor material, insulation, and length
 - 1) Prior to equipment purchase, utilize conservative lengths (up/down included) based on planned conduit routing to validate equipment ratings. Final study to utilize contractor provided as-built lengths to confirm equipment ratings.
- f. Busway ampacity and impedance
- g. Motor horsepower and code letter designation according to NEMA MG 00001
- h. Load current that is the basis for sizing continuous ratings of circuits for cables and equipment
- 4. Data sheets to supplement power riser diagrams, cross-referenced with tag numbers on diagrams, showing the following:
 - a. Special load considerations, including starting inrush currents and frequent starting and stopping
 - b. Transformer characteristics, including primary protective device, magnetic inrush current, and overload capability
 - c. Motor full-load current, locked-rotor current, service factor, starting time, type of start, and thermal-damage curve
 - d. Generator thermal-damage curve
 - e. Ratings, types, and settings of utility company's overcurrent protective devices
 - f. Special overcurrent protective device settings or types stipulated by utility company
 - g. Time-current characteristic curves of devices indicated to be coordinated
 - h. Manufacturer, frame size, interrupting rating in amperes rms symmetrical, ampere or current sensor rating, long-time adjustment range, short-time adjustment range, and instantaneous adjustment range for circuit breakers
 - i. Manufacturer and type, ampere-tap adjustment range, time-delay adjustment range, instantaneous attachment adjustment range, and current transformer ratio for overcurrent relays
 - j. Panelboards, switchboards, automatic transfer switch, ampacity, and interrupting rating in amperes rms symmetrical
 - 1) Automatic transfer switch withstand rating to comply with UL 1008. Equivalent trip curves are not accepted for specific breaker rated equipment – exact breaker and associated trip unit must be listed on UL certification.

3.3 FAULT CURRENT STUDY

- A. Calculate maximum available short circuit current in amperes rms symmetrical at circuit breaker positions of electrical power distribution system. Provide calculation for a current immediately after initiation and for a three-phase bolted short circuit at the following:
 - 1. Branch circuit panelboard
 - 2. Disconnect switches
 - 3. Automatic transfer switch
 - 4. Equipment fed by Variable Frequency Drive (VFD)

- B. For standard non-bypass Pulse Width Modulation VFDs, a line short circuit condition may be ignored.
- C. Verify mechanical equipment served meets or exceeds maximum short circuit available.
- D. Study electrical distribution system from normal and alternate power sources throughout electrical distribution system for project. Include studies of system switching configurations and alternate operations that could result in maximum fault conditions.
 - 1. Model the entire electrical distribution system from utility company point of connection to circuit breakers in 208 V distribution panels at secondary side of distribution transformers. Include mechanical HVAC equipment, motor driven equipment feeder circuits, and elevator feeder circuits.
 - 2. Model shall include components of the distribution system which would be exposed to fault current levels of 10,000 A symmetrical on a calculated basis.
- E. Calculate momentary and interrupting duties on basis of maximum available fault current.
- F. Perform calculations to verify interrupting ratings of overcurrent protective devices in compliance with IEEE 241 and IEEE 242.
 - 1. Transformers:
 - a. IEEE C57.12.10
 - b. ANSI C57.12.22
 - c. ANSI C57.12.40
 - d. IEEE C57.12.00
 - e. IEEE C57.96
 - 2. Low-Voltage Circuit Breakers: IEEE 1015 and IEEE C37.20.1
 - 3. Low-Voltage Fuses: ANSI C37.46
- G. Study Report:
 - 1. Show calculated X/R ratios and equipment interrupting rating (5-cycle) fault currents on power riser diagrams in report. List other output values from computer analyses, including momentary (1/2-cycle), interrupting (5-cycle), and 30-cycle fault current values for 3-phase, 2-phase, and phase-to-ground faults.
 - 2. Show interrupting (5-cycle) and time-delayed currents (6 cycles and above) on medium- and high voltage circuit breakers to set relays and assess the sensitivity of overcurrent relays.
- H. Equipment Evaluation Report:
 - 1. Prepare report on adequacy of overcurrent protective devices and conductors by comparing fault current ratings of devices with calculated fault current momentary and interrupting duties.
 - 2. For 600V overcurrent protective devices, ensure interrupting ratings are equal to or higher than calculated 1/2-cycle symmetrical fault current.
 - 3. For devices and equipment rated for asymmetrical fault current, apply multiplication factors listed in standards to 1/2-cycle symmetrical fault current.

4. Verify adequacy of phase conductors at maximum 3-phase bolted fault currents; verify adequacy of equipment grounding conductors and grounding electrode conductors at maximum ground-fault currents. Ensure short circuit withstand ratings are equal to or higher than calculated 1/2-cycle symmetrical fault current.
5. Notify Owner/Engineer promptly of discrepancies, problem areas, or inadequacies and provide recommendations for problem resolution.

3.4 COORDINATION STUDY

- A. Perform coordination study using approved computer software program. Prepare a written report using results of fault current study. Comply with IEEE 3002.3.
 1. Calculate maximum and minimum 1/2-cycle short circuit currents.
 2. Calculate maximum and minimum interrupting duty (5 cycles to 2 seconds) short circuit currents.
 3. Calculate maximum and minimum ground-fault currents.
- B. Comply with NFPA 70 for overcurrent protection of circuit elements and devices.
- C. Comply with IEEE 241 recommendations for fault currents and time intervals.
- D. Transformer Primary Overcurrent Protective Devices:
 1. Devices non-operational in response to the following:
 - a. Inrush current when first energized
 - b. Self-cooled, full-load current or forced-air-cooled, full-load current, whichever is specified for that transformer.
 - c. Permissible transformer overloads according to IEEE C57.96 if required by unusual loading or emergency conditions.
 2. Protect transformers according to IEEE C57.12.00, for fault currents by device settings.
- E. Protect motors served by voltages more than 600 V according to IEEE 620.
- F. Conductor Protection: Protect cables against damage from fault currents according to ICEA P-32-382, ICEA P-45-482, and conductor melting curves in IEEE 242. Demonstrate equipment withstands the maximum short circuit current for a time equivalent to tripping time of primary relay protection or total clearing time of fuse. To determine temperatures that damage insulation, use curves from cable manufacturers or from listed standards indicating conductor size and short circuit current. Verify adequacy of phase conductors at maximum three-phase bolted fault currents, equipment grounding conductors, and grounding electrode conductors at maximum ground-fault currents.
- G. Include voltage classes of equipment from utility's incoming line protective device down to and including each panelboard. The phase and ground overcurrent protection shall be included as well as settings for other adjustable protective devices.

- H. Selective Coordination: Overcurrent devices installed upstream and downstream of automatic transfer switches and/or associated with NEC Article 700 Emergency and 701 Legally Required loads shall be selectively coordinated from source of supply (both normal and emergency sources) through final device. Change specific circuit breakers (type, frame, trip-unit, etc.) and equipment bus rating as necessary to meet this requirement. Selective coordination of the system indicated on the one-line diagram is based on (list manufacturer) equipment.
1. Additionally, provide selective coordination for ground fault through-out.
- I. Coordination Study Report: Prepare a written report indicating results of coordination study:
1. Tabular Format of Settings Selected for Overcurrent Protective Devices:
 - a. Device tag
 - b. Relay-current transformer ratios; and tap, time-dial, and instantaneous-pickup values
 - c. Circuit breaker sensor rating; and long-time, short-time, and instantaneous settings
 - d. Fuse-current rating and type
 - e. Ground-fault relay-pickup and time-delay settings
 - f. Manufacturer and type of device
 - g. Range of adjustments and recommended settings
 2. Coordination Curves: Determine settings of overcurrent protective devices to achieve selective coordination. Graphically illustrate adequate time separation exists between devices installed in series, drawn to show the boundaries of device operation on log-log scale graphs, including power utility company's upstream devices. Where time current curves do not explicitly illustrate selective coordination but breakers have been tested and documented as being selectively coordinated, submit manufacturer's literature to substantiate device coordination. Include on curve sheet a title and legend identifying portion of the system covered. Prepare separate sets of curves for the switching schemes and for emergency periods where the power source is local generation. Show the following information:
 - a. Device tag
 - b. Voltage and current ratio for curves
 - c. Three-phase and single-phase damage points for each transformer
 - d. No damage, melting, and clearing curves for fuses
 - e. Cable damage curves
 - f. Transformer inrush points, full-load amps, and damage curves
 - g. Maximum fault current cutoff point
 - h. Generator decrement curve and full-load amps
 3. Plot characteristics where applicable:
 - a. Low-voltage fuses including minimum melt, total clearing and damage bands
 - b. Low-voltage circuit breaker trip devices
 - c. Transformer full-load current, magnetizing inrush current, and ANSI transformer withstand parameters
 - d. Ground-fault protective devices
 - e. Motor starting characteristics and motor damage points
 - f. Generator short circuit decrement curve and generator damage point

- g. Conductor damage curves
- 4. Notify Owner/Engineer promptly of discrepancies, problem areas, or inadequacies and provide recommendations for problem resolution. Propose approaches to effectively protect the underrated equipment. Present technical evaluation with discussion of logical compromises for best coordination.

J. Completed data sheets for setting of overcurrent protective devices.

3.5 ARC FLASH STUDY

- A. Perform arc flash calculations for Arc Flash Incident Energy (AFIE) levels and flash protection boundary distances. Utilize short circuit rating of equipment identified in fault current study – note infinite bus fault current alone is not acceptable.
- B. Model worst-case arc flash conditions .
 - 1. Equipment with PPE rating greater than 2 shall be investigated. Investigation shall include adjustment of upstream overcurrent device settings to determine if PPE rating can be reduced with minimal compromise to coordination with other overcurrent devices.
- C. Arc Flash Study Report: Provide study results in tabular form and include:
 - 1. Device or bus name
 - 2. Bolted fault and arcing fault current levels
 - 3. Arc Flash Incident Energy (AFIE) level at 455 mm expressed in cal/cm²
 - 4. Flash protection boundary distances including:
 - a. Limited shock approach boundary
 - b. Restricted shock approach boundary
 - 5. Trip/Delay time
 - 6. Breaker opening time
 - 7. Working distance
 - 8. Equipment class and bus gap
 - 9. Personal protective equipment class (PPE)
- D. Provide recommendation for reducing AFIE levels and enhancing worker safety.

3.6 FIELD QUALITY CONTROL

- A. Inspect, set, test, and calibrate the protective relays, circuit breakers, fuses, and other applicable devices per requirements in Section 260812 - Power Distribution Acceptance Tests and Section 260813 - Power Distribution Acceptance Test Tables.
- B. Upon final approval of study, provide weatherproof vinyl or polyester arc flash label for all electrical equipment defined above. Label shall include calculated flash protection boundary, incident energy in cal/cm² at working distance (mm working distance based on equipment class, per IEEE 1584), required PPE level, limited approach, restricted approach, equipment name, company name/logo who performed the study, available fault current, overcurrent device settings if applicable, and date label was produced.

1. Label to comply with ANSI Z535. Use "WARNING" (ANSI safety orange background with an orange exclamation point safety symbol) for all arc flash levels.
2. Per NFPA 70E 130.5(C), due to use of incident energy analysis method, PPE categories shall not be provided on labels unless site specific standard PPE categories differ from incident energy levels noted in NFPA 70E Table 130.7(C)(16).
3. Include verbiage on label noting study should be re-examined once every (5) years or upon modifications to electrical system.

3.7 ADJUSTING

- A. Make modifications to equipment, as required, to accomplish conformance with equipment evaluation study.
- B. Adjust relay and overcurrent protective device settings according to recommended settings table provided by overcurrent protective device coordination study.
- C. Notify Owner/Engineer in writing of any required major modifications.

3.8 INSTALLATION

- A. Install arc labels on each piece of equipment prior to energizing equipment.
- B. Arc flash labels shall be protected by clear plastic cover, weatherproof type material, or laminated and mounted on front of equipment. Taping of PPE label to front of equipment is unacceptable.
- C. Arc flash label shall be clearly visible upon approach to equipment.
- D. For large pieces of equipment, label shall be placed near main overcurrent device or incoming feeder to equipment. For equipment such as switchboards, UPS, or switchgear, with multiple vertical sections, provide (1) label per vertical section.
- E. Contractor to provide one-line diagrams (meet IEEE/ANSI standard 141), mounted on 24"x36" (minimum) Styrofoam backboard. These one-line diagrams shall be mounted in each electrical room.
- F. Label shall be mounted at a minimum of 42" to bottom and maximum 66" to top above finished floor.

END OF SECTION

**SECTION 260593
ELECTRICAL SYSTEMS FIRESTOPPING**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 078413 – Penetration Firestopping
- B. Section 260533 - Raceway and Boxes for Electrical Systems

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes through-penetration firestop systems for penetrations through fire-resistance-rated constructions (walls, partitions, floors, and ceilings) including both empty openings and openings containing electrical penetrating items, including raceways, cables, cable trays, busways, and wireways.

1.4 REFERENCE STANDARDS

- A. ASTM E814 – Standard Test Method for Fire Tests of Through-Penetration Firestops
- B. UL 1479 - Fire Tests of Through-Penetration Firestops

1.5 PERFORMANCE REQUIREMENTS

- A. Provide firestop system to resist spread of fire, resist passage of smoke and other gases, and maintain original fire-resistance rating of construction penetrated.
- B. Firestop systems shall be UL Classified for the application and correspond to those indicated by reference to designations listed by UL Fire Resistance Directory.
- C. Conform to applicable Code requirements of Authority Having Jurisdiction.

1.6 SUBMITTALS

- A. Product Data: For each type of product indicated.

- B. Shop Drawings: For each through-penetration firestop system, show each type of construction condition penetrated, relationships to adjoining construction, and type of penetrating item. Include firestop design designation of qualified testing and inspecting agency that evidences compliance with requirements for each condition indicated.
 - 1. Submit documentation, including illustrations, from a qualified testing and inspecting agency that is applicable to each through-penetration firestop system configuration for construction and penetration items, including documentation of UL certification for firestop systems.
- C. Manufacturer's Installation Instructions: Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, and installation of product.
- D. Material Safety Data Sheets provided with product delivered to job site.
- E. Certification of compliance with Building Codes of the State of Illinois.
- F. Inspection reports
- G. Onsite Training Letter: Firestop manufacturer to provide and contractor to submit letter stating the names(s) of the companies, person(s) in attendance and date of onsite training as required in section 1.7.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications: A firm experienced in installing through-penetration firestop systems similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful performance.
- B. Firestopping tests shall be performed by a qualified testing and inspecting agency, or another agency performing testing and follow-up inspection services for firestop systems acceptable to Authorities Having Jurisdiction.
- C. Manufacturer's representative shall be on-site during initial installation of firestop systems to train appropriate Contractor personnel in proper selection and installation procedures.

1.8 DELIVERY, STORAGE AND HANDLING

- A. Deliver through-penetration firestop system products to Project site in original, unopened containers or packages with intact and legible manufacturers' labels identifying product, type and manufacturer, and UL Label where applicable.
- B. Store and handle materials for through-penetration firestop systems to prevent their deterioration or damage due to moisture, temperature changes, contaminants, or other causes.

- C. Handle in accordance with recommended procedures, precautions, or remedies described in material safety data sheets as applicable.

1.9 PROJECT CONDITIONS

- A. Do not install through-penetration firestop systems when ambient or substrate temperatures are outside limits permitted by through-penetration firestop systems' manufacturers or when substrates are wet due to rain, frost, condensation, or other causes.
- B. Ventilate through-penetration firestop systems per manufacturers' written instructions by natural means or, where this is inadequate, forced-air circulation.

1.10 COORDINATION

- A. Coordinate construction of openings and penetrating items to ensure that through-penetration firestop systems are installed according to specified requirements.
- B. Coordinate sizing of sleeves, openings, core-drilled holes, or cut openings to accommodate through-penetration firestop systems.
- C. Notify Owner's inspecting agency at least 7 days in advance of through-penetration firestop system installations; confirm dates and times on days preceding each series of installations.
- D. Do not cover up through-penetration firestop system installations that will become concealed behind other construction until each installation has been examined by Owner's inspecting agency and building inspector, if required by Authorities Having Jurisdiction.

1.11 SEQUENCING

- A. Sequence work to avoid interferences with building finishes and installation of other products.

1.12 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. 3M (Fire Protection Products Division), Hilti Inc, Tremco (Sealant/Weatherproofing Division), Nelson Firestop Products, Specified Technologies Inc, RectorSeal Corporation, , approved equal.

2.2 MATERIALS

- A. Firestop Products: UL 1479, ASTM E814 tested for specific fire-rated construction conditions conforming to construction assembly type, penetrating item type, annular space requirements and fire-rating involved for each separate instance; materials shall not contain flammable solvents.
- B. Firestop Systems: Produced by the same manufacturer.
- C. VOC Content: Penetration firestopping sealants and sealant primers shall comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
 - 1. Sealants: 250 g/L
 - 2. Sealant Primers for Nonporous Substrates: 250 g/L
 - 3. Sealant Primers for Porous Substrates: 775 g/L

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and conditions for compliance with requirements for opening configurations, penetrating items and other conditions affecting performance of firestopping.
- B. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Clean out openings immediately prior to installing through-penetration firestop system to comply with firestop system manufacturer's written instructions.
- B. Provide masking and temporary covering to prevent soiling of adjacent surfaces by firestopping materials.
- C. Comply with manufacturer's recommendations for temperature and humidity conditions before, during and after installation of firestopping.

3.3 INSTALLATION

- A. Comply with "System Performance Requirements" Article in Part 1 and with firestop system manufacturer's written installation instructions and drawings for products and applications indicated.
- B. Install forming/damming/backing materials and other accessories of types required to support fill materials during application as required. After installing fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of firestop systems.
- C. Each conduit penetration through a fire rated assembly is to be dedicated to a single conduit. Multiple conduits penetrating a single opening is not acceptable unless a UL listed, multi-conduit assembly is used.

3.4 IDENTIFICATION

- A. Identify through-penetration firestop systems with pressure-sensitive, self-adhesive, preprinted vinyl labels. Attach labels permanently to surfaces of penetrated construction on both sides of each firestop system installation where labels will be visible to anyone seeking to remove penetrating items or firestop systems. Include label(s) complying with 1 or 2 below.
 - 1. Custom label with the following information:
 - a. The words: "Warning—Through-Penetration Firestop System—Do Not Disturb. Notify Building Management of Any Damage."
 - b. Contractor's name, address, and phone number
 - c. Through-penetration firestop system designation of applicable testing and inspecting agency
 - d. Date of installation
 - e. Through-penetration firestop system manufacturer's name
 - f. Installer's name
 - 2. Manufacturer's preprinted labels with similar information per 1 above.

3.5 FIELD QUALITY CONTROL

- A. Inspecting Agency: Owner will engage a qualified independent inspecting agency to inspect through-penetration firestop systems and to prepare test reports.
 - 1. Inspecting agency will state in each report whether inspected through-penetration firestop systems comply with or deviate from requirements.
- B. Provide certification by Installer that all through-penetration firestop systems have been firestopped in accordance with applicable Building Codes of the State of Illinois.
- C. Proceed with enclosing through-penetration firestop systems with other construction only after inspection reports are issued and firestop installations comply with requirements.

- D. Where deficiencies are found, repair or replace through-penetration firestop systems so they comply with specifications.

3.6 CLEANING

- A. Clean surfaces adjacent to sealed holes and joints to be free of excess firestop materials and soiling as work progresses.

END OF SECTION

SECTION 260812
POWER DISTRIBUTION ACCEPTANCE TESTS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 260573 - Power System Studies
- D. Section 260813 - Power Distribution Acceptance Test Tables
- E. Section 262816 - Enclosed Switches and Circuit Breakers
- F. Section 262913 - Enclosed Controllers
- G. Section 263213 - Engine Generators
- H. Section 263623 - Automatic Transfer Switches

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes acceptance testing requirements for assessing the suitability for service and reliability of the power distribution system.
- B. Contractor to ensure all tested electrical equipment, both Contractor and Owner supplied, is operational and within industry and manufacturer's tolerances and is installed in accordance with design specifications.
- C. Tests and inspections shall be performed after installation.
- D. Tests and inspections shall determine suitability for energization.
- E. Electrical systems shall pass tests prior to substantial completion or Owner occupancy.
- F. This specification requires Contractor to engage services of testing agency.

G. All tests tables referenced in this specification provided in Section 260813 – Power Distribution Acceptance Test Tables.

H. Items to be tested and inspected as follows:

1. 600-volt conductors and cables
2. Engine generator
3. Low-voltage power circuit breakers
4. Low-voltage disconnect switches
5. Automatic transfer switches
6. Ground fault protection systems
7. Grounding systems
8. Thermographic survey

1.4 REFERENCE STANDARDS

- A. ANSI/IEEE C2 – National Electrical Safety Code
- B. ANSI/IEEE C37 – Guides and Standards for Circuit Breakers, Switchgear, Relays, Substations, and Fuses
- C. ANSI/IEEE C37.04 – Standard Rating Structure for AC High Voltage Circuit Breaker
- D. ANSI/IEEE C57 – Distribution, Power, and Regulating Transformers
- E. ANSI/IEEE C57.13.1 – Guide for Field Testing of Relaying Current Transformers
- F. ANSI/IEEE C57.13.3 – Grounding of Instrument Transformer Secondary Circuits and Gases
- G. ANSI/IEEE c57.104 – Guide for the Interpretation of Gases Generated in Oil-immersed Transformers
- H. ANSI/IEEE C62 – Surge Protection
- I. ANSI/IEEE 43 – IEEE Recommended Practice for Testing Insulation Resistance of Rotating Machinery
- J. ANSI/IEEE 48 – Standard Test Procedure and Requirements for High-Voltage Alternating-Current Cable Terminations
- K. ANSI/IEEE 81 – Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
- L. ANSI/IEEE 141 – IEEE Recommended Practice for Electrical/Power Distribution for Industrial Plants (IEEE Red Book)
- M. ANSI/IEEE 142 – IEEE Recommended Practice for Grounding of Industrial and Commercial Power Systems (IEEE Green Book)

- N. ANSI/IEEE 241– IEEE Recommended Practice for Electrical Power Systems in Commercial Buildings (IEEE Gray Book)
- O. ANSI/IEEE 242 – IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems (IEEE Buff Book)
- P. ANSI/IEEE 399 – IEEE Recommended Practice for Power Systems Analysis (IEEE Brown Book)
- Q. ANSI/IEEE 400 – Guide for Making High-Direct-Voltage Tests on Power Cable Systems in the Field
- R. ANSI/IEEE 446 – IEEE Recommended Practice for Emergency and Standby Power Systems for Industrial and Commercial Applications (IEEE Orange Book)
- S. ANSI/IEEE 493 – IEEE Recommended Practice for the Design of Reliable Industrial and Commercial Power Systems (IEEE Gold Book)
- T. ANSI/IEEE 1100 – IEEE Recommended Practice for Powering and Grounding Sensitive Electronic Equipment (IEEE Emerald Book)
- U. ASTM D877/D877M – Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Disk Electrodes
- V. ASTM D823 – Standard Practices for Sampling Electrical Insulating Liquids
- W. ASTM D924 – Standard Test Method for Dissipation Factor (or Power Factor) and Relative Permittivity (Dielectric Constant) of Electrical Insulating Liquids
- X. ASTM D971 – Standard Test Method for Interfacial Tension of Oil Against Water by the Ring Methods
- Y. ASTM D974 – Standard Test Method for Acid and Base Number by Color-Indicator Titration
- Z. ASTM D1298 – Standard Test Method for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method
- AA. ASTM D1500 – Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)
- BB. ASTM D1524 – Standard Test Method for Visual Examination of Used Electrical Insulating Oils of Petroleum Origin in the Field
- CC. ASTM D1816 – Standard Test Method for Dielectric Breakdown Voltage of Insulating Oils of Petroleum Origin Using VDE Electrodes

- DD. ASTM D2285 – Standard Test Method for Interfacial Tension of Electrical Insulating Oils of Petroleum Origin Against Water by the Drop-Weight Method
- EE. ASTM D3612 – Standard Test Method for Analysis of Gases Dissolved in Electrical Insulating Oil by Gas Chromatography
- FF. ASTM D3613 – Standard Practice for Sampling Insulating Liquids for Gas Analysis and Determination of Water Content
- GG. NETA ATS – Acceptance Testing Specifications for Electrical Power Distribution Equipment and Systems
- HH. NEMA AB 4 – Guidelines for Inspection and Preventive Maintenance of Molded-Case Circuit Breakers Used in Commercial and Industrial Applications
- II. NEMA MG 00001 – Motors and Generators
- JJ. NFPA 70 – National Electrical Code
- KK. NFPA 70B – Recommended Practice for Electrical Equipment Maintenance
- LL. NFPA 70E – Electrical Safety Requirements for Employee Workplaces
- MM. NFPA 101 – Life Safety Code
- NN. NFPA 110 – Emergency and Standby Power Systems
- OO. NIST – National Institute of Standards and Technology
- PP. OSHA – Part 1910 – Subpart S – 1910.308 – Special Systems

1.5 SUBMITTALS

- A. Test Reports: Include the following:
 - 1. Summary of project
 - 2. Description of equipment tested
 - 3. Equipment used to conduct the test
 - 4. Description of test
 - 5. Test results, as compared to manufacturers' or industry accepted standards and tolerances
 - 6. Conclusions and recommendations
 - 7. Signature of responsible test organization authority
- B. List of equipment used to perform tests. Identify the following:
 - 1. Type
 - 2. Manufacturer
 - 3. Model number
 - 4. Serial number

5. Date of last calibration
6. Documentation of calibration leading to NIST standards

1.6 QUALITY ASSURANCE

A. Qualifications of Testing Agency:

1. Testing firm shall be a corporately and financially independent testing organization that can function as an unbiased testing authority, professionally independent of the manufacturer, supplier, and installers of equipment or system evaluated by the testing firm.
2. Testing firm shall be regularly engaged in testing of electrical equipment, devices, installations and systems.
3. Testing firm shall meet Federal Occupational Safety and Health Administration (OSHA) requirements for accreditation of independent testing laboratories.
4. On-site technical person shall be currently certified by the International Electrical Testing Association in electrical power distribution system testing.
5. Testing firm shall use technicians who are regularly employed by the firm for testing services.
6. Testing firm shall submit proof of above qualifications with bid documents when requested.

PART 2 - PRODUCTS

2.1 NOT APPLICABLE TO THIS SECTION.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Documentation: Deliver the following to testing firm, minimum two weeks prior to commencement of testing:
 1. Complete set of electrical plans and specifications, with available short circuit indicated on power riser diagrams.
 2. Approved submittals and shop drawings of equipment being tested.
 3. Pertinent change orders.
 4. Evaluation, overcurrent protective device coordination and arc flash studies, per requirements in Section 260573 - Power System Studies.
- B. Schedule: Notify Owner and Engineer 10 working days prior to performance of any tests.
- C. Coordination: Coordinate with Construction Manager/Owner/Engineer the testing schedule and availability of equipment ready for testing.
- D. Test Power: Provide test power (including specialized) for equipment testing before and after service energizing.

3.2 FIELD QUALITY CONTROL

A. Inspection and Test Procedures: Comply with NETA.

1. 600 V Conductors and Cables:
 - a. Visual and Mechanical Inspection:
 - 1) Compare cable data with drawing and specifications.
 - 2) Inspect exposed sections of cables for physical damage.
 - 3) Verify tightness of accessible bolted electrical connections by calibrated torque wrench in accordance with manufacturer's published data or Table 12.
 - 4) Perform thermographic survey of bolted electrical connections in accordance with paragraph "Thermographic Survey."
 - 5) Inspect compression-applied connectors for correct cable match and indentation.
 - 6) Verify visible cable bends meet or exceed ICEA and manufacturer's minimum allowable bending radius.
 - 7) For cables are terminated through window-type current transformers, provide an inspection to verify neutral and ground conductors are correctly placed for operation of protective devices.
 - 8) Inspect for correct identification and arrangements.
 - 9) Inspect jacket and insulation condition.
 - b. Electrical Tests:
 - 1) Perform insulation-resistance test using megohm meter. Applied potential to be 1000 VDC. Individually test each conductor with other conductors grounded. Test duration shall be one minute.
 - 2) Perform continuity tests to ensure correct cable connection.
 - c. Test Values:
 - 1) Insulation-resistance values should not be less than 50 megohms.
2. Engine Generator:
 - a. Visual and Mechanical Inspection:
 - 1) Compare equipment nameplate data with drawings and specifications.
 - 2) Inspect physical and mechanical condition.
 - 3) Inspect correct anchorage and grounding.
 - 4) Inspect air baffles, filter media, and cooling fans.
 - 5) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 12.
 - 6) Confirm correct application of manufacturer's recommended lubricants.
 - b. Electrical and Mechanical Tests:
 - 1) Perform insulation-resistance test on generator winding with respect to ground in accordance with ANSI/IEEE 43. Calculate polarization index.
 - 2) Test protective relay devices in accordance with paragraph "Protective Relays."
 - 3) Perform phase-rotation test to determine compatibility with load requirements.

- 4) Functionally test engine shutdown for low oil pressure, overtemperature, overspeed, and other protection features as applicable.
 - 5) Perform vibration baseline test. Plot amplitude versus frequency for each main bearing cap.
 - 6) Conduct performance test in accordance with NFPA 110, Section 7-13 (Installation Acceptance).
 - 7) Verify correct functioning of governor and regulator.
 - 8) Verify function and temperature regulation for battery and engine heaters.
 - c. Test Values:
 - 1) Polarization index values shall be in accordance with ANSI/IEEE 43.
 - 2) Vibration levels shall be in accordance with manufacturer's published data.
 - 3) Performance tests shall conform to manufacturer's published data and NFPA 110.
 - 4) Vibration amplitudes shall not exceed values shown in Table 10.
 - 5) Bolt-torque levels shall be in accordance with Table 12, unless otherwise specified by manufacturer.
3. Low-Voltage Power Circuit Breakers:
 - a. Visual and Mechanical Inspection:
 - 1) Compare nameplate data with drawings and specifications.
 - 2) Inspect physical and mechanical conditions.
 - 3) Confirm correct application of manufacturer's recommended lubricants.
 - 4) Inspect anchorage, alignment, and grounding.
 - 5) Inspect arc chutes.
 - 6) Inspect moving and stationary contacts for condition, wear, and alignment.
 - 7) Verify maintenance devices are available for serving and operating breaker.
 - 8) Verify primary and secondary contact wipe and other dimensions vital to satisfactory operation of breaker are correct.
 - 9) Perform mechanical operator and contact alignment tests on breaker and its operating mechanism.
 - 10) Verify tightness of accessible bolted bus connections by calibrated torque-wrench method. Refer to manufacturer's instructions or Table 12 for correct torque levels.
 - 11) Perform thermographic survey of accessible bolted bus connections in accordance with paragraph "Thermographic Survey."
 - 12) Check cell fit and element alignment.
 - 13) Check racking mechanism.
 - 14) Record as-found and as-left operation-counter readings.
 - b. Electrical Tests:
 - 1) Perform contact-resistance test.
 - 2) Perform insulation-resistance test at 1000 VDC from pole-to-pole and from each pole-to-ground with breaker closed and across open contacts of each phase. Test duration shall be one minute. Use a minimum test voltage in accordance with Table 1 or manufacturer's published data.

- 3) Perform insulation-resistance test at 1000 VDC on control wiring. Test duration shall be one minute. Do not perform test on wiring connected to solid-state components. Follow manufacturer's recommendation.
- 4) Make adjustments for final trip settings in accordance with overcurrent protective device coordination study.
- 5) Determine minimum pickup current by primary current injection.
- 6) Determine long-time delay by primary current injection.
- 7) Determine short-time pickup and delay by primary current injection.
- 8) Determine ground-fault pickup and delay by primary current injection.
- 9) Determine instantaneous pickup value by primary current injection.
- 10) Verify trip unit calibrations by secondary injection.
- 11) Activate auxiliary protective devices, such as ground-fault or undervoltage relays, to ensure operation of shunt trip devices. Check operation of electrically operated breakers in their cubicles. Perform minimum operation voltage on shunt trip and close coils in accordance with Table 20.
- 12) Verify correct operation of auxiliary features such as trip and pickup indicators, zone interlocking, electrically close and trip operations, trip-free, anti-pump function, and trip unit battery condition. Reset all trip logs and indicators.
- 13) Check charging mechanism.
- 14) Determine minimum operation voltage on shunt trip and close coils in accordance with Table 20.
- c. Test Values:
 - 1) Bolt-torque levels shall be in accordance with Table 12, unless otherwise specified by manufacturer.
 - 2) Compare microhm or millivolt drop values to adjacent poles or similar breakers. Investigate deviations of more than 50% of the lowest value. Investigate any value exceeding manufacturer's recommendations.
 - 3) Circuit breaker insulation resistance shall be in accordance with Table 1.
 - 4) Control wiring insulation-resistance shall comply with manufacturer's published data. In the absence of manufacturer's published data, use Table 1. Values of insulation-resistance less than this table or manufacturer's minimum shall be investigated.
 - 5) Trip characteristics of breakers shall fall within manufacturer's published time-current tolerance bands.
 - 6) Minimum operation voltages on shunt trip and close coils shall be in accordance with manufacturer's published data. In the absence of manufacturer's data, refer to Table 20.
4. Low-Voltage Disconnect Switches:
 - a. Visual and Mechanical Inspection:
 - 1) Compare equipment nameplate data with drawings and specifications.
 - 2) Inspect physical and mechanical condition.
 - 3) Inspect anchorage, alignment, grounding, and required clearances.
 - 4) Verify correct blade alignment, blade penetration, travel stops, and mechanical operation.

- 5) Verify that fuse sizes and types are in accordance with drawings, short-circuit and overcurrent protective device coordination studies.
- 6) Verify that each fuse has adequate mechanical support and contact integrity.
- 7) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 12.
- 8) Perform thermographic survey of accessible bolted electrical connection in accordance with paragraph "Thermographic Survey."
- 9) Verify operation and sequencing of interlocking systems.
- 10) Verify correct phase barrier installation.
- 11) Verify correct operation of all indicating and control devices.
- 12) Confirm correct application of manufacturer's recommended lubricants.
- b. Electrical Tests:
 - 1) Measure contact resistance across each switchblade and fuseholder.
 - 2) Perform insulation-resistance test at 1000 VDC from pole-to-pole and from each pole-to-ground. Test duration shall be one minute. Use a minimum test voltage in accordance with Table 1 or manufacturer's published data.
 - 3) Measure fuse resistance.
 - 4) Perform ground fault test, if applicable.
- c. Test Values:
 - 1) Compare bolted connection resistances to values of similar connections.
 - 2) Bolt-torque levels should be in accordance with Table 12, unless otherwise specified by the manufacturer.
 - 3) Compare microhm or millivolt drop values to adjacent poles or similar switches. Investigate deviations of more than 50% of lowest value. Investigate any value exceeding manufacturer's recommendations.
 - 4) Minimum insulation-resistance shall be in accordance with manufacturer's published data or Table 1.
 - 5) Investigate fuse-resistance values that deviate from each other by more than 15%.
5. Automatic Transfer Switches:
 - a. Visual and Mechanical Inspection:
 - 1) Compare equipment nameplate data with drawings and specifications.
 - 2) Inspect physical and mechanical condition.
 - 3) Confirm correct application of manufacturer's recommended lubricants.
 - 4) Verify manual transfer warnings are attached and visible.
 - 5) Verify tightness of control connections.
 - 6) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 12.
 - 7) Perform thermographic survey of accessible bolted electrical connections in accordance with paragraph "Thermographic Survey."
 - 8) Perform manual transfer operation.
 - 9) Verify positive mechanical interlocking between normal and alternative sources.

- 10) Inspect anchorage, alignment, grounding and required clearances.
- b. Electrical Tests:
 - 1) Measure contact resistance.
 - 2) Perform insulation-resistance tests, phase-to-phase and phase-to-ground, with switch in both source positions. Test duration shall be one minute. Use a test voltage in accordance with Table 1 or manufacturer's published data. For control devices that cannot tolerate test voltage, follow manufacturer's recommendation.
 - 3) Verify settings and operation of control devices.
 - 4) Calibrate and set relays and timers in accordance with paragraph "Protective Relays."
 - 5) Verify phase rotation, phasing and synchronized operation as required by the application.
 - 6) Perform automatic transfer tests:
 - a) Simulate loss of normal power.
 - b) Return to normal power.
 - c) Simulate loss of emergency power.
 - d) Simulate all forms of single-phase conditions.
 - 7) Verify correct operation and timing of following functions:
 - a) Normal source voltage-sensing relays.
 - b) Engine start sequence.
 - c) Time delay upon transfer.
 - d) Alternate source voltage-sensing relays.
 - e) Automatic transfer operation.
 - f) Interlocks and limit switch function.
 - g) Time delay and retransfer upon normal power restoration.
 - h) Engine cool down and shutdown feature.
- c. Test Values:
 - 1) Bolt-torque levels shall be in accordance with Table 12, unless otherwise specified by manufacturer.
 - 2) Insulation-resistance test voltage and minimum values shall be in accordance with Table 1.
 - 3) Compare microhm values to adjacent poles or similar switches. Investigate deviations of more than 50% of lowest value. Investigate any value exceeding manufacturer's recommendations.
6. Ground Fault Protection Systems:
 - a. Visual and Mechanical Inspection:
 - 1) Compare equipment nameplate data with drawings and specifications.
 - 2) Visually inspect components for damage and errors in polarity or conductor routing:
 - a) Verify ground connection is made ahead of neutral disconnect link and on line side of any ground fault sensor.
 - b) Verify neutral sensors are connected with correct polarity on both primary and secondary.
 - c) Verify all phase conductors and neutral pass through sensor in same direction for zero sequence systems.
 - d) Verify grounding conductors do not pass through zero sequence sensors.
 - e) Verify grounded conductor is solidly grounded.

- 3) Verify tightness of accessible bolted electrical connections, including control circuits, by calibrated torque-wrench method in accordance with manufacturer's published data or Table 12.
- 4) Verify correct operation of self-test panel.
- 5) Set pickup and time-delay settings in accordance with settings provided on drawings and in specifications. Record operation and test sequences as required by NFPA 70.
- 6) Verify the control power transformer has adequate capacity for the system.
- b. Electrical Tests:
 - 1) Measure system neutral-to-ground insulation resistance with neutral disconnect link temporarily removed. Replace neutral disconnect link after testing.
 - 2) Perform insulation-resistance test of control wiring at 1000 VDC for one minute. Do not perform this test on wiring connected to solid-state components. Follow manufacturer's recommendations.
 - 3) Perform the following pickup tests using primary injection:
 - a) Verify relay does not operate at 90% of pickup setting.
 - b) Verify pickup is less than 125% of setting or 1200 amp, whichever is smaller.
 - 4) For summation type systems using phase-neutral current transformers, verify correct polarities by applying current to each phase-neutral current transformer pair. This test also applies to molded-case breakers using external neutral current transformer.
 - a) Relay should operate when current direction is the same relative to polarity marks in the two current transformers.
 - b) Relay should not operate when current direction is opposite relative to polarity marks in the two current transformers.
 - 5) Measure time delay of the relay at 150% or greater of pickup.
 - 6) Verify reduced voltage tripping capability: 55% for AC systems and 80% for DC systems.
- c. Test Values:
 - 1) System neutral-to-ground insulation shall be minimum of one megohm.
 - 2) Insulation resistance values shall be in accordance with Table 1.
 - 3) Relay timing shall be in accordance with manufacturer's specifications but must also be no longer than one second at 3000 amp.
 - 4) Bus bolt-torque levels shall be in accordance with Table 12, unless otherwise specified by manufacturer.
7. Grounding Systems:
 - a. Visual and Mechanical Inspection:
 - 1) Verify ground system is in compliance with drawings, specifications, and NFPA 70.
 - 2) Inspect physical and mechanical condition.
 - 3) Inspect anchorage.
 - b. Electrical Tests:
 - 1) Perform fall-of-potential or alternative test in accordance with ANSI/IEEE 81 on the main grounding electrode or system.

- 2) Perform point-to-point tests to determine the resistance between the main grounding system and all major electrical equipment frames, system neutral, and/or derived neutral points.
 - 3) Make resistance measurements in dry weather not earlier than 48 hours after rainfall.
 - c. Test Values:
 - 1) The resistance between the main grounding electrode and ground should be no greater than five ohms. (Reference ANSI/IEEE 142.) Investigate any values above five ohms and notify Engineer immediately for further instructions.
 - 2) Investigate point-to-point resistance values that exceed 0.5 ohm.
 8. Thermographic Survey:
 - a. Visual and Mechanical Inspection:
 - 1) Inspect physical, electrical, and mechanical conditions.
 - 2) Remove all necessary covers prior to thermographic inspection.
 - 3) Equipment to be inspected shall include all current-carrying devices. Provide report including the following:
 - a) Discrepancies.
 - b) Temperature difference between area of concern and reference area.
 - c) Cause of temperature difference.
 - d) Areas inspected. Identify inaccessible and unobservable areas and equipment.
 - e) Identify load conditions at time of inspection.
 - f) Provide photographs and thermogram of deficient area.
 - b. Test Parameters:
 - 1) Inspect distribution systems with imaging equipment capable of detecting minimum temperature difference of 2°F at 86°F.
 - 2) Equipment shall detect emitted radiation and convert detected radiation to visual signal.
 - 3) Thermographic surveys should be performed during periods of maximum possible loading but not less than 40% of rated load of the electrical equipment being inspected. Refer to NFPA 70B, Section 20.17 (Infrared Inspection).
 - c. Test Results:
 - 1) Temperature differences of 2°F to 5°F indicate possible deficiency and warrant investigation.
 - 2) Temperature differences of 7°F to 27°F indicate deficiency; repair as time permits.
 - 3) Temperature differences of 29°F and above indicate major deficiency; repair immediately.
 - 4) Suggested actions based on temperature rise can be found in Table 18.
- B. Test Reports:
 1. Testing firm shall do the following:
 - a. Prepare test report, including description of equipment tested, description of test, test results, conclusions and recommendations, retesting results, list of test equipment used and calibration date.
 - b. Show test results in comparison to industry and manufacturer's values and tolerances.

- c. Interpret test results in writing and give recommendations for acceptance or rejection upon consultation with Engineer and prior to energizing equipment.
 - d. Assure electrical equipment is operational and within industry and manufacturer's tolerances, and is installed in accordance with contract documents.
 - e. Assure suitability of energization.
 - f. Report to the Owner and Engineer any system, material, or workmanship that is found defective on the basis of acceptance tests.
 - g. Retest equipment when required.
 - h. Maintain written record of tests.
 - i. Utilize safety practices during the tests in accordance with:
 - 1) Acceptable state and local safety operating procedures
 - 2) Owner's safety practices
 - 3) OSHA
 - 4) NFPA 70E
 - j. Perform tests with apparatus de-energized and grounded, except where otherwise specifically required ungrounded by test procedures.
 - k. Assemble and certify final test report.
 - l. Provide 4 copies of complete test report.
 - m. Attach label to all tested equipment with indication of date tested and testing firm name.
 - 2. Contractor shall do the following:
 - a. Investigate, replace, or repair any fault in material or in any part of the installation revealed by the tests.
 - b. Deliver one copy of each test report directly to Engineer within 30 days after completion of testing, unless directed otherwise. Insert a copy of each test report in the equipment operation and maintenance manuals.
- C. Test Equipment:
- 1. Test Instrument Calibration:
 - a. Testing firm shall have calibration program that assures test instruments are maintained with rated accuracy.
 - b. Instruments shall be calibrated in accordance with the following frequency schedule:
 - 1) Field instruments: Analog, 6 months maximum; Digital, 12 months maximum
 - 2) Laboratory instruments: 12 months
 - 3) Leased specialty equipment: 12 months where accuracy is guaranteed by lessor
 - c. Dated calibration labels shall be visible on test equipment.
 - d. Records, which show date and results of instruments calibrated or tested, must be kept up-to-date.
 - e. Up-to-date instrument calibration instructions and procedures shall be maintained for test instrument.
 - f. Equipment used for field testing shall be more accurate than instrument being tested.
 - g. Calibrating standard applied to testing equipment shall be of higher accuracy than instrument tested.

END OF SECTION

**SECTION 260813
POWER DISTRIBUTION ACCEPTANCE TEST TABLES**

**TABLE 1
INSULATION RESISTANCE TEST VALUES
ELECTRICAL APPARATUS AND SYSTEMS**

Voltage Rating	Minimum DC Test Voltage	Recommended Minimum Insulation Resistance In Megohms
250	500	25
600	1,000	100
1,000	1,000	100
2,500	1,000	500
5,000	2,500	1,000
8,000	2,500	2,000
15,000	2,500	5,000
25,000	5,000	20,000
34,500 and above	15,000	100,000

See Table 14 for temperature correction factors.

**TABLE 2
SWITCHGEAR LOW-FREQUENCY WITHSTAND TEST VOLTAGES**

Type of Switchgear	Rated kV	Maximum Test Voltage kV	
		AC	DC
Low-Voltage Power Circuit Breaker Switchgear	.254	1.6	2.3
	.508	1.6	2.3
	.635	1.6	2.3
MC (Metal-Clad Switchgear)	4.76	14.0	20.0
	8.25	27.0	37.0
	15.0	27.0	37.0
	27.0	45.0	+
	38.0	60.0	+
SC (Station-Type Cubicle Switchgear)	15.5	37.0	+
	38.0	60.0	+
	72.5	120.0	+
MEI	4.76	14.0	20.0
	8.25	19.0	27.0

Type of Switchgear	Rated kV	Maximum Test Voltage kV	
		AC	DC
(Metal-Enclosed Interrupter Switchgear)	15.0	27.0	37.0
	15.5	37.0	52.0
	25.8	45.0	+
	38.0	60.0	+

Derived from ANSI/IEEE C37.20.2-1993, Paragraph 5.5, Metal-Clad and Station-Type Cubicle Switchgear and C37.20.3-1993, Paragraph 5.5, Metal-Enclosed Interrupter Switchgear, and includes 0.75 multiplier with fraction rounded down.

The column headed “DC, Maximum Test Voltage kV” is given as a reference only for those using DC tests to verify the integrity of connected cable installations without disconnecting the cables from the switchgear. It represents values believed to be appropriate and approximately equivalent to the corresponding power frequency withstand test values specified for voltage rating of switchgear. The presence of this column in no way implies any requirement for a DC withstand test on AC equipment or that a DC withstand test represents an acceptable alternative to the low-frequency withstand tests specified in this specification, either for design tests, production tests, conformance tests, or field tests. When making DC tests, the voltage should be raised to the test value in discrete steps and held for a period of one minute.

Because of the variable voltage distribution encountered when making DC withstand tests, the manufacturer should be contacted for recommendations before applying DC withstand tests to the switchgear. Voltage transformers above 34.5kV should be disconnected when testing with DC. Refer to ANSI/IEEE C57.13-1993 (IEEE Standard Requirements for Instrument Transformers) paragraph 8.8.2.

+ Consult Manufacturer

TABLE 3

RECOMMENDED DISSIPATION FACTOR/POWER FACTOR AT 20°C

LIQUID FILLED TRANSFORMERS, REGULATORS, AND REACTORS

ACCEPTANCE TEST VALUES

Oil, Silicone, and Less-Flammable Hydrocarbon Maximum Value (Percent)	
New Power Transformers and Reactors	0.5%
New Distribution Transformers and Regulators	1.0%
Remanufactured Power Transformers and Reactors	1.0%
Remanufactured Distribution Transformers and Regulators	1.5%

TABLE 4

INSULATING FLUID LIMITS

Table 4.1 Test Limits for New Insulating Oil Received in New Equipment		
Mineral Oil		
Test	ASTM Method	# 69 kV and Below
Test	ASTM Method	# 69 kV and Below
Dielectric breakdown, kV minimum	D877	30
Dielectric breakdown, kV minimum @ 0.04" gap	D1816	25
Dielectric breakdown, kV minimum @ 0.08" gap	D1816	45
Interfacial tension mN/m minimum	D971 or D2285	38
Neutralization number, mg KOH/g maximum	D974	0.015
Water content, (ppm) maximum	D1533	20
Power factor at 25°C, %	D924	0.05
Power factor at 100°C, %	D924	0.40
Color	D1500	1.0
Visual condition	D1524	Bright and clear
Table 4.2 Test Limits for Silicone Insulating Liquid in New Transformers		
Test	ASTM Method	Acceptable Values
Dielectric breakdown, kV minimum	D877	30
Visual	D2129	clear, free of particles
Water content, (ppm) maximum	D1533	50
Dissipation/power factor, 60 Hz, % max. @ 25°C	D924	0.1
Viscosity, cSt @ 25°C	D445	47.5 – 52.5
Fire point, °C, minimum	D92	340
Neutralization number, mg KOH/g max.	D974	0.01

ANSI/IEEE C57.111-1989 (R1995), Guide for Acceptance of Silicone Insulating Fluid and Its Maintenance in Transformers, Table 2.

TABLE 4 (CONT.)

INSULATING FLUID LIMITS

Table 4.3 Typical Values for Less-Flammable Hydrocarbon Insulating Liquid Received in New Equipment			
ASTM Method	Test	Results	
		Minimum	Maximum
D1816	Dielectric breakdown voltage for 0.08" gap, kV	34.5 kV 40 class and below	---
		60 Desirable	
D1816	Dielectric breakdown voltage for 0.04" gap, kV	34.5 kV 20 class and below	---
		30 Desirable	
D974	Neutralization number, mg KOH/g	----	0.03
D877	Dielectric breakdown voltage kV	30	----
D924	AC loss characteristic (dissipation factor), %		
	25°C	----	0.1
	100°C	----	1
D1533B	Water content, (ppm)	----	25
D1524	Condition-visual	Clear	
D92	Flash point (°C)	275	----
D92	Fire point (°C)	300a	----
D971	Interfacial tension, mN/m, 25°C	38	----
D445	Kinematic viscosity, mm ² /s. (cSt), 40°C	1.0 X 10 ² (100)	1.3 X 10 ² (130)
D1500	Color	----	L2.5

ANSI/IEEE C57.121-1998, IEEE Guide for Acceptance and Maintenance of Less-Flammable Hydrocarbon Fluid in Transformers, Table 3.

The test limits shown in this table apply to less-flammable hydrocarbon fluids as a class. Specific typical values for each brand of fluid should be obtained from each fluid manufacturer.

If the purpose of the HMWH installation is to comply with the NFPA 70 National Electrical Code, this value is the minimum for compliance with NEC Article 450.23.

TABLE 5
TRANSFORMER INSULATION-RESISTANCE

ACCEPTANCE TEST VOLTAGE AND MINIMUM RESULTS

Transformer Coil Rating Type in Volts	Minimum DC Test Voltage	Recommended Minimum Insulation Resistance in Megohms	
		Liquid Filled	Dry
0 - 600	1000	100	500
601 - 5000	2500	1000	5000
5001 - 15000	5000	5000	25000

See Table 14 for Temperature Correction Factors.

NOTE: Since insulation resistance depends on insulation rating (kV) and winding capacity (kVA), values obtained should be compared to manufacturer's test data.

TABLE 6
MEDIUM-VOLTAGE CABLES

ACCEPTANCE TEST VALUES

Table 6.1 DC Test Voltages					
Rated Voltage Phase-to-Phase kV	Conductor Sizes AWG or kcmil (mm)	Nominal Insulation Thickness mils (mm)		Maximum DC Field Test Voltages, kV During/After Installation	
		100% Insulation Level	133% Insulation Level	100% Insulation Level	133% Insulation Level
5	8-1000 (8.4-507)	90 (2.29)	115 (2.92)	28	36
	Above 1000 (507)	140 (3.56)	140 (3.56)	28	36
8	6-1000 (13.3- 507)	115 (2.92)	140 (3.56)	36	44
	Above 1000 (507)	175 (4.45)	175 (4.45)	36	44
15	2-1000 (33.6- 507)	175 (4.45)	220 (5.59)	56	64
	Above 1000 (507)	220 (5.59)	220 (5.59)	56	64
25	1-2000 (42.4- 1013)	260 (6.60)	320 (8.13)	80	96
28	1-2000 (42.4- 1013)	280 (7.11)	345 (8.76)	84	100
35	1/0-2000 (53.5- 1013)	345 (8.76)	420 (10.7)	100	124

Tables derived from ANSI/ICEA S-93-639/NEMA WC 74-2000, 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy; ANSI/ICEA S-94-649-2000, Standard for Concentric Neutral Cables Rated 5,000 – 46,000 Volts; ANSI/ICEA S-97-682-2000, Standard for Utility Shielded Power Cables Rated 5,000 – 46,000 Volts; and The Okonite Company, High-Voltage Proof Testing.

The DC field test voltages listed above are intended for cable designed in accordance with ICEA specifications. When older cables or other types/classes of cables or accessories are connected to the system, voltages lower than those shown may be necessary. Consult the manufacturers of the cables and/or accessories before applying the test voltage.

TABLE 6 (CONT.)

MEDIUM-VOLTAGE CABLES

ACCEPTANCE TEST VALUES

Table 6.2 AC Test Voltages					
Rated Voltage Phase-to-Phase kV	Conductor Sizes AWG or kcmil (mm)	Nominal Insulation Thickness mils (mm)		AC Test Voltage, kV	
		100% Insulation Level	133% Insulation Level	100% Insulation Level	133% Insulation Level
Rated Voltage Phase-to-Phase kV	Conductor Sizes AWG or kcmil (mm)	Nominal Insulation Thickness mils (mm)		AC Test Voltage, kV	
		100% Insulation Level	133% Insulation Level	100% Insulation Level	133% Insulation Level
5 kV	8-1000	90 (2.29)	115 (2.92)	18	23
	1001-3000	140 (3.56)	140 (3.56)	28	28
8 kV	6-1000	115 (2.92)	140 (3.56)	23	28
	1001-3000	175 (4.45)	175 (4.45)	35	35
15 kV	2-1000	175 (4.45)	220 (5.59)	35	44
	1001-3000	220 (5.59)	220 (5.59)	44	44
25 kV	1-3000	260 (6.60)	320 (8.13)	52	64
28 kV	1-3000	280 (7.11)	345 (8.76)	56	69
35 kV	1/0-3000	345 (8.76)	420 (10.7)	69	84

Tables derived from ANSI/ICEA S-93-639/NEMA WC 74-2000, 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy; ANSI/ICEA S-94-649-2000, Standard for Concentric Neutral Cables Rated 5,000 – 46,000 Volts; ANSI/ICEA S-97-682-2000, Standard for Utility Shielded Power Cables Rated 5,000 – 46,000 Volts.

All AC voltages are RMS values.

TABLE 6 (CONT.)

MEDIUM-VOLTAGE CABLES

ACCEPTANCE TEST VALUES

Table 6.3 Partial Discharge Requirements for Semiconducting Coating and Tape Designs Only		
Rated Circuit Voltage Phase-to-Phase Volts	Minimum Partial Discharge Extinction Level, kV	
	100% Insulation Level	133% Insulation Level
2001-5000	4	5
5001-8000	6	8
8001-15000	11	15

ANSI/ICEA S-93-639/NEMA WC 74-2000, 5-56 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy.

Table 6.4 Very Low Frequency Testing Levels 0.1 Hz Test Voltage (RMS)	
System Voltage Phase-to-Phase(kV) (RMS)	Proof Phase-to-Ground(kV) (RMS)
5	10
15	22
25	33
35	47

TABLE 7
MOLDED-CASE CIRCUIT BREAKERS
VALUES FOR INVERSE TIME TRIP TEST

(AT 300% OF RATED CONTINUOUS CURRENT OF CIRCUIT BREAKER)

Range of Rated Continuous Current Amperes	Maximum Trip Time in Seconds For Each Maximum Frame Rating ¹	
	250V	251 - 600V
0-30	50	70
31-50	80	100
51-100	140	160
101-150	200	250
151-225	230	275
226-400	300	350
401-600	-----	450
601-800	-----	500
801-1000	-----	600
1001-1200	-----	700
1201-1600	-----	775
1601-2000	-----	800
2001-2500	-----	850
2501-5000	-----	900

Reproduction of Table 5-3 from NEMA AB 4-1996.

¹ For integrally-fused circuit breakers, trip times may be substantially longer if tested with the fuses replaced by solid links (shorting bars).

TABLE 8

**INSTANTANEOUS TRIP SETTING TOLERANCES FOR FIELD TESTING
OF MARKED ADJUSTABLE TRIP CIRCUIT BREAKERS**

	Tolerances of High and Low Settings	
Ampere Rating	High	Low
Adjustable	+40%	-30%
Non-adjustable	+25%	-25%

For circuit breakers with nonadjustable instantaneous trips, tolerances apply to the manufacturer's published trip range, i.e., +40% on high side, -30% on low side.

TABLE 9
INSTRUMENT TRANSFORMER DIELECTRIC TESTS

FIELD ACCEPTANCE

Nominal System (kV)	BIL(kV)	Applied Potential Tests Field Test Voltage (kV)	
		AC	DC ¹
0.6	10	3	4
1.2	30	7.5	10
2.4	45	11.25	15
5.0	60	14.25	19
8.7	75	19.5	26
15	95	25.5	34
15	110	25.5	34
25	125	30	40
25	150	37.5	50
34.5	150	37.5	50
34.5	200	52.5	70

Derived from Paragraph 8.8.2 and Tables 2 and 7 of ANSI/IEEE C57.13-1993
(Standard Requirements for Instrument Transformers).

¹ DC potential tests are not recommended for transformers rated higher than 200 kV BIL. DC tests may prove beneficial as a reference for future testing. In such cases the test direct voltage should not exceed the original factory test RMS alternating voltages.

TABLE 10

MAXIMUM ALLOWABLE VIBRATION AMPLITUDE

RPM @60 Hz	Velocity in/s peak	Velocity mm/s	RPM @60 Hz	Velocity in/s peak	Velocity mm/s
3600	0.15	3.8	3000	0.15	3.8
1800	0.15	3.8	1500	0.15	3.8
1200	0.15	3.8	1000	0.13	3.3
900	0.12	3.0	750	0.10	2.5
720	0.09	2.3	600	0.08	2.0
600	0.08	2.0	500	0.07	1.7

Derived from NEMA publication MG 1-1998, Section 7.8.1, Table 7-1. Table is unfiltered vibration limits for resiliently mounted machines. For machines with rigid mounting, multiply the limiting values by 0.8.

TABLE 11

OVERPOTENTIAL TEST VOLTAGES FOR ELECTRICAL APPARATUS OTHER THAN INDUCTIVE EQUIPMENT

Nominal System (Line) Voltage¹ (kV)	Insulation Class	AC Factory Test (kV)	Maximum Field Applied AC Test (kV)	Maximum Field Applied DC Test (kV)
1.2	1.2	10	6.0	8.5
2.4	2.5	15	9.0	12.7
4.8	5.0	19	11.4	16.1
8.3	8.7	26	15.6	22.1
14.4	15.0	34	20.4	28.8
18.0	18.0	40	4.0	33.9
25.0	25.0	50	30.0	42.4
34.5	35.0	70	42.0	59.4
46.0	46.0	95	57.0	80.6
69.0	69.0	140	84.0	118.8

¹ Intermediate voltage ratings are placed in the next higher insulation class.

TABLE 12

U.S. STANDARD

BOLT TORQUES FOR BUS CONNECTIONS

HEAT-TREATED STEEL – CADMIUM OR ZINC PLATED

Grade	SAE 1 & 2	SAE 5	SAE 7	SAE 8
Minimum Tensile (psi)	64K	105K	133K	150K
Bolt Diameter In Inches	Torque (Foot Pounds)			
1/4	4	6	8	8
5/16	7	11	15	18
3/8	12	20	27	30
7/16	19	32	44	48
1/2	30	48	68	74
9/16	42	70	96	105
5/8	59	96	135	145
3/4	96	160	225	235
7/8	150	240	350	380
1.0	225	370	530	570

BOLT TORQUES FOR BUS CONNECTIONS

SILICON BRONZE FASTENERS¹

TORQUE (FOOT POUNDS)

Bolt Diameter in (Inches)	Nonlubricated	Lubricated
5/16	15	10
3/8	20	14
1/2	40	25
5/8	55	40
3/4	70	60

¹ Bronze alloy bolts shall have a minimum tensile strength of 70,000 psi.

TABLE 12 (CONT.)
BOLT TORQUES FOR BUS CONNECTIONS
ALUMINUM ALLOY FASTENERS²

TORQUE (FOOT POUNDS)

Bolt Diameter in Inches	Lubricated
5/16	8.0
3/8	11.2
1/2	20.0
5/8	32.0
3/4	48.0

² Aluminum alloy bolts shall have a minimum tensile strength of 55,000 psi.

BOLT TORQUES FOR BUS CONNECTIONS

TORQUE (FOOT POUNDS)

Bolt Diameter in Inches	Uncoated
5/16	14
3/8	25
1/2	45
5/8	60
3/4	90

³ Bolts, cap screws, nuts, flat washers, locknuts: 18-8 alloy.

Belleville washers: 302 alloy.

TABLE 13

SF₆ GAS TESTS

Test	Method	Serviceability Limits^a
Moisture	Hygrometer	Per manufacturer or 200 ppm ^b
SF ₆ decomposition byproducts	ASTM D 2685	500 ppm
Air	ASTM D 2685	5000 ppm ^c
Dielectric breakdown Hemispherical contents	0.10" gap at atmospheric pressure	11.5 – 13.5 kV ^d

^a In the absence of consensus standards dealing with SF₆ gas tests, the NETA Standards Review Council suggests the above representative values.

^b According to some manufacturers.

^c Dominelli, N. and Wylie, L., Analysis of SF₆ Gas as a Diagnostic Technique for GIS, Electric Power Research Institute, Substation Equipment Diagnostics Conference IV, February 1996.

^d Per Even, F.E., and Mani, G. Sulfur Fluorides, Kirk, Othmer Encyclopedia of Chemical Technology, 4th ed., 11,428, 1994.

Reference: IEC 61634 High-Voltage Switchgear and Controlgear – Use and Handling of Sulfur Hexafluoride (SF₆) in High-Voltage Switchgear and Controlgear.

TABLE 14
INSULATION RESISTANCE CONVERSION FACTORS FOR
CONVERSION OF TEST TEMPERATURE TO 20°C

Temperature		Multiplier	
°C	°F	Apparatus Containing Immersed Oil Insulations	Apparatus Containing Solid Insulations
0	32	0.25	0.40
5	41	0.36	0.45
10	50	0.50	0.50
15	59	0.75	0.75
20	68	1.00	1.00
25	77	1.40	1.25
30	86	1.98	1.58
35	95	2.80	2.00
40	104	3.95	2.50
45	113	5.60	3.15
50	122	7.85	3.98
55	131	11.20	5.00
60	140	15.85	6.30
65	149	22.40	7.90
70	158	31.75	10.00
75	167	44.70	12.60
80	176	63.50	15.80

TABLE 15 (NOT USED)

HIGH-POTENTIAL TEST VOLTAGE

AUTOMATIC CIRCUIT RECLOSERS

Nominal Voltage Class, kV	Maximum Voltage, kV	Rated Impulse Withstand Voltage, kV	Maximum Field Test Voltage, kV, AC
14.4	15.0	95	35
14.4	15.5	110	50
24.9	27.0	150	60
34.5	38.0	150	70
46.0	48.3	250	105
69.0	72.5	350	160

Derived from ANSI/IEEE C37.61-1973(R1992), Standard Guide for the Application, Operation, and Maintenance of Automatic Circuit Reclosers and from C37.60-1981(R1992), Standard Requirements for Overhead, Pad-Mounted, Dry-Vault, and Submersible Automatic Circuit Reclosers and Fault Interrupters for AC Systems.

TABLE 16 (NOT USED)

HIGH-POTENTIAL TEST VOLTAGE

FOR ACCEPTANCE TEST OF LINE SECTIONALIZERS

Nominal Voltage Class, kV	Maximum Voltage, kV	Rated Impulse Withstand Voltage, kV	Maximum Field Test Voltage, kV, AC	DC 15 Minute Withstand (kV)
14.4 (1 Ø)	15.0	95	35	53
14.4 (1 Ø)	15.0	125	42	53
14.4 (3 Ø)	15.5	110	50	53
24.9 (1 Ø)	27.0	125	60	78
34.5 (3 Ø)	38.0	150	70	103

Derived from ANSI/IEEE C37.63-1984(R1990) Table 2 (Standard Requirements for Overhead, Pad-Mounted, Dry-Vault, and Submersible Automatic Line Sectionalizers of AC Systems).

NOTE: Values of AC voltage given are dry test one-minute factory test values.

TABLE 17
DIELECTRIC WITHSTAND TEST VOLTAGES

METAL-ENCLOSED BUS

Type of Bus	Rated kV	Maximum Test Voltage, kV	
		AC	DC
Isolated Phase for Generator Leads	24.5	37.0	52.0
	29.5	45.0	--
	34.5	60.0	--
Isolated Phase for Other than Generator Leads	15.5	37.0	52.0
	25.8	45.0	--
	38.0	60.0	--
Nonsegregated Phase	0.635	1.6	2.3
	4.76	14.2	20.0
	15.0	27.0	37.0
	25.8	45.0	63.0
	38.0	60.0	--
Segregated Phase	15.5	37.0	52.0
	25.8	45.0	63.0
	38.0	60.0	--
DC Bus Duct	0.3	1.6	2.3
	0.8	2.7	3.9
	1.2	3.4	4.8
	1.6	4.0	5.7
	3.2	6.6	9.3

Derived from ANSI/IEEE C37.23-1987, Tables 3A, 3B, 3C, 3D, and paragraph 6.4.2.
The table includes a 0.75 multiplier with fractions rounded down.

NOTE:

The presence of the column headed "DC" does not imply any requirement for a DC withstand test on AC equipment. This column is given as a reference only for those using DC tests and represents values believed to be appropriate and

approximately equivalent to the corresponding power frequency withstand test values specified for each class of bus.

Direct current withstand tests are recommended for flexible bus to avoid the loss of insulation life that may result from the dielectric heating that occurs with rated frequency withstand testing.

Because of the variable voltage distribution encountered when making DC withstand tests and variances in leakage currents associated with various insulation systems, the manufacturer should be consulted for recommendations before applying DC withstand tests to this equipment.

TABLE 18
THERMOGRAPHIC SURVEY

SUGGESTED ACTIONS BASED ON TEMPERATURE RISE

Temperature difference (TD) based on comparisons between similar components under similar loading	Temperature difference (TD) based upon comparisons between component and ambient air temperatures	Recommended action
1°C to 3°C	1°C to 10°C	Possible deficiency; warrants investigation
4°C to 15°C	11°C to 20°C	Indicates probably deficiency; repair as time permits
-- -- --	21°C to 40°C	Monitor until corrective measures can be accomplished
>15°C	>40°C	Major discrepancy; repair immediately

Temperature specifications vary depending on the exact type of equipment. Even in the same class of equipment (i.e., cables) there are various temperature ratings. Heating is generally related to the square of the current; therefore, the load current will have a major impact on ΔT . In the absence of consensus standards for TD, the values in this table will provide reasonable guidelines.

An alternative method of evaluation is the standards-based temperature rating system as discussed in Chapter 8.9.2, Conducting an IR Thermographic Inspection, Electrical Power Systems Maintenance and Testing, by Paul Gill, PE, 1998.

It is a necessary and valid requirement that the person performing the electrical inspection be thoroughly trained and experienced concerning the apparatus and systems being evaluated as well as knowledgeable of thermographic methodology.

TABLE 19
OVERPOTENTIAL TEST VOLTAGES

ELECTRICAL APPARATUS OTHER THAN INDUCTIVE EQUIPMENT

Nominal System (Line) Voltage^a (kV)	Insulation Class	AC Factory Test (kV)	Maximum Field Applied AC Test (kV)	Maximum Field Applied DC Test (kV)
1.2	1.2	10	6.0	8.5
2.4	2.5	15	9.0	12.7
4.8	5.0	19	11.4	16.1
8.3	8.7	26	15.6	22.1
14.4	15.0	34	20.4	28.8
18.0	18.0	40	24.0	33.9
25.0	25.0	50	30.0	42.4
34.5	35.0	70	42.0	59.4

^a Intermediate voltage ratings are placed in the next higher insulation class.

TABLE 20
RATED CONTROL VOLTAGES AND THEIR RANGES
FOR CIRCUIT BREAKERS

The maximum voltage is measured at the point of user connection to the circuit breaker (see Notes (9) (10)) with no operating current flowing, and the minimum voltage is measured with maximum operating current flowing.

Rated Control Voltages and their Ranges for Circuit Breakers					
Rated Control Voltage (8)	Direct Current Voltage Ranges (1)(2) Volts, DC		Opening Functions All Types	Rated Control Voltage (60 Hz)	Alternating Current (1)(2)(3)(5) Closing, Tripping, and Auxiliary Functions
	Closing and Auxiliary Functions				
	Indoor Circuit Breakers	Outdoor Circuit Breakers		Single Phase	SINGLE PHASE
24	---	---	14-28	120	104-127 (4)
48	38-56	36-56	28-56	240	208-254 (4)
125	100-140	90-140	70-140	Polyphase	Polyphase
250	200-280	180-280	140-280	208Y/120	180Y/104-
---	---	---	---		220Y/127
---	---	---	---		208-254

Derived from Table 8, ANSI C37.06-2000, AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis – Preferred Ratings and Related Required Capabilities.

Rated Control Voltages and their Ranges for Circuit Breakers Solenoid-Operated Devices	
Rated Voltage	Closing Voltage Ranges for Power Supply
125 dc	90 – 115 or 105 – 130
250 dc	180 – 230 or 210 – 260
230 ac	190 – 230 or 210 - 260

Notes:

- Electrically operated motors, contactors, solenoids, valves, and the like need not carry a nameplate voltage rating that corresponds to the control voltage rating

- shown in the table as long as these components perform the intended duty cycle (usually intermittent) in the voltage range specified.
2. Relays, motors, or other auxiliary equipment that function as a part of the control for a device shall be subject to the voltage limits imposed by this standard, whether mounted at the device or at a remote location.
 3. Includes supply for pump or compressor motors. Note that rated voltages for motors and their operating ranges are covered by ANSI/NEA MG-1-1978.
 4. Includes heater circuits.
 5. Voltage ranges apply to all closing and auxiliary devices when cold. Breakers using standard auxiliary relays for control functions may not comply at lower extremes of voltage ranges when relay coils are hot, as after repeated or continuous operation.
 6. Direct current control voltage sources, such as those derived from rectified alternating current, may contain sufficient inherent ripple to modify the operation of control devices to the extent that they may not function over the entire specified voltage ranges.
 7. This table also applies for circuit breakers in gas-insulation substation installations.
 8. For an outdoor circuit breaker, the point of user connection to the circuit breaker is the secondary terminal block point at which the wires from the circuit breaker operating mechanism components are connected to the user's control circuit wiring.
 9. In cases where other operational ratings are a function of the specific control voltage applied, tests in C37.09 may refer to the "Rated Control Voltage." In these cases, tests shall be performed at the levels in this column.
 10. For an indoor circuit breaker, the point of user connection to the circuit breaker is either the secondary disconnecting contact (where the control power is connected from the stationary housing to the removable circuit breaker), or the terminal block point in the housing nearest to the secondary disconnecting contact.
 11. Some solenoid operating mechanisms are not capable of satisfactory performance over the range of voltage specified in the standard; moreover, two ranges of voltage may be required for such mechanisms to achieve an acceptable standard of performance.
 12. The preferred method of obtaining the double range of closing voltage is by use of tapped coils. Otherwise, it will be necessary to designate one of the two closing voltage ranges listed above as representing the condition existing at the device location due to battery or lead voltage drop or control power transformer regulation. Also, caution should be exercised to ensure that the maximum voltage of the range used is not exceeded.

END OF SECTION

**SECTION 260923
LIGHTING CONTROL DEVICES**

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260000 - General Electrical Requirements
- B. Section 265000 - Lighting

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION OF SYSTEM

- A. Provide devices such as wall box dimmers, wall and ceiling-mounted occupancy sensors, ambient light sensors, sensor power packs, etc., as shown on drawings.
- B. Openings shall be covered with devices and matching plates.
- C. Devices of same type shall be from same manufacturer.

1.4 REFERENCE STANDARDS

- A. UL 20 - General Use Snap Switches.
- B. UL 773A - Non-Industrial Photoelectric Switches for Lighting Control.
- C. UL 924 - Emergency Lighting and Power Equipment
- D. NEMA WD 7 - Occupancy Motion Sensors.

1.5 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings shall include:
 - 1. Bill of material
 - 2. Schematic diagrams
 - 3. Suggested manufacturer layouts of all devices including overlays of product range.

- C. Samples: One for each type of device and wall plate specified, in each color specified upon request.
- D. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, installation, and starting of product.
- E. Test Reports: Indicate field test and inspection procedures and interpret test results and corrective action taken for compliance with specification requirements.
- F. Closeout Submittals:
 - 1. Project Record Documents:
 - a. Record actual locations and type of devices.
 - 2. Operation and Maintenance Data:
 - a. Include in manufacturers' packing label warnings and instruction manuals with labeling conditions.
 - b. Include source and current prices of replacement parts and supplies.

1.6 QUALITY ASSURANCE

- A. Obtain devices from one source and by single manufacturer.
- B. Regulatory Requirements:
 - 1. Comply with NFPA 70 for components and installation.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory unopened packaging until ready for installation.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers:

1. Wall Box Dimmers: Acuity Brands Controls, Cooper Controls, Hubbell, Leviton, Lutron, Philips (Sunrise Series), Wattstopper
2. Low Voltage Switches: Acuity Brands Controls, Cooper Controls, Hubbell, Leviton, Lutron, Philips, Wattstopper
3. Sensors and Power Packs: Acuity Brands Controls, Cooper Controls, Hubbell, Leviton, Lutron, Philips, Wattstopper
4. UL 924 Emergency Bypass/Control Device: Acuity Brands Controls, Hubbell, Leviton, Wattstopper, LVS
5. Exterior Photocells: Cooper Controls, Hubbell, Intermatic, Leviton, Paragon, Tork
6. Timeclocks: Intermatic, Paragon, Tork
7. Self-Contained Automatic Timer Switches: Acuity Brands Controls, Cooper Controls, Hubbell, Leviton, Philips, Wattstopper

- B. It is the responsibility of Electrical Contractor to ensure devices submitted meet or exceed functional intent and design quality standards.

2.2 FABRICATION AND MANUFACTURE

- A. Devices shall be UL listed for loads and voltages as indicated in contract drawings and specifications.

2.3 WALL BOX DIMMERS

A. Dimmers shall:

1. Operate in ambient temperature range of 32°F to 104°F.
2. Be linear slide or pushbutton preset or programmable dimmers with power-failure memory.
3. Incorporate separate control of intensity and ON/OFF.
4. Include voltage compensation circuitry that adjusts firing angle of dimmer to compensate light output for variations in AC line voltage. Dimmers in which firing angle is held constant with varying AC line voltage shall not be acceptable.
5. Provide smooth and continuous IESNA Square Law Dimming Curve throughout entire dimming range.
6. Incorporate filter network to minimize interference (RFI) with radio, audio, and video equipment.
7. Incorporate air-gap switch to meet requirements of UL 20 for air-gap switches in incandescent dimmers.

B. LED dimmers shall:

1. Be approved for use with luminaire and driver.
2. Provide smooth non-flicker dimming of controlled luminaires.
3. Be 0-10V type, unless otherwise noted on drawings.
4. Provide at least 10 steps for continuously dimmed luminaires.

5. Refer to Section 265000 - Lighting for solid state dimming ballast/driver specification.

2.4 LOW-VOLTAGE SWITCHES

- A. Low voltage switches shall:
 1. Mount in a single or double gang box.
 2. Be capable of multi-way switching.

2.5 OCCUPANCY AND VACANCY SENSORS

- A. Sensors shall:
 1. Operate with all lamp and ballast combinations; including magnetic, hybrid, and solid-state ballasts/drivers.
 2. Operate with ultrasonic, microphonic, passive infrared or presence technologies as indicated on drawings.
 3. Have visible LED to indicate occupant detection.
 4. Have adjustable time delay or remote relay with a maximum setting of 30 minutes and adjustable sensitivity.
 5. Contain isolated relay with normally open, normally closed, and common outputs for use with HVAC system, data logging, controlled receptacles or other system control options where indicated in contract documents.
 6. Be provided with ceiling, wall or wall switch style mounting as indicated on drawings.
 7. Have daylight filter to ensure PIR sensor is insensitive to short-wavelength waves emitted by the sun.
 8. Incorporate by-pass switch to enable lighting to be turned on if sensor fails.
- B. Occupancy Sensor shall:
 1. Provide automatic ON, automatic OFF operation where indicated on drawings.
- C. Vacancy Sensor shall:
 1. Provide manual ON, automatic OFF operation where indicated on drawings.

2.6 AMBIENT LIGHT SENSORS

- A. Ambient light sensors shall:
 1. Incorporate photoconductive cell to measure light levels between 1 and 1,000 footcandles.
 2. Be adjustable with deadband feature to prevent cycling of lighting from minor changes in cloud cover.
 3. Have adjustable time delay range from 3 to 5 minutes.
 4. Not permit lighting systems to be turned on if enough daylight is present.
 5. Incorporate by-pass switch to enable lighting to be turned on if sensor fails.

2.7 POWER PACKS

- A. Sensor power packs shall:
 1. Be self-contained transformer relay modules.
 2. Have universal rated voltage inputs 120-277 VAC, 60 Hz.

3. Have normally closed dry contacts rated for switching 120-277 volts, 60 Hz. 20 amp loads. Provide 24VDC output capable of controlling low-voltage occupancy sensors.

2.8 EXTERIOR PHOTOCELLS

- A. Photocells shall:
 1. Have universal rated voltage inputs 120-277 VAC, 60 Hz.
 2. Be rated for up to 2,000 watts.
 3. Have cadmium sulfide, 1" diameter cell.
 4. Have SPST normally closed contacts.
 5. Have a minimum delay of 3 minutes to prevent false switching.
- B. ON/OFF adjustment shall be done by moving light selector with range from 2 to 50 footcandles.
- C. Operational temperature range shall be -40°F to 140°F.
- D. Enclosure shall be die cast zinc, gasketed for maximum weatherproofing.
- E. Enclosure shall include positioning lug on top.
- F. Mounting shall be for 1/2" conduit nipple.

2.9 TIMECLOCKS

- A. Timeclocks shall:
 1. Be multi-purpose, 7-day, 365-day advance single and skip a day, combination 2-channel electronic astronomical time clock with SPDT switching configuration.
 2. Have universal rated voltage inputs 120-277 VAC, 60 Hz.
 3. Be capable of programming in AM/PM or 24-hour format by jumper selection or digital setting, in one-minute resolution, using 2 buttons for basic settings.
 4. Have 365-day and/or holiday selection capabilities, with 16 single date and 5 holiday selection options and user selectable daylight savings/standard time functions.
 5. Have 72-hour memory backup with rechargeable battery and charger.
 6. Have manual override, ON/OFF to the next scheduled event, using one button for each channel.
 7. Have operational temperature range of -40°F to 150°F.
 8. Have a maximum allowed over-ride period no greater than 2 hours.
- B. Contacts shall be rated 10 amp resistive at 120/250 VAC, 7.5 amps inductive at 120/250 VAC, 5 amps inductive at 30 VDC and up to 1/2 hp at 250 VAC.
- C. Display shall be LCD type.
- D. Enclosure shall be rated for installation location.

2.10 FINISHES

- A. Color:
 - 1. Wall box dimmers, low-voltage switches, occupancy sensors, ambient light sensors and device cover plates: white.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install devices at heights scheduled, and as indicated on drawings.
- B. Install wall devices vertically on latch side of door within 6" of frame edge, unless otherwise noted.
- C. Install ceiling devices as shown on drawings and as recommended by device manufacturer.
- D. Ceiling mounted occupancy sensors shall be located minimum of 6 ft from supply air diffusers.
- E. Install devices plumb, level with finished surfaces and free from blemishes.
- F. Verify device locations prior to rough in.
- G. Control wiring shall be low voltage, Class II wiring, electrically isolated from power wiring by a Class II transformer.
- H. Provide separate neutral conductor for each dimmer.
- I. Wiring shall be in conduit.
- J. Electrical Contractor shall be responsible for final adjustment and testing of all devices.

3.2 TESTING

- A. Verify proper location and operation of all devices.
- B. Verify dimmers function without:
 - 1. Producing lamp flicker or audible noise.
 - 2. Interference of audio and visual equipment.
- C. Adjust occupancy sensors for a 15 minute time delay.
- D. Adjust occupancy sensor sensitivity such that movement outside range of coverage shall not trigger sensor.

- E. Adjust ambient light sensor to maintain illuminance level equal to light level from controlled lighting in the space when no daylight is present or as indicated per drawings. Demonstrate ambient light sensor(s) control lighting as specified.

END OF SECTION

SECTION 262416.13
LIGHTING AND APPLIANCE PANELBOARDS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 260529 - Hangers and Supports for Electrical Systems
- D. Section 260533 - Raceway and Boxes for Electrical Systems
- E. Section 260553 - Electrical Systems Identification
- F. Section 260573 - Power System Studies
- G. Section 260812 - Power Distribution Acceptance Tests
- H. Section 260813 - Power Distribution Acceptance Test Tables
- I. Section 262813 - Fuses
- J. Section 264300 - Surge Protective Devices (SPD)

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes circuit breaker type lighting and appliance branch circuit panelboards as shown on drawings and as scheduled.

1.4 REFERENCE STANDARDS

- A. NECA 407 - Recommended Practice for Installing and Maintaining Panelboards
- B. NEMA EN 10250 - Enclosures for Electrical Equipment (1000 Volts Maximum)
- C. NEMA AB 1 - Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit Breaker Enclosures

- D. NEMA FU 1 - Low-Voltage Cartridge Fuses
- E. NEMA PB 1 - Panelboards
- F. NEMA PB 1.1 - General Instructions for Proper Installation, Operation, and Maintenance of Panelboards Rated 600 Volts or Less
- G. NFPA 70 - National Electrical Code
- H. UL 50 - Enclosures for Electrical Equipment
- I. UL 67 - Panelboards
- J. UL 248-1 - Low-Voltage Fuses
- K. UL 486A-486B - Wire Connectors
- L. UL 489 - Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit Breaker Enclosures
- M. UL 869A - Reference Standard for Service Equipment

1.5 SUBMITTALS

- A. Product Data:
 - 1. Submit catalog data showing specified features of standard products. Eliminate extraneous catalog data.
- B. Shop Drawings:
 - 1. Submit for review prior to manufacture. Include complete description, front view, dimensions, voltage, main bus ampacity, circuit breaker arrangement and sizes, short circuit current rating, and factory settings of individual protective devices.
 - 2. Submit 1/4" scale electrical room floor plans with panelboard locations.
- C. Partial Submittals:
 - 1. Panelboards shall be submitted for review together. Partial submittals of panelboards are not acceptable and will be rejected.
- D. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, installation, and starting of product.
- E. Test Reports:
 - 1. Indicate field test and inspection procedures and interpret test results and corrective action taken for compliance with specification requirements.

- F. Closeout Submittals:
 - 1. Project Record Documents:
 - a. Record actual locations of panelboards and record actual circuiting arrangements.
 - 2. Operation and Maintenance Data:
 - a. Include manufacturer's recommended operating instructions, maintenance procedures and intervals, and preventive maintenance instructions.
 - b. Include manufacturer's written instructions for testing and adjusting overcurrent protective devices.
 - c. Include time-current curves and selectable ranges for each type of overcurrent protective device.
 - d. Include spare parts data listing, source, and current prices of replacement parts and supplies.
 - e. Include manufacturer's Seismic Qualification Certification and Installation Seismic Qualification Certification.

1.6 QUALITY ASSURANCE

- A. Obtain panelboards, overcurrent protective devices, components, and accessories from one source and by single manufacturer.
- B. Regulatory Requirements:
 - 1. Comply with NFPA 70.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory wrapping or provide additional canvas or plastic cover to protect from dirt, water, construction debris, and traffic.
- B. Comply with NEMA PB 1.1 and manufacturer's written instructions.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr written warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

1.9 MAINTENANCE

- A. Extra Materials:
 - 1. Furnish Owner with two keys per panelboard.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Schneider Square D
- B. ABB
- C. Eaton Cutler Hammer
- D. Siemens

2.2 LIGHTING AND APPLIANCE BRANCH CIRCUIT PANELBOARDS

- A. NEMA PB 1, UL 67 [98]
- B. Fabrication:
 - 1. Factory assembled.
 - 2. door-in-door construction.
 - 3. Incoming feeder lugs: copper conductors.
 - 4. Multiple lugs to match number of conductors per phase.
 - 5. Sub-feed (double) lugs, or feed-through lugs where indicated.
 - 6. Filler plates.
 - 7. Wiring terminals for field installed conductors: Pressure wire connectors, except wire-binding screws for #10 AWG or smaller conductors.
- C. Panelboard Buses:
 - 1. Copper
 - 2. Ampere rating as scheduled
 - 3. Ground bus: uninsulated, bonded to panelboard cabinet
 - 4. Insulated neutral bus: 100% of phase bus rating
- D. Molded-Case Circuit Breakers:
 - 1. NEMA AB 1, UL 489
 - 2. Bolt-on type, labeled for 75°C copper and aluminum conductors
 - 3. Quick-make, quick-break, with thermal-magnetic trip.
 - 4. Common internal trip on multi-pole breakers. Handle-ties are not permitted.
 - 5. Ampere rating as scheduled
 - 6. Listed as Type SWD for lighting circuits
 - 7. Listed as Type HACR for air conditioning equipment circuits
 - 8. Bussing, device mounting hardware, and steel knockouts in dead front where "space" is indicated
 - 9. Tandem circuit breakers are not acceptable
 - 10. Ground fault equipment protection (GFEP), rated 30 mA trip, to provide equipment protection for branch circuits feeding electrical heat tracing, where indicated
 - 11. Ground fault circuit interrupter (GFCI), rated at 4-6 mA trip for protection of personnel, where indicated

E. Cabinet

1. NEMA EN 10250, UL 50
2. NEMA Type 1, Type 3R (outdoor locations) enclosure.
3. Front (trim) surface mounted with door in front with concealed self-adjusting trim clamps, and complete with cylinder-type lock and catch.
4. Same height matching trim, where two cabinets are mounted adjacent to one another in finished areas.
5. All sections of panelboards have the same size, where oversize cabinets are required for one section of multi-section panelboard.
6. Boxes and fronts made of code-gauge galvanized steel.
7. Manufacturer's standard gray enamel finish over prime coat.

2.3 SHORT CIRCUIT CURRENT RATING

- A. Each panelboard with minimum short circuit current rating as indicated on drawings.
- B. Panelboards marked with their maximum short circuit current rating at supply voltage.
- C. Panelboards: Fully rated

2.4 SURGE PROTECTIVE DEVICES (SPD)

- A. Furnished under 264300 - Surge Protective Devices (SPD)

2.5 SPARE CONDUITS

- A. Spare conduits per requirements in Section 260533 - Raceway and Boxes for Electrical Systems.

PART 3 - EXECUTION

3.1 COORDINATION WITH MANUFACTURER

- A. Instruct manufacturer about the location of additional wiring gutter space when required (i.e., top, bottom, right, left, or combination).
- B. Instruct manufacturer about the location of main lugs or main circuit breaker (i.e., top or bottom feed based on incoming feeder entrance location).
- C. Instruct manufacturer to provide multiple lugs where conductors in parallel or sub-feed (double) lugs or feed-through lugs are indicated.
- D. Instruct manufacturer on the size of cross-connection cables for panelboards fed via sub-feed (double) lugs or feed-through lugs. Make cable size with ampacity equal to incoming feeder.
- E. Verify that "touch-up" paint kit is available for repainting.

- F. Coordinate painting of cabinets in finished areas with work performed under Division 09 - Finishes.

3.2 EXAMINATION

- A. Verify that space indicated for panelboard mounting meets code-required working clearances and dedicated equipment space.
- B. Notify Architect/Engineer of any discrepancies prior to submittal of product data and shop drawings.

3.3 INSTALLATION

- A. Install panelboards in accordance with NECA 407 and NEMA PB 1.1.
- B. Install panelboards plumb and rigid without distortion of box, in accordance with manufacturer's written instructions, and in compliance with recognized industry practices.
- C. Panelboard mounting and seismic restraints:
 - 1. Fasten panelboards firmly to walls and structural surfaces, ensuring they are permanently and mechanically anchored.
 - 2. Anchor and fasten panelboards and their supports to building structural elements (wood, concrete, masonry, hollow walls and nonstructural building surfaces) by the methods described in Section 260529 - Hangers and Supports for Electrical Systems.
 - 3. Install two rows of steel slotted channel, with a minimum of 4 attachment points, for each panelboard section.
 - 4. When not located directly on wall, provide support frame of steel slotted channel anchored to floor and ceiling structure.
- D. Install top breaker handle a maximum of 6'-7" above finished floor or working platform with handle in its highest position.
- E. Tighten electrical connectors and terminals according to equipment manufacturer's published torque-tightening values. Where manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- F. Install as-built typewritten circuit directory in directory frame (to indicate installed circuit loads) mounted inside each panelboard door. Include description of connected loads, room number, room name, area, or item served for each branch circuit. Indicate motor names and horsepower as applicable. Cover circuit directory with colorless plastic. Coordinate with Owner and Architect to ensure that room numbers used in panel directory are final numbers assigned by Owner.

- G. Install engraved plastic nameplates under provisions of Section 260553 - Electrical Systems Identification. Attach nameplate to exterior of each panelboard using small metal screws or rivets. Do not use contact adhesive.
 - 1. Include panelboard name, amperage, voltage, phase, and number of wires.
- H. Label spare circuits as SPARE. Leave spare breakers in OFF position.
- I. Room numbers used shall be those used by Owner except as otherwise directed by Architect.
- J. Install panelboard in dedicated electrical space per NFPA 70 and as shown on drawings. Coordinate with miscellaneous trades for equipment foreign to the electrical installation to be outside of dedicated electrical space.
- K. Install filler plates in unused spaces.
- L. Install three 3/4" spare conduits stubbed into accessible ceiling space or space designated to be ceiling space in the future for all flush-mounted panelboards. Install conduits in accordance with requirements in Section 260533 - Raceway and Boxes for Electrical Systems.
- M. Install three 3/4" spare conduits stubbed into ceiling space above and below for panelboards that serve loads on levels other than that where the panelboard is located. Install conduits in accordance with requirements in Section 260533 - Raceway and Boxes for Electrical Systems.

3.4 CONNECTIONS

- A. Ground panelboards according to Section 260526 - Grounding and Bonding for Electrical Systems.
- B. Connect wiring according to Section 260519 - Low-Voltage Electrical Power Conductors and Cables.

3.5 FIELD QUALITY CONTROL

- A. Inspect for physical damage, proper alignment, anchorage, and grounding.
- B. Maintain proper phasing for multi-wire circuits.
- C. Test main circuit breakers in accordance with requirements in Sections 260812 - Power Distribution Acceptance Tests and 260813 - Power Distribution Acceptance Test Tables.

3.6 REPAINTING

- A. Remove paint splatters or other marks from surface of panelboards.

- B. Touch-up chips, scratches, or marred finishes to match original finish, using manufacturer-supplied paint kit. Leave remaining paint with Owner.

3.7 ADJUSTING

- A. Adjust fronts, covers, hinges, and locks.

3.8 CLEANING

- A. Clean panelboard interiors and exteriors prior to final inspection. Remove paint splatters and other spots, dirt and debris.

END OF SECTION

SECTION 262726 WIRING DEVICES

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260526 - Grounding and Bonding for Electrical Systems
- B. Section 260553 - Electrical Systems Identification

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes general-use snap switches, wall-box dimmers, fan speed controls, receptacles, hazardous (classified) location receptacles, pendant cord-connector devices, cord and plug sets and device cover plates.

1.4 REFERENCE STANDARDS

- A. IEEE C62.41.2 – Characterization of Surges in Low-Voltage (1000V and less) AC Power Circuits
- B. IEEE C62.45 – Surge Testing for Equipment Connected to Low-Voltage (1000V and less) AC Power Circuits
- C. NECA 1 – Good Workmanship in Electrical Contracting
- D. NFPA 70 – National Electrical Code
- E. NFPA 99 – Health Care Facilities
- F. NEMA FB 11 – Plugs, Receptacles, and Connectors of the Pin and Sleeve Type for Hazardous Locations
- G. NEMA WD 1 – General Color Requirements for Wiring Devices
- H. NEMA WD 6 – Wiring Devices - Dimensional Requirements
- I. NEMA EN 10250 – Enclosures for Electrical Equipment (1000 Volts Maximum)

- J. UL 20 – General-Use Snap Switches
- K. UL 498 – Attachment Plugs and Receptacles
- L. UL 943 – Ground-Fault Circuit-Interrupters
- M. UL 1203 – Safety Explosion-Proof and Dust-Ignition Proof Electrical Equipment for Use in Hazardous (Classified) Locations
- N. UL 1436 – Outlet Circuit Testers and Similar Indicating Devices
- O. UL 1449 – Transient Voltage Surge Suppressors
- P. UL 1472 – Solid-State Dimming Controls
- Q. UL 1917 – Solid-State Fan Speed Controls

1.5 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings: List of legends and description of materials and process used for premarking wall plates.
- C. Samples: One for each type of device and wall plate specified, in each color specified.
- D. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, installation, and starting of product.
- E. Test Reports: Indicate field test and inspection procedures and interpret test results and corrective action taken for compliance with specification requirements.
- F. Closeout Submittals:
 - 1. Project Record Documents:
 - a. Record actual locations and ratings of wiring devices.
 - 2. Operation and Maintenance Data:
 - a. Include in manufacturers' packing label warnings and instruction manuals with labeling conditions.
 - b. Include source and current prices of replacement parts and supplies.

1.6 QUALITY ASSURANCE

- A. Obtain wiring devices from one source and by single manufacturer.
- B. Regulatory Requirements:
 - 1. Comply with NFPA 70 for components and installation.

2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory unopened packaging until ready for installation.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

1.9 MAINTENANCE

- A. Extra Materials: Furnish extra materials described below that match products installed, are packaged with protective covering for storage, and are identified with labels describing contents.
 1. TVSS Receptacles: One for every 10 of each type installed, but no fewer than two of each type.
- B. Furnish 1 key for each switch.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Cooper Wiring Devices; a division of Cooper Industries, Inc.
- B. Hubbell Incorporated; Wiring Device-Kellems
- C. Leviton Manufacturing Company, Inc.
- D. Pass & Seymour/Legrand; Wiring Devices & Accessories

2.2 GENERAL-USE SNAP SWITCHES

- A. Comply with NEMA WD 1 and UL 20.
- B. Switches: Heavy-duty (specification grade); back and side wired; flush or surface mounting; Body and Handle: thermoplastic with toggle handle; for connection to copper or copper-clad conductors:
 1. Ratings:

- a. Voltage: 120-277V, AC
- b. Current: 20 A
2. Single pole
3. Double pole
4. Three-way
5. Four-way
6. Locator Light: Lighted handle type switch (single pole with red neon-lighted handle, illuminated when switch is "OFF.")
7. Pilot Light: Indicator light switch (single pole with red neon-lighted handle, illuminated when switch is "ON.")
8. Locking Type: Designed to prevent tampering and unauthorized switching.
9. Key-Operated: Single pole, with factory-supplied key in lieu of switch handle.
10. Single-Pole, Double-Throw, Momentary Contact, Center-Off: For use with mechanically held lighting contactors.
11. Key-Operated, Single-Pole, Double-Throw, Momentary Contact, Center-Off: For use with mechanically held lighting contactors, with factory-supplied key in lieu of switch handle.
12. Pendant and Through-cord: For field installation on flexible cord and provided with one "ON" and one "OFF" position.
13. Weatherproof: Toggle switch
14. Dimmers are permitted to control lighting fixtures but not motorized loads or receptacles. Fan speed controls are listed for that application.
15. Special devices may be desirable to minimize buzzing at lights, radio frequency interference, or in the case of multiwire circuits with dimmers on multiple phases, and variation in brightness due to interaction through the grounded conductor.
16. Fluorescent luminaires, unless specifically marked otherwise, require special dimmers and dimming ballasts.

2.3 WALL-BOX DIMMERS

- A. Comply with NEMA WD 1 and UL 1472.
- B. Dimmer Switches: Modular, full-wave, solid-state units with integral, quiet on-off switches, with audible frequency and EMI/RFI suppression filters.
 1. Power Rating: Match load shown on drawings watts As indicated on drawings
 2. Body: Thermoplastic
 3. Control: Continuously adjustable slider; with single-pole or three-way switching.
 4. Incandescent Lamp Dimmers: 120 V; control shall follow square-law dimming curve. On-off switch positions shall bypass dimmer module.
 5. Fluorescent Lamp Dimmer Switches: Modular; compatible with dimmer ballasts; trim potentiometer to adjust low-end dimming; dimmer-ballast combination capable of consistent dimming with low end not greater than 20% of full brightness; 277 V.

2.4 FAN SPEED CONTROLS

- A. Comply with UL 1917

- B. Modular, 120V, full-wave, solid-state units with integral, quiet on-off switches and audible frequency and EMI/RFI filters.
 - 1. Continuously adjustable rotary knob, 5A.
 - 2. Three-speed adjustable rotary knob, 1.5A.

2.5 RECEPTACLES

- A. Comply with NEMA WD 1, NEMA WD 6 configuration 5-20R, and UL 498.
- B. Receptacles: 125 V, 20A, heavy-duty (specification grade); back and side wired; flush or surface mounted; straight blade; 2 pole, 3 wire grounding; thermoplastic body; duplex and as indicated on drawings.
 - 1. Ground Fault Circuit Interrupter (GFCI):
 - a. Additional compliance with UL 943 Class A.
 - b. Leakage current trip level: 4 to 6 mA.
 - c. Trip time: .025 seconds nominal.
 - d. Non-feed through type
 - e. Reverse line-load function to prevent GFCI from functioning if wired incorrectly.
 - f. Indicator Light: Lighted when device is tripped.
 - g. Due to numerous issues with nuisance tripping, Hubbell model# GFRST83W / GFRST83R analog self-test GFCI receptacles shall not be approved. Hubbell model # GFR83W will be acceptable.
 - 2. Isolated Ground (IG):
 - a. Ground strap isolated from mounting strap.
 - b. Ground screw connected directly to ground contacts.
 - 3. Transient Voltage Surge Suppression (TVSS):
 - a. Additional compliance with UL 1449, with integral TVSS in line to ground, line to neutral, and neutral to ground.
 - b. TVSS Components: Multiple metal-oxide varistors; with a nominal clamp-level rating of 400 volts and minimum single transient pulse energy dissipation of 240 J, according to IEEE C62.41.2 and IEEE C62.45.
 - c. Active TVSS Indication: Visual and audible, with light visible in face of device to indicate device is "active" or "no longer in service."
 - 4. Tamper Resistant (TR):
 - a. Requires insertion of object in both left and right contacts to energize.
 - b. 2- or 3-prong plug.
 - 5. Twist-locking:
 - a. NEMA WD 6 configuration L5-20R
 - 6. Switched: Upper half switched and lower half not switched.
 - 7. Dedicated: Labeled "Dedicated."
 - 8. Special Purpose Receptacles: Specification grade, rated for voltage, amperage and NEMA configuration as noted on drawings.

2.6 HAZARDOUS (CLASSIFIED) LOCATION RECEPTACLES

- A. Comply with NEMA FB 11 and UL 1203

- B. Manufacturers:
 - 1. Cooper Crouse-Hinds
 - 2. EGS/Appleton Electric
 - 3. Killark; a division of Hubbell, Inc.

2.7 PENDANT CORD-CONNECTOR DEVICES

- A. Description: Matching, locking-type plug and receptacle body connector; NEMA WD 6 configuration L5-20P and L5-20R, heavy-duty grade.
 - 1. Body: Nylon with screw-open cable-gripping jaws and provision for attaching external cable grip.
 - 2. External Cable Grip: Woven wire-mesh type made of high-strength galvanized-steel wire strand, matched to cable diameter, and with attachment provision designed for corresponding connector.

2.8 CORD AND PLUG SETS

- A. Description: Match voltage and current ratings and number of conductors to requirements of equipment being connected.
 - 1. Cord: Thermoset-insulated, stranded-copper conductors, with Type SOOW jacket; with green-insulated grounding conductor and equipment-rating ampacity plus a minimum of 30%.
 - 2. Plug: Nylon body and integral cable-clamping jaws. Match cord and receptacle type for connection.

2.9 DEVICE COVER PLATES

- A. Single and combination types to match corresponding wiring devices:
 - 1. Attachment: Metal screws with head color to match plate finish.
 - 2. Material for Finished Spaces: Steel with white baked enamel, suitable for field painting.
 - 3. Material for Unfinished Spaces: Smooth, high-impact thermoplastic.
 - 4. Material for Damp Locations: Thermoplastic with while-in-use hinged cover, and listed and labeled for use in "wet locations".
- B. Weatherproof Cover Plates: NEMA EN 10250, complying with type 3R weather-resistant, die-cast aluminum with weatherproof while-in-use hinged cover.
- C. Lockable Cover:
 - 1. Hinged steel cover with cylinder lock.
 - 2. Keyed all the same.
- D. Tamper Resistant (TR):
 - 1. Slide cover over receptacle.

2.10 FINISHES

- A. Color:
 - 1. Switch handles, receptacle faceplates, and device cover plates: white, except as follows:

PART 3 - EXECUTION

3.1 COORDINATION

- A. Special Purpose Receptacles: Coordinate final selections of NEMA configuration (locking, straight, blade, etc.) with configuration of plug on utilization equipment.
- B. Receptacles for Owner-furnished equipment and equipment furnished under other divisions of specifications: Match plug configurations.
- C. Cord and Plug Sets: Match equipment requirements.
- D. Coordination with Other Trades:
 - 1. Take steps to ensure that devices and their boxes are protected. Do not place wall finish materials over device boxes and do not cut holes for boxes with routers guided by riding against outside of the boxes.
 - 2. Keep outlet boxes free of plaster, drywall joint compound, mortar, cement, concrete, dust, paint, and other material that may contaminate the raceway system, conductors, and cables.
 - 3. Install device boxes in brick or block walls so that the device cover plate does not cross a joint unless the joint is troweled flush with the face of the wall.
 - 4. Install wiring devices after all wall preparation, excluding painting, is complete. Install device cover plates after painting is complete.

3.2 EXAMINATION

- A. Verify location of wiring devices with architectural interior elevation drawings, prior to rough-in.
- B. Verify outlet boxes are installed at proper height.
- C. Verify wall openings are neatly cut and completely covered by wall plates.
- D. Verify branch circuit wiring installation is completed, tested, and ready for connection to wiring devices.

3.3 PREPARATION

- A. Clean debris from outlet boxes.

3.4 INSTALLATION

- A. Comply with NECA 1, including the mounting heights listed in that standard, unless otherwise scheduled or indicated on drawings. Indicated dimensions are to center of device.
- B. Conductors:
 - 1. Do not strip insulation from conductors until just before they are spliced or terminated on devices.
 - 2. Strip insulation evenly around the conductor using tools designed for the purpose. Avoid scoring or nicking of solid wire or cutting strands from stranded wire.
 - 3. Length of free conductors at outlets for devices shall meet provisions of NFPA 70, Article 300, without pigtails.
 - 4. Do not place bare stranded conductors directly under device screws. Use crimp on fork terminals for device terminations.
- C. Device Installation:
 - 1. Replace all devices that have been in temporary use during construction or show signs of installation prior to completion of building finishing operations.
 - 2. Keep each wiring device in its package or otherwise protected until it is time to connect conductors.
 - 3. Do not remove surface protection, such as plastic film and smudge covers, until last possible moment.
 - 4. Connect devices to branch circuits using pigtails that are not less than 6" in length.
 - 5. When there is a choice, use side wiring with binding-head screw terminals. Wrap solid conductor tightly clockwise, 2/3 to 3/4 of the way around terminal screw.
 - 6. Use a torque screwdriver when a torque is recommended or required by the manufacturer.
 - 7. When conductors larger than #12 AWG are installed on 15A or 20A circuits, splice #12 AWG pigtails for device connections.
 - 8. Tighten unused terminal screws on the device.
 - 9. When mounting into metal boxes, remove fiber or plastic washers used to hold device mounting screws in yokes, allowing metal-to-metal contact.
 - 10. Install devices plumb, level with finished surfaces and free from blemishes.
 - 11. Install lighting switches vertically on latch side of door within 6" of frame edge.
 - 12. Install devices above counters, 2" to the bottom of device above countertop or backsplash. Install all devices at same height above any one counter or fixed cabinet.
 - 13. Install special purpose receptacles and switches according to shop and rough-in drawings furnished by trade(s) producing such equipment. Verify locations prior to rough-in.
 - 14. Install weatherproof GFCI receptacles:
 - a. Within 25'-0" of roof-mounted mechanical equipment
 - b. Outdoors
 - c. As indicated on drawings
 - 15. Group adjacent switches under single, multigang wall plates.

16. Connect wiring device grounding terminal to branch circuit equipment grounding conductor. Ground per requirements in Section 260526 - Grounding and Bonding for Electrical Systems.

D. Installation Orientations:

1. Install ground pin of vertically mounted receptacles up, and on horizontally mounted receptacles to the left.
2. Install hospital-grade receptacles in patient-care areas with the ground pin or neutral blade at the top.
3. Install switches with handle operating vertically, with "ON" position up.
4. Unless otherwise indicated or where space problem occurs, mount devices flush, with long dimension vertical.

- E. Device Cover Plates: Do not use oversized or extra-deep plates. Repair wall finishes and remount outlet boxes when standard device plates do not fit flush or do not cover rough wall opening.

F. Wall-Box Dimmers:

1. Install dimmers within terms of their listing.
2. Verify that dimmers used for fan speed control are listed for that application.
3. Install unshared neutral conductors on line and load side of dimmers according to manufacturers' device listing conditions in the written instructions.

G. Arrangement of Devices:

1. Unless otherwise indicated or where space problem occurs, mount flush, with long dimension vertical and with grounding terminal of receptacles on top. Group adjacent switches under single, multigang wall plates.

3.5 IDENTIFICATION

A. Comply with Section 260553 - Electrical Systems Identification.

1. Switches and Receptacles: Use hot, stamped or engraved machine printing with black-filled lettering on face of cover plate, and durable wire markers or tags inside outlet boxes.
 - a. Receptacles: Label shall indicate receptacle voltage, phase, and amperage for receptacles other than 20A, 120 V, at top of cover plate, and panel and circuit number at bottom of cover plate.
 - b. Switches: Label shall indicate switch voltage, phase, and amperage at top of cover plate, and panel, circuit number and switch designation at bottom of cover plate.
2. Engrave cover plates on all Owner-furnished equipment and equipment furnished under other divisions of these specifications with panelboard, circuit number and "emergency" (where applicable) as specified in this section. This includes headwalls, gas columns and booms, patient consoles, medical rail systems, custom casework with electrical devices, etc.

3.6 FIELD QUALITY CONTROL

- A. Inspect wiring devices for defects.

- B. Operate wall switches with circuits energized and verify proper operation.
 - C. Verify receptacle device is energized.
 - D. Perform tests and prepare test reports:
 - 1. Test receptacle devices for proper polarity:
 - a. Test every receptacle with receptacle circuit tester. Tester shall test for open ground, reverse polarity, open hot, open neutral, hot and ground reversed, hot or neutral and hot open. Rewire receptacles with faults and retest.
 - 2. Test each GFCI receptacle device for proper operation:
 - a. Perform testing using an instrument specifically designed and manufactured for testing ground-fault circuit interrupters. Apply the test to the receptacle. "TEST" button operation will not be acceptable as a substitute for this test. Replace receptacles that do not shut off power with 5/1000 A within 1/40 second and retest.
 - 3. In healthcare facilities, prepare reports that comply with recommendations in NFPA 99.
 - 4. Test Instruments: Use instruments that comply with UL 1436.
 - 5. Test Instrument for Convenience Receptacles: Digital wiring analyzer with digital readout or illuminated LED indicators of measurement.
 - E. Tests for Convenience Receptacles:
 - 1. Line Voltage: Acceptable range is 105 V to 132 V.
 - 2. Percent Voltage Drop under 15A Load: A value of 5% or higher is not acceptable.
 - 3. Ground Impedance: Values of up to 2 ohms are acceptable.
 - 4. GFCI Trip: Test for tripping values specified in UL 1436 and UL 943.
 - 5. Using the test plug, verify that the device and its outlet box are securely mounted.
 - 6. The tests shall be diagnostic, indicating damaged conductors, high resistance at the circuit breaker, poor connections, inadequate fault current path, defective devices, or similar problems. Correct circuit conditions, remove malfunctioning units and replace with new ones, and retest as specified above.
 - F. Operational Tests: Demonstrate the operation of each switch with the systems fully energized and operating. Each switch shall be demonstrated three times.
 - G. Interpret test results in writing and submit to Engineer.
- 3.7 ADJUSTING
- A. Adjust devices and wall plates to be flush and level.
- 3.8 CLEANING
- A. Remove excess plaster from interior of outlet boxes.
 - B. Clean devices and cover plates after painting is complete. Replace stained or improperly painted devices and cover plates.

END OF SECTION

SECTION 262813 FUSES

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260573 - Power System Studies
- B. Section 262816 - Enclosed Switches and Circuit Breakers
- C. Section 262913 - Enclosed Controllers

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and section under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes nonrenewable cartridge fuses, rated 600V and less, for use in low-voltage power distribution system and spare fuse cabinet.

1.4 REFERENCE STANDARDS

- A. NEMA FU 1 - Low Voltage Cartridge Fuses
- B. UL 248-1 - Low Voltage Fuses - Part 1: General Requirements
- C. UL 248-4 - Low-Voltage Fuses - Part 4: Class CC Fuses
- D. UL 248-5 - Low-Voltage Fuses - Part 5: Class G Fuses
- E. UL 248-8 - Low-Voltage Fuses - Part 8: Class J Fuses
- F. UL 248-10 - Low-Voltage Fuses - Part 10: Class L Fuses
- G. UL 248-12 - Low-Voltage Fuses - Part 12: Class R Fuses
- H. UL 248-15 - Low-Voltage Fuses - Part 15: Class T Fuses
- I. UL 512 – Fuseholders

1.5 SUBMITTALS

- A. Product Data:
 - 1. Submit the following for each fuse type and size indicated:
 - a. Manufacturer's technical data on features, performance, electrical characteristics, ratings, and dimensions.
 - b. Time-current curves, coordination charts and tables, and related data.
 - c. Let-through current curves for fuses with current-limiting characteristics.
 - d. Fuse size for each elevator disconnect switch.
- B. Closeout Submittals:
 - 1. Project Record Documents:
 - a. Record actual class, size, and location of fuses.

1.6 QUALITY ASSURANCE

- A. Obtain fuses from one source and by single manufacturer.
- B. Comply with NFPA 70 for components and installation.

1.7 MAINTENANCE

- A. Extra Materials:
 - 1. Furnish to the Owner a quantity of spare fuses equal to 20% of the total quantity of each fuse class and size installed, minimum of 3 of each fuse class and size.
 - 2. Furnish 2 fuse pullers for each size fuse.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Cooper Bussmann
- B. Mersen
- C. Littelfuse
- D. Edison Fusegear

2.2 CARTRIDGE FUSES

- A. NEMA FU 1, UL 248-1.
- B. Characteristics: nonrenewable current-limiting cartridge fuse; current rating and class, as specified or indicated, and voltage rating consistent with circuit voltage.

C. Miscellaneous data:

UL Standard	Class	Volts	Amperage	Interrupting Rating (Amp RMS Sym.)
248-4	CC	600	0-30	200,000
248-5	G	600	0-20	100,000
248-5	G	480	25-60	100,000
248-8	J	600	0-600	200,000
248-10	L	600	601-6000	200,000
248-12	RK1	250 or 600	0-600	200,000
248-12	RK5	250 or 600	0-600	200,000
248-15	T	300	0-1200	200,000
248-15	T	600	0-800	200,000

2.3 FUSEBLOCKS

- A. UL 512
- B. Thermoplastic base with UL flammability 94VO
- C. Clip reinforcing springs – 100A and above
- D. 200,000 A RMS Sym withstand rating
- E. Copper or aluminum connections

2.4 TOUCH SAFE FUSEHOLDERS

- A. UL 512
- B. Thermoplastic base with UL flammability 94VO
- C. Cover over fuses
- D. Neon indicator lamp: “ON” when fuse opens

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine utilization equipment nameplates and installation instructions. Install fuses of sizes and with characteristics appropriate for each piece of equipment.

3.2 INSTALLATION

- A. Verify proper fuse locations, sizes, and characteristics.

- B. Install fuses in fusible devices (specified in Sections 262416.13 - Lighting and Appliance Panelboards, 262816 - Enclosed Switches and Circuit Breakers, and 262913 – Enclosed Controllers) at job site.
- C. Arrange fuses so manufacturer, class, and size are readable without removing fuse.
- D. Install typewritten labels on inside door of each fused device, indicating fuse replacement information.

3.3 APPLICATION

- A. Motor Branch Circuits: Class RK1.

3.4 CLEANING

- A. Clean fuses and tighten connections prior to energizing of equipment.

END OF SECTION

SECTION 262816
ENCLOSED SWITCHES AND CIRCUIT BREAKERS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 260529 - Hangers and Supports for Electrical Systems
- D. Section 260553 - Electrical Systems Identification
- E. Section 260573 - Power System Studies
- F. Section 260812 - Power Distribution Acceptance Tests
- G. Section 260813 - Power Distribution Acceptance Test Tables
- H. Section 262813 - Fuses

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes fusible and non-fusible disconnect switches and circuit breakers in individual enclosures.

1.4 REFERENCE STANDARDS

- A. ANSI/NECA 1 - Standard Practices for Good Workmanship in Electrical Contracting
- B. NEMA AB 1 - Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit Breakers Enclosures
- C. NEMA BS 31047 - Enclosed and Miscellaneous Distribution Equipment Switches (600 Volts Maximum)
- D. NFPA 70 - National Electrical Code

- E. UL 98 - Enclosed and Dead Front Switches
- F. UL 486A-486B - Wire Connectors
- G. UL 489 - Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit Breaker Enclosures
- H. UL 869A - Reference Standard for Service Equipment

1.5 SUBMITTALS

- A. Product Data:
 - 1. Submit catalog cut sheet indicating voltage, amperage, HP ratings, enclosure type, and dimension, fuse clip features, terminal lugs and all accessories including interlock devices, short circuit current ampere rating and factory settings of individual protective devices.
- B. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, installation, and starting of product.
- C. Test Reports:
 - 1. Indicate field test and inspection procedures and interpret test results and corrective action taken for compliance with specification requirements.
- D. Closeout Submittals:
 - 1. Project Record Documents:
 - a. Record actual locations of disconnect switches and ratings of installed fuses.
 - b. Record actual locations and continuous current ratings of enclosed circuit breakers.
 - 2. Operation and Maintenance Data:
 - a. Include manufacturer's recommended operating instructions, maintenance procedures and intervals, and preventive maintenance instructions.
 - b. Include spare parts data listing, source, and current prices of replacement parts and supplies.
 - c. Include Manufacturer's Seismic Qualification Certification and Installation Seismic Qualification Certification.

1.6 QUALITY ASSURANCE

- A. Obtain disconnect switches and enclosed circuit breakers from one source and by single manufacturer.
- B. Regulatory Requirements:
 - 1. Comply with NFPA 70 for components and installation.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory wrapping or provide additional canvas or plastic cover to protect from dirt, water, construction debris, and traffic.
- B. Comply with manufacturer's written instructions.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr written warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Schneider Square D
- B. ABB-GE Industrial Solutions
- C. Eaton Cutler-Hammer
- D. Siemens

2.2 DISCONNECT SWITCHES

- A. NEMA BS 31047, UL 98
- B. Load interrupter enclosed knife switch, heavy-duty type.
- C. Fusible or non-fusible type as indicated.
- D. Switch Interiors:
 - 1. Switch blades that are visible in "OFF" position when switch door is open.
 - 2. Plated current carrying parts.
 - 3. Removable arc suppressors to permit easy access to line side lugs.
- E. Switch Mechanism:
 - 1. Quick-make, quick-break, with visible blades and externally operable handle.
 - 2. Lockable only in "OFF" position and accept three industrial type, heavy-duty padlocks.
 - 3. Dual cover interlock to prevent unauthorized opening of switch door when handle is in "ON" position, and to prevent closing of switch mechanism with door open.

4. Defeater mechanism to bypass interlock.
5. Operating handle integral part of enclosure.
6. Handle to physically indicate "ON" and "OFF" position.

F. Ratings:

1. Ampacity as indicated on drawings.
2. Horsepower rated.

G. Fusible Switches:

1. Rejection clips for Class R fuses specified.
2. Provisions for Class J or Class L fuses, as applicable.
3. Fuses: Per requirements in Section 262813 - Fuses.

2.3 ENCLOSED CIRCUIT BREAKERS

A. NEMA AB 1, UL 489.

B. Enclosed molded-case circuit breakers:

1. Tripped indication clearly shown on breaker handle taking position between "ON" and "OFF".
2. 225A frame size and below: thermal-magnetic trip.
3. 250A frame size and above: electronic (solid-state microprocessor-based) trip units interchangeable in the field within the frame size and field-adjustable long time pick-up, long time delay, short time pick-up, short time delay, and instantaneous current settings. Each adjustment shall have discrete settings and shall be independent of other adjustments.

C. Breaker Mechanism:

1. Quick-make, quick-break.

D. Ratings:

1. Ampacity as indicated on drawings.
2. Listed as Type HACR for air conditioning equipment circuits.
3. Listed as Type SWD for lighting circuits.

2.4 LUGS

A. Front removable lugs.

B. Labeled for 75°C copper and aluminum conductors.

C. Multiple lugs to match number of conductors per phase.

D. Termination of field installed conductors: Pressure wire connectors, except wire-binding screws for #10 AWG or smaller conductors.

2.5 ACCESSORIES:

A. Solid neutral assembly, where required.

- B. Equipment ground kit.
- C. Blown fuse indicators on fused disconnect switches.
- D. Factory installed fuse puller on fused disconnect switches.
- E. One set of normally open (NO) auxiliary contacts, where disconnect switch is installed at a remote motor served by variable frequency drive (VFD).

2.6 ENCLOSURES

- A. NEMA BS 31047, NEMA AB 1, UL 98, UL 489, as applicable.
- B. NEMA Type 1, Type 3R (outdoor locations) enclosure.
- C. Code-gauge galvanized steel.
- D. Manufacturer's standard gray enamel finish over prime coat.
- E. Surface-mounted. .

2.7 SHORT CIRCUIT CURRENT RATING

- A. Each circuit breaker shall have minimum short circuit current rating as indicated on drawings.

PART 3 - EXECUTION

3.1 COORDINATION WITH MANUFACTURER

- A. Instruct manufacturer about the location of incoming lugs, i.e., top or bottom feed based on incoming feeder entrance location.
- B. Verify that "touch-up" paint kit is available for repainting.

3.2 EXAMINATION

- A. Examine areas and surface to receive disconnect switches and enclosed circuit breakers for compliance with requirements, installation tolerances, and other conditions affecting performance. Proceed with installation only after unsatisfactory conditions have been corrected.
- B. Verify that space indicated for disconnect switches and enclosed circuit breakers mounting meets code-required working clearances.
- C. Notify Architect/Engineer of any discrepancies prior to submittal of product data and shop drawings.

3.3 INSTALLATION

- A. Install disconnect switches and/or enclosed circuit breakers in accordance with ANSI/NECA 1.
- B. Install disconnect switches and/or enclosed circuit breakers level and plumb, in accordance with manufacturer's written instruction.
- C. Disconnect switches and enclosed circuit breakers mounting and seismic restraints:
 - 1. Fasten disconnect switches and enclosed circuit breakers firmly to walls and structural surfaces, ensuring they are permanently and mechanically anchored.
 - 2. Anchor and fasten disconnect switches and enclosed circuit breakers and their supports to building structural elements (wood, concrete, masonry, hollow walls and nonstructural building surfaces) by the methods described in Section 260529 - Hangers and Supports for Electrical Systems.
 - 3. Install two rows of steel slotted channel, with a minimum of four attachment points, for each disconnect switch and enclosed circuit breaker.
 - 4. When not located directly on wall, install support frame of steel slotted channel anchored to floor and ceiling structure.
- D. Do not support disconnect switches and/or enclosed circuit breakers by raceway.
- E. Install top disconnect switch and/or enclosed circuit breaker handle a minimum of 3'-6" and maximum of 6' 6" above finished floor.
- F. Tighten electrical connectors and terminals according to equipment manufacturer's published torque-tightening values. Where manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- G. Install engraved plastic nameplates under provisions of Section 260553 - Electrical Systems Identification. Attach nameplate to exterior of each switch and/or enclosed circuit breaker using small corrosion-resistant metal screws or rivets. Do not use contact adhesive.
 - 1. Include switch and/or enclosed circuit breaker name, amperage, voltage, phase, and number of wires.
- H. Install fuses in fusible switches at job site per requirements in Section 262813 - Fuses.

3.4 CONNECTIONS

- A. Ground equipment according to Section 260526 - Grounding and Bonding for Electrical Systems.
- B. Connect wiring according to Section 260519 - Low-Voltage Electrical Power Conductors and Cables.

3.5 FIELD QUALITY CONTROL

- A. Inspect for physical damage, proper alignment connections, anchorage, and grounding.
- B. Correct malfunctioning units on-site and retest to demonstrate compliance. Remove and replace with new units and retest.
- C. Test disconnect switches and/or enclosed circuit breakers per requirements in Sections 260812 - Power Distribution Acceptance Tests.
- D. Interpret test results in writing and submit to Engineer.

3.6 REPAINTING

- A. Remove paint splatters and other marks from surface of equipment.
- B. Touch-up chips, scratches, or marred finishes to match original finish, using manufacturer-supplied paint kit. Leave remaining paint with Owner.

3.7 ADJUSTING

- A. Circuit Breakers: Set field-adjustable trip settings or change the trip settings recommended by the overcurrent protective device coordination study per Section 260573 - Power System Studies.

3.8 CLEANING

- A. Vacuum dirt and construction debris from interior and exterior of equipment; do not use compressed air to assist in cleaning.

END OF SECTION

SECTION 262913 ENCLOSED CONTROLLERS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 260529 - Hangers and Supports for Electrical Systems
- D. Section 260553 - Electrical Systems Identification
- E. Section 260812 - Power Distribution Acceptance Tests
- F. Section 260813 - Power Distribution Acceptance Test Tables
- G. Section 262813 - Fuses

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Section includes enclosed manual and magnetic motor controllers and enclosed contactors.
- B. Motors shown on the drawings or specified in other Divisions of these specifications shall be provided with motorized equipment and connected under this section. Provide motor controllers and power circuit disconnect devices for all motors, unless shown or specified to be furnished with motorized equipment under other Divisions of these specifications, and by others, for installation by this contract.
- C. Variable-frequency controllers furnished by Division 20 for installation by Division 26.
- D. Motor Voltage Information:
 - 1. Voltages available are: 208 and 480 V, 3-phase and 120, 208 and 277V single phase. Circuits are designed for motors with voltage ratings as follows:
 - a. Smaller than 1/2 hp motors: 115 V, single phase.
 - b. 1/2 hp motors and larger: 230 and 460 V, 3-phase.

1.4 REFERENCE STANDARDS

- A. ANSI/NECA 1 – Standard Practices for Good Workmanship in Electrical Contracting
- B. NEMA AB 1 – Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit Breakers Enclosures
- C. NEMA EN 10250 – Enclosures for Electrical Equipment (1000 V Maximum)
- D. NEMA ICS 2 – Industrial Control and Systems: Controllers, Contactors and Overload Relays, Rated Not More Than 2000 VAC or 750 VDC
- E. NEMA ICS 4 – Industrial Control and Systems: Terminal Blocks
- F. NEMA IA 10039 – Industrial Control and Systems: Control Circuit and Pilot Devices
- G. NEMA IA 10030 – Industrial Control and Systems: Enclosures
- H. NEMA BS 31047 – Enclosed and Miscellaneous Distribution Equipment Switches (600 V Maximum)
- I. NEMA MG 00001 – Motors and Generators
- J. NFPA 70 – National Electrical Code
- K. UL 98 – Enclosed and Dead Front Switches
- L. UL 486A-486B – Wire Connectors
- M. UL 489 – Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit Breakers Enclosures
- N. UL 508 – Industrial Control Equipment

1.5 SUBMITTALS

- A. Product Data:
 - 1. Motor controllers: Submit catalog cut sheets showing voltage, size, rating and size of switching and overcurrent protective devices, dimensions, and enclosure details.
 - 2. Contactors: Submit catalog cut sheets showing voltage, size, current rating, dimensions, and enclosure details.
 - 3. Factory settings and time-current curves of individual protective devices.
 - 4. Confirm motor sizes and voltages with submittals of other Divisions of specifications, and/or by others, prior to Section submittals.

- B. Manufacturer's Installation Instructions:
 - 1. Indicate application conditions and limitations of use stipulated by product testing agency. Include instructions for storage, handling, protection, examination, preparation, installation, and/or starting of product.
- C. Test Reports: Indicate field test and inspection procedures and interpret test results and corrective action taken for compliance with specification requirements.
- D. Closeout Submittals:
 - 1. Project Record Documents:
 - a. Record actual locations and ratings of enclosed motor controllers and enclosed contactors.
 - 2. Operation and Maintenance Data:
 - a. Include manufacturer's recommended operating instructions, maintenance procedures and intervals, and preventive maintenance instructions.
 - b. Include spare parts data listing, source, and current prices of replacement parts and supplies.
 - c. Include Manufacturer Seismic Qualification Certification and Installation Seismic Qualification Certification.

1.6 QUALITY ASSURANCE

- A. Obtain motor controllers, and contactors from one source and by single manufacturer.
- B. Regulatory Requirements:
 - 1. Comply with NFPA 70 for components and installation.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory wrapping or provide additional canvas or plastic cover to protect units from dirt, water, construction debris, and traffic.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

1.9 MAINTENANCE

- A. Extra Materials: Furnish extra materials described below that match product installed, are packaged with protective covering for storage, and are identified with labels describing contents.

1. Spare pilot lights: Furnish 1 spare lamp for every 5 installed units, but not less than 1 set of 3 of each kind.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Schneider Square D
- B. ABB-GE Industrial Solutions
- C. Eaton Cutler-Hammer
- D. Siemens
- E. Allen Bradley

2.2 MANUAL MOTOR CONTROLLERS

- A. Description: NEMA ICS 2, AC general-purpose, Class A, manually operated, full-voltage controller for small motors, with bimetal type overload relay, red pilot light, and pushbutton operator.

2.3 FRACTIONAL-HORSEPOWER MANUAL CONTROLLERS

- A. Description: NEMA ICS 2, AC general-purpose, Class A, manually operated, full-voltage controller for fractional horsepower induction motors, with bimetal type overload relay, red pilot light, and toggle operator.

2.4 MOTOR STARTING SWITCHES

- A. Description: NEMA ICS 2, AC general-purpose Class A manually operated, full-voltage controller for fractional horsepower induction motors, without thermal overload unit, with red pilot light ,and key operator.

2.5 FULL-VOLTAGE NON-REVERSING MAGNETIC MOTOR CONTROLLERS

- A. Description: NEMA ICS 2, AC general-purpose, Class A, magnetic controller for induction motors rated in horsepower, three-phase and single-phase, as scheduled, except where single-phase motors scheduled to be provided with built-in overload elements:
 1. Size 1 minimum
 2. Control Voltage: 120 V, 60 Hz
 3. Overload Relays: NEMA ICS 2, solid-state bimetal, 1 overload relay per phase:
 - a. Solid-state type:
 - 1) Class 10 inverse-time tripping characteristics.
 - 2) Non-volatile operating memory.

- 3) 3:1 current adjustment range.
 - 4) Phase loss/phase unbalance protection.
 - 5) Ambient temperature insensitive.
 - 6) Self-powered.
 - 7) Manual reset. Automatic recent not acceptable.
 - 8) Manual trip.
 - 9) Visible trip indication.
 - b. Bimetallic type:
 - 1) Class 10 inverse-time tripping characteristics.
 - 2) Manual reset.
- 4. Features:
 - a. Auxiliary Contacts: NEMA ICS 2, 2 each field-convertible contacts in addition to seal-in contact.
 - b. Pushbuttons: Recessed type.
 - c. Pilot Lights NEMA IA 10039: push-to-test LED resistor, neon type.
 - d. Hand-Off-Auto (H-O-A) Selector Switches: Rotary type switch with fully wired normally open and normally closed contacts for switch position indication.
 - e. Control Power Transformers: 120V secondary, adequate capacity to operate connected pilot, indicating and control devices, plus 100% spare capacity in each motor controller, but not less than 100VA. Fused primary and secondary, and unfused leg of secondary bonded to enclosure.
 - f. Terminals: NEMA ICS 4.
 - g. Other accessories detailed or required by drawings.

2.6 COMBINATION CONTROLLERS

- A. Factory-assembled motor controllers with externally operable disconnect, fusible switch type, in common enclosure; means for locking disconnect handle and means for defeating cover interlock.
 - 1. Fusible Switch: NEMA BS 31047 and UL 98; enclosed knife switch, heavy-duty type, external operable handle, clips or pads to accommodate specified fuses:
 - a. Rejection clips for Class R fuses.
 - b. Provisions for Class J or Class L fuses, as applicable.
 - c. Fuses: Per requirements in Section 262813 - Fuses.

2.7 FULL VOLTAGE NON-REVERSING MULTI-SPEED MAGNETIC MOTOR CONTROLLERS

- A. Description: Same as full voltage non-reversing single-speed magnetic motor controllers with addition of the following:
 - 1. Speed selector switch
 - 2. Auto-Off-Low-High selector switch

2.8 REDUCED VOLTAGE MOTOR CONTROLLERS

- A. Motor controllers for NEMA rated 230 V motors 25 hp and above.
- B. Motor controllers for NEMA rated 460 V motors 60 hp and above.

2.9 MOTOR CONTROLLER ACCESSORIES

- A. Factory installed devices in controller enclosure, unless otherwise indicated, as follows:
 - 1. "On-Off" and "Start-Stop" pushbutton stations, pilot lights, selector switches: NEMA ICS 2, heavy duty, oiltight type.
 - 2. 120 V control circuits and pilot light, unless otherwise noted.
 - 3. Red pilot light to indicate motor operation.
 - 4. Green pilot light to indicate motor stopped.
 - 5. Minimum wire size for control circuits: #14 AWG.
 - 6. Stop and Lockout Pushbutton Station: Momentary-break pushbutton station with a factory-applied hasp arranged so a padlock can be used to lock pushbutton in depressed position with control circuit open, where indicated.
- B. Control services: As scheduled on motor schedule or indicated.

2.10 GENERAL PURPOSE MAGNETIC CONTACTORS

- A. Description: NEMA ICS 2, same as magnetic controllers, except without overload protection.
- B. Poles: To match circuit configuration and control function.
- C. Configuration: Electrically held
- D. Contact Rating: Match branch circuit overcurrent protection.

2.11 LIGHTING MAGNETIC CONTACTORS

- A. Description: NEMA ICS 2, same as magnetic controller, except without overload protection.
- B. Poles: To match circuit configuration and control function.
- C. Configuration: Electrically held
- D. Contact Rating: Match branch circuit overcurrent protection.

2.12 LUGS

- A. Labeled for 75°C copper and aluminum conductors.
- B. Multiple lugs to match number of conductors per phase.
- C. Termination of field installed conductors: Pressure wire connectors, except wire-binding screws for #10 AWG or smaller conductors.
- D. For equipment specified in this section and for equipment furnished under other Divisions of this specification and/or by others.

2.13 MOTOR CONTROLLERS AND CONTACTOR ENCLOSURES

- A. NEMA EN 10250, NEMA IA 10030.
- B. NEMA Type 1, Type 3R (outdoor locations) enclosure.
- C. Code-gauge galvanized steel.
- D. Manufacturer's standard gray enamel finish over prime coat.
- E. Surface-mounted. Flush-mounted where indicated.

PART 3 - EXECUTION

3.1 COORDINATION

- A. Coordinate motor control wiring with Division 23 of these specifications.
- B. Coordinate motor sizes and voltages with submittals of other Divisions of these specifications and/or by others.
- C. Verify with manufacturer that "touch-up" paint kit is available for repainting.

3.2 EXAMINATION

- A. Examine areas and surface to receive motor controllers and contactors for compliance with requirements, installation tolerances, and other conditions affecting performance. Proceed with installation only after unsatisfactory conditions have been corrected.
- B. Verify that space indicated for motor controllers and contactors mounting meets code-required working clearances.
- C. Notify Architect/Engineer of any discrepancies prior to submittal of product data.

3.3 INSTALLATION

- A. Install motor controllers and contactors in accordance with ANSI/NECA 1.
- B. Install level and plumb, in accordance with manufacturer's written instruction.
- C. Motor controllers and contactors mounting and seismic restraints:
 - 1. Fasten motor controllers and contractors firmly to walls and structural surfaces, ensuring they are permanently and mechanically anchored.
 - 2. Anchor and fasten motor controllers and contactors and their supports to building structural elements (wood, concrete, masonry, hollow walls and nonstructural building surfaces) by the methods described in Section 260529 - Hangers and Supports for Electrical Systems.

3. Install two rows of steel slotted channel, with minimum of four attachment points, for each motor controller and contactor.
 4. When not located directly on wall, install support frame of steel slotted channel anchored to floor and ceiling structure.
 5. Do not support motor controllers and contactors only by raceway.
- D. Tighten electrical connectors and terminals according to equipment manufacturer's published torque-tightening values. Where manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- E. Install engraved plastic nameplates under provisions of Section 260553 - Electrical Systems Identification. Attach nameplate to exterior of each motor controller and contactor, using small corrosion resistant metal screws or rivets. Do not use contact adhesive:
1. Indicate motor served, nameplate horsepower, full load amperes, code letter, service factor, voltage/phase rating, and fuse size and type, when applicable.
- F. Connect each motor terminal box to rigid conduit system with maximum 18" of flexible liquid-tight metal conduit. Install conduit per requirements in Section 26 0533 – Raceway and Boxes for Electrical Systems.
- G. Check for proper rotation and phase relationship of each motor.
- H. Install fuses in fusible switch at job site per requirements in Section 262813 - Fuses.
- I. Control Wiring Installation:
1. Install wiring between motor control devices according to Section 260519 - Low-Voltage Electrical Power Conductors and Cables.
 2. Install motor control wiring in accordance with control wiring diagrams and in raceways where indicated or required by contract drawings.
 3. Bundle, train, and support wiring in enclosures.
 4. Connect hand-off-automatic switch and other automatic-control devices where applicable.
 - a. Connect selector switches to bypass only manual- and automatic-control devices that have no safety functions when switch is in hand position.
 - b. Connect selector switches with motor-control circuit in both hand and automatic positions for safety-type control devices such as low- and high-pressure cutouts, high-temperature cutouts, and motor overload protectors.

3.4 APPLICATION

- A. Select features of each motor controller to coordinate with ratings and characteristics of supply circuit and motor; required control sequence; duty cycle of motor, and configuration of pilot device and control circuit affecting controller functions.

3.5 CONNECTIONS

- A. Provide green wire ground through flexible conduit to interconnect motor frame and rigid conduit system.
- B. Ground and bond motor controller and contactor enclosures according to Section 260526 - Grounding and Bonding for Electrical Systems.
- C. Connect power and control wiring according to Section 260519 - Low-Voltage Electrical Power Conductors and Cables.
- D. Connect control wiring for operation, control and supervision of motorized equipment as shown on drawings and/or specified in this and other Divisions of these specifications.

3.6 FIELD QUALITY CONTROL

- A. Inspect motor controllers and contactors for physical damage, proper alignment, connections, anchorage, seismic restraints and grounding.
- B. Correct malfunctioning motor controllers and contactors on-site and retest to demonstrate compliance. Remove and replace with new units and retest.
- C. Test continuity of each circuit.
- D. Test motor controllers per requirements in Sections 260812 - Power Distribution Acceptance Tests and 260813 - Power Distribution Acceptance Test Tables.
- E. Interpret test results in writing and submit to Engineer.

3.7 REPAINTING

- A. Remove paint splatters and other marks from surface of equipment.
- B. Touch-up chips, scratches or marred finishes to match original finish, using manufacturer-supplied paint kit. Leave remaining paint with Owner.

3.8 ADJUSTING

- A. Set field-adjustable circuit breakers trip settings or change the trip settings as indicated on drawings.
- B. Adjust motor circuit protectors.

3.9 CLEANING

- A. Vacuum dirt and construction debris from interior and exterior of equipment; do not use compressed air to assist in cleaning.

END OF SECTION

SECTION 263213 ENGINE GENERATORS

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 220700 - Plumbing Systems Insulation
- B. Section 233114 - Ductwork
- C. Section 233314 - Ductwork Specialties
- D. Section 260000 - General Electrical Requirements
- E. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- F. Section 260526 - Grounding and Bonding for Electrical Systems
- G. Section 260812 - Power Distribution Acceptance Tests
- H. Section 260813 - Power Distribution Acceptance Test Tables
- I. Section 263623 - Automatic Transfer Switches

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION OF SYSTEM

- A. Section describes complete package generator set, unit-mounted radiator cooling system, heat exchanger, microprocessor based control and monitoring panel, battery and charger, drop over sound attenuated enclosure.
- B. Package generator set rated for standby duty.
- C. Engine fuel system:
 - 1. Sub-base fuel tank

1.4 REFERENCE STANDARDS

- A. NEMA MG 00001 – Motors and Generators

- B. IEEE 446 – Recommended Practice for Emergency and Standby Power Systems for Commercial and Industrial Applications
- C. NFPA 37 - Standard for Installation and Use of Stationary Combustion Engines and Gas Turbines
- D. NFPA 110 – Standard for Emergency and Standby Power Systems
- E. UL 2200 – Stationary Engine Generator Assemblies
- F. IEC8528 Part 4 – Control Systems for Generator Sets
- G. UL 142 – Steel Aboveground Tanks for Flammable and Combustible Liquids
- H. UL 2085 – Protected Aboveground Tanks for Flammable and Combustible Liquids

1.5 SUBMITTALS

- A. Shop Drawings
 - 1. Submit for engineering review and approval prior to production release. Include the following for engine-generator:
 - a. Outline drawings of equipment showing weights
 - b. Overall dimensions including bolting template and earthquake restraints
 - c. Right hand, left hand, end, and top views of proposed assembly
 - d. Battery, battery rack, battery charger, and wiring diagrams
 - e. Vibration isolation bases, mounts, and hangers
 - f. Exhaust silencer and flexible fittings
 - g. Stub ups for fuel
 - h. Power and control wiring entrance locations
 - i. Main circuit breaker size, location, and required clearance
 - j. Lug sizes and locations
 - k. Engine-generator control panel drawings showing devices to be provided, with each device referenced to material list with complete description for device.
 - l. Weather protective enclosure installation drawings, structural calculations, lighting fixture catalog cut, conduit, and wiring.
 - m. Enclosure sound performance data
 - n. Muffler characteristics
 - o. Calculations for starting based on step loads outlined in Paragraph 2.2, B.5.
 - p. Factory certified prototype test report indicating fuel efficiency and emission levels
 - 2. Information on engine characteristics:
 - a. Make, type, and number of cylinders
 - b. Brake horsepower (bhp) available
 - c. Jacket water heat rejection
 - d. Cooling pump characteristics
 - e. Exhaust flow rate and temperature at 25, 50, 75, and 100% rated load
 - f. Ventilation requirements

- g. Combustion air requirements
 - h. Fuel consumption rates at 25, 50, 75, and 100% rated load
 - i. Liquid refill capacities
 - j. Exhaust backpressure limitation
 - k. Type and manufacturer of governor
 - l. Alternator size to limit voltage dip to 10%
 - 3. Information on generator characteristics:
 - a. Make and type
 - b. Type of construction and overspeed capabilities
 - c. Temperature rise
 - d. Regulation characteristics
 - e. Ventilation requirements
 - f. Type of winding insulation
 - g. KW power factor
 - h. Type of exciter and voltage regulator
 - B. Interconnection detail drawing showing control and power connections in complete standby system. Control connections between components are to be labeled with identical nomenclature. Coordinate with generator manufacturer.
 - C. Accessories including fuel lines, flexible exhaust couplings, exhaust flange, and other exhaust system components.
 - D. Complete review of this specification, noting for each paragraph whether or not proposed equipment complies with project specifications, or deviates in some fashion. Justification must be provided for each deviation.
 - E. Complete test specification detailing testing procedure to be used to verify performance of equipment provided.
 - F. Recommended spare parts lists.
 - G. Test Reports:
 - 1. Submit certified factory tests report on engine-generator delivery. Alarms, sensors, and meters must be tested and certified.
 - 2. Submit, upon completion of installation and testing of engine-generator sets, certified test reports from load tests for each engine-generator.
- 1.6 DELIVERY, STORAGE, AND HANDLING
- A. Handle equipment in accordance with manufacturer's written instructions. One copy of instructions is to be included with equipment at time of shipment. Maintain factory bracing, packaging, and wrapping.
- 1.7 OPERATION AND MAINTENANCE MANUALS
- A. Refer to Section 017700 - Closeout Procedures and herein below.

- B. Submit Operation and Maintenance (O&M) manuals to Engineer for review 60 days prior to acceptance of unit.
- C. Installation, maintenance, and operating instruction manuals shall include the following:
 - 1. 100% accurate system "as-installed" drawings, interconnect diagrams, schematic diagrams, wiring diagrams, individual sub-system component manuals, operation procedures, system description with theory of operation, maintenance schedules and procedures, original programmed settings and parameters, and other information necessary for the Owner to maintain, operate, test, and troubleshoot system.
 - 2. The O&M manual shall contain step-by-step instructions for startup and shutdown. The first page shall contain name, address, and phone number of local representative to be called for service or parts. Follow with complete parts lists by actual ordering catalog numbers. O&M manual also shall contain four copies each of test record forms and service record forms for Owner use. Forms shall show proper interval for testing, servicing, and replacing of components, lubrication, filters, antifreeze, etc., including recommended specifications and fluid levels for lubricants.
 - 3. Recommended spare parts list (with pricing) for 2 yrs of operation.
- D. O&M manuals shall not solely rely on sub-component manuals. Thorough consolidation of operating and maintenance information shall be available in system overview guide. Include major components of system in overview.
- E. Turn final reviewed manuals over to Owner prior to conducting training of Owner personnel.
- F. Seal single copy of service record forms, recommended operation and service practices for unit in plastic and wall mount in weather-protective enclosure.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Acceptable Manufacturers:
 - 1. Engine Generator Set - Caterpillar, Cummins, MTU Onsite Energy, Kohler
 - 2. Exhaust Silencer – Maxim, Nelson, or approved equal
 - 3. Isolation equipment
 - 4. Battery charger – Sens, La Marche, Charles Industries

2.2 RATINGS AND PERFORMANCE

- A. Engine Generator Set
 - 1. Generator kW Output: As shown on drawings
 - 2. Altitude 800 ft above sea level in ambient temperature of 0 - 100°F.
 - 3. Stable frequency regulation

- B. Alternator
 - 1. As shown on drawings kVA, .8 Power Factor
 - 2. 480 V, 3 Ph, 60 Hz, 4 Wire Y
 - 3. Stable voltage regulation 0-full load less than or equal to $\pm .5\%$.
- C. Transient Performance
 - 1. Engine
 - a. Start and load in 10 seconds per NFPA 110
 - b. Accept 100% block load per NFPA 110
 - 2. Frequency regulation $\pm 5\%$ no load to full load. $\pm .5\%$ steady state.
 - 3. Alternator
 - a. 15% Voltage dip
 - b. AC waveform output contains $<5\%$ total harmonic distortion (THD) at full linear load when measured from line to neutral with $<3\%$ in any single harmonic, and no third-order harmonics or their multiples.
 - c. Telephone influence factor < 40
 - d. Telephone harmonic factor < 3
 - e. Fault current capability
- D. Factory Prototype Test Certified
 - 1. Harmonic Distortion Levels
 - a. Demonstrate
 - 2. Airflow Restriction tests
 - a. Demonstrate controlled shutdown after overheating
 - 3. Unit tested with enclosure
 - a. UL 2200 listed
 - 4. 30 Degree Water Spray Unit Rain Test
 - a. Demonstrate no water leakage into electrical boxes
 - 5. Overload Test
 - a. Demonstrate 10% overload with no damage to engine
 - 6. Air Filter Test
 - a. Demonstrate engine contains engine backfire explosion
- E. Factory Production Test Certified
 - 1. Alternator Impedance to Ground
 - 2. Dielectric Testing
 - a. At 1000 V and 2 times rated voltage
 - 3. Maximum kW Rating
 - 4. Engine Response Time
 - 5. Alternator Construction Testing
 - a. Impedance Balance Tested
 - 6. Alternator Insulation Testing
 - a. Surge Tested

2.3 FABRICATION AND MANUFACTURER

- A. Engine:
 - 1. Type: Inline or vee
 - 2. Four-stroke cycle at 1800 RPM consistent with engine durability.
 - 3. Aspiration: Natural

4. Horsepower
 - a. Brake Horsepower Rule of thumb = $1.6 \times \text{kW}$
 5. air cooled
 6. Fuel Type: No. 2 domestic diesel fuel oil.
 7. Engine accessories:
 - a. Fuel filter
 - b. Lube oil filter
 - c. Intake air filter
 - d. Lube oil cooler
 - 1) Suitable for operation of generator set at full rated load in ambient temperature specified.
 - e. Fuel transfer pump
 - f. Fuel priming pump
 - 1) Engine driven positive displacement, mechanical, full pressure
 - g. Gear driven water pump
 - h. Electronic direct fuel injection or have suitable emission control equipment
 - i. Electric speed sensing governor capable of isochronous regulation.
 - j. Safety-shut-offs for:
 - 1) High water temperature
 - 2) Low oil pressure
 - 3) Overspeed
 - 4) Overcranking
 8. EPA Certified Tier 2
- B. Cooling System:
1. Engine skid mounted, engine-driven radiator with blower type fan, sized to maintain safe operation at 122°F ambient temperature.
 2. Vertical discharge remote mounted radiator sized to maintain safe operation at 122°F ambient temperature.
 3. Arrange liquid-cooled prime movers for NFPA level 1 applications for closed-loop cooling.
 4. Provide radiator with:
 - a. Motor-driven fan with voltage same as generator
 - b. Motor Starter
 - c. Initiating contacts to actuate on generator startup
 - d. Connect to generator distribution system
 - e. Core guard
 - f. Fan guard
 - g. Mounting hardware
 - h. Direct adapter flange. Ductwork with flexible connection between radiator and exhaust plenum to be provided by Division 23. Coordinate with Division 23.
 - i. Flexible pipe connections at engine and radiator.
 - j. Supply power for fans and pumps on remote radiators from a tap at generator output terminals or ahead of first load circuit overcurrent protective device.
 - k. Heat exchangers
 5. Block Heater
 - a. Water Jacket Heater:
 - b. Maintain engine jacket water to 110°F in ambient temperature of 30°F
 - c. Heater to be equipped with thermostatic switch.

- d. Single phase 208 V
- 6. Fill engine cooling system with solution of 50/50 mix ethylene glycol at initial fill.
- C. Exhaust System:
 - 1. Furnish industrial type exhaust silencer:
 - a. Sized according to manufacturer's recommendations
 - b. Mount so weight is not supported by engine
 - c. Flexible exhaust fitting
 - d. Installation inside drop over enclosure
 - 2. Condensate Traps
 - a. Drain plug at low point of muffler
 - 3. Thermal Expansion
 - a. Stainless steel exhaust flex to accommodate thermal growth and vibration isolation
 - 4. Exhaust Blankets
 - a. 1" high temperature fiberglass cloth wrap
 - b. Coordinate with weather protective enclosure.
 - 5. Thimble
 - a. Pipe and wall of compatible construction
 - 6. Acceptable Back Pressure
 - a. Size silencer and exhaust pipe so exhaust back pressure does not exceed maximum limitations specified by generator set manufacturer.
 - 7. Exhaust clearing area
- D. Starting System
 - 1. Provide DC electric starting system with positive engagement drive. Provide DC voltage recommended by manufacturer.
 - 2. Provide fully automatic start-stop controls.
 - 3. Provide cycle cranking to open and lock out start circuit after 3 attempts to start failed engine start.
 - 4. Batteries
 - a. Provide sealed lead-acid storage battery set:
 - 1) Heavy duty diesel starting type
 - 2) Voltage compatible with starting system voltage
 - 3) Capacity to provide for 1-1/2 minutes total cranking time at 0°F without recharging. In accordance with NFPA Level 1.
 - b. Provide vinyl coated steel battery rack.
 - c. Provide starting battery heater:
 - 1) Heater plate under battery
 - 2) Heater type blanket around battery case
 - 3) Thermal switch - heater control relay
 - 4) 120 VAC input
 - d. Battery cables and clamps
 - 5. Battery Charger
 - a. Four Rate Battery Charger
 - 1) Constant current, constant voltage, high rate taper, and float equalized.
 - b. Dual Rate Battery Charger
 - 1) Constant current, and float equalized
 - c. Charger Accessories:
 - 1) Overload protection

- 2) $\pm 0.5\%$ line and load regulation
- 3) Electronic current limit output 105%
- 4) DC ammeter and voltmeter .
- 5) UL 1236 listed and meets NFPA 110 requirements
- 6) Output protection
- 7) Temperature compensation
- 8) Enclosed in NEMA 1 aluminum or stainless steel enclosure
- 9) Form C contacts for the following alarms
 - a) AC fail
 - b) Low battery volts
 - c) High battery volts
 - d) Charger fail
 - e) Battery fault
6. AC input voltage: 208 V
7. When installed on the engine generator set, mount on vibration isolators.

E. Speed Control

1. Electronic: Isochronous

F. Alternator:

1. Maximum temperature rise 275°F at 104°F ambient
2. Synchronous type
3. Self ventilated
4. Drip-proof construction
5. Directly connected to engine flywheel housing with a flex coupling
6. Capable of sustaining 300% overcurrent for 10 seconds under a 3 Ph symmetrical short circuit
7. 120 V Anti-Condensation heater
8. Subtransient Reactance limited to 12%
9. Insulation
 - a. Complies with NEMA (MG1-33.4)
 - b. Class H Insulation Systems
 - 1) UL 1449 recognized
 - 2) Vacuum impregnated with epoxy varnish
 - 3) Fungus resistant
10. Permanent magnet brushless excitation (PMG).
 - a. PMG shall derive excitation current from pilot exciter mounted on the rotor shaft. It is to be able to sustain 300% of rated current for ten seconds during a fault condition.
11. Rotor
 - a. 4 pole
 - b. Winding
 - 1) Random
 - c. Varnish process
 - 1) Epoxy based material applied to each layer of magnet wire
 - d. Coil supports
 - 1) Driven through flexible coupling to ensure permanent alignment.
 - e. End winding spacing
 - f. Amortisseur windings
 - g. Bearings
 - 1) Sealed

- 2) Single
12. Stator
 - a. 3 Ph winding
 - b. Laminations
 - c. Cooling air passages and fan
 - 1) Provide space heater to keep alternator free of moisture. Space heater to be 1500 W, 120 VAC, 1 Ph.
 - d. Welded laminations to prevent cutting of wires
 - e. Skewed stack to minimize slot ripple on output voltage and produce smooth voltage waveform.
 - f. Pitch – Skewed design to optimize efficiency and minimize total harmonic distortion.
 - g. Varnish process
 - 1) 2 dips and bakes using Class A impregnating varnish
13. Alternator Components
 - a. Solid state design digital voltage regulator:
 - 1) Performance
 - a) Microprocessor based.
 - b) Programmable
 - c) Regulation: $\pm .25\%$ at any constant load for any load from 0% to 100% of pf rating.
 - d) 3 Ph, true RMS sensing
 - e) PMG input, engine unloading
 - f) Design insensitive to severe, load induced wave shape distortion from SCR or thyrister circuits such as those used in battery charging, UPS, and motor speed control equipment loads.
 - g) Controls to limit build-up of AC generator voltage to provide a linear rise and limit overshoot.
 - h) Digital adjustments for out voltage adjustment gain, damping and frequency rate-off. .
 - i) System setup controls and fault alarms.
 - 2) Protection
 - a) Over-excitation protection
 - b) Electronic voltage buildup protection
 - c) Loss of sensing protection
 - d) Temperature compensation
 - e) Limitation of voltage overshoot on startup
 - 3) Features
 - a) Parallel support
 - b) VAR/PF control
 - 4) Environmentally sealed
 - 5) UL 508A listing
 - b. Output Circuit Breaker(s)
 - 1) 1 100% thermal magnetic rated circuit breaker
 - 2) Adjustable long time, long time delay, short time, and short time delay curve shaping elements
 - 3) Shunt Trip for integration with load bank controls
 - 4) Solid state trip fixed mounted insulated case generator mounted circuit breaker
 - 5) NEC required access in front of breaker

- 6) Ground fault alarm only: Monitoring relay for breaker 1000A and above. Relay to be adjustable from 3.8 – 1200A and include an adjustable time delay of 0-10S.

G. Controls:

1. NFPA 110 listed
2. Micro-processor based solid state controls to automatically start, protect and monitor engine-generator set with panel illuminating lighting and digital display.
3. Control panel includes:
 - a. Solid state trip main circuit breaker
 - b. Motor starting switch
 - c. Electrically operated fuel control
 - d. Relay to disconnect battery charger during cranking
 - e. Switching lamps and meters to be oil tight and dust tight. All active components to be installed within a NEMA 3R enclosure. There shall be no exposed components with door open operating 750 V.
 - f. Protective relays to open main circuit breaker and shut down and lockout engine on abnormal conditions including:
 - 1) Overspeed
 - 2) Operation of Remote Stop
 - 3) Overcrank
 - 4) Low lube oil pressure
 - 5) High Engine Temp
 - 6) Low coolant level
 - 7) Fail to crank
 - 8) Dead battery
 - g. Monitoring items shall include but is not limited to the following items and control:
 - 1) Coolant temperature
 - 2) Oil pressure
 - 3) Battery voltage
 - 4) RPM
 - a) Voltmeter, 3-1/2" dual type, 0.5% accuracy with selector switch
 - b) Ammeter, 3-1/2" dual type, .05% accuracy with selector switch
 - 5) Frequency meter, 55-65 Hz ± 0.125 Hz.
 - 6) Running Time Meter (hours and 1/10 hours)
 - 7) AC power metering to be 0.5% accuracy and include frequency, phase, selector switch with real time power metering including, kW, kVA, kVAR, kWh, PF, % of rated load.
 - h. Control Items:
 - 1) Voltage level adjustment rheostat
 - 2) Overspeed level adjustment
 - 3) Overvoltage level adjustment
 - 4) Undervoltage level adjustment
 - 5) Overfrequency level adjustment
 - 6) Underfrequency level adjustment
 - 7) Position function switch(es) marked AUTO, MANUAL RUN, OFF/RESET and STOP
 - 8) 4 NO and 4 NC dry contacts for local and remote alarms, wired to terminal strips.
 - 9) Emergency off mushroom button

- 10) Automatic remote start capability. Engine cranking system to permit minimum 4 cranking attempts of 10 seconds (adjustable) duration with rest of periods of 10 seconds (adjustable).
 - a) Overcrack lockout shall occur after 4 cranking attempts.
 - i. In accordance with NFPA 110, Level 1, control panel shall furnish battery-powered individual visual alarm indicator functions at battery voltage and visual and audible pre-alarm:
 - 1) Overcrank
 - 2) Low water temperature
 - 3) High engine temperature pre-alarm
 - 4) High engine temperature
 - 5) Low lube oil pressure pre-alarm
 - 6) Low lube oil pressure
 - 7) Overspeed
 - 8) Low fuel main tank
 - 9) Low coolant level
 - 10) EPS supplying load
 - 11) Control switch not in automatic position
 - 12) High battery voltage
 - 13) Low cranking voltage
 - 14) Low voltage in battery
 - 15) Battery charger ac failure
 - 16) Lamp test
 - 17) Contacts for local and remote common alarm
 - 18) Low starting air pressure
 - 19) Low starting hydraulic pressure
 - j. Engine shut down, with audible alarm:
 - 1) Low oil pressure
 - 2) High engine temperature
 - 3) Overcrank
 - 4) Overspeed
 - k. Status report:
 - 1) Engine running
 - 2) Circuit breaker open
 - 3) Circuit breaker closed
 4. Visual alarm resettable only after fault condition has been corrected.
 5. Audible alarm shall include silencing circuit, which, after activation, will permit annunciation of subsequent failures.
 6. Control Panel mounting:
 - a. Mounted on engine generator set in NEMA 1 enclosure on shock isolators
 - b. Wall mounted in NEMA 1 enclosure
 - c. Free standing in NEMA 1 enclosure
 7. Provide remote annunciator panel
 - a. Compliant with NFPA Level 1 requirements.

H. Isolate engine generator set from building structure and from connecting services.

 1. Separately derived grounding system. Connect generator ground as shown on drawings to grounding electrode system.

I. Termination Bars and Connections:

 1. Silver- or tin-plated copper bus bars for terminating cables.

2. Standard NEMA standard bolt hole spacing, for 3 Ph and neutral, within generator connection box with gasketed bolt on cover.
3. Engine-generator set control interfaces to other system components to be made on a permanently labeled terminal block assembly. Provide labels describing connection points.
4. Connections to engine-generator set: Flexible or isolation type connections. Include electrical, fuel, exhaust, and ventilation connections.

J. Equipment Bases:

1. Mount complete unit on a structural steel sub-base, rectangular in shape, with sufficient rigidity to maintain alignment of generator set. Provide perimeter beams with minimum depth equal to 1/10 of longest dimension of base, except beam depth need not exceed 14" provided that deflection and misalignment are kept within acceptable limits as determined by manufacturer. Engine-generator set to be statically and dynamically balanced at factory. Peak-to-peak amplitude of vibration velocity in horizontal, vertical, and axial direction shall not exceed 0.65" per second at main structural components.
2. Engine-generator set weight distribution is to be considered to provide uniform deflections.
3. Bases shall provide equipment alignment and assure uniform weight distribution. Provide side brackets on bases to contain isolating mounts and reduce total installed heights of equipment.

K. Vibration Isolators:

1. Prevent equipment vibrations from being transmitted to enclosure.
2. Required between the structural steel sub-base and concrete housekeeping pad.
3. Steel or cast iron top and bottom housings incorporating 1 or more steel springs with built-in leveling bolts and built-in resilient chocks to control oscillation and withstand lateral forces in all directions.

L. Outdoor Weather Protective Drop-Over Enclosure:

1. Rated NEMA 3R
2. Attenuation:
 - a. Provide engine-generator set with sound-attenuated enclosure. Enclosure will reduce sound level of engine-generator set while operating at full rated load and ventilation running to maximum of 85 dBA at 3 ft or 77 dBA at 21 ft from engine-generator set in free-field environment.
3. Overall Size:
 - a. Drawings show generator footprint, which is maximum allowed for available space.
 - b. Air intake requirements are to be taken into consideration and shall not prevent enclosure from operating within space limitations indicated above.
4. Construction:
 - a. Construction to be welded, 14 ga galvanized steel to ASTM A446/A446M.
 - b. Package listed to UL 2200
 - c. Lockable doors

- d. Include panelboard wired to the following:
 - 1) 6 battery-powered fluorescent fixtures, each with (2) 32W T-8 lamps, battery and wire guard; battery units shall have a minimum run-time of 90 minutes; Provide Columbia KL4-2 SLR, equipped with an electronic .9PF ballast and a parallel Bodine B33 ballast or approved equal.
 - 2) (2) 3-way switches
 - 3) 2 GFCI duplex receptacles
 - 4) 2 block heaters wired to a junction box local to front of generator set
 - 5) Battery charger
 - 6) Battery heater
 - 7) Space heater
 - 8) 120 V circuit for day tank reverse pump
 - 9) 120 V circuit for day tank controls
 - e. Interior lights
 - f. Remote mounted emergency stop button
 - g. Lifting lugs
Refer to Paragraph 2.3.N for fuel system.
5. Ventilation:
- a. Intake: Complete with gravity damper, hood with silencer, flex connections, and 1/2" x 1/2", 16 ga galvanized bird screen to protect against ice and snow.
 - b. Discharge: Complete with gravity-type damper with discharge duct, hood with silencer, flex connector, and 1" x 1/2", 16 ga galvanized bird screen.
6. Paint:
- a. Clean surfaces to SSPC-SP1, seal seams, prime with industrial phenolic primer to 1.5 mils D.F.T. Top coat exterior with Clovacoat epoxy enamel to 2.0 mils D.F.T.
7. Power Coat Paint. Selections to include white, beige, ASA61 gray, and manufacturer factory standard. Architect to select.

M. Fuel System

- 1. Sub-base Fuel Tank and Appurtenances
 - a. Capacity: 24 hours at full load
 - b. Rating: UL 142 Listed
 - c. Double Wall Construction
 - d. Flex fuel lines (NFPA 110 construction)
 - e. Epoxy lined to prevent corrosion
 - f. Tank Exterior: Chemically treated, primed, and painted in factory standard.
 - g. Tank Accessories
 - 1) Overfill prevention valve on tank fill inlet
 - 2) Visual dial type fuel level gauge
 - 3) Manual gauge port with stick gauge
 - 4) Fuel supply and return ports
 - 5) Atmospheric vent port and mushroom type UL-listed flame arrester vent to be installed onto vent pipe by others
 - 6) Containment tank emergency vent port and pressure relief cap.
 - 7) Primary tank emergency vent port and pressure relief cap.

- 8) Digital tank monitor/overflow control panel with pilot warning lights, audible alarm, output contactors, and press to silence/test/reset feature equal to Pneumercator TMS 2000. Provide with analog output option for use by Building Automation System. (Mounted near tank indoors.)
- 9) Overflow panel with pilot warning light, audible alarm and press to silence/test/reset feature (mounted near tank indoors). Panel shall interface to digital tank monitor.
- 10) Fill spill box
- 11) Leak detection
- 12) Required internal piping and wiring
- 13) Decals and signs required by code and authority having jurisdiction.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install engine-generator set and associated equipment as indicated. Coordinate final location of equipment with General Contractor. Final location of equipment to be reviewed with Engineer prior to installation.
- B. Install equipment in accordance with manufacturers recommendations. Provide equipment protection during and subsequent to installation.

3.2 ACCEPTANCE TESTS

- A. Perform Acceptance Testing in accordance with Section 260812 - Power Distribution Acceptance Tests and Section 260813 - Power Distribution Acceptance Test Tables.

3.3 LOAD TEST

- A. Conduct load testing of engine-generator set, under direct supervision of factory-authorized representatives of manufacturers of engine-generator set and auto-transfer switch.
- B. Installation is NFPA 110, Level 2.
- C. Testing to include cold start, 25, 50, 75, and 100% step loads (slow addition of load to confirm operation), 50% and 100% block loads (instantaneous load percentages), and 4 hour testing at rated nameplate.
- D. Provide certified results of testing, including frequency and voltage regulation at 25, 50, 75, and 100% of rated load, fuel consumption, exhaust temperature, and exhaust emissions at the above load ratings, actual measured values for pickup and drop out relays for ATS, measured values for time delay relays.

- E. Engine-generator set test results are to be certified to comply with specification parameters or necessary corrective actions implemented and tests repeated until compliance is certified at no additional cost to Owner.
- F. At conclusion of testing, service engine-generator set including replacing air, oil and fuel filters, changing lubrication oil, checking and refilling batteries, adjusting fan belts for proper tightness, and refilling of cooling system as required.
- G. Provide fuel for load testing of engine-generator set.

END OF SECTION

SECTION 263353
STATIC UNINTERRUPTIBLE POWER SUPPLY WITH BATTERY ENERGY STORAGE

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260000 - General Electrical Requirements

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION OF SYSTEM

- A. Provide an on-line, double conversion, solid-state uninterruptible power system (UPS) to furnish continuous, regulated alternating current (AC) power with battery backup. The UPS shall consist of a rectifier, charger, batteries, inverter, static bypass transfer switch, synchronizing equipment, maintenance bypass switch protective devices, and accessories. Refer to the one line diagram for specific configuration of devices.

1.4 REFERENCE STANDARDS

- A. ANSI C62.41 – IEEE Recommended Practice for Surge Voltages in Low-Voltage AC Power Circuits
- B. NEMA PE 1 - Uninterruptible Power System Specification and Performance Verification.
- C. NECA 411 – Standard for Installing and Maintaining Uninterruptible Power Supplies.
- D. UL 1778 - Uninterruptible Power Supply Equipment.

1.5 SUBMITTALS

- A. Provide cover letter with compliance table listing each specification section and indicating compliance "C", deviation for alternate "D", or exception with explanation "E". Any deviation or exception shall be accompanied with detailed explanation of how design intent is being upheld.

- B. For each UPS being furnished, submit the following information:
 - 1. Product Data:
 - a. Manufacturer's technical datasheet(s) indicating system performance characteristics, including guaranteed efficiency at 25%, 50%, 75%, and 100% load.
 - 2. Shop Drawings:
 - a. General Arrangement:
 - 1) Front, plan, and side views; access requirements (front, side, rear); overall dimensions and components list; shipping splits and weights.
 - b. Wiring Diagrams:
 - 1) Showing input, output, and bypass power flow within the UPS.
 - 2) Identifying terminal numbers / labels for all external connections.
 - c. Unit Data:
 - 1) Input power requirements
 - 2) Heat dissipation
 - 3. Closeout Submittals:
 - a. Operations and Maintenance (O&M) Manual:
 - 1) Installation instructions
 - 2) Operating instructions
 - 3) Maintenance instructions including recommended maintenance intervals
 - b. As-built shop drawings

1.6 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory wrapping or provide additional canvas or plastic cover to protect units from dirt, fumes, water, corrosive substances, construction debris, and traffic. Provide temporary heaters as required to prevent condensation.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.
- C. Extended factory warranty shall include replacement coverage for UPS parts for a period of 12 months from Owner system acceptance after commissioning.

- D. Service protection package shall include on-site repair/replacement labor for UPS parts and batteries and technical support coverage. Standard response time shall be 48 hours from receipt of call.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. UPS System:
 - 1. ABB
 - 2. Eaton
 - 3. General Electric
 - 4. Liebert
 - 5. Mitsubishi
 - 6. Schneider/APC
 - 7. Toshiba

- B. VRLA Batteries:
 - 1. C&D
 - 2. EnerSys
 - 3. Exide

2.2 SYSTEM OPERATION

- A. Normal Mode:
 - 1. UPS shall supply continuous power to load via double-conversion process. Rectifier/charger shall convert input AC to DC. Inverter output shall convert DC to AC.
 - 2. Inverter output shall be synchronized with AC power source provided that AC power source is within specified frequency range.
 - 3. Rectifier/charger shall convert AC input power to DC power for continuous float charging the battery.
- B. Loss of Normal AC Input Power:
 - 1. Battery plant, via DC-DC converter, shall supply DC power to inverter (without any switching) so that there is no interruption of AC power to critical load whenever AC input power source of UPS module deviates from specified tolerances or fails completely.
 - 2. Battery shall continue to supply power to inverter for specified protection time.
- C. Return of AC Power Source:
 - 1. Rectifier/charger shall assume DC load from battery when AC power source returns.
 - 2. Rectifier/charger shall supply inverter with DC power and recharge battery.

- D. Internal Static Bypass Mode:
 - 1. If UPS is taken out of service for maintenance or repair, or should inverter capacity be exceeded, static bypass transfer switch shall perform transfer of load from inverter to AC power source via internal bypass pathway.
 - 2. Transfer back to Normal Mode shall be accomplished with no interruption in power to critical loads.
- E. Retransfer to Inverter:
 - 1. Static transfer switch shall automatically retransfer load back to inverter after inverter has returned to normal voltage and synchronizes with alternate bypass source.
- F. External Maintenance Bypass Mode:
 - 1. Provide distribution panel to completely bypass UPS system with no interruption to critical load when UPS is taken out of service for maintenance or repair. Provide means for safety interlocks between UPS and external maintenance bypass to ensure safe closed transition transfer between internal static bypass pathway and external maintenance bypass.
- G. Recharge Mode:
 - 1. Upon restoration of utility AC source, rectifier shall derive power from utility AC source and supply DC power to inverter and DC-DC converter, which simultaneously float charges battery. This shall be an automatic function and shall cause no interruption to critical AC load.
- H. Battery Unavailable Mode:
 - 1. If battery is unavailable, UPS shall continue to function and meet specified steady-state performance criteria for power outage backup time capability.

2.3 EQUIPMENT AND RATINGS

- A. Design, manufacture, and test UPS in accordance with applicable portions of the following standards:
 - 1. UL 1778 - Uninterruptible Power Supply Equipment
 - 2. NEMA PE 1 - Uninterruptible Power System Specification and Performance Verification
- B. Electrical Input (Normal):
 - 1. Nominal Voltage: As indicated on the Drawings
 - 2. Voltage Range: +10%, -15% of Nominal
 - 3. Frequency Range: +60 Hz, ± 5 Hz
 - 4. Input Power Factor: Minimum 0.99 lagging at full load
 - 5. Current Limit: 170% of nominal input full load current if load is connected to UPS module, or to bypass line by static transfer switch, or if load is disconnected. If load is connected to maintenance bypass line, input current limit to UPS module is to automatically reduce to 25% of normal full load input current.
 - 6. Current Distortion: Maximum 5% reflected input total current harmonic distortion at nominal input voltage and 100% load.
 - 7. Input Surge Withstand Rating: Complying with ANSI C62.41, Category A and B.

- C. Electrical Input (Bypass):
 - 1. Nominal Voltage: As indicated on the Drawings
 - 2. Sync Voltage Range: +10%, -10% of Nominal
 - 3. Voltage Range: +15%, -20% of Nominal
 - 4. Sync Frequency Range: +60 Hz, ± 5 Hz
 - 5. Frequency Range: +20%, -20% of Nominal
- D. Electrical Output
 - 1. Nominal Voltage: As indicated on the Drawings
 - 2. Voltage Regulation: $\pm 1\%$ for balanced load
 - a. $\pm 2\%$ for unbalanced load
 - 3. Sync Voltage Regulation: $\pm 10\%$ of nominal
 - 4. Frequency Regulation: 60 Hz $\pm 0.25\%$
 - 5. Voltage Transient Response: 100% Load Step: $\pm 2\%$
 - a. 10-100% or 100-10% Load Step: $\pm 5\%$
 - b. Loss or return of AC input power: $\pm 1\%$
 - c. Retransfer from bypass to inverter: $\pm 3\%$
 - 6. Overload Capability: 150% for 1 minute
 - a. 125% for 10 minutes
 - b. 110% for 60 minutes
 - 7. Voltage Transient Recovery: To within 1% of output voltage rating within 20 milliseconds.
 - 8. Distortion: 5% THDv maximum at 100% non-linear load
 - a. Maximum 2% THDv for linear load
 - b. Maximum 5% THDv for 100% non-linear load
- E. Environmental Conditions:
 - 1. Operating ambient temperature:
 - a. 32° to 104°F
 - b. 77° $\pm 9^\circ$ F
 - 2. Non-operating and storage ambient temperature -4° to 158°F.
 - 3. Relative humidity 0% to 95% for temperatures in operating range without condensation.
- F. Audible Noise:
 - 1. Noise generated by UPS under any condition of normal operation shall not exceed 69 dB measured at 5 ft from nearest surface of cabinet.
- G. Efficiency:
 - 1. Efficiency measured under following conditions:
 - a. Battery fully charged and floating on system
 - b. Input voltage within specified range.

- c. In fully-online double-conversion mode, minimum efficiency shall meet requirements in the table below:

UPS Rating (kVA)	25% Load	50% Load	75% Load	100% Load
10	86.5	91.7	93.0	93.9
20	92.9	94.5	94.6	94.3
30	91.0	92.6	92.8	92.3
40	93.5	95.4	95.2	94.5
50	93.1	95.3	95.6	95.6
80	93.9	95.1	95.0	94.7
100	95.2	96.3	96.3	96.1
150	96.4	96.9	96.8	96.6
225	95.6	96.3	96.3	96.2
250	95.9	96.6	96.4	96.1
500	93.5	95.0	95.3	95.2
750	95.5	96.5	96.8	96.6
1000	95.5	96.5	96.8	96.5
1200	95.5	96.5	96.8	96.5

2.4 COMPONENTS

A. UPS Module

1. Rectifier/Charger:
 - a. Rectifier/charger shall include:
 - 1) Circuit to limit AC input current to an adjustable level from 100% of nominal up to rated maximum input current rating.
 - 2) Circuit to limit battery charging current to an adjustable level of 2% to 25% of maximum battery discharge current.
 - a) A second circuit shall provide greater limiting when signaled by an external contact (i.e., operation of generator).
 - b) Battery charge current limit to be factory set at 10% for normal operation and 2% for generator operation.
 - 3) Output filter minimize ripple current to battery. Ripple current to battery shall not exceed 2% RMS. Filter shall ensure that DC output of rectifier/charger will meet input requirements of inverter.
2. Inverter:
 - a. Solid-state, capable of accepting rectifier/charger or battery output and providing rated output power.
 - b. Insulated gate bipolar transistor (IGBT) phase-controlled, pulse width modulated (PWM) design.
 - c. Able to sustain an overload of up to 150% with + 2% output voltage regulation.
 - d. Capable of 300% current for short circuit conditions. If short circuit is sustained, inverter shall transfer load to bypass.
 - e. With input current walk-in feature to gradually transfer load from battery to AC input source upon restoration of AC input power, over a minimum period of 5 seconds.

3. DC-DC Converter:
 - a. Includes controls to regulate output of rectifier to levels appropriate for charging battery and to boost battery voltage to level required to operate inverter.
 4. Momentary Overloads:
 - a. In event of load current inrush or branch load circuit fault, static transfer switch to pulse on for at least 40 milliseconds allowing up to 1000% load current to flow from bypass line to clear overload.
 5. Automatic Closing:
 - a. When load is transferred to bypass line by static transfer switch, bypass switch to automatically close removing static transfer switch from power flow.
 6. Manual Transfers:
 - a. Manual load transfer between UPS system and bypass line can be initiated from control panel.
- B. Static Bypass Section
1. System bypass section:
 - a. Shall contain devices to electromechanically transfer load from the UPS system to bypass source and vice versa.
 - b. Devices shall provide isolation of UPS system from critical load when utilizing bypass as source
 - c. Shall provide system overload protection for UPS system.
 2. Input contactor with fused protection
 3. Static transfer switch:
 - a. Solid-state device that automatically transfers load to bypass without interruption if UPS cannot supply continuous power.
 4. Bypass switch:
 - a. Motor operated circuit breaker that connects load to bypass.
 - b. Automatic load transfers to be initiated when a sustained system overload occurs or when an essential (non-redundant) UPS module is taken off line.
 5. System bypass shall be an in-phase non-interruptive bypass.
 6. Operation of bypass switch shall cause transfer of load to AC input power without deviation of output bus beyond acceptable limits. Simultaneously, bypass switch shall isolate load bus from UPS output bus.
 7. System bypass shall have appropriate phase, voltage and frequency monitors to prevent synchronous transfer or transfer to bypass source with unacceptable voltage, frequency or phase rotation.
- C. Controls
1. Monitoring functions to be displayed in numbers and words on digital display.
 2. Operator controls and monitors to be located on front of UPS module cabinet.
 - a. Controls:
 - 1) UPS On
 - 2) Bypass On
 - 3) Emergency Power Off
 - 4) Alarm Silence
 - b. Meters:
 - 1) AC Input Voltage, 3 phase, L-L, and L-N for each phase
 - 2) Bypass Input Voltage, 3 phase, L-L and L-N for each phase
 - 3) AC Input Current, each phase

- 4) AC Output Voltage, 3 phase, L-L and L-N for each phase
- 5) AC Output Current, each phase
- 6) AC Output Frequency
- 7) Battery Voltage
- 8) Battery Charge/Discharge Current
- 9) Percent of Rated Load Being Supplied by UPS
- 10) Battery Time Left During Battery Operation
- c. Alarms to be displayed:
 - 1) UPS on-line
 - 2) Input power failed
 - 3) Alternate source power failed
 - 4) Battery discharging
 - 5) Low battery
 - 6) Overload
 - 7) Overload shutdown
 - 8) Load on bypass
 - 9) Equipment over temperature
 - 10) Over temperature shutdown
 - 11) Fan failure
 - 12) DC overvoltage
 - 13) Control power failed
 - 14) Output overvoltage
 - 15) Output undervoltage
 - 16) Fuse open
 - 17) Rectifier/charger failure
 - 18) Static transfer switch failure
 - 19) Emergency power off
 - 20) On maintenance bypass
 - 21) On static bypass line
 - 22) Sync with line
 - 23) Not sync with line
 - 24) Out of frequency range
 - 25) Rectifier/charger in current limit
 - 26) Battery circuit breaker open
- d. Pushbuttons to test and reset visual and audio alarms.

2.5 BATTERY SYSTEM

- A. Battery System:
 1. Battery Units:
 - a. Type: Valve-regulated lead acid (VRLA)
 - b. Design Life: 10 years, as guaranteed by the battery manufacturer
 - c. Quantity:
 - 1) As required to deliver voltage specified for UPS system.
 - 2) Strings: Single string.
 - d. Capacity Rating:
 - 1) As required to deliver runtime specified for UPS system, as indicated on Drawings at full load at end of expected battery life.

- e. Insulation:
 - 1) Insulated to withstand ± 1000 V DC high potential test applied to either positive or negative terminals, while connected to UPS system in operable condition with DC disconnect open.
- f. Self-resealing flame-arresting caps
- g. Interconnecting Components:
 - 1) Provided with intercell and inter-tier connectors.
- h. At end of discharge, cell voltage shall not be less than 1.67 EVPC and 1.250 specific gravity.
- 2. Battery Enclosures/Racks:
 - a. Type: Vented NEMA Type 1 Enclosure
- 3. Overcurrent Protection:
 - a. Circuit breaker between each battery string and UPS.
 - 1) Located in battery enclosure
 - 2) With (2) Form C auxiliary contacts indicating breaker position.
 - 3) Shunt trip to disconnect battery from UPS module when battery reaches minimum discharge voltage level or when signaled by other control functions.
- 4. Thermal runaway protection:
 - a. Provide International Fire Code compliant thermal runaway protection to trip battery breaker when thermal runaway condition is detected.

2.6 BATTERY MONITORING SYSTEM (BASIC):

- A. Monitor the following parameters:
 - 1. Jar voltage (for each jar)
 - 2. Jar resistance (for each jar)
 - 3. String voltage (for each string)
 - 4. String current (for each string)
 - 5. Temperature (for each string - cabinet ambient or pilot cell)
- B. Alarm generation:
 - 1. Configure alarms based on the specific batteries being monitored, with parameters set for mission critical application.
 - 2. Local indicators for alarms and status
 - 3. Two form C alarm contacts for communicating a summary alarm condition to external systems.
- C. Communications:
 - 1. Relay Output Card: (4) Programmable form C contacts
- D. Electrolyte neutralization:
 - 1. Provide a kit suitable to neutralize a spill of 3% of the largest cell in UPS system to a pH between 5.0 and 9.0.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install UPS in accordance with NECA 411 – Standard for Installing and Maintaining Uninterruptible Power Supplies.
- B. Perform manufacturer's recommended start-up procedure using the services of a factory-trained representative of UPS manufacturer.

3.2 OWNER TRAINING

- A. Provide training in start-up, operation, and maintenance of equipment supplied using the services of a factory-certified representative of UPS manufacturer. Provide (1) training session(s) of (2) hour(s) minimum on separate days.
- B. Submit training plan for Owner's approval. Training plan to include duration of training sessions, suggested class size/attendees, locations, lesson objectives, and outline of training topics.

3.3 TESTING REQUIREMENTS

- A. Factory Testing:
 - 1. UPS system shall be tested during manufacturing process to assure that equipment is fully functioning and its performance meets specifications. Upon request, manufacturer shall describe testing program and supply sample test procedures and test data sheets to document testing program.
 - 2. Provide factory witness testing for Owner, Designer, and Commissioning Agent.
- B. Site Testing:
 - 1. Provide acceptance testing and thermographic scanning with loadbanks per Section 260812.
 - 2. UPS equipment shall be fully tested after installation. Testing shall be part of start-up service provided by manufacturer. Site testing shall include complete inspection of installation to assure it was performed in accordance with manufacturer's recommendations.

END OF SECTION

SECTION 263623
AUTOMATIC TRANSFER SWITCHES

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260000 - General Electrical Requirements
- B. Section 260529 - Hangers and Supports for Electrical Systems
- C. Section 263213 - Engine Generators

1.2 REFERENCE

- A. Work under this section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION OF SYSTEM

- A. Provide automatic transfer switch, 3 phase, 60 Hz, 4 pole, with overlapping neutral for voltage and current as indicated on drawings.

1.4 REFERENCE STANDARDS

- A. NEMA IA 10042-1 Industrial Control and Systems Part 1: Electromechanical AC Transfer Switch Equipment
- B. UL 1008 Automatic Transfer Switches

1.5 SUBMITTALS

- A. Submit shop Drawings for equipment provided under this Section.

1.6 QUALITY ASSURANCE

- A. Obtain automatic transfer switches from one source and by single manufacturer.
- B. Regulatory Requirements:
 - 1. Comply with NFPA 70 for components and installation.
 - 2. Furnish products listed and classified by Underwriters Laboratories, Inc., as suitable for purpose specified and indicated.

- C. Factory Test
 - 1. Test system in accordance at the factory in accordance with Section 260812 - Power Distribution Acceptance Tests and Demonstration of Transfer Functions.
 - 2. Provide factory test report.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store in clean, dry space. Maintain factory wrapping or provide additional canvas or plastic cover to protect units from dirt, fumes, water, corrosive substances, construction debris, and traffic. Provide temporary heaters in switchgear as required to prevent condensation.
- B. Deliver in shipping splits of length that can be moved in delivery path, as indicated, individually wrapped for protection, and mounted on shipping skids. Mark crates, boxes, and cartons clearly to identify equipment. Show crate, box, or carton identification number on shipping invoices.
- C. Use factory-installed lifting provisions. Handle carefully to avoid damage to internal components, enclosure, and finish.

1.8 WARRANTY

- A. Refer to Division 01 and Section 260000 - General Electrical Requirements for general warranty requirements.
- B. Manufacturer shall provide standard 1 yr warranty against defects in materials and workmanship for products specified in this Section. Warranty period shall begin on date of substantial completion.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Acceptable Manufacturers: Asco by Schneider Electric, Cummins, Russelectric by Siemens, or Zenith by ABB

2.2 AUTO-TRANSFER SWITCH

- A. Auto-transfer switch:
 - 1. Mechanically held, electrically operated type
 - 2. Interlocked to ensure only 2 possible positions, normal and standby.
 - 3. Rated for continuous duty in unventilated sheet metal enclosure
 - 4. Suitable for all classes of loads at maximum rated voltages
 - 5. Withstand rating that meets or exceeds withstand rating of transfer switch feeder breakers.
 - 6. Shall be open transition type as specified on drawings.

- B. Provide main contacts with silver alloy wiping action type protected by arcing contacts.
- C. Provide switch components accessible from front of enclosure.
- D. Provide 3 cycle short circuit rating to guarantee contact opening and no damage when transfer switch is served by molded case circuit breakers.
- E. Provide 10 cycle short circuit rating to guarantee contact opening and no damage when transfer switch is served by power circuit breakers.
- F. Provide switch with the following items:
 - 1. Adjustable 1 to 3 second time delay to override momentary voltage dips and outages.
 - 2. Time delay on transfer to emergency. Adjustable from 1 to 300 seconds (factory set at 3 seconds).
 - 3. Time delay on retransfer to normal. Adjustable from 2 seconds to 30 minutes.
 - 4. Full phase protection consisting of 2 phase relays and one close differential relay. Phase relays shall be set to 70% drop out, 90% pick up, and differential relays set for 92 to 95% pick-up and 83 to 85% drop-out.
 - 5. Pushbutton reset to normal.
 - 6. Pushbutton to bypass time delay on retransfer to normal.
 - 7. Pilot light to indicate normal position.
 - 8. Pilot light to indicate emergency position.
 - 9. Auxiliary contact to close when normal power fails.
 - 10. Auxiliary contact to open when normal power fails.
 - 11. Auxiliary contact on same shaft as main contacts (closed on normal).
 - 12. Auxiliary contact on same shaft as main contacts (closed on emergency).
 - 13. Pushbutton, orselector switch, or graphical display input to provide "Test," "Auto," "OFF", and "Engine Start."
 - 14. Contacts to start engine-generator when normal power fails.
 - 15. Time delay engine start, adjustable from 0 to 5 seconds.
 - 16. Adjustable time delay on retransfer to normal source with 5 minute unloaded running time of standby plant:
 - a. Minimum delay 2 minutes
 - b. Maximum delay 25 minutes
 - c. Built in circuitry to nullify time delay if emergency source fails and power is available at normal source.
 - 17. Relay to prevent transfer to emergency until voltage and frequency of generating plant have reached 90% of rated value.
 - 18. Provide bi-direction in-phase monitor or dual motor operator with programmed neutral to allow voltage decay in motor and transformer circuits.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install equipment in accordance with manufacturer's recommendations.
- B. Provide equipment protection during and subsequent to installation.

- C. Provide wiring between transfer switch and elevator controller. Final connections at elevator controller by Elevator Contractor.

3.2 OPERATION

- A. Parallel “start engine-generator” contacts of automatic transfer switches, such that failure of normal source at any switch shall start engine.
- B. Engine generator cool down cycle shall not start until transfer switch has timed out back to normal source.

3.3 ACCEPTANCE TESTING

- A. Perform acceptance testing in accordance with Section 260812 - Power Distribution Acceptance Tests.
- B. Adjust or replace equipment as needed to comply with manufacturer's specifications and resubmit corrected test reports.

END OF SECTION

SECTION 264300
SURGE PROTECTIVE DEVICES (SPD)

PART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260000 - General Electrical Requirements
- B. Section 260526 - Grounding and Bonding for Electrical Systems
- C. Section 262416.13 - Lighting and Appliance Panelboards
- D. Section 262726 - Wiring Devices

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents, including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION

- A. Provide Type 1 Surge Protective Devices (SPD) for the protection of AC electrical circuits formerly known as Transient Voltage Surge Suppression (TVSS) System. Provide high energy surge current diversion and be suitable for application in Type 1 environments.
- B. Modes of Protection:
 - 1. Line to Ground, Line to Neutral, and Neutral to Ground for services with a neutral
 - 2. For Services without a neutral, Line to Line and Line to Ground
- C. Provide common and normal modes of protection.

1.4 REFERENCE STANDARDS

- A. ANSI/UL 1449 Surge Protective Devices Fourth Edition
- B. IEEE C62.41.1 – Recommended Practice on Surge Voltages in Low Voltage AC Power Circuits FIPS PUB 94
- C. IEEE C62.11 – Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (> 1 kV)
- D. IEEE C62.41.1 Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits

- E. IEEE C62.41.2 Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits
- F. IEEE C62.45 IEEE Recommended Practice on Surge Testing for Equipment Connected to Low-Voltage (1000 V and less) AC Power Circuits
- G. IEEE C62.62 IEEE Standard Test Specifications for Surge Protective Devices (SPDs) for Use on the Load Side of the Service Equipment in Low Voltage (1000V and less) AC Power Circuits
- H. National Electrical Code – Article 285
- I. NEMA LA 1 – Surge Arresters
- J. National Fire Protection Association – NFPA 20, 70, 75, and 780
- K. UL 96A Installation Requirements for Lightning Protection Systems

1.5 SUBMITTALS

- A. Submit Shop Drawings for equipment provided under this Section.
- B. Submit shop drawings and product information for approval and final documentation in quantities listed according to Conditions of the Contract. Identify customer name, customer location, and customer order number.
- C. Submit ANSI/UL 1449 Listing documentation to indicate the following:
 - 1. Short Circuit Current Rating (SCCR)
 - 2. Voltage Protection Ratings (VPRs) for all modes
 - 3. Maximum Continuous Operating Voltage rating (MCOV)
 - 4. Nominal Discharge Current rating (I-n)
 - 5. Type 1 device
 - 6. VPR, MCOV, I-n, and Type 1 information is posted at www.UL.com under Certifications; search using UL Category Code: VZCA. SCCR's are posted in manufacturer's published documentation.
 - 7. UL data and visual inspection takes precedence over manufacturer's published documentation.
- D. Provide shop drawings including manufacturer installation instruction manual and line drawings detailing dimensions and weight of enclosure, internal wiring diagram illustrating all modes of protection in each type of SPD required, wiring diagram showing field connections, and manufacturer's recommended wire and breaker sizes (if required).
- E. Upon request, modules using encapsulation shall be presented without encapsulation for visual inspection, proprietary technology included. MOV type and quantity shall reflect kA ratings on cutsheets, verification of diagnostic monitoring, thermal and overcurrent protection, etc.

PART 2 - PRODUCTS

2.1 MATERIAL

- A. Acceptable manufacturers:
 - 1. Externally-mounted unit by the power distribution equipment manufacturer.
 - 2. Erico

2.2 PERFORMANCE CHARACTERISTICS

- A. SPD shall bear the UL Mark and shall be Listed to Third Edition of ANSI/UL 1449. "Manufactured in accordance with" is not equivalent to UL Listing and does not meet intent of specification.
- B. Post SPD and performance parameters at www.UL.com under Category Code: VZCA. Products or parameter without posting at UL.com are not approved.
- C. Minimum surge current capacity for Service Entrance units based on 8 x 20 microsecond current waveform:
 - 1. 200,000 A between each phase for line-to-line mode
 - 2. 200,000 A each phase for line-to-ground mode
 - 3. 200,000 A each phase for line-to-neutral mode
 - 4. 200,000 A for neutral-to-ground mode
- D. Minimum surge current capacity for panelboard units based on 8 x 20 microsecond current waveform:
 - 1. 80,000 A between each phase for line-to-line mode
 - 2. 80,000 A each phase for line-to-ground mode
 - 3. 80,000 A each phase for line-to-neutral mode
 - 4. 80,000 A for neutral-to-ground mode
- E. Sequential Surge Current Survivability:
 - 1. 1,000 sequential category surges without failure.
- F. Current Rating:
 - 1. Rated for continuous current and AIC rating of equipment protected.
 - 2. Mark SPD Short-Circuit Current Rating on the SPD label.

2.3 OPERATING CONDITIONS

- A. Temperature range: -40°F to 122°F
- B. Relative humidity range: 0 to 95%, non-condensing
- C. Audible noise level: > 40 dBA at 5 ft
- D. SPD Surface Temperature: less than 131°F

2.4 FABRICATION

A. SPD Modules:

1. UL Labeled as Type 1 (verifiable at www.UL.com), intended for use without need for external or supplemental overcurrent controls. Protect suppression component of every mode, including N-G, by internal overcurrent and thermal overtemperature controls. SPDs relying on external or supplementary installed safety disconnects do not meet intent of specification.
2. Suppression components: Heavy-duty MOVs, selenium cells, or combination of both.
3. Provide surge current diversion paths for all modes of protection: L-N, L-G, N-G in WYE systems, and L-L, L-G in DELTA systems.
4. Provide service entrance SPD audible diagnostic monitoring by way of audible alarm.
5. Provide service entrance SPD with 1 set of NO/NC dry contacts for alarm conditions.
6. Provide visual LED diagnostics including a minimum of 1 green LED indicator per phase, and 1 red service LED. Include an audible alarm with on/off silence function and diagnostic test function (excluding branch).
7. If a dedicated breaker for the SPD is not provided in the switchgear, switchboard, or panelboard include an integral UL Recognized disconnect switch. Dedicated breaker to serve as a means of disconnect for distribution SPDs.
8. Meet or exceed the following criteria:
 - a. ANSI/UL 1449 Listed Voltage Protection Ratings (VPRs) for 6kV 3000A testing as follows:

VOLTAGE	L-N L-G N-G	L-L
208Y/120V	≤800V	≤1200V
480Y/277V	≤1200V	≤2000V

9. ANSI/UL 1449 Listed Maximum Continuous Operating Voltage (MCOV) (verifiable at UL.com):

System Voltage	Allowable System Voltage Fluctuation (%)	MCOV
208Y/120	15%	150V
480Y/277	15%	350V

10. Provide serviceable, replaceable modules (excluding Branch).
11. Provide warranty for a period of 10 yrs, incorporating unlimited replacements of suppressor parts if they are destroyed by transients during the warranty period.

B. Panelboards:

1. Install 1 suppressor external to each designated panelboard.
2. Install surge suppression device in accordance with manufacturer instructions.

C. SPD Low-Impedance Interconnect Cable:

1. Provide low-impedance cable specifically listed for SPD installations.

2. Low impedance approximately 25% of conventional pipe and wire for improved clamping voltage.

PART 3 - EXECUTION

3.1 APPLICATION OF SPD

- A. Provide UL approved disconnect switch at Service Entrance or Transfer Switch as a means of service disconnect if a breaker sized per manufacturer's recommendations is not available.
- B. Provide independent means of servicing disconnect at Distribution, MCC, and Branch such that the protected panel remains energized. A 30A breaker (or larger based on manufacturer's recommendations) may serve this function.
- C. Provide SPD for each panel associated with the optional standby branch of power.

3.2 INSTALLATION

- A. Install per manufacturer's recommended practices.
- B. Provide short and straight conductors not exceeding 3 ft in length. Manufacturer-approved cables may be used that allow conductor length to extend beyond 3 ft in length without affecting capability of unit.
- C. Input conductors twisted together to reduce inductance.
- D. Avoid 90-degree bends in cable.

3.3 QUALITY ASSURANCE

- A. Factory test system before shipment. Include quality control check, "Hi-Pot" tests at 2 times rated voltage plus 1,000 V, ground leakage tests, and calibration.
- B. Manufacturer Qualifications: Engage a firm with at least 5 yrs experience in manufacturing surge protective devices.
- C. Manufacturer of equipment shall have produced similar electrical equipment for a minimum period of 5 yrs. When requested by Engineer, an acceptable list of installations with similar equipment shall be provided demonstrating compliance with requirement.
- D. Provide SPD compliant with the Restriction of Hazardous Substances (RoHS) Directive 2002/95/EC.

3.4 FIELD QUALITY CONTROL

- A. Inspections before SPD startup:
 - 1. Visual Inspection:
 - a. Verify installation per drawings.
 - b. Verify phase, neutral, and ground conductors are properly sized and configured.
 - 2. Mechanical Inspection:
 - a. Check connections for tightness.
 - b. Check terminal screws, nuts and/or connectors for tightness.
 - 3. Electrical Inspection:
 - a. Confirm input voltage.
 - b. Confirm phase, neutral and ground connections are proper.

3.5 WARRANTY

- A. Provide 10 yr manufacturer warranty.

END OF SECTION

SECTION 265000 LIGHTING

ART 1 - GENERAL

1.1 RELATED WORK

- A. Section 260000 - General Electrical Requirements
- B. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
- C. Section 260526 - Grounding and Bonding for Electrical Systems
- D. Section 260533 - Raceway and Boxes for Electrical Systems
- E. Section 260923 - Lighting Control Devices

1.2 REFERENCE

- A. Work under this Section is subject to requirements of Contract Documents including General Conditions, Supplementary Conditions, and sections under Division 01 General Requirements.

1.3 DESCRIPTION OF WORK

- A. Provide complete and fully operational lighting system per Contract Drawings and Specifications.
- B. Luminaires shall be provided complete with necessary accessories for proper installation.
- C. Catalog numbers shown in luminaire schedule are basic luminaire types. Additional features, accessories and options specified, scheduled or necessary for proper installation shall be included.
- D. Specifications and drawings convey the features and functions of luminaires only and do not show every item or detail necessary for the work.
- E. Work includes final aiming and focusing of luminaires under direction of the Architect /Engineer.

1.4 REFERENCE STANDARDS

- A. NECA/IESNA 500 - Standard for Installing Indoor Commercial Lighting Systems (ANSI)
- B. NECA/IESNA 501 - Standard for Installing Exterior Lighting Systems (ANSI)

- C. NECA/IESNA 502 - Standard for Installing Industrial Lighting Systems (ANSI)
- D. NEMA LE 4 - Recessed Luminaires, Ceiling Compatibility
- E. UL 676 – Underwater Luminaires and Submersible Junction Boxes
- F. UL 773 - Plug-in Photocontrols for use with area lighting
- G. UL 924 - Emergency Lighting and Power Equipment
- H. UL 1574 – Track Lighting
- I. UL 1598 – Luminaires
- J. UL 1838 – Low Voltage Landscape Lighting Systems
- K. UL 2108 – Low Voltage Lighting Systems
- L. UL 2388 – Flexible Lighting Products
- M. UL 2562 – Pendant Cable
- N. UL 8750 – LED Light Sources for use in Lighting Products
- O. ANSI C78.377 – Chromaticity
- P. IESNA LM-79 – Approved Method: Electrical and Photometric Measurements of Solid-State Lighting Products
- Q. IESNA LM-80 - Approved Method: Testing Lumen Maintenance of LED Light Sources
- R. IESNA TM21-11 - Projecting Long Term Lumen Maintenance of LED Light Sources including Addendum A
- S. IEEE STD 1789-2015 - Recommended Practices for Modulating Current in High-Brightness LEDs for Mitigating Health Risks to Viewers

1.5 QUALITY ASSURANCE

- A. Luminaire and accessory components shall be constructed of materials appropriate for their use. Luminaires intended for use in marine environments shall be constructed of materials resistant to corrosion.
- B. Luminaires, drivers, other components shall meet the requirements of all applicable State and Municipal codes and energy codes.
- C. Provide luminaires listed and labeled by UL or other testing lab acceptable to local jurisdiction for their indicated use and installation conditions.

- D. Contractor shall coordinate installation of lighting systems with all trades.
1. Manufacturers listed in the luminaire schedule shall be assumed capable of supplying listed luminaires. Any such exceptions shall immediately be brought to the attention of Architect, Engineer, or Lighting Designer.
 2. Multiple Name Specification:
 - a. When multiple manufacturers are listed, Electrical Contractor shall choose which of the listed products are to be provided.
 - b. Products of the same type shall be of same manufacturer.
 3. Single Name Specification:
 - a. When only one product is suitable for the application and/or no other known acceptable products exist, only one manufacturer/product is listed in the Luminaire Schedule. For such instances, Electrical Contractor shall provide the listed product with no exceptions.
 4. Contractor shall coordinate and verify compatibility of luminaires with lighting control system
 - a. Control protocol indicated for luminaires matches protocol of lighting control system specified. Contractor shall coordinate and verify compatibility of all dimming luminaires with control system to ensure that dimming is flicker free, continuous dimming through the dimming range noted on the luminaire schedule.
- E. Substitution requests:
1. Will be evaluated prior to Bid.
 2. Shall follow procedures set forth in this Section under paragraph 1.7 and in Section 012500 - Substitution Procedures.
 3. Shall be made not less than 10 days prior to bid date.
 4. Shall include the following information indicating that the proposed substitution is of similar construction quality and assembly, lumen output and distribution, color temperature, color consistency, and controllability:
 - a. Specified and proposed manufacturer's product data sheet, noting options and features.
 - b. Provide dimensioned drawing of luminaire.
 - c. Provide photometric data in form of an electronic IES file on USB or via email.
 5. Electrical Contractor shall be responsible for all costs incurred by substitution request sample and/or mockup production and review.
 6. Equipment delivery lead time shall not be held as a valid reason for requesting luminaire substitution unless luminaire lead time from specified manufacturer is in excess of 14 weeks. It shall be sole responsibility of Electrical Contractor to determine necessary equipment lead times, deliver submittals for review in a timely fashion, and place orders accordingly to ensure timely delivery.
 7. When requesting a substitution, Electrical Contractor shall provide unit and extended pricing for specified luminaire, unit and extended pricing for proposed alternate, and unit and extended delta savings to Owner to be realized by accepting proposed alternate. If requested, provide unit pricing for each luminaire type specified to provide a baseline comparison for substitution request.
 8. Electrical Contractor shall guarantee pricing on all luminaire types for which a substitution request has been granted. This price guarantee shall be per unit and shall be maintained through the end of construction, regardless of quantity purchased.

9. For all luminaire types using an LED light source, provide independently tested, IESNA LM-79 compliant photometry testing data and IESNA LM-80 Lumen Maintenance data.

1.6 WARRANTY

- A. Exit Signs Utilizing LED Technology: Provide manufacturer's warranty for a period of not less than five years from the date of substantial completion including parts and labor for full replacement of defective product.
- B. LED Luminaires: Provide Manufacturer's warranty for a period of not less than five years from the date of substantial completion or the specified warranty period greater than five years for repair or replacement of defective electrical parts, including light source and driver.
- C. Luminaires without integral LED sources: Provide manufacturer's warranty for a period of not less than one year from the date of substantial completion including parts and labor for full replacement of defective product.

1.7 SUBMITTALS

- A. Electronic submittal format shall be limited to the greater of 500 pages or 30 MB to ensure that all pages load correctly. Bookmarks by luminaire type are required for ease of navigation. Submittals exceeding these limits should be broken in a logical fashion into multiple volumes for separate review. Measures to reduce file size should not compromise legibility or any other factors affecting ease of review.
- B. Upon award of Contract, submit complete list of lighting products to be furnished, with manufacturer and catalog designations, including current unit cost. Unit price shall be for equipment only and not include installation or miscellaneous electrical costs.
- C. Upon award of Contract, submit complete list of lighting products to be furnished, with manufacturer and catalog designations, including currently quoted lead times for product delivery. Should Electrical Contractor anticipate delivery schedule of any specified product may adversely impact construction schedule, they shall bring it to the attention of Owner/Architect /Lighting Designer at this time.
- D. In addition to complying with requirements of Section 260000 - General Electrical Requirements, submittals shall include the following:
 1. Manufacturer's product data
 2. Installation instructions
 3. Maintenance data
 4. Parts list for each luminaire accessory
 5. Photometric Data: photometric data for luminaire, including optical performance as follows:
 - a. Coefficients of utilization
 - b. Luminance table

- c. Candela distribution data
 - d. Zonal lumens
 - e. Area and roadway luminaires shall include Isocandela Charts, IES Roadway Distribution Classification and IES BUG (Backlight – Uplight – Glare) ratings.
- 6. Driver schedule indicating manufacturer, type, and catalog number for each luminaire
- 7. Driver cut sheet for each driver used, referencing luminaire type(s)
- 8. Product color/finish
 - a. Where specific finish or color is not specified and options exist, submit color or finish samples to Architect/Engineer/Lighting Designer for selection.
- E. Shop Drawings for equipment provided under this Section shall include the following:
 - 1. Overall submittal drawings indicating luminaire size, mounting (including ceiling type), light source, shielding, and voltage attributes, as well as manufacturer's product data, installation instructions, maintenance data, and parts list for each luminaire.
 - 2. Catalog cutsheets lacking sufficient detail will not be accepted.
 - 3. Detailed drawings of linear pendant mounted and suspended luminaires including dimensions, support spacing, suspension type, power feed type and locations, driver locations, wiring and controls configuration, luminaire joint locations and end plates. Provide canopy details that indicate coordination with the ceiling system provided.
 - 4. Detailed drawings for each cove and linear wall system configuration including dimensions, power feed locations, driver locations, luminaire joint locations, extension plates for end and corner sections and end plates.
 - a. LED strip luminaires mounted in architectural coves, provide dimensioned drawings and sections and include accessory cut sheets as specified. Within coves, all luminaires are to be mounted end to end with no more than 12" unlit split evenly between ends
 - 5. Detailed drawings for LED systems including LED color, color consistency, rated life, warranty, and scale plans with luminaire layout, number, type and location for drivers, and a complete bill of materials.
 - 6. Detailed drawings for continuous recessed or continuous surface mounted luminaires including dimensions, power feed locations, driver locations/quantity, luminaire joint locations, extension plates for end and corner sections and end plates as applicable.
 - 7. Detailed drawings for custom LED handrail systems including dimensions, power feed locations, driver locations/quantity, luminaire joint locations as applicable.
 - 8. Submit documentation that indicates specified products have been tested, or will be tested, for compatibility with the lighting controls being procured and will perform as specified. Control devices or system shall be able to control luminaires with flicker free, continuous dimming, in range specified. Electrical Contractor, luminaire manufacturer and lighting control manufacturer shall be financially responsible for any incompatibilities.
 - 9. Detailed drawings for nonstandard/custom luminaires indicating dimensions, weights, method of field assembly, components, features, and accessories. Details shall be scaled to a legible size.

10. Detailed drawings for fiber optic systems including scaled plans with cable layout number and type of fiber bundles, illuminator quantity and location, and a complete bill of materials.
 11. Drawings for site lighting shall include pole data with wind loading, complete dimensions and finish, pertinent physical characteristics and accessories including mounting details, driver type and location and any specified control options.
 12. Photometric Data: Where indicated on luminaire schedule and Contract Drawings, supply complete photometric data for luminaire, including optical performance rendered by independent testing laboratory developed according to methods of the Illuminating Engineering Society of North America as follows:
 - a. Coefficients of utilization
 - b. Luminance table with data presented numerically, showing maximum luminaire luminance at shielding angles. Readings should be taken both crosswise and lengthwise in case of linear luminaire or luminaire with an asymmetric distribution.
 - c. Candela distribution data, presented graphically and numerically, in 5° increments (5°, 10°, 15°, etc.). Data developed for up and down quadrants normal, parallel, and at 11-1/2°, 45°, 67-1/2° to source if light output is asymmetric.
 - d. Zonal lumens stated numerically in 10° increments (5°, 15°, etc.) as above.
 13. No variation from the general arrangement and details indicated on drawings shall be made on shop drawings unless required by actual conditions. All variations shall be marked on drawings submitted for approval.
- F. Provide luminaires with factory or field finish as directed by Architect/Engineer/Lighting Designer. Verify final finish requirements before releasing luminaires for fabrication.
- G. Where specific finish or color is not specified and options exist, submit color or finish samples to Architect for selection. Luminaires not having color or finish acceptable to Architect shall be replaced at no additional cost.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Luminaires:
1. As shown on Luminaire Schedule
- B. LED Drivers:
1. Shall be manufacturer recommended compatible driver.
 2. All LED drivers shall be dimming type standard unless otherwise noted. Refer to construction documents for control per application.
 3. Manufacturers must be compatible with lighting control system(s) provided and control luminaires to specified minimum output per Luminaire Schedule and controls intent documents.
 4. LED Drivers "At power levels between 100% to 1%, shall perform within the frequency (Hz) and Modulation (%) of IEEE STD 1789-2015 level.

- C. Emergency LED Battery Pack:
 - 1. Bodine, Dual-Lite, Iota or as specified in the Luminaire Schedule
- D. Low Voltage Transformers:
 - 1. Q-Tran or as specified in the Luminaire Schedule
- E. LED Modules:
 - 1. Philips Lumileds, Xicato, Cree, GE, Nichia, Osram Sylvania, Bridgelux, Citizen or as specified in the Luminaire Schedule

2.2 FABRICATION AND MANUFACTURER

- A. Luminaires:
 - 1. Construction
 - a. Luminaires shall bear label indicating circuit voltage. Labels shall not be visible from normal viewing angles.
 - b. Luminaires shall be constructed with joints made by means of welded, brazed, screwed, or bolted construction methods.
 - c. Housings shall be so constructed that all electrical components are accessible and replaceable without removing luminaires from their mountings.
 - d. Surface temperatures of luminaires with integral drivers shall not exceed 90°C in 30°C ambient.
 - e. Luminaires recessed in ceilings utilized as air handling plenums shall be certified as suitable for the purpose and conform to NEC Article 300.
 - f. Miter cuts shall be accurate, joints shall be flush and without burrs.
 - g. Troffers with doors shall have spring-loaded door cam latches unless otherwise noted for special environments and applications.
 - h. Luminaires shall be free of light leaks and designed to provide sufficient ventilation of source to provide the photometric performance documented. Low voltage transformers and drivers shall be vented per manufacturer's specifications.
 - i. Provide inscription for exit and stairway signs to conform to applicable codes. Provide battery back-up as specified.
 - j. Verify types of ceiling construction with General Contractor prior to releasing luminaires for fabrication and delivery and provide luminaires adapted to ceiling construction used.
 - k. Coordinate recessed luminaire mounting appurtenances, flanges and trims with construction of ceiling in which luminaire is to be recessed. Provide correct luminaire mounting assembly.
 - l. Luminaire frames shall be manufactured of non-ferrous metal or be painted after fabrication.
 - 2. LED Luminaires are considered a lighting system with dependent components that must be evaluated as a complete system. Each LED luminaire includes a light emitting source, provisions for heat transfer, electrical control, optical control, mechanical support and protection, as well as aesthetic design elements. All LED luminaires shall:
 - a. Be NRTL 3rd party tested to applicable UL standards. Where remote drivers are specified, all drivers shall also have UL listing or equivalent and comply with code requirements.

- b. Be tested to IESNA LM-79-19 testing using absolute photometry criteria.
 - c. Be reported greater or equal to 70% lumen maintenance at 50,000 hours of operation.
 - d. Be rapid cycle stress tested.
 - e. Have integral LED modules with a minimum operating temperature of - 20°C.
 - f. Have modules that are capable of being easily replaced upon failure with a manufacturer provided replacement module without voiding the UL listing of the luminaire.
 - g. Have driver housings easily accessible for ease of maintenance.
 - h. Have a maximum operating temperature at LED junction to not exceed 90°C over the expected operating range of the luminaire.
 - i. Be RoHS compliant, lead and mercury free.
 - j. Have an LED operating frequency of + or – 120 Hz.
 - k. Must meet the appropriate Federal Communications Commission (FCC) requirements for FCC 47 CFR 15 (consumer use) and/or FCC 47 CFR Part 18 (industrial use)
 - l. Be Class A Sound rated.
 - m. Be supplied with power supply that complies with IEEE C62.41-1991.
 - n. Operate at 120 or 277 volts, $\pm 10\%$.
 - o. Have reverse polarity protected at all hardwired connections and have high voltage protection in the event connections are reversed or shorted during the installation process.
3. Lenses, Reflectors and Diffusers
- a. All lenses or louvers shall be removable but held so that normal motion will not cause them to drop out.
 - b. All glass used in luminaires shall be made from thermal shock resistant borosilicate glass.
 - c. Optical lenses shall be free from spherical and chromatic aberrations.
 - d. Acrylic lenses shall be 100% virgin acrylic material.
 - e. Diffuser materials shall be UV stabilized in applications exposed to sunlight.
 - f. Troffer lenses shall be 0.125" thick, unless otherwise noted.
 - g. Alzak reflectors and louvers shall be low iridescent equivalent to Coil Anodizers. All Alzak parabolic cones shall be guaranteed against discoloration for a minimum of ten years.
 - h. Reflector cones shall not have visible source flashing in the cone.
4. Optics and Adjustments
- a. Adjustable Angle Luminaire: Luminaires with adjustment beam angle shall contain reliable angle locking devices.
5. Finishes
- a. Provide luminaires with finish as shown in the luminaire schedule. Verify final finish requirements before releasing luminaires for fabrication.
 - b. Painted luminaires shall be painted after fabrication or "post painted".
 - c. Ferrous parts and supports shall be rust proofed after fabrication.
 - d. For weatherproof or vaportight installations, painted finishes of luminaires and accessories shall be weather resistant using proper primers or galvanized and bonderized epoxy, so that entire assembly is completely corrosion resistant for service intended and rated for an outdoor life expectancy of not less than 20 years.

6. Wiring
 - a. Luminaires shall be completely wired at the factory and as required by code.
 - b. Internal wiring shall contain no splices.
 - c. Connections shall be made with insulated "wire nut" type mechanical connectors except that driver connections shall comply with NEC Article 410.
 - d. Luminaires shall be provided with flexible conduit, pigtails, and equipment for external connections.
 - e. Recessed luminaires installed in inaccessible ceilings shall be UL listed for through wiring with the junction box accessible from the luminaire opening.
 - f. Cords shall be fitted with proper strain reliefs and watertight entries where required by application.
7. Ceiling Coordination
 - a. Verify type of ceiling construction prior to releasing luminaires for fabrication and delivery.
 - b. Provide mounting appurtenance, flanges, sloped ceiling adaptors where required.
 - c. Provide mounting assembly, clips or other mechanical mounting lugs as required for support of luminaires.
8. Track-Lighting Systems:
 - a. A lighting track system is defined as a manufactured assembly designed to support and energize luminaires that are capable of being readily repositioned on the track. Its length may be altered by addition or subtraction of sections of track. Lighting track may be either flexible or rigid depending on the specific application.
 - b. Provide lighting track types as specified in Luminaire Schedule, in lengths as indicated on lighting plans.
 - c. Lighting track system includes current carrying conductors which may convey either line voltages (120V or 277V) or low voltages (12V or 24V). Characteristics of lighting track that conveys line voltages are different than a lighting track system that conveys low voltages and as such are governed by different requirements. Therefore, they are considered individually in these Specifications.
 - 1) Line voltage (120V or 277V) Lighting Track systems:
 - a) Provide components, including track, fittings, and luminaires from the same manufacturer as recommended by manufacturer for the intended use. All components shall be UL Listed and comply with the National Electric Code Standards for Lighting Track.
 - b) Maintain continuity of conductors through feeds, splice fittings and boxes. Relative positions of live and neutral conductors must always be maintained along continuous run so that track fittings connect into the track in a consistent manner.
 - c) Support lighting track at intervals recommended by the track manufacturer.
 - d) One or two circuit Lighting Track shall be supplied with separate neutral busbars and have the ability to have each circuit separately dimmed as required when using standard voltage and low voltage luminaires with either magnetic or electronic transformers.

- e) Lighting Track shall have the ability to be dimmed or switched in selected sections in addition to dimming or switching an entire track configuration or track run.
 - f) One and two circuit 120 volt Lighting Track shall be rated at 120/250 volt, 60 Hz, 2,400 watts maximum each circuit. Neutral busbar(s) shall be oversized and comparable to #10 AWG 30 amp wire to reduce the possibility of overheating due to non-linear loads and harmonics.
 - g) One and two circuit 277 volt Lighting Track shall be rated at 277 volt, 50/60 Hz, 5,540 watts maximum each circuit. Neutral busbar(s) shall be oversized and comparable to #10 AWG 30 amp wire to reduce the possibility of overheating due to non-linear loads and harmonics.
 - h) A separate grounding busbar shall be integral in all track lengths.
 - i) 277 volt Track fittings shall be identified by a red rotor and a 277 volt label.
- 2) Low voltage (12V or 24V) Lighting Track systems:
- a) Provide components, including track conductors, remote mounted transformers, fittings, and luminaires from the same manufacturer as recommended by the manufacturer for the intended use. Components shall be UL Listed as applicable for low voltage use.
 - b) Maintain continuity of conductors through feeds, splice fittings and boxes. Relative positions of conductors must always be maintained along continuous run so that track fittings connect into track in a consistent manner.
 - c) Support lighting track at intervals recommended by track manufacturer.
 - d) One and two circuit low voltage Lighting Track shall be supplied with three conductors and have the ability to have each circuit separately switched with either magnetic or electronic transformers provided by the track manufacturer. Two circuit low voltage Lighting Track can only be dimmed if both circuits are fed from the same transformer and as a result, separate circuit dimming shall not be attempted or permitted.
 - e) All transformers shall be supplied with both primary and secondary voltage over-current protection devices that shall remain readily accessible for maintenance and testing purposes.
 - f) Lighting Track shall have the ability to be dimmed or switched in selected sections in addition to dimming or switching an entire track configuration or track run. Separate, single circuit transformers are required for each independently controlled circuit with the use of electrically isolated couplers.
 - g) Conductors used in low voltage Lighting Track shall be, at minimum, equivalent to #10 AWG 30 amp wire or heavier and be capable of carrying a 300 watt load (at 12 volts) up to 32 ft from transformer feed within range of luminaire voltage tolerance. At 24 volts, conductors shall be capable of supplying

- a 600 watt load up to 60 ft from transformer feed within range of luminaire voltage tolerance.
 - h) If taut strung cable conductors are used as low voltage Lighting Track system, they shall have a Kevlar core to prevent strain on outer current carrying conductors.
 - i) Only insulated type taut strung cable conductors shall be used in order to comply with local electrical codes governing installation.
9. Outdoor Lighting Systems:
- a. Provide luminaires, mounting arms, brackets, poles, hand-hole covers, base components, and all other accessories for a complete assembly. Manufacturers shall be responsible for proper fitting of elements and structural integrity of unit
 - b. Exterior Luminaires:
 - 1) Shall operate at a minimum ambient temperature of 0°F.
 - 2) Shall be fully gasketed, with UL wet location label.
 - 3) Shall have approved wire mesh screens for ventilation openings.
 - 4) Anodized aluminum reflectors shall have minimum of 0.00079" anodizing thickness.
 - c. All castings and extrusions shall be given minimum one coat of baked-on clear lacquer, unless painted finish is specified.
 - d. Aluminum surfaces shall receive a duronodic or polyester powder paint finish.
 - e. Cast-in Luminaire housings installed directly in concrete shall be fabricated of hot dip galvanized steel or cast aluminum or composite.
 - f. Where cast aluminum housings are used, give two coats of asphaltum paint prior to installation.
 - g. Provide 1/8" thick x 2" diameter solid neoprene grommets at each point light luminaire surfaces are mounted to concrete structure.

2.3 LED SOURCES

- A. Static white LED sources shall be:
- 1. Minimum CRI of 85 unless otherwise noted on Luminaire Schedule
 - 2. Less than 5% flicker
 - 3. Within 0.004 on the CIE 1976 diagram for color spatial uniformity
 - 4. Within 0.007 on the CIE 1976 diagram for color maintenance over the rated lifetime of the source
 - 5. Binned within a 3-step MacAdam ellipse minimum, or as indicated in Luminaire Schedule
 - 6. Color temperature as noted on Luminaire Schedule
 - 7. Have a published life rating based on the point at which LED sources reach L70 lumen maintenance and tested in accordance with IES LM80-08 Approved Method: Testing Lumen Maintenance of LED light sources and IES TM-21-11: Projecting Long Term Lumen Maintenance of LED Light Sources
 - 8. L70 rated life shall be a minimum of 50,000 hours.
 - 9. LED modules, unless otherwise noted, shall be provided by light fixtures manufacturer and integral to luminaire.
 - 10. Screw-base LED replacement lamps, where required and listed on the luminaire schedule, shall meet dimming, output, CCT and CRI as specified. 25,000 hours is acceptable L70 rated life for these products.

2.4 DRIVERS

- A. Drivers for use in cold or freezer rooms, parking structures, loading docks, and outdoors shall be low temperature type.
 - 1. Driver shall have lowest temperature rating available in standard manufacture for its type.
- B. Drivers shall be located in luminaire they serve, unless otherwise noted.
- C. LED Drivers and Power Supplies shall:
 - 1. Operate system LEDs within the current limit specification of the LED manufacturer.
 - 2. Be supplied with over-temperature protection circuitry.
 - 3. Be programmable via manufacturer-provided means to allow for LED replacement modules to be "tuned" to match the output of remaining adjacent modules in the event that some time has passed and there has been lumen depreciation.
 - 4. Be within a NEMA enclosure.
 - 5. Be equipped with knockouts to accommodate standard conduit sizes
 - 6. Have a Power Factor to be = or > than 0.9
 - 7. Dimmable LED drivers must be compatible with dimming system(s) provided and control luminaires per luminaire schedule and controls documentation.
 - 8. ETL certified, CBM and UL Listed, high power factor, and meet or exceed NEMA and ANSI Standards.
 - 9. Be Class A sound rated
 - 10. be Equipped with resetting thermal sensitive device.
 - 11. For operation at with input power of 60 Hz and voltage as scheduled.
 - 12. Meet or exceed all ANSI or NEMA standards
 - 13. Be Capable of operating LEDs with less than 5% flicker
 - 14. Be DMX compatible in Color changing LED luminaires.
 - 15. At power levels between 100% to 1%, shall perform within the frequency (Hz) and Modulation (%) of IEEE STD 1789-2015 level.
- D. Emergency LED Drivers shall:
 - 1. Be UL 924 listed
 - 2. Operate LED luminaire at 10W minimum output for 90 minutes with efficacy equal to or greater than the normal power efficacy.
 - 3. Have high temperature nickel-cadmium battery. Field replaceable
 - 4. Be installed on top of or inside luminaires
 - 5. Have solid state charging
 - 6. Battery to be recharged within 24 h
 - 7. Remote battery test switch

2.5 LOW VOLTAGE TRANSFORMERS

- A. Transformers and power supplies shall be:
 - 1. Sized to compensate for voltage drop over indicated distances
 - 2. Locally fused
- B. Transformers shall have line voltage switch within reach.

- C. Provide adequate ventilation to meet code and manufacturers requirements.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Marking:

1. Voltage identification: Luminaires designed for voltages other than 110-125 volt circuits shall be clearly marked with rated voltage.
2. Markings must be clear and shall be located to be readily visible to service personnel but invisible from normal viewing angles when luminaires are in place.

B. Installation of Luminaires:

1. Housing, glassware, reflectors and refractors shall be clean and free of chips, cracks and scratches.
2. Install decorative luminaires, reflector cones, baffles, aperture plates, lenses, trims, and decorative elements of recessed luminaires after completion of ceiling tile, plastering, painting, and general cleanup is completed. Where luminaire location or construction does not permit sequential installation, all reflectors, lenses, flanges and other visible surfaces shall be carefully protected.
3. Light leaks between ceiling trim of recessed luminaires and ceiling are not allowed.
4. Locations
 - a. Install luminaires at locations and heights as indicated.
 - b. Do not scale electrical drawings for locations of luminaires.
 - c. Architectural reflected ceiling plans show locations of luminaires.
 - d. Where noted on the drawings, the exact location of luminaires shall be confirmed (in the field) with the Architect/Engineer prior to installation.
 - e. Where luminaires are to be concealed, or surface mounted in highly visible public spaces, a small sampling of luminaires shall be installed, adjusted and aimed for Architect/Engineer's review approval, prior to installing remaining luminaire of same type.
 - f. Mount all luminaires so as to maintain full range of motion.
 - g. Install luminaires plumb, square, and level with ceilings and walls.
 - h. Coordinate stem, rod, chain, or aircraft cable hanger lengths with job conditions. Provide extra length of adjustable supports where diffusers are mounted directly above light fixtures to facilitate air balancing efforts.
 - i. Industrial type luminaires in unfinished areas, which are near obstructions such as ducts and pipes, shall be:
 - 1) Suspended so that bottom of luminaire is no higher than bottom of obstruction
 - 2) Located at height of lowest luminaire
 - 3) Minimum height: 8'-0"
 - 4) Shall not be located until locations of obstructions are determined.
 - 5) Where a minimum height of 8'-0" is unachievable, wall mounted luminaires shall be utilized.
5. Support
 - a. Support surface mount luminaires from building structure.
 - b. Metal decking shall not be pierced for luminaire support.

- c. Provide luminaires and/or luminaire outlet boxes with hangers to support luminaire weight.
 - d. Troffers shall be held in place by support clips.
 - e. Provide plaster frames for recessed luminaires in plaster ceilings.
 - f. Rigid metallic pipe stems shall be utilized for the support of pendant mounted luminaires, unless otherwise noted.
 - g. Stem hangers shall be equipped with aligner box covers or canopies so that stems hang vertically, irrespective of the angle of the surface they are mounted from.
 - h. Wherever a luminaire or its hanger canopy is attached to a surface mounted outlet box, a finishing ring shall conceal the outlet box.
 - i. Yokes, brackets and supplementary supporting members needed to mount luminaires to suitable ceiling members shall be furnished and installed by Contractor. Verify mounting hardware required prior to installation.
 - j. Recessed luminaires shall be supported with 12 ga wire hangers, 2 per luminaire, at diagonally opposite corners.
 - k. Troffers and luminaires over 55 lbs, such as 4x4 ft shall be supported with 12 ga wire hangers, 4 per luminaire, 2 at 45 degree diagonals, and two perpendicular to structure. Wire hangers and attachment to structure shall be capable of supporting 4 times luminaires weight.
 - l. In areas with seismic requirements, suspended or pendant mounted luminaires shall be able to swing 45 degrees in any direction without hitting an obstruction. In the event hitting an obstruction is unavoidable, guy wires will be used to secure the luminaire in place.
 - m. Surface luminaires installed in grid ceilings shall be supported by independent support clips and 12 ga wire.
 - n. Exit signs installed in grid ceilings shall be supported by electrical box hanger and additional 12 ga wire installed from box to structure.
 - o. Support surface mounted luminaires greater than 2 ft in length at a minimum of each additional 2 ft, or as recommended by manufacturer.
 - p. Brace suspended luminaires installed near ducts or other constructions with solid pendants or threaded rods.
 - q. Rigidly align continuous rows of luminaires.
 - r. Luminaire types with remote mounted driver shall have:
 - 1) Proper support for driver weight.
 - 2) Mounting distance from remote driver to luminaire per manufacturer's recommendations.
6. Mounting and Enclosures
- a. Install flush mounted luminaires to eliminate light leakage.
 - b. For luminaires mounted adjacent to insulation, provide barrier to prevent insulation from coming in contact with luminaire, unless luminaire is approved for installation in contact with such insulation.
 - c. Provide approved fire rated enclosures around luminaires in fire rated ceilings.
7. Conduit and Wiring
- a. Wire for connections to modules and auxiliaries shall be suitable for temperature, current, and voltage conditions.
 - b. Recessed luminaires shall have final connections made with flexible metal conduit, not in excess of 72", with THHN conductors and green wire ground conductor.

- c. Conduit shall be hidden from normal view in all possible cases. In public areas where surface mounted conduit must be used, Contractor shall install conduit as unobtrusively as possible. Contractor shall obtain field approval by the Architect for all exposed conduit runs prior to rough in.
 - 8. In-Grade Luminaires:
 - a. Where installed in tree grates, furnish burial light lens and louver to tree grate manufacturer for coordination of opening.
 - b. Provide adequate drainage system per manufacturer's recommendations.
- C. Grounding:
 - 1. Ground luminaires and metal poles according to Division 26 Section "Grounding and Bonding for Electrical Systems".
 - 2. Poles:
 - a. Install 10 ft driven ground rod at each pole.
 - 3. Nonmetallic Poles:
 - a. Ground metallic components of lighting unit and foundations. Connect luminaires to grounding system with #10 AWG conductor.

3.2 SUBSTANTIAL COMPLETION

- A. Quality Control:
 - 1. At Date of Substantial Completion, replace LED modules/LED luminaires which are not operating properly.
 - 2. Protection wrapping on lensed or louvered luminaires shall be removed before installation of furniture, but after finish work is complete.
 - 3. Deliver spare equipment to Owner's representative.
- B. Tests:
 - 1. Give advance notice of dates and times for field tests.
 - 2. Provide instruments to make and record test results.
 - 3. Verify normal operation of each luminaire after luminaires have been installed and circuits have been energized.
 - 4. Verify operation of luminaires with lighting control system and daylight harvesting systems. Any dimmed fixtures shall exhibit no signs of flickering.
 - 5. Replace or repair malfunctioning luminaires and components, then retest. Repeat procedure until all units operate properly.
 - 6. Report results of tests.
- C. Adjusting and Cleaning:
 - 1. Clean luminaires of handling marks, dust and dirt.
 - 2. Cleaning and touch-up work shall be performed in accordance with luminaire manufacturer's recommendations.
 - 3. Damaged luminaires or components shall be replaced with new.
 - 4. Keep luminaires clean and protected for remainder of construction period.
 - 5. Verify orientation of directional luminaires prior to installation.
 - a. This includes wall washers, cove lighting, floodlights, exterior area lights and adjustable accent luminaires. Contractor shall provide electrician's services to aim, adjust, and focus luminaires, as required, at direction of Architect/Engineer. These electricians shall be available at times designated by Architect/Engineer and shall be provided at no extra charge

to Owner over base bid. Contractor shall provide equipment for luminaires' focus including ladders and mechanical lifting systems.

6. Program preset dimming system lighting levels.
7. Program ambient light sensors integral to luminaires for appropriate illumination levels as indicated in control narrative or in lighting control specifications.
8. Program occupancy sensors integral luminaires for appropriate time delay as indicated in control narrative or in lighting control specifications.
9. Exterior poles, bollards, bases and other exterior luminaires shall be painted to match factory color where finish has been damaged.
10. No light leaks shall be permitted at ceiling line from any visible part or joint.

D. Training

1. Contractor shall provide Owner with 1 complete hardcopy and (1) digital copy of Operations and Maintenance manuals.
 - a. All "Approved as Noted" comments shall be corrected/picked-up in this record manual set.
 - b. Each manual shall contain specific information pertaining to the equipment installed. Each manual shall contain at a minimum:
 - 1) Detailed as built shop drawings for all lighting equipment installed.
 - 2) Manufacturer's product cut sheets for all equipment installed keyed by type as to as built drawings.
 - 3) Manufacturer's complete installation instructions for all equipment installed keyed by type to as built drawings.
 - 4) Equipment maintenance requirements and schedules.
 - 5) Equipment manufacturer contacts.
 - 6) Equipment manufacturer warranties.
 - a) Luminaires
 - b) Drivers
 - c) LED modules
2. Contractor shall provide qualified personnel onsite to provide a minimum of one day of training to Owner's representatives.
3. This training shall cover:
 - a. Luminaire use and maintenance
 - b. Architectural lighting system use and maintenance

END OF SECTION